

Investments & Portfolio Management

Concept-First Notes + Exam Q&A + MCQ Bank + Mock Test — CA Intermediate

One complete file: concepts, worked Q&A, revision cheat-sheet, MCQs & a full mock test

Contents

1. Chapter 01 — Introduction to Investments and the Risk-Return Trade-off

↳ Q&A: Chapter 01 — Introduction to Investments and the Risk-Return Trade-off

3. Chapter 02 — Measuring Risk and Return

↳ Q&A: Chapter 02 — Measuring Risk and Return

5. Chapter 03 — Modern Portfolio Theory (Markowitz)

↳ Q&A: Chapter 03 — Modern Portfolio Theory (Markowitz)

7. Chapter 04 — The Efficient Frontier and Diversification

↳ Q&A: Chapter 04 — The Efficient Frontier and Diversification

9. Chapter 05 — The Capital Asset Pricing Model (CAPM)

↳ Q&A: Chapter 05 — The Capital Asset Pricing Model (CAPM)

11. Chapter 06 — APT and Multifactor Models

↳ Q&A: Chapter 06 — APT and Multifactor Models

13. Chapter 07 — Asset Allocation

↳ Q&A: Chapter 07 — Asset Allocation

15. Chapter 08 — The Portfolio Construction Process

↳ Q&A: Chapter 08 — The Portfolio Construction Process

17. Chapter 09 — Equity Portfolio Management

↳ Q&A: Chapter 09 — Equity Portfolio Management

19. Chapter 10 — Fixed Income Portfolio Management

↳ Q&A: Chapter 10 — Fixed Income Portfolio Management

21. Chapter 11 — Performance Measurement and Attribution

↳ Q&A: Chapter 11 — Performance Measurement and Attribution

23. Chapter 12 — Market Efficiency and the Active-Passive Debate

↳ Q&A: Chapter 12 — Market Efficiency and the Active-Passive Debate

25. Chapter 13 — Behavioral Finance

↳ Q&A: Chapter 13 — Behavioral Finance

27. Chapter 14 — The Investment Policy Statement

↳ Q&A: Chapter 14 — The Investment Policy Statement

29. Chapter 15 — Alternatives in Portfolios

↳ Q&A: Chapter 15 — Alternatives in Portfolios

31. Chapter 16 — Factor Investing and Smart Beta

↳ Q&A: Chapter 16 — Factor Investing and Smart Beta

33. Chapter 17 — Passive vs Active Management

↳ Q&A: Chapter 17 — Passive vs Active Management

35. **Chapter 18 — Portfolio Risk Management**

↳ Q&A: Chapter 18 — Portfolio Risk Management

Chapter 01 — Introduction to Investments and the Risk-Return Trade-off

1. The Problem / Need

Every rupee you hold today faces a quiet erosion. If inflation runs at 6% a year, ₹100 kept idle in a drawer buys only ₹94 worth of goods next year, ₹88 the year after, and so on. Money that does not work loses purchasing power. So the first problem investment solves is **preservation** — staying ahead of inflation so that future consumption is not silently taxed away.

But preservation alone is not the real prize. Most people have a mismatch between **when they earn** and **when they need to spend**. A 25-year-old software engineer earns more than she needs today, but at 60 she will earn nothing and still need income for perhaps 25 more years. A business earns cash in bursts but must fund a factory that takes a decade to pay back. The deeper problem is **inter-temporal transfer of consumption** — moving surplus purchasing power from periods of plenty to periods of need, and being *rewarded* for the wait.

That reward is not free. To get it, the saver must part with money now and accept **uncertainty** about what comes back. The whole discipline of investments exists to answer one question rigorously:

Given that higher expected reward can only be bought by accepting more uncertainty, how much uncertainty should I take, in what form, and what is a fair reward for it?

This is not a philosophical question — it is a numerical, model-driven one. Interviewers at asset management companies (AMCs), equity research desks, and portfolio roles probe exactly this: *Do you understand that return is compensation for risk, and can you quantify both?* This chapter builds that foundation.

2. The Core Idea

An investment is the current commitment of money (or other resources) for a period of time, in the expectation of receiving future payments that compensate the investor for (a) the time the funds are committed, (b) the expected rate of inflation over that time, and (c) the uncertainty of the future payments.

That single definition (the classic Reilly–Brown formulation) already contains the risk-return trade-off. The required return has three layers:

$$\text{Required Return} = \underbrace{\text{Real risk-free rate}}_{\text{(a) time value}} + \underbrace{\text{Inflation premium}}_{\text{(b) purchasing-power protection}} + \underbrace{\text{Risk premium}}_{\text{(c) uncertainty compensation}}$$

The first two layers together form the **nominal risk-free rate** (roughly what a government treasury bill pays). The third layer — the **risk premium** — is where the entire craft of investing lives. Everything in portfolio theory is ultimately a debate about how big that premium *should* be and how it should be measured.

The core idea of the trade-off is stark and non-negotiable in efficient markets:

You cannot earn a higher expected return without accepting more risk. Anyone promising high return with low risk is either mistaken, lucky, or lying.

3. Why / How It Works

Why must reward scale with risk? Because of **investor preferences and market clearing**.

Most investors are **risk-averse**: given two assets with the *same* expected return, they prefer the one with less uncertainty. Equivalently, to persuade a risk-averse investor to hold a *riskier* asset, you must offer a *higher* expected return. This is not an assumption pulled from thin air — it follows from **diminishing marginal utility of wealth**. The pain of losing ₹1 lakh is felt more sharply than the pleasure of gaining ₹1 lakh, because each additional rupee of wealth adds a little less happiness than the one before.

Now let markets clear. Suppose a risky stock and a safe bond offered the same expected return. Risk-averse investors would all sell the stock and buy the bond. The stock's price would fall until its *expected* future return (future payoff ÷ today's lower price) rose enough to compensate holders for its risk. Prices adjust until every asset offers a risk premium proportional to the risk that *cannot be diversified away*. That equilibrium is exactly what the Capital Asset Pricing Model formalises (Chapter on CAPM), but the intuition is here: **the risk premium is the price the market sets to persuade someone to bear risk nobody wants for free**.

A crucial refinement — and a favourite interview trap — is *which* risk gets rewarded. Only risk you **cannot escape** earns a premium. Risk that can be **diversified away** by combining many assets earns nothing, because you could have eliminated it for free. Hence:

- **Systematic (market) risk** — recessions, interest-rate shocks, wars — hits everything at once and cannot be diversified. **It is rewarded.**
- **Unsystematic (specific) risk** — a factory fire, a CEO scandal, a failed product — is idiosyncratic and cancels out across a large portfolio. **It is not rewarded.**

This distinction is the reason diversification is called "the only free lunch in finance": it removes risk *without* removing expected return.

4. Full Content — Definitions, Frameworks, Formulas

4.1 Investment vs Speculation vs Gambling

These three are often conflated in casual speech; being precise about them signals maturity in an interview.

Dimension	Investment	Speculation	Gambling
Basis of decision	Fundamental analysis, expected cash flows, intrinsic value	Market timing, price momentum, information edge	Chance; no underlying economic value
Time horizon	Medium to long (years)	Short (days to months)	Instant / event-based
Risk taken	Calculated, commensurate with return	High, but analysed and deliberate	Artificially <i>created</i> risk, not economically necessary
Expected value	Positive (risk premium earned)	Positive if edge is real; zero otherwise	Negative (house edge / zero-sum)
Return source	Income + capital appreciation from real economic value	Price changes	Redistribution among players
Example	Buying an index fund for retirement	Buying a stock ahead of an expected earnings surprise	Betting on a coin toss / roulette

Key nuances interviewers look for:

- **Speculation is not a dirty word.** A speculator who provides liquidity and takes on risk others avoid performs a genuine economic function. The line between "investor" and "speculator" is really a spectrum of *holding period and analytical basis*, not good vs evil.
- **Gambling creates risk where none needed to exist.** Investment and speculation deal with *pre-existing* economic risk (a business's fortunes); gambling *manufactures* risk purely for the wager. Its expected value is typically negative — the house always keeps a cut.
- **Benjamin Graham's test:** "An investment operation is one which, upon thorough analysis, promises safety of principal and an adequate return. Operations not meeting these requirements are speculative."

4.2 The Investment Management Process

Portfolio management is a **continuous, cyclical process**, not a one-shot decision. The standard five-step framework:

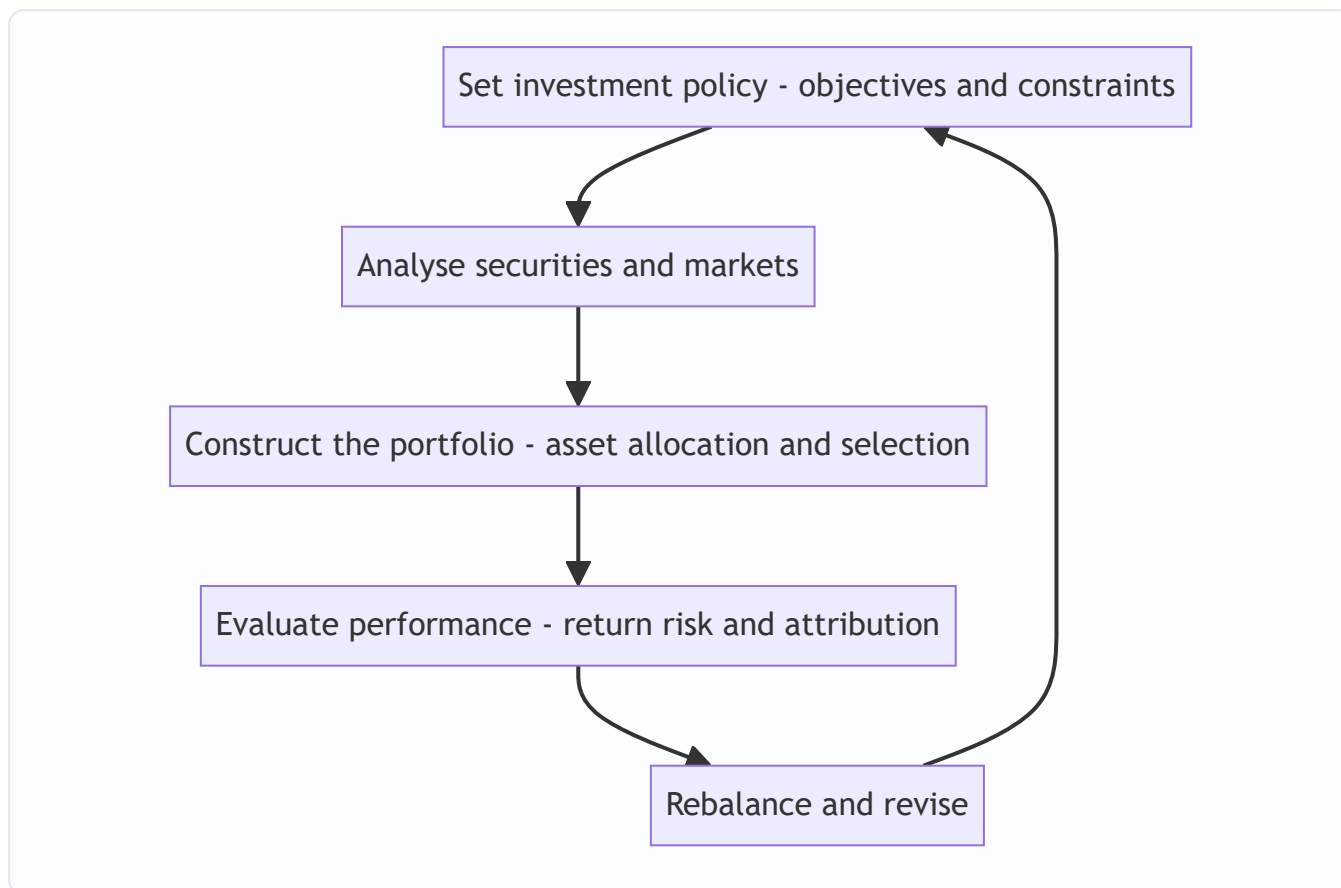


Figure 4.2 — The investment management cycle is a feedback loop; performance evaluation feeds back into policy revision.

1. **Setting the Investment Policy (the IPS).** Define objectives (return and risk) and constraints (liquidity, time horizon, taxes, legal/regulatory, unique circumstances). This is captured in an **Investment Policy Statement (IPS)** — the governing document.
2. **Security and market analysis.** Identify mispriced securities (fundamental / technical analysis) and form capital-market expectations (expected returns, volatilities, correlations across asset classes).
3. **Portfolio construction.** Decide the **strategic asset allocation** (the long-run mix across equities, bonds, cash, alternatives) and then **security selection** within each class. Empirically, asset allocation explains the large majority of the *variability* of portfolio returns over time.
4. **Performance evaluation.** Measure realised return and risk; compare against a benchmark; decompose results via **attribution** (how much came from allocation vs selection vs timing).
5. **Rebalancing / revision.** As prices move and circumstances change, drift the portfolio back to target weights and update the IPS.

4.3 Measuring Return

Holding Period Return (HPR) — total return over one period:

$$HPR = \frac{(P_1 - P_0) + D_1}{P_0} = \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{capital gain yield}} + \underbrace{\frac{D_1}{P_0}}_{\text{income yield}}$$

where P_0 = price at start, P_1 = price at end, D_1 = income (dividend/coupon) received.

Expected return of a single asset over discrete scenarios:

$$E(R) = \sum_{i=1}^n p_i R_i$$

where p_i is the probability of state i and R_i the return in that state.

Arithmetic vs Geometric mean (multi-period). For realised returns $R_1, \dots, R_T$:

$$\bar{R}_{\text{arith}} = \frac{1}{T} \sum_{t=1}^T R_t, \quad \bar{R}_{\text{geom}} = \left(\prod_{t=1}^T (1 + R_t) \right)^{1/T} - 1$$

The **geometric mean** is the true *compound* growth rate (what your money actually did). The **arithmetic mean** is the best *estimate of next period's* return. Geometric $\leq$ arithmetic always, with the gap widening as volatility rises (approximately $\bar{R}_{\text{geom}} \approx \bar{R}_{\text{arith}} - \frac{1}{2} \sigma^2$).

4.4 Measuring Risk

Risk is the **dispersion of possible outcomes around the expected value**. The standard statistical measures:

Variance and **standard deviation** (using probabilities):

$$\sigma^2 = \sum_{i=1}^n p_i [R_i - E(R)]^2, \quad \sigma = \sqrt{\sigma^2}$$

Standard deviation σ (volatility) is in the same units as return (%), which is why it is the workhorse risk measure.

Coefficient of Variation (CV) — risk *per unit* of return, for comparing assets of different scale:

$$CV = \frac{\sigma}{E(R)}$$

Lower CV = better risk-adjusted profile.

Other risk lenses used in practice (deeper chapters expand these):

- **Beta (β)** — sensitivity to market movements; measures *systematic* risk only.
- **Downside / semi-deviation** — dispersion of only the *bad* outcomes (many argue volatility unfairly penalises upside surprises).
- **Value at Risk (VaR)** — the loss threshold not exceeded with a given confidence over a horizon (e.g. "5% chance of losing more than ₹10 lakh in a day").
- **Maximum drawdown** — the largest peak-to-trough fall; captures the pain of the worst stretch.

4.5 The Risk-Return Trade-off, Formalised

Plotting expected return against risk gives the foundational picture of finance:



Figure 4.5 — The risk-return spectrum: moving right, both expected return and volatility rise together. The ordering is the whole point.

The trade-off is captured cleanly by the **Sharpe ratio**, the single most-asked risk-adjusted performance number:

$$\text{Sharpe Ratio} = \frac{E(R_p) - R_f}{\sigma_p}$$

It measures **excess return earned per unit of total risk taken**. A portfolio earning 12% with 15% volatility when T-bills pay 5% has a Sharpe of $(12-5)/15 = 0.47$. Higher is better; it is how you compare a "high return, high risk" fund against a "modest return, low risk" one on a level field.

4.6 Types of Investors — Objectives and Constraints

Every IPS is built on two objectives and (classically) five or six constraints. The standard mnemonic frames objectives as **R-R** (Return, Risk) and constraints as **L-L-T-T-U**: **L**iquidity, **L**egal/regulatory, **T**ime horizon, **T**axes, **U**nique circumstances.

Objectives

Objective	Question it answers
Return requirement	What return does the investor <i>need</i> to meet goals? (income vs growth vs total return)
Risk tolerance	<i>Ability</i> (financial capacity to absorb losses) + <i>Willingness</i> (psychological comfort). Use the lower of the two.

Constraints

Constraint	Meaning	Example
Liquidity	Need for ready cash without loss of value	Pension paying monthly benefits needs liquidity
Time horizon	How long until funds are needed	A 30-year-old's retirement corpus has a 30-yr horizon
Taxes	Tax treatment of income and gains	Taxable individual vs tax-exempt endowment
Legal / regulatory	Rules governing the investor	Insurers face solvency limits on equity holdings
Unique circumstances	Anything else (ethics, concentration, ESG)	Investor barring tobacco/alcohol stocks

How investor types differ

Investor type	Return objective	Risk tolerance	Time horizon	Liquidity	Tax status
Young individual	Growth (capital appreciation)	High (long horizon, human capital)	Long	Low	Taxable
Retiree	Income + capital preservation	Low	Short–medium	High	Taxable
Pension fund (defined benefit)	Match liabilities; total return	Moderate	Very long	Depends on maturity of workforce	Tax-exempt
Endowment / foundation	Preserve real value + spending rate	Moderate–high	Perpetual	Low–moderate	Tax-exempt
Life insurer	Spread over guaranteed rate; ALM	Low	Long	Moderate (claims)	Taxed
Bank	Positive net interest margin	Low	Short	High (deposits callable)	Taxed
Mutual fund	Beat/track stated benchmark	Per mandate	Per mandate	High (daily redemptions)	Pass-through

Two ideas worth stating explicitly because interviewers test them:

- **Human capital shapes financial risk-taking.** A young person's future salary is a large, bond-like asset. With so much "safe" human capital, she can afford more equity risk in her financial portfolio. As human capital depletes with age, the financial portfolio should tilt toward bonds — the intuition behind lifecycle/target-date funds.
- **Ability vs willingness conflict.** If a wealthy investor (high ability) is terrified of losses (low willingness), the adviser generally respects the *lower* one but educates the client. Never assume high wealth equals high risk tolerance.

4.7 The Risk-Return Spectrum of Asset Classes

Long-run history (developed-market data, illustrative nominal figures) roughly orders the major classes:

Asset class	Typical role	Long-run real return	Volatility (σ)	Primary risks
Cash / T-bills	Liquidity, safety	~0–1%	~1%	Inflation (purchasing-power) risk
Government bonds	Capital preservation, income	~1–2%	~4–7%	Interest-rate, inflation
Investment-grade corporate bonds	Income	~2–3%	~6–9%	Credit + interest-rate
High-yield bonds	Income + some growth	~3–5%	~10–12%	Default/credit, illiquidity
Large-cap equity	Growth	~5–7%	~15–20%	Market, business cycle
Small-cap equity	Growth	~6–8%	~20–25%	Market + liquidity + size
Emerging-market equity	Growth	~6–9%	~22–30%	Political, currency, market
Real estate	Income + inflation hedge	~3–5%	~10–15%	Illiquidity, leverage, local
Alternatives (PE/VC/hedge/crypto)	Diversification, high return	wide range	very high	Illiquidity, model, extreme

The ordering matters more than the exact numbers (which vary by period and source). The takeaways:

- **Cash is not "riskless" in real terms** — it reliably loses to inflation. Its risk is subtle but real.
- Bonds trade one risk for another: safer on *default* than equity, but exposed to *interest-rate* moves.
- The **equity risk premium** — the long-run excess of stocks over bills, historically ~4–6% — is the single most important number in investing, and the compensation for enduring equity's gut-wrenching volatility.

5. Worked Examples

Example 1 — Decomposing return, and expected return under uncertainty

Part A — Holding Period Return. You buy a share of Infosys-like Co. at ₹1,400. A year later it trades at ₹1,540 and paid a ₹35 dividend.

$$\text{HPR} = \frac{(1540 - 1400) + 35}{1400} = \frac{140 + 35}{1400} = \frac{175}{1400} = 0.125 = 12.5\%$$

Split: capital gain yield $= 140/1400 = 10.0\%$; income yield $= 35/1400 = 2.5\%$. Check: $10.0\% + 2.5\% = 12.5\%$. ✓

Part B — Expected return and risk under scenarios. Next year's return depends on the economy:

State	Probability p_i	Return R_i
Boom	0.25	+30%
Normal	0.50	+12%
Recession	0.25	−10%

Expected return:

$$E(R) = 0.25(30) + 0.50(12) + 0.25(-10) = 7.5 + 6.0 - 2.5 = 11.0\%$$

Variance:

$$\sigma^2 = 0.25(30-11)^2 + 0.50(12-11)^2 + 0.25(-10-11)^2 = 0.25(361) + 0.50(1) + 0.25(441) = 90.25 + 0.50 + 110.25 = 201.0$$

$$\text{Standard deviation: } \sigma = \sqrt{201.0} = 14.18\%$$

Coefficient of variation: $CV = 14.18 / 11.0 = 1.29$. So this asset carries 1.29 units of risk (%) per unit of expected return (%).

Example 2 — Comparing two assets with the Sharpe ratio

The risk-free rate is 5%. You must choose between:

Fund	$E(R)$	σ
A — balanced fund	10%	8%
B — aggressive equity fund	16%	20%

At first glance B "returns more." But adjust for risk:

$$\text{Sharpe}_A = \frac{10 - 5}{8} = \frac{5}{8} = 0.625 \quad \text{Sharpe}_B = \frac{16 - 5}{20} = \frac{11}{20} = 0.55$$

Fund A is superior on a risk-adjusted basis ($0.625 > 0.55$): it delivers more excess return per unit of risk. This is the exact reasoning an AMC analyst uses — raw returns flatter risky funds; the Sharpe ratio levels the field. (Which fund a *client* should hold still depends on their risk tolerance, but A is the more *efficient* engine.)

Reconciliation check: if you could borrow/lend at 5% and lever Fund A up to B's 20% volatility (a factor of $20/8 = 2.5$), A's excess return scales to $\$2.5 \times 5\% = 12.5\%$, giving a total return of $5\% + 12.5\% = 17.5\%$ — *more* than B's 16% at the same risk. That confirms A dominates. ✓

Example 3 — Arithmetic vs geometric mean (why the distinction bites)

A fund returns **+50% in year 1** and **−40% in year 2**.

Arithmetic mean: $(50 + (-40))/2 = +5\%$ — sounds like a gain.

Geometric (actual) mean:

$$\bar{R}_{\text{geom}} = \sqrt{(1+0.50)(1-0.40)} - 1 = \sqrt{1.50 \times 0.60} - 1 = \sqrt{0.90} - 1 = 0.9487 - 1 = -5.13\%$$

Reconciliation: ₹100 grows to ₹150, then falls 40% to ₹90 — a **loss** of ₹10 over two years, i.e. $\sqrt{0.90} - 1 = -5.13\%$ per year compounded. ✓ The arithmetic mean (+5%) overstated reality; the geometric mean

(−5.13%) tells the truth about wealth. The gap ($\approx 10\%$ points) is large precisely because volatility (a 90-point swing) is large — consistent with $\bar{R}_{\text{geom}} \approx \bar{R}_{\text{arith}} - \frac{1}{2}\sigma^2$.

6. Connections

- **Portfolio theory (Markowitz):** This chapter's single-asset risk (σ) becomes *portfolio* risk once we add **covariance** — the insight that combining imperfectly-correlated assets lowers risk without lowering expected return. The risk-return trade-off graduates into the **efficient frontier**.
 - **CAPM and the SML:** The idea that only *systematic* risk is rewarded becomes the formal equation $E(R_i) = R_f + \beta_i[E(R_m) - R_f]$. Beta replaces σ as the priced risk measure.
 - **Fixed income:** Bond pricing is the pure study of the "time value + inflation" layers of required return, plus credit risk in the premium.
 - **Behavioural finance:** Risk aversion and the ability/willingness split connect to loss aversion and prospect theory — why real investors deviate from the rational model.
 - **Corporate finance:** A firm's cost of equity is an investor's required return. The risk premium you demand as an investor is the hurdle rate the CFO must beat.
-

7. Key Terms

- **Investment** — current commitment of resources for expected future compensation for time, inflation, and uncertainty.
 - **Risk premium** — expected return in excess of the risk-free rate, compensating for uncertainty.
 - **Risk aversion** — preference for certainty; demanding higher expected return for higher risk.
 - **Systematic (market) risk** — undiversifiable, economy-wide risk; the only risk rewarded with a premium.
 - **Unsystematic (specific) risk** — asset-specific, diversifiable risk; earns no premium.
 - **Holding Period Return (HPR)** — total return (income + capital gain) over one period.
 - **Standard deviation (σ)** — volatility; dispersion of returns around the mean.
 - **Sharpe ratio** — excess return per unit of total risk.
 - **Coefficient of variation (CV)** — risk per unit of return.
 - **Investment Policy Statement (IPS)** — governing document listing objectives and constraints.
 - **Risk tolerance** — ability (capacity) plus willingness (psychology) to bear risk.
 - **Equity risk premium** — long-run excess return of stocks over risk-free assets.
 - **Human capital** — present value of future earnings; a bond-like asset shaping financial risk capacity.
-

8. Common Confusions

1. **"Risk means losing money."** No — in finance risk is *dispersion / uncertainty of outcomes*, measured by σ . An asset can be volatile yet have positive expected return. Upside surprises count as "risk" too under the σ definition (a valid critique that motivates downside measures).
2. **"Higher risk guarantees higher return."** Risk buys higher *expected* return, not *realised* return. If risk always paid off, it wouldn't be risk. Realised outcomes can and do disappoint.

3. **"Cash is risk-free."** Only in *nominal* terms. In *real* terms, cash reliably loses to inflation. Every asset trades one risk for another.
 4. **"Diversification always reduces return."** It reduces *unsystematic* risk without touching expected return — the free lunch. It does not eliminate systematic risk.
 5. **"Arithmetic average = what I earned."** Your compounded wealth follows the *geometric* mean, always $\leq$ arithmetic. Confusing them overstates performance (Example 3).
 6. **"High net worth = high risk tolerance."** Ability $\neq$ willingness. Use the *lower* of the two.
 7. **"Speculation is just gambling."** Speculation analyses *pre-existing* economic risk and can have positive expected value; gambling *manufactures* risk with negative expected value.
 8. **"Beta and standard deviation are the same risk."** σ is *total* risk; β is only the *systematic* portion. A stock can have high σ but low β (lots of diversifiable noise).
-

9. First-Principles Recap

Strip everything back and the logic chain is:

1. Money now is worth more than money later (time value) and inflation erodes idle cash — so we invest.
2. Future payoffs are uncertain, and people dislike uncertainty (risk aversion from diminishing marginal utility).
3. Therefore, to hold risky assets, investors *demand* extra expected return — a **risk premium**.
4. Markets clear by adjusting prices until each asset's premium matches the *undiversifiable* risk it carries. Diversifiable risk earns nothing because it can be removed for free.
5. Hence **expected return rises with (systematic) risk** — the trade-off — and this ordering is visible across the asset-class spectrum from T-bills to venture capital.
6. We quantify return with HPR/expected value, risk with σ , and the trade-off with the Sharpe ratio.
7. *How much* risk any given investor should take is not universal — it flows from their **objectives** (return need, risk tolerance) and **constraints** (liquidity, horizon, taxes, legal, unique), written into an **IPS**.

Everything later in the guide — portfolios, CAPM, factor models, valuation, fixed income, derivatives — is an elaboration of steps 3 to 6.

10. Quick-Reference / Interview Points

Formulas to have cold:

Concept	Formula
Required return	$R_f + \text{inflation premium} + \text{risk premium}$ (with R_f real; or nominal R_f + risk premium)
Holding period return	$HPR = \frac{(P_1 - P_0) + D_1}{P_0}$
Expected return (scenarios)	$E(R) = \sum p_i R_i$
Variance	$\sigma^2 = \sum p_i [R_i - E(R)]^2$
Geometric mean	$\left[\prod (1 + R_t) \right]^{1/T} - 1$
Coefficient of variation	$CV = \sigma / E(R)$
Sharpe ratio	$[E(R_p) - R_f] / \sigma_p$
Geo vs arith approx	$\bar{R}_{geom} \approx \bar{R}_{arith} - \frac{1}{2} \sigma^2$

What interviewers actually ask:

- "What's the difference between investing and speculating?" → Analytical basis, horizon, and expected value; cite Graham's definition.
- "Why is there a risk-return trade-off at all?" → Risk aversion + market clearing; only undiversifiable risk is priced.
- "A fund returned 20% last year. Is it good?" → "Compared to what risk and what benchmark? Give me its volatility and I'll give you the Sharpe ratio." (Never judge return without risk.)
- "Two funds: 10% at 8% vol, or 16% at 20% vol — which is better?" → Compute Sharpe (0.625 vs 0.55); the lower-return fund is more efficient. (Example 2.)
- "Is cash risk-free?" → Nominally yes, really no — inflation risk.
- "Walk me through building a portfolio for a 60-year-old retiree vs a 28-year-old." → Objectives + constraints (LLTTU); horizon and human capital drive the equity/bond split.
- "Why doesn't diversification eliminate all risk?" → It kills unsystematic risk; systematic (market) risk remains and is what earns the premium.
- "Arithmetic or geometric mean for reporting past performance?" → Geometric (compounded wealth); arithmetic for forecasting next period.

One-liners to sound sharp: - "Return is the compensation; risk is the price you pay for it." - "Diversification is the only free lunch in finance." - "The market only pays you for risk you *can't* avoid." - "Judge no return without its risk, and no risk without its horizon."

Q&A — Introduction to Investments and the Risk-Return Trade-off

A practice bank for Chapter 01. Every question is followed by a full answer. Formulas match the chapter; numerical answers are self-verified.

Section A — Concept-Check Questions

A1. State the classic (Reilly–Brown) definition of an investment and name the three things the future payments must compensate for.

An investment is the current commitment of money (or other resources) for a period of time in the expectation of future payments that compensate the investor for (a) the *time* the funds are committed (time value), (b) the *expected inflation* over that time (purchasing-power protection), and (c) the *uncertainty* of the future payments (risk). The first two together form the nominal risk-free rate; the third is the risk premium.

A2. Break the required return into its layers and identify which layer is "where the craft of investing lives".

Required Return = Real risk-free rate + Inflation premium + Risk premium. The real risk-free rate compensates for pure time; the inflation premium protects purchasing power; together they equal the nominal risk-free rate (roughly a T-bill yield). The **risk premium** — compensation for uncertainty — is where the entire craft lives, because portfolio theory is fundamentally a debate about how big that premium should be and how it should be measured.

A3. Why must expected reward scale with risk? Give the two-part economic argument.

(1) *Risk aversion*: most investors, faced with two assets of equal expected return, prefer the less uncertain one — a consequence of the diminishing marginal utility of wealth (losing ₹1 lakh hurts more than gaining ₹1 lakh pleases). (2) *Market clearing*: if a risky and a safe asset offered the same expected return, risk-averse investors would dump the risky one; its price falls until its expected return rises enough to compensate holders. Prices adjust until every asset offers a premium proportional to its undiversifiable risk.

A4. Distinguish systematic from unsystematic risk and state which one is rewarded, and why.

Systematic (market) risk — recessions, rate shocks, wars — hits all assets at once and cannot be diversified away; it is **rewarded** with a premium. Unsystematic (specific) risk — a factory fire, a CEO scandal — is idiosyncratic and cancels out across a large portfolio; it earns **no** premium, because an investor could have eliminated it for free through diversification. The market never pays you for risk you could have avoided at no cost.

A5. Why is diversification called "the only free lunch in finance"?

Because combining many imperfectly-correlated assets removes unsystematic risk *without* reducing expected return. You lower total volatility for free — no return is sacrificed. It is a genuine improvement in the risk-return profile, unlike almost everything else in finance where more return demands more risk.

A6. Distinguish investment, speculation, and gambling.

Investment rests on fundamental analysis and intrinsic value over a medium-to-long horizon, taking calculated risk for a positive expected value (the risk premium). Speculation rests on

timing/momentum/information edge over a short horizon; its expected value is positive only if the edge is real. Gambling rests on chance, has an instant horizon, and *manufactures* risk that need not exist, with a typically negative expected value (the house edge). Key nuance: investment and speculation deal with *pre-existing* economic risk; gambling creates risk purely for the wager.

A7. State Benjamin Graham's test for distinguishing an investment from a speculation.

"An investment operation is one which, upon thorough analysis, promises safety of principal and an adequate return. Operations not meeting these requirements are speculative." The two pillars are *thorough analysis* and *safety of principal plus adequate return*.

A8. Is cash risk-free? Explain the subtlety.

Only in *nominal* terms. In *real* terms cash reliably loses purchasing power to inflation, so its risk is subtle but real — inflation (purchasing-power) risk. Every asset trades one risk for another; cash minimises price volatility but maximises exposure to inflation erosion.

A9. List the five steps of the investment management process and explain why it is a loop.

(1) Set the investment policy (the IPS — objectives and constraints); (2) analyse securities and markets; (3) construct the portfolio (asset allocation, then selection); (4) evaluate performance (return, risk, attribution); (5) rebalance/revise. It is a continuous feedback loop because performance evaluation feeds back into policy revision — prices move, circumstances change, so the cycle returns to step 1.

A10. What are the two objectives and the classic constraints in an IPS? Give the mnemonic.

The two objectives are Return requirement and Risk tolerance (mnemonic R-R). The constraints are Liquidity, Legal/regulatory, Time horizon, Taxes, and Unique circumstances (mnemonic L-L-T-T-U).

A11. Risk tolerance has two components. Name them and state the rule when they conflict.

Ability (the financial capacity to absorb losses) and *Willingness* (psychological comfort with losses). When they conflict, the adviser generally respects the **lower** of the two, while educating the client. High net worth (high ability) does not imply high willingness.

A12. What is human capital and how does it shape a young investor's asset mix?

Human capital is the present value of future earnings. For a young person it is large and bond-like (a relatively safe stream of future salary). Because so much of her total wealth is already "safe", she can afford more equity risk in her financial portfolio. As human capital depletes with age, the financial portfolio should tilt toward bonds — the logic behind lifecycle/target-date funds.

A13. Explain the difference between the arithmetic and geometric mean and when to use each.

The arithmetic mean is the simple average of periodic returns; it is the best estimate of *next period's* return. The geometric mean is the compound growth rate — what your money actually did over the whole span. Geometric $\leq$ arithmetic always, with the gap widening as volatility rises ($\bar{R}_{geom} \approx \bar{R}_{arith} - \frac{1}{2}\sigma^2$). Use geometric to report realised past performance (compounded wealth); use arithmetic to forecast a single future period.

Section B — Numerical Problems (Full Step-by-Step Solutions)

B1. Holding Period Return and its decomposition. You buy a share at ₹800. A year later it trades at ₹880 and paid a ₹24 dividend. Compute the HPR and split it into capital-gain yield and income yield.

Solution. $HPR = \frac{(P_1 - P_0) + D_1}{P_0} = \frac{(880 - 800) + 24}{800} = \frac{80 + 24}{800} = \frac{104}{800} = 0.13 = 13.0\%$ Capital-gain yield = $80/800 = 10.0\%$; income yield = $24/800 = 3.0\%$.
Reconciliation: $10.0\% + 3.0\% = 13.0\%$. ✓

B2. Expected return, variance, standard deviation, and CV under scenarios. A stock's next-year return depends on the economy:

State	Probability	Return
Boom	0.30	+25%
Normal	0.50	+10%
Recession	0.20	-15%

Compute $E(R)$, σ^2 , σ , and the coefficient of variation.

Solution. $E(R) = 0.30(25) + 0.50(10) + 0.20(-15) = 7.5 + 5.0 - 3.0 = 9.5\%$ Deviations from mean: Boom $25 - 9.5 = 15.5$; Normal $10 - 9.5 = 0.5$; Recession $-15 - 9.5 = -24.5$. $\sigma^2 = 0.30(15.5)^2 + 0.50(0.5)^2 + 0.20(-24.5)^2 = 0.30(240.25) + 0.50(0.25) + 0.20(600.25) = 72.075 + 0.125 + 120.05 = 192.25$
 $\sigma = \sqrt{192.25} = 13.87\%$ $CV = \frac{\sigma}{E(R)} = \frac{13.87}{9.5} = 1.46$ So the stock carries about 1.46 units of risk per unit of expected return. *Reconciliation:* probabilities sum to $0.30 + 0.50 + 0.20 = 1.00$. ✓ $\sqrt{192.25} = 13.87$ since $13.87^2 = 192.4$ (rounding). ✓

B3. Comparing two funds with the Sharpe ratio. The risk-free rate is 6%. Fund X returns 11% with $\sigma = 10\%$; Fund Y returns 18% with $\sigma = 22\%$. Which is more efficient?

Solution. $\text{Sharpe}_X = \frac{11 - 6}{10} = \frac{5}{10} = 0.50$ $\text{Sharpe}_Y = \frac{18 - 6}{22} = \frac{12}{22} = 0.545$ Fund Y has the higher Sharpe ratio ($0.545 > 0.50$), so **Y** delivers more excess return per unit of total risk and is the more efficient engine. *Reconciliation via leverage:* lever X up to Y's 22% volatility, a factor of $22/10 = 2.2$. X's excess return scales to $2.2 \times 5\% = 11\%$, giving a total of $6\% + 11\% = 17\%$ — less than Y's 18% at the same risk. This confirms Y dominates X here. ✓ (Which fund a given client holds still depends on risk tolerance; Y is merely the more efficient engine.)

B4. Arithmetic vs geometric mean. A fund returns +40% in year 1 and -30% in year 2. Compute both means and reconcile against actual wealth.

Solution. $\bar{R}_{arith} = \frac{40 + (-30)}{2} = \frac{10}{2} = +5\%$ $\bar{R}_{geom} = \sqrt{(1+0.40)(1-0.30)} - 1 = \sqrt{1.40 \times 0.70} - 1 = \sqrt{0.98} - 1 = 0.98995 - 1 = -1.01\%$ *Reconciliation:* ₹100 → ₹140 (up 40%) → ₹140 × 0.70 = ₹98 (down 30%). Ending wealth ₹98 is a loss, so the true compound rate must be negative: $\sqrt{0.98} - 1 = -1.01\%$ per year. ✓ The arithmetic mean (+5%) overstates reality; the geometric mean (-1.01%) tells the truth about wealth. The gap arises because volatility is high.

B5. Required return from its layers. The real risk-free rate is 2%, expected inflation is 5%, and the equity risk premium for a stock is 6%. What nominal return should the investor require, using both the simple (additive) and the exact (multiplicative) Fisher approach for the risk-free portion?

Solution. Simple/additive: Required return = $2\% + 5\% + 6\% = 13\%$. Exact nominal risk-free rate via Fisher: $(1 + 0.02)(1 + 0.05) - 1 = 1.071 - 1 = 7.1\%$, then add the 6% premium → $7.1\% + 6\% = 13.1\%$. *Reconciliation:* the additive approximation (7% nominal risk-free) understates the exact figure (7.1%) by the cross-term $0.02 \times 0.05 = 0.1\%$. The two agree to first order. ✓

B6. Two-scenario probability weighting (breakeven check). A ₹1,000 investment will be worth ₹1,300 with probability 0.6 or ₹700 with probability 0.4 in one year. Find the expected value, the expected return, and state whether it beats a 6% risk-free alternative.

Solution. $E(\text{Value}) = 0.6(1300) + 0.4(700) = 780 + 280 = ₹1,060$
 $E(R) = \frac{1060 - 1000}{1000} = 6.0\%$
The expected return equals the 6% risk-free rate exactly. Since this asset carries risk (outcomes range from -30% to +30%) yet offers *no* excess return over the risk-free rate, a risk-averse investor should **reject** it — it fails to pay a risk premium. *Reconciliation:* probabilities $0.6 + 0.4 = 1.0$. ✓ Expected value ₹1,060 → 6% return, matching the risk-free rate, so risk premium = 6% – 6% = 0. ✓

Section C — Interview-Style Questions with Model Answers

C1. "A fund returned 20% last year. Is that good?"

Model answer: "I can't judge return without risk and a benchmark. Twenty percent is impressive if it came with low volatility and beat a relevant index; it's unremarkable if the fund took 35% volatility in a year the index itself rose 25%. Give me its standard deviation and the risk-free rate and I'll compute its Sharpe ratio — excess return per unit of risk — and compare it to its benchmark. The rule is: never judge a return without its risk, and never judge a risk without its horizon."

C2. "Why is there a risk-return trade-off at all? Couldn't markets just misprice it away?"

Model answer: "It survives because it's an equilibrium, not an accident. Investors are risk-averse due to diminishing marginal utility of wealth, so they demand extra expected return to hold riskier assets. If a risky asset ever offered the same expected return as a safe one, investors would sell it, its price would fall, and its expected return would rise until it again compensated for its risk. Crucially, only *undiversifiable* (systematic) risk is priced — diversifiable risk earns nothing because it can be removed for free. So mispricing gets arbitrated back toward a premium proportional to systematic risk."

C3. "Walk me through building a portfolio for a 28-year-old software engineer versus a 62-year-old retiree."

Model answer: "I'd start from objectives and constraints — the IPS. The 28-year-old has a long time horizon, large bond-like human capital, low near-term liquidity needs, and can absorb drawdowns, so her return objective is growth and her risk tolerance is high — an equity-heavy allocation. The retiree has a short-to-medium horizon, depleted human capital, high liquidity needs for living expenses, and low tolerance for losses he can't recover from, so his objective is income plus capital preservation — a bond-heavy, more conservative mix. The horizon and human capital do most of the work in setting the equity/bond split; taxes and any unique circumstances fine-tune it."

C4. "Is speculation just a polite word for gambling?"

Model answer: "No. Speculation analyses *pre-existing* economic risk — a company's fortunes, a commodity's supply — and can carry positive expected value if the speculator has a real edge; it also performs a genuine function by providing liquidity and bearing risk others avoid. Gambling *manufactures* risk that needn't exist purely for the wager, and typically has negative expected value because of the house edge. The line between investor and speculator is really a spectrum of holding period and analytical basis, not a moral one."

C5. "You keep saying cash is risky. Explain that to a client who thinks cash is the safest thing they own."

Model answer: "Cash is the safest in *nominal* terms — the number in the account won't fall. But in *real* terms it reliably loses. At 6% inflation, ₹100 today buys only about ₹94 of goods next year and ₹88 the year after. So cash quietly guarantees a loss of purchasing power. It minimises price volatility but maximises inflation risk. Every asset trades one risk for another; cash's trade is low volatility for guaranteed real erosion, which is exactly why we don't hold a retirement corpus entirely in cash."

C6. "Should I report a fund's past performance using the arithmetic or the geometric mean?"

Model answer: "Geometric, for reporting realised performance — it's the compound growth rate that reflects what actually happened to invested wealth, and it's always less than or equal to the arithmetic mean. The classic trap: +50% then −40% averages to +5% arithmetically but is actually −5.1% geometrically, because ₹100 becomes ₹90. Use the arithmetic mean only when you're forecasting a single future period's expected return, not when you're describing multi-period compounded results."

Section D — Multiple-Choice Questions (with Reasoning)

D1. The risk premium compensates an investor primarily for: (a) the time value of money (b) expected inflation (c) the uncertainty of future payments (d) transaction costs

Answer: (c). Time value and inflation are captured by the real risk-free rate and the inflation premium respectively (together the nominal risk-free rate). The risk premium is the third, separate layer compensating for *uncertainty*. (a) and (b) are the other two layers; (d) is not a component of required return in this framework.

D2. Which risk is rewarded with a premium in an efficient market? (a) unsystematic risk (b) systematic risk (c) total risk (d) all risk equally

Answer: (b). Only systematic (undiversifiable, market-wide) risk earns a premium. Unsystematic risk (a) can be diversified away for free, so the market pays nothing for it. Total risk (c) includes the unrewarded unsystematic portion, so it is not the correct priced measure.

D3. Two assets have equal standard deviation. Asset P has $E(R) = 12\%$, Asset Q has $E(R) = 9\%$. A risk-averse investor prefers: (a) Q, because lower return means lower risk (b) P, because for equal risk it offers more return (c) is indifferent (d) cannot say without beta

Answer: (b). With equal risk (σ), the risk-averse investor prefers the higher expected return, so Asset P dominates. (a) confuses return with risk; risk here is already equal. Beta (d) isn't needed because total risk is explicitly stated equal and the question compares on that basis.

D4. A fund earns 14% with $\sigma = 16\%$; the risk-free rate is 6%. Its Sharpe ratio is: (a) 0.875 (b) 0.50 (c) 0.36 (d) 1.14

Answer: (b). Sharpe = $(14 - 6)/16 = 8/16 = 0.50$. (a) wrongly divides return by σ without subtracting R_f ($14/16$). (c) and (d) don't correspond to the correct excess-return-over-risk calculation.

D5. The geometric mean return is: (a) always greater than the arithmetic mean (b) equal to the arithmetic mean when volatility is zero (c) the best estimate of next period's return (d) unaffected by volatility

Answer: (b). When every period's return is identical (zero volatility), the two means coincide. Otherwise geometric < arithmetic, so (a) is false. (c) describes the arithmetic mean. (d) is false — the gap between the means widens with volatility ($\approx \frac{1}{2}\sigma^2$).

D6. When ability and willingness to take risk conflict, the adviser should generally: (a) use ability, since it is objective (b) use willingness, since the client is always right (c) use the lower of the two (d) average the two

Answer: (c). The prudent default is the lower of ability and willingness (while educating the client), so the plan doesn't exceed either the client's financial capacity or psychological comfort. Averaging (d) or picking one unconditionally (a, b) can leave the investor over-exposed relative to one dimension.

D7. Which statement about diversification is correct? (a) it reduces expected return (b) it eliminates all risk (c) it reduces unsystematic risk without reducing expected return (d) it reduces systematic risk

Answer: (c). Diversification removes idiosyncratic (unsystematic) risk while leaving expected return intact — the "free lunch." It does not touch systematic (market) risk (d, false) and does not reduce expected return (a, false); no strategy eliminates *all* risk (b, false).

D8. An asset returns ₹1,200 with probability 0.5 and ₹800 with probability 0.5 on a ₹1,000 stake. Its expected return is: (a) 0% (b) 10% (c) 20% (d) –10%

Answer: (a). $E(\text{Value}) = 0.5(1200) + 0.5(800) = 600 + 400 = ₹1,000$, so $E(R) = (1000 - 1000)/1000 = 0\%$. The symmetric $\pm 20\%$ payoffs cancel. Since the asset carries real risk for zero expected excess return, a risk-averse investor would reject it.

End of Q&A bank. Every formula matches Chapter 01; all numerical answers reconciled.

Chapter 02 — Measuring Risk and Return

1. The Problem / Need

Every investment decision is a trade between two things: what you expect to *earn* and what you might *lose along the way*. But "expect to earn" and "might lose" are vague, emotional words. A portfolio manager cannot allocate capital, an analyst cannot compare two stocks, and a risk officer cannot set limits on the basis of vague feelings. They need **numbers** — a single, defensible figure for return and a single, defensible figure for risk, computed the same way every time so that a bank deposit, an Infosys share, a government bond and a gold ETF can all be laid side by side on the same ruler.

This is harder than it sounds, and the difficulties are exactly where interviewers probe:

- **Return is not one number.** If a stock goes from ₹100 to ₹150 over two years, is the "return" 50%, or 25% a year, or 22.5% a year? All three are defensible; each answers a different question. Get the wrong one into a compounding calculation and you overstate performance — a classic way funds flatter themselves.
- **Return is random, not fixed.** Next year's Nifty return is unknown today. So "expected return" is a probability-weighted guess, not a promise. We must model the *distribution* of possible returns, not a point.
- **Risk needs a definition.** Intuitively, risk is "bad things happening." Mathematically, we usually proxy it by **dispersion** — how widely returns scatter around their average. That choice (variance/standard deviation) is powerful and tractable, but it treats upside surprises as "risky" too, and it quietly assumes the world is roughly bell-shaped. Both assumptions bite in real markets.
- **Assets move together.** The risk of a *portfolio* is not the average of the risks of its parts, because holdings partly cancel and partly reinforce each other. To capture that we need **covariance** and **correlation** — the single most important idea that makes diversification work (Chapter 03 builds the whole theory on it).

This chapter builds the measurement toolkit — return measures, expected return, variance, standard deviation, covariance, correlation, and the normal-distribution scaffolding — with the actual arithmetic worked end to end. Everything later (CAPM, the efficient frontier, Sharpe ratios, VaR) is assembled from these bricks.

2. The Core Idea

Return is *reward*, summarised by an average. **Risk** is *uncertainty about that reward*, summarised by the spread of outcomes around the average. And because assets are held in combinations, we also measure **how assets' returns move together**, because co-movement — not individual jumpiness — determines portfolio risk.

Three quantities do most of the work:

Quantity	Symbol	Plain meaning	Units
Expected return	$E(R)$ or μ	The centre of the distribution — reward you anticipate	% (return)
Variance / standard deviation	σ^2 / σ	The width of the distribution — how far outcomes stray	% ² / %
Covariance / correlation	σ_{ij} / ρ_{ij}	The tendency of two assets to move together	% ² / unitless (–1 to +1)

The mental model: picture a histogram of an asset's possible annual returns. Its **peak/centre** is $E(R)$. Its **width** is σ . Line up two such histograms and ask whether their good and bad years coincide — that co-movement is ρ . Master these three and you can quantify almost any single-asset or portfolio risk-return question an interviewer throws at you.

3. Why / How It Works

Why an average for return? Because a probability distribution is fully "located" by its mean. If you had to bet on one number, the expected value is the one that is unbiased in the long run — over many repetitions the realised average converges to $E(R)$ (law of large numbers). It is the natural reward figure.

Why dispersion for risk? Two investments can share the same expected return yet feel utterly different: a T-bill returning "5% for sure" versus a stock that averages 5% but swings between –30% and +40%. What separates them is *how tightly outcomes cluster around 5%*. Standard deviation measures exactly that clustering. Squaring the deviations (to get variance) does two jobs: it stops positive and negative deviations from cancelling, and it makes the maths additive and differentiable — essential when we later minimise portfolio variance with calculus.

Why co-movement, not just individual risk? Here is the crux of all of modern portfolio theory. Suppose two assets each have $\sigma = 20\%$. Hold them 50/50. If they move in lockstep ($\rho = +1$), portfolio σ stays 20% — no benefit. If they are unrelated ($\rho = 0$), portfolio σ drops to about 14%. If they move perfectly opposite ($\rho = -1$), portfolio σ can fall to **zero**. Same two assets, same individual risks — but the *interaction* changes portfolio risk dramatically. That is why a good analyst obsesses over correlations, not just volatilities: risk that cancels in a portfolio is not risk you should demand to be paid for.

Why the normal distribution? Because it is fully described by just two numbers — μ and σ — which is precisely the pair we already compute. If returns were exactly normal, mean and standard deviation would tell you *everything*: the probability of any outcome, the size of a "1-in-20" loss, all of it. That is the elegant bargain of mean-variance analysis. Real returns are only *approximately* normal, and the places they deviate (fat tails, skew) are where models fail and money is lost — a theme we return to in Section 8.

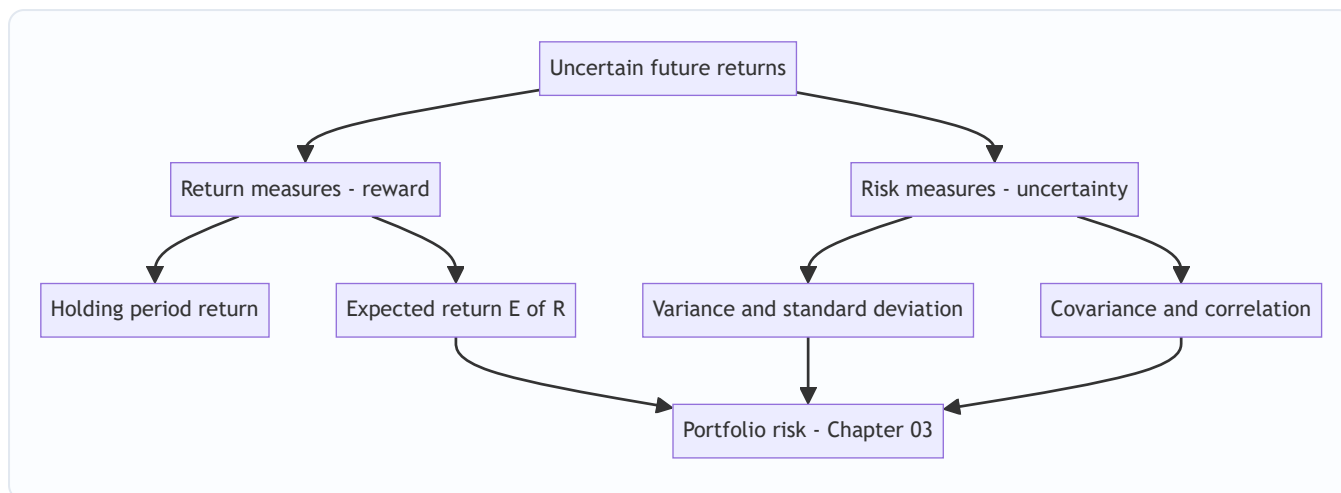


Figure 1 — The measurement toolkit: reward on the left, uncertainty on the right, both feeding portfolio construction.

4. Full Content — Formulas and Models

4.1 Return measures

Holding-Period Return (HPR). The total return over one holding period, capturing both income and price change:

$$HPR = \frac{(P_1 - P_0) + D_1}{P_0} = \underbrace{\frac{D_1}{P_0}}_{\text{income yield}} + \underbrace{\frac{P_1 - P_0}{P_0}}_{\text{capital gain}}$$

where P_0 = purchase price, P_1 = ending price, D_1 = cash income (dividend/coupon) received. HPR is period-agnostic — the "period" can be a day, month or five years — which is exactly why you must annualise before comparing across horizons.

Arithmetic mean return (AM). The simple average of period returns:

$$\bar{R}_{AM} = \frac{1}{n} \sum_{t=1}^n R_t$$

It answers: "In a *typical single* period, what return should I expect?" It is the right input for forecasting the *next* period's expected return.

Geometric mean return (GM). The compounded average that actually grew the money:

$$\bar{R}_{GM} = \left[\prod_{t=1}^n (1 + R_t) \right]^{1/n} - 1$$

It answers: "What constant per-period return would have turned my starting capital into my ending capital?" It is the right measure for *realised, historical* performance over multiple periods.

Key relationship: $GM \leq AM$ always, with equality only if every period's return is identical. The gap widens with volatility, approximated by:

$$\bar{R}_{GM} \approx \bar{R}_{AM} - \frac{\sigma^2}{2}$$

This "volatility drag" (variance drain) is why a fund can have a cheerful arithmetic average yet a mediocre compounded track record. **Interview trap:** quoting AM as if it were the compounded return overstates growth.

Annualising. Convert a per-period return to an annual-equivalent by compounding over the number of periods per year (m):

$$R_{\text{annual}} = (1 + R_{\text{period}})^m - 1$$

For example a monthly return annualises with $m = 12$; a quarterly return with $m = 4$. (Volatility annualises differently — see 4.3.)

Real vs nominal return (Fisher relation). Nominal returns ignore inflation; real returns strip it out to show change in *purchasing power*:

$$1 + R_{\text{real}} = \frac{1 + R_{\text{nominal}}}{1 + i} \quad \Rightarrow \quad R_{\text{real}} = \frac{1 + R_{\text{nominal}}}{1 + i} - 1$$

where i = inflation rate. The common shortcut $R_{\text{real}} \approx R_{\text{nominal}} - i$ is only a first-order approximation; use the exact ratio when rates are high (India's inflation has often run 5–7%, so the approximation error is material).

4.2 Expected return (forward-looking)

When the future is described by scenarios with probabilities p_s and returns R_s :

$$E(R) = \sum_{s=1}^S p_s R_s$$

This is the probability-weighted mean — the distribution's centre of mass. It is a *forecast*, distinct from a historical average (which is one realised sample from the true distribution).

4.3 Risk — variance and standard deviation

From scenarios (expected/ex-ante):

$$\sigma^2 = \sum_{s=1}^S p_s (R_s - E(R))^2, \quad \sigma = \sqrt{\sigma^2}$$

From historical data (sample, ex-post):

$$s^2 = \frac{1}{n-1} \sum_{t=1}^n (R_t - \bar{R})^2, \quad s = \sqrt{s^2}$$

Note the **(n – 1)** divisor — *Bessel's correction*. We divide by $n - 1$, not n , because using the sample mean (itself estimated from the data) "uses up" one degree of freedom; dividing by n would systematically underestimate the true variance. (For a full population, or scenario probabilities, use n .)

Standard deviation σ is preferred for reporting because it is in the same units as return (%), whereas variance is in $\%^2$ and hard to interpret directly.

Annualising volatility (the square-root-of-time rule):

$$\sigma_{\text{annual}} = \sigma_{\text{period}} \times \sqrt{m}$$

So a monthly σ scales by $\sqrt{12} \approx 3.46$, not by 12. This holds when returns are serially uncorrelated (independent across periods): variances add over time, so standard deviation grows with the *square root* of time. **Interview favourite:** "Daily vol is 1%, roughly annual vol?" $\rightarrow 1\% \times \sqrt{252} \approx 15.9\%$ (252 trading days).

4.4 Co-movement — covariance and correlation

Covariance measures whether two assets' deviations from their means share the same sign:

$$\text{Ex-ante (scenarios): } \sigma_{AB} = \text{Cov}(R_A, R_B) = \sum_s p_s (R_{A,s} - E(R_A))(R_{B,s} - E(R_B))$$

$$\text{Ex-post (historical sample): } s_{AB} = \frac{1}{n-1} \sum_{t=1}^n (R_{A,t} - \bar{R}_A)(R_{B,t} - \bar{R}_B)$$

Positive covariance → they tend to be above/below their means together; negative → they offset. The problem: covariance units are %² and its magnitude depends on the assets' own volatilities, so you cannot judge "how strong" the relationship is from covariance alone.

Correlation standardises covariance onto a clean −1 to +1 scale:

$$\rho_{AB} = \frac{\sigma_{AB}}{\sigma_A \sigma_B} \quad \text{in } [-1, +1]$$

- $\rho = +1$: perfect positive linear co-movement (no diversification benefit).
- $\rho = 0$: no *linear* relationship.
- $\rho = -1$: perfect negative co-movement (risk can be fully hedged away).

Rearranged, $\sigma_{AB} = \rho_{AB} \cdot \sigma_A \cdot \sigma_B$, the form you plug into the portfolio-variance formula.

4.5 Two-asset portfolio return and risk (the payoff)

Portfolio expected return is just the weighted average (weights w_i sum to 1):

$$E(R_p) = w_A E(R_A) + w_B E(R_B)$$

But portfolio **variance is not** a weighted average — the covariance term is what matters:

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \sigma_{AB} = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho_{AB} \sigma_A \sigma_B$$

The cross term carries ρ . When $\rho < 1$, portfolio σ is *less* than the weighted average of the individual σ 's — that shortfall **is** the diversification benefit. This single equation is the seed of the entire efficient-frontier theory in Chapter 03.

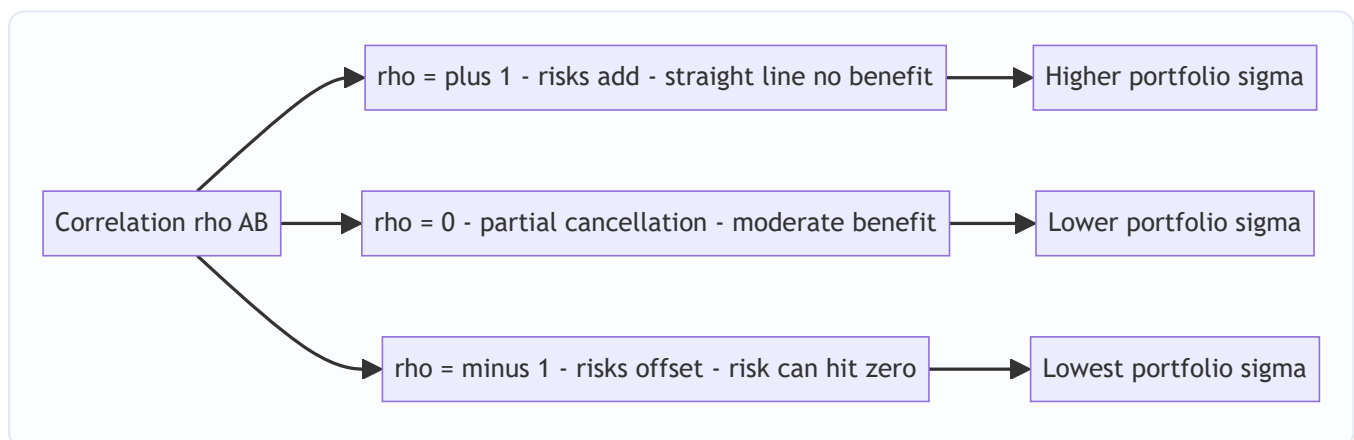


Figure 2 — Why correlation drives diversification: the lower the correlation, the more portfolio risk falls below the weighted-average risk.

5. Worked Examples

Example 1 — HPR, arithmetic vs geometric, volatility drag, real return

You buy a stock at ₹100. Annual data over three years:

Year	Start price	End price	Dividend	HPR
1	100	120	2	$(120 - 100 + 2)/100 = +22\%$
2	120	108	3	$(108 - 120 + 3)/120 = -7.5\%$
3	108	135	3	$(135 - 108 + 3)/108 = +27.78\%$

Arithmetic mean: $(22 - 7.5 + 27.78)/3 = 42.28/3 = 14.09\%$.

Geometric mean: $[(1.22)(0.925)(1.2778)]^{(1/3)} - 1 = [1.22 \times 0.925 \times 1.2778]^{(1/3)} - 1 = [1.44215]^{(1/3)} - 1$. Cube root of 1.44215: $1.44215^{(1/3)} = 1.1299$, so GM = **12.99%** $\approx 13.0\%$.

Reconciliation check. Verify GM actually reproduces the compounded growth. Ignoring the timing of dividends for a clean check, the product of growth factors is 1.44215, i.e. total three-year growth of +44.2%. A constant 12.99% for three years gives $1.1299^3 = 1.4422$ ✓ — matches. GM (13.0%) < AM (14.1%), exactly as theory requires.

Volatility-drag approximation. Sample σ of the three HPRs: deviations from AM (14.09%) are +7.91, -21.59, +13.69. Squared: 62.6, 466.1, 187.4; sum = 716.1; $\div (n-1 = 2) = 358.0$; $\sigma = \sqrt{358.0} = 18.9\%$. Then $AM - \sigma^2/2 = 14.09\% - 0.0358/2 \dots$ careful with units: $\sigma^2 = 0.0358$ (in decimal²), $\sigma^2/2 = 0.0179 = 1.79\%$. So $AM - \sigma^2/2 \approx 14.09 - 1.79 = 12.3\%$, in the neighbourhood of the exact GM 13.0% (the approximation is rough for large swings, but confirms the direction and rough size of the drag). ✓

Real return. If cumulative inflation ran 6% per year, the real geometric return = $(1.1299/1.06) - 1 = 1.0659 - 1 = 6.6\%$. The nominal 13.0% shrinks to 6.6% of genuine purchasing-power growth — inflation ate roughly half. The crude shortcut $(13.0 - 6.0 = 7.0\%)$ overstates it by 0.4 ppt.

Annualising a sub-period return. Suppose instead you only had a 6-month HPR of 8%. Annualised = $(1.08)^2 - 1 = 16.64\%$ (not 16%), because the second half-year compounds on the first.

Example 2 — Expected return, variance, standard deviation from scenarios

Analyst's one-year scenarios for **Stock X**:

State	Probability p_s	Return R_s
Boom	0.30	+30%
Normal	0.50	+10%
Recession	0.20	-20%

Expected return: $E(R) = 0.30(30) + 0.50(10) + 0.20(-20) = 9 + 5 - 4 = 10\%$.

Variance (probability-weighted squared deviations from 10%):

State	$R_s - E(R)$	$(R_s - E(R))^2$	$p_s \times (\dots)^2$
Boom	+20	400	$0.30 \times 400 = 120$
Normal	0	0	$0.50 \times 0 = 0$
Recession	-30	900	$0.20 \times 900 = 180$
		$\Sigma = \sigma^2$	300 (%²)

$\sigma^2 = 300 \%$, so $\sigma = \sqrt{300} = 17.32\%$.

Interpretation. Stock X offers a 10% expected return with a 17.3% standard deviation. If returns were normal, roughly two-thirds of outcomes would fall within $10\% \pm 17.3\%$, i.e. between -7.3% and $+27.3\%$ — a wide band that quantifies the risk in one number.

Example 3 — Covariance, correlation, and the diversification payoff

Same scenarios, now with **Stock Y** alongside X:

State	p_s	R_X	R_Y
Boom	0.30	+30%	−5%
Normal	0.50	+10%	+12%
Recession	0.20	−20%	+25%

$E(R_Y)$: $0.30(-5) + 0.50(12) + 0.20(25) = -1.5 + 6 + 5 = 9.5\%$.

σ_Y : deviations from 9.5 → Boom -14.5 , Normal $+2.5$, Recession $+15.5$. Weighted squares: $0.30(210.25) + 0.50(6.25) + 0.20(240.25) = 63.075 + 3.125 + 48.05 = 114.25 \%$. $\sigma_Y = \sqrt{114.25} = 10.69\%$. (From Example 2, $E(R_X)=10\%$, $\sigma_X = 17.32\%$.)

Covariance = $\sum p_s \cdot (R_X - 10)(R_Y - 9.5)$:

State	$(R_X - 10)$	$(R_Y - 9.5)$	product	$\times p_s$
Boom	+20	−14.5	−290	$0.30 \times -290 = -87.0$
Normal	0	+2.5	0	0
Recession	−30	+15.5	−465	$0.20 \times -465 = -93.0$
			Cov	−180 (%²)

$\sigma_{XY} = -180 \%$ — negative, so X and Y offset each other (X loves booms, Y loves recessions).

Correlation: $\rho_{XY} = -180 / (17.32 \times 10.69) = -180 / 185.15 = -0.972$. Strongly negative — Y is close to a natural hedge for X.

Portfolio payoff (50/50 weights):

Expected return: $E(R_p) = 0.5(10) + 0.5(9.5) = 9.75\%$.

Variance: $\sigma_p^2 = (0.5)^2(300) + (0.5)^2(114.25) + 2(0.5)(0.5)(-180) = 0.25(300) + 0.25(114.25) + 0.5(-180) = 75 + 28.5625 - 90 = 13.56 \%$. $\sigma_p = \sqrt{13.56} = 3.68\%$.

Reconciliation — the diversification magic. The weighted-average of the two standard deviations would be $0.5(17.32) + 0.5(10.69) = 14.01\%$. Yet the actual portfolio σ is just **3.68%** — a collapse of over 10 percentage points, entirely due to the negative covariance term (-90) in the variance equation. We gave up almost nothing in return (9.75% vs X's 10%) while cutting risk by roughly 79% versus holding X alone (3.68% vs 17.32%). **This is the whole point of portfolio theory in one number:** risk that cancels is not risk you carry.

Sanity cross-check via scenario returns of the 50/50 portfolio directly: - Boom: $0.5(30) + 0.5(-5) = 12.5\%$ - Normal: $0.5(10) + 0.5(12) = 11.0\%$ - Recession: $0.5(-20) + 0.5(25) = 2.5\%$

$E(R_p) = 0.30(12.5) + 0.50(11.0) + 0.20(2.5) = 3.75 + 5.5 + 0.5 = \mathbf{9.75\%}$ ✓. σ_p^2 : deviations from 9.75 → +2.75, +1.25, -7.25; weighted squares $0.30(7.5625) + 0.50(1.5625) + 0.20(52.5625) = 2.269 + 0.781 + 10.5125 = \mathbf{13.56\%}^2$ ✓. Both methods agree — the covariance formula reconciles exactly with direct computation.

6. Connections

- **Chapter 03 (Portfolio Theory / Efficient Frontier).** The two-asset variance formula generalises to N assets via the covariance matrix; minimising σ_p for each target $E(R_p)$ traces the efficient frontier. Everything there is built on Section 4.5.
- **CAPM and the SML (later chapter).** Once diversifiable risk is netted out, only *systematic* risk is priced. Beta — $\beta = \text{Cov}(R_i, R_{\text{market}}) / \sigma^2_{\text{market}}$ — is literally a covariance from this chapter, rescaled. $E(R_i) = R_f + \beta[E(R_m) - R_f]$.
- **Performance measurement.** The Sharpe ratio $(E(R) - R_f) / \sigma$ puts *this chapter's* return and σ into one score; Treynor uses β instead of σ .
- **Risk management / VaR.** Value-at-Risk reads a loss quantile straight off the (assumed normal) return distribution: $95\% \text{ VaR} \approx \mu - 1.645\sigma$. Its failures trace directly to the fat-tail limits in Section 8.
- **Fixed income & real assets.** Real-vs-nominal (Fisher) reappears in bond yields (real yield, breakeven inflation) and in TIPS/inflation-linked bonds.
- **Behavioural finance.** The mismatch between σ (which penalises upside) and how investors actually feel loss motivates downside measures (semi-variance, Sortino ratio).

7. Key Terms

Term	Definition
Holding-period return (HPR)	Total return (income + capital gain) over one holding period, as a fraction of starting value.
Arithmetic mean return	Simple average of period returns; best estimate of a <i>single future</i> period's return.
Geometric mean return	Compounded per-period return that reproduces actual multi-period growth; $\leq$ arithmetic mean.
Volatility drag (variance drain)	The gap $AM - GM \approx \sigma^2/2$ by which compounding erodes the arithmetic average.
Annualising	Restating a sub-period return/vol to a yearly basis: returns compound (m), vol scales by $\sqrt{m}$.
Nominal vs real return	Before vs after stripping inflation; $1+R_{\text{real}} = (1+R_{\text{nom}})/(1+i)$.
Expected return $E(R)$	Probability-weighted mean of possible returns; the distribution's centre.
Variance σ^2	Probability- (or sample-) weighted average squared deviation from the mean.
Standard deviation σ	Square root of variance; risk in the same % units as return.
Bessel's correction	Dividing sample variance by $(n-1)$ to remove downward bias from an estimated mean.
Covariance σ_{AB}	Co-movement of two assets' deviations; sign shows direction, magnitude is scale-dependent.
Correlation ρ_{AB}	Covariance standardised to $[-1, +1]$; scale-free strength of linear co-movement.
Diversification benefit	Reduction in portfolio σ below the weighted-average σ , arising when $\rho < 1$.
Normal distribution	Bell curve fully described by μ and σ ; the mean-variance framework's convenience assumption.
Fat tails (leptokurtosis)	Extreme outcomes more likely than normal predicts; why crashes are underestimated.

8. Common Confusions

- **Arithmetic vs geometric mean.** Use AM to *forecast next period*; use GM to *describe realised multi-period growth*. Never compound an arithmetic average — it overstates the money you actually made.
- **Annualising returns vs volatility.** Returns compound: $(1+r)^m - 1$. Volatility scales by $\sqrt{m}$. Using $\sqrt{m}$ on returns, or m on vol, is a frequent and costly slip.
- **Variance is NOT a weighted average of variances.** Portfolio variance has the covariance cross-term. Averaging the parts' variances ignores the entire reason diversification works.
- **Covariance magnitude $\neq$ strength.** A covariance of 500 %² could be a weak or strong relationship depending on the assets' vols; only correlation (-1 to $+1$) tells you strength. Always standardise before judging.
- **Correlation captures only linear co-movement.** $\rho = 0$ means no *linear* relation, not "independent." Two assets can be nonlinearly linked (e.g. an option and its underlying) yet show low ρ .

- **n vs n-1.** Sample data → divide by n-1. Full population or given probabilities → divide by n (or weight by p_s). Getting this wrong biases every downstream risk estimate.
- **σ treats upside as risk.** Standard deviation penalises a +40% surprise exactly like a -40% one. Investors don't fear gains — hence downside measures (semi-variance, Sortino). Know this critique; it's a favourite interview follow-up.
- **"Returns are normal."** They are *approximately* normal at best. Real equity returns are negatively skewed with **fat tails** — big crashes occur far more often than a bell curve predicts (1987, 2008, March 2020). Mean-variance and VaR systematically *underprice* tail risk. Kurtosis of real return series routinely exceeds the normal's value of 3.
- **Real vs nominal shortcut.** $R_{\text{real}} \approx R_{\text{nom}} - i$ is only first-order. At Indian inflation levels (5–7%), use the exact Fisher ratio for accuracy.
- **Expected (ex-ante) vs historical (ex-post).** A historical average is *one sample* drawn from the true distribution — an *estimate* of E(R), not E(R) itself. Small samples give noisy, unstable estimates; this "estimation error" plagues real portfolio optimisation.

9. First-Principles Recap

Strip everything back and rebuild:

1. The future return of any asset is **uncertain** → describe it with a **probability distribution**.
2. Summarise that distribution's **centre** with the mean → **expected return E(R)**, our measure of *reward*.
3. Summarise its **width** with average squared deviations → **variance σ^2** , and its square root **σ**, our measure of *risk*. We square so deviations don't cancel and the maths stays additive; we take the root to return to return-units.
4. Because assets are held together, we need how pairs **co-move** → **covariance**, standardised to **correlation $\rho \in [-1, 1]$** .
5. Combine: portfolio reward is a simple weighted average, but portfolio risk $\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$ shrinks below the weighted average whenever **$\rho < 1$** . That shrinkage is diversification — the free lunch of finance.
6. If the distribution were **normal**, μ and σ would tell us everything — the seductive bargain behind mean-variance. Reality's **fat tails and skew** are where the model, and unprepared investors, break.

From "the future is uncertain" to "diversification lowers risk" in six logical steps — no memorisation, just consequences.

10. Quick-Reference / Interview Points

Core formulas

Concept	Formula
Holding-period return	$HPR = [(P_1 - P_0) + D_1] / P_0$
Arithmetic mean	$\bar{R} = (1/n) \sum R_t$
Geometric mean	$\bar{R}_{GM} = [\prod (1+R_t)]^{(1/n)} - 1$
Volatility drag	$\bar{R}_{GM} \approx \bar{R}_{AM} - \sigma^2/2$
Annualise return	$(1 + R_{period})^m - 1$
Annualise volatility	$\sigma_{period} \times \sqrt{m}$
Real return (Fisher)	$R_{real} = (1+R_{nom})/(1+i) - 1$
Expected return	$E(R) = \sum p_s R_s$
Variance (scenarios)	$\sigma^2 = \sum p_s [R_s - E(R)]^2$
Variance (sample)	$s^2 = \sum (R_t - \bar{R})^2 / (n-1)$
Covariance	$\sigma_{AB} = \sum p_s [R_{A,s} - E(R_A)][R_{B,s} - E(R_B)]$
Correlation	$\rho_{AB} = \sigma_{AB} / (\sigma_A \sigma_B)$
Portfolio return	$E(R_p) = w_A E(R_A) + w_B E(R_B)$
Portfolio variance	$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho_{AB} \sigma_A \sigma_B$

Rules of thumb - $GM \leq AM$, gap $\approx \sigma^2/2$; higher vol $\rightarrow$ bigger drag. - Daily $\sigma \rightarrow$ annual: $\times \sqrt{252} \approx \times 15.9$. Monthly $\rightarrow$ annual: $\times \sqrt{12} \approx \times 3.46$. - Normal: $\sim 68\%$ within $\pm 1\sigma$, $\sim 95\%$ within $\pm 2\sigma$ (precisely 1.96σ), $\sim 99.7\%$ within $\pm 3\sigma$. - $95\% \text{ VaR} \approx \mu - 1.645\sigma$; $99\% \text{ VaR} \approx \mu - 2.33\sigma$.

What interviewers ask - "Why geometric not arithmetic mean for past performance?" $\rightarrow$ Compounding; AM overstates realised growth; gap = volatility drag $\approx \sigma^2/2$. - "Two assets, $\sigma = 20\%$ each, $\rho = 0$, equal weights — portfolio σ ?" $\rightarrow \sqrt{(0.25 \cdot 400 + 0.25 \cdot 400 + 0)} = \sqrt{200} \approx \mathbf{14.1\%}$. Then vary ρ : at $+1 \rightarrow 20\%$, at $-1 \rightarrow 0\%$. - "Why do we care about covariance more than individual volatility?" $\rightarrow$ Portfolio variance is driven by the cross-term; risk that cancels shouldn't be compensated $\rightarrow$ basis of CAPM's systematic-only pricing. - "What's wrong with standard deviation as a risk measure?" $\rightarrow$ Penalises upside; assumes symmetry/normality; ignores fat tails and skew $\rightarrow$ motivates semi-variance, Sortino, VaR/ES. - "Returns aren't normal — so what?" $\rightarrow$ Fat tails/negative skew mean extreme losses are underpriced; mean-variance and Gaussian VaR understate tail risk (2008, 2020). - "Annualise a 1% daily vol." $\rightarrow 1\% \times \sqrt{252} \approx \mathbf{15.9\%}$. - "Difference between expected and historical return?" $\rightarrow$ Ex-ante forecast vs one ex-post sample estimate; historical averages carry estimation error and are unstable inputs to optimisers. - "How does correlation drive diversification?" $\rightarrow$ Lower $\rho \rightarrow$ larger negative/zero cross-term $\rightarrow \sigma_p$ falls further below weighted-average σ ; $\rho = -1$ can eliminate risk entirely.

One-line summary to close an answer: Return is the mean of the distribution, risk is its spread, and portfolio risk is governed less by how volatile assets are than by how they move together.

Q&A — Measuring Risk and Return

A practice bank for Chapter 02. Work each item before reading the answer. Formulas match the chapter; every numerical answer is reconciled by an independent route where possible.

Section A — Concept Checks

A1. Why do we square the deviations when computing variance instead of just averaging them? Two reasons. First, raw deviations from the mean sum to exactly zero (positives and negatives cancel), so their average is uninformative. Squaring removes the sign so all dispersion counts positively. Second, squared terms are additive and differentiable, which lets us minimise portfolio variance with calculus later. We then take the square root (σ) to return to return-units (%).

A2. State when you would report the arithmetic mean and when the geometric mean. Use the arithmetic mean to forecast a *single next period's* expected return — it is the unbiased estimate of one draw from the distribution. Use the geometric mean to describe *realised multi-period growth*, because it is the constant compounded rate that turns starting capital into ending capital. $GM \leq AM$ always.

A3. What is "volatility drag" and roughly how large is it? The gap between the arithmetic and geometric mean, approximately $\sigma^2/2$. Because losses require larger percentage gains to recover, higher volatility compounds to a lower terminal value; the arithmetic average flatters performance. A fund averaging 14% arithmetically with $\sigma = 19\%$ loses roughly 1.8 percentage points to drag.

A4. Why is portfolio variance *not* the weighted average of the component variances? Because returns co-move. The correct formula carries a covariance cross-term: $\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\rho_{AB}\sigma_A\sigma_B$. When $\rho < 1$ the cross-term is smaller than a naive weighting implies, so portfolio σ falls below the weighted-average σ . That shortfall *is* diversification — ignoring the cross-term erases the entire reason to diversify.

A5. A covariance of +500 %² — is that a strong relationship? Unknowable from covariance alone. Covariance magnitude scales with the two assets' own volatilities, so it has no fixed benchmark. Only correlation, $\rho = \sigma_{AB}/(\sigma_A\sigma_B)$, which is bounded in $[-1, +1]$, tells you the *strength* of the linear relationship.

A6. Why divide sample variance by (n – 1) rather than n? Bessel's correction. The sample mean is itself estimated from the same data, consuming one degree of freedom; dividing by n would systematically underestimate the true variance. With given probabilities or a full population, divide by n (weight by p_s) instead.

A7. Give two distinct criticisms of standard deviation as a risk measure. (1) It is symmetric — it penalises a +40% surprise identically to a –40% loss, yet investors do not fear gains; this motivates downside measures (semi-variance, Sortino). (2) It fully describes risk only if returns are normal, but real equity returns are negatively skewed with fat tails, so σ underprices extreme-loss probability.

A8. Returns compound but volatility scales by $\sqrt{\text{time}}$ — why the difference? A multi-period return is the product of period gross returns, so it compounds: $(1+r)^m - 1$. Variance, however, *adds* across independent (serially uncorrelated) periods, so total variance = $m \times \text{period variance}$, and standard deviation = $\sqrt{m} \times \text{period } \sigma$. Applying $\sqrt{m}$ to returns or m to volatility is a classic error.

Section B — Numerical Problems (full working, reconciled)

B1. Holding-period return with income. Buy at ₹250; one year later price is ₹280 and a ₹6 dividend was paid. Find HPR, splitting income and capital-gain yield. $HPR = [(280 - 250) + 6] / 250 = 36 / 250 = 14.4\%$. Split: capital-gain yield = $30/250 = 12.0\%$; income yield = $6/250 = 2.4\%$. Sum = $12.0\% + 2.4\% = 14.4\%$ ✓ (reconciles with the total).

B2. Arithmetic vs geometric mean and the drag. Annual returns: +40%, -20%, +15%. Arithmetic mean = $(40 - 20 + 15)/3 = 35/3 = 11.67\%$. Geometric mean = $[(1.40)(0.80)(1.15)]^{(1/3)} - 1 = [1.288]^{(1/3)} - 1$. $1.288^{(1/3)}$: since $1.088^3 = 1.088 \times 1.088 \times 1.088 = 1.1837 \times 1.088 = 1.288$, the cube root is **1.088**, so GM = **8.80%**. Reconciliation: growing ₹1 at a constant 8.80% for three years gives $1.088^3 = 1.288$, the same terminal factor as the actual path ✓. GM (8.80%) < AM (11.67%) as required; the ~2.9 ppt gap reflects the -20% year's large drag.

B3. Real (inflation-adjusted) return via Fisher. Nominal return 13.0%, inflation 6.0%. Exact real return = $(1.13/1.06) - 1 = 1.06604 - 1 = 6.60\%$. The crude shortcut $13 - 6 = 7.0\%$ overstates the real return by 0.40 ppt — material at Indian inflation levels, so prefer the exact ratio.

B4. Expected return, variance, standard deviation from scenarios. Scenarios for Stock M:

State	p_s	R_s
Boom	0.25	+24%
Normal	0.50	+8%
Bust	0.25	-16%

$E(R) = 0.25(24) + 0.50(8) + 0.25(-16) = 6 + 4 - 4 = 8\%$. Variance = $\sum p_s[R_s - E(R)]^2$: - Boom: $(24 - 8)^2 = 256$; $\times 0.25 = 64$ - Normal: $(8 - 8)^2 = 0$; $\times 0.50 = 0$ - Bust: $(-16 - 8)^2 = 576$; $\times 0.25 = 144$ $\sigma^2 = 64 + 0 + 144 = 208$ $\%^2$; $\sigma = \sqrt{208} = 14.42\%$. Reconciliation: $E(R)$ sits symmetrically (Boom and Bust are ± 16 from Normal with equal probability), consistent with the 8% centre; the two equal tail contributions (64 and 144 differ only because deviations differ) sum cleanly. Interpretation: 8% expected return, $\pm 14.4\%$ one- σ spread.

B5. Annualising a sub-period return and a daily volatility. (a) A 6-month HPR of 8%. Annualised = $(1.08)^2 - 1 = 1.1664 - 1 = 16.64\%$ (not 16% — the second half compounds on the first). (b) Daily $\sigma = 1\%$, 252 trading days. Annual $\sigma = 1\% \times \sqrt{252} = 1\% \times 15.875 = 15.87\% \approx 15.9\%$.

B6. Covariance, correlation, and the two-asset portfolio (the payoff). Stocks A and B over the scenarios below:

State	p_s	R_A	R_B
Boom	0.30	+25%	-10%
Normal	0.40	+10%	+8%
Recession	0.30	-15%	+20%

Step 1 — expected returns. $E(R_A) = 0.30(25) + 0.40(10) + 0.30(-15) = 7.5 + 4 - 4.5 = 7.0\%$. $E(R_B) = 0.30(-10) + 0.40(8) + 0.30(20) = -3 + 3.2 + 6 = 6.2\%$.

Step 2 — variances. σ_A^2 : $(25-7)^2=324 \rightarrow \times 0.30=97.2$; $(10-7)^2=9 \rightarrow \times 0.40=3.6$; $(-15-7)^2=484 \rightarrow \times 0.30=145.2$. Sum = **246 $\%^2$** ; $\sigma_A = \sqrt{246} = 15.68\%$. σ_B^2 : $(-10-6.2)^2=262.44 \rightarrow \times 0.30=78.732$;

$(8-6.2)^2=3.24 \rightarrow \times 0.40=1.296$; $(20-6.2)^2=190.44 \rightarrow \times 0.30=57.132$. Sum = **137.16 %²**; $\sigma_B = \sqrt{137.16} = \mathbf{11.71\%}$.

Step 3 — covariance = $\sum p_s(R_A-7)(R_B-6.2)$: - Boom: $(18)(-16.2) = -291.6$; $\times 0.30 = -87.48$ - Normal: $(3)(1.8) = 5.4$; $\times 0.40 = 2.16$ - Recession: $(-22)(13.8) = -303.6$; $\times 0.30 = -91.08$ $\sigma_{AB} = -87.48 + 2.16 - 91.08 = \mathbf{-176.4 \%^2}$.

Step 4 — correlation. $\rho_{AB} = -176.4 / (15.68 \times 11.71) = -176.4 / 183.61 = \mathbf{-0.961}$. Strongly negative — B hedges A.

Step 5 — 50/50 portfolio. $E(R_p) = 0.5(7.0) + 0.5(6.2) = \mathbf{6.6\%}$. $\sigma_p^2 = (0.5)^2(246) + (0.5)^2(137.16) + 2(0.5)(0.5)(-176.4) = 0.25(246) + 0.25(137.16) + 0.5(-176.4) = 61.5 + 34.29 - 88.2 = \mathbf{7.59 \%^2}$; $\sigma_p = \sqrt{7.59} = \mathbf{2.76\%}$.

Reconciliation via direct portfolio returns (50/50 each state): - Boom: $0.5(25) + 0.5(-10) = 7.5\%$ - Normal: $0.5(10) + 0.5(8) = 9.0\%$ - Recession: $0.5(-15) + 0.5(20) = 2.5\%$ $E(R_p) = 0.30(7.5) + 0.40(9.0) + 0.30(2.5) = 2.25 + 3.6 + 0.75 = \mathbf{6.6\%}$ $\checkmark$. σ_p^2 : devs from 6.6 $\rightarrow +0.9, +2.4, -4.1$; weighted squares = $0.30(0.81) + 0.40(5.76) + 0.30(16.81) = 0.243 + 2.304 + 5.043 = \mathbf{7.59 \%^2}$ $\checkmark$. Both routes agree. The weighted-average σ would have been $0.5(15.68) + 0.5(11.71) = 13.70\%$, yet the portfolio σ is only 2.76% — the negative covariance term (-88.2) collapsed risk by nearly 11 ppt while return barely moved (6.6% vs A's 7.0%). That is diversification in one number.

B7. Interview-style portfolio- σ drill across correlations. Two assets, $\sigma_A = \sigma_B = 20\%$, equal weights.

Compute σ_p at $\rho = +1, 0, -1$. $\sigma_p^2 = 0.25(400) + 0.25(400) + 2(0.5)(0.5)\rho(20)(20) = 100 + 100 + 200\rho = 200 + 200\rho$. - $\rho = +1$: $\sigma_p^2 = 400 \rightarrow \sigma_p = \mathbf{20\%}$ (no benefit; risks add linearly). - $\rho = 0$: $\sigma_p^2 = 200 \rightarrow \sigma_p = \sqrt{200} = \mathbf{14.14\%}$. - $\rho = -1$: $\sigma_p^2 = 0 \rightarrow \sigma_p = \mathbf{0\%}$ (risk fully hedged). Reconciliation at $\rho = +1$: σ_p should equal the weighted-average $\sigma = 0.5(20) + 0.5(20) = 20\%$ $\checkmark$; at $\rho = -1$ with equal σ and equal weights, offsetting is complete, giving 0% $\checkmark$.

Section C — Interview-Style Questions (model answers)

C1. "Why use the geometric, not the arithmetic, mean to report a fund's past performance?" Model answer: Because investors experience compounding, and only the geometric mean reproduces the actual growth of capital. The arithmetic mean averages independent single-period returns and systematically overstates realised growth whenever returns vary — the gap is the volatility drag, roughly $\sigma^2/2$. Example: returns of +50% then -50% average to 0% arithmetically, but ₹100 becomes ₹150 then ₹75 — a -25% cumulative, i.e. a geometric mean of about -13.4% per year. Quoting the arithmetic 0% would be a lie about what the money did. So: arithmetic for forecasting the next period, geometric for describing history.

C2. "Why do you care more about covariance than about an asset's own volatility?" Model answer: Because portfolio variance is driven by the covariance cross-term, not by individual variances alone. A highly volatile asset that is uncorrelated — or negatively correlated — with the rest of the book can *lower* total portfolio risk. Conversely, adding a low-volatility asset that moves in lockstep with everything else adds little diversification. The logical consequence is that risk which cancels within a portfolio should not be compensated by the market — only non-diversifiable, systematic covariance risk is priced. That is the seed of CAPM, where beta is exactly a rescaled covariance with the market.

C3. "Standard deviation — what's wrong with it as a risk measure?" Model answer: Two structural problems. First, it is symmetric: it treats a large gain as just as "risky" as an equal loss, which does not match how investors define risk. Downside measures like semi-variance and the Sortino ratio address this by penalising only below-target returns. Second, σ fully characterises risk only under normality, but real returns

are negatively skewed and fat-tailed (leptokurtic) — crashes like 1987, 2008 and March 2020 occur far more often than a bell curve predicts. So mean-variance analysis and Gaussian VaR systematically underprice tail risk. σ remains useful as a tractable, additive first approximation, but a good analyst names its limits.

C4. "Returns aren't normally distributed. Practically, so what?" Model answer: It matters most in the tails, which is where portfolios blow up. Fat tails mean extreme losses are more likely than a normal distribution implies, and negative skew means the big surprises tend to be on the downside. Any tool that reads risk off a Gaussian — mean-variance optimisation, a 95%/99% VaR of $\mu - 1.645\sigma$ or $\mu - 2.33\sigma$ — will understate the probability and size of large drawdowns. The fixes are to stress-test beyond the model (scenario analysis, expected shortfall, historical simulation) and to treat optimiser outputs with scepticism because they assume a distribution reality does not honour.

C5. "Explain expected versus historical return to a client." Model answer: Expected return is a forward-looking, probability-weighted forecast — the centre of the distribution of what *could* happen next. A historical average is a single realised sample drawn from that unknown distribution; it is an *estimate* of the expected return, not the truth. Because it is one sample, it carries estimation error and is unstable, especially over short windows — which is why feeding raw historical means into an optimiser produces fragile, extreme allocations. We use history to inform the forecast, not to equal it.

C6. "Annualise a 1% daily volatility, and explain the rule." Model answer: Roughly $1\% \times \sqrt{252} \approx 15.9\%$, using ~ 252 trading days. Volatility scales with the square root of time because variances add across independent periods while returns compound. If daily returns are serially uncorrelated, annual variance = $252 \times$ daily variance, so annual $\sigma = \sqrt{252} \times$ daily σ . The common mistake is multiplying the daily figure by 252 (that annualises variance, not volatility) or applying the $\sqrt{\cdot}$ -rule to returns, which should compound instead.

Section D — Multiple-Choice Questions (with reasoning)

D1. A stock returns +30% then −30% in two years. Its geometric mean annual return is closest to: A) 0% B) −4.6% C) −9% D) +4.6% **Answer: B.** $(1.30)(0.70) = 0.91$, terminal factor for two years. $GM = 0.91^{(1/2)} - 1 = 0.9539 - 1 = -4.61\%$. The arithmetic mean is 0%, but compounding a symmetric up/down pair always loses money — the drag. A (0%) is the arithmetic-mean trap.

D2. Which pair of correlations gives, respectively, the *most* and *least* portfolio-risk reduction? A) $\rho = 0$ most, $\rho = +1$ least B) $\rho = -1$ most, $\rho = +1$ least C) $\rho = +1$ most, $\rho = -1$ least D) $\rho = -1$ most, $\rho = 0$ least **Answer: B.** Diversification benefit rises as ρ falls. $\rho = -1$ permits the largest reduction (risk can reach zero with suitable weights); $\rho = +1$ gives none (portfolio σ equals the weighted-average σ). C reverses the logic; D wrongly ranks $\rho = 0$ as worst.

D3. Sample variance uses an $(n - 1)$ divisor because: A) it makes σ smaller B) it corrects downward bias from using an estimated sample mean C) the population is always unknown D) it matches the probability-weighted formula **Answer: B.** Bessel's correction offsets the degree of freedom consumed by estimating the mean from the same data; dividing by n would bias variance downward. A is a side effect, not the reason; D is false — the probability-weighted (population) form divides by n , not $n - 1$.

D4. Monthly volatility is 5%. The annualised volatility is closest to: A) 60% B) 17.3% C) 5% D) 8.7% **Answer: B.** $\sigma_{\text{annual}} = 5\% \times \sqrt{12} = 5\% \times 3.464 = 17.32\%$. A ($\times 12$) wrongly annualises variance rather than volatility; it is the classic error.

D5. Two assets each have $\sigma = 25\%$, held 60/40, with $\rho = 0.20$. Portfolio σ is closest to: A) 25.0% B) 19.6% C) 15.8% D) 22.4% **Answer: B.** $\sigma_p^2 = 0.6^2(625) + 0.4^2(625) + 2(0.6)(0.4)(0.20)(25)(25) = 0.36(625) + 0.16(625) + 0.48(0.20)(625) = 225 + 100 + 60 = 385$. $\sigma_p = \sqrt{385} = 19.62\%$. Cross-term check: $2(0.6)(0.4) = 0.48$; $\times 0.20 = 0.096$; $\times 625 = 60$ ✓. So $\sigma_p \approx \mathbf{19.6\%}$. Option A (25%) is the "no-diversification" distractor — the weighted-average σ , which holds only at $\rho = 1$.

D6. Expected return differs from a historical average because expected return is: A) always larger B) a realised outcome C) a forward-looking probability-weighted forecast D) computed with $(n - 1)$ **Answer: C.** Expected return is ex-ante, $E(R) = \sum p_s R_s$; a historical average is one ex-post sample estimate of it. Neither is systematically larger (A false), it is not a realised outcome (B false), and the divisor point (D) is irrelevant to the ex-ante/ex-post distinction.

One-line close: return is the mean of the distribution, risk is its spread, and portfolio risk is governed less by how volatile assets are than by how they move together.

Chapter 03 — Modern Portfolio Theory (Markowitz)

1. The Problem / Need

Every investor faces the same uncomfortable truth: **higher returns come bundled with higher uncertainty**. A government T-bill pays you 6% and you will get exactly that. An equity might average 14%, but any given year could deliver +45% or -30%. So how much of your money should sit in each? And once you own several risky assets, how do you even measure the risk of the *whole* pile?

Before 1952, the practical answer was folklore. Analysts picked stocks they liked one at a time, on their individual merits — "this is a good company at a fair price." Risk was treated as a property of each security in isolation. The portfolio was just an unexamined bag of individually-chosen bets. Nobody had a rigorous way to answer two questions that turn out to be the whole game:

1. **How risky is a portfolio**, given the riskiness of its components? Is it just the average of their risks? (Spoiler: no.)
2. Given a target return, **what is the least-risky way to achieve it** — the optimal *mix*, not the optimal single stock?

Harry Markowitz, then a 25-year-old PhD student at Chicago, noticed the missing piece. Investors don't care about return alone (else everyone borrows to the hilt and buys the single highest-return asset) and they don't care about risk alone (else everyone holds cash). They care about **both, jointly**. And crucially, the risk of a portfolio is *not* the weighted average of the individual risks — because assets don't move in lockstep. When one zigs, another zags, and the zigs and zags partly cancel. That cancellation is **diversification**, and it is, in Markowitz's own later phrase, "the only free lunch in finance."

His 1952 paper *Portfolio Selection* turned this intuition into mathematics and won him the 1990 Nobel Prize. It is the foundation on which CAPM, index funds, risk budgeting, and the entire asset-management industry rest.

The need in one line: we need a framework that treats a portfolio as a *system* of interacting assets, quantifies its risk correctly (accounting for co-movement), and identifies the mixes that give the most return per unit of risk.

2. The Core Idea

Modern Portfolio Theory (MPT) rests on a **mean–variance framework**. Reduce every asset — and every portfolio — to just two numbers:

- **Mean** (expected return, $E(R)$): the reward you expect.
- **Variance** (or its square root, **standard deviation**, σ): the risk, i.e., the dispersion of outcomes around that mean.

A rational investor, MPT assumes, wants **more mean and less variance**. Plot every possible portfolio on a graph with risk (σ) on the x-axis and return on the y-axis, and you get a cloud of points. The investor wants to be as far **up-and-to-the-left** as possible.

The core insight has three moves:

1. **A portfolio's expected return is a simple weighted average** of its components' returns. Nothing surprising here.
2. **A portfolio's risk is NOT a simple weighted average.** It depends on how the assets *co-move* — their **covariance / correlation**. Because correlations are usually below 1, portfolio risk comes out *lower* than the weighted-average risk. This is the diversification benefit, and it is the heart of the theory.
3. Therefore, for any target return there exists a **minimum-variance mix**, and the set of all such best mixes forms the **efficient frontier** — the menu of intelligently-constructed portfolios no rational investor should ever fall below.

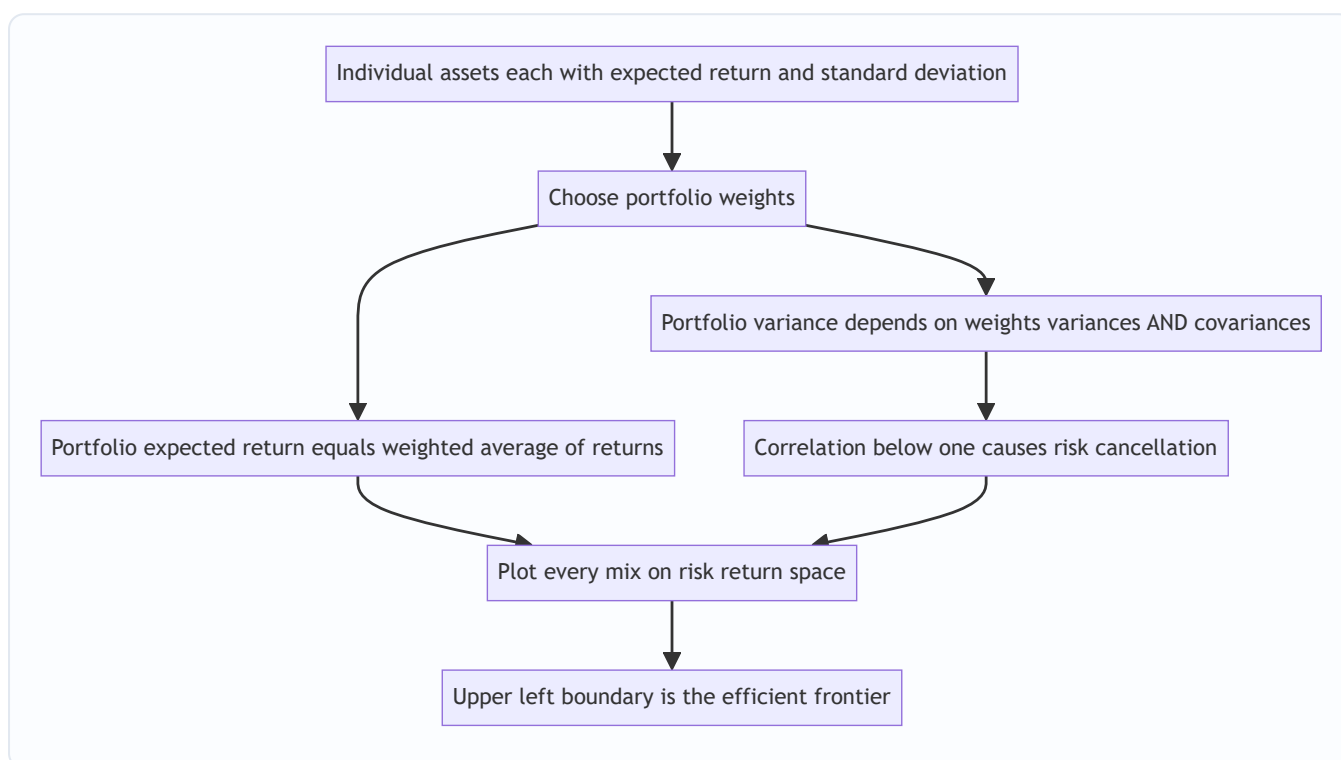


Figure 1 — The logic of MPT: return averages, but risk under-averages because of correlation, producing an efficient frontier.

3. Why / How It Works

Why risk under-averages

Imagine two assets, each with standard deviation 20%. Naively you might think a 50/50 blend also has 20% risk. It doesn't — unless the two assets move *perfectly together*.

Think of two shopkeepers: one sells umbrellas, one sells sunscreen. Each business is volatile month to month — a rainy month is terrible for sunscreen and great for umbrellas. But own **both** and your combined revenue is remarkably stable: whatever the weather, one leg pays off. The individual volatilities didn't vanish; they *offset* because the two revenues are **negatively correlated**.

Mathematically, this offsetting lives in the **covariance** term of the variance formula. When you square a sum, $(a + b)^2 = a^2 + b^2 + 2ab$, you get cross-terms. The $2ab$ cross-term is the covariance. If assets move oppositely, that term is negative and *subtracts* from total risk. Even if they're merely *imperfectly* positively correlated (the usual case for stocks), the cross-term is smaller than it would be under perfect correlation, so risk still falls below the weighted average.

The role of correlation (ρ)

Correlation ρ ranges from -1 to $+1$ and is the dial that controls how much diversification you get:

Correlation ρ	What it means	Diversification benefit
+1	Assets move perfectly together	None — portfolio σ is the weighted average of σ 's
0	Movements unrelated	Substantial — risk falls meaningfully
-1	Perfect opposites	Maximum — risk can be driven to <i>zero</i> with the right weights

The key takeaway interviewers love: **diversification benefit exists for any $\rho < +1$** . You do not need negative correlation. You just need assets that are less than perfectly correlated — which is virtually all real asset pairs.

Why this is the "only free lunch"

Normally in finance you pay for what you get: want more return, take more risk. Diversification is different — it lets you *reduce* risk **without sacrificing expected return**, purely by combining assets intelligently. You're not forecasting better or taking a view; you're exploiting the mathematical fact that imperfectly-correlated risks partly cancel. That's why it's "free."

Real market context

This is not a textbook curiosity — it is *the* organising principle of professional money management:

- **The 60/40 portfolio.** The classic balanced fund (60% equities, 40% bonds) is a direct application of two-asset MPT. Bonds historically carried low-to-negative correlation with equities, so the blend smoothed the ride. Notably, in 2022 that assumption broke — stocks and bonds fell *together* as inflation spiked, ρ went positive, and 60/40 had one of its worst years ever. A vivid live demonstration that **the diversification benefit is only as good as the correlation input**, and that correlations are not constant.
- **Asset allocation vs stock picking.** A famous (if debated) finding from Brinson, Hood & Beebower is that the *allocation* decision — how you split across asset classes — explains the vast majority of a portfolio's return variability, far more than individual security selection. MPT is what makes allocation a rigorous, quantitative discipline rather than guesswork.
- **Why AMCs hold hundreds of names.** An equity mutual fund holding 50–100 stocks isn't hedging its bets out of timidity; it's deliberately pushing unsystematic risk toward zero so that the fund's risk is dominated by systematic (market) risk it is being paid to bear. This is MPT's n -asset logic in action.
- **Crisis correlation spikes.** The cruel irony practitioners quote: "diversification works until you need it most." In 2008 and March 2020, correlations across risky assets jumped toward $+1$ as everything sold off together, temporarily gutting the diversification benefit. MPT's static-correlation assumption is precisely what this violates — a key limitation to flag in interviews.

4. Full Content — Formulas & Models

4.1 Expected return of a single asset

Given probability-weighted scenarios:

$$E(R) = \sum_{i=1}^n p_i R_i$$

where p_i is the probability of state i and R_i the return in that state. (For historical data, this is just the arithmetic mean of past returns.)

4.2 Variance and standard deviation of a single asset

$$\sigma^2 = \sum_{i=1}^n p_i (R_i - E(R))^2 \quad \sigma = \sqrt{\sigma^2}$$

Variance is in "squared return" units (awkward), so we take the square root to get **standard deviation**, expressed in the same % units as return — the standard risk measure.

4.3 Covariance and correlation between two assets

Covariance measures how two assets move together:

$$\text{Cov}(A,B) = \sigma_{AB} = \sum_i p_i (R_{A,i} - E(R_A))(R_{B,i} - E(R_B))$$

Covariance's magnitude is hard to interpret (it depends on the assets' scales), so we standardise it into **correlation**:

$$\rho_{AB} = \frac{\sigma_{AB}}{\sigma_A \sigma_B} \quad \Longleftrightarrow \quad \sigma_{AB} = \rho_{AB} \sigma_A \sigma_B$$

Correlation is unit-free and bounded in $[-1, +1]$. The right-hand rearrangement is the form you'll plug into the variance formula.

4.4 Two-asset portfolio — expected return

$$E(R_p) = w_A E(R_A) + w_B E(R_B)$$

with weights summing to one, $w_A + w_B = 1$. A simple weighted average — this is the part that *does* average.

4.5 Two-asset portfolio — variance (the central formula)

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho_{AB} \sigma_A \sigma_B$$

and portfolio standard deviation $\sigma_p = \sqrt{\sigma_p^2}$.

Read it as three pieces: the two **own-variance** terms ($w^2 \sigma^2$) plus the **cross-term** $2 w_A w_B \rho \sigma_A \sigma_B$. That cross-term is where correlation does its work. Because ρ can be less than 1, the whole expression comes in below the weighted-average-of- σ 's benchmark.

Special cases (set A and B weights and vary ρ):

- $\rho = +1$: $\sigma_p = w_A \sigma_A + w_B \sigma_B$ (perfect positive $\rightarrow \sigma$ is a straight weighted average, *no* benefit).
- $\rho = -1$: $\sigma_p = |w_A \sigma_A - w_B \sigma_B|$ (perfect negative $\rightarrow$ a specific weight makes $\sigma_p = 0$, a riskless combination).

The zero-risk weight when $\rho = -1$: set $w_A \sigma_A = w_B \sigma_B$, giving $w_A = \sigma_B / (\sigma_A + \sigma_B)$.

4.6 The minimum-variance portfolio (two assets)

The weight in A that minimises portfolio variance (found by differentiating σ_p^2 w.r.t. w_A and setting to zero):

$$w_A^{\min} = \frac{\sigma_B^2 - \rho_{AB}\sigma_A\sigma_B}{\sigma_A^2 + \sigma_B^2 - 2\rho_{AB}\sigma_A\sigma_B}$$

This gives the leftmost point of the two-asset curve — the **Global Minimum Variance (GMV) portfolio** for that pair.

4.7 The n-asset generalisation

For a portfolio of n assets with weights w_i :

Expected return: $E(R_p) = \sum_{i=1}^n w_i E(R_i)$

Variance (double-sum form): $\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} = \sum_i w_i^2 \sigma_i^2 + \sum_i \sum_{j \neq i} w_i w_j \sigma_{ij}$

where $\sigma_{ii} = \sigma_i^2$. In compact **matrix form**, with weight vector $\mathbf{w}$ and covariance matrix $\mathbf{\Sigma}$:

$$\sigma_p^2 = \mathbf{w}^{\top} \mathbf{\Sigma} \mathbf{w}$$

The counting insight (why covariance dominates): an n -asset portfolio's variance is a sum of n variance terms and $n^2 - n$ covariance terms. As n grows, covariances vastly outnumber variances. For a 50-stock portfolio: 50 variance terms vs 2,450 covariance terms. This is the mathematical proof that **in a large portfolio, what matters is not each stock's own risk but how stocks co-move**.

Push this to the limit with an equally-weighted portfolio ($w_i = 1/n$):

$$\sigma_p^2 = \frac{1}{n} \overline{\sigma_i^2} + \left(1 - \frac{1}{n}\right) \overline{\sigma_{ij}}$$

As $n \rightarrow \infty$, the first term (average own-variance, the **diversifiable / unsystematic** risk) vanishes, and σ_p^2 converges to the **average covariance** — the **systematic risk** that no amount of diversification can remove. This decomposition is the bridge to CAPM (next chapter).

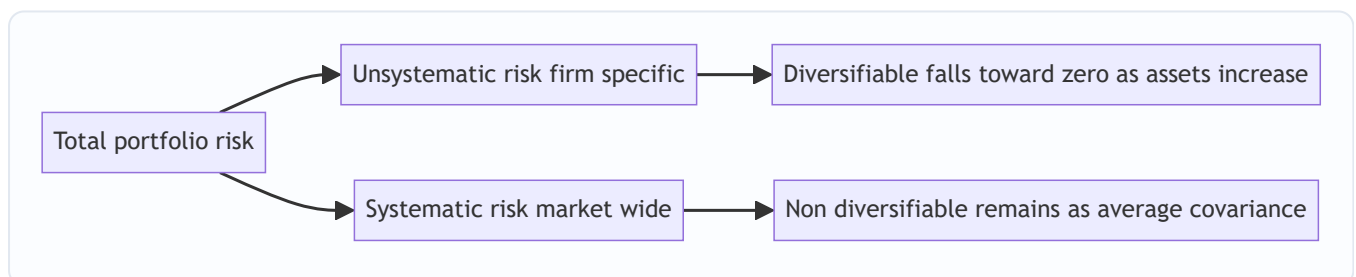


Figure 2 — Diversification eliminates unsystematic risk but leaves systematic risk as a floor.

4.8 The efficient frontier & assumptions

Plot every feasible portfolio in $(\sigma, E(R))$ space. The **feasible set** is a solid region; its upper-left boundary is the **efficient frontier** — portfolios offering **maximum return for a given risk** (equivalently, minimum risk for a given return). Any portfolio *below* the frontier is **dominated** (you could get more return for the same risk). The frontier's leftmost tip is the **GMV portfolio**; the segment below the GMV point is the *inefficient* lower boundary.

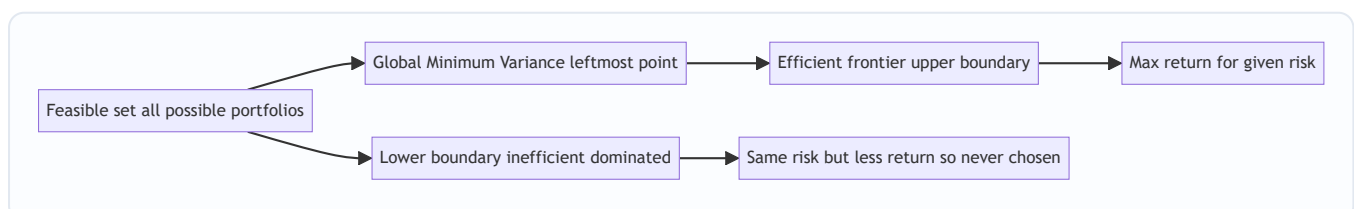


Figure 4 — The efficient frontier is the upper-left edge of the feasible set, anchored at the GMV point; everything below it is dominated.

Markowitz's assumptions: - Investors are **rational** and **risk-averse** (prefer less variance for equal return). - Decisions are made purely on **mean and variance** over a single period (either returns are normally distributed, or investors have quadratic utility). - Markets are frictionless (no taxes/costs) in the basic model. - Investors agree on inputs (in the equilibrium extension).

4.9 The shape of the frontier and how a third asset helps

Why does the two-asset combination trace a *curve* (a hyperbola) rather than a straight line? Because return moves linearly with weight while risk moves *sub*-linearly (the square-root of a quadratic). The gap between the straight line you'd get under $\rho = +1$ and the bowed-in curve you get under $\rho < +1$ is a geometric picture of the diversification benefit — **the more the curve bows to the left, the lower the correlation and the bigger the free lunch**.

Adding a **third asset** expands the feasible set from a curve into a two-dimensional region, and the efficient frontier becomes the upper-left envelope of *all* pairwise and multi-way combinations. The practical lesson: a new asset improves your frontier not because it has high return or low risk on its own, but because it has **low correlation with what you already hold**. This is why gold, commodities, and real estate earn a place in institutional portfolios despite mediocre standalone returns — their diversifying (low- ρ) property shifts the whole frontier up-and-left.

4.10 A note on the risk measure itself

MPT uses **variance** (symmetric dispersion) as its risk proxy. Critics point out that investors don't fear *upside* volatility — only downside. This motivates alternatives like **semi-variance** (dispersion below the mean only) and, later, **Value at Risk (VaR)** and downside-deviation-based ratios (Sortino). Markowitz himself acknowledged semi-variance was arguably more logical but chose variance for tractability — the covariance algebra above simply doesn't work as cleanly for downside measures. Know this trade-off: variance is chosen for **mathematical convenience and the clean covariance decomposition**, at the cost of treating good and bad surprises identically.

5. Worked Examples

Example 1 — Full two-asset risk/return (the core numerical)

Setup. You are blending an equity fund **E** and a bond fund **B**.

Asset	Expected return	Std. dev. (σ)	Weight
Equity (E)	14%	20%	60%
Bond (B)	6%	8%	40%
Correlation $\rho(E,B)$			+0.20

Step 1 — Portfolio expected return. $E(R_p) = 0.60(14\%) + 0.40(6\%) = 8.4\% + 2.4\% = \mathbf{10.8\%}$

Step 2 — Portfolio variance. Use $\sigma_p^2 = w_E^2 \sigma_E^2 + w_B^2 \sigma_B^2 + 2 w_E w_B \rho \sigma_E \sigma_B$. Work in decimals (20% = 0.20).

- Term 1: $w_E^2 \sigma_E^2 = 0.60^2 \times 0.20^2 = 0.36 \times 0.04 = 0.014400$
- Term 2: $w_B^2 \sigma_B^2 = 0.40^2 \times 0.08^2 = 0.16 \times 0.0064 = 0.001024$
- Term 3: $2 w_E w_B \rho \sigma_E \sigma_B = 2 \times 0.60 \times 0.40 \times 0.20 \times 0.20 \times 0.08$
- $= 2 \times 0.60 \times 0.40 = 0.48$; then $\times 0.20 (\rho) = 0.096$; then $\times 0.20 \times 0.08 = 0.096 \times 0.016 = 0.001536$

$$\sigma_p^2 = 0.014400 + 0.001024 + 0.001536 = 0.016960$$

Step 3 — Portfolio standard deviation. $\sigma_p = \sqrt{0.016960} = 0.13023 = \mathbf{13.02\%}$

Step 4 — Reconcile against the no-diversification benchmark. If ρ were +1, risk would be the weighted average of σ 's: $\sigma_{\{\text{no-benefit}\}} = 0.60(20\%) + 0.40(8\%) = 12\% + 3.2\% = 15.2\%$

But our actual portfolio σ is only **13.02%** — a **2.18 percentage-point reduction** in risk, achieved *purely* by combining, while still earning 10.8%. **That gap is the diversification benefit.** Self-check: $13.02\% < 15.2\%$ ✓, and it sits below equity's 20% and above bond's 8%, as a blend should. ✓

Example 2 — Same assets, vary the correlation

Hold weights (60/40) and all else fixed; change only ρ . Recompute σ_p to see correlation's power. Only Term 3 changes: $2 w_E w_B \sigma_E \sigma_B = 0.48 \times 0.016 = 0.00768$, then multiplied by ρ .

ρ	Term 3 = $0.00768 \times \rho$	$\sigma_p^2 = 0.015424 + \text{Term 3}$	σ_p
+1.0	+0.007680	0.023104	15.20%
+0.5	+0.003840	0.019264	13.88%
+0.2	+0.001536	0.016960	13.02%
0.0	0.000000	0.015424	12.42%
-0.5	-0.003840	0.011584	10.76%
-1.0	-0.007680	0.007744	8.80%

(Here $0.015424 = \text{Term 1} + \text{Term 2} = 0.014400 + 0.001024$.)

Reading it: as correlation falls, portfolio risk falls monotonically — from 15.20% at $\rho=+1$ down to 8.80% at $\rho=-1$ — while expected return stays pinned at 10.8% throughout. **Same reward, less and less risk, just from lower correlation.** At $\rho = +1$ you get 15.20%, exactly the weighted average from Example 1 — confirming the "no benefit under perfect positive correlation" special case. ✓

Note: at $\rho = -1$, the 60/40 mix doesn't reach zero risk because the zero-risk weight is $w_E = \sigma_B / (\sigma_E + \sigma_B) = 8 / (20 + 8) = 28.6\%$, not 60%.

Example 3 — The minimum-variance portfolio

Using the same E and B with $\rho = +0.20$, find the mix that *minimises* risk. Apply the GMV weight formula with $\sigma_E = 0.20$, $\sigma_B = 0.08$, $\sigma_{\{EB\}} = \rho \sigma_E \sigma_B = 0.20 \times 0.20 \times 0.08 = 0.0032$.

$$w_E^{\{\min\}} = \frac{\sigma_B^2 - \sigma_{\{EB\}}}{\sigma_E^2 + \sigma_B^2 - 2\sigma_{\{EB\}}} = \frac{0.0064 - 0.0032}{0.04 + 0.0064 - 2(0.0032)} = \frac{0.0032}{0.0400} = \mathbf{0.08}$$

So the GMV portfolio holds **8% equity, 92% bonds**. Its return: $0.08(14\%) + 0.92(6\%) = 1.12\% + 5.52\% = 6.64\%$. Its variance:

- T1: $0.08^2 \times 0.04 = 0.0064 \times 0.04 = 0.000256$
- T2: $0.92^2 \times 0.0064 = 0.8464 \times 0.0064 = 0.005417$
- T3: $2 \times 0.08 \times 0.92 \times 0.0032 = 0.1472 \times 0.0032 = 0.000471$
- $\sigma_p^2 = 0.006144 \rightarrow \sigma_p = 7.84\%$

Reconcile: 7.84% is *below* bond's own 8% standalone risk — a striking result. Even adding a *riskier* asset (equity) can lower total risk, because equity's imperfect correlation with bonds provides offset. This is diversification at its most counterintuitive, and a favourite interview "gotcha." ✓

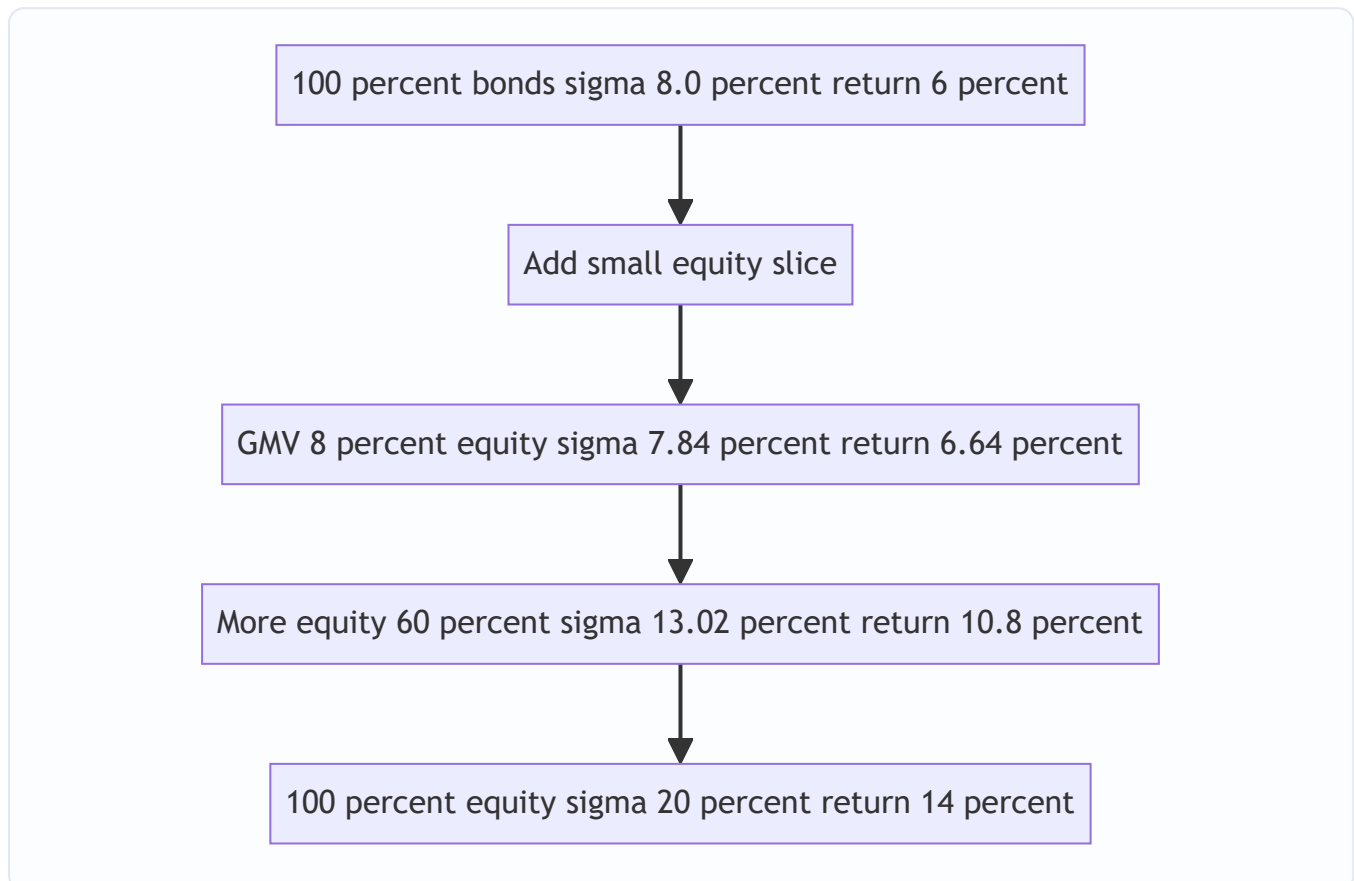


Figure 3 — Moving along the two-asset frontier: adding equity first lowers risk to the GMV point, then raises both risk and return.

6. Connections

- → **CAPM & the SML (Ch. 4):** MPT's decomposition of risk into diversifiable vs systematic leads directly to CAPM, which says only *systematic* risk (beta) is rewarded because the rest is diversifiable and thus "free" to remove. The efficient frontier plus a risk-free asset produces the **Capital Market Line** and the tangency (market) portfolio.
- → **Index investing:** If diversification removes unsystematic risk for free and no one can beat the market portfolio on a risk-adjusted basis, hold the broadest, cheapest index. MPT is the intellectual backbone of Vanguard-style passive investing.
- → **Sharpe ratio (Ch. 5):** MPT ranks portfolios by return *per unit of risk*; the Sharpe ratio $(R_p - R_f) / \sigma_p$ formalises "up and to the left" into a single number. The tangency portfolio is the max-Sharpe portfolio.

- → **Risk parity & asset allocation:** Modern multi-asset funds allocate by *risk contribution* using the same covariance-matrix machinery ($w' \Sigma w$).
- → **Behavioural critique (Ch. later):** MPT assumes rational mean-variance investors; behavioural finance documents where this breaks (loss aversion, correlation spikes in crises).

7. Key Terms

Term	Meaning
Mean–variance framework	Describing assets/portfolios by expected return (mean) and risk (variance/ σ) only
Expected return $E(R)$	Probability-weighted average of possible returns
Variance / Std. dev. (σ)	Dispersion of returns around the mean; σ is the standard risk measure
Covariance (σ_{ij})	Unstandardised measure of how two assets co-move
Correlation (ρ)	Standardised covariance, bounded $[-1, +1]$
Diversification	Risk reduction from combining imperfectly-correlated assets
Systematic risk	Market-wide, non-diversifiable risk (the covariance floor)
Unsystematic risk	Firm-specific, diversifiable risk
Efficient frontier	Set of portfolios with max return per unit of risk
GMV portfolio	Global Minimum Variance — the least-risky feasible portfolio
Feasible / dominated set	All achievable portfolios; dominated ones lie below the frontier

8. Common Confusions

- **"Portfolio risk is the average of the assets' risks."** *No.* Return averages; risk does not, because of the covariance term. $\sigma_p \leq$ weighted-average σ , with equality only when $\rho = +1$.
- **"You need negatively-correlated assets to diversify."** *No.* Any $\rho < +1$ gives a benefit. Negative correlation is *ideal* (can reach zero risk), but positive-but-imperfect still helps.
- **"Adding a risky asset always raises portfolio risk."** *No.* Example 3 showed adding equity to a bond portfolio *lowered* σ below the bond's own. Correlation, not standalone risk, governs the effect.
- **"Diversification eliminates all risk."** *No.* It removes only unsystematic (diversifiable) risk. Systematic risk — the average covariance — remains as a floor no matter how many assets you add.
- **"Covariance and correlation are the same."** *Related, not identical.* Correlation = covariance scaled by the two σ 's; it's covariance made comparable and bounded.
- **"More assets is always better."** Benefits taper sharply — most diversification is captured by ~20–30 stocks; beyond that the marginal reduction is tiny (you're near the systematic floor).
- **Units error.** Always convert percentages to decimals *before* squaring in the variance formula, or square the % consistently. Mixing the two is the #1 exam mistake.

9. First-Principles Recap

Strip it to bedrock:

1. Investors want **more return and less uncertainty** — two dimensions, judged jointly.
2. A portfolio's **return is a weighted average** of its parts. Boring but true.
3. A portfolio's **risk is less than the weighted average** of its parts' risks, because assets don't move perfectly together — the **covariance cross-term** captures this and, being sub-maximal whenever $\rho < 1$, drags total risk down.
4. This risk reduction is **free** — you sacrifice no expected return to get it. Hence "the only free lunch in finance."
5. The reduction has a **limit**: firm-specific (unsystematic) risk washes out, but market-wide (systematic) risk — the average covariance — cannot be diversified away.
6. For each target return there's a **minimum-risk mix**; the collection of these is the **efficient frontier**, the rational investor's entire menu.

Everything downstream — CAPM, beta, index funds, Sharpe ratios, risk budgeting — is a corollary of these six lines.

10. Quick-Reference / Interview Points

Formula sheet

Quantity	Formula
Expected return (asset)	$E(R) = \sum p_i R_i$
Variance (asset)	$\sigma^2 = \sum p_i (R_i - E(R))^2$
Covariance	$\sigma_{AB} = \sum p_i (R_{A,i} - E(R_A))(R_{B,i} - E(R_B))$
Correlation	$\rho_{AB} = \sigma_{AB} / (\sigma_A \sigma_B)$
Portfolio return (2-asset)	$E(R_p) = w_A E(R_A) + w_B E(R_B)$
Portfolio variance (2-asset)	$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$
Portfolio variance (n-asset)	$\sigma_p^2 = \sum \sum w_i w_j \sigma_{ij} = w' \Sigma w$
GMV weight (2-asset)	$w_A = (\sigma_B^2 - \rho \sigma_A \sigma_B) / (\sigma_A^2 + \sigma_B^2 - 2\rho \sigma_A \sigma_B)$
Zero-risk weight ($\rho = -1$)	$w_A = \sigma_B / (\sigma_A + \sigma_B)$

What interviewers ask (and the crisp answer)

- "Why isn't portfolio risk the average of the individual risks?" → Because of the covariance term; with $\rho < 1$ the cross-term is sub-maximal, so $\sigma_p < \text{weighted-avg } \sigma$.
- "Do you need negative correlation to diversify?" → No — any $\rho < +1$ helps; negative is just the strongest case.
- "Can adding a riskier asset reduce portfolio risk?" → Yes, if its correlation with the portfolio is low enough (show the bond+equity GMV example).
- "What risk can't diversification remove?" → Systematic (market) risk — the average-covariance floor; this motivates CAPM's beta.
- "How many stocks to diversify?" → Most benefit by ~20–30; diminishing returns thereafter.

- "What's the efficient frontier?" → The upper-left boundary of feasible portfolios — max return per unit risk; below it, portfolios are dominated.
- "State the assumptions." → Rational risk-averse investors, single-period, decisions on mean & variance only (normal returns or quadratic utility), agreed inputs, frictionless markets.
- "Limitations?" → Relies on estimated inputs (garbage-in), assumes stable correlations (they spike in crises), single-period, ignores higher moments (skew/kurtosis / fat tails).

One-liner to remember: *Return averages; risk under-averages — and that gap is the free lunch.*

Q&A — Modern Portfolio Theory (Markowitz)

Practice bank for Chapter 03. Every question is followed by a full answer. Work numericals yourself first, then check the step-by-step reconciliation.

Section A — Concept Check

A1. In one sentence, what gap did Markowitz's 1952 framework fill that stock-by-stock analysis could not?

It treated the portfolio as a *system* of interacting assets and gave a rigorous way to quantify a portfolio's risk from its components' risks *and their co-movement*, so investors could find the least-risky mix for any target return — rather than picking "good" securities one at a time in isolation.

A2. A portfolio's expected return is a weighted average of its assets' returns, but its risk is not a weighted average of their risks. Why the asymmetry?

Because expected return is *linear* in the weights, so it averages cleanly. Variance is *quadratic*: squaring the weighted sum produces cross-terms — the covariance terms $2 w_A w_B \rho \sigma_A \sigma_B$. Whenever $\rho < +1$ that cross-term is sub-maximal, so portfolio σ comes out *below* the weighted average of the individual σ 's. Return averages; risk under-averages.

A3. Define covariance and correlation and state the relationship between them.

Covariance $\sigma_{AB} = \sum p_i (R_{A,i} - E(R_A))(R_{B,i} - E(R_B))$ measures how two assets move together, but its magnitude depends on the assets' scales, so it is hard to interpret. Correlation standardises it: $\rho_{AB} = \sigma_{AB} / (\sigma_A \sigma_B)$, giving a unit-free number bounded in $[-1, +1]$. The rearrangement $\sigma_{AB} = \rho_{AB} \sigma_A \sigma_B$ is the form plugged into the variance formula.

A4. True or false: you need negatively correlated assets to gain a diversification benefit. Explain.

False. A benefit exists for *any* $\rho < +1$. Negative correlation is merely the strongest case (at $\rho = -1$ risk can be driven to zero with the right weights). Since virtually all real asset pairs are imperfectly correlated, diversification is almost always available — you do not need the assets to move oppositely, only *not perfectly together*.

A5. Distinguish systematic from unsystematic risk and state which one diversification removes.

Unsystematic (firm-specific, diversifiable) risk is idiosyncratic to individual assets and washes out as you add more names. Systematic (market-wide, non-diversifiable) risk is the common risk all assets share.

Diversification removes only the unsystematic part; systematic risk remains as a floor — mathematically, the *average covariance* that portfolio variance converges to as $n \rightarrow \infty$.

A6. Why do covariances "dominate" variance in a large portfolio?

An n -asset portfolio's variance has n own-variance terms but $n^2 - n$ covariance terms. For 50 stocks that is 50 variance terms versus 2,450 covariance terms. As n grows, covariances vastly outnumber variances, so what drives total risk is not each stock's standalone volatility but how the stocks co-move.

A7. What is the efficient frontier, and what does it mean for a portfolio to be "dominated"?

The efficient frontier is the upper-left boundary of the feasible set of portfolios — those offering maximum return for a given risk (equivalently minimum risk for a given return). A portfolio is *dominated* if another

feasible portfolio offers more return for the same risk (or less risk for the same return); dominated portfolios lie below the frontier and no rational investor holds them. The frontier's leftmost tip is the Global Minimum Variance (GMV) portfolio.

A8. Why does MPT use variance as its risk measure, and what is the main criticism?

Variance is chosen for mathematical tractability: it yields the clean covariance decomposition that makes the whole theory work. The criticism is that variance is *symmetric* — it penalises upside surprises the same as downside — whereas investors only fear the downside. This motivates alternatives like semi-variance, VaR, and the Sortino ratio. Markowitz himself conceded semi-variance was arguably more logical but chose variance for the cleaner algebra.

Section B — Numerical Problems (full step-by-step)

B1. Two-asset return and risk. Stock A: $E(R) = 12\%$, $\sigma = 18\%$. Stock B: $E(R) = 8\%$, $\sigma = 10\%$. Correlation $\rho = 0.30$. Weights 50/50. Find the portfolio's expected return and standard deviation, and quantify the diversification benefit.

Step 1 — Expected return. $E(R_p) = 0.50(12\%) + 0.50(8\%) = 6\% + 4\% = \mathbf{10.0\%}$

Step 2 — Variance (work in decimals; $\sigma_A = 0.18$, $\sigma_B = 0.10$): - Term 1: $w_A^2 \sigma_A^2 = 0.50^2 \times 0.18^2 = 0.25 \times 0.0324 = 0.008100$ - Term 2: $w_B^2 \sigma_B^2 = 0.50^2 \times 0.10^2 = 0.25 \times 0.0100 = 0.002500$ - Term 3: $2 w_A w_B \rho \sigma_A \sigma_B = 2 \times 0.50 \times 0.50 \times 0.30 \times 0.18 \times 0.10 = 2 \times 0.50 \times 0.50 = 0.50$; $\times 0.30 = 0.15$; $\times (0.18 \times 0.10 = 0.018) = 0.15 \times 0.018 = 0.002700$

$\sigma_p^2 = 0.008100 + 0.002500 + 0.002700 = \mathbf{0.013300}$

Step 3 — Standard deviation. $\sigma_p = \sqrt{0.013300} = \mathbf{11.53\%}$

Step 4 — Reconcile. The no-benefit ($\rho = +1$) benchmark is the weighted average of σ 's: $0.50(18\%) + 0.50(10\%) = \mathbf{14.0\%}$. Our actual σ_p is 11.53%, a **2.47 pp** reduction earned purely by combining, while still returning 10.0%. Self-check: $11.53\% < 14.0\%$ ✓ and it lies between B's 10% and A's 18% ✓.

B2. Covariance and correlation from scenarios. Three states of the economy:

State	Prob	R_A	R_B
Boom	0.30	25%	5%
Normal	0.50	10%	8%
Recession	0.20	-15%	12%

Find $E(R_A)$, $E(R_B)$, σ_A , σ_B , covariance and correlation.

Step 1 — Expected returns. $E(R_A) = 0.30(25) + 0.50(10) + 0.20(-15) = 7.5 + 5 - 3 = \mathbf{9.5\%}$ $E(R_B) = 0.30(5) + 0.50(8) + 0.20(12) = 1.5 + 4 + 2.4 = \mathbf{7.9\%}$

Step 2 — Variances (in $\%^2$, using deviations from the mean): - σ_A^2 : $0.30(15.5)^2 + 0.50(0.5)^2 + 0.20(-24.5)^2 = 0.30(240.25) + 0.50(0.25) + 0.20(600.25) = 72.075 + 0.125 + 120.05 = 192.25 \rightarrow \sigma_A = \mathbf{13.87\%}$ - σ_B^2 : $0.30(-2.9)^2 + 0.50(0.1)^2 + 0.20(4.1)^2 = 0.30(8.41) + 0.50(0.01) + 0.20(16.81) = 2.523 + 0.005 + 3.362 = 5.89 \rightarrow \sigma_B = \mathbf{2.43\%}$

Step 3 — Covariance (product of deviations, probability-weighted): - Boom: $(15.5)(-2.9) = -44.95$, $\times 0.30 = -13.485$ - Normal: $(0.5)(0.1) = 0.05$, $\times 0.50 = 0.025$ - Recession: $(-24.5)(4.1) = -100.45$, $\times 0.20 = -20.090$
 $\sigma_{AB} = -13.485 + 0.025 - 20.090 = \mathbf{-33.55 (\%^2)}$

Step 4 — Correlation. $\rho_{AB} = -33.55 / (13.87 \times 2.43) = -33.55 / 33.66 = \mathbf{-0.997}$

Reconcile: the near -1 correlation is intuitive — A falls in recessions while B rises (a defensive/counter-cyclical pattern). Almost perfectly negatively correlated assets are diversification gold: a well-chosen blend could push portfolio risk close to zero.

B3. Global Minimum Variance portfolio. Two assets: $\sigma_A = 30\%$, $\sigma_B = 10\%$, $\rho = 0$. Find the GMV weights, and its standard deviation.

Step 1 — GMV weight in A: $w_A^{\min} = (\sigma_B^2 - \rho\sigma_A\sigma_B) / (\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B) = (0.10^2 - 0) / (0.30^2 + 0.10^2 - 0) = 0.0100 / (0.0900 + 0.0100) = 0.0100 / 0.1000 = \mathbf{0.10}$

So GMV = **10% in A, 90% in B.**

Step 2 — GMV standard deviation ($\rho = 0$ kills Term 3): $\sigma_p^2 = 0.10^2(0.30^2) + 0.90^2(0.10^2) = 0.01(0.09) + 0.81(0.01) = 0.000900 + 0.008100 = 0.009000$ $\sigma_p = \sqrt{0.009000} = \mathbf{9.49\%}$

Reconcile: $\sigma_p = 9.49\%$ is *below* B's own standalone risk of 10% — even though we added 10% of an asset three times riskier (30%). This is the counterintuitive heart of MPT: because A is uncorrelated with B, a small slice of A offsets some of B's variance. Correlation, not standalone risk, governs the effect. ✓

B4. Zero-risk weight under perfect negative correlation. $\sigma_A = 20\%$, $\sigma_B = 12\%$, $\rho = -1$. What weight in A drives portfolio risk to zero, and verify it?

Step 1 — Zero-risk weight: $w_A = \sigma_B / (\sigma_A + \sigma_B) = 12 / (20 + 12) = 12 / 32 = \mathbf{0.375} \rightarrow 37.5\% \text{ A, } 62.5\% \text{ B.}$

Step 2 — Verify using the $\rho = -1$ special case $\sigma_p = |w_A \sigma_A - w_B \sigma_B|$: $|0.375 \times 20 - 0.625 \times 12| = |7.5 - 7.5| = \mathbf{0\%}$ ✓

Reconcile: at $\rho = -1$ the two assets' movements exactly offset when their weighted volatility contributions are equal ($w_A \sigma_A = w_B \sigma_B$). This is the only correlation value that allows a genuinely riskless combination of two risky assets.

Section C — Interview-Style Questions (model answers)

C1. "Walk me through why diversification is called the only free lunch in finance."

Normally in finance you pay for reward with risk: more return demands more risk. Diversification is the exception. By combining imperfectly correlated assets, you *reduce* portfolio risk without giving up any expected return — because return is a weighted average (so a blend keeps the average return) while risk under-averages thanks to the covariance term. You aren't forecasting better or taking a directional view; you're exploiting the mathematical fact that less-than-perfectly-correlated risks partly cancel. Reward held constant, risk lowered — that's free.

C2. "Can adding a riskier asset ever lower a portfolio's total risk? Convince me."

Yes, and it's a favourite counterintuitive result. Take a bond portfolio with $\sigma = 8\%$ and add equity with $\sigma = 20\%$ but low correlation. In the chapter's example the GMV mix (8% equity, 92% bonds, $\rho = 0.20$) has $\sigma = 7.84\%$ — *below* the bond's own 8%. The reason: what matters is the new asset's *correlation* with what you

already hold, not its standalone volatility. If the correlation is low enough, the offsetting (negative or small positive) cross-term outweighs the extra own-variance the asset brings, and total risk falls.

C3. "Your firm runs a 60/40 stock-bond fund. In 2022 it had one of its worst years. What went wrong, and what does it teach about MPT?"

The 60/40 works because bonds historically carried low-to-negative correlation with equities, smoothing the ride. In 2022, inflation spiked, central banks hiked aggressively, and stocks *and* bonds fell together — correlation flipped positive. The diversification benefit is only as good as the correlation input, and correlations are not constant. The lesson is a core limitation of MPT: it assumes stable correlations, but in regime shifts (and crises generally) correlations across risky assets spike toward +1 exactly when you most need the offset — "diversification works until you need it most."

C4. "What are the assumptions behind Markowitz's model, and where do they break?"

Assumptions: investors are rational and risk-averse; they decide on mean and variance alone over a single period (which holds if returns are normally distributed *or* utility is quadratic); markets are frictionless (no taxes/costs); and, in the equilibrium extension, investors agree on inputs. They break in practice because: inputs (expected returns, variances, correlations) are *estimated* and noisy — garbage in, garbage out, and optimisers amplify estimation error; correlations are unstable and jump in crises; it's single-period while real investing is multi-period; and variance ignores higher moments — real returns have fat tails and negative skew that mean-variance simply doesn't see.

C5. "How many stocks do I need to be diversified, and why doesn't more always help?"

Most of the diversifiable risk is removed by roughly 20–30 well-chosen stocks; beyond that the marginal risk reduction is tiny. The reason is the decomposition $\sigma_p^2 = (1/n) \cdot (\text{avg own-variance}) + (1 - 1/n) \cdot (\text{avg covariance})$. The first term — unsystematic risk — shrinks fast as n rises, but the second term converges to the *average covariance*, which is the systematic floor. Adding names past ~30 chips at an already-small first term while the floor stays put, so you approach the floor with diminishing returns rather than eliminating risk entirely.

C6. "Why does the two-asset combination trace a curve rather than a straight line in risk-return space?"

Because return moves *linearly* with the weight while risk moves *sub-linearly* — σ_p is the square root of a quadratic in the weights. Under $\rho = +1$ the risk relation collapses to a straight weighted average (a straight line). For any $\rho < +1$ the curve bows in toward the left of that line; the size of the bow *is* the diversification benefit. The lower the correlation, the more the curve bows left, and the bigger the free lunch.

Section D — Multiple Choice (with reasoning)

D1. Two assets have σ of 15% and 9%, weighted 50/50. The resulting portfolio σ is 10.5%. The correlation between them is: (a) exactly +1 (b) less than +1 (c) exactly –1 (d) cannot be determined

Answer: (b). The $\rho = +1$ benchmark is the weighted average of the σ 's: $0.5(15) + 0.5(9) = 12\%$. The actual portfolio σ of 10.5% is *below* 12%, which can only happen when $\rho < +1$ — the covariance cross-term is sub-maximal and drags risk below the weighted average. (a) would require σ_p to *equal* 12%; (c) would push σ_p far lower (toward the $|w_A\sigma_A - w_B\sigma_B| = 3\%$ level); (d) is wrong because the sub-benchmark σ pins down $\rho < +1$.

D2. Diversification eliminates: (a) all portfolio risk (b) systematic risk (c) unsystematic (firm-specific) risk (d) correlation

Answer: (c). Diversification removes only firm-specific, diversifiable risk. Systematic (market-wide) risk remains as a floor — the average covariance that variance converges to as assets increase. (a) is wrong because the floor cannot be removed; (b) is exactly backwards; (d) confuses a statistical input with risk itself.

D3. Two assets have $\rho = -1$. A specific weighting can make portfolio σ equal to: (a) the average of the two σ 's (b) zero (c) the larger σ (d) infinity

Answer: (b). At $\rho = -1$ the movements perfectly offset, and the weight $w_A = \sigma_B/(\sigma_A + \sigma_B)$ makes $w_A \sigma_A = w_B \sigma_B$, so $\sigma_p = |w_A \sigma_A - w_B \sigma_B| = 0$. This is the only correlation value permitting a riskless two-risky-asset combination. (a) describes $\rho = +1$; (c) and (d) are impossible for a diversified blend.

D4. In a 50-stock portfolio, the number of distinct covariance terms relative to variance terms is: (a) equal (b) fewer covariances (c) far more covariances (d) exactly double

Answer: (c). There are $n = 50$ variance terms and $n^2 - n = 2,450$ covariance terms. Covariances vastly outnumber variances, which is the mathematical proof that co-movement, not standalone volatility, dominates a large portfolio's risk.

D5. The Global Minimum Variance portfolio is best described as: (a) the highest-return portfolio on the frontier (b) the leftmost point of the feasible set (c) always 100% in the lower-risk asset (d) the tangency portfolio with the risk-free asset

Answer: (b). The GMV portfolio is the least-risky feasible portfolio — the leftmost tip of the frontier. (a) is the opposite corner; (c) is false because an imperfectly correlated riskier asset usually earns a positive GMV weight (recall the 8%-equity GMV that beat 100% bonds); (d) is the max-Sharpe portfolio, a CAPM concept, not the GMV.

D6. Which is the single biggest practical weakness of applying MPT? (a) the return formula is wrong (b) it needs estimated inputs that are noisy and unstable (c) it cannot handle more than two assets (d) it ignores expected return

Answer: (b). MPT's algebra is exact; the problem is empirical. Expected returns, variances and especially correlations must be *estimated* from data, they are unstable (correlations spike in crises), and optimisers amplify estimation error — garbage in, garbage out. (a) and (d) are false (return is a simple, correct weighted average and is central to the model); (c) is false — the n -asset matrix form $w'\Sigma w$ generalises freely.

Self-verification note: All numericals were recomputed in decimals and cross-checked against the $\rho = +1$ weighted-average benchmark. Key results — B1 $\sigma_p = 11.53\%$ ($< 14\%$ benchmark), B3 GMV $\sigma_p = 9.49\%$ ($< 10\%$ low-risk asset), B4 zero-risk weight 37.5% verified to $\sigma_p = 0$ — all reconcile with the chapter's principle that risk under-averages whenever $\rho < +1$.

Chapter 04 — The Efficient Frontier and Diversification

1. The Problem / Need

An investor never holds a single asset in a vacuum. Even the most concentrated equity-research analyst who "loves" one stock must answer a portfolio-level question: *given all the securities in the world, and given that I dislike risk, which combinations should I even consider owning, and which should I reject outright?* Buying the highest-return stock is naive because return comes bundled with risk, and risk does not simply add up when you hold many assets — it partially cancels.

The core problem is therefore twofold:

1. **Which portfolios are "good"?** Out of the infinite combinations of weights across thousands of securities, most are dominated — you can find another portfolio with the same risk but higher return, or the same return with lower risk. We need a rule to throw away the junk and keep only the survivors.
2. **How much risk can I destroy for free?** If I spread money across many stocks, some of the wobble in individual names cancels out. But not all of it. There is a floor. We need to understand *what kind* of risk vanishes through diversification and what kind is irreducible — because the market only pays you for the irreducible part.

Harry Markowitz's 1952 paper *Portfolio Selection* turned this from folk wisdom ("don't put all your eggs in one basket") into a precise optimisation problem. The output is the **efficient frontier**: the set of portfolios that are not dominated. Layer on a risk-free asset and you get the **Capital Allocation Line** and the **Capital Market Line**, which tell you that the entire investment problem collapses into two decisions: *what mix of risky assets to hold* (the same for everyone) and *how much of your wealth to put in it* (personal to your risk appetite). This is the intellectual backbone of modern asset management, index funds, and CAPM.

2. The Core Idea

Risk and return are the two axes of every investment decision. Plot every possible portfolio on a graph with **standard deviation (risk) on the x-axis** and **expected return on the y-axis**. The cloud of achievable portfolios forms a region. Its left-hand boundary is a bullet-shaped curve — the **minimum-variance frontier**. The upper half of that boundary, from the **global minimum-variance portfolio (GMVP)** upward, is the **efficient frontier**: for each level of risk it gives the maximum return, and no rational investor would hold anything below it.

Two ideas power everything:

- **Diversification is a free lunch on risk.** Because assets are less than perfectly correlated, the risk of a portfolio is *less* than the weighted average of individual risks. The lower the correlation, the more the frontier bows to the left (toward lower risk).
- **Total risk splits into two parts.** *Unsystematic* (firm-specific) risk can be diversified away to nearly zero. *Systematic* (market) risk cannot. The market rewards you only for bearing systematic risk — a principle that flows directly into CAPM in the next chapter.

Add a risk-free asset (a T-bill), draw a straight line from it that just touches the efficient frontier, and the tangency point is the single best risky portfolio for *everybody*. This is the **Tobin separation theorem**, and the

line is the Capital Allocation / Capital Market Line.

3. Why / How It Works

Why diversification reduces risk

Portfolio variance depends not just on individual variances but on **covariances** — how assets move together. When two assets are imperfectly correlated, on days when one disappoints, the other may not, so their combined swings are damped. Mathematically, the cross terms in the variance formula carry the correlation coefficient ρ , and any $\rho < 1$ pulls portfolio variance below the weighted-average variance.

The intuition sharpens in a large portfolio. As you add more and more stocks with equal weights, the contribution of each stock's *own* variance shrinks like $1/n$, while the *average covariance* between pairs of stocks does not shrink. So as n grows, portfolio variance converges to the **average covariance** — the residual, irreducible, systematic risk. Firm-specific noise (a factory fire, a CEO scandal, a surprise earnings beat) is idiosyncratic and averages out; economy-wide shocks (interest rates, recessions, oil prices) hit every stock and remain.

Why only the upper boundary matters

Given two portfolios with identical risk, a rational (non-satiated, risk-averse) investor always prefers the one with higher return. So every portfolio strictly below the top edge of the feasible region is *dominated* and discarded. What survives is the efficient frontier. It is concave (bows up-and-left) precisely because of the diversification effect.

Why a risk-free asset creates a straight line

If you split money between a risk-free asset (zero variance) and a risky portfolio P , the expected return is linear in the weight, and — because the risk-free asset has zero variance and zero covariance with P — the standard deviation is *also* linear in the weight. A linear return and a linear risk trace a **straight line** in risk-return space, running from the risk-free rate through P . To get the best such line, you rotate it upward until it is tangent to the efficient frontier. Steeper is better (more return per unit of risk), so the tangency portfolio is optimal.

4. Full Content — Formulas, Models, Derivations

4.1 Two-asset portfolio

Expected return (weights w_1, w_2 with $w_1 + w_2 = 1$):

$$E(R_p) = w_1 E(R_1) + w_2 E(R_2)$$

Variance:

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

where the covariance is $\text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$ and $-1 \leq \rho_{12} \leq 1$.

Standard deviation $\sigma_p = \sqrt{\sigma_p^2}$.

Special correlation cases (illuminating):

Correlation ρ	Portfolio SD formula	Meaning
$\rho = +1$	$\sigma_p = w_1\sigma_1 + w_2\sigma_2$	No diversification — risk is a weighted average; frontier is a straight line
$\rho = 0$	$\sigma_p = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2}$	Meaningful diversification
$\rho = -1$	$\sigma_p = w_1\sigma_1 - w_2\sigma_2 $	Perfect hedge — risk can be driven to zero

With $\rho = -1$, setting $\sigma_p = 0$ gives the risk-eliminating weight $w_1 = \frac{\sigma_2}{\sigma_1 + \sigma_2}$.

4.2 Minimum-variance weights (two assets)

Differentiate σ_p^2 with respect to w_1 and set to zero:

$$w_1^{\min} = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$$

This is the **minimum-variance portfolio** for two assets — the leftmost tip of the two-asset curve.

4.3 n-asset portfolio (matrix form)

$$E(R_p) = \sum_{i=1}^n w_i E(R_i) = \mathbf{w}^{\top} \boldsymbol{\mu}$$

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_{ij} = \mathbf{w}^{\top} \boldsymbol{\Sigma} \mathbf{w}$$

where Σ is the $n \times n$ covariance matrix. Note the double sum contains n variance terms and $n(n-1)$ covariance terms — as n grows, covariances dominate, which is the mathematical root of undiversifiable risk.

4.4 The large-portfolio limit (equal weights)

For an equally weighted portfolio, $w_i = 1/n$:

$$\sigma_p^2 = \frac{1}{n} \overline{\sigma_i^2} + \left(1 - \frac{1}{n}\right) \overline{\sigma_{ij}}$$

As $n \rightarrow \infty$: $\sigma_p^2 \rightarrow \overline{\sigma_{ij}}$ (the **average covariance**). The own-variance term vanishes (that is the unsystematic risk you diversify away); the average-covariance term is the **systematic floor**.

4.5 Total risk decomposition

$$\underbrace{\sigma_i^2}_{\text{total risk}} = \underbrace{\beta_i^2 \sigma_M^2}_{\text{systematic}} + \underbrace{\sigma_{\epsilon_i}^2}_{\text{unsystematic}}$$

from the single-index (market) model $R_i = \alpha_i + \beta_i R_M + \epsilon_i$, where $\beta_i = \sigma_{iM} / \sigma_M^2$ measures sensitivity to the market and ϵ_i is diversifiable firm-specific noise.

4.6 The efficient frontier

The **minimum-variance frontier** is the solution, for every target return, to:

$$\min_{\mathbf{w}} \sqrt{\mathbf{w}^T \boldsymbol{\Sigma} \mathbf{w}} \quad \text{s.t.} \quad \mathbf{w}^T \boldsymbol{\mu} = R^*, \mathbf{w}^T \mathbf{1} = 1$$

The upper branch (returns $\geq$ the GMVP return) is the **efficient frontier**. The GMVP itself minimises variance ignoring the return target.

4.7 The Capital Allocation Line (CAL)

Combining a risk-free asset R_f (with y = weight in risky portfolio P):

$$E(R_c) = R_f + y[E(R_P) - R_f], \quad \sigma_c = y\sigma_P$$

Eliminating y gives the straight-line CAL:

$$\boxed{E(R_c) = R_f + \frac{E(R_P) - R_f}{\sigma_P} \sigma_c}$$

The slope is the **Sharpe ratio** of P:

$$S_P = \frac{E(R_P) - R_f}{\sigma_P}$$

The **optimal risky portfolio** is the P that maximises this slope — the tangency of the CAL with the efficient frontier.

4.8 Optimal risky-portfolio weights (two risky assets + risk-free)

$$w_1^* = \frac{[E(R_1) - R_f]\sigma_2^2 - [E(R_2) - R_f]\sigma_{12}}{[E(R_1) - R_f]\sigma_2^2 + [E(R_2) - R_f]\sigma_1^2 - [E(R_1) - R_f + E(R_2) - R_f]\sigma_{12}}$$

with $w_2^* = 1 - w_1^*$. This gives the tangency (maximum-Sharpe) portfolio.

4.9 The Capital Market Line (CML)

When *all* investors face the same inputs and the same R_f , the tangency portfolio must be the **market portfolio M** (everything, value-weighted). The CAL that runs through M is elevated to the **Capital Market Line**:

$$\boxed{E(R_p) = R_f + \frac{E(R_M) - R_f}{\sigma_M} \sigma_p}$$

The CML prices *efficient* portfolios using **total risk** σ . (CAPM's Security Market Line, next chapter, prices *individual* assets using **beta**.)

4.10 Optimal complete portfolio (utility)

An investor with risk-aversion coefficient A maximising $U = E(R_c) - \frac{1}{2}A\sigma_c^2$ chooses:

$$y^* = \frac{E(R_P) - R_f}{A\sigma_P^2}$$

Higher expected excess return $\rightarrow$ invest more in risky P; higher risk or higher aversion $\rightarrow$ invest less. $y^* > 1$ means borrowing at R_f to leverage P.

Diagram — the efficient frontier and the GMVP

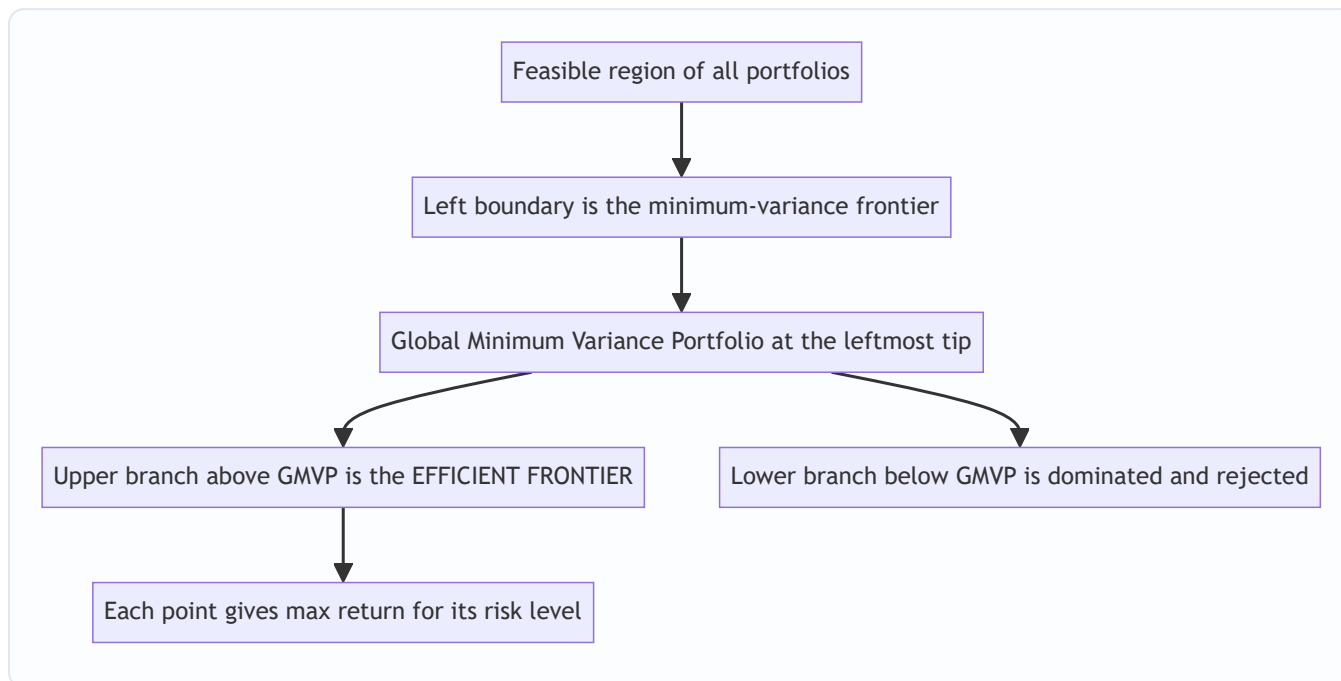


Figure 1 — The bullet-shaped feasible set. Only the upper edge above the GMVP survives as efficient.

Diagram — how correlation bows the two-asset curve

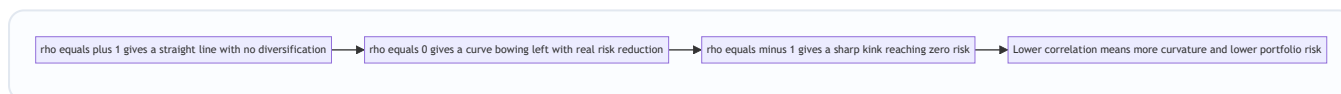


Figure 2 — As correlation falls, the risk-return curve bends further left, delivering more free risk reduction.

Diagram — diversification and the two risks

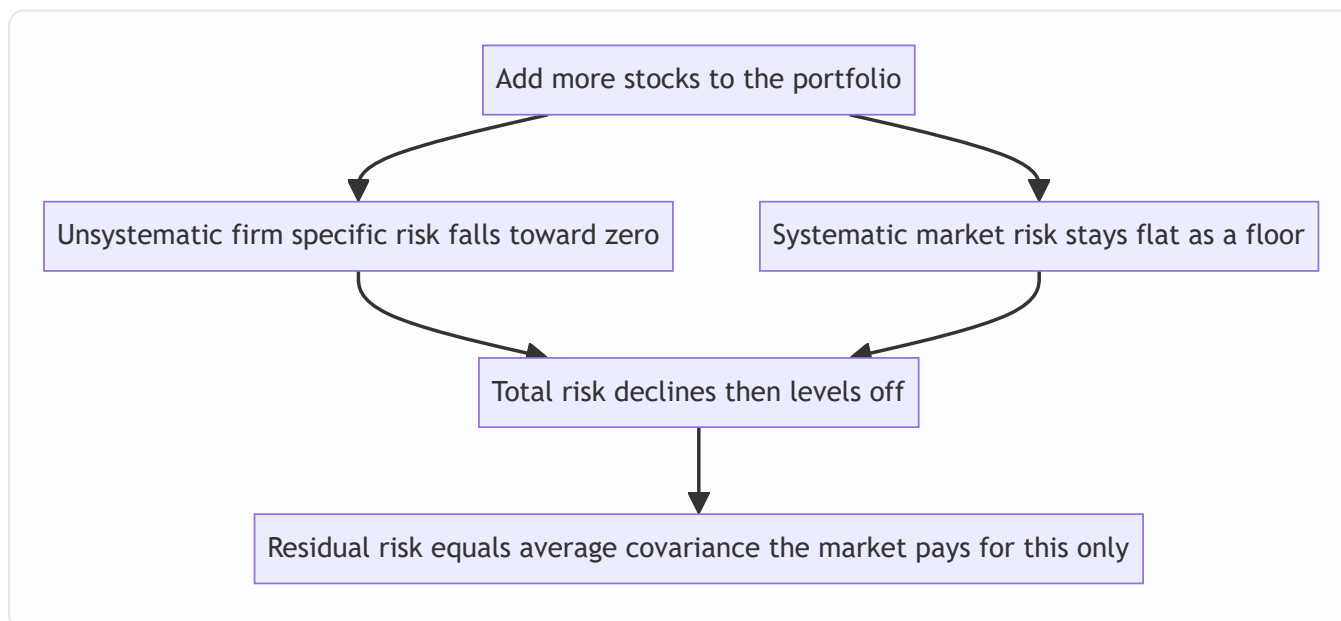


Figure 3 — Total risk falls with the number of holdings but flattens at the systematic floor.

Diagram — from CAL to CML via the tangency portfolio

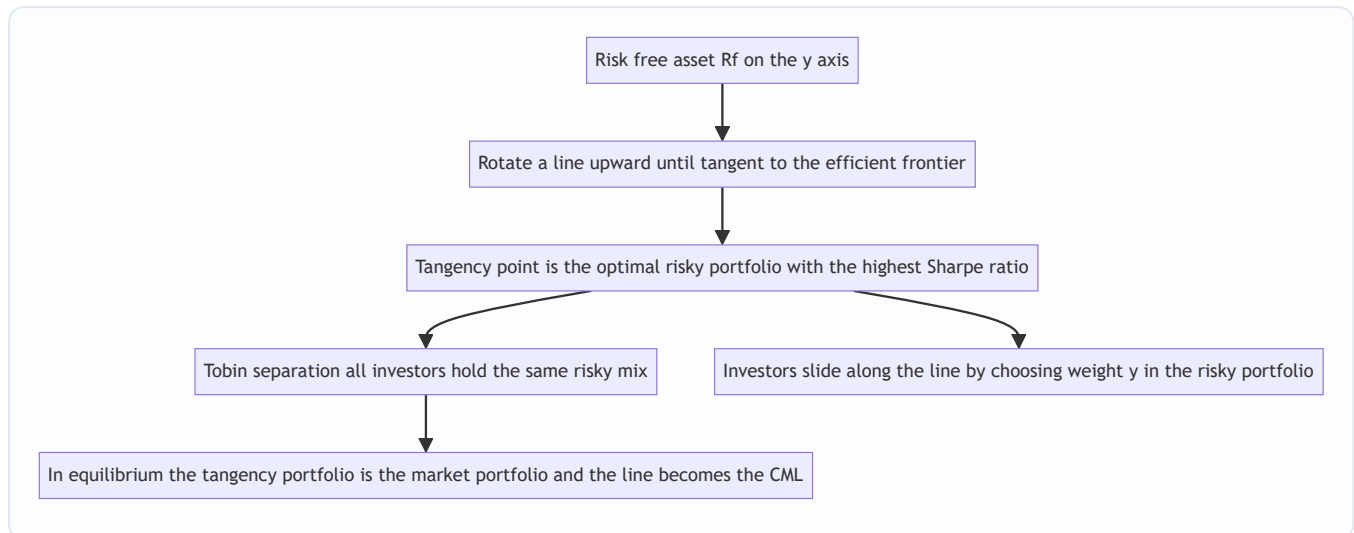


Figure 4 — The best CAL is tangent to the frontier; in market equilibrium it is the Capital Market Line.

5. Worked Examples

Example 1 — Two-asset risk and the diversification effect

Asset A: $E(R_A)=10\%$, $\sigma_A=20\%$. Asset B: $E(R_B)=16\%$, $\sigma_B=30\%$. Invest 60% in A, 40% in B. Compute portfolio return and risk for $\rho = +1$, $\rho = 0$, $\rho = -1$.

Expected return (independent of ρ): $E(R_p) = 0.6(10\%) + 0.4(16\%) = 6\% + 6.4\% = 12.4\%$

Weighted-average SD (the $\rho = +1$ benchmark): $0.6(20) + 0.4(30) = 12 + 12 = 24\%$.

$\rho = +1$: $\sigma_p^2 = 0.6^2(20^2) + 0.4^2(30^2) + 2(0.6)(0.4)(1)(20)(30) = 0.36(400) + 0.16(900) + 2(0.24)(600) = 144 + 144 + 288 = 576 \rightarrow \sigma_p = 24\%$ Exactly the weighted average — **no diversification benefit**.

$\rho = 0$: $\sigma_p^2 = 144 + 144 + 0 = 288 \rightarrow \sigma_p = \sqrt{288} = 16.97\%$ Risk fell from 24% to **16.97%** for the *same* 12.4% return — that gap is the free lunch.

$\rho = -1$: $\sigma_p = |0.6(20) - 0.4(30)| = |12 - 12| = 0\%$ Perfect hedge: this exact weighting **eliminates all risk**. (Check the risk-free weight formula: $w_A = \sigma_B/(\sigma_A + \sigma_B) = 30/50 = 0.6 \checkmark$, matching our 60/40.)

Reconciliation: return is fixed at 12.4% across all three, while risk collapses $24\% \rightarrow 16.97\% \rightarrow 0\%$ as correlation drops from +1 to 0 to -1 — exactly what Figure 2 predicts.

Example 2 — Minimum-variance portfolio

Using Example 1's data with **$\rho = 0.30$** , so $\sigma_{AB}=0.30(20)(30)=180$. Find the GMVP.

$$w_A^{\min} = \frac{\sigma_B^2 - \sigma_{AB}}{\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}} = \frac{900 - 180}{400 + 900 - 360} = \frac{720}{940} = 0.766$$

So $w_A = 76.6\%$, $w_B = 23.4\%$.

Check the variance: $\sigma_p^2 = 0.766^2(400) + 0.234^2(900) + 2(0.766)(0.234)(180) = 0.587(400) + 0.0548(900) + 2(0.1792)(180) = 234.7 + 49.3 + 64.5 = 348.5 \rightarrow \sigma_p = 18.67\%$

Return: $E(R_p) = 0.766(10) + 0.234(16) = 7.66 + 3.74 = 11.40\%$.

Verification that this is the minimum: try the 60/40 mix instead. $\sigma_p^2 = 144 + 144 + 2(0.24)(180) = 144 + 144 + 86.4 = 374.4 \rightarrow \sigma_p = 19.35\%$. Higher than 18.67% ✓. The GMVP formula genuinely found a lower-risk mix.

Example 3 — Optimal risky portfolio, Sharpe ratio, and the complete portfolio

Same assets, $\rho = 0.30$ ($\sigma_{AB} = 180$), risk-free rate $R_f = 5\%$. Excess returns: A $\rightarrow 10 - 5 = 5\%$; B $\rightarrow 16 - 5 = 11\%$.

Tangency weights (Section 4.8): $w_A^* = \frac{(5)(900) - (11)(180)}{(5)(900) + (11)(400) - (5+11)(180)}$
 $= \frac{4500 - 1980}{4500 + 4400 - 2880} = \frac{2520}{6020} = 0.4186$

So the optimal risky portfolio P is $w_A = 41.9\%$, $w_B = 58.1\%$.

Its return: $E(R_P) = 0.4186(10) + 0.5814(16) = 4.186 + 9.302 = 13.49\%$.

Its variance: $\sigma_P^2 = 0.4186^2(400) + 0.5814^2(900) + 2(0.4186)(0.5814)(180)$
 $= 0.1752(400) + 0.3380(900) + 2(0.2434)(180) = 70.1 + 304.2 + 87.6 = 461.9 \rightarrow \sigma_P = 21.49\%$

Sharpe ratio of P: $S_P = \frac{13.49 - 5}{21.49} = \frac{8.49}{21.49} = 0.395$

Check it beats the GMVP's Sharpe: GMVP had 11.40% return, 18.67% SD $\rightarrow S = (11.40 - 5)/18.67 = 0.343 < 0.395$ ✓. The tangency portfolio genuinely has the steepest CAL.

Complete portfolio for an investor with A = 4: $y^* = \frac{E(R_P) - R_f}{\sigma_P^2} = \frac{0.0849}{4(0.04619)} = \frac{0.0849}{0.1848} = 0.459$

Put **45.9%** in P and **54.1%** in T-bills. The complete portfolio's stats: - Return: $5\% + 0.459(8.49\%) = 5\% + 3.90\% = 8.90\%$ - Risk: $0.459(21.49\%) = 9.86\%$ - Sharpe (unchanged, on the CAL): $(8.90 - 5)/9.86 = 0.395$ ✓

Reconciliation: the complete portfolio sits *on the CAL* (same Sharpe as P, lower risk/return because it is between R_f and P), confirming that mixing the risk-free asset with the single optimal risky portfolio just slides you along one straight line — the essence of two-fund separation.

6. Connections

- **CAPM / Security Market Line (Chapter 05):** The efficient frontier + risk-free asset produces the market portfolio and the CML. CAPM extends this to price *individual* assets: since only systematic risk survives diversification, expected return depends on β , not total σ . $E(R_i) = R_f + \beta_i[E(R_M) - R_f]$ is the direct heir of this chapter.
- **Single-index / multi-factor models:** The variance decomposition (4.5) underlies the market model, and later APT and Fama-French factors, which generalise "systematic risk" to multiple sources.
- **Sharpe ratio & performance evaluation:** The slope of the CAL is the Sharpe ratio — the workhorse metric for ranking funds. Maximising Sharpe = finding the tangency portfolio.
- **Index investing:** Two-fund separation is the theoretical charter for passive investing — if everyone should hold the market portfolio, buy a low-cost index and adjust risk with cash/leverage.
- **Bond & multi-asset allocation:** The same frontier math drives strategic asset allocation across equities, bonds, and alternatives — low cross-asset correlations are exactly why diversified 60/40 portfolios exist.

- **Value at Risk & risk budgeting:** Covariance-driven portfolio variance is the input to VaR and to risk-parity construction.

7. Key Terms

Term	Meaning
Efficient frontier	Set of portfolios giving maximum return for each risk level; upper edge of the feasible set above the GMVP
Minimum-variance frontier	The full left boundary (both branches) of the feasible set
Global Minimum Variance Portfolio (GMVP)	The single lowest-risk portfolio; leftmost tip of the frontier
Systematic risk	Market-wide, undiversifiable risk; measured by β ; the only risk that earns a premium
Unsystematic risk	Firm-specific / idiosyncratic risk; diversifiable to ~ 0 (also called residual or specific risk)
Diversification	Combining imperfectly correlated assets to lower portfolio risk below the weighted average
Covariance / correlation	Co-movement of two assets; $\rho \in [-1, +1]$ drives the diversification benefit
Capital Allocation Line (CAL)	Straight line of risk-free + a risky portfolio combinations; slope = Sharpe ratio
Optimal risky portfolio	Tangency portfolio maximising the Sharpe ratio
Capital Market Line (CML)	The CAL through the market portfolio; prices efficient portfolios via total risk σ
Sharpe ratio	Excess return per unit of total risk, $\frac{E(R_p) - R_f}{\sigma_p}$
Two-fund / Tobin separation	Everyone holds the same risky portfolio; risk preference only sets the risk-free/risky split
Market portfolio	Value-weighted portfolio of all risky assets; the equilibrium tangency portfolio

8. Common Confusions

- **"More stocks always means less risk."** Only *unsystematic* risk falls; risk asymptotes to the systematic floor (average covariance). Beyond ~ 20 – 30 well-chosen stocks the marginal benefit is tiny.
- **Standard deviation vs. beta.** Total risk (σ) matters for a *standalone* portfolio and defines the CML. But for an *individual asset held inside a diversified portfolio*, only its systematic contribution (β) matters and defines the SML. Confusing the two is the classic exam trap.
- **CAL vs. CML.** The CAL is any risk-free-plus-risky line for a given investor/portfolio. The CML is the *specific* CAL through the market portfolio in equilibrium. Every CML is a CAL; not every CAL is the CML.
- **Efficient frontier vs. minimum-variance frontier.** The lower branch (below the GMVP) is part of the minimum-variance frontier but is *not* efficient — it is dominated.

- **Correlation vs. covariance sign.** They share a sign, but only correlation is scale-free and bounded in $[-1, 1]$; diversification power is judged by ρ , not raw covariance.
- **"The GMVP is the best portfolio."** No — the GMVP minimises risk but usually has a lower Sharpe ratio than the tangency portfolio. Once a risk-free asset exists, you want the tangency portfolio, not the GMVP.
- **Diversification eliminates all risk.** Only in the $\rho = -1$ knife-edge case for two assets. In real markets $\rho > 0$ across equities, so a floor always remains.
- **Adding a high-risk asset always raises portfolio risk.** Not necessarily — a high- σ asset with low/negative correlation can *lower* total portfolio risk (why gold or long bonds earn a place despite volatility).

9. First-Principles Recap

Start from three primitives: investors like return, dislike variance, and assets do not move in lockstep.

1. Because assets are imperfectly correlated, the risk of a combination is *less* than the weighted average of individual risks — the cross terms carry $\rho < 1$. This is diversification, and it is free.
2. In a large portfolio, each asset's own variance contribution shrinks like $1/n$ but average covariance does not. So firm-specific (unsystematic) risk washes out and only market-wide (systematic) risk survives. The market can only pay you for what you *cannot* diversify away.
3. Plot all portfolios in σ – $E(R)$ space; discard every dominated one. What remains is the concave efficient frontier, tipped at the global minimum-variance portfolio.
4. Introduce a risk-free asset. Mixing it with any risky portfolio traces a straight line whose slope is the Sharpe ratio. Rotate to the steepest line tangent to the frontier — that tangency point is the one optimal risky portfolio for everyone.
5. Your personal risk appetite only decides *how far along that line* you sit (how much cash vs. risky mix) — not *which* risky mix. That is two-fund separation.
6. In equilibrium the tangency portfolio must be the market portfolio, and the line becomes the Capital Market Line — the launchpad for CAPM.

Everything above is just variance arithmetic plus the refusal to hold a dominated portfolio.

10. Quick-Reference / Interview Points

Formula sheet

Quantity	Formula
Portfolio return	$E(R_p) = \sum w_i E(R_i)$
2-asset variance	$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho \sigma_1 \sigma_2$
n-asset variance	$\sigma_p^2 = \mathbf{w}^T \Sigma \mathbf{w}$
GMVP weight (2-asset)	$w_1 = \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}$
Large-n limit	$\sigma_p^2 \rightarrow \overline{\sigma_{ij}}$ (average covariance)
Risk decomposition	$\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma_{\epsilon}^2$
Beta	$\beta_i = \frac{\sigma_{iM}}{\sigma_M^2}$
Sharpe ratio	$S = (E(R_p) - R_f) / \sigma_p$
CAL / CML	$E(R_p) = R_f + \frac{E(R_M) - R_f}{\sigma_M} \sigma_p$
Optimal risky-asset split	$y^* = \frac{E(R_P) - R_f}{\sigma_P^2}$

What interviewers actually ask

- **"Why don't you just buy the highest-return stock?"** Because risk isn't rewarded linearly and idiosyncratic risk is uncompensated — diversify to kill it, then hold the highest-Sharpe mix.
- **"Difference between systematic and unsystematic risk?"** Market-wide vs. firm-specific; only systematic (β) earns a premium; unsystematic diversifies away.
- **"How many stocks to be diversified?"** Most unsystematic risk gone by ~20–30 low-correlation names; you can never remove systematic risk.
- **"Explain the efficient frontier / why is it concave?"** Set of non-dominated portfolios; concavity comes from $\rho < 1$ diversification bowing the curve left.
- **"CAL vs. CML vs. SML?"** CAL = any R_f +risky line (slope = Sharpe); CML = CAL through the market, prices efficient portfolios by σ ; SML = CAPM line pricing *all* assets by β .
- **"Why is the tangency portfolio special?"** It maximises Sharpe; two-fund separation means everyone holds it and only varies the cash weighting.
- **"If two stocks have $\rho = -1$, what's the min-variance portfolio's risk?"** Zero — with weights $\frac{\sigma_2}{\sigma_1 + \sigma_2}$.
- **"Would you add a volatile asset to a portfolio?"** Yes if its correlation is low/negative — marginal contribution to portfolio risk, not standalone σ , is what matters.

Numbers to remember

- Equity pairwise correlations run roughly 0.3–0.7 → real but incomplete diversification.
- Sharpe ratio of the broad equity market is historically ~0.4–0.5.
- Adding assets past ~30 gives diminishing risk reduction — the systematic floor.

Q&A — The Efficient Frontier and Diversification

Investments | Portfolio Theory | Companion to Chapter 04 | All returns and standard deviations in % p.a. unless stated | Variances quoted in %² (i.e. "400" means (20%)²)

SECTION A — Concept Check (Short Answer)

A1. Why can't an investor simply buy the single highest-expected-return security? Because return arrives bundled with risk, and risk is not rewarded linearly. Part of a single stock's risk — its firm-specific (unsystematic) component — is uncompensated: the market pays nothing for risk you could have diversified away for free. Holding one name forces you to bear that unpaid risk. The rational move is to diversify the idiosyncratic risk to near zero, then hold the combination with the best return-per-unit-of-risk (highest Sharpe ratio).

A2. Define the efficient frontier and distinguish it from the minimum-variance frontier. The minimum-variance frontier is the *entire* left boundary of the feasible set of portfolios — the bullet shape giving the lowest risk for every level of return. The efficient frontier is only its *upper* branch, from the Global Minimum Variance Portfolio (GMVP) upward. The lower branch (below the GMVP) is dominated: for any portfolio there, another portfolio directly above it offers the same risk with higher return, so no rational investor holds it.

A3. Why is the efficient frontier concave (bowed up and to the left) rather than a straight line?

Concavity is the geometric signature of diversification. Because assets are imperfectly correlated ($\rho < 1$), the risk of a combination is *less* than the weighted average of the component risks. The cross-term in the variance formula carries ρ , and any $\rho < 1$ pulls portfolio variance below the linear ($\rho = +1$) benchmark, bending the curve leftward. Only if every pair had $\rho = +1$ would the frontier collapse to a straight line.

A4. Total risk splits into two parts. Name them and state which one is priced. Total risk = systematic (market-wide, undiversifiable) risk + unsystematic (firm-specific, diversifiable) risk. Formally $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma^2(\epsilon_i)$. Only systematic risk is priced — it cannot be diversified away, so the market pays a premium for bearing it. Unsystematic risk earns nothing because a diversified investor has already eliminated it.

A5. As you add equally weighted stocks, what does portfolio variance converge to, and why? It

converges to the *average covariance* between pairs. With $w_i = 1/n$, $\sigma_P^2 = (1/n) \cdot (\text{avg variance}) + (1 - 1/n) \cdot (\text{avg covariance})$. The own-variance term is scaled by $1/n$ and vanishes as $n \rightarrow \infty$ (this is the unsystematic risk washing out), while the average-covariance term survives as the systematic floor. Firm-specific noise averages out; economy-wide shocks hit everything and remain.

A6. Why does combining a risk-free asset with a risky portfolio produce a straight line? The risk-free asset has zero variance and zero covariance with the risky portfolio P. So when you put weight y in P, the complete portfolio's return is linear in y — $E(R_c) = R_f + y[E(R_P) - R_f]$ — and its standard deviation is *also* linear, $\sigma_c = y \cdot \sigma_P$ (no cross term survives). A linear return plotted against a linear risk traces a straight line, the Capital Allocation Line, running from R_f through P.

A7. What is the slope of the CAL, and why do we rotate the line to be tangent to the frontier? The slope is the Sharpe ratio of P, $S_P = [E(R_P) - R_f] / \sigma_P$ — extra return per unit of total risk. A steeper CAL dominates a flatter one (more reward per unit risk at every point), so we rotate the line upward until it just touches the efficient frontier. That tangency point is the maximum-Sharpe risky portfolio, optimal for every investor regardless of risk appetite.

A8. State the Tobin (two-fund) separation theorem and its practical consequence. Every investor, whatever their risk aversion, holds the *same* risky portfolio — the tangency portfolio — and adjusts only the split between it and the risk-free asset. The investment decision (which risky mix) is separated from the financing decision (how much cash vs. risky). Practically, this is the theoretical charter for index investing: hold one broad market fund and dial risk up or down with cash or leverage.

A9. Distinguish the CAL, the CML, and (looking ahead) the SML. The CAL is *any* risk-free-plus-risky-portfolio line for a given investor; its slope is that portfolio's Sharpe ratio. The CML is the *specific* CAL through the market portfolio in equilibrium — it prices *efficient* portfolios using total risk σ . The SML (CAPM, next chapter) prices *all individual assets* using β , not σ . Every CML is a CAL; not every CAL is the CML.

A10. Is the GMVP the best portfolio to hold once a risk-free asset exists? Why or why not? No. The GMVP minimises risk but usually has a lower Sharpe ratio than the tangency portfolio. Once a risk-free asset exists, the optimal risky holding is the tangency (max-Sharpe) portfolio; you reach any desired risk level by mixing it with the risk-free asset along the CAL, which dominates the frontier everywhere except at the single tangency point.

SECTION B — Numerical Problems (Full Step-by-Step)

Base data for B1–B4: Asset A — $E(R_A) = 10\%$, $\sigma_A = 20\%$. Asset B — $E(R_B) = 16\%$, $\sigma_B = 30\%$.

B1 — The diversification effect across correlations

Invest 60% in A, 40% in B. Find portfolio return and risk for $\rho = +1, 0$, and -1 .

Return (independent of ρ): $E(R_p) = 0.6(10) + 0.4(16) = 6 + 6.4 = 12.4\%$.

Weighted-average SD benchmark: $0.6(20) + 0.4(30) = 12 + 12 = 24\%$.

$\rho = +1$: $\sigma_p^2 = 0.36(400) + 0.16(900) + 2(0.6)(0.4)(1)(20)(30) = 144 + 144 + 288 = 576 \rightarrow \sigma_p = 24\%$. Equals the weighted average — no diversification.

$\rho = 0$: $\sigma_p^2 = 144 + 144 + 0 = 288 \rightarrow \sigma_p = \sqrt{288} = 16.97\%$. Risk fell from 24% to 16.97% for the *same* 12.4% return — the free lunch.

$\rho = -1$: $\sigma_p = |0.6(20) - 0.4(30)| = |12 - 12| = 0\%$. Perfect hedge.

Reconciliation: return is pinned at 12.4% throughout while risk collapses $24\% \rightarrow 16.97\% \rightarrow 0\%$ as ρ falls $+1 \rightarrow 0 \rightarrow -1$. Lower correlation $\Rightarrow$ more free risk reduction. The $\rho = -1$ result checks against the risk-eliminating weight $w_A = \sigma_B/(\sigma_A + \sigma_B) = 30/50 = 0.60$, exactly our 60% ✓.

B2 — Global Minimum Variance Portfolio ($\rho = 0.30$)

Covariance $\sigma_{AB} = \rho\sigma_A\sigma_B = 0.30(20)(30) = 180$. Find the GMVP weights, return, and risk.

Weights: $w_A = (\sigma_B^2 - \sigma_{AB}) / (\sigma_A^2 + \sigma_B^2 - 2\sigma_{AB}) = (900 - 180) / (400 + 900 - 360) = 720 / 940 = 0.766$. So $w_A = 76.6\%$, $w_B = 23.4\%$.

Risk: $\sigma_p^2 = 0.766^2(400) + 0.234^2(900) + 2(0.766)(0.234)(180) = 0.587(400) + 0.0548(900) + 2(0.1792)(180) = 234.7 + 49.3 + 64.5 = 348.5 \rightarrow \sigma_p = 18.67\%$.

Return: $E(R_p) = 0.766(10) + 0.234(16) = 7.66 + 3.74 = 11.40\%$.

Verification it is the minimum: test the 60/40 mix at $\rho = 0.30$. $\sigma^2_P = 144 + 144 + 2(0.24)(180) = 144 + 144 + 86.4 = 374.4 \rightarrow \sigma_P = 19.35\% > 18.67\% \checkmark$. The formula genuinely found a lower-risk mix.

B3 — Optimal risky portfolio, Sharpe ratio, complete portfolio ($R_f = 5\%$, $\rho = 0.30$)

Excess returns: $A \rightarrow 10 - 5 = 5\%$; $B \rightarrow 16 - 5 = 11\%$. $\sigma_{AB} = 180$.

Tangency weights: $w_A = [(5)(900) - (11)(180)] / [(5)(900) + (11)(400) - (5 + 11)(180)] = (4500 - 1980) / (4500 + 4400 - 2880) = 2520 / 6020 = 0.4186^*$. So $P = 41.9\% A, 58.1\% B$.

Return of P: $0.4186(10) + 0.5814(16) = 4.186 + 9.302 = \mathbf{13.49\%}$.

Risk of P: $\sigma^2_P = 0.4186^2(400) + 0.5814^2(900) + 2(0.4186)(0.5814)(180) = 0.1752(400) + 0.3380(900) + 2(0.2434)(180) = 70.1 + 304.2 + 87.6 = 461.9 \rightarrow \sigma_P = \mathbf{21.49\%}$.

Sharpe ratio of P: $S_P = (13.49 - 5) / 21.49 = 8.49 / 21.49 = \mathbf{0.395}$.

Verify P beats the GMVP on Sharpe: $GMVP \text{ Sharpe} = (11.40 - 5) / 18.67 = 6.40 / 18.67 = 0.343 < 0.395 \checkmark$. The tangency portfolio has the steepest CAL, as required.

Complete portfolio for A = 4: $y = [E(R_P) - R_f] / (A \cdot \sigma^2_P) = 0.0849 / (4 \times 0.04619) = 0.0849 / 0.18476 = 0.459$. Put 45.9% in P, 54.1% in T-bills. - Return: $5 + 0.459(8.49) = 5 + 3.90 = 8.90\%$. - Risk: $0.459(21.49) = 9.86\%$. - Sharpe (must be unchanged on the CAL): $(8.90 - 5) / 9.86 = 3.90 / 9.86 = 0.395^* \checkmark$.

Reconciliation: the complete portfolio sits on the same CAL as P — identical Sharpe, but lower risk and return because it lies between R_f and P. Mixing the risk-free asset with the one optimal risky portfolio just slides you along a single straight line — two-fund separation in action.

B4 — Large-portfolio limit (systematic floor)

Suppose 50 stocks each have $\sigma = 30\%$ (variance 900) and every pair has covariance 100 (average correlation $\approx 100/900 \approx 0.11$). Equal weights. Find portfolio variance for $n = 50$ and the limit as $n \rightarrow \infty$.

$n = 50$: $\sigma^2_P = (1/n) \cdot (\text{avg variance}) + (1 - 1/n) \cdot (\text{avg covariance}) = (1/50)(900) + (49/50)(100) = 18 + 98 = 116 \rightarrow \sigma_P = \sqrt{116} = \mathbf{10.77\%}$.

$n \rightarrow \infty$: $\sigma^2_P \rightarrow \text{average covariance} = 100 \rightarrow \sigma_P = \sqrt{100} = \mathbf{10\%}$.

Reconciliation: a single stock carried $\sigma = 30\%$; diversifying into 50 cut it to 10.77%, and the theoretical floor is 10%. The extra 0.77% is the last sliver of unsystematic risk (the 18 own-variance term shrinking as n grows). No amount of diversification breaks below 10% — that is the systematic floor equal to the average covariance.

SECTION C — Interview-Style Questions (Model Answers)

C1. "Walk me through why diversification is called a free lunch." It is free because you give up nothing in expected return to get it. Portfolio return is just the weighted average of component returns, so mixing assets never lowers expected return below its components' average. But portfolio risk is below the weighted average whenever assets are imperfectly correlated, because the covariance cross-terms carry $\rho < 1$. So you buy risk reduction at zero cost in return. In B1 the 60/40 mix held return fixed at 12.4% while risk dropped from 24% to 16.97% simply by moving from $\rho = +1$ to $\rho = 0$. That gap is the free lunch, and it is the only such gift in finance.

C2. "A stock is very volatile in isolation. Would you ever add it to a diversified portfolio?" Yes — standalone σ is the wrong lens. What matters is the asset's *marginal contribution* to portfolio risk, which is driven by its covariance with what you already hold, not its own variance. A high- σ asset with low or negative correlation can actually *lower* total portfolio risk. That is exactly why long-duration bonds or gold earn a place in equity portfolios despite being volatile: their low or negative correlation with equities damps the combined swings. The extreme case is B1's $\rho = -1$, where adding the volatile asset in the right weight drove risk to zero.

C3. "How many stocks does it take to be diversified?" Most unsystematic risk is gone by roughly 20–30 well-chosen, low-correlation names; beyond that the marginal benefit is tiny because you are already close to the systematic floor. The math is that own-variance falls like $1/n$ while average covariance does not, so portfolio variance asymptotes to the average covariance — you can never diversify below it. In B4, going from one stock (30%) to fifty (10.77%) captured almost all the benefit, and the infinite-stock floor was still 10%. Adding the 200th stock barely moves the needle.

C4. "Explain CAL versus CML to a client without jargon." A CAL is your personal menu line: pick any risky portfolio, mix it with cash, and you get a straight risk-return line whose steepness is that portfolio's reward-to-risk ratio. There are infinitely many CALs — one per risky portfolio you might pick. The CML is the single best one: the line through the *market portfolio*, which in an efficient market is the steepest line achievable and therefore the one everyone should ride. So the CML is just the winning CAL. The client's only real choice is where to sit on it — more cash for safety, more market (or borrowing) for growth.

C5. "Why does modern finance say only systematic risk is rewarded?" Because rewards must reflect risk you cannot avoid. Unsystematic risk is avoidable for free through diversification, so a competitive market will not pay a premium for bearing it — anyone demanding compensation for diversifiable risk is undercut by an investor who has already diversified it away. What is left, systematic risk, hits every asset and cannot be escaped by any investor, so it commands a premium. This is precisely why CAPM prices assets on β (systematic sensitivity) rather than total σ , and it flows directly from the variance decomposition $\sigma_i^2 = \beta_i^2 \sigma_M^2 + \sigma^2(\epsilon_i)$.

C6. "What is the significance of the tangency portfolio being the market portfolio in equilibrium?" If every investor faces the same information and the same risk-free rate, they all compute the same tangency portfolio and all want to hold it. For markets to clear, that common portfolio must contain every risky asset in proportion to its market value — otherwise some asset has no buyer. So the tangency portfolio is the value-weighted market portfolio, and its CAL becomes the Capital Market Line. This is the hinge between Markowitz's optimisation and CAPM: it converts a personal optimisation into an equilibrium pricing statement about the whole market.

SECTION D — Multiple Choice (with Reasoning)

D1. The lower branch of the minimum-variance frontier (below the GMVP) is: (a) part of the efficient frontier (b) the Capital Market Line (c) dominated and not efficient (d) where the tangency portfolio lies

Answer: (c). For any portfolio on the lower branch, a portfolio directly above it has the same risk and higher return, so it is dominated. Only the upper branch above the GMVP is efficient. (b) and (d) involve the risk-free asset, not the risky-only frontier.

D2. Two assets have $\rho = -1$, $\sigma_1 = 20\%$, $\sigma_2 = 30\%$. The weight in asset 1 that eliminates all risk is: (a) 40% (b) 50% (c) 60% (d) risk cannot be eliminated **Answer: (c).** With $\rho = -1$, $\sigma_p = |w_1\sigma_1 - w_2\sigma_2|$; setting it to

zero gives $w_1 = \sigma_2/(\sigma_1 + \sigma_2) = 30/50 = 0.60$. (d) is wrong precisely because $\rho = -1$ is the knife-edge case where a perfect hedge exists.

D3. As the number of equally weighted, imperfectly correlated stocks grows without bound, portfolio variance approaches: (a) zero (b) the average variance of the individual stocks (c) the average covariance between pairs (d) the market variance σ^2_M **Answer: (c).** $\sigma^2_p = (1/n)(\text{avg variance}) + (1 - 1/n)(\text{avg covariance})$; the own-variance term vanishes as $n \rightarrow \infty$, leaving the average covariance as the systematic floor. It only reaches zero (a) if all pairwise covariances are zero, which real equities never satisfy.

D4. The slope of the Capital Allocation Line equals: (a) beta of the portfolio (b) the Sharpe ratio of the risky portfolio (c) the risk-free rate (d) portfolio variance **Answer: (b).** The CAL is $E(R_c) = R_f + [(E(R_P) - R_f)/\sigma_P]\sigma_c$, so its slope is $(E(R_P) - R_f)/\sigma_P$, the Sharpe ratio. Maximising this slope is what identifies the tangency portfolio.

D5. Which statement about the CML and the SML is correct? (a) both price individual assets using beta (b) the CML uses total risk σ for efficient portfolios; the SML uses β for all assets (c) the CML uses β ; the SML uses σ (d) they are the same line **Answer: (b).** The CML relates expected return of *efficient* portfolios to total risk σ . The SML (CAPM) relates expected return of *any* asset to systematic risk β . Confusing which risk measure applies to which line is the classic exam trap.

D6. An investor with risk aversion $A = 4$ faces an optimal risky portfolio with excess return 8.49% and variance 0.04619. The optimal weight in the risky portfolio is closest to: (a) 0.23 (b) 0.46 (c) 0.92 (d) 1.15 **Answer: (b).** $y = [E(R_P) - R_f]/(A \cdot \sigma^2_P) = 0.0849/(4 \times 0.04619) = 0.0849/0.18476 = 0.459 \approx 0.46$. Matches B3. Since $y < 1$ the investor lends some money at R_f rather than borrowing to leverage.

D7. Adding a risk-free asset to the set of risky portfolios changes the optimal risky holding so that: (a) every investor holds the GMVP (b) every investor holds the same tangency (max-Sharpe) portfolio (c) risk-averse investors hold a different risky mix from risk-tolerant ones (d) the efficient frontier disappears **Answer: (b).** This is two-fund (Tobin) separation: all investors hold the identical tangency portfolio and differ only in the risk-free/risky split. (a) is wrong because the GMVP is not max-Sharpe; (c) contradicts separation.

Self-check note: every numerical answer in Section B was reconciled against an independent route — the $\rho = -1$ hedge weight (B1), a higher-risk comparison mix (B2), the Sharpe ratio recomputed on the complete portfolio and compared to the GMVP (B3), and the single-stock-versus-floor bracket (B4). All formulas match Chapter 04 §4.1–4.10.

Chapter 05 — The Capital Asset Pricing Model (CAPM)

1. The Problem / Need

By the end of the Markowitz–Sharpe story (Chapter 04) we knew *how* a rational investor should build a portfolio: diversify until only **systematic risk** remains, then hold the tangency portfolio combined with the risk-free asset along the Capital Market Line. That is a beautiful prescription, but it leaves a huge practical question unanswered:

What is the *right* expected return for an individual security?

Consider you are an equity analyst at an AMC. You have forecast that Infosys will return 14% next year. Is that good or bad? You cannot answer without a benchmark — a "fair" required return that compensates you for the risk you are bearing. If the fair return is 11%, Infosys is a screaming buy (you earn 3% of *alpha*). If the fair return is 16%, it is overpriced and you should sell.

The problem is deeper than picking a number. Total risk (standard deviation) is the *wrong* yardstick for a single stock, because a well-diversified investor has already thrown away the diversifiable part. A stock that is wildly volatile on its own but whose swings cancel against the rest of the portfolio adds almost no risk to that portfolio. So the market should **not** pay you for volatility it can diversify away. We need a pricing rule that rewards **only the risk that survives diversification**.

CAPM is the answer. It converts the geometric insight of the CML into a formula for *every* asset — traded or not, efficient or not — using a single risk measure: **beta**. It is the workhorse of the finance profession: cost of equity in every DCF, the discount rate in valuation, the benchmark in performance attribution, the hurdle rate in capital budgeting. Interviewers assume you can derive it, use it, and criticise it in your sleep.

2. The Core Idea

CAPM makes one sharp claim:

The expected return of an asset is a linear function of a single risk factor — its sensitivity to the overall market, called beta. Investors are compensated only for systematic (non-diversifiable) risk, and the price of one unit of that risk is the market risk premium.

Formally:

$$E(R_i) = R_f + \beta_i [E(R_m) - R_f]$$

Read it as a sentence: *the return you should expect equals the risk-free rate (reward for waiting) plus your share of market risk (beta) times the reward the market pays per unit of that risk (the equity risk premium).*

Three moves make this work:

1. **Everyone holds the market.** If all investors follow Markowitz and share the same inputs, the single tangency portfolio they all want *is* the market portfolio (every asset held in proportion to its market value). In equilibrium supply must equal demand, so the optimal risky portfolio is the market itself.

2. **Only covariance with the market matters.** When you already hold the market, the risk a new stock adds is not its own variance but how much it *co-moves* with your existing holdings. That co-movement, scaled, is beta.
3. **Reward is linear in beta.** In equilibrium, no asset can offer more or less reward per unit of market covariance than any other — otherwise investors would pile in or flee until prices adjust. The result is a straight line: the **Security Market Line (SML)**.

3. Why / How It Works

From diversification to beta

Recall total risk splits in two:

$$\underbrace{\sigma_i^2}_{\text{total}} = \underbrace{\beta_i^2 \sigma_m^2}_{\text{systematic}} + \underbrace{\sigma_{\epsilon_i}^2}_{\text{unsystematic}}$$

As you add stocks, the unsystematic (idiosyncratic) terms average out toward zero because they are largely uncorrelated. What refuses to diversify away is the part every stock shares — its response to market-wide forces (interest rates, GDP, oil, sentiment). A diversified investor is therefore exposed **only** to systematic risk, so that is the *only* risk the market rewards. Idiosyncratic risk earns nothing, because bearing it is a choice (you could have diversified), not a service to the market.

Why covariance, then beta, is the risk measure

Take a large, well-diversified portfolio and ask: what does adding a tiny slice of stock i do to portfolio risk? The marginal contribution of stock i to portfolio variance is proportional to its **covariance with the portfolio**, not its own variance. Since in equilibrium the relevant portfolio is the market, the risk that "counts" is $\text{Cov}(R_i, R_m)$. Normalising by market variance gives a clean, unit-free sensitivity:

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2} = \rho_{i,m} \frac{\sigma_i}{\sigma_m}$$

Beta answers: *if the market moves 1%, how much does stock i move, on average?* A beta of 1.3 means the stock amplifies market moves by 30%; a beta of 0.7 means it dampens them.

Why the relationship must be linear (the arbitrage-style argument)

Suppose two assets had the same beta but different expected returns. A diversified investor could buy the higher-return one and short the lower-return one, earning a return with *no additional market risk*. Everyone would do this; prices would adjust until the two returns matched. The only configuration where no such "free lunch per unit of beta" exists is a straight line through $(0, R_f)$ and $(1, E(R_m))$. The slope of that line — the extra return per unit of beta — is the market risk premium $E(R_m) - R_f$. That line is the SML, and CAPM is simply its equation.

CML vs SML — a crucial distinction

Students constantly blur these. Both are straight lines, but their axes and scope differ:

Feature	Capital Market Line (CML)	Security Market Line (SML)
X-axis (risk)	Total risk, σ	Systematic risk, β
Y-axis	Expected return	Expected return
Applies to	Efficient portfolios only	All assets and portfolios
Slope	Sharpe ratio of market $= \frac{E(R_m) - R_f}{\sigma_m}$	Market risk premium $= E(R_m) - R_f$
A fairly-priced asset sits...	On the line only if efficient	Always on the line

The CML prices *portfolios* by total risk; the SML prices *anything* by systematic risk. Every asset lies on the SML in equilibrium; only efficient portfolios lie on the CML.

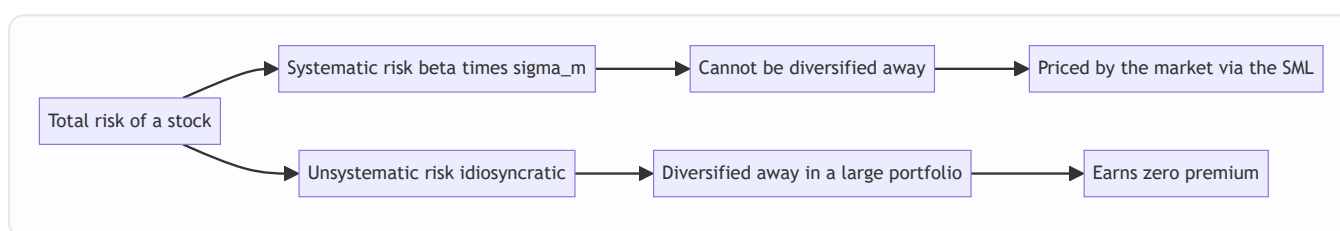


Figure 1 — Only the systematic slice of total risk survives diversification and earns a premium.

4. Full Content — Formulas, Models, Derivations

4.1 The CAPM equation

$$E(R_i) = R_f + \beta_i [E(R_m) - R_f]$$

- R_f = risk-free rate (typically the yield on a government bond matching the horizon; in India, the 10-year G-Sec).
- $E(R_m)$ = expected return on the market portfolio (a broad index proxy — Nifty 50, S&P 500).
- $E(R_m) - R_f$ = **Equity Risk Premium (ERP)** / market risk premium — the reward for holding market risk.
- β_i = sensitivity of asset i to the market.

4.2 Beta — definitions and estimation

$$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)} = \rho_{i,m} \frac{\sigma_i}{\sigma_m}$$

In practice beta is estimated by regressing the stock's excess returns on the market's excess returns (the **market model / characteristic line**):

$$R_i - R_f = \alpha_i + \beta_i (R_m - R_f) + \epsilon_i$$

The slope of that regression is beta; the intercept α_i (Jensen's alpha) measures return unexplained by market risk.

Beta benchmarks

Beta	Meaning	Example type
$\beta > 1$	Aggressive; amplifies market	Autos, banks, tech, cyclicals
$\beta = 1$	Moves with market	The index itself
$0 < \beta < 1$	Defensive; dampens market	FMCG, utilities, pharma
$\beta = 0$	Uncorrelated with market	Risk-free asset (in theory)
$\beta < 0$	Moves opposite	Gold at times, some hedges

Portfolio beta is the value-weighted average of component betas — a linear, additive property that makes beta enormously convenient:

$$\beta_p = \sum_{i=1}^n w_i \beta_i$$

4.3 Levered vs unlevered (asset) beta — the Hamada relation

Observed **equity beta** reflects both business risk and financial leverage. To compare firms with different capital structures, strip out leverage to get the **asset (unlevered) beta**:

$$\beta_L = \beta_U \left[1 + (1 - t) \frac{D}{E} \right]$$

Rearranged to unlever:

$$\beta_U = \frac{\beta_L}{1 + (1 - t) \frac{D}{E}}$$

This is essential in practice: to value a private firm or a division, you take a listed peer's equity beta, **unlever** it (remove the peer's leverage), then **relever** it at the target's own D/E. More debt → higher equity beta → higher cost of equity.

4.4 The Security Market Line

The SML is the graph of CAPM: expected return on the Y-axis, beta on the X-axis.

- **Intercept** = R_f (at $\beta = 0$).
- **Slope** = ERP = $E(R_m) - R_f$.
- At $\beta = 1$, expected return = $E(R_m)$.

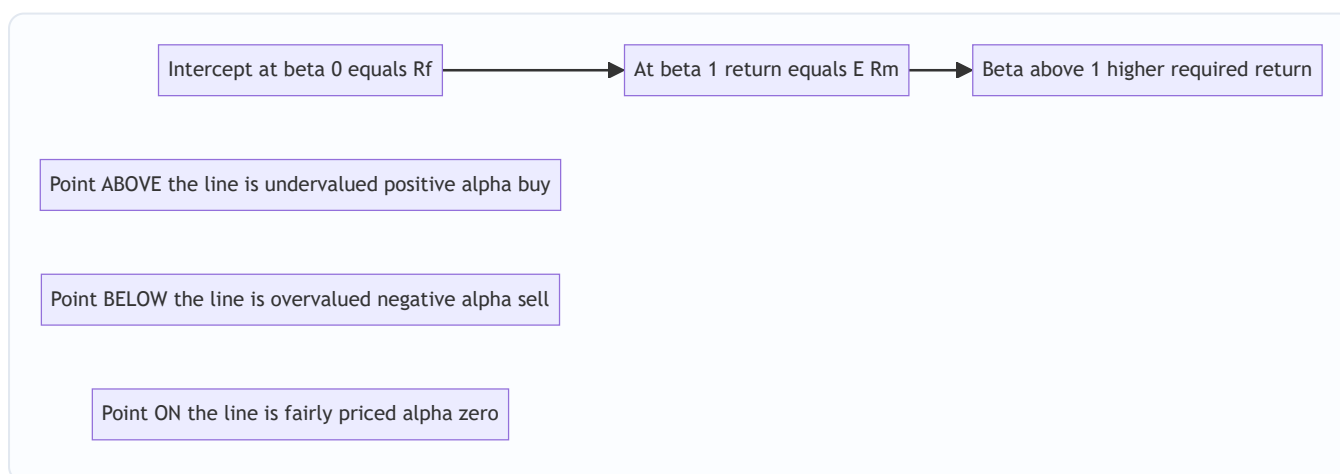


Figure 2 — The Security Market Line: fairly-priced assets sit on it; alpha is vertical distance from it.

4.5 Alpha — the mispricing signal

Alpha is the difference between the return you actually *expect* (your analyst forecast) and the return CAPM says is *fair*:

$$\alpha_i = E(R_i)^{\text{forecast}} - [R_f + \beta_i(E(R_m) - R_f)]$$

- $\alpha > 0$: the stock plots **above** the SML — undervalued, buy.
- $\alpha < 0$: plots **below** the SML — overvalued, sell/short.
- $\alpha = 0$: fairly priced, on the line.

Active managers exist to find positive alpha. In an efficient market, alpha should be zero on average — which is exactly why CAPM is the null hypothesis against which active skill is judged.

4.6 CAPM as cost of equity

CAPM's single most common professional use is supplying the **cost of equity** (k_e) in valuation:

$$k_e = R_f + \beta(E(R_m) - R_f)$$

This k_e then feeds the **Weighted Average Cost of Capital (WACC)**:

$$\text{WACC} = \frac{E}{V}k_e + \frac{D}{V}k_d(1 - t)$$

which is the discount rate in a DCF. So a chain of logic runs: $\beta \rightarrow \text{CAPM} \rightarrow \text{cost of equity} \rightarrow \text{WACC} \rightarrow \text{firm value}$. Get β wrong and the whole valuation moves.

4.7 The assumptions

CAPM is an equilibrium model resting on idealised assumptions:

1. **Rational, mean-variance investors** who care only about expected return and variance over one period.
2. **Homogeneous expectations** — everyone shares the same estimates of returns, variances, covariances (so everyone derives the *same* efficient frontier and tangency portfolio).
3. **A single-period horizon.**
4. **Risk-free borrowing and lending** at the same rate, unlimited.
5. **Frictionless markets** — no taxes, no transaction costs, infinitely divisible assets.
6. **All assets tradable and markets in equilibrium**; investors are price-takers.
7. **Information is free and simultaneously available.**

These are obviously false in detail. The defence is Milton Friedman's: a model is judged by the usefulness of its predictions, not the literal truth of its premises. CAPM's predictions are approximately useful, tractable, and — crucially — give a common language for risk. That is why it endures despite failing formal statistical tests.

5. Worked Examples

Example 1 — Basic required return and mispricing (alpha)

Given: $R_f = 6\%$, expected market return $E(R_m) = 13\%$, stock beta $\beta = 1.4$. Your research forecasts the stock will return **17%**.

Step 1 — Equity risk premium: $\text{ERP} = 13\% - 6\% = 7\%$

Step 2 — CAPM required (fair) return: $E(R) = 6\% + 1.4 \times 7\% = 6\% + 9.8\% = 15.8\%$

Step 3 — Alpha: $\alpha = 17\% - 15.8\% = +1.2\%$

Interpretation: The stock is expected to beat its fair return by 1.2%. It plots **above** the SML → **undervalued** → **buy**.

Self-check: If instead your forecast were 15.8%, $\alpha = 0$ and the stock would sit exactly on the SML (fairly priced). A forecast of 14% would give $\alpha = -1.8\%$ → overvalued → sell. The logic is internally consistent.

Example 2 — Portfolio beta and required return

Given: $R_f = 5\%$, $E(R_m) = 12\%$. A portfolio:

Stock	Weight	Beta
A	40%	0.80
B	35%	1.20
C	25%	1.60

Step 1 — Portfolio beta (weighted average): $\beta_p = 0.40(0.80) + 0.35(1.20) + 0.25(1.60) = 0.320 + 0.420 + 0.400 = 1.14$

Step 2 — Portfolio required return via CAPM: $E(R_p) = 5\% + 1.14 \times (12\% - 5\%) = 5\% + 1.14 \times 7\% = 5\% + 7.98\% = 12.98\%$

Cross-check via component returns. Compute each stock's CAPM return, then weight them: - A: $5\% + 0.80(7\%) = 10.60\%$ - B: $5\% + 1.20(7\%) = 13.40\%$ - C: $5\% + 1.60(7\%) = 16.20\%$

Weighted: $0.40(10.60) + 0.35(13.40) + 0.25(16.20) = 4.24 + 4.69 + 4.05 = 12.98\%$. ✓

Both routes give **12.98%** — because CAPM is linear, averaging betas then applying CAPM equals applying CAPM then averaging returns.

Example 3 — Unlevering and relevering beta for a valuation

Task: Value a private cement firm. A listed comparable, *PeerCo*, has equity beta 1.30, D/E = 0.60, tax rate 30%. Your target firm has D/E = 0.25, tax rate 30%. Use $R_f = 7\%$, ERP = 6%.

Step 1 — Unlever PeerCo's beta (remove its leverage): $\beta_U = \frac{1.30}{1 + (1 - 0.30)(0.60)} = \frac{1.30}{1 + 0.42} = \frac{1.30}{1.42} = 0.9155$

Step 2 — Relever at the target's capital structure: $\beta_L = 0.9155[1 + (1 - 0.30)(0.25)] = 0.9155[1 + 0.175] = 0.9155 \times 1.175 = 1.0757$

Step 3 — Cost of equity for the target: $k_e = 7\% + 1.0757 \times 6\% = 7\% + 6.45\% = 13.45\%$

Interpretation: The target carries less debt than PeerCo, so its equity beta (1.08) is lower than PeerCo's (1.30), and its cost of equity is a moderate 13.45%. Had we lazily used PeerCo's raw beta of 1.30, we would have got $k_e = 7\% + 1.30(6\%) = 14.8\%$ — over-discounting the target and **undervaluing** it by ignoring its lighter leverage. This is exactly the mistake unlevering prevents.

Example 4 — Solving for an implied variable

Given: A stock's fair (CAPM) return is known to be 14%, $R_f = 6\%$, $\beta = 1.6$. What ERP does this imply?

$$8\% = 6\% + 1.6 \times \text{ERP} \Rightarrow \text{ERP} = \frac{2\%}{1.6} = 1.25\%$$

And the implied market return: $E(R_m) = R_f + \text{ERP} = 6\% + 1.25\% = 7.25\%$. Interviewers love making you rearrange CAPM for any of its four inputs — practise solving for each.

Example 5 — Ranking managers with Treynor (CAPM-family)

Given: $R_f = 5\%$. Two funds:

Fund	Realised return	Beta	Std dev
X	14%	1.30	22%
Y	12%	0.80	15%

Fund X earned more in absolute terms, but is that skill or just more market risk? Use **Treynor** (reward per unit of *systematic* risk):

- X: $\$(14 - 5)/1.30 = 9/1.30 = 6.92\%$
- Y: $\$(12 - 5)/0.80 = 7/0.80 = 8.75\%$

Fund Y wins — it delivered more return per unit of beta despite the lower headline number. This is the CAPM lens in action: raw returns mislead; risk-adjusted returns reveal skill. Note Treynor uses beta (systematic risk) because it assumes the investor is diversified; the Sharpe ratio would use σ instead and is the right choice for an *undiversified* investor. Confirm the intuition: Y's excess return of 7% on beta 0.80 is a steeper reward slope than X's 9% on beta 1.30. ✓

5.6 A note on estimating the ERP

Two schools exist. The **historical** approach averages realised (market – risk-free) returns over decades — simple but backward-looking and sensitive to the window. The **forward-looking / implied** approach backs the ERP out of current prices via a dividend-discount or earnings model — it asks what premium today's index level implies. Professionals often blend both and sanity-check against survey data. For India, higher country and growth risk pushes the commonly-cited ERP above developed-market levels; always state your source and reasoning rather than quoting a single "textbook" number.

6. Connections

- **Chapter 04 (Portfolio Theory):** CAPM is the equilibrium endpoint of Markowitz–Sharpe. The CML (Ch. 04) prices efficient portfolios by total risk; the SML (this chapter) generalises to all assets by systematic risk.
- **Chapter 03 (Risk & diversification):** the split of total risk into systematic + unsystematic is the *foundation* — CAPM prices only the systematic part.
- **Valuation / DCF (later chapters):** CAPM → cost of equity → WACC → discount rate. Terminal value, intrinsic value, and every buy/sell target price trace back to a beta.
- **Efficient Market Hypothesis:** CAPM is the natural null. If markets are efficient, $\alpha \approx 0$ and CAPM prices are "fair"; active management's job is to find deviations.
- **Multifactor models (APT, Fama-French):** these extend CAPM by adding factors (size, value, momentum, quality) after research showed a single market beta explains too little. CAPM is the one-factor special case.

- **Performance measurement:** Sharpe (total risk), Treynor (beta), and Jensen's alpha (CAPM residual) are all CAPM-family metrics. Treynor ratio $\frac{R_p - R_f}{\beta_p}$ is literally "reward per unit of SML risk".
- **Capital budgeting:** a project's hurdle rate should reflect the project's *own* beta, not the firm's — a low-beta utility should not use a high-beta parent's WACC.

7. Key Terms

Term	Definition
CAPM	Model giving required return as $R_f + \beta(E(R_m) - R_f)$.
Beta (β)	Sensitivity of an asset's return to market return; systematic-risk measure.
Systematic risk	Market-wide, non-diversifiable risk; the only risk CAPM rewards.
Unsystematic risk	Firm-specific, diversifiable risk; earns no premium.
Equity Risk Premium (ERP)	$E(R_m) - R_f$; reward per unit of beta; slope of the SML.
Security Market Line (SML)	Graph of CAPM; expected return vs beta; all assets lie on it in equilibrium.
Capital Market Line (CML)	Return vs total risk for efficient portfolios only.
Alpha (α)	Return above (or below) the CAPM-fair return; mispricing / skill measure.
Characteristic line	Regression of stock excess return on market excess return; slope = beta.
Asset (unlevered) beta	Beta stripped of financial leverage; reflects pure business risk.
Homogeneous expectations	Assumption that all investors share the same inputs.
Market portfolio	Value-weighted portfolio of all assets; the tangency portfolio in equilibrium.

8. Common Confusions

1. **"Beta measures total risk."** No — beta measures *systematic* risk only. Standard deviation measures total risk. A stock can have high total volatility but low beta if its swings are idiosyncratic.
2. **CML vs SML.** CML uses σ (total risk) and applies only to efficient portfolios; SML uses β and applies to everything. Do not plot individual stocks on the CML.
3. **"Higher beta always means higher realised return."** CAPM is about *expected* return in equilibrium, not guaranteed outcomes. High-beta stocks are *expected* to earn more as compensation, but can and do underperform — indeed the empirical "low-beta anomaly" shows low-beta stocks have historically delivered better risk-adjusted returns than CAPM predicts.
4. **Confusing alpha's sign for direction of price.** Positive alpha = undervalued = *buy* (price should *rise*). Some students invert this. Above the SML = good = *buy*.
5. **Using raw peer beta without unlevering.** Leverage inflates equity beta. Always unlever a comparable and relever at your target's capital structure.
6. **Treating R_f and $E(R_m)$ as the same across horizons.** Match the risk-free rate to the investment horizon; use a consistent, forward-looking ERP.
7. **"Negative beta is impossible."** Rare but real — assets that hedge the market (some gold positions, certain derivatives) can carry negative beta and, by CAPM, a *required return below the risk-free rate*

because they reduce portfolio risk.

8. **Expecting alpha to persist.** In competitive markets alpha is arbitrated away; CAPM's baseline is that alpha averages zero.

9. First-Principles Recap

Strip everything back and rebuild:

1. Rational investors diversify. Diversification erases firm-specific risk for free.
2. Therefore the market should reward *only* the risk that cannot be diversified — systematic risk.
3. When everyone holds the optimal diversified portfolio, that portfolio is the market itself.
4. The risk a single stock contributes to the market portfolio is its *covariance* with the market, scaled to a unit-free number: **beta**.
5. In equilibrium, no asset can pay a different reward per unit of beta than another (else arbitrage). So expected return is a **straight line** in beta.
6. That line starts at R_f (zero-beta reward) and rises at slope = ERP. Its equation is $E(R_i) = R_f + \beta_i(E(R_m) - R_f)$.
7. Assets on the line are fairly priced; above it undervalued (positive alpha); below overvalued.

Everything else — cost of equity, WACC, Treynor, Jensen's alpha, unlevering — is application of these seven steps.

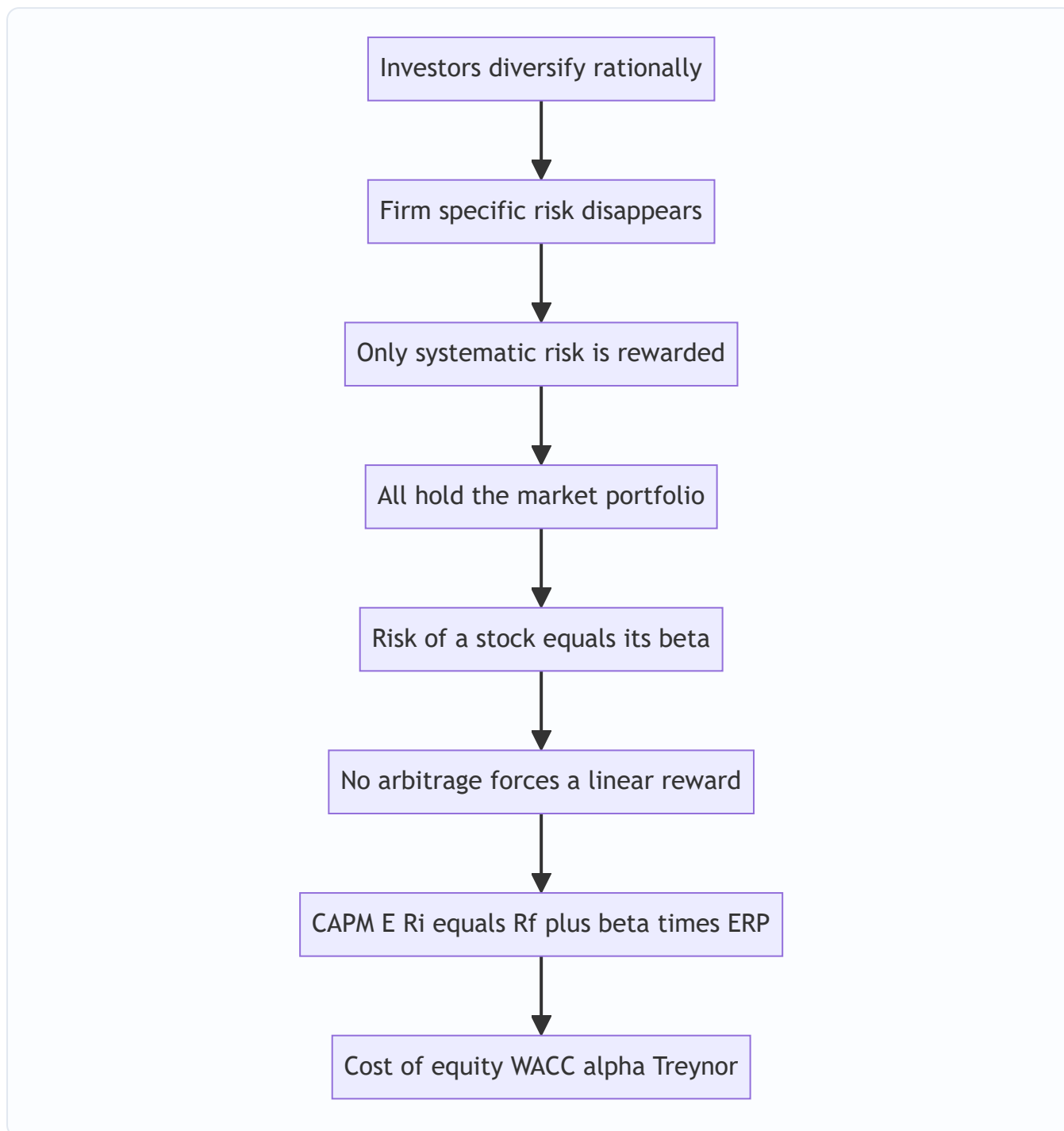


Figure 3 — The logical chain from diversification to the CAPM formula and its applications.

10. Criticisms & Quick-Reference / Interview Points

10.1 Criticisms (know these cold)

- **Roll's Critique (1977):** the true market portfolio (all assets globally — stocks, bonds, property, human capital) is unobservable. Any test of CAPM is really a joint test of CAPM *and* whether your chosen proxy (e.g., Nifty) is efficient. So CAPM is arguably untestable.
- **Empirical failure of the security market line:** the real relationship between beta and return is **flatter** than CAPM predicts. Low-beta stocks earn more, and high-beta stocks less, than the model says — the **low-beta anomaly**, exploited by "betting-against-beta" strategies.

- **Missing factors:** Fama–French showed **size** (small caps) and **value** (high book-to-market) explain returns that beta cannot. Later factors — momentum, profitability, investment — followed. A single market beta is too coarse. This spawned APT and multifactor models.
- **Unstable, noisy beta:** beta estimates depend on the estimation window, return frequency, and index chosen; they drift over time (hence Blume's adjustment toward 1.0).
- **Unrealistic assumptions:** no single borrowing/lending rate, taxes and frictions exist, expectations are heterogeneous, horizons are multi-period.
- **Single period:** ignores that risk premia and betas change through the cycle (addressed by the Intertemporal CAPM and Consumption CAPM).

Despite all this, CAPM survives because it is simple, gives a defensible discount rate, and no successor is decisively better in practice. The pragmatic view: *use CAPM for the cost of equity, sanity-check with a multifactor model.*

10.2 Formula cheat-sheet

Quantity	Formula
CAPM required return	$E(R_i) = R_f + \beta_i(E(R_m) - R_f)$
Beta	$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2} = \rho_{i,m} \frac{\sigma_i}{\sigma_m}$
Portfolio beta	$\beta_p = \sum w_i \beta_i$
Alpha (Jensen's)	$\alpha_i = R_i^{\text{actual/forecast}} - [R_f + \beta_i(E(R_m) - R_f)]$
Levered beta (Hamada)	$\beta_L = \beta_U [1 + (1-t)\frac{D}{E}]$
Cost of equity	$k_e = R_f + \beta(E(R_m) - R_f)$
Treynor ratio	$\frac{R_p - R_f}{\beta_p}$
SML slope / ERP	$E(R_m) - R_f$

10.3 What interviewers actually ask

- **"Walk me through CAPM."** State the formula, define each term, explain that beta is systematic risk, and that only systematic risk is priced because the rest is diversifiable.
- **"Difference between CML and SML?"** Axes (σ vs β), scope (efficient portfolios vs all assets), slope (Sharpe vs ERP).
- **"A stock plots above the SML — what do you do?"** Positive alpha → undervalued → buy; expect price to rise until it returns to the line.
- **"How do you get cost of equity for a private company?"** Take a listed peer's beta, unlever it, relever at the target's D/E, plug into CAPM.
- **"What's wrong with CAPM?"** Roll's critique, empirical SML too flat / low-beta anomaly, size and value factors (Fama-French), unstable beta, unrealistic assumptions.
- **"Why doesn't the market reward unsystematic risk?"** Because it can be diversified away for free; bearing it is a choice, not a service.
- **"What is a reasonable ERP for India?"** Typically discussed around 6–8% (higher than developed markets' ~4–5% due to higher country/growth risk) — be ready to justify a number, not memorise one.

- **Numerical drills:** solve CAPM for any of R_f , $E(R_m)$, β , or $E(R_i)$; compute portfolio beta; compute alpha; unlever/relever beta. Practise until reflexive.

One-line summary to leave them with: *CAPM prices risk by saying you are paid the risk-free rate plus beta times the market risk premium — rewarded only for the market risk you cannot diversify away.*

Q&A — The Capital Asset Pricing Model (CAPM)

Practice bank for Chapter 05. Work each question before reading the answer. All figures are in percentages unless stated; betas are unit-free.

Section A — Concept Checks

A1. In one sentence, what does CAPM claim? The expected (required) return on any asset equals the risk-free rate plus its beta times the market risk premium — i.e. investors are paid only for systematic risk, priced at the market's risk premium per unit of beta: $E(R_i) = R_f + \beta_i[E(R_m) - R_f]$.

A2. Why does the market refuse to reward unsystematic risk? Because unsystematic (firm-specific) risk can be diversified away for free. A diversified investor bears none of it, so bearing it is a *choice*, not a service to the market. Only systematic risk survives diversification, so only it earns a premium.

A3. Define beta two different ways. Statistically, $\beta_i = \frac{\text{Cov}(R_i, R_m)}{\sigma_m^2}$. Equivalently, $\beta_i = \rho_{i,m} \frac{\sigma_i}{\sigma_m}$. Intuitively, beta answers: if the market moves 1%, how much does the stock move on average?

A4. Why is covariance with the market — not the stock's own variance — the relevant risk? When you already hold the market portfolio, the marginal risk a new stock adds is proportional to how it co-moves with your existing holdings, i.e. its covariance with the market, not its standalone variance. Its own variance is mostly diversified away.

A5. Distinguish the CML from the SML on four dimensions. X-axis: CML uses total risk σ ; SML uses systematic risk β . Scope: CML prices *efficient portfolios only*; SML prices *all* assets and portfolios. Slope: CML slope is the market Sharpe ratio $\frac{E(R_m) - R_f}{\sigma_m}$; SML slope is the ERP $E(R_m) - R_f$. Equilibrium: every asset lies on the SML; only efficient portfolios lie on the CML.

A6. Why must the risk–return relationship be linear in beta? No-arbitrage. If two assets shared the same beta but different expected returns, investors would buy the higher and short the lower, earning return with no extra market risk. Prices adjust until reward per unit of beta is identical for all assets — a straight line through $(0, R_f)$ and $(1, E(R_m))$.

A7. What does a point above the SML mean, and what action follows? Positive alpha → the asset is expected to beat its fair return → it is undervalued → buy. Its price should rise until it falls back onto the line.

A8. Give the sign and meaning of a negative beta. A negative-beta asset moves opposite to the market (e.g. certain hedges/gold at times). By CAPM its required return is *below* the risk-free rate, because it reduces portfolio risk — investors pay for that insurance by accepting a lower return.

A9. Why is portfolio beta so convenient? It is a simple value-weighted average: $\beta_p = \sum w_i \beta_i$. Beta is linear and additive, so you can aggregate portfolio systematic risk directly without re-estimating covariances.

A10. State CAPM's key assumptions in brief. Rational mean-variance investors; homogeneous expectations; single-period horizon; unlimited risk-free borrowing and lending at one rate; frictionless markets (no taxes/costs, divisible assets); all assets tradable, markets in equilibrium, investors are price-takers; free simultaneous information.

A11. Why does CAPM survive despite failing statistical tests? Friedman's defence: a model is judged by the usefulness of its predictions, not the literal truth of its assumptions. CAPM gives a tractable, defensible discount rate and a common language for risk, and no successor is decisively better in practice.

A12. What is alpha, precisely? The gap between the return you *forecast* and the return CAPM says is *fair*: $\alpha_i = E(R_i)^{\text{forecast}} - [R_f + \beta_i(E(R_m) - R_f)]$. It measures mispricing or manager skill; in an efficient market it averages zero.

Section B — Numerical Problems (full working)

B1. Required return and alpha. Given $R_f = 6\%$, $E(R_m) = 13\%$, $\beta = 1.4$; your forecast return is 17%. Find the fair return and alpha.

- $ERP = 13 - 6 = 7\%$.
- Fair return $= 6 + 1.4(7) = 6 + 9.8 = 15.8\%$.
- Alpha $= 17 - 15.8 = +1.2\%$.

Interpretation: positive alpha $\rightarrow$ plots above the SML $\rightarrow$ undervalued $\rightarrow$ buy. **Reconcile:** if the forecast were exactly 15.8%, alpha = 0 (on the line); a 14% forecast gives alpha $= -1.8\% \rightarrow$ overvalued $\rightarrow$ sell. Internally consistent.

B2. Portfolio beta and return — two routes. $R_f = 5\%$, $E(R_m) = 12\%$. Portfolio: A (40%, β 0.80), B (35%, β 1.20), C (25%, β 1.60).

Route 1 — beta first: $\beta_p = 0.40(0.80) + 0.35(1.20) + 0.25(1.60) = 0.320 + 0.420 + 0.400 = 1.14$. $E(R_p) = 5 + 1.14(12 - 5) = 5 + 1.14(7) = 5 + 7.98 = 12.98\%$.

Route 2 — returns first: A: $\$5 + 0.80(7) = 10.60\%$; B: $\$5 + 1.20(7) = 13.40\%$; C: $\$5 + 1.60(7) = 16.20\%$. Weighted: $\$0.40(10.60) + 0.35(13.40) + 0.25(16.20) = 4.24 + 4.69 + 4.05 = 12.98\%$. ✓

Reconcile: both routes give 12.98% because CAPM is linear — averaging betas then applying CAPM equals applying CAPM then averaging returns.

B3. Unlever and relever beta for a valuation. PeerCo: equity beta 1.30, D/E 0.60, tax 30%. Target: D/E 0.25, tax 30%. $R_f = 7\%$, ERP 6%.

- Unlever: $\beta_U = \frac{1.30}{1 + (1-0.30)(0.60)} = \frac{1.30}{1.42} = 0.9155$.
- Relever at target: $\beta_L = 0.9155[1 + (1-0.30)(0.25)] = 0.9155(1.175) = 1.0757$.
- Cost of equity: $k_e = 7 + 1.0757(6) = 7 + 6.45 = 13.45\%$.

Reconcile: the lazy shortcut (use PeerCo's raw 1.30) gives $k_e = 7 + 1.30(6) = 14.8\%$ — over-discounting and undervaluing the target, which carries less debt. Unlevering corrects for the different capital structures.

B4. Solve CAPM for the ERP. A stock's fair return is 14%, $R_f = 6\%$, $\beta = 1.6$. Find the implied ERP and market return. $14 = 6 + 1.6 \times \text{ERP} \rightarrow \text{ERP} = \frac{8}{1.6} = 5\%$. Then $E(R_m) = 6 + 5 = 11\%$. **Reconcile:** plug back: $\$6 + 1.6(5) = 6 + 8 = 14\%$. ✓

B5. Solve CAPM for beta. $R_f = 4\%$, $E(R_m) = 11\%$; a stock's required return is 15.1%. Find beta. $15.1 = 4 + \beta(11 - 4) \rightarrow \beta = \frac{11.1}{7} = 1.586$. **Reconcile:** $\$4 + 1.586(7) = 4 + 11.1 = 15.1\%$. ✓ A beta above 1 is consistent with a required return above the market's 11%.

B6. Beta from correlation and volatilities. $\rho_{i,m} = 0.6$, $\sigma_i = 30\%$, $\sigma_m = 18\%$. Find beta, then required return with $R_f = 5\%$, $E(R_m) = 12\%$. $\beta = 0.6 \times \frac{30}{18} = 0.6 \times$

$1.667 = 1.0$. $E(R) = 5 + 1.0(7) = 12\%$. **Reconcile:** despite the stock's much higher standalone volatility (30% vs 18%), its modest correlation pulls beta to exactly 1, so it earns the market return. This shows why total risk (σ) is the wrong yardstick — only the co-moving part is priced.

B7. Treynor ranking (CAPM-family). $R_f = 5\%$. Fund X: return 14%, β 1.30. Fund Y: return 12%, β 0.80. - Treynor X $= (14 - 5)/1.30 = 9/1.30 = 6.92\%$. - Treynor Y $= (12 - 5)/0.80 = 7/0.80 = 8.75\%$.

Y wins — more reward per unit of systematic risk despite the lower headline return. **Reconcile:** X's higher 14% is largely compensation for taking more market risk (β 1.30), not skill. Treynor uses beta because it assumes a diversified investor; Sharpe (using σ) would be the right lens for an undiversified investor.

B8. Jensen's alpha for a fund. $R_f = 6\%$, $E(R_m) = 13\%$. A fund realised 15% with β 1.1. Fair return $= 6 + 1.1(7) = 13.7\%$. Jensen's alpha $= 15 - 13.7 = +1.3\%$. **Reconcile:** positive alpha means the fund beat its risk-adjusted benchmark by 1.3 points — genuine outperformance after accounting for the market risk it took.

B9. Decomposing total risk. A stock has β 1.2, $\sigma_m = 20\%$, and total variance $\sigma_i^2 = 0.09$ ($\sigma = 30\%$). Find systematic and unsystematic variance. Systematic $= \beta^2 \sigma_m^2 = 1.2^2 (0.20)^2 = 1.44 \times 0.04 = 0.0576$. Unsystematic $= 0.09 - 0.0576 = 0.0324$. **Reconcile:** systematic is 64% of total variance ($0.0576/0.09$), diversifiable is 36%. Only the 0.0576 slice is priced by the SML; the rest earns no premium.

Section C — Interview-Style Questions (model answers)

C1. "Walk me through CAPM." CAPM says the required return on an asset is the risk-free rate plus beta times the market risk premium: $R_f + \beta(E(R_m) - R_f)$. R_f compensates you for waiting; the second term compensates you for market risk. Beta measures systematic risk — sensitivity to the market. The model's core claim is that only systematic risk is priced, because idiosyncratic risk is diversifiable and bearing it earns nothing. It emerges from Markowitz portfolio theory: when everyone holds the tangency portfolio, in equilibrium that portfolio is the market, and no-arbitrage forces expected return to be linear in beta — the Security Market Line.

C2. "What's the difference between the CML and the SML?" The CML plots expected return against total risk (σ) and applies only to efficient portfolios — combinations of the risk-free asset and the market. Its slope is the market Sharpe ratio. The SML plots expected return against beta and applies to every asset and portfolio, efficient or not. Its slope is the equity risk premium. A key error is plotting individual stocks on the CML — they belong on the SML because their relevant risk is systematic, not total.

C3. "A stock plots above the SML. What do you do and why?" Above the line means its expected return exceeds the CAPM-fair return — positive alpha, so it is undervalued. I'd buy it. As other investors recognise the mispricing and buy, the price rises, expected return falls, and the stock moves back down onto the line. In an efficient market this alpha is arbitrated away, which is exactly why CAPM is the null hypothesis for judging active skill.

C4. "How do you estimate the cost of equity for a private company?" The company has no traded beta, so I take a listed comparable, unlever its equity beta to strip out its financial leverage (getting the asset/business-risk beta), then relever at the target's own D/E using the Hamada relation $\beta_L = \beta_U[1 + (1-t)\frac{D}{E}]$. Plug that target beta into CAPM with an appropriate R_f and ERP. This

isolates pure business risk and then adds back the target's specific leverage, avoiding the error of importing the peer's capital structure.

C5. "What's wrong with CAPM?" Several things. Roll's critique: the true market portfolio (all global assets including human capital) is unobservable, so any test is a joint test of CAPM and the proxy's efficiency — arguably it's untestable. Empirically the SML is too flat: low-beta stocks earn more and high-beta stocks less than predicted (the low-beta anomaly). Fama–French showed size and value explain returns beta cannot, spawning multifactor models. Beta itself is unstable and noisy, depending on window, frequency, and index. And the assumptions — one borrowing rate, no frictions, homogeneous expectations, single period — are unrealistic. Still, I'd use it for cost of equity and sanity-check with a multifactor model.

C6. "Why doesn't the market reward unsystematic risk?" Because it can be eliminated for free through diversification. If the market paid a premium for diversifiable risk, an arbitrageur could build a diversified portfolio that captures the premium while the risk itself cancels out — a free lunch. Competition removes it, so in equilibrium only non-diversifiable (systematic) risk carries a premium.

C7. "Two stocks have the same standard deviation but different betas. Which has the higher required return?" The one with the higher beta. CAPM prices systematic risk, not total risk. Equal σ means equal total volatility, but the higher-beta stock co-moves more with the market — it contributes more non-diversifiable risk to a diversified portfolio — so it commands a higher required return. The lower-beta stock's volatility is more idiosyncratic and largely diversified away.

C8. "What's a reasonable equity risk premium for India, and how would you estimate it?" Commonly discussed around 6–8%, higher than developed markets' roughly 4–5% because of greater country and growth risk. I'd estimate it two ways: a historical average of realised (market – risk-free) returns over a long window, and a forward-looking implied premium backed out of current index levels via a dividend-discount or earnings model. I'd blend them, sanity-check against surveys, and always state my source rather than quoting a single textbook number.

Section D — MCQs (with reasoning)

D1. The slope of the Security Market Line equals: (a) the market Sharpe ratio (b) $E(R_m) - R_f$ (c) $\frac{E(R_m) - R_f}{\sigma_m}$ (d) beta **Answer: (b).** The SML's slope is the equity risk premium $E(R_m) - R_f$. The Sharpe ratio $\frac{E(R_m) - R_f}{\sigma_m}$ is the slope of the CML, not the SML.

D2. A stock with beta 0 has a CAPM required return equal to: (a) 0 (b) $E(R_m)$ (c) R_f (d) the ERP **Answer: (c).** At $\beta = 0$, $E(R) = R_f + 0 \times \text{ERP} = R_f$. Zero systematic risk earns only the risk-free rate.

D3. Which risk measure belongs on the x-axis of the SML? (a) standard deviation (b) variance (c) beta (d) covariance **Answer: (c).** The SML prices assets by systematic risk, measured by beta. Standard deviation (total risk) is the CML's axis.

D4. Increasing a firm's financial leverage, holding business risk constant, will: (a) lower its equity beta (b) raise its equity beta (c) leave equity beta unchanged (d) lower its asset beta **Answer: (b).** By Hamada, $\beta_L = \beta_U [1 + (1-t)\frac{D}{E}]$ — more debt (higher D/E) raises the multiplier and thus the equity beta and cost of equity. Asset (unlevered) beta reflects business risk and is unaffected.

D5. A stock plotting below the SML is: (a) undervalued, buy (b) fairly priced (c) overvalued, sell (d) risk-free **Answer: (c).** Below the line means the expected return is less than the fair return — negative alpha — so it is

overvalued; sell or short.

D6. Portfolio beta is computed as: (a) the simple average of component betas (b) the value-weighted average of component betas (c) the square root of weighted variances (d) covariance with the market

Answer: (b). $\beta_p = \sum w_i \beta_i$, a value-weighted average — beta is linear and additive. A simple average only coincides when weights are equal.

D7. Roll's critique argues that CAPM is problematic because: (a) beta is always negative (b) the true market portfolio is unobservable (c) the risk-free rate is unknown (d) alpha is always positive **Answer: (b).** The true market portfolio includes all assets globally (including human capital) and cannot be observed, so any empirical test is a joint test of CAPM and whether the chosen proxy is efficient.

D8. The Treynor ratio measures reward per unit of: (a) total risk (b) unsystematic risk (c) systematic risk (beta) (d) variance **Answer: (c).** Treynor $= (R_p - R_f) / \beta_p$ — excess return per unit of systematic risk. Sharpe uses σ (total risk) instead and suits an undiversified investor.

D9. Under CAPM, an asset with negative beta should have a required return that is: (a) above the market return (b) equal to R_f (c) below the risk-free rate (d) always zero **Answer: (c).** A negative beta reduces portfolio risk (acts as a hedge), so investors accept a required return below R_f : $R_f + (\text{negative}) \times \text{ERP} < R_f$.

D10. Which of the following would earn NO risk premium under CAPM? (a) systematic risk (b) market risk (c) unsystematic (firm-specific) risk (d) beta risk **Answer: (c).** Firm-specific risk is diversifiable, so the market pays nothing for it. Options (a), (b), and (d) all describe the same priced, non-diversifiable risk.

End of Chapter 05 Q&A. Drill Section B until CAPM rearrangement for any of its four inputs — R_f , $E(R_m)$, β , $E(R_i)$ — is reflexive, and be able to unlever/relever beta on demand.

Chapter 06 — APT and Multifactor Models

1. The Problem / The Need

The Capital Asset Pricing Model (CAPM) from the previous chapter gave us something beautiful: a single equation that says an asset's expected return depends on exactly one thing — its sensitivity to the market portfolio (beta). Clean, testable, Nobel-winning. But when researchers took CAPM to real data, it started leaking.

Three problems piled up:

1. **Empirically, beta alone doesn't explain returns.** When Eugene Fama and Kenneth French ran the numbers in 1992, they found that the relationship between beta and average return was almost *flat* over 1963–1990. Two stocks with very different betas earned roughly similar returns, while stocks that CAPM said should be identical earned very different returns. The single-factor story was incomplete.
2. **Persistent "anomalies" appeared.** Small-cap stocks systematically beat large-caps. Cheap "value" stocks (high book-to-market) systematically beat expensive "growth" stocks. High-momentum stocks kept winning. Low-volatility stocks earned more than their beta predicted. None of this should happen in a world where beta is the *only* priced risk.
3. **CAPM rests on heroic assumptions.** It needs a mean-variance-efficient *market portfolio* that includes every risky asset (equities, bonds, real estate, human capital) — something we can never actually observe or measure (Roll's Critique). It also assumes everyone holds the same portfolio, cares only about one period, and agrees on all inputs.

The practical need is sharp. If you are an equity research analyst or a portfolio manager at an AMC, you need a return-generating model that (a) actually fits the data, (b) tells you *which* risks you are being paid to bear, and (c) lets you decompose a fund's performance into "did the manager pick good stocks, or did they just tilt toward small/value factors that anyone could have bought cheaply?" CAPM cannot do this. **Arbitrage Pricing Theory (APT)** and the **multifactor models** it inspired can.

The core question this chapter answers: *if a single market factor isn't enough, what set of systematic risk factors actually drives expected returns — and how do we price them?*

2. The Core Idea

There are two intellectual roots here, and it helps to keep them separate.

Arbitrage Pricing Theory (APT), developed by Stephen Ross in 1976, is a *theory of how assets must be priced* if no arbitrage is possible. It says: an asset's return is driven by a handful of common **systematic factors** plus firm-specific noise. If you build a portfolio that has *zero* exposure to every factor and costs nothing to assemble, it must earn zero return — otherwise you'd have a money machine. That single no-arbitrage condition forces expected returns into a linear relationship with factor exposures. APT does *not* tell you what the factors are; it tells you the *structure* the pricing must take.

Multifactor models are the empirical descendants. They say: "Fine, if returns load on several factors, let's go find them." Researchers identified specific, measurable factors — market, size, value, profitability, investment,

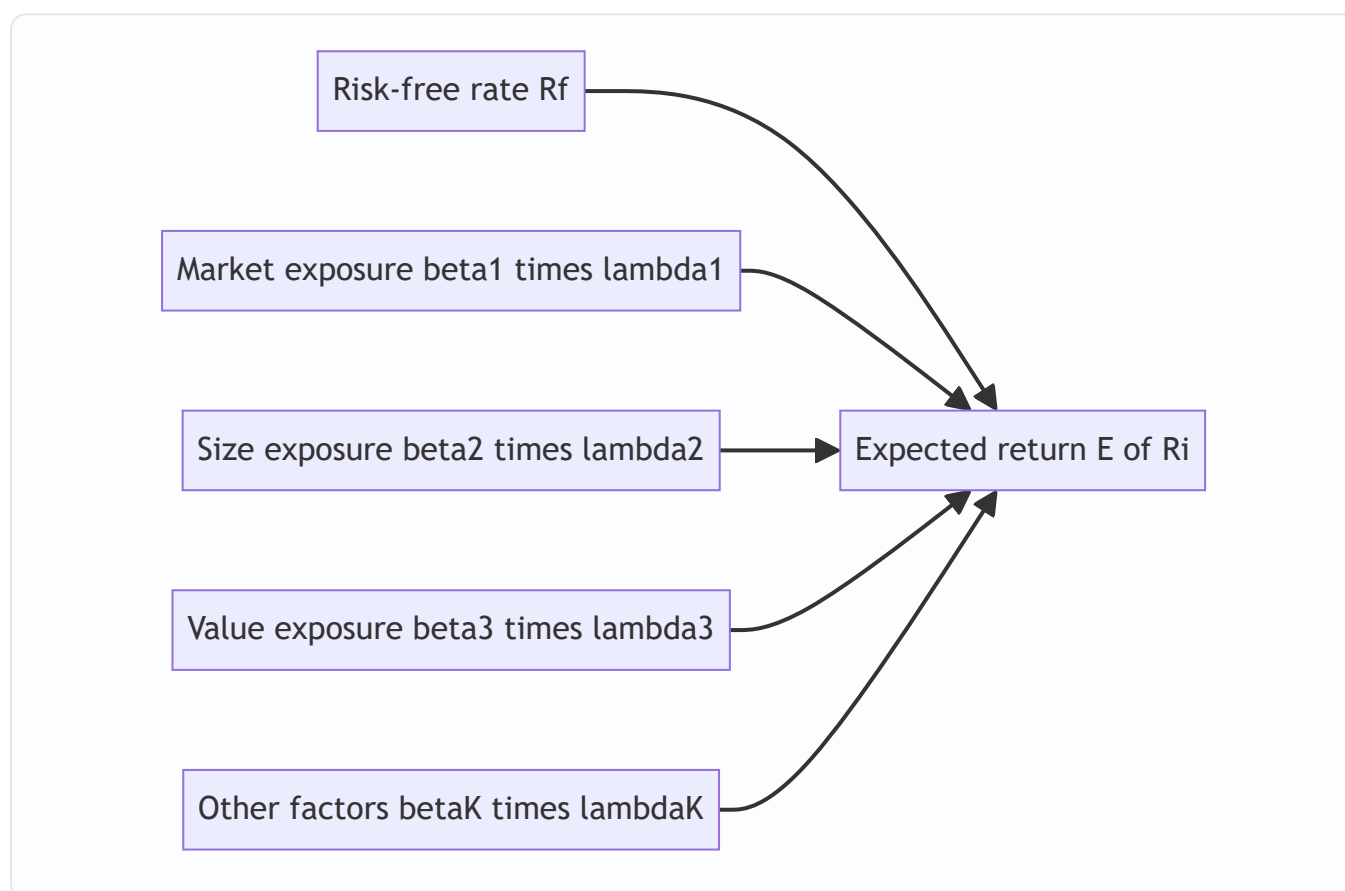
momentum, and macroeconomic variables like inflation and interest-rate shocks — and built models such as the **Fama-French three-factor** and **five-factor** models.

The unifying idea:

$$E(R_i) = R_f + \beta_{i,1}\lambda_1 + \beta_{i,2}\lambda_2 + \dots + \beta_{i,K}\lambda_K$$

An asset's expected excess return is the sum, across all K factors, of (**how exposed** the asset is to each factor, the beta) times (**how much reward** the market pays per unit of that exposure, the risk premium λ). CAPM is just this equation with K = 1. Multifactor models let K be 3, 5, or more — and crucially, they let the factors be things *other* than the market.

Think of it as a nutrition label: CAPM listed one ingredient (market risk). APT/multifactor models reveal the full ingredient list — and tell you the price of each.



Expected return builds up as the risk-free rate plus each factor exposure multiplied by that factor's price of risk.

3. Why / How It Works

The no-arbitrage engine behind APT

Start with a **return-generating process**. Assume each stock's realized return is a linear function of common factors:

$$R_i = E(R_i) + \beta_{i,1}F_1 + \beta_{i,2}F_2 + \dots + \beta_{i,K}F_K + \varepsilon_i$$

where the F's are unexpected *shocks* to the factors (surprise inflation, surprise GDP, etc., each with mean zero) and ε_i is firm-specific noise (mean zero, uncorrelated across firms, uncorrelated with the factors).

Now the magic. Because ϵ_i is *diversifiable*, a well-diversified portfolio of many stocks has essentially zero ϵ . Its return is driven *only* by factor exposures:

$$R_p \approx E(R_p) + \beta_{p,1}F_1 + \dots + \beta_{p,K}F_K$$

Consider building an **arbitrage portfolio** that: - costs nothing (long positions funded by short positions, net investment = 0), - has zero exposure to every factor (all $\beta_p = 0$), - and is well-diversified ($\epsilon \approx 0$).

Such a portfolio has *no risk of any kind* and *no capital at stake*. By the law of one price it must therefore earn zero return. If it earned a positive return, arbitrageurs would scale it up infinitely — free money. This one requirement, applied across all possible portfolios, mathematically forces expected returns to be a **linear combination of the factor betas**:

$$E(R_i) = R_f + \beta_{i,1}\lambda_1 + \dots + \beta_{i,K}\lambda_K$$

That is the APT. Notice what it did *not* need: no market portfolio, no assumption that investors are mean-variance optimizers, no assumption of normally distributed returns, no requirement that everyone agrees. It only needs (i) a factor structure for returns and (ii) enough assets to diversify away idiosyncratic risk so that arbitrage is possible. This is why APT is considered more *robust* and *general* than CAPM — it is built on the weakest possible assumption (no free lunch) rather than the strongest (everyone holds the efficient market portfolio).

Why multiple factors beat one

CAPM compresses all systematic risk into a single number. But real economic risk is multidimensional. An oil producer, a bank, and a software firm all have market exposure — yet the oil firm is uniquely exposed to energy-price shocks, the bank to interest-rate and credit-spread shocks, the software firm to nothing much beyond the market and growth sentiment. A one-factor model *cannot* distinguish these. It will misprice all three unless their betas happen to capture everything, which they don't.

Multifactor models add axes of risk. Each factor is a distinct source of *systematic, non-diversifiable* risk that commands its own premium because you cannot diversify it away — every asset that loads on it moves together. If value stocks all crash together in a recession, "value" is a systematic risk, and risk-averse investors demand extra return to hold it. That extra return is the value premium λ_{HML} .

The improvement is both statistical (higher R^2 , smaller pricing errors) and economic (you learn which risks you're actually being paid for).

4. Full Content — Formulas, Models, Derivations

4.1 APT vs CAPM at a glance

Dimension	CAPM	APT
Number of factors	Exactly 1 (the market)	K factors (unspecified by theory)
Key assumption	All investors hold the mean-variance-efficient market portfolio	No arbitrage; factor structure of returns
Requires market portfolio?	Yes (unobservable — Roll's Critique)	No
Investor behavior	Mean-variance optimizers, homogeneous expectations	Only needs a few arbitrageurs
Holds for	All assets exactly (in theory)	Well-diversified portfolios exactly; individual assets approximately
Tells you the factors?	Yes — the market	No — must be found empirically
Return equation	$E(R) = R_f + \beta(E(R_m) - R_f)$	$E(R) = R_f + \sum \beta_k \lambda_k$

APT is more general: **CAPM is a special case of APT** where the single factor is the market and its risk premium is $E(R_m) - R_f$.

4.2 The general K-factor pricing equation

$$E(R_i) = R_f + \sum_{k=1}^K \beta_{i,k} \lambda_k$$

- **$\beta_{i,k}$ (factor loading / exposure):** sensitivity of asset i's return to a one-unit shock in factor k. Estimated by multivariate time-series regression of the asset's excess returns on the factor returns.
- **λ_k (factor risk premium):** the expected excess return earned per unit of exposure to factor k. It is the return on a portfolio with a beta of 1 to factor k and 0 to all others, in excess of R_f . Can be positive or negative.

Each λ_k is itself the expected return of a **factor-mimicking portfolio** (a portfolio constructed to track one factor with unit exposure and zero exposure to the rest), minus R_f .

4.3 Two families of factors

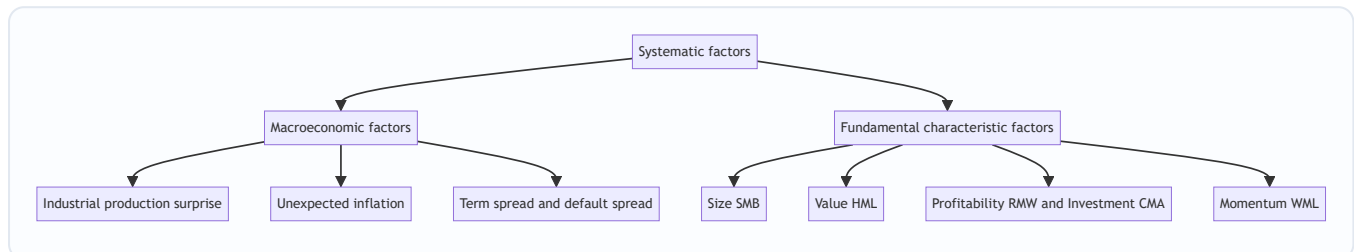
(a) Macroeconomic factors — factors are surprises in observable economic variables. The classic implementation is **Chen, Roll & Ross (1986)**, who found these priced factors:

Macro factor	Economic meaning
Industrial production growth (surprise)	Real economic activity / business cycle
Unexpected inflation	Erosion of real cash flows and discount-rate shock
Change in expected inflation	Shifts in future inflation expectations
Term spread (long minus short govt yields)	Slope of yield curve; growth expectations
Default/credit spread (Baa minus Aaa)	Risk appetite and credit conditions

Betas here are *macro sensitivities* — how much a stock moves when, say, inflation surprises to the upside.

(b) Fundamental / characteristic factors — factors are built from firm characteristics (size, value, profitability). The Fama-French models are the flagship. These factors are *long-short portfolios* sorted on characteristics, so the "factor return" is directly observable each month.

In practice, fundamental factor models (Fama-French, BARRA, Axioma) dominate industry use because the factors are cleaner and more stable to estimate than noisy macro surprises.



The two families of priced factors: macroeconomic surprises versus firm-characteristic long-short portfolios.

4.4 The Fama-French Three-Factor Model (1993)

$$R_i - R_f = \alpha_i + \beta_i(R_m - R_f) + s_i \cdot \text{SMB} + h_i \cdot \text{HML} + \varepsilon_i$$

Three factors:

1. **Market ($R_m - R_f$):** same excess market return as CAPM.
2. **SMB — Small Minus Big (the size factor):** the return on a portfolio *long* small-cap stocks and *short* large-cap stocks. Captures the historical tendency of small firms to outperform. A positive s_i means the stock behaves like a small-cap.
3. **HML — High Minus Low (the value factor):** the return on a portfolio *long* high book-to-market (value) stocks and *short* low book-to-market (growth) stocks. A positive h_i means the stock behaves like a value stock.

Expected-return form:

$$E(R_i) - R_f = \beta_i \lambda_{\text{Mkt}} + s_i \lambda_{\text{SMB}} + h_i \lambda_{\text{HML}}$$

How SMB and HML are built (the 2×3 sort). Fama and French sort all stocks into 2 size buckets (Small, Big) and 3 book-to-market buckets (Low, Medium, High), forming 6 portfolios:

- **SMB** = average return of the 3 Small portfolios – average return of the 3 Big portfolios.
- **HML** = average return of the 2 High-B/M portfolios – average return of the 2 Low-B/M portfolios.

Why these factors? Empirically they mop up the CAPM anomalies. But there is also an economic story: small and value firms tend to be *distressed* or *riskier* — value firms often have depressed prices because of poor prospects, and their earnings are more sensitive to bad economic times — so their extra return is *compensation for risk* (the risk-based interpretation). The competing behavioral interpretation says investors *overreact*, over-pricing glamorous growth stocks and under-pricing dull value stocks, creating a mispricing premium. The debate is unsettled, but the factors work either way.

4.5 The Fama-French Five-Factor Model (2015)

Fama and French added two factors motivated by the **dividend discount model** — firms with higher profitability and more conservative investment should, all else equal, deliver higher expected returns:

$$R_i - R_f = \alpha_i + \beta_i(R_m - R_f) + s_i \text{ SMB} + h_i \text{ HML} + r_i \text{ RMW} + c_i \text{ CMA} + \varepsilon_i$$

Two new factors:

4. **RMW — Robust Minus Weak (profitability):** long high-profitability firms (robust operating profitability), short low-profitability (weak) firms.
5. **CMA — Conservative Minus Aggressive (investment):** long firms that invest conservatively (low asset growth), short firms that invest aggressively (high asset growth). Aggressive over-investors historically underperform.

Factor	Long leg	Short leg	Captures
Mkt-RF	Market	Risk-free	Equity market risk
SMB	Small caps	Large caps	Size premium
HML	High B/M (value)	Low B/M (growth)	Value premium
RMW	Robust profitability	Weak profitability	Quality/profitability premium
CMA	Conservative investment	Aggressive investment	Investment/discipline premium

Notable wrinkle: in the five-factor model, HML becomes partly *redundant* — RMW and CMA absorb much of the value premium, because profitability and investment discipline are correlated with value. Fama and French found that for US data, HML added little once RMW and CMA were included. This is an active, ongoing debate.

A common sixth factor: Momentum (WML / UMD) — "Winners Minus Losers" (or Up Minus Down), from Carhart (1997). Long the past 12-month winners, short the past losers. Momentum is *not* in the Fama-French models (they dislike it because it has no risk-based story and high turnover), but the **Carhart four-factor model** (Mkt, SMB, HML, WML) is the industry standard for evaluating mutual fund performance.

4.6 Alpha in a multifactor world

The intercept α_i from the regression is the model's verdict on skill/mispricing:

$$\alpha_i = (R_i - R_f) - [\beta_i(R_m - R_f) + s_i \text{ SMB} + h_i \text{ HML} + \dots]$$

If a fund earns high raw returns *purely* by tilting toward small and value stocks, a CAPM regression would show a big positive alpha (looks like skill), but a Fama-French regression would reveal $\alpha \approx 0$ — the returns were just **factor exposure ("smart beta")**, **not stock-picking skill**. This is the single most important practical use of multifactor models in asset management: **separating true alpha from cheap, replicable beta**.

5. Worked Examples

Example 1 — Two-factor APT pricing

Suppose a two-factor APT holds. The factors are (F1) industrial-production surprise and (F2) inflation surprise. Given:

- Risk-free rate $R_f = 5\%$

- Factor risk premiums: $\lambda_1 = 6\%$, $\lambda_2 = -2\%$ (inflation surprise carries a *negative* premium — assets that pay off when inflation surprises up are valuable as hedges, so they earn *less*)
- Stock A's factor betas: $\beta_{\{A,1\}} = 1.2$, $\beta_{\{A,2\}} = 0.5$

Expected return:

$$E(R_A) = 5\% + (1.2)(6\%) + (0.5)(-2\%) = 5\% + 7.2\% - 1.0\% = 11.2\%$$

Now suppose Stock A is *actually* trading to give an expected return of 13%. APT says it is **underpriced** — it offers 13% but only 11.2% is warranted by its risk. The 1.8% gap is a positive alpha.

Arbitrage: build a portfolio with the *same* factor betas ($\beta_1 = 1.2$, $\beta_2 = 0.5$) out of the factor-mimicking portfolios and R_f — that "fair" portfolio yields 11.2%. Go **long Stock A** (13%) and **short the replicating portfolio** (11.2%). Net factor exposure = 0, net cost = 0, yet you earn $13\% - 11.2\% = 1.8\%$ **riskless**.

Arbitrageurs pile in, bidding A's price up until its expected return falls to 11.2% and the alpha vanishes. *This is the enforcement mechanism that makes the APT equation hold.*

Example 2 — Fama-French three-factor expected return, reconciled against CAPM

A small-cap value stock, "Stock V." Regression estimates:

- β (market) = 1.10, s (SMB) = 0.80, h (HML) = 0.60
- $R_f = 4\%$; market risk premium $\lambda_{\text{Mkt}} = 5.5\%$; $\lambda_{\text{SMB}} = 2.5\%$; $\lambda_{\text{HML}} = 3.5\%$

Fama-French expected return:

$$E(R_V) = 4\% + (1.10)(5.5\%) + (0.80)(2.5\%) + (0.60)(3.5\%) = 4\% + 6.05\% + 2.0\% + 2.1\% = 14.15\%$$

CAPM expected return (same market beta):

$$E(R_V)_{\text{CAPM}} = 4\% + (1.10)(5.5\%) = 4\% + 6.05\% = 10.05\%$$

Reconciliation / interpretation. CAPM says this stock "should" earn 10.05%. Fama-French says 14.15%. The **4.10% difference** is precisely the reward for the stock's small-cap tilt (2.0%) and value tilt (2.1%) — risks CAPM is blind to. Now suppose the stock *actually* delivered 14.15% on average. Under CAPM you'd record a **+4.10% alpha** and congratulate the manager for genius stock-picking. Under Fama-French, $\alpha = 14.15\% - 14.15\% = 0\%$ — the manager simply harvested well-known size and value premia. The two models reconcile: the "alpha" under CAPM is exactly the "factor return" under Fama-French. *That is the whole point of the exercise.*

Example 3 — Attributing fund performance (four-factor Carhart)

A hedge fund returns 18% over a year; $R_f = 4\%$, so excess return = 14%. A Carhart regression gives:

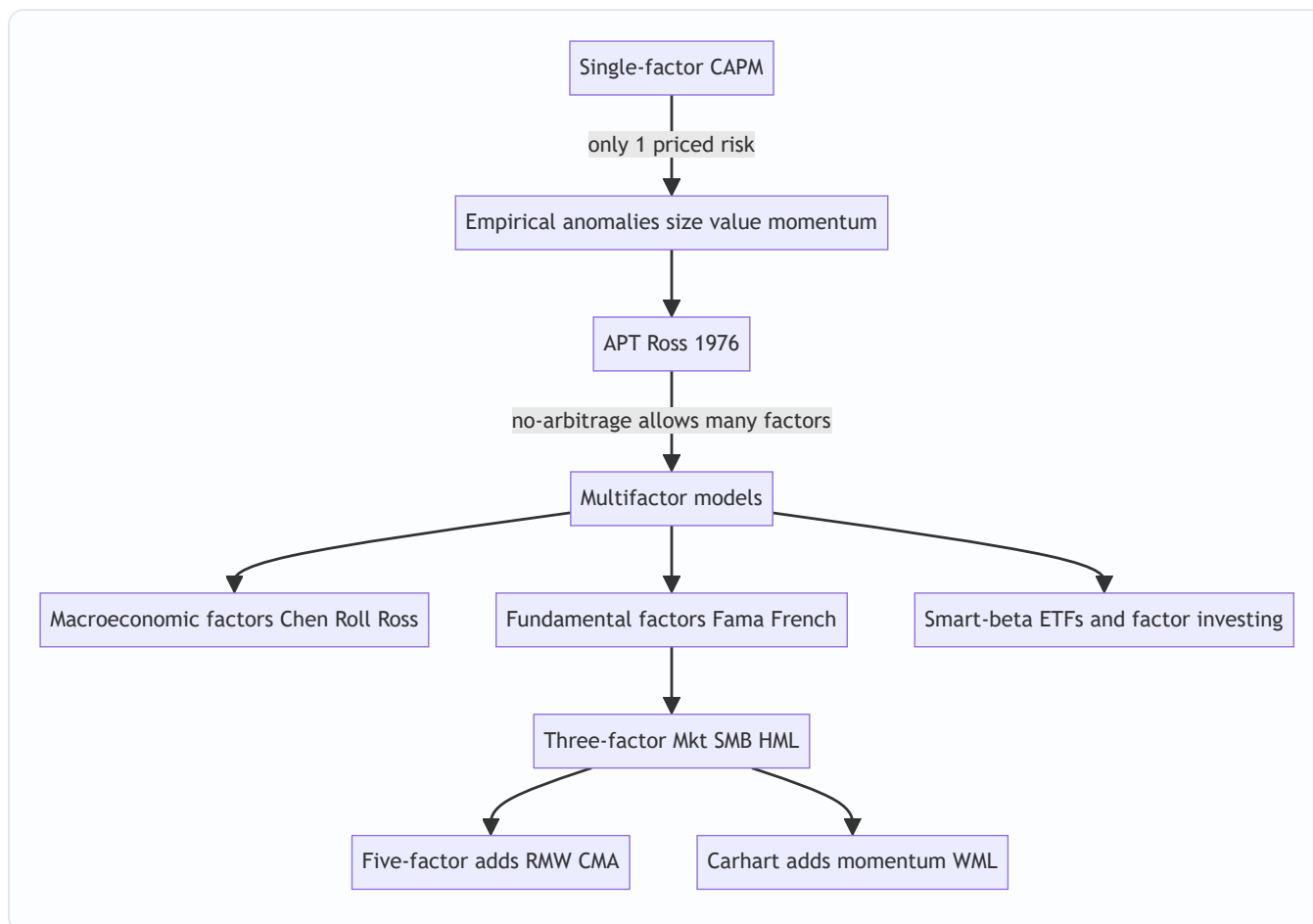
Factor	Loading	Factor premium (realized)	Contribution
Market	1.00	8%	8.0%
SMB	0.50	3%	1.5%
HML	0.40	2%	0.8%
WML (momentum)	0.60	4%	2.4%
Sum of factor contributions			12.7%

Alpha = 14% (excess return) – 12.7% (explained by factors) = **1.3%**.

Reading it: of the 14% excess return, 12.7% came from bearing systematic factor risks the investor could have bought cheaply through index products (a levered market position plus small, value, and momentum tilts). Only **1.3% is genuine alpha** — the manager's skill beyond factor exposure. If the fund charges "2 and 20" fees justified by "alpha," an allocator now knows most of the return was replicable beta. *This decomposition is exactly what an AMC's product/manager-selection team runs before allocating capital.*

6. Connections

- **To CAPM (Ch. 05):** APT *nests* CAPM. Set $K = 1$ with the market as the only factor and you recover the SML. Multifactor models generalize the Security Market Line from a line into a **hyperplane** in factor-exposure space.
- **To portfolio theory (Ch. 04):** Factors are the axes of *systematic* risk that survive diversification. Idiosyncratic ϵ washes out in a diversified portfolio — the same diversification logic that underpins Markowitz — which is precisely why only factor risk gets *priced*.
- **To performance evaluation:** Multifactor alpha is the modern successor to Jensen's alpha. Fund ratings, manager due diligence, and "closet indexing" detection all run on factor regressions.
- **To smart beta / factor investing:** The entire ETF factor-investing industry (value, momentum, quality, low-vol, size ETFs) is applied multifactor theory — packaging factor premia as cheap, rules-based products. What was once "alpha" became "smart beta."
- **To cost of capital (corporate finance):** Firms can use a Fama-French model instead of CAPM to estimate the discount rate for valuation, especially for small or value-oriented firms where CAPM systematically understates the required return.
- **To fixed income:** Bond returns decompose into *level, slope, and curvature* factors — a multifactor model for the yield curve, same spirit as APT.



From the cracks in CAPM to the modern factor-investing industry.

7. Key Terms

- **Arbitrage Pricing Theory (APT):** Ross's no-arbitrage model deriving a linear multifactor relationship between expected return and factor exposures.
- **Factor:** a common source of systematic (non-diversifiable) risk that moves many assets together and commands a risk premium.
- **Factor loading / beta ($\beta_{i,k}$):** sensitivity of an asset's return to a one-unit move in factor k .
- **Factor risk premium (λ_k):** expected excess return per unit of exposure to factor k ; the return of that factor's mimicking portfolio over R_f .
- **Factor-mimicking portfolio:** a portfolio built to have unit exposure to one factor and zero to all others; its excess return is the factor premium.
- **Return-generating process:** the assumed linear equation $R_i = E(R_i) + \sum \beta_k F_k + \epsilon_i$ describing how returns are produced.
- **Idiosyncratic (residual) risk (ϵ_i):** firm-specific, diversifiable, unpriced risk.
- **SMB (Small Minus Big):** size factor; long small-cap, short large-cap.
- **HML (High Minus Low):** value factor; long high book-to-market, short low.
- **RMW (Robust Minus Weak):** profitability factor.
- **CMA (Conservative Minus Aggressive):** investment factor.
- **WML / UMD (momentum):** Winners Minus Losers; Carhart's fourth factor.

- **Alpha (α):** return unexplained by the factor model; the measure of skill or mispricing.
 - **Smart beta:** rules-based products that deliver factor premia cheaply — factor exposure repackaged from "alpha."
-

8. Common Confusions

- **"APT tells you the factors."** No. APT is silent on *what* the factors are — it only says pricing *must* be linear in *some* set of factors. Identifying them is an empirical exercise, which is where Fama-French, Chen-Roll-Ross, and BARRA come in. This is APT's great strength (generality) and great weakness (untestable without specifying factors).
 - **Confusing factor loading (β) with factor premium (λ).** β is a property of the *asset* (how much it moves with the factor). λ is a property of the *market* (how much reward the factor earns). Expected return needs *both*, multiplied.
 - **Thinking more factors always means a better model.** Adding factors mechanically raises R^2 , but risks *data mining* — the "factor zoo" of 300+ published factors, most of which fail out-of-sample. A factor should have (i) an economic rationale, (ii) robustness across time/markets, and (iii) low correlation with existing factors.
 - **Assuming a positive Fama-French alpha = skill, always.** Only if the model is *correctly specified*. If a genuine priced factor is missing, its premium leaks into alpha and masquerades as skill. Alpha is always *relative to the chosen model*.
 - **Believing APT requires a market portfolio.** It doesn't — that's CAPM's problem (Roll's Critique). APT sidesteps the unobservable market portfolio entirely, needing only enough assets to diversify.
 - **Thinking SMB/HML premia are guaranteed.** They are *risk premia* — long-run averages that can vanish or go negative for a decade (value famously underperformed 2007–2020). Factor investing carries factor risk; that's *why* it pays.
 - **Sign of λ must be positive.** Not necessarily. A factor that pays off in bad states (a hedge, like the inflation-surprise factor in Example 1) can carry a *negative* premium — investors accept lower returns for insurance.
-

9. First-Principles Recap

1. **Diversifiable risk isn't rewarded.** Only risk you *cannot* escape by diversifying — systematic, common-factor risk — earns a premium. This is the bedrock, shared with CAPM.
2. **There is more than one systematic risk.** The economy has many independent shocks (growth, inflation, credit, size, value, profitability). Compressing them into one "market" number loses information and misprices assets.
3. **No arbitrage forces linear pricing.** If returns are generated by K factors, then the impossibility of a free lunch *mathematically requires* expected returns to be a linear function of factor exposures: $E(R) = R_f + \sum \beta_k \lambda_k$. No behavioral assumptions needed.
4. **Expected return = exposure $\times$ price of risk, summed.** For each factor, multiply how exposed you are (β) by how much that risk pays (λ), and add them up, on top of the risk-free rate.
5. **Whatever the model can't explain is alpha.** Alpha is the residual — skill or mispricing — *relative to the factors you chose to include*. Choose better factors, and yesterday's alpha becomes today's beta.

From these five ideas, everything in the chapter follows — APT, Chen-Roll-Ross, Fama-French three and five factors, Carhart, and the entire smart-beta industry are just different answers to the question "*which factors, and what do they pay?*"

10. Quick Reference / Interview Points

Formula sheet

Concept	Formula
General APT / multifactor	$E(R_i) = R_f + \sum_k \beta_{i,k} \lambda_k$
Return-generating process	$R_i = E(R_i) + \sum_k \beta_{i,k} F_k + \varepsilon_i$
CAPM (special case, $K=1$)	$E(R_i) = R_f + \beta_i(E(R_m) - R_f)$
Fama-French 3-factor	$R_i - R_f = \alpha + \beta(R_m - R_f) + s \cdot \text{SMB} + h \cdot \text{HML} + \varepsilon$
Fama-French 5-factor	$+ r \cdot \text{RMW} + c \cdot \text{CMA}$
Carhart 4-factor	3-factor + $w \cdot \text{WML}$ (momentum)
Alpha	$\alpha = (R_i - R_f) - \sum_k \beta_{i,k} \lambda_k$

The five (six) canonical factors

SMB (size), HML (value), RMW (profitability), CMA (investment), WML (momentum), plus the Market. Know how each long-short portfolio is constructed.

What interviewers actually ask

- **"How does APT differ from CAPM?"** — APT allows multiple factors and rests on no-arbitrage, not on everyone holding the market portfolio; it doesn't require the unobservable market portfolio (dodges Roll's Critique); CAPM is a one-factor special case of APT. Trade-off: APT doesn't tell you the factors.
- **"What are the Fama-French factors and why do they exist?"** — Market, SMB, HML (+RMW, CMA in the 5-factor). They exist because size and value (and profitability/investment) tilts explained returns that CAPM couldn't; interpretations split between risk-based (compensation) and behavioral (mispricing).
- **"A fund returned 20% — is the manager skilled?"** — Run a multifactor regression. Decompose into factor contributions vs. alpha. If the 20% is just leveraged market + small + value + momentum exposure, there's no skill — it's replicable smart beta. Only the alpha reflects skill.
- **"Why is HML sometimes redundant in the 5-factor model?"** — Because RMW and CMA (profitability and investment) are correlated with the value characteristic and absorb much of the value premium in US data.
- **"Can a factor premium be negative?"** — Yes, for hedging factors (e.g., inflation-surprise): assets that pay off in bad states are valuable and thus earn *lower* expected returns.
- **"What's the danger of multifactor models?"** — Data mining / the factor zoo; overfitting; factors that vanish out-of-sample; alpha is only meaningful relative to the model, so an omitted true factor masquerades as skill.

- **"Macro vs. fundamental factors?"** — Macro (Chen-Roll-Ross): surprises in inflation, IP growth, term/default spreads; economically interpretable but noisy to estimate. Fundamental (Fama-French, BARRA): characteristic-sorted long-short portfolios; observable, stable, and dominant in industry practice.

The one-liner to remember

CAPM says "how much market risk?"; APT and multifactor models say "how much of each kind of risk — and what does each one pay?" Expected return is just your exposures dotted with the prices of risk.

Q&A — APT and Multifactor Models

Scope: Investments — Chapter 06 (APT and Multifactor Models). Every question is followed by a full model answer. All rates are annual and in percent unless stated. Work every numerical problem yourself before checking the solution. Sections: **A** concept-check · **B** numerical (full step-by-step, reconciling) · **C** interview-style · **D** MCQs with reasoning.

The chapter in one line

$$E(R_i) = R_f + \sum_{k=1}^K \beta_{i,k} \lambda_k$$

One-line statement: Expected return equals the risk-free rate plus, for each systematic factor, your exposure (β) times that factor's price of risk (λ); CAPM is just the one-factor case, and no-arbitrage — not a market portfolio — is what forces the pricing to be linear.

Section A — Concept Check

A1. What single assumption does APT rest on, and why does that make it more general than CAPM?

The **no-arbitrage** condition (no free lunch / law of one price). If a portfolio costs nothing to build and carries no factor risk, it must earn zero. That is a far weaker assumption than CAPM's requirement that *every* investor is a mean-variance optimiser holding the true market portfolio. Because APT needs only a factor structure plus enough assets to diversify, it does not need the (unobservable) market portfolio, homogeneous expectations, normal returns, or single-period preferences. Hence it is more robust and more general — CAPM is a special case.

A2. Distinguish a factor loading (β) from a factor risk premium (λ). β is a property of the **asset** — how much the asset's return moves per one-unit shock to the factor (estimated by regression). λ is a property of the **market** — the expected excess return earned per unit of exposure to that factor (the return of the factor-mimicking portfolio over R_f). Expected return needs *both*, multiplied together: exposure $\times$ price of risk.

A3. Why is only systematic (factor) risk priced, while idiosyncratic risk ϵ_i is not? ϵ_i is firm-specific and uncorrelated across firms, so it washes out in a well-diversified portfolio — you can eliminate it for free by diversifying. The market pays you only for risk you *cannot* escape. Factor risk is common to many assets (they all move together), so it survives diversification and therefore commands a premium.

A4. Does APT tell you what the factors are? No. This is its defining feature. APT proves pricing *must* be linear in *some* set of K factors, but is silent on their identity. Finding the factors is an empirical exercise — that is where Chen-Roll-Ross (macro factors) and Fama-French (characteristic factors) come in. This is both APT's strength (generality) and its weakness (not directly testable without specifying factors).

A5. Name the Fama-French three factors and say what each long-short portfolio holds. (1) **Market ($R_m - R_f$)** — excess return on the market. (2) **SMB, Small Minus Big** — long small-cap stocks, short large-caps (size premium). (3) **HML, High Minus Low** — long high book-to-market (value) stocks, short low book-to-market (growth) stocks (value premium).

A6. What two factors did the 2015 five-factor model add, and what motivated them? RMW (Robust Minus Weak) — long high-profitability firms, short low-profitability — and **CMA (Conservative Minus Aggressive)** — long firms that invest conservatively (low asset growth), short aggressive investors. They were motivated by the dividend discount model: holding valuation fixed, higher profitability and more disciplined investment imply higher expected returns.

A7. Which factor is *not* in the Fama-French models but defines the Carhart four-factor model, and why do Fama-French exclude it? Momentum — WML / UMD (Winners Minus Losers), long past-12-month winners, short past losers. Fama and French exclude it because it lacks a clean risk-based story and has high turnover, but Carhart (1997) added it, and the four-factor model (Mkt, SMB, HML, WML) is the industry standard for mutual-fund performance evaluation.

A8. In a multifactor world, what is alpha? Alpha is the part of return the factor model *cannot* explain: $\alpha = (R_i - R_f) - \sum \beta_{i,k} \lambda_k$. It is the measure of genuine skill or mispricing — but only *relative to the factors you chose*. An omitted true factor leaks its premium into alpha and masquerades as skill.

A9. Can a factor risk premium λ be negative? Give the intuition. Yes. A factor that pays off in *bad* states of the world is a hedge — investors value the insurance and accept a *lower* expected return to hold assets exposed to it. Classic example: an inflation-surprise factor. Assets that rise when inflation surprises upward protect you, so they can carry a negative premium.

A10. Why can adding more factors be dangerous rather than simply "better"? More factors mechanically raise R^2 , but risk **data mining** — the "factor zoo" of 300+ published factors, most of which fail out-of-sample. A legitimate factor needs (i) an economic rationale, (ii) robustness across time and markets, and (iii) low correlation with existing factors.

A11. How does APT nest CAPM geometrically? CAPM's Security Market Line is a *line* in (β , expected-return) space. A multifactor model generalises it into a **hyperplane** in K-dimensional factor-exposure space. Set $K = 1$ with the market as the factor and $\lambda = E(R_m) - R_f$, and the hyperplane collapses back to the SML.

A12. Why did HML become partly redundant in the five-factor model? Because RMW (profitability) and CMA (investment) are correlated with the value characteristic and absorb much of the value premium. In US data, once RMW and CMA are included, HML adds little explanatory power — an active, unsettled debate.

A13. Distinguish macroeconomic factors from fundamental (characteristic) factors. Macro factors (Chen-Roll-Ross) are *surprises* in observable economic variables — industrial-production growth, unexpected inflation, term spread, default spread. Economically interpretable but noisy to estimate. **Fundamental factors** (Fama-French, BARRA) are long-short portfolios sorted on firm characteristics (size, value, profitability); their returns are directly observable and stable, which is why industry practice is dominated by them.

Section B — Numerical Problems

B1. Two-factor APT pricing. A two-factor APT holds. $R_f = 5\%$; λ_1 (industrial-production surprise) = 6% ; λ_2 (inflation surprise) = -2% . Stock A has $\beta_1 = 1.2$, $\beta_2 = 0.5$. Find its required return.

Solution. $E(R_A) = 5\% + (1.2)(6\%) + (0.5)(-2\%)$ Step by step: factor-1 reward = $1.2 \times 6\% = 7.2\%$; factor-2 reward = $0.5 \times (-2\%) = -1.0\%$. $E(R_A) = 5\% + 7.2\% - 1.0\% = \mathbf{11.2\%}$ The negative λ_2 reduces the required return: exposure to a hedging factor is rewarded with less return, not more.

B2. Arbitrage from mispricing. Continue B1. Stock A actually offers an expected return of 13%. Is it mis-priced? Construct the arbitrage and state the profit.

Solution. Required = 11.2% (B1); offered = 13%. Since $13\% > 11.2\%$, A is **underpriced** — a positive alpha of $13\% - 11.2\% = +1.8\%$. Arbitrage: build a replicating portfolio from R_f and the two factor-mimicking portfolios with the *same* exposures ($\beta_1 = 1.2, \beta_2 = 0.5$); it fairly yields 11.2%. Go **long Stock A** (+13%) and **short the replicating portfolio** (−11.2%). - Net cost = 0 (short funds the long). - Net factor exposure: $\beta_1 = 1.2 - 1.2 = 0; \beta_2 = 0.5 - 0.5 = 0$. Riskless. - Net return = $13\% - 11.2\% = +1.8\%$ **riskless**. Arbitrageurs scale this up, bidding A's price up until its expected return falls to 11.2% and the alpha vanishes. This is the enforcement mechanism behind the APT equation.

B3. Fama-French three-factor return, reconciled against CAPM. Small-cap value "Stock V": $\beta = 1.10, s = 0.80, h = 0.60$. $R_f = 4\%; \lambda_{Mkt} = 5.5\%; \lambda_{SMB} = 2.5\%; \lambda_{HML} = 3.5\%$. Find the FF expected return and the CAPM expected return, and reconcile.

Solution — Fama-French: $E(R_V) = 4\% + (1.10)(5.5\%) + (0.80)(2.5\%) + (0.60)(3.5\%)$ Contributions: market $1.10 \times 5.5\% = 6.05\%$; size $0.80 \times 2.5\% = 2.00\%$; value $0.60 \times 3.5\% = 2.10\%$. $E(R_V) = 4\% + 6.05\% + 2.00\% + 2.10\% = \mathbf{14.15\%}$ **CAPM (same market beta):** $E(R_V)_{CAPM} = 4\% + (1.10)(5.5\%) = \mathbf{10.05\%}$ **Reconciliation.** The gap = $14.15\% - 10.05\% = \mathbf{4.10\%}$, exactly the sum of the size (2.00%) and value (2.10%) rewards CAPM is blind to. If V actually earns 14.15%, a CAPM regression records a fake **+4.10% alpha** ("genius stock-picking"), whereas Fama-French shows **$\alpha = 0$** — the return was just harvested size and value premia. The CAPM "alpha" is the FF "factor return."

B4. Performance attribution (Carhart four-factor). A fund returns 18%; $R_f = 4\%$. Loadings and realised factor premia: Market $\beta = 1.00$ (8%), SMB = 0.50 (3%), HML = 0.40 (2%), WML = 0.60 (4%). Decompose the excess return and find alpha.

Solution. Excess return = $18\% - 4\% = 14\%$. | Factor | Loading | Realised premium | Contribution | |---|---|---|---|
 --| | Market | 1.00 | 8% | 8.0% | | SMB | 0.50 | 3% | 1.5% | | HML | 0.40 | 2% | 0.8% | | WML | 0.60 | 4% | 2.4% | |
Sum | | | **12.7%** |

Alpha = $14\% - 12.7\% = +1.3\%$. Interpretation: of the 14% excess return, 12.7% is replicable factor beta (a full market position plus small, value, and momentum tilts an investor could buy cheaply); only **1.3% is genuine skill**. This is precisely the check an allocator runs before paying "2 and 20."

B5. Detecting closet indexing / negative alpha. Fund X returns 11%; $R_f = 3\%$. A three-factor regression gives $\beta = 0.95$ ($\lambda_{Mkt} = 7\%$), $s = 0.20$ ($\lambda_{SMB} = 2\%$), $h = 0.10$ ($\lambda_{HML} = 3\%$). Compute the factor-implied return and the alpha, and comment.

Solution. Factor contributions to excess return: $0.95 \times 7\% = 6.65\%; 0.20 \times 2\% = 0.40\%; 0.10 \times 3\% = 0.30\%$. Sum = **7.35%**. Factor-implied total return = $R_f + 7.35\% = 3\% + 7.35\% = \mathbf{10.35\%}$. Alpha = actual – implied = $11\% - 10.35\% = +0.65\%$ (small positive). But note the loadings are close to the market ($\beta \approx 0.95$, tiny size/value tilts): the fund is *almost* a levered index fund, so the 0.65% alpha is thin and, after typical active fees of ~1%, the *net* alpha is **negative**. Multifactor regression exposes "closet indexing" — near-index exposure sold at active prices.

B6. Solving for a factor premium by no-arbitrage. Two well-diversified portfolios load on one factor. Portfolio P: $\beta = 1.5, E(R) = 17\%$. Portfolio Q: $\beta = 0.5, E(R) = 9\%$. Find R_f and the factor premium λ , then price a portfolio with $\beta = 1.0$.

Solution. APT: $E(R) = R_f + \beta\lambda$. Two equations: - $17\% = R_f + 1.5\lambda$ - $9\% = R_f + 0.5\lambda$

Subtract: $8\% = 1.0\lambda \rightarrow \lambda = 8\%$. Back-substitute into the second: $9\% = R_f + 0.5(8\%) = R_f + 4\% \rightarrow R_f = 5\%$. A $\beta = 1.0$ portfolio should price at $E(R) = 5\% + 1.0 \times 8\% = 13\%$. If any $\beta = 1$ portfolio traded away from 13%, you could arbitrage it against a mix of P and Q with net $\beta = 1$.

Section C — Interview-Style Questions

C1. "How does APT differ from CAPM — and what does APT give up in exchange for its generality?"

APT allows *many* systematic factors and rests on **no-arbitrage** rather than on everyone holding the mean-variance-efficient market portfolio. Consequently it does not require the unobservable market portfolio, so it sidesteps Roll's Critique, and it needs only a handful of arbitrageurs rather than universally rational optimisers. CAPM is simply the one-factor special case. The trade-off: APT does *not* tell you what the factors are or how many — you must find them empirically, and the theory is hard to test without committing to a factor set. So APT buys robustness at the cost of specificity.

C2. "A fund returned 20% last year. Was the manager skilled?" You cannot say from the raw number — decompose it. Run a multifactor regression (Carhart or five-factor) of the fund's excess returns on the factors. Split the 20% into (a) factor contributions — the return explained by the fund's market, size, value, momentum, profitability tilts, all of which an investor can buy cheaply through index/ETF products — and (b) alpha, the residual. If most of the 20% is levered market plus small/value/momentum exposure, there is little skill; it is replicable **smart beta**. Only the alpha, and only if it is statistically significant and survives fees, reflects genuine skill.

C3. "Explain the no-arbitrage argument that produces the APT equation." Assume returns follow a factor model: $R_i = E(R_i) + \sum \beta_{i,k} F_k + \epsilon_i$, where the F 's are mean-zero factor shocks and ϵ_i is diversifiable noise. Form a portfolio that (i) costs nothing (longs funded by shorts), (ii) has zero exposure to every factor (all $\beta_p = 0$), and (iii) is well-diversified so $\epsilon \approx 0$. Such a portfolio has no risk and no capital at stake, so by the law of one price it must earn zero. If it earned anything positive, arbitrageurs would scale it infinitely — free money. Imposing "zero return for zero-cost, zero-risk portfolios" across all portfolios forces expected returns to be a linear function of the betas: $E(R_i) = R_f + \sum \beta_{i,k} \lambda_k$. No behavioural or market-portfolio assumptions are needed.

C4. "Why do the size and value premia exist — risk or mispricing?" Two camps. **Risk-based (Fama-French):** small and value firms are more distressed and their earnings are more sensitive to bad economic times, so their extra return is fair compensation for bearing that systematic risk. **Behavioural:** investors over-extrapolate — they over-price glamorous growth stocks and under-price dull value stocks — so the premium is a slowly-correcting mispricing. The debate is unsettled, but the factors have worked under either interpretation. Crucially, if they are risk premia they can, and do, disappear for long stretches (value underperformed roughly 2007–2020) — factor investing carries factor risk, which is *why* it pays on average.

C5. "When would you use a Fama-French model instead of CAPM to estimate a firm's cost of equity?"

When the firm has strong size or value characteristics that CAPM's single beta misses. CAPM systematically *understates* the required return for small-cap and value (high book-to-market) firms because it ignores the priced size and value risks they carry. Using a three- or five-factor model gives a discount rate that reflects those exposures, producing a more accurate valuation. The cost is added estimation error from more betas and the need for factor-premium inputs.

C6. "What is the single biggest practical use of multifactor models in asset management?" Separating **true alpha from replicable beta** ("smart beta"). Before a multifactor lens, a manager tilting toward small

and value stocks looked like a genius under CAPM (big positive alpha). A factor regression reveals that "alpha" was just cheap, rules-based factor exposure. This drives manager due diligence, fee negotiation, closet-indexing detection, and the entire factor-ETF industry — packaging what used to be alpha as low-cost beta.

Section D — MCQs with Reasoning

D1. APT differs from CAPM primarily because APT — A) requires the market portfolio B) assumes normally distributed returns C) permits multiple priced factors and rests on no-arbitrage D) ignores systematic risk **Answer: C.** APT allows K factors and is built on no-arbitrage; A and B are (partly) CAPM's requirements, and D is false — APT prices systematic risk, just across several factors.

D2. In $E(R_i) = R_f + \sum \beta_k \lambda_k$, the term λ_k represents — A) the asset's sensitivity to factor k B) the expected excess return per unit of exposure to factor k C) idiosyncratic risk D) the risk-free rate **Answer: B.** λ is the factor's *price of risk* (a market property). A describes β (an asset property); C and D are unrelated.

D3. A factor risk premium λ can be negative when the factor — A) has a high beta B) pays off in bad states of the world (acts as a hedge) C) is diversifiable D) is the market factor **Answer: B.** Assets that pay off in bad states provide insurance; investors accept lower returns to hold them, so the premium is negative. A confuses β with λ ; C describes unpriced risk; D (the market) carries a positive premium.

D4. In the Fama-French three-factor model, HML is long — A) large-cap, short small-cap B) high book-to-market (value), short low book-to-market (growth) C) winners, short losers D) high-profitability, short low-profitability **Answer: B.** HML = High Minus Low book-to-market = value minus growth. A is (reversed) size, C is momentum, D is RMW.

D5. The Carhart four-factor model adds which factor to the Fama-French three? A) RMW B) CMA C) WML (momentum) D) Term spread **Answer: C.** Carhart adds momentum (Winners Minus Losers). RMW and CMA belong to the five-factor model; term spread is a Chen-Roll-Ross macro factor.

D6. A fund's entire excess return is explained by its factor loadings (alpha = 0). This means — A) the manager has strong skill B) the return was replicable factor exposure, i.e. smart beta C) the fund took no risk D) the model is misspecified **Answer: B.** Zero alpha means the return came entirely from bearing priced factor risks an investor could buy cheaply — no skill beyond beta. It does not imply zero risk (C) or, by itself, misspecification (D).

D7. Idiosyncratic risk ϵ_i is not priced because it — A) is larger than systematic risk B) can be diversified away for free C) has a negative premium D) equals the risk-free rate **Answer: B.** Firm-specific risk vanishes in a diversified portfolio, so the market pays nothing to bear it — only non-diversifiable factor risk earns a premium.

D8. The main danger of building models with many factors is — A) R^2 becomes too low B) data mining / the factor zoo — factors that fail out-of-sample C) alpha becomes impossible to compute D) betas can no longer be estimated **Answer: B.** Adding factors always *raises* in-sample R^2 (so A is backwards), but risks overfitting spurious factors that do not survive out-of-sample. Alpha and betas remain computable.

D9. CAPM is best described within APT as — A) a contradiction of APT B) the special case where $K = 1$ and the single factor is the market C) a five-factor model D) a model with no factors **Answer: B.** Set one factor (the market) with $\lambda = E(R_m) - R_f$ and the APT equation reduces to the CAPM/SML.

D10. HML is often found to be "redundant" in the five-factor model because — A) value stocks no longer exist B) RMW and CMA are correlated with the value characteristic and absorb its premium C) HML has a negative beta D) the market factor replaces it **Answer: B.** Profitability (RMW) and investment (CMA) overlap with value, so in US data they soak up most of HML's explanatory power.

End of Q&A — APT and Multifactor Models.

Chapter 07 — Asset Allocation

1. The Problem / The Need

Imagine two investors, each with ₹1 crore to invest for the next twenty years. The first spends her weekends poring over quarterly results, switching between HDFC Bank and ICICI Bank, timing her entry into IT stocks before earnings. The second makes exactly one decision — "60% of my money in equities, 30% in bonds, 10% in gold" — writes it on a card, and rebalances once a year without ever picking a single stock.

The uncomfortable truth of professional portfolio management is that the second investor has made the *more consequential* decision. The single choice of **how much to put in equities versus bonds versus cash versus alternatives** — the **asset allocation** — swamps almost everything else the first investor obsesses over.

This is not intuition; it is one of the most replicated findings in finance. When your job is to build portfolios for a mutual fund, a pension, an endowment, or a wealthy family, the first question is never "which stock?" It is "what is the right *mix*?" A portfolio that is 100% equity and one that is 100% government bonds can hold identically excellent securities and still deliver wildly different outcomes — one might fall 40% in a crash, the other 4%. The mix, not the contents, determines the ride.

Asset allocation exists because:

1. **Risk and return are inseparable, and the asset-class choice sets both.** You cannot earn equity-like returns from a cash portfolio, and you cannot survive a bear market with an all-equity portfolio if you need the money next year.
2. **Diversification across asset classes is the only genuinely "free lunch."** Combining assets that don't move in lockstep reduces risk without proportionally reducing return.
3. **Investors differ.** A 28-year-old accumulating for retirement and a 68-year-old drawing income from the same corpus need fundamentally different mixes. Allocation is the lever that maps a portfolio to a *person*.
4. **Behaviour needs a rulebook.** A pre-committed policy mix is the discipline that stops investors from selling equities at the bottom and buying at the top.

This chapter builds the full machinery: why allocation dominates returns, the distinction between strategic and tactical allocation, the "policy portfolio," how to allocate across the four building blocks (equities, bonds, cash, alternatives), how and when to rebalance, and how horizon and risk tolerance drive the whole exercise.

2. The Core Idea

Asset allocation is the deliberate division of a portfolio among broad asset classes — equities, fixed income, cash, and alternatives — chosen to match an investor's return objective, risk tolerance, and time horizon.

Three layered ideas sit underneath:

- **The policy portfolio (strategic asset allocation, SAA)** is the long-run "home base" mix. It is set from the investor's objectives and capital-market expectations, and it is where the portfolio lives most of the time. Think of it as the *default* the portfolio always drifts back toward.

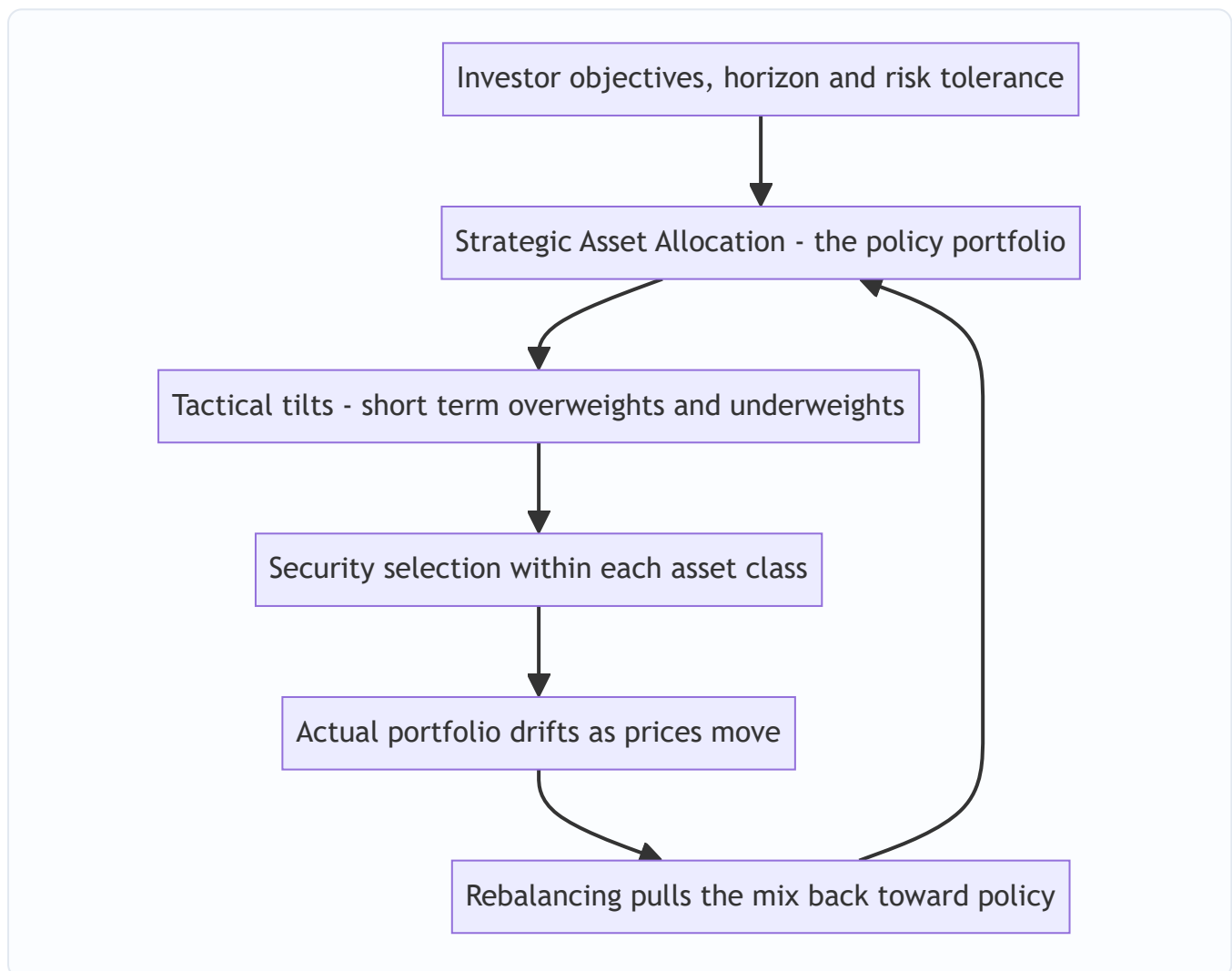
- **Tactical asset allocation (TAA)** is the set of short-to-medium-term *tilts* away from the policy mix to exploit temporary opportunities — going overweight equities when they look cheap, underweight when they look stretched. It is active bets around the anchor.
- **Rebalancing** is the maintenance mechanic: because asset prices drift, the actual mix wanders away from the target. Rebalancing periodically sells what has grown and buys what has shrunk, restoring the intended risk profile — and, as a side effect, mechanically enforcing "buy low, sell high."

The mental model to hold:

Policy portfolio = the anchor. Tactical tilts = deliberate, temporary drift you *choose*. Rebalancing = correcting the *unintended* drift the market imposes on you.

Everything else — security selection, market timing — operates *inside* the box that allocation draws.

Figure 1 — The three-layer structure of the allocation decision, from long-run anchor to day-to-day maintenance.



3. Why / How It Works

3.1 Why allocation dominates returns

The famous claim — often mangled in interviews — traces to **Brinson, Hood and Beebower (1986)** and its 1991 follow-up, studying large US pension plans. Their headline finding: **investment-policy allocation explained about 93.6% of the *variability* (variance) of a typical plan's quarterly returns over time.** Security selection and market timing accounted for the rest.

The single most important interview nuance is **what "93%" actually measures:**

- Brinson measured the **variability of returns *through time for a single fund*** — i.e., how much of a fund's ups and downs are explained by its policy mix. Answer: almost all of it. This makes sense: if you are 60% equity, your quarter-to-quarter swings are overwhelmingly driven by what equities did.
- **Ibbotson and Kaplan (2000)** decomposed the question into three distinct sub-questions and got three different answers: 1. *How much of the variability of a single fund's returns over time is explained by policy?* $\approx 90\%$ (confirms Brinson). 2. *How much of the variation in returns across different funds is explained by their differing policies?* $\approx 40\%$. The rest is explained by *how* each manager implemented — selection, timing, fees, costs. 3. *What fraction of the return level does policy explain?* $\approx 100\%$ on average — because active management is roughly a zero-sum game before costs and negative after, so the average fund's return level is essentially its policy return minus costs.

The disciplined statement for an interview:

"Asset allocation explains the vast majority — around 90% — of the *variability of a portfolio's returns over time*, and roughly all of the *average level* of returns. It explains a smaller share — about 40% — of the *differences between funds*, where selection, timing, and cost do the work. So allocation is the dominant driver of the risk you take and the return you can expect, but it is *not* the whole story of why one manager beats another."

Why is this mechanically true? Because asset classes have very different long-run risk-return signatures, and the *weights* you assign linearly determine the portfolio's expected return and, together with correlations, its variance:

$$E(R_p) = \sum_{i=1}^n w_i E(R_i)$$

$$\sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n w_i w_j \sigma_i \sigma_j \rho_{ij}$$

The expected-return equation shows the level is *purely* a weighting of asset-class returns. The variance equation shows risk depends on weights **and** correlations ρ_{ij} — which is where diversification enters.

3.2 Why diversification across classes is the "free lunch"

When two assets are imperfectly correlated ($\rho < 1$), the portfolio's risk is *less* than the weighted average of the individual risks. For two assets:

$$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2 w_1 w_2 \sigma_1 \sigma_2 \rho_{12}}$$

If $\rho_{12} = 1$, risk is just the weighted average — no benefit. As ρ falls, the cross term shrinks, and total risk drops below the weighted average *while expected return stays the weighted average*. You keep the

average return but shed some risk. That asymmetry is the entire economic case for holding multiple asset classes rather than concentrating in the highest-return one.

This is why equities + bonds + gold beats equities alone on a *risk-adjusted* basis even though equities have the highest standalone return: bonds and gold often zig when equities zag.

3.3 Why a written policy beats improvisation

Markets punish emotion. The average investor's *realised* return lags the funds they invest in — the "behaviour gap" — because they buy after rallies and sell after crashes. A pre-committed policy portfolio plus a mechanical rebalancing rule removes discretion at exactly the moments discretion destroys value. The policy is a **commitment device**.

4. Full Content — Formulas, Models and Frameworks

4.1 The two (really three) types of allocation

Dimension	Strategic (SAA)	Tactical (TAA)	Dynamic / Insured (e.g. CPPI)
Horizon	Long run (5-10+ yrs)	Short-medium (weeks-months)	Ongoing, rules-based
Basis	Long-run capital-market assumptions + investor objectives	Short-term views on relative value / momentum	Formula tied to a floor / cushion
Question answered	"What mix should I hold <i>on average</i> ?"	"Where should I <i>tilt</i> right now?"	"How do I protect a floor while participating?"
Turnover	Very low	Moderate-high	Can be high (buys strength, sells weakness)
Risk profile	Fixed target risk	Varies around target	Risk rises with wealth above floor
Nickname	The policy portfolio	Active over/underweights	Constant Proportion Portfolio Insurance

Strategic Asset Allocation (SAA) sets the long-term target weights and permissible ranges. Example policy statement: "*Equity 60% (range 50-70%), Fixed income 30% (20-40%), Alternatives 7%, Cash 3%.*" The ranges are the *tactical latitude* — TAA operates only within them.

Tactical Asset Allocation (TAA) deviates from policy to add value, then reverts. If the policy equity weight is 60% and a manager believes equities are cheap, she might go to 68% (still inside the 50-70% band). TAA is a *mean-reverting* bet: it assumes the tilt will pay off and then the position closes back to policy.

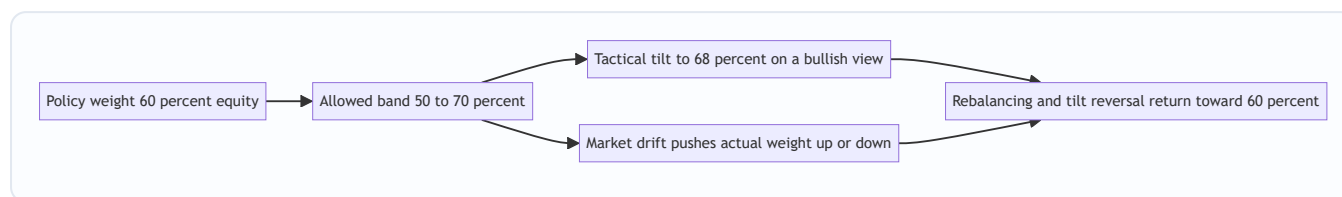
Dynamic strategies change weights according to a formula tied to market moves rather than to forecasts. The classic is **CPPI (Constant Proportion Portfolio Insurance)**:

$$\text{Equity exposure} = m \times (\text{Portfolio value} - \text{Floor})$$

where m is the multiplier and (Portfolio – Floor) is the "cushion." As the portfolio rises, exposure rises; as it falls toward the floor, exposure is cut toward zero. It is the mirror image of rebalancing (it *buys winners, sells*

losers) and is used when protecting a minimum value matters more than mean reversion.

Figure 2 — How SAA, TAA and rebalancing relate: TAA and drift both move the actual mix, but only within the strategic bands.



4.2 Setting the strategic mix — the objective function

The formal engine behind SAA is **mean-variance optimisation (Markowitz)**: choose weights w_i to maximise a risk-adjusted objective (utility):

$$U = E(R_p) - \frac{1}{2} \lambda \sigma_p^2$$

where λ is the investor's risk-aversion coefficient. A high- λ (risk-averse) investor's optimum sits at lower σ_p — hence more bonds and cash; a low- λ investor accepts more equity. The optimiser traces the **efficient frontier**; the chosen SAA is the frontier point matching the investor's λ , subject to constraints (no shorting, liquidity needs, etc.).

In practice, pure optimisers are notoriously sensitive to input errors (tiny changes in expected returns produce wildly different weights), so professionals temper them with: constraints and ranges, **Black-Litterman** blending of market-equilibrium weights with the manager's views, resampling, or simple **heuristic rules** (see 4.6).

4.3 The four building blocks

Asset class	Role in portfolio	Typical long-run real return	Risk (volatility)	Behaves well when...
Equities	Growth engine; long-run compounding	High (~6-8% real)	High (15-20% p.a.)	Economy expands, earnings grow
Fixed income (bonds)	Income + ballast; diversifier	Low-moderate (~1-3% real)	Low-moderate (4-8%)	Growth slows, rates fall, deflation fear
Cash / money market	Liquidity, capital preservation, dry powder	≈0 real	Very low	Everything else falls; crises
Alternatives (real estate, gold, commodities, PE, hedge funds, infra)	Diversification, inflation hedge, illiquidity premium	Varies	Varies	Inflation, regime shifts, low equity-bond correlation

The art of allocation is combining these so their *correlations* work for you. Historically equities and high-quality government bonds have often been **negatively or weakly correlated** — bonds rally in the "flight to safety" when equities crash — which is why the 60/40 portfolio became the industry default. Gold is a classic crisis and inflation hedge with near-zero long-run correlation to equities. (Caveat: correlations are *not* stable

— in 2022, rising inflation drove *both* equities and bonds down together, breaking the 60/40 diversification for a year. Alternatives earned their keep that year.)

4.4 The policy portfolio in practice

A **policy portfolio** is documented in the **Investment Policy Statement (IPS)** and specifies:

- Target weights for each asset class and permissible ranges.
- The **benchmark** for each class (e.g., Nifty 50 TRI for Indian large-cap equity, CRISIL Composite Bond Index for fixed income).
- Rebalancing rules and tolerances.
- Constraints: liquidity, time horizon, tax, legal/regulatory, unique circumstances (the "RRTTLLU" framework — Return, Risk, Time, Taxes, Liquidity, Legal, Unique).

The **policy-portfolio return is the benchmark against which the *whole* active process is judged**. Total portfolio return decomposes as:

$$R_{\text{portfolio}} = R_{\text{policy}} + R_{\text{TAA}} + R_{\text{selection}} + R_{\text{interaction}}$$

The first term is the passive result of just holding the policy mix. The remaining terms are what active management (allocation tilts + stock picking) *added or subtracted* — the subject of **performance attribution** (Chapter on attribution). This is exactly why the policy portfolio matters so much: it is both the strategy *and* the yardstick.

4.5 Rebalancing — the maintenance engine

Left alone, a 60/40 portfolio in which equities outperform becomes 70/30, then 75/25 — silently taking *more* risk than intended precisely when equities are most expensive. Rebalancing restores the target. Two dominant disciplines:

(a) Calendar rebalancing — rebalance on a fixed schedule (monthly, quarterly, annually) regardless of drift. - *Pros*: simple, predictable, low monitoring cost. - *Cons*: may trade when unnecessary (small drift) or fail to react between dates to a large move.

(b) Threshold (percentage-of-portfolio) rebalancing — rebalance whenever any asset class drifts beyond a tolerance band (e.g., ± 5 percentage points, or $\pm 25\%$ of the target weight). - *Pros*: responds to actual risk; only trades when it matters; captures more of the "buy low/sell high" premium. - *Cons*: requires continuous monitoring; can trigger many trades in volatile markets.

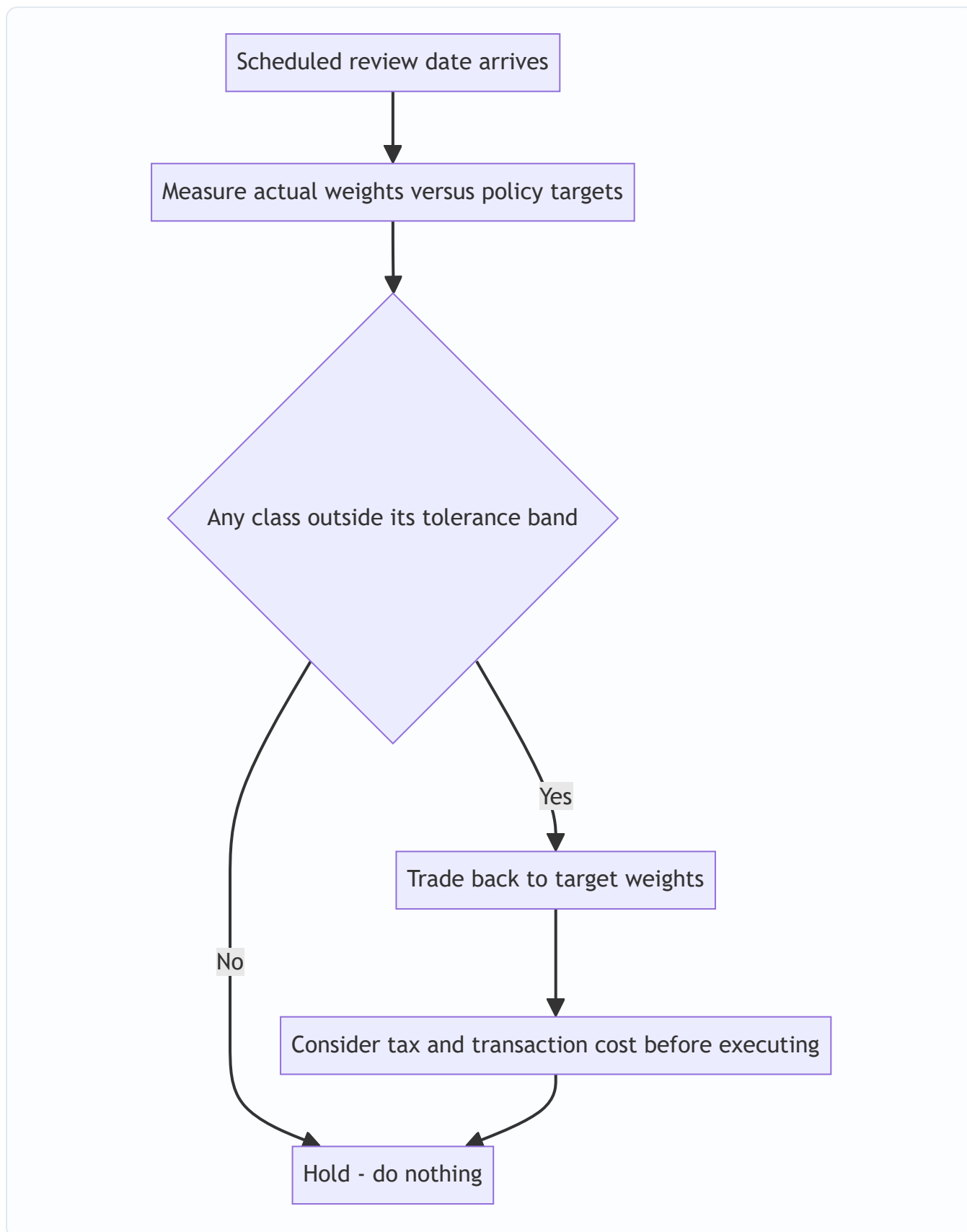
(c) Hybrid (calendar-and-threshold) — check on a schedule (say quarterly) but only trade if drift exceeds the band. This is the professional default: it caps monitoring cost while avoiding needless small trades.

Key design trade-offs:

Factor	Push toward <i>tighter</i> bands / <i>more frequent</i>	Push toward <i>wider</i> bands / <i>less frequent</i>
Transaction costs	Low costs → rebalance more	High costs → tolerate drift
Taxes	Tax-sheltered account → rebalance freely	Taxable account → wider bands, defer gains
Volatility of asset	High vol → tighter to control risk	—
Correlation with rest	Low correlation → wider bands acceptable	—
Risk tolerance	Low tolerance → tighter bands	High tolerance → wider

The rebalancing bonus (why it can add return, not just control risk): because asset returns are somewhat mean-reverting and volatile, systematically selling the asset that rose and buying the one that fell harvests a small "**volatility / diversification return.**" It is not guaranteed — in a strong sustained trend, rebalancing *drags* (you keep selling the winner too early). Its dependable benefit is **risk control**; the return bonus is a bonus.

Figure 3 — Decision flow for a hybrid rebalancing policy.



4.6 Horizon and risk tolerance — the two master inputs

Everything in SAA ultimately flows from two investor characteristics:

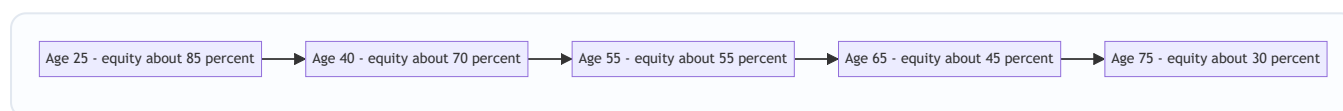
(1) Investment horizon. Longer horizons justify more equity because: - **Time diversification of shortfall risk (in a horizon sense):** while annual equity volatility does *not* shrink, a long horizon lets an investor *ride*

out drawdowns without being forced to sell at the bottom, and lets compounding dominate. (Careful nuance for interviews: the *dispersion of terminal wealth* actually *widens* with horizon in absolute terms — equities are not "safe if held long enough" in a strict variance sense. What a long horizon buys you is the *ability to wait*, i.e., low liquidity risk, not the elimination of risk.) - Human capital: a young worker's future earnings are a large, bond-like asset, so she can hold more equities in her financial portfolio to balance the total.

(2) Risk tolerance, which has two components: - **Ability to take risk** (objective): horizon, wealth relative to needs, income stability, liquidity needs. A person with 30 years and stable income has high *ability*. - **Willingness to take risk** (psychological): how much volatility the investor can stomach without panic-selling. - **Rule:** when ability and willingness conflict, the prudent adviser anchors to the *lower* of the two (or educates the client), because a mathematically optimal portfolio the client abandons at the bottom is worthless.

A common heuristic (crude but interview-worthy) is the **"110 minus age"** rule: equity weight $\approx 110 - \text{age}$. A 30-year-old $\rightarrow$ 80% equity; a 65-year-old $\rightarrow$ 45%. It encodes both horizon shortening and rising risk-aversion with age, and underpins **target-date / lifecycle funds**, whose equity "glide path" declines automatically as the target retirement year approaches.

Figure 4 — The glide path: equity share falls as horizon shortens toward retirement.



5. Worked Examples

Example 1 — Expected return and risk of a strategic mix (and the diversification effect)

An AMC proposes a policy portfolio of **60% equity, 30% bonds, 10% gold**. Capital-market assumptions:

Class	Expected return	Volatility (σ)
Equity	12%	18%
Bonds	7%	6%
Gold	8%	15%

Correlations: Equity-Bonds = **-0.20**, Equity-Gold = **0.10**, Bonds-Gold = **0.30**.

Step 1 — Portfolio expected return.

$$E(R_p) = 0.60(12\%) + 0.30(7\%) + 0.10(8\%) = 7.20\% + 2.10\% + 0.80\% = \mathbf{10.10\%}$$

Step 2 — Portfolio variance. Weights $w_E=0.6$, $w_B=0.3$, $w_G=0.1$. Use $\sigma_p^2 = \sum \sum w_i w_j \sigma_i \sigma_j \rho_{ij}$ (in decimals: $\sigma_E=0.18$, $\sigma_B=0.06$, $\sigma_G=0.15$).

Own-variance terms: - Equity: $0.6^2 \times 0.18^2 = 0.36 \times 0.0324 = 0.011664$ - Bonds: $0.3^2 \times 0.06^2 = 0.09 \times 0.0036 = 0.000324$ - Gold: $0.1^2 \times 0.15^2 = 0.01 \times 0.0225 = 0.000225$

Cross terms (each counted twice $\rightarrow$ use $2w_i w_j \sigma_i \sigma_j \rho_{ij}$): - Equity-Bonds: $2(0.6)(0.3)(0.18)(0.06)(-0.20) = 2(0.18)(0.0108)(-0.20) = -0.0003888$; $\times 2 = \mathbf{-0.0007776}$ - Equity-Gold: $2(0.6)(0.1)(0.18)(0.15)(0.10) = 2(0.06)(0.027)(0.10) = 0.000324$

$\times 0.10 = 0.000162$; $\times 2 = \mathbf{0.000324}$ - Bonds-Gold: $\$2(0.3)(0.1)(0.06)(0.15)(0.30) = 2(0.03)(0.009)(0.30)\$$.
 $\$0.03 \times 0.009 = 0.00027$; $\times 0.30 = 0.000081$; $\times 2 = \mathbf{0.000162}$

Sum: $\sigma_p^2 = 0.011664 + 0.000324 + 0.000225 - 0.0007776 + 0.000324 + 0.000162 = 0.0129214$

$\sigma_p = \sqrt{0.0129214} = 0.11367 \approx \mathbf{11.37\%}$

Step 3 — Verify the diversification benefit. The *weighted-average* volatility (what you'd get with all $\rho=1$) is:
 $0.6(18\%) + 0.3(6\%) + 0.1(15\%) = 10.8\% + 1.8\% + 1.5\% = 14.1\%$ Actual portfolio risk is **11.37% < 14.1%** — diversification shaved off ~2.7 percentage points of volatility *while keeping the 10.10% return. This is the free lunch, quantified.* (Sanity check: because the equity-bond correlation is negative, the biggest cross term is negative, pulling risk well below the weighted average — consistent with the result.)

Example 2 — Rebalancing: calendar vs "do nothing," and the buy-low/sell-high mechanic

Start: ₹10,00,000 at target **60% equity / 40% bonds** → ₹6,00,000 equity, ₹4,00,000 bonds.

Year 1: equities **+25%**, bonds **+4%**. - Equity → $6,00,000 \times 1.25 = \mathbf{₹7,50,000}$ - Bonds → $4,00,000 \times 1.04 = \mathbf{₹4,16,000}$ - Total = **₹11,66,000**. Actual weights: equity $7,50,000/11,66,000 = \mathbf{64.3\%}$, bonds 35.7%.

Equity has drifted from 60% to 64.3% — the portfolio is now *riskier* than policy.

Rebalance back to 60/40 (target equity = $0.60 \times 11,66,000 = ₹6,99,600$): - Sell equity ₹7,50,000 – ₹6,99,600 = **₹50,400** (sell the winner). - Buy bonds to reach $0.40 \times 11,66,000 = ₹4,66,400$, i.e. +₹50,400 (buy the laggard).

Year 2 (a reversal): equities **–15%**, bonds **+5%**. Compare two paths on the ₹11,66,000.

Path A — Rebalanced (starts Yr2 at 6,99,600 / 4,66,400): - Equity → $6,99,600 \times 0.85 = ₹5,94,660$ - Bonds → $4,66,400 \times 1.05 = ₹4,89,720$ - **Total A = ₹10,84,380**

Path B — Never rebalanced (starts Yr2 at 7,50,000 / 4,16,000): - Equity → $7,50,000 \times 0.85 = ₹6,37,500$ - Bonds → $4,16,000 \times 1.05 = ₹4,36,800$ - **Total B = ₹10,74,300**

Result: the rebalanced portfolio ends at **₹10,84,380** vs **₹10,74,300** unrebalanced — **₹10,080 richer**, because rebalancing trimmed the over-weighted equity *before* it fell. **Verification of the mechanic:** by selling ₹50,400 of equity at the top, Path A had less capital exposed to the –15% equity move; the difference in outcomes $\approx 50,400 \times (\text{return gap between bonds +5\% and equity –15\%} = 20\%) = 50,400 \times 0.20 = \mathbf{₹10,080}$ — matches exactly. This is the rebalancing bonus when markets mean-revert. (Had equities continued *up* in Year 2, Path B would have won — confirming rebalancing's benefit is risk-control and reversal-capture, not a free return.)

Example 3 — Mapping risk tolerance to a mix via the utility function

Two clients, same expected-return/volatility menu as Example 1's *portfolios* along an efficient frontier. Suppose two candidate policy mixes:

Mix	E(R)	σ
Aggressive (80/20)	11.5%	15.0%
Balanced (60/40)	10.1%	11.4%

Use utility $U = E(R) - \frac{1}{2}\lambda\sigma^2$ (returns and σ in decimals).

Client X — high risk aversion, $\lambda = 4$: - Aggressive: $\$0.115 - 0.5(4)(0.15^2) = 0.115 - 2(0.0225) = 0.115 - 0.045 = \mathbf{\$0.070}$ - Balanced: $\$0.101 - 0.5(4)(0.114^2) = 0.101 - 2(0.012996) = 0.101 - 0.025992 = \mathbf{\$0.0750}$ - **Balanced wins (0.0750 > 0.070)**. The risk-averse client should hold 60/40.

Client Y — low risk aversion, $\lambda = 1.5$: - Aggressive: $\$0.115 - 0.5(1.5)(0.0225) = 0.115 - 0.016875 = \mathbf{\$0.0981}$ - Balanced: $\$0.101 - 0.5(1.5)(0.012996) = 0.101 - 0.009747 = \mathbf{\$0.0913}$ - **Aggressive wins (0.0981 > 0.0913)**. The risk-tolerant client should hold 80/20.

Interpretation and verification: the *same* two portfolios flip ranking purely because of λ . The break-even λ (where the two mixes tie) solves $\$0.115 - 0.5\lambda(0.0225) = \$0.101 - 0.5\lambda(0.012996) \rightarrow \$0.014 = 0.5\lambda(0.0225 - 0.012996) = 0.5\lambda(0.009504) = 0.004752\lambda \rightarrow \lambda = 2.95$. So investors with λ below ~ 2.95 prefer aggressive, above ~ 2.95 prefer balanced — and indeed $\lambda=1.5$ chose aggressive, $\lambda=4$ chose balanced, consistent. This is *exactly* how risk tolerance selects the strategic mix.

6. Connections

- **Modern Portfolio Theory & the efficient frontier (earlier chapters):** SAA is the act of choosing a point on the efficient frontier that matches the investor's risk aversion. Asset allocation is MPT applied at the asset-class level.
 - **CAPM and the Capital Market Line:** the "two-fund separation" theorem (hold the market portfolio + risk-free asset, mixed by risk tolerance) is the theoretical purest form of asset allocation — the risk-free/risky split is literally a cash-vs-equity allocation.
 - **Performance attribution:** the policy portfolio is the baseline; attribution decomposes active return into *allocation effect* (TAA) and *selection effect*. You cannot attribute performance without first defining the policy mix.
 - **Risk-adjusted performance (Sharpe, Treynor, Jensen's alpha):** these evaluate whether tactical tilts and selection *added value per unit of risk* over just holding the policy mix. A manager who beats the policy return but with far more volatility may have a *worse* Sharpe.
 - **Behavioural finance:** the policy portfolio + mechanical rebalancing is the institutional answer to loss aversion, recency bias, and the behaviour gap.
 - **Fixed income & equity valuation chapters:** capital-market assumptions (expected returns per class) that feed the optimiser come from bond-yield math (yield $\approx$ expected return for bonds) and equity risk-premium models.
 - **Liability-driven investing (pensions/insurers):** for these investors, allocation is set against *liabilities*, not an absolute benchmark — a specialised form of SAA.
-

7. Key Terms

Term	Definition
Asset allocation	Division of a portfolio among broad asset classes to match objectives, horizon and risk tolerance.
Strategic asset allocation (SAA)	The long-run target mix; the policy portfolio.
Tactical asset allocation (TAA)	Short-term deliberate tilts away from the policy mix to exploit opportunities.
Policy portfolio	The documented long-run target weights, benchmarks and ranges; the yardstick for active performance.
Investment Policy Statement (IPS)	The governing document capturing objectives, constraints, the policy mix and rebalancing rules.
Rebalancing	Trading back to target weights after price drift.
Calendar rebalancing	Rebalancing on a fixed schedule.
Threshold / tolerance-band rebalancing	Rebalancing only when a class drifts beyond a set band.
Rebalancing bonus	The small return earned from systematically selling winners/buying losers when markets mean-revert.
Glide path	The pre-set schedule by which equity weight declines as a target date approaches.
Target-date / lifecycle fund	A fund that automatically follows a glide path.
Efficient frontier	Set of portfolios offering the highest return for each level of risk.
Risk tolerance (ability vs willingness)	Objective capacity vs psychological comfort for bearing risk.
CPPI	Constant Proportion Portfolio Insurance — dynamic strategy keeping exposure = $m \times$ cushion above a floor.
Human capital	Present value of future labour income; a bond-like asset justifying more financial-portfolio equity when young.
RRTLLU	Return, Risk, Time horizon, Taxes, Liquidity, Legal, Unique — the IPS constraint checklist.

8. Common Confusions

1. **"Asset allocation explains 90% of returns."** *Wrong wording.* It explains ~90% of the **variability of a single portfolio's returns over time**, and ~100% of the average **level**, but only ~40% of the **variation across funds**. Never conflate "variability over time" with "how one fund beats another."
2. **SAA vs TAA vs rebalancing.** SAA = the *target*. TAA = *deliberate* short-term deviation from it. Rebalancing = *correcting unintended* deviation caused by price drift. Rebalancing pushes *back toward* policy; TAA

pushes *away* from it (on purpose).

3. **"Time diversification makes equities safe in the long run."** Loosely true in the sense that you can *wait out* drawdowns, but strictly false: the absolute dispersion of terminal wealth *grows* with horizon. Long horizon buys you the *ability to hold*, not the elimination of risk.
4. **Rebalancing always adds return.** No. Its reliable benefit is **risk control**. The return "bonus" appears only when markets mean-revert; in a sustained trend, rebalancing *drags* by selling winners too soon.
5. **More frequent rebalancing is better.** Not once you count **transaction costs and taxes**. In taxable accounts, tight bands can trigger avoidable capital-gains tax that dwarfs the risk-control benefit. Hybrid (check often, trade rarely) usually dominates.
6. **The 60/40 is dead / always diversified.** Correlations are regime-dependent. Equity-bond correlation is usually negative (great diversification) but turned *positive* in 2022's inflation shock, so both fell together. Allocation must be stress-tested, not assumed.
7. **Diversification reduces return.** It reduces *risk* for a given return; the portfolio's expected return is still the weighted average. You give up the *chance* of the single best asset's upside, not expected return per unit of risk.
8. **Ability and willingness to take risk are the same.** They diverge constantly. A wealthy retiree may have high ability but low willingness. Anchor to the more conservative and educate.

9. Recap

- **Asset allocation is the dominant decision** in portfolio management: it explains ~90% of the variability of a portfolio's returns over time and essentially all of the average return level. Security selection and timing matter for *beating peers*, but the mix sets the fundamental risk-return signature.
- **Diversification across imperfectly correlated classes is the only free lunch** — it cuts risk below the weighted average while preserving weighted-average return (Example 1: 11.4% risk vs 14.1% weighted average).
- **Strategic asset allocation (the policy portfolio)** is the long-run anchor, derived from objectives, horizon, risk tolerance and capital-market assumptions, documented in the IPS, and used as the benchmark for all active management.
- **Tactical asset allocation** is deliberate, temporary tilting around the policy mix within permitted bands; **rebalancing** corrects the unintended drift the market imposes, enforcing buy-low/sell-high and controlling risk (Example 2: +₹10,080 in a reversal).
- **Rebalancing choices** — calendar vs threshold vs hybrid — trade off risk control against transaction costs and taxes; the hybrid rule is the professional default.
- **Horizon and risk tolerance are the master inputs.** Longer horizon and higher risk tolerance → more equity; the utility function $U = E(R) - \frac{1}{2}\lambda\sigma^2$ formally maps risk aversion to the chosen mix (Example 3), and the glide path automates this over a lifetime.

10. Quick-Reference / Interview Points

One-liners to have loaded:

- "Asset allocation drives roughly 90% of the variability of a portfolio's returns over time and ~100% of the average return level; it's ~40% of the cross-fund variation — selection and cost do the rest (Brinson 1986; Ibbotson-Kaplan 2000)."
- "Policy portfolio = anchor and benchmark; TAA = deliberate tilt; rebalancing = correcting unintended drift."
- "Rebalancing's dependable payoff is risk control; the return bonus only shows up when markets mean-revert."

Core formulas:

Concept	Formula
Portfolio expected return	$E(R_p) = \sum w_i E(R_i)$
Portfolio variance	$\sigma_p^2 = \sum_i \sum_j w_i w_j \sigma_i \sigma_j \rho_{ij}$
Two-asset risk	$\sigma_p = \sqrt{w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \sigma_1 \sigma_2 \rho_{12}}$
Utility (mix selection)	$U = E(R_p) - \frac{1}{2\lambda} \sigma_p^2$
Return decomposition	$R_{\text{port}} = R_{\text{policy}} + R_{\text{TAA}} + R_{\text{selection}} + R_{\text{interaction}}$
CPPI exposure	$\text{Exposure} = m \times (\text{Value} - \text{Floor})$
Equity heuristic	Equity % $\approx 110 - \text{age}$

Rapid-fire Q&A:

- Q: SAA vs TAA? A: SAA is the long-run policy mix from objectives; TAA is short-term tactical tilts within the policy's ranges to exploit relative value.
- Q: Calendar vs threshold rebalancing — which is better? A: Threshold responds to actual risk and captures more rebalancing premium but needs monitoring and can trade often; calendar is simple but blind to drift between dates. Hybrid (scheduled check, trade only if outside band) is best in practice, especially after tax.
- Q: Why does a longer horizon justify more equity? A: It lets you ride out drawdowns without forced selling and lets compounding dominate; young investors also have bond-like human capital to offset. Nuance: absolute terminal-wealth dispersion still grows — horizon buys the *ability to wait*, not risk elimination.
- Q: Ability vs willingness to take risk? A: Ability is objective (horizon, wealth, income stability); willingness is psychological. Anchor the mix to the lower of the two.
- Q: Does diversification lower returns? A: No — expected return is the weighted average; it lowers *risk* for that return, improving Sharpe.
- Q: What is the policy portfolio's second job? A: It's the benchmark for performance attribution — active return is measured against just holding the policy mix.
- Q: When does rebalancing hurt? A: In sustained trends — you sell the winner too early. And in taxable accounts, over-frequent rebalancing triggers tax drag.

Q&A — Asset Allocation

Scope: Investments — Chapter 07 (Asset Allocation). Every question is followed by a full model answer. All rates are annual and in percent unless stated. Work every numerical problem yourself before checking the solution. Sections: **A** concept-check · **B** numerical (full step-by-step, reconciling) · **C** interview-style · **D** MCQs with reasoning.

The chapter in one line

$$E(R_p) = \sum_i w_i E(R_i), \quad \sigma_p^2 = \sum_i \sum_j w_i w_j \sigma_i \sigma_j \rho_{ij}, \quad U = E(R_p) - \frac{1}{2} \lambda \sigma_p^2$$

One-line statement: The mix of asset classes — not the securities inside them — sets the portfolio's fundamental risk-return signature; the strategic mix is the long-run anchor, tactical tilts are deliberate deviations from it, and rebalancing corrects the unintended drift the market imposes.

Section A — Concept Check

A1. Define asset allocation and explain why it is described as the "dominant" portfolio decision. Asset allocation is the deliberate division of a portfolio among broad asset classes — equities, fixed income, cash, and alternatives — chosen to match the investor's return objective, risk tolerance, and time horizon. It dominates because the weights linearly fix the portfolio's expected return and, together with correlations, its variance. A 100% equity and a 100% government-bond portfolio can hold identically excellent securities yet deliver completely different rides — one may fall 40% in a crash, the other 4%. The mix, not the contents, determines the outcome, which is why the first professional question is never "which stock?" but "what mix?"

A2. Distinguish strategic asset allocation (SAA), tactical asset allocation (TAA), and rebalancing. SAA is the long-run *target* mix — the policy portfolio — derived from objectives and capital-market assumptions; it is the anchor the portfolio lives at most of the time. TAA is a *deliberate*, short-to-medium-term tilt *away* from the policy mix to exploit temporary opportunities, operating only within permitted bands. Rebalancing *corrects the unintended* drift caused by price moves, pulling the actual mix *back toward* policy. Key contrast: TAA pushes away from the target on purpose; rebalancing pushes back toward it mechanically.

A3. Explain precisely what the Brinson "93%" finding does and does not say. Brinson, Hood and Beebower (1986) found that investment-policy allocation explained about 93.6% of the *variability (variance) of a single plan's returns through time*. It does **not** say allocation explains 93% of the *level* of returns, nor 93% of the *differences between funds*. Ibbotson–Kaplan (2000) separated the questions: policy explains ~90% of a single fund's variability over time, ~100% of the average return *level* (because active management is roughly zero-sum before costs), but only ~40% of the *variation across funds*, where selection, timing, and cost do the work. The interview trap is conflating "variability over time" with "how one fund beats another."

A4. Why is diversification across asset classes called the only "free lunch"? Because when assets are imperfectly correlated ($\rho < 1$), portfolio risk falls *below* the weighted average of the individual risks, while expected return remains *exactly* the weighted average. You keep the average return but shed risk — an

asymmetry no other technique offers for free. As ρ falls, the cross term in the variance formula shrinks (and goes negative if $\rho < 0$), pulling total risk down.

A5. What is the policy portfolio, and what are its two jobs? The policy portfolio is the documented long-run target mix (weights, benchmarks, ranges) held in the Investment Policy Statement. Its first job is *strategy* — it is where the portfolio lives and the risk profile the investor signed up for. Its second job is to be the *benchmark*: total return decomposes as $R_{\text{portfolio}} = R_{\text{policy}} + R_{\text{TAA}} + R_{\text{selection}} + R_{\text{interaction}}$, so active value-add can only be measured against the passive result of just holding the policy mix. It is both the strategy and the yardstick.

A6. Distinguish ability and willingness to take risk, and state the rule when they conflict. Ability is *objective* capacity — horizon, wealth relative to needs, income stability, liquidity needs. Willingness is *psychological* — how much volatility the investor can stomach without panic-selling. When they conflict, anchor to the *lower* of the two (or educate the client), because a mathematically optimal portfolio the client abandons at the bottom is worthless. A wealthy retiree may have high ability but low willingness — the prudent mix respects the willingness.

A7. Why does a longer horizon justify more equity — and what is the interview nuance? A long horizon lets the investor *ride out* drawdowns without being forced to sell at the bottom and lets compounding dominate; a young worker also holds large bond-like *human capital*, so more financial-portfolio equity balances the total. The nuance: annual equity volatility does not shrink, and the absolute *dispersion of terminal wealth actually widens* with horizon. Equities are not "safe if held long enough" in a strict variance sense — a long horizon buys the *ability to wait* (low liquidity risk), not the elimination of risk.

A8. Contrast calendar, threshold, and hybrid rebalancing. *Calendar* rebalances on a fixed schedule (e.g., quarterly) regardless of drift — simple and predictable but blind to moves between dates. *Threshold* rebalances whenever a class breaches a tolerance band (e.g., ± 5 pp) — it responds to actual risk and captures more of the buy-low/sell-high premium but needs continuous monitoring and can trade often. *Hybrid* checks on a schedule but trades only if outside the band — the professional default, capping monitoring cost while avoiding needless small trades.

A9. When does rebalancing add return, and when does it hurt? Its *dependable* benefit is risk control. A return "bonus" (the volatility/diversification return) appears only when markets *mean-revert* — systematically selling the risen asset and buying the fallen one harvests the reversal. In a strong *sustained trend*, rebalancing *drags* because you keep selling the winner too early. In taxable accounts, over-frequent rebalancing can also trigger capital-gains tax that dwarfs the risk-control benefit.

A10. What is CPPI and how is it the mirror image of rebalancing? Constant Proportion Portfolio Insurance sets risky exposure = $\$m \times (\text{Portfolio value} - \text{Floor})$, where the bracket is the "cushion." As the portfolio rises, exposure rises; as it falls toward the floor, exposure is cut toward zero. It therefore *buys winners and sells losers* — the opposite of rebalancing, which sells winners and buys losers. CPPI is used when protecting a minimum value matters more than mean reversion.

A11. What does the RRTLLU checklist capture, and where does it live? It is the IPS constraint framework: **R**eturn objective, **R**isk tolerance, **T**ime horizon, **T**axes, **L**iquidity, **L**egal/regulatory, **U**nique circumstances. It lives in the Investment Policy Statement and ensures the strategic mix is feasible and tailored to the specific investor before any optimisation.

Section B — Numerical Problems

B1. Expected return and risk of a strategic mix. A policy portfolio is 60% equity, 30% bonds, 10% gold. $E(R)$: equity 12%, bonds 7%, gold 8%. σ : equity 18%, bonds 6%, gold 15%. Correlations: $E-B = -0.20$, $E-G = 0.10$, $B-G = 0.30$. Find $E(R_p)$ and σ_p , and quantify the diversification benefit.

Step 1 — Expected return. $E(R_p) = 0.60(12) + 0.30(7) + 0.10(8) = 7.20 + 2.10 + 0.80 = \mathbf{10.10\%}$

Step 2 — Variance (decimals: $\sigma_E = 0.18$, $\sigma_B = 0.06$, $\sigma_G = 0.15$). Own-variance terms: - Equity: $0.6^2(0.18^2) = 0.36(0.0324) = 0.011664$ - Bonds: $0.3^2(0.06^2) = 0.09(0.0036) = 0.000324$ - Gold: $0.1^2(0.15^2) = 0.01(0.0225) = 0.000225$

Cross terms ($2w_iw_j\sigma_i\sigma_j\rho_{ij}$): - E-B: $2(0.6)(0.3)(0.18)(0.06)(-0.20) = -0.0007776$ - E-G: $2(0.6)(0.1)(0.18)(0.15)(0.10) = 0.000324$ - B-G: $2(0.3)(0.1)(0.06)(0.15)(0.30) = 0.000162$

Sum: $\sigma_p^2 = 0.011664 + 0.000324 + 0.000225 - 0.0007776 + 0.000324 + 0.000162 = 0.0129214$
 $\sigma_p = \sqrt{0.0129214} = \mathbf{11.37\%}$

Step 3 — Reconcile. Weighted-average σ (the all- $\rho = 1$ case) = $0.6(18) + 0.3(6) + 0.1(15) = 10.8 + 1.8 + 1.5 = 14.1\%$. Actual $11.37\% < 14.1\%$ — diversification shaved ~2.7 pp of volatility while keeping the 10.10% return. The negative E-B correlation drives the largest cross term negative, pulling risk well below the weighted average — the free lunch, quantified.

B2. Rebalancing versus doing nothing in a reversal. Start ₹10,00,000 at 60/40 → ₹6,00,000 equity, ₹4,00,000 bonds. Year 1: equity +25%, bonds +4%. Rebalance back to 60/40. Year 2: equity –15%, bonds +5%. Compare rebalanced vs never-rebalanced ending wealth.

Year 1 growth. Equity → $6,00,000 \times 1.25 = ₹7,50,000$; bonds → $4,00,000 \times 1.04 = ₹4,16,000$; total = ₹11,66,000. Equity weight = $7,50,000 / 11,66,000 = \mathbf{64.3\%}$ (drifted above 60%, now riskier).

Rebalance. Target equity = $0.60 \times 11,66,000 = ₹6,99,600$ → sell ₹50,400 equity, buy ₹50,400 bonds (bonds to ₹4,66,400).

Year 2, Path A (rebalanced). Equity $6,99,600 \times 0.85 = ₹5,94,660$; bonds $4,66,400 \times 1.05 = ₹4,89,720$; **total = ₹10,84,380.**

Year 2, Path B (never rebalanced). Equity $7,50,000 \times 0.85 = ₹6,37,500$; bonds $4,16,000 \times 1.05 = ₹4,36,800$; **total = ₹10,74,300.**

Result & reconciliation. Rebalanced wins by $₹10,84,380 - ₹10,74,300 = \mathbf{₹10,080}$. Check the mechanic: Path A had ₹50,400 less equity exposed to the Year-2 return gap (bonds +5% vs equity –15% = 20 pp). $₹50,400 \times 0.20 = \mathbf{₹10,080}$ — exact match. The bonus exists because markets reversed; had equity kept rising, Path B would have won.

B3. Mapping risk aversion to a mix via utility. Two candidate mixes — Aggressive (80/20): $E(R) = 11.5\%$, $\sigma = 15.0\%$; Balanced (60/40): $E(R) = 10.1\%$, $\sigma = 11.4\%$. Using $U = E(R) - \frac{1}{2}\lambda\sigma^2$, which does an investor with $\lambda = 4$ choose, and which with $\lambda = 1.5$?

$\lambda = 4$. - Aggressive: $0.115 - 0.5(4)(0.15^2) = 0.115 - 2(0.0225) = 0.115 - 0.045 = \mathbf{0.0700}$ - Balanced: $0.101 - 0.5(4)(0.114^2) = 0.101 - 2(0.012996) = 0.101 - 0.025992 = \mathbf{0.0750}$ - **Balanced wins** ($0.0750 > 0.0700$).

$\lambda = 1.5$. - Aggressive: $0.115 - 0.5(1.5)(0.0225) = 0.115 - 0.016875 = \mathbf{0.0981}$ - Balanced: $0.101 - 0.5(1.5)(0.012996) = 0.101 - 0.009747 = \mathbf{0.0913}$ - **Aggressive wins** ($0.0981 > 0.0913$).

Break-even λ (reconcile). Set utilities equal: $\$0.115 - 0.5\lambda(0.0225) = 0.101 - 0.5\lambda(0.012996)\$ \rightarrow \$0.014 = 0.5\lambda(0.009504) = 0.004752\lambda\$ \rightarrow \lambda = 2.95\$$. So λ below 2.95 $\rightarrow$ aggressive, above $\rightarrow$ balanced. $\lambda = 1.5$ (below) chose aggressive; $\lambda = 4$ (above) chose balanced — consistent.

B4. Threshold rebalancing trigger. Policy equity weight is 55% with a ± 5 -percentage-point band. Current portfolio: equity ₹6,30,000, bonds ₹3,70,000. Is a trade required, and if so how much equity is bought/sold? Total = ₹10,00,000; equity weight = $6,30,000 / 10,00,000 = 63\%$. Band is 50%–60%; $63\% > 60\%$, so the band is **breached** and a trade is required. Target equity = $0.55 \times 10,00,000 = ₹5,50,000 \rightarrow$ **sell ₹80,000 of equity** and buy ₹80,000 of bonds. Check: new equity $5,50,000 / 10,00,000 = 55\% \checkmark$, bonds $4,50,000 = 45\% \checkmark$.

B5. The "110 minus age" heuristic and a glide path. An investor is 35 today. Using equity % $\approx 110 - \text{age}$, what is the target equity weight now and at 65, and what is the average annual reduction in equity weight along the glide path?

Now: $110 - 35 = 75\%$ equity. At 65: $110 - 65 = 45\%$ equity. Change over 30 years = $75 - 45 = 30$ pp $\rightarrow$ **1 percentage point per year**. This mechanical decline is exactly what a target-date/lifecycle fund automates as it de-risks toward the retirement year.

B6. CPPI exposure. Portfolio value ₹20,00,000, floor ₹15,00,000, multiplier $m = 3$. What is the target equity exposure, and what happens to it if the portfolio falls to ₹16,00,000?

At ₹20,00,000: cushion = $20,00,000 - 15,00,000 = ₹5,00,000$; exposure = $3 \times 5,00,000 = ₹15,00,000$ (75% of the portfolio). At ₹16,00,000: cushion = ₹1,00,000; exposure = $3 \times 1,00,000 = ₹3,00,000$ (18.75%). As the portfolio fell toward the floor, exposure was cut sharply — CPPI de-risks into weakness, the mirror image of rebalancing.

Section C — Interview-Style Questions

C1. "Your client's adviser says asset allocation determines 90% of returns. Is that right?" Model answer: It is right only with careful wording. The Brinson evidence is that policy allocation explains $\sim 90\%$ of the *variability of a single portfolio's returns over time* — sensible, because if you are 60% equity your quarter-to-quarter swings track equities. Ibbotson–Kaplan showed it also explains $\sim 100\%$ of the *average level* of returns, since active management is roughly zero-sum before costs. But it explains only $\sim 40\%$ of the *differences between funds* — there, selection, timing, and fees decide. So allocation dominates the risk you take and the return you can expect, but it is not the whole story of why one manager beats another. I would never say "90% of returns" unqualified.

C2. "Walk me through how you would set a strategic asset allocation for a 40-year-old with a stable salary and a 25-year horizon." Model answer: Start with the IPS and RRTLLU. Return objective and risk tolerance come first — I separate *ability* (high: long horizon, stable income, large bond-like human capital) from *willingness* (I would probe with drawdown scenarios) and anchor to the lower. Then capital-market assumptions per asset class feed a mean-variance optimiser to trace the efficient frontier, and I pick the frontier point matching the client's risk-aversion λ via $U = E(R) - \frac{1}{2}\lambda\sigma^2$. Because raw optimisers are hyper-sensitive to input errors, I temper them with constraints/ranges, Black–Litterman blending, or a heuristic anchor like "110 – age" ($\sim 70\%$ equity here). I document target weights, bands, benchmarks, and a rebalancing rule. Given his profile I would land near 70–75% equity with a diversified fixed-income and modest alternatives sleeve, then de-risk along a glide path.

C3. "In 2022 both stocks and bonds fell. Doesn't that prove the 60/40 and diversification are dead?"

Model answer: No — it proves correlations are *regime-dependent*, which is a known caveat, not a refutation. Equity and high-quality government bonds are usually negatively or weakly correlated because bonds rally in flights to safety, which is why 60/40 became the default. But 2022 was an inflation/rate shock that drove both down together, so the equity-bond diversification broke for a year — and alternatives like commodities and gold earned their keep. The lesson is to stress-test correlation assumptions and hold genuinely differentiated diversifiers, not to abandon diversification. Over long horizons the imperfect correlation still delivers the free lunch; you just cannot assume it holds in every regime.

C4. "Why not just rebalance daily to keep risk perfectly on target?" Model answer: Because rebalancing has costs that eventually swamp its benefit. Its dependable payoff is risk control, and a modest return bonus when markets mean-revert — but each trade incurs transaction costs, and in taxable accounts realises capital-gains tax. Daily rebalancing multiplies both while the marginal risk-control gain over, say, a quarterly hybrid rule is tiny. The professional default is hybrid: check on a schedule but trade only when a class breaches its tolerance band. That captures nearly all the risk control while trading rarely. Tighter bands make sense only for high-volatility assets, low-cost or tax-sheltered accounts, and low-risk-tolerance clients.

C5. "Explain the difference between tactical asset allocation and rebalancing to a client who thinks they are the same." Model answer: Both move the actual mix, but in opposite spirit. Rebalancing is *maintenance*: the market pushes your weights away from target as prices drift, and rebalancing trims what grew and tops up what shrank to restore the intended risk — it pulls you *back toward* policy and requires no forecast. TAA is a *deliberate bet*: I intentionally tilt *away* from policy — say to 68% equity from a 60% target — because I judge equities temporarily cheap, then revert once the view plays out. Rebalancing corrects drift you did not choose; TAA is drift you chose. Both stay inside the strategic bands.

Section D — Multiple Choice (with reasoning)

D1. The Brinson (1986) result that policy explains ~93.6% refers to: (a) the level of a fund's return · (b) the variability of a single fund's return over time · (c) the differences in returns across funds · (d) after-fee alpha **Answer: (b).** It measured variance of one plan's returns through time. Level is ~100% (Ibbotson–Kaplan); cross-fund variation is ~40%.

D2. Rebalancing reliably provides: (a) higher return in all markets · (b) risk control, plus a return bonus only when markets mean-revert · (c) tax savings · (d) protection of a floor value **Answer: (b).** Risk control is dependable; the return bonus needs mean reversion and reverses in sustained trends. Floor protection is CPPI's job (d).

D3. An investor with high ability but low willingness to take risk should be given a mix that is: (a) aggressive, matching ability · (b) conservative, anchored to the lower (willingness) · (c) the average of the two · (d) 100% cash **Answer: (b).** Anchor to the lower of ability and willingness; an optimal portfolio the client abandons in a panic is worthless.

D4. A portfolio's expected return relative to the weighted average of asset-class returns is: (a) lower, due to diversification · (b) higher, due to diversification · (c) exactly the weighted average · (d) unpredictable **Answer: (c).** Expected return is linear in weights — diversification reduces *risk*, not expected return. That asymmetry is the free lunch.

D5. CPPI with multiplier m and floor F sets risky exposure equal to: (a) $m \times F$ · (b) $m \times (\text{Value} - F)$ · (c) $(\text{Value} - F)/m$ · (d) $m \times \text{Value}$ **Answer: (b).** Exposure = $m \times$

cushion, where cushion = Value – Floor. Exposure falls to zero as value approaches the floor.

D6. Under $U = E(R) - \frac{1}{2}\lambda\sigma^2$, raising the risk-aversion coefficient λ moves the chosen strategic mix toward: (a) more equity · (b) more bonds and cash · (c) no change · (d) more leverage

Answer: (b). Higher λ penalises variance more heavily, so the optimum sits at lower σ — more fixed income and cash.

D7. Which rebalancing method is generally the professional default? (a) daily calendar · (b) pure threshold with no schedule · (c) hybrid: scheduled check, trade only if outside the band · (d) never rebalance

Answer: (c). Hybrid caps monitoring cost while avoiding needless small trades — best risk control per unit of cost and tax.

D8. The "110 minus age" rule and target-date fund glide paths primarily encode: (a) tax optimisation · (b) declining equity weight as horizon shortens and risk-aversion rises with age · (c) tactical timing · (d) currency hedging
Answer: (b). Equity share falls automatically as the target date approaches — de-risking over the lifecycle.

Self-check log

B1: σ_p 11.37% < weighted-avg 14.1% ✓ (negative E–B corr drives it). B2: ₹10,080 gap reconciled via $50,400 \times 0.20$ ✓. B3: $\lambda=1.5 \rightarrow$ aggressive, $\lambda=4 \rightarrow$ balanced, break-even $\lambda=2.95$ consistent ✓. B4: post-trade 55/45 hits target ✓. B5: 75% $\rightarrow$ 45% over 30 yrs = 1 pp/yr ✓. B6: exposure 15,00,000 $\rightarrow$ 3,00,000 as value falls toward floor ✓.

Chapter 08 — The Portfolio Construction Process

1. The Problem / The Need

Modern portfolio theory (Chapters 5–7) gives you the *destination*: an efficient frontier, an optimal risky portfolio, a capital allocation line. But a frontier is a mathematical object. A client's money is real, arrives in a specific currency, must obey a tax regime, cannot short certain names, must stay above a liquidity floor, and has to be turned into actual buy and sell tickets sent to a broker before 3:30 p.m.

The **portfolio construction process** is the discipline that bridges that gap. It answers a deceptively simple chain of questions:

- *What* is this money for, and what constraints bind it? (the mandate)
- *Which* securities express our views? (selection)
- *How much* of each do we hold? (sizing and weighting)
- *How* do we get from today's holdings to the target without bleeding away returns in costs? (implementation)
- *When* and *how* do we course-correct as prices, views and cash flows change? (monitoring and rebalancing)

Get any link wrong and the theoretical alpha evaporates. A brilliant stock pick sized at 0.3% of the book barely moves the needle. A correct tilt implemented with 80 bps of market impact underperforms the index it was supposed to beat. A portfolio that drifts un-rebalanced for three years is no longer the portfolio the client signed up for. Construction is where "being right" is converted into "getting paid for being right" — or not.

The two dominant failure modes this process is designed to prevent:

1. **The un-investable ideal.** An optimiser output that violates the mandate — 40% in one illiquid small-cap, a short position the fund is not allowed to hold, a currency exposure the client explicitly excluded.
2. **The silent decay.** A perfectly good portfolio at inception that, through neglect of costs and drift, delivers a return far below its paper backtest. Studies of the "backtest-to-live" gap routinely find 1–3% per year lost to implementation and slippage.

Portfolio construction is the industrial process that turns a research view and a client mandate into a live, monitored, cost-aware book of positions.

2. The Core Idea

Think of portfolio construction as a **pipeline with a feedback loop**. Money and constraints flow in at the top; a live portfolio flows out at the bottom; and information from the live portfolio flows *back up* to trigger the next cycle. It is never "done" — it is a cycle that repeats for the life of the mandate.

The spine of the pipeline is the **Investment Policy Statement (IPS)**. The IPS is the contract that encodes *return objective, risk tolerance, and the five constraints* (Liquidity, Time horizon, Taxes, Legal/regulatory, Unique circumstances — the mnemonic **RR-LTTLU**, or more commonly "return, risk, and LTTLU"). Every

downstream decision must be traceable to the IPS. If a position cannot be justified against the IPS, it does not belong in the book.

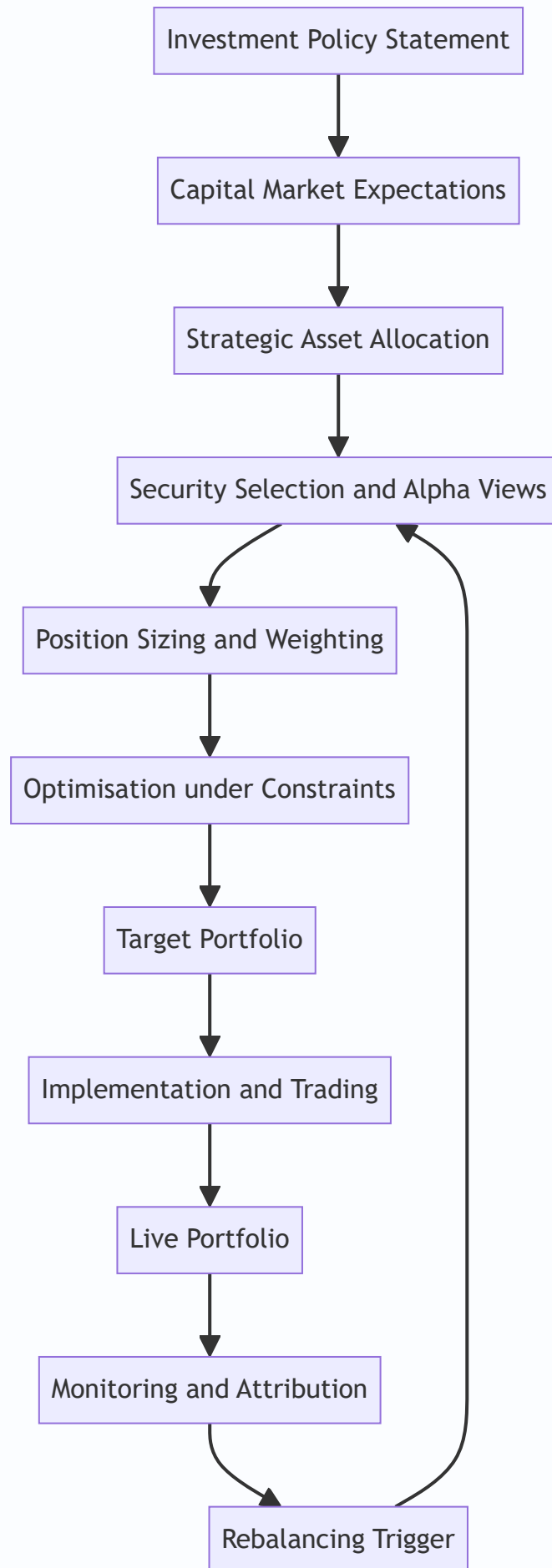


Figure 8.1 — The portfolio management cycle. The IPS sets the frame; the loop from monitoring back to selection is what makes it a process rather than a one-off construction.

The core insight for an interview: **allocation decides most of your risk; selection and construction decide most of your alpha; implementation decides how much of that alpha survives.** These three layers are separable, and a good process treats them as distinct decisions with distinct owners.

3. Why / How It Works

Why does structuring the problem this way actually improve outcomes? Four reasons.

(a) Separation of concerns controls error. Asset allocation (how much equity vs. bonds) and security selection (which equities) have very different error characteristics. Allocation errors are systematic and large in magnitude but few in number; selection errors are numerous but individually small and partially diversifiable. Handling them in separate stages lets you apply the right risk budget to each. The classic (and often misquoted) Brinson-Hood-Beebower finding is that *policy* allocation explains the vast majority of the **variability of returns over time** across balanced funds — not that it explains the level of return or cross-sectional differences. The correct reading motivates spending real governance effort on the strategic allocation, then treating selection as an active-risk overlay on top.

(b) Constraints are not annoyances — they are the definition of the feasible set. An optimiser without constraints will happily concentrate, lever, and short its way to a paper Sharpe ratio that cannot be traded. Constraints (position caps, sector bounds, turnover limits, no-short rules) are what make the output *investable*. The art is imposing enough constraints to keep the portfolio sane without so many that you strangle the alpha.

(c) Costs are deterministic drag; alpha is uncertain. A basis point of transaction cost is *certain*; the alpha you trade to capture is *probabilistic*. Rational construction therefore treats trading as an economic decision — you only trade when expected alpha exceeds the round-trip cost plus the opportunity cost of the tracking error you were carrying. This is why "no-trade bands" and cost-aware rebalancing exist.

(d) The feedback loop compounds learning. Attribution (Chapter on performance measurement) feeds back into selection: if your alpha is coming from a factor tilt you did not intend, you correct it next cycle. Without the loop, you cannot distinguish skill from luck, and you repeat mistakes.

The engine that formalises (a)–(c) is **mean-variance optimisation with a transaction-cost and constraint overlay**, which we build up next.

4. Full Content — Formulas and Models

4.1 From IPS to strategic asset allocation

The IPS produces a **required return** and a **risk budget**. A simple required-return build-up for an individual:

$$R_{\text{required}} = \frac{\text{Annual spending need} - \text{Other income}}{\text{Investable assets}} + \text{Inflation}$$

The strategic asset allocation (SAA) is the long-run policy mix — e.g., 60% equity / 40% bonds — chosen so the portfolio's expected return meets R_{required} at acceptable risk. SAA is set using **capital market expectations (CME)**: forward-looking estimates of asset-class returns, volatilities and correlations.

4.2 The mean-variance optimisation core

For N assets with weight vector w , expected-return vector μ , and covariance matrix Σ , portfolio expected return and variance are:

$$E(R_p) = w^T \mu \quad \sigma_p^2 = w^T \Sigma w$$

The **maximum-Sharpe (tangency) portfolio** solves, subject to $w^T \mathbf{1} = 1$:

$$\max_w \frac{w^T \mu - r_f}{\sqrt{w^T \Sigma w}}$$

The unconstrained tangency weights (when shorting is allowed) have the closed form:

$$w^* \propto \Sigma^{-1}(\mu - r_f \mathbf{1})$$

then rescaled to sum to 1. The **utility-based** formulation used in practice maximises mean-variance utility for a risk-aversion coefficient λ :

$$\max_w w^T \mu - \frac{\lambda}{2} w^T \Sigma w$$

Higher $\lambda \Rightarrow$ more risk-averse $\Rightarrow$ portfolio slides down the frontier toward the minimum-variance point.

4.3 Constraints — the investability layer

Real optimisations bolt constraints onto the objective above:

Constraint type	Form	Purpose
Budget (fully invested)	$\sum w_i = 1$	No idle cash / no leverage
Long-only	$w_i \geq 0$	No shorting (typical for long-only funds)
Position cap	$w_i \leq c$	Single-name concentration limit
Sector bounds	$L_s \leq \sum_{i \in s} w_i \leq U_s$	Diversification across sectors
Tracking-error cap	$\sqrt{(w - w_b)^T \Sigma (w - w_b)} \leq TE_{\max}$	Stay close to benchmark
Turnover cap	$\sum_i w_i - w_i^{\text{old}} \leq T$	Limit trading cost
Factor neutrality	$\beta_f^T (w - w_b) = 0$	Isolate stock-specific alpha

Constrained MVO has no closed form; it is solved numerically (quadratic programming).

4.4 Position sizing methods

Beyond full optimisation, practitioners use structured sizing rules:

- **Equal weight:** $w_i = 1/N$. Naïve but robust — no estimation error, strong diversification, implicit small-cap/value tilt and mean-reversion. The "1/N" benchmark that many optimisers fail to beat out-of-sample.
- **Market-cap weight:** $w_i = \text{cap}_i / \sum \text{cap}_j$. The index-neutral default; zero active bet.
- **Risk parity:** size so each asset contributes equal risk. Asset i 's marginal risk contribution is $w_i (\Sigma w)_i / \sigma_p$; set all equal.

- **Volatility targeting:** $w_i \propto 1/\sigma_i$ (inverse-vol), scaling exposure so riskier names get less capital.
- **Conviction / signal weighting:** tilt weights toward the strength of the alpha signal, often as active weights around benchmark: $w_i = w_{b,i} + k \cdot \text{score}_i$.
- **Kelly / fractional Kelly:** optimal growth-maximising bet size $f^* = \mu_{\text{excess}}/\sigma^2$ for a single bet; practitioners use a fraction* (1/2 or 1/4 Kelly) to cut volatility.

The **active weight** framing is central to how PMs actually think:

$$w_i^{\text{active}} = w_i^{\text{portfolio}} - w_i^{\text{benchmark}}$$

Your bets *are* your active weights. Overweights and underweights, not absolute holdings, generate active return and active (tracking) risk.

4.5 The Grinold–Kahn "Fundamental Law of Active Management"

The link between skill, breadth and construction:

$$IR \approx IC \times \sqrt{BR} \times TC$$

where IR = information ratio, IC = information coefficient (correlation of forecasts to outcomes, i.e. skill), BR = breadth (number of *independent* bets per year), and TC = transfer coefficient (how much of the ideal portfolio survives constraints, 0–1). The transfer coefficient is precisely the *construction/implementation* term: constraints and costs shrink TC , and a low TC throws away the alpha your research produced.

4.6 Transaction costs and implementation shortfall

Total trading cost decomposes into:

Component	Nature	Rough size (liquid equity)
Commissions & fees	Explicit, fixed-ish	1–5 bps
Bid-ask spread	Explicit/implicit	half-spread per side
Market impact	Implicit, size-driven	grows with order/ADV
Delay / timing cost	Opportunity	varies
Opportunity cost (unfilled)	Implicit	price move on missed shares

Implementation shortfall (IS) is the master metric — the gap between a paper portfolio (traded instantly at the decision price) and the real one:

$$IS = (\text{Paper return}) - (\text{Actual return}) = \text{Execution cost} + \text{Opportunity cost} + \text{Fees}$$

A common **market-impact** model is the square-root law:

$$\text{Impact} \approx \eta \cdot \sigma \sqrt{\frac{Q}{V}}$$

where σ is daily volatility, Q the order size, V the daily volume (ADV), and η a stock-specific constant of order 1. Impact rises with the *square root* of participation — trading twice the size costs roughly $\sqrt{2} \approx 1.41\times$ the impact, which is why large orders are sliced over time.

4.7 Rebalancing rules

Three canonical disciplines:

Method	Rule	Trade-off
Calendar	Rebalance every period (monthly/quarterly)	Simple, predictable; ignores actual drift
Threshold (tolerance band)	Rebalance when a weight breaches $\pm b\%$	Cost-efficient; needs monitoring
Calendar-and-threshold	Check on calendar, act only if band breached	Best of both; industry standard

The optimal no-trade band widens with transaction costs and narrows with the risk (tracking error) of drift — you tolerate more drift when trading is expensive and the drift is cheap in risk terms.

5. Worked Examples

Example 1 — Two-asset optimal weights, then reconcile the Sharpe ratio

Setup. Equities: $\mu_E = 12\%$, $\sigma_E = 20\%$. Bonds: $\mu_B = 5\%$, $\sigma_B = 8\%$. Correlation $\rho = 0.20$. Risk-free $r_f = 3\%$.

Step 1 — covariance. $\sigma_{EB} = \rho \sigma_E \sigma_B = 0.20 \times 0.20 \times 0.08 = 0.0032$.

Step 2 — tangency weight in equities (two-asset closed form using excess returns $e_E = 0.09$, $e_B = 0.02$):

$$w_E = \frac{e_E \sigma_B^2 - e_B \sigma_{EB}}{e_E \sigma_B^2 + e_B \sigma_E^2 - (e_E + e_B) \sigma_{EB}}$$

Numerator $= 0.09(0.0064) - 0.02(0.0032) = 0.000576 - 0.000064 = 0.000512$. Denominator $= 0.09(0.0064) + 0.02(0.04) - (0.11)(0.0032) = 0.000576 + 0.000800 - 0.000352 = 0.001024$.

$$w_E = \frac{0.000512}{0.001024} = 0.50, \quad w_B = 0.50$$

Step 3 — portfolio stats at (0.5, 0.5). $E(R_p) = 0.5(12\%) + 0.5(5\%) = 8.5\%$. $\sigma_p^2 = 0.5^2(0.04) + 0.5^2(0.0064) + 2(0.5)(0.5)(0.0032) = 0.010 + 0.0016 + 0.0016 = 0.0132 \rightarrow \sigma_p = 11.49\%$.

Step 4 — Sharpe. $S = (8.5\% - 3\%) / 11.49\% = 5.5 / 11.49 = \mathbf{0.479}$.

Reconciliation — verify it is truly the max. Try 60/40: $E(R) = 9.2\%$, $\sigma^2 = 0.36(0.04) + 0.16(0.0064) + 2(0.24)(0.0032) = 0.0144 + 0.001024 + 0.001536 = 0.01696$, $\sigma = 13.02\%$, $S = (9.2 - 3) / 13.02 = 0.476$. Try 40/60: $E(R) = 7.8\%$, $\sigma^2 = 0.16(0.04) + 0.36(0.0064) + 2(0.24)(0.0032) = 0.0064 + 0.002304 + 0.001536 = 0.01024$, $\sigma = 10.12\%$, $S = (7.8 - 3) / 10.12 = 0.474$. Both neighbours give a *lower* Sharpe (0.476 and $0.474 < 0.479$), confirming 50/50 is the tangency portfolio. ✓

Example 2 — Sizing to a tracking-error budget with active weights

Setup. A long-only equity PM runs a \$500m book against an index. She has three high-conviction overweights, funded by an equal underweight spread across the rest of the book. She wants active risk (tracking error) $\approx 3\%$.

Assume the three active bets are mutually independent, each with active volatility (of the active-weight $\times$ stock idiosyncratic vol) contributing as below. Active weights (portfolio – benchmark) and each name's

residual volatility:

Name	Active weight	Residual vol	Variance contribution $(w^a)^2 \sigma^2$
A	+2.0%	25%	$0.02^2 \times 0.25^2 = 0.000025$
B	+1.5%	30%	$0.015^2 \times 0.30^2 = 0.00002025$
C	+1.0%	35%	$0.01^2 \times 0.35^2 = 0.00001225$

Sum of variance contributions (independence $\Rightarrow$ add) $= 0.000025 + 0.00002025 + 0.00001225 = 0.0000575$.

Tracking error from these three bets $= \sqrt{0.0000575} = 0.758\%$.

Interpretation. Three names give only $\sim 0.76\%$ TE — far below the 3% budget. To reach 3% she must either *scale up* the active weights or *add breadth*. Scaling all three active weights by factor k scales TE linearly: to hit 3% she needs $k = 3/0.758 = 3.96$, i.e. active weights of roughly +7.9%, +5.9%, +4.0% — likely breaching a single-name position cap. **The lesson (Fundamental Law):** with only 3 bets you cannot reach the risk budget without dangerous concentration. Breadth, not bigger bets, is the safe route to a target IR. ✓

Reconcile with the Fundamental Law. If her $IC = 0.05$ and she wants $IR = 0.5$ with $TC = 0.8$: required breadth $BR = (IR/(IC \cdot TC))^2 = (0.5/(0.05 \times 0.8))^2 = (12.5)^2 \approx 156$ independent bets/year. Three concentrated positions cannot deliver that IR — the math agrees with the sizing intuition. ✓

Example 3 — Implementation shortfall and the trade/no-trade decision

Setup. A model says stock X is 1.2% cheap (expected alpha to be captured = 120 bps). Current active weight is 0; target is +1.5% of a \$500m book $\Rightarrow$ **\$7.5m** to buy. Stock X: price ₹800, ADV = 400,000 shares, daily vol $\sigma = 2\%$, impact constant $\eta = 1$. Half-spread = 4 bps; commission = 2 bps.

Step 1 — order size in shares. $\$7,500,000 / 800 = 9,375$ shares. Participation $Q/V = 9,375/400,000 = 2.34\%$ of ADV.

Step 2 — market impact (square-root law). $\text{Impact} = \eta \sigma \sqrt{Q/V} = 1 \times 2\% \times \sqrt{0.0234} = 2\% \times 0.1531 = 0.306\% \approx \mathbf{31 \text{ bps}}$.

Step 3 — total round-trip-ish entry cost. Impact 31 + half-spread 4 + commission 2 = **37 bps** to establish the position.

Step 4 — net expected value. Expected alpha 120 bps – entry cost 37 bps = **+83 bps** net. Trade is clearly worth doing. ✓

Now the reconciling twist — a smaller edge. Suppose the model edge were only 40 bps and the order were 4 $\times$ larger (participation 9.36% of ADV). New impact $= 2\% \times \sqrt{0.0936} = 2\% \times 0.306 = 0.612\% \approx \mathbf{61 \text{ bps}}$. Plus 6 bps explicit $\Rightarrow$ 67 bps cost against 40 bps alpha $\Rightarrow$ **–27 bps net: do not trade today**, or slice the order over several days to cut participation. Slicing to 2.34%/day again drops impact back to ~ 31 bps/tranche, turning the trade positive — which is exactly *why* execution algorithms exist. ✓

Notice the square-root law in action: quadrupling participation (2.34% $\rightarrow$ 9.36%) exactly *doubles* impact (31 $\rightarrow$ 61 bps), since $\sqrt{4} = 2$. Internal consistency confirmed. ✓

6. Connections

- **To Markowitz / MPT (Ch. 5–6):** MVO is the optimisation engine of construction. Construction adds the constraint and cost layers MPT ignores.
- **To CAPM and factor models (Ch. 7):** Factor exposures are what sector/beta neutrality constraints control; residual (idiosyncratic) risk is what stock selection bets on. Attribution decomposes returns along these same factors.
- **To performance measurement (later chapters):** Sharpe, Treynor, Jensen's alpha and the information ratio are the *scorecards* that feed the monitoring stage; Brinson attribution splits realised return into allocation vs. selection effects that loop back into the process.
- **To behavioural finance:** Rebalancing is mechanically *contrarian* — it sells winners and buys losers, counteracting the disposition effect and momentum-chasing that hurt undisciplined investors.
- **To fixed income & multi-asset:** The same pipeline governs bond ladders (duration/convexity constraints replace beta/sector) and multi-asset funds (SAA + tactical asset allocation overlay).
- **To risk management:** Position caps, TE budgets, VaR limits and liquidity floors are the guardrails inside the optimiser.

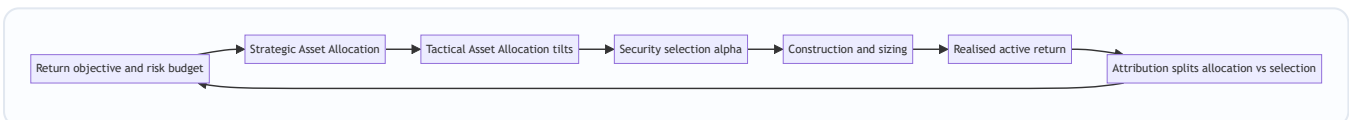


Figure 8.2 — Where value is added and measured. Attribution closes the loop by telling you which layer earned or lost the active return.

7. Key Terms

Term	Definition
Investment Policy Statement (IPS)	Governing document specifying return objective, risk tolerance, and constraints (Liquidity, Time horizon, Taxes, Legal/regulatory, Unique).
Strategic Asset Allocation (SAA)	Long-run policy mix of asset classes; the single biggest driver of return <i>variability</i> .
Tactical Asset Allocation (TAA)	Short-term deviations from SAA to exploit near-term views.
Capital Market Expectations (CME)	Forward-looking estimates of asset-class returns, vols and correlations feeding the SAA.
Active weight	Portfolio weight minus benchmark weight; the actual bet.
Tracking error (active risk)	Standard deviation of active return; the risk budget for active management.
Information ratio (IR)	Active return $\div$ tracking error; risk-adjusted skill metric.
Information coefficient (IC)	Correlation between forecasts and realised returns; a measure of skill.
Transfer coefficient (TC)	Fraction of the ideal (unconstrained) portfolio that survives constraints and costs.
Implementation shortfall	Gap between paper (decision-price) return and actual realised return; the master cost metric.
Market impact	Adverse price move caused by your own trading; scales $\sim\sqrt{(\text{order}/\text{ADV})}$.
ADV	Average daily volume — the liquidity denominator for sizing trades.
No-trade band	Tolerance range around target weights within which drift is left alone.
Risk parity	Sizing so each asset contributes equal risk to the portfolio.
Rebalancing	Trading back toward target weights after drift; mechanically contrarian.

8. Common Confusions

"Asset allocation determines 90% of returns." *Misquote.* Brinson-style studies show policy allocation explains $\sim 90\%$ of the *variability of returns over time* for a diversified fund — not 90% of the *level* of return, and not 90% of the *cross-sectional difference* between funds (selection and timing dominate there). Say it precisely in an interview.

Optimal weights $\neq$ good weights. An unconstrained optimiser is an "error maximiser" — it loads onto assets with the highest *estimated* return, which are often the ones with the largest *estimation error*. This is why constrained, shrinkage-based, or resampled optimisation (and even plain equal-weight) frequently beats naïve MVO out of sample.

Rebalancing back to target is not "doing nothing." It is an active, contrarian trade that harvests a small "rebalancing premium" from mean reversion but incurs real cost. The decision to rebalance is itself a cost-benefit calculation, not an automatic reflex.

Turnover is not the enemy — *uncompensated* turnover is. Trading that captures alpha exceeding its cost is value-adding. The goal is not zero turnover; it is high *return-per-unit-turnover*.

Tracking error is symmetric risk, not downside. A fund can have high TE from *outperforming*. TE measures deviation from benchmark in both directions; it is not the same as drawdown or downside risk.

Position size $\neq$ conviction alone. Size should reflect conviction *scaled by* the position's marginal risk contribution and liquidity. A high-conviction, high-vol, illiquid name may deserve a *smaller* weight than a medium-conviction liquid one.

Gross vs. net exposure. In long-short books, adding a long and a short can raise gross exposure (and financing cost, borrow cost) while leaving net market exposure unchanged — construction must budget both.

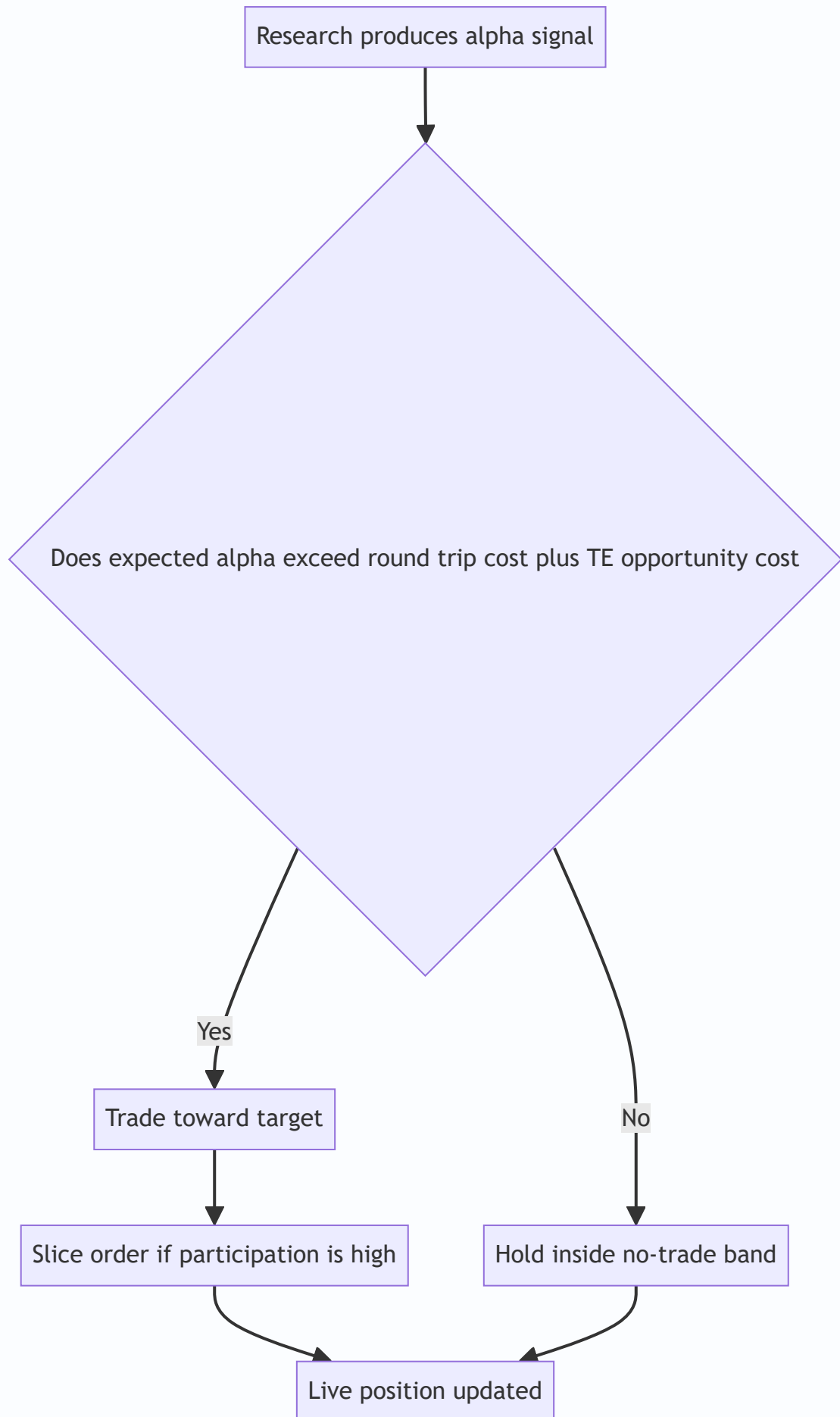


Figure 8.3 — The trade / no-trade gate. Every rebalancing decision passes through this cost-benefit test.

9. Recap

- Portfolio construction is a **cycle**, anchored on the **IPS**, that converts objectives and constraints into a live, monitored book.
- The pipeline runs **IPS** → **CME** → **strategic allocation** → **security selection** → **sizing/weighting** → **constrained optimisation** → **implementation** → **monitoring** → **rebalancing**, then loops.
- **Allocation drives risk; selection and sizing drive alpha; implementation decides how much alpha survives.** These are separable decisions.
- **Optimisation** (MVO / utility max) is the engine; **constraints** make the output investable; the **transfer coefficient** measures how much alpha the constraints leave intact.
- **Sizing** ranges from equal-weight and cap-weight to risk parity, inverse-vol, conviction and (fractional) Kelly — expressed through **active weights** against the benchmark.
- The **Fundamental Law** ($\$IR \approx IC \sqrt{BR} \backslash TC\$$) ties skill, breadth and implementation together — reaching a risk budget safely requires breadth, not oversized bets.
- **Transaction costs** (spread, impact $\sim \sqrt{\text{participation}}$, delay, opportunity) are captured by **implementation shortfall**; you trade only when expected alpha beats cost.
- **Rebalancing** is a cost-aware, contrarian discipline governed by calendar-and-threshold rules and no-trade bands.
- Worked examples confirmed: the 50/50 tangency portfolio maximised Sharpe (0.479 vs. 0.476/0.474 neighbours); three bets give only 0.76% TE so breadth is needed; and the square-root impact law makes slicing large orders the rational response.

10. Quick-Reference / Interview Points

Formulas to have cold:

Concept	Formula
Portfolio return / variance	$E(R_p) = w^{\text{top}} \mu$; $\sigma_p^2 = w^{\text{top}} \Sigma w$
Tangency (unconstrained)	$w^* \propto \Sigma^{-1} (\mu - r_f \mathbf{1})$
Utility objective	$\max w^{\text{top}} \mu - \frac{\lambda}{2} w^{\text{top}} \Sigma w$
Active weight	$w^a_i = w_i - w_{b,i}$
Sharpe / Information ratio	$S = \frac{R_p - r_f}{\sigma_p}$; $IR = \frac{R_p - R_b}{TE}$
Fundamental Law	$IR \approx IC \sqrt{BR} \backslash TC$
Market impact	$\approx \eta \backslash \sigma \sqrt{Q/V}$
Implementation shortfall	Paper return – actual return = exec + opportunity + fees
Required breadth	$BR = (IR / (IC \cdot TC))^2$

One-liners that land:

- "I think in *active weights and tracking error*, not absolute holdings — the benchmark is the origin of my coordinate system."
- "Constraints aren't the enemy of alpha; the *transfer coefficient* tells me how much alpha they cost, and I manage that number explicitly."
- "Allocation explains return *variability*, not return *level* — the 90% stat is about time-series variance of a diversified fund."
- "Market impact scales with the *square root* of participation, so I slice orders — quadrupling size only doubles impact, but doing it all at once still costs 2×."
- "I only trade when expected alpha exceeds round-trip cost plus the opportunity cost of the tracking error I'm currently carrying."
- "Rebalancing is mechanically contrarian — it's the discipline that sells winners and buys losers when my behavioural instincts say the opposite."
- "The Fundamental Law tells me to get IR from *breadth*, not from oversizing a handful of bets into concentration risk."

The 30-second narrative of the whole chapter: *Start with the IPS — objective, risk budget, constraints. Set capital market expectations, then a strategic allocation that meets the required return. Layer security-selection views as active weights, size them to a tracking-error budget, and run a constrained optimisation so the output is actually investable. Implement carefully — every trade must clear its cost hurdle, and big orders get sliced because impact scales with the square root of participation. Then monitor: run attribution, watch for drift past the no-trade bands, and rebalance in a cost-aware, contrarian way. Feed the attribution back into selection. Repeat for the life of the mandate.*

Q&A — The Portfolio Construction Process

Scope: Investments — Chapter 08 (The Portfolio Construction Process). Every question is followed by a full model answer. All rates are annual and in percent unless stated. Work every numerical problem yourself before checking the solution. Sections: **A** concept-check · **B** numerical (full step-by-step, reconciling) · **C** interview-style · **D** MCQs with reasoning.

The chapter in one line

$\text{IPS} \rightarrow \text{CME} \rightarrow \text{SAA} \rightarrow \text{selection} \rightarrow \text{sizing} \rightarrow \text{constrained MVO} \rightarrow \text{implementation} \rightarrow \text{monitoring} \rightarrow \text{rebalance} \leftarrow$

One-line statement: Portfolio construction is the disciplined, looping industrial process — anchored on the Investment Policy Statement — that converts a research view and a client mandate into a live, cost-aware, monitored book of positions, where allocation drives risk, selection and sizing drive alpha, and implementation decides how much alpha survives.

Section A — Concept Check

A1. What is the Investment Policy Statement (IPS) and why is it the spine of the whole process? The IPS is the governing contract encoding the **return objective**, **risk tolerance**, and the **five constraints** — Liquidity, Time horizon, Taxes, Legal/regulatory, Unique circumstances (LTTLU). It is the spine because every downstream decision must be *traceable* to it: a position that cannot be justified against the IPS does not belong in the book. It converts "grow my money" into a concrete required return and risk budget the optimiser can target.

A2. State the three-layer insight about where risk and alpha come from. Allocation decides most of your risk; selection and construction decide most of your alpha; implementation decides how much of that alpha survives. The three layers are separable, each with distinct error characteristics and distinct owners, which lets you apply the right risk budget to each.

A3. What does the Brinson-Hood-Beebower result actually say — and what is the common misquote? It says policy (strategic) asset allocation explains ~90% of the **variability of returns over time** for a diversified balanced fund. The misquote: that allocation explains 90% of the *level* of return or of the *cross-sectional difference* between funds — it does not; selection and timing dominate those. The correct reading motivates real governance effort on the strategic allocation.

A4. Why are constraints "the definition of the feasible set" rather than annoyances? An unconstrained optimiser will concentrate, lever, and short its way to a paper Sharpe ratio that cannot be traded. Constraints (position caps, sector bounds, no-short rules, turnover limits) are what make the output *investable*. The art is imposing enough to keep the portfolio sane without so many that you strangle the alpha.

A5. Define active weight and explain why it is the PM's natural unit of thought. Active weight = portfolio weight – benchmark weight ($w_i^a = w_i - w_{(b,i)}$). It is the natural unit because a PM's *bets are*

her *active weights* — overweights and underweights, not absolute holdings, generate active return and active (tracking) risk. The benchmark is the origin of the coordinate system.

A6. Why is an unconstrained mean-variance optimiser called an "error maximiser"? Because it loads most heavily onto assets with the highest *estimated* returns, which are frequently the ones carrying the largest *estimation error* — it mistakes noisy high estimates for genuine opportunity. This is why constrained, shrinkage-based, or resampled optimisation — and even plain equal-weight (1/N) — frequently beats naïve MVO out of sample.

A7. State the Fundamental Law of Active Management and name each term. $IR \approx IC \times \sqrt{BR} \times TC$ IR = information ratio (active return ÷ tracking error); IC = information coefficient (correlation of forecasts to outcomes, i.e. skill); BR = breadth (number of *independent* bets per year); TC = transfer coefficient (fraction of the ideal portfolio that survives constraints and costs, 0–1). The TC term is precisely the construction/implementation term.

A8. What is the transfer coefficient and why should a PM manage it explicitly? TC is the correlation between the *ideal* (unconstrained) active positions and the *actual* implemented positions — how much of the paper alpha survives the constraint and cost layer. It should be managed explicitly because a low TC silently throws away the alpha the research produced; constraints that look prudent can quietly halve your realised IR.

A9. Distinguish strategic asset allocation (SAA) from tactical asset allocation (TAA). SAA is the **long-run policy mix** (e.g., 60/40) set from capital market expectations to meet the required return at acceptable risk — the single biggest driver of return *variability*. TAA is **short-term, deliberate deviation** from SAA to exploit near-term views; it is an active overlay measured and risk-budgeted against the SAA.

A10. Why is trading treated as an economic decision rather than a reflex? Because a basis point of transaction cost is *certain* while the alpha you trade to capture is *probabilistic*. Rational construction only trades when expected alpha exceeds round-trip cost plus the opportunity cost of the tracking error currently carried — which is why no-trade bands and cost-aware rebalancing exist.

A11. What is implementation shortfall and why is it the "master" cost metric? Implementation shortfall (IS) is the gap between a *paper* portfolio (traded instantly at the decision price) and the *real* one: $IS = \text{Execution cost} + \text{Opportunity cost} + \text{Fees}$. It is the master metric because it captures *all* slippage — explicit and implicit, filled and unfilled — in one number, directly measuring the backtest-to-live gap.

A12. State the square-root market-impact law and its key implication. $\text{Impact} \approx \eta \sigma \sqrt{Q/V}$, where σ is daily volatility, Q order size, V average daily volume, η a stock-specific constant of order 1. The implication: impact scales with the *square root* of participation, so trading twice the size costs only $\sim \sqrt{2} \approx 1.41\times$ the impact — which is why large orders are sliced over time to cut participation per tranche.

A13. Why is rebalancing described as "mechanically contrarian"? Because it sells assets that have risen above target weight (winners) and buys those that have fallen below (losers) to restore the policy mix. It harvests a small rebalancing premium from mean reversion and counteracts the behavioural disposition effect — it is the discipline that acts against the momentum-chasing instinct.

A14. Give the three canonical rebalancing disciplines and the industry standard. (1) **Calendar** — rebalance every fixed period; simple but ignores actual drift. (2) **Threshold (tolerance band)** — rebalance

only when a weight breaches $\pm b\%$; cost-efficient but needs constant monitoring. (3) **Calendar-and-threshold** — check on a calendar but act only if a band is breached; this hybrid is the **industry standard**.

A15. Why is tracking error not the same as downside risk? Tracking error is the standard deviation of *active* return — deviation from the benchmark in *both* directions. A fund can post high TE purely by *outperforming*. It is a symmetric measure of active risk, not a measure of drawdown or downside loss.

Section B — Numerical Problems

B1. Required return build-up. A retiree has ₹2,00,00,000 (₹2 crore) of investable assets, needs ₹9,00,000 per year of spending, receives ₹1,00,000 of other income, and faces 4% expected inflation. Find the required nominal return.

Solution. Net spending drawn from the portfolio = $9,00,000 - 1,00,000 = ₹8,00,000$. Real return needed = $8,00,000 / 2,00,00,000 = 0.04 = 4.0\%$. Add inflation: $R_{\text{required}} = 4.0\% + 4.0\% = \mathbf{8.0\%}$.

Reconcile. At 8% the ₹2 cr earns ₹16,00,000; of that, ₹8,00,000 (the 4% real part) funds spending and ₹8,00,000 (the 4% inflation part) preserves real capital. Spending need is exactly met and purchasing power is maintained. ✓

B2. Constrained sizing to a position cap. An optimiser wants active weights of +7.9%, +5.9%, +4.0% on three names, but the mandate caps any single active weight at $\pm 3.0\%$. What is the constrained active-weight vector, and what does this imply for the transfer coefficient?

Solution. Cap each at +3.0% $\Rightarrow$ constrained vector = **(+3.0%, +3.0%, +3.0%)**. The ideal vector totalled 17.8% of active exposure; the constrained one totals 9.0% — roughly half — and the cap also *flattens the ranking* (the optimiser wanted $A > B > C$; now they're equal). Both effects lower the correlation between ideal and actual positions, so TC falls well below 1. *Reconcile.* Since $IR \approx IC \sqrt{BR} \cdot TC$, halving the effective transfer roughly halves realised IR for the same skill and breadth — the position cap has a direct, quantifiable cost in IR. ✓

B3. Tracking error from independent active bets. A PM holds three independent active bets: A (active weight +2.5%, residual vol 24%), B (+2.0%, 30%), C (+1.5%, 28%). Find the total tracking error.

Solution. Variance contribution of each = $(w^a)^2 \sigma^2$: A: $0.025^2 \times 0.24^2 = 0.000625 \times 0.0576 = 0.000036$. B: $0.020^2 \times 0.30^2 = 0.0004 \times 0.09 = 0.000036$. C: $0.015^2 \times 0.28^2 = 0.000225 \times 0.0784 = 0.00001764$. Independence $\Rightarrow$ variances add: $0.000036 + 0.000036 + 0.00001764 = 0.00008964$. $TE = \sqrt{0.00008964} = 0.00947 = \mathbf{0.95\%}$. *Reconcile.* Three modest bets deliver under 1% TE — far short of a typical 3% budget. To reach 3% the PM must scale bets by $3/0.95 \approx 3.2\times$ (risking position caps) or add breadth. The math confirms the Fundamental Law's lesson: breadth, not oversized bets, is the safe route to a risk budget. ✓

B4. Required breadth from the Fundamental Law. A PM has skill $IC = 0.06$, faces a transfer coefficient $TC = 0.75$, and targets an information ratio $IR = 0.6$. How many independent bets per year does she need?

Solution. Rearranging $IR = IC \sqrt{BR} \cdot TC$: $BR = \left(\frac{IR}{IC \cdot TC} \right)^2 = \left(\frac{0.6}{0.06 \times 0.75} \right)^2 = \left(\frac{0.6}{0.045} \right)^2 = (13.33)^2 \approx \mathbf{178 \text{ bets/year}}$. *Reconcile.* Forward-check: $IC \sqrt{BR} \cdot TC = 0.06 \times \sqrt{178} \times 0.75 = 0.06 \times 13.34 \times 0.75 = 0.600$. ✓ Matches the target exactly. Note how sensitive this is: if constraints cut TC to 0.5, required breadth jumps to $(0.6/0.03)^2 = 400$ bets — the implementation term dominates the workload.

B5. Market impact and the trade / no-trade gate. A model says stock Y is 0.90% cheap (90 bps expected alpha). Target trade = ₹6,00,00,000 (₹6 cr). Y: price ₹1,000, ADV = 500,000 shares, daily vol $\sigma = 2.5\%$, impact constant $\eta = 1$. Half-spread = 5 bps, commission = 3 bps. Should she trade today?

Solution. Step 1 — shares: $6,00,00,000 / 1,000 = 60,000$ shares. Participation $\$Q/V\$ = 60,000/500,000 = 0.12 = 12\%$ of ADV. Step 2 — impact: $\eta \sigma \sqrt{Q/V} = 1 \times 2.5\% \times \sqrt{0.12} = 2.5\% \times 0.3464 = 0.866\% \approx 87$ bps. Step 3 — total entry cost: $87 + 5 + 3 = 95$ bps. Step 4 — net: 90 bps alpha – 95 bps cost = **–5 bps**. *Do not trade in one clip today. Reconcile via slicing.* Slice over, say, 4 days at 3% ADV/day: impact per tranche = $2.5\% \times \sqrt{0.03} = 2.5\% \times 0.1732 = 0.433\% \approx 43$ bps. Blended entry cost $\approx 43 + 5 + 3 = 51$ bps $\Rightarrow$ net = $90 - 51 = +39$ bps. Slicing turns a losing trade into a winner — exactly why execution algorithms exist. ✓ *Consistency check on the law:* cutting participation from 12% to 3% is a $4\times$ reduction; impact fell from 87 to 43 bps, a factor of $\sqrt{4} = 2$. ✓

B6. Rebalancing tolerance band. A 60/40 policy portfolio uses $\pm 5\%$ relative tolerance bands on the equity weight. Equities rally and the mix drifts to 68/32. Has the band been breached, and what is the rebalancing trade on a ₹1 cr book?

Solution. A $\pm 5\%$ relative band around a 60% target means the allowable range is $60\% \times (1 \pm 0.05) = [57\%, 63\%]$. Current equity weight $68\% > 63\% \Rightarrow$ **band breached**, rebalance triggered. Trade: target equity = $60\% \times ₹1 \text{ cr} = ₹60,00,000$; current = ₹68,00,000. **Sell ₹8,00,000 of equities, buy ₹8,00,000 of bonds.** *Reconcile.* The action is contrarian — it sells the asset that just rose — and restores the 60/40 policy mix. A $\pm 10\%$ relative band ($[54\%, 66\%]$) would *still* be breached by 68%; only $\pm 13.3\%$ or wider would leave the drift alone. ✓

Section C — Interview-Style (with model answers)

C1. "Walk me through your portfolio construction process in 30 seconds." *Model answer.* "Start with the IPS — objective, risk budget, and the five constraints. Set capital market expectations, then a strategic allocation that meets the required return. Layer security-selection views as *active weights*, size them to a tracking-error budget, and run a *constrained* optimisation so the output is actually investable. Implement carefully — every trade must clear its cost hurdle, and big orders get sliced because impact scales with the square root of participation. Then monitor: run attribution, watch for drift past the no-trade bands, and rebalance in a cost-aware, contrarian way. Feed attribution back into selection, and repeat for the life of the mandate."

C2. "A junior analyst hands you an optimiser output with 40% in one small-cap. What's wrong and how do you fix it?" *Model answer.* "It's un-investable — that breaches any sane position cap and the liquidity floor; you couldn't build or exit it without huge impact. It's also the classic *error-maximiser* symptom: 40% in one name usually means that name had the highest *estimated* return, i.e. the largest estimation error. Fix: add constraints (position caps, sector bounds, turnover limit), apply shrinkage to the return inputs or resample the frontier, and sanity-check against equal-weight. I'd compare the transfer coefficient before and after to see what the constraints actually cost in alpha."

C3. "How do you decide whether to act on a signal that says a stock is 0.5% cheap?" *Model answer.* "Through the trade/no-trade gate: expected alpha versus *round-trip cost plus the opportunity cost of the tracking error I'm currently carrying*. I estimate impact with the square-root law given my order as a fraction of ADV, add spread and commissions, and compare. Fifty bps is a thin edge — if participation pushes impact

near that, I either don't trade or slice over several days. Cost is certain, alpha is probabilistic, so a marginal edge doesn't automatically justify a trade."

C4. "Your fund has 1% tracking error but the client wants 4%. What are your options and their risks?"

Model answer. "By the Fundamental Law, $\$IR \approx IC\sqrt{BR}\sqrt{TC}$, and TE is the scale of my active bets. Three levers: *scale up existing bets* — fast, but drives concentration, may breach caps, and adds risk without improving IR; *add breadth* — more independent bets — which raises TE *and* preserves IR, the safe route but needs genuinely uncorrelated skilled forecasts; and *raise the transfer coefficient* by relaxing over-tight constraints. I'd favour breadth plus a TC review, and be honest that manufacturing TE by concentrating is buying risk without buying skill."

C5. "Is rebalancing worth the cost? Convince me either way." *Model answer.* "It's a cost-benefit decision, not a reflex. Benefit: it keeps risk aligned with the IPS — an un-rebalanced 60/40 drifts to 80/20 after a bull run, no longer the client's risk — and harvests a mean-reversion premium while counteracting behavioural biases. Cost: every rebalance incurs impact, spread, and possibly taxes. The reconciliation is no-trade bands: tolerate more drift when trading is expensive and the drift is cheap in risk terms, using a calendar-and-threshold rule. Worth it — but only inside a disciplined band, never on a hair-trigger."

C6. "Why not just maximise Sharpe with an unconstrained optimiser?" *Model answer.* "Because that portfolio is usually un-tradable and fragile. It shorts freely, concentrates into high-estimation-error names, and its weights swing wildly with tiny input changes — it maximises estimation error as much as return. Real construction adds constraints for investability and shrinkage or resampling for robustness. I'm not after the theoretically optimal paper portfolio; I want the best one *that survives constraints and costs and holds up out of sample* — and I track the transfer coefficient to know what that trade-off costs."

Section D — MCQs (with reasoning)

D1. The Brinson-Hood-Beebower studies are best summarised as: policy asset allocation explains ~90% of...
A) the level of a fund's return B) the cross-sectional difference between funds C) the variability of a diversified fund's returns over time D) the alpha generated by security selection

Answer: C. The result is about time-series *variability* of a diversified fund's returns, not the level (A) or cross-fund differences (B), where selection and timing dominate. The most-misquoted stat in the chapter.

D2. Market impact scales with the square root of participation. Quadrupling the order size (as a fraction of ADV) multiplies impact by approximately: A) 4× B) 2× C) 1.41× D) 16×

Answer: B. $\sqrt{4} = 2$. The square-root law means impact grows *sublinearly* — quadruple the size, only double the impact. (Doubling size would give $\sqrt{2} \approx 1.41\times$, option C, which is the distractor for a 2× order.)

D3. In the Fundamental Law $\$IR \approx IC\sqrt{BR}\sqrt{TC}$, the term most directly controlled by portfolio construction and implementation is: A) IC B) BR C) TC D) IR

Answer: C. The transfer coefficient measures how much of the ideal portfolio survives constraints and trading costs — that is precisely the construction/implementation layer. IC is skill, BR is the research opportunity set, IR is the output.

D4. An investor's active weight in a stock is +2%. This means: A) the stock is 2% of the portfolio B) the portfolio holds 2% more of the stock than the benchmark does C) the stock returned 2% more than the benchmark D) the stock has a beta 2% above the market

Answer: B. Active weight = portfolio weight – benchmark weight. It is a positioning (overweight) statement, not an absolute holding (A), a realised return (C), or a beta (D).

D5. The optimal no-trade band around a target weight should be *wider* when: A) transaction costs are low and drift risk is high B) transaction costs are high and drift risk is low C) both costs and drift risk are low D) tracking error must be minimised at all costs

Answer: B. You tolerate more drift when trading is expensive (high cost of acting) and the drift adds little risk (low cost of *not* acting). Cheap trading or risky drift (A) argues for a *narrow* band; option D would force a near-zero band regardless of cost.

D6. Which statement about the unconstrained mean-variance optimiser is correct? A) It is robust because it uses all available information B) It tends to concentrate in assets with the largest estimation error C) It always beats equal-weight out of sample D) It requires no expected-return inputs

Answer: B. The optimiser overweights the highest *estimated* returns, which are disproportionately the highest *estimation-error* assets — hence "error maximiser." It is fragile not robust (A), often *loses* to equal-weight out of sample (C), and depends heavily on return inputs (D).

D7. A PM wants to raise tracking error from 1% to 3% while preserving her information ratio. The soundest approach is to: A) triple the size of her three largest bets B) add more independent, equally-skilled bets (increase breadth) C) remove all position caps D) increase portfolio turnover

Answer: B. Adding independent bets raises TE while holding IC and TC roughly constant, preserving IR (the Fundamental Law). Tripling three bets (A) buys risk through concentration without improving skill; removing caps (C) risks un-investability; turnover (D) is a cost, not a source of active risk.

Self-check note: All numericals reconciled — B1 spending funded with real capital preserved; B3 three bets give 0.95% TE (under budget); B4 forward-check reproduces $IR = 0.60$; B5 slicing cuts impact by exactly $\sqrt{4} = 2\times$; B6 rebalance restores 60/40.

Chapter 09 — Equity Portfolio Management

1. The Problem / The Need

You have a mandate: a client hands you ₹1,000 crore and says "grow my equity money." That single sentence hides a dozen decisions. Do you try to *beat* the market or simply *own* it cheaply? If you try to beat it, beat it by owning what — cheap stocks, fast-growing stocks, high-quality compounders? Do you start from the economy and drill down to stocks, or start from individual companies and build up? How many stocks do you hold — 25 or 250? How far do you dare drift from the benchmark, and how do you measure that drift so the client (and your risk manager) can see it?

Equity portfolio management is the discipline of turning a vague growth mandate into a *specific, monitorable, repeatable* set of holdings. The problem is not "which stock will go up." The problem is designing a **process** that:

1. Has a defensible reason to exist (why should anyone pay you a fee?).
2. Takes *deliberate*, sized bets rather than accidental ones.
3. Can be measured against a fair yardstick.
4. Survives the manager having a bad year without being abandoned.

A portfolio that outperforms by luck is indistinguishable, ex-ante, from one that will blow up. The whole apparatus of this chapter — benchmarks, tracking error, active share, factor tilts, style discipline — exists to separate *skill* from *noise* and to make risk **intentional**. An equity book is a collection of active bets. If you cannot name your bets and size them, you are not managing a portfolio; you are collecting stocks.

2. The Core Idea

Every equity portfolio is the benchmark *plus a set of deliberate deviations*. Write it as an identity:

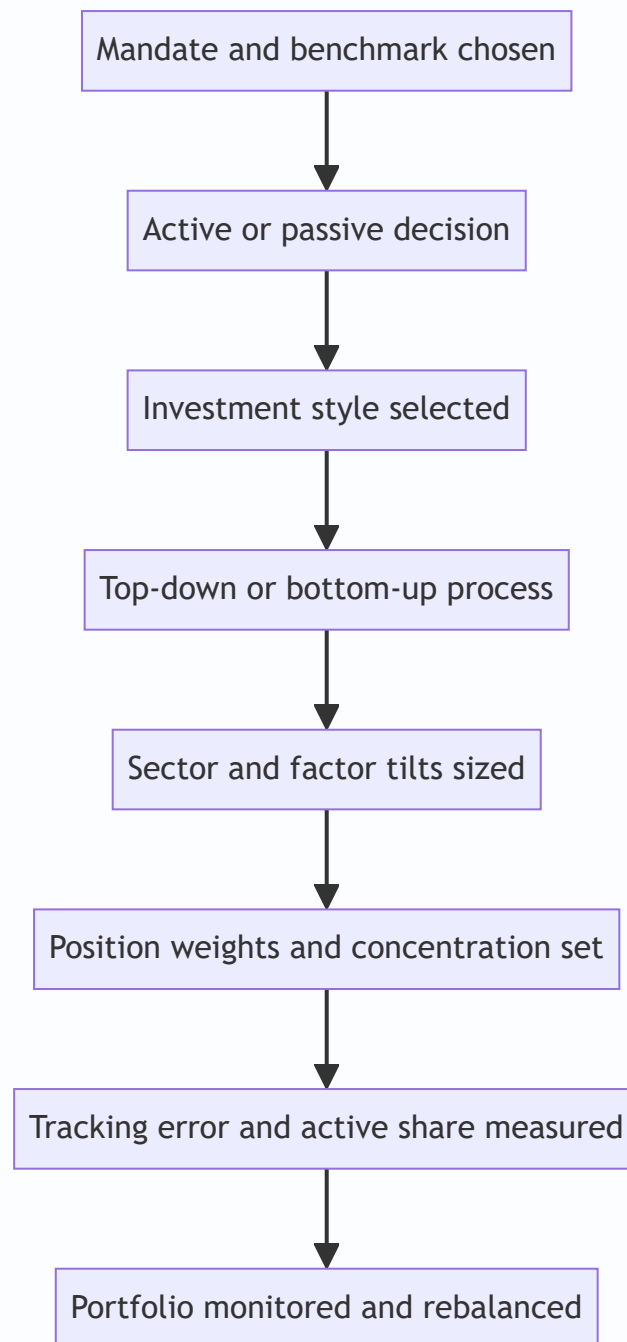
$$\text{Portfolio} = \text{Benchmark} + \text{Active Positions}$$

The active positions are the *only* thing that can make you beat — or lag — the benchmark. Everything you get paid for lives in those deviations. So the craft breaks into four linked choices:

- **Active vs passive** — do you take active positions at all, and if so how aggressively?
- **Style** — what *kind* of stocks do your bets systematically favour (value, growth, quality, GARP)?
- **Process** — how do you generate the bets (top-down macro, bottom-up stock picking, or both)?
- **Structure** — how concentrated are the bets, and how far do they push you from the benchmark (tracking error, active share)?

The unifying mental model: **a portfolio's excess return comes entirely from where its weights differ from the index**. If your weight on every stock equals the index weight, your active return is mathematically zero before fees, and negative after. Value is created only by the *active weight vector*, and risk is created by the same vector. Return and risk share one source — the bets you chose to take.

Figure 1 — The active-management decision stack, from mandate down to individual position.



3. Why / How It Works

Why active management is hard (the arithmetic of the loser's game)

William Sharpe's "Arithmetic of Active Management" is the starting point every analyst must internalise. Before costs, the average actively managed rupee earns exactly the market return, because in aggregate active managers *are* the market — one manager's overweight is another's underweight, and they net to the index. After fees and trading costs, the average active rupee must therefore **underperform** the passive rupee. This is not a claim about efficiency; it is accounting.

The implication is brutal and clarifying: active management is a *zero-sum game before costs and negative-sum after*. To win it, you must be better than the average participant, and the average participant is a

professional. So your edge must come from a genuine, structural source: an informational edge, an analytical edge, or a *behavioural/structural* edge (patience, willingness to hold through drawdowns, capacity to be contrarian when others are forced sellers).

Why passive works

If the average active rupee can't beat the index net of costs, then the index — captured at near-zero fee — beats most active managers over long horizons. Passive investing harvests the equity risk premium cheaply and reliably. It doesn't try to win the loser's game; it refuses to play. The market-cap-weighted index is also *self-rebalancing* (winners' weights rise automatically) and has minimal turnover, so its costs are tiny.

How active managers try to win anyway

Active return is decomposed by the **Fundamental Law of Active Management** (Grinold):

$$IR \approx IC \times \sqrt{BR}$$

where IR is the information ratio (skill-adjusted excess return), IC is the information coefficient (correlation between your forecasts and outcomes — your raw skill), and BR is breadth (the number of *independent* bets per year). The insight: you can build a strong track record either by being very skilled on a few bets or by being modestly skilled across many *independent* bets. A macro manager with 4 big calls a year needs enormous IC ; a quant harvesting factor premia across 500 stocks needs only tiny IC per name because breadth is huge. This single equation explains why concentrated stock-pickers and diversified quants are *both* viable — they sit at different points on the same skill-breadth curve.

4. Full Content — Formulas, Models, and Frameworks

4.1 Active vs Passive — the spectrum

It is not binary. There is a continuum:

Approach	Active share	Tracking error	Fee (typical)	What you're paying for
Pure index fund	~0%	<0.2%	0.05–0.20%	Cheap market beta
Enhanced index	5–20%	0.5–2%	0.2–0.5%	Small, controlled tilts
Factor / smart beta	30–60%	2–5%	0.3–0.7%	Rules-based factor premia
Active core	40–70%	3–6%	0.7–1.2%	Genuine stock selection
High-conviction / focused	70–95%	6–12%	1.0–2.0%	Concentrated manager skill

The "closet indexer" is the value-destroyer here: a fund charging 1% for 20% active share is selling you 80% index at 20x the index price. Active share (Section 4.6) is precisely the tool that exposes this.

4.2 Investment styles

Value — buy stocks trading cheap relative to fundamentals (low P/E, low P/B, high dividend yield, low EV/EBITDA). Thesis: the market over-extrapolates bad news; cheapness mean-reverts. Risk: the "value trap" — cheap because the business is dying, not mispriced. Classic metrics: P/B < peers, P/E in bottom quintile, FCF yield high.

Growth — buy companies with rapid, sustainable earnings/revenue growth, accepting high multiples because the *future* earnings justify today's price. Thesis: the market under-appreciates the duration and magnitude of growth. Risk: multiple compression when growth disappoints — high-multiple stocks fall hardest.

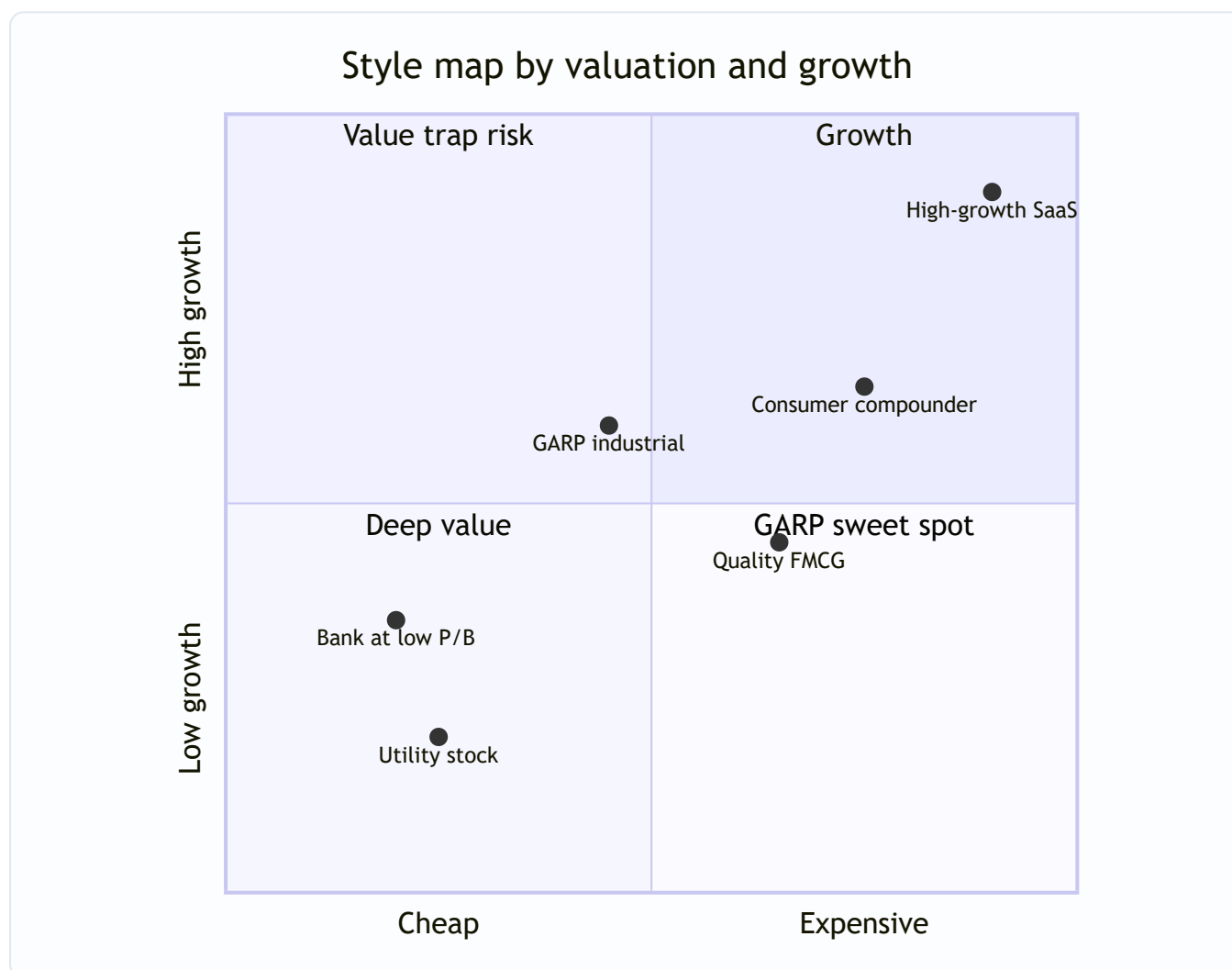
GARP (Growth at a Reasonable Price) — a hybrid. Buy growth but refuse to overpay. The signature tool is the **PEG ratio**:

$$PEG = \frac{P/E}{\text{expected earnings growth rate (\%)}}$$

A PEG near 1.0 is "fair," below 1.0 is attractive. Peter Lynch popularised this. GARP screens for, say, 15–25% earnings growth at a PEG under 1.2.

Quality — buy durable, profitable, low-leverage businesses regardless of the value/growth axis. Metrics: high and stable ROE/ROCE, high gross margins, low debt/equity, high interest coverage, consistent FCF conversion, low earnings volatility. Thesis: quality compounds and protects capital in downturns; the market chronically under-prices durability. Quality is the style that "wins in the down years."

Figure 2 — Styles placed on the two axes that matter: how cheap and how fast-growing.



4.3 Top-down vs bottom-up

Top-down starts with the macro picture — GDP, rates, inflation, currency — then picks countries, then sectors, then stocks within favoured sectors. The dominant bets are *sector and factor allocations*. Suits macro-

driven markets and asset allocators.

Bottom-up starts with individual companies — you find great businesses at good prices one at a time, and the sector weights are a *residual* of where you found value. Warren Buffett is the archetype: "we have no idea what the economy will do." Suits stock-selection-driven alpha.

Most real processes are hybrids: bottom-up stock picking within top-down risk guardrails (e.g., "no sector more than 10% over benchmark").

Dimension	Top-down	Bottom-up
Starting point	Macro / economy	Individual company
Primary bet	Sector & country allocation	Stock selection
Sector weights	Chosen deliberately	Residual outcome
Key skill	Macro forecasting	Business & valuation analysis
Vulnerability	Macro calls are low-breadth	Can accidentally over-concentrate in a sector

4.4 Sector and factor tilts

A **sector tilt** is an active weight on a GICS sector: if IT is 15% of the index and you hold 20%, you have a +5% active overweight to IT. Sector active return contribution:

$$\text{Sector contribution} = (w_{\{p,i\}} - w_{\{b,i\}}) \times (r_i - r_b)$$

A **factor tilt** is a systematic bias toward a *characteristic* that has historically earned a premium. The canonical equity factors:

Factor	Definition / proxy	Premium rationale
Value	Low P/B, low P/E	Risk premium + behavioural over-reaction
Size	Small-cap over large-cap	Illiquidity & distress premium
Momentum	Past 12-1 month winners	Under-reaction, herding
Quality / profitability	High ROE, low accruals	Mispriced durability
Low volatility	Low-beta stocks	Leverage constraints, lottery-seeking

These map to multi-factor models. The Fama-French-Carhart four-factor model:

$$R_p - R_f = \alpha + \beta_{\{mkt\}}(R_m - R_f) + \beta_{\{SMB\}}SMB + \beta_{\{HML\}}HML + \beta_{\{WML\}}WML + \epsilon$$

Here \$SMB\$ (small minus big) is the size factor, \$HML\$ (high minus low book-to-market) is value, \$WML\$ (winners minus losers) is momentum. Regressing your fund returns on these tells you *how much of your return is factor tilt versus true stock-selection alpha* (\$\alpha\$). This is the tool that catches a "star manager" who is really just a leveraged small-cap value bet.

4.5 The benchmark

The benchmark is the yardstick that defines "the market" for your mandate — Nifty 50, Nifty 500, S&P 500, MSCI India. A valid benchmark (per the CFA "SAMURAI" criteria) must be **S**pecified in advance, **A**ppropriate,

Measurable, **U**nambiguous, **R**eflective of the manager's opinions, **A**ccountable (owned by the manager), and **I**nvestable. Choosing the benchmark is itself a decision — benchmark a small-cap fund against Nifty 50 and every metric below is meaningless.

4.6 Tracking error and active share — the two rulers

These measure *different* things and both matter.

Tracking error (TE) — the volatility of active return. Let $R_{A,t} = R_{p,t} - R_{b,t}$ be period- t active return. Then:

$$TE = \sigma(R_A) = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (R_{A,t} - \bar{R}_A)^2}$$

TE is *return-based* and *backward-looking*. It tells you how bumpy the ride relative to the index has been. Annualise monthly TE by $\times \sqrt{12}$.

Information Ratio (IR) — active return per unit of tracking error, the headline skill metric:

$$IR = \frac{\bar{R}_A}{TE} = \frac{R_p - R_b}{TE}$$

An IR above 0.5 is good, above 1.0 is excellent and rare.

Active share (AS) — the fraction of the portfolio that *differs* from the benchmark by holdings. It is *holdings-based* and *forward-looking*:

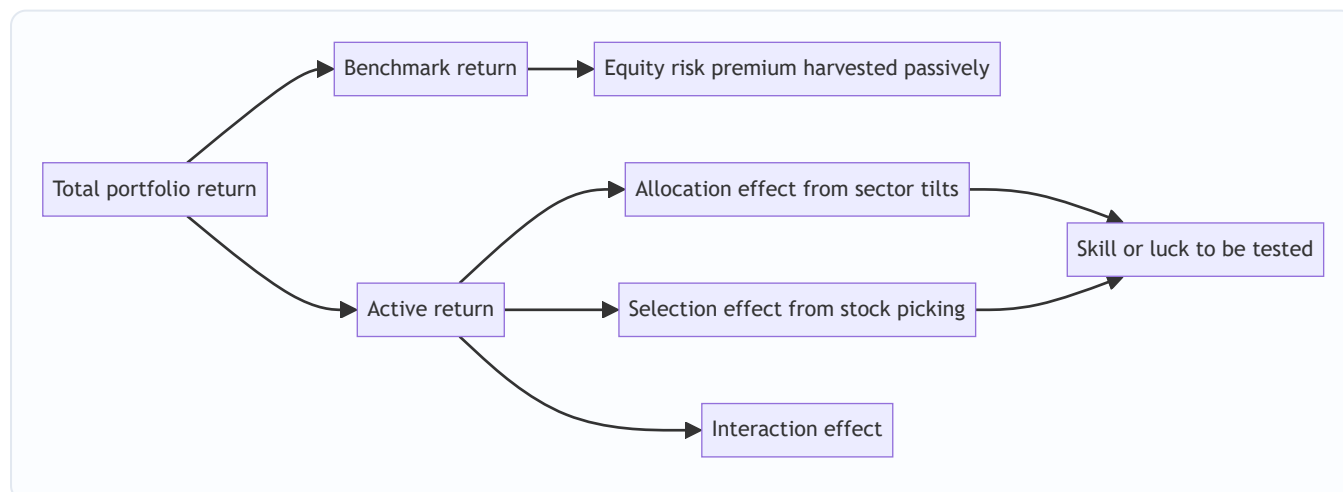
$$AS = \frac{1}{2} \sum_{i=1}^N |w_{p,i} - w_{b,i}|$$

AS ranges 0% (pure index) to 100% (no overlap). The one-half is because every overweight is matched by an underweight, so the absolute deviations double-count.

The crucial insight (Cremers & Petajisto): **TE and AS capture different bets**. High AS with *low* TE = a diversified stock-picker taking many small idiosyncratic bets that wash out at the portfolio level. High TE = concentrated factor/sector bets. You need *both* to characterise a manager:

	Low active share	High active share
Low TE	Closet indexer	Diversified stock picker
High TE	Factor / sector bettor	Concentrated stock picker

Figure 3 — Portfolio return decomposed into what you cannot control and what you chose.



4.7 Performance and attribution metrics

Once the portfolio runs, you decompose *where* return came from.

Sharpe ratio — total-risk-adjusted return, for the whole portfolio:

$$\text{Sharpe} = \frac{R_p - R_f}{\sigma_p}$$

Treynor ratio — systematic-risk-adjusted return (uses beta, for diversified sub-portfolios):

$$\text{Treynor} = \frac{R_p - R_f}{\beta_p}$$

Jensen's alpha — return above what CAPM predicted for the beta taken:

$$\alpha = R_p - [R_f + \beta_p (R_m - R_f)]$$

Brinson-Hood-Beebower attribution splits active return into allocation (sector-weight bets) and selection (stock-picking within sectors):

- Allocation effect (per sector): $(w_{p,i} - w_{b,i})(r_{b,i} - r_b)$
- Selection effect (per sector): $w_{b,i}(r_{p,i} - r_{b,i})$
- Interaction effect (per sector): $(w_{p,i} - w_{b,i})(r_{p,i} - r_{b,i})$

Summed across sectors, these three exactly reconcile to total active return $R_p - R_b$. This is the machinery that answers "was I right about *sectors* or about *stocks*?"

4.8 Concentration vs diversification

Diversification reduces *idiosyncratic* (stock-specific) risk. Portfolio variance:

$$\sigma_p^2 = \sum_i w_i^2 \sigma_i^2 + \sum_i \sum_{j \neq i} w_i w_j \sigma_i \sigma_j \rho_{ij}$$

As N rises with roughly equal weights, the first (own-variance) term shrinks like $1/N$, leaving only average covariance — the systematic floor you cannot diversify away. Empirically ~20–30 stocks remove most diversifiable risk; beyond ~50 the marginal benefit is tiny.

The tension: **diversification lowers risk but also dilutes your best ideas**. If your edge is real, your 5 best ideas have higher expected alpha than your 50th. Concentration concentrates alpha *and* idiosyncratic risk. The Fundamental Law resolves it: concentrate only to the degree your IC (per-bet skill) justifies losing breadth. A high-conviction fund of 20 names is a bet that skill > breadth; a quant fund of 400 names is the opposite bet.

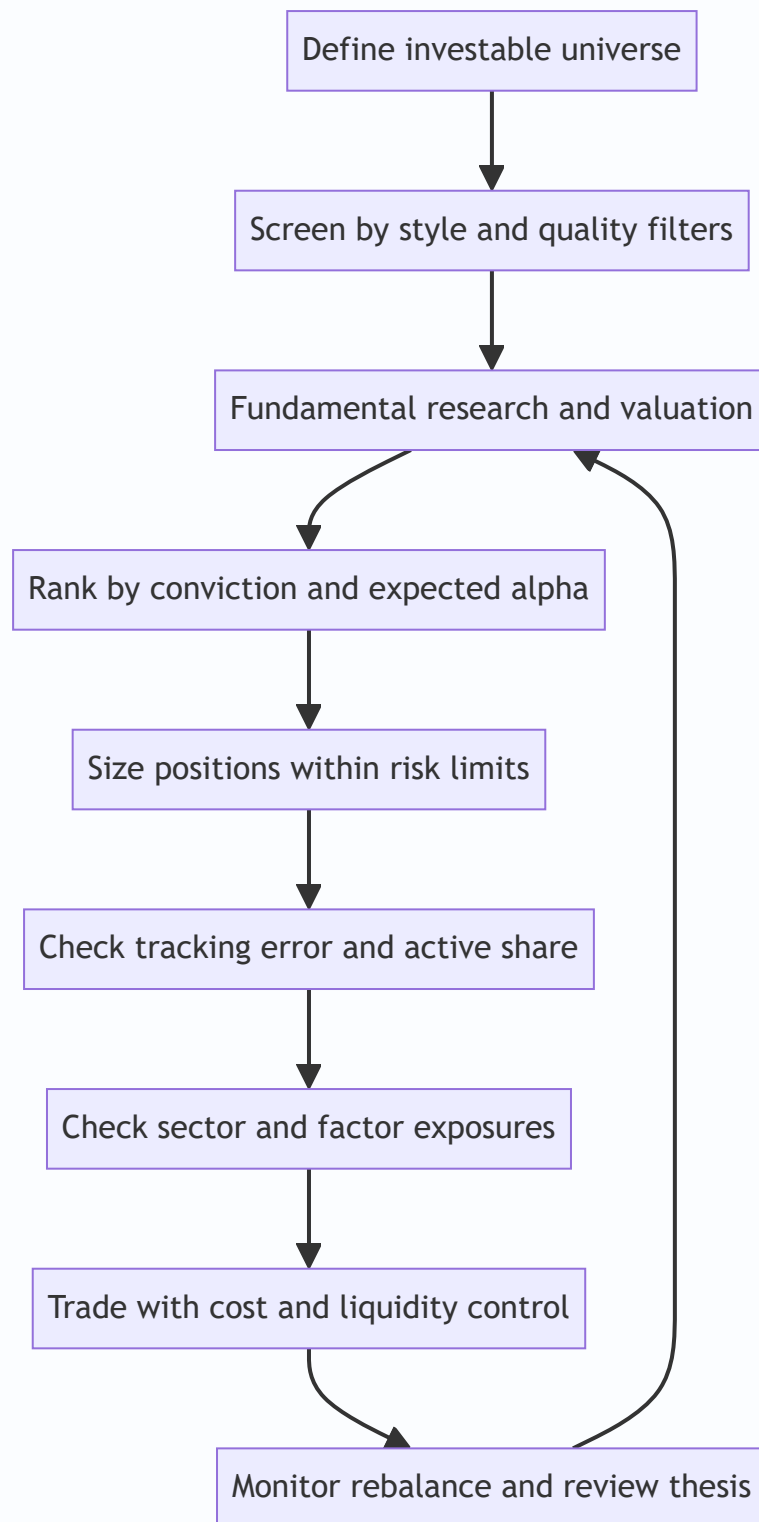
A common concentration gauge is the effective number of stocks via the Herfindahl index:

$$N_{\text{eff}} = \frac{1}{\sum_i w_i^2}$$

A 50-stock portfolio that is 40% in its top 5 names might have N_{eff} of only ~15 — far less diversified than the raw count suggests.

4.9 Building the portfolio — the process

Figure 4 — The end-to-end construction loop, from universe to monitored book.



Position sizing typically blends conviction with liquidity and risk limits. A simplified risk-budgeted sizing rule caps each name's contribution to tracking error; a heuristic version is:

$$w_i = w_{b,i} + k \cdot \frac{\text{score}_i}{\sigma_i}$$

subject to constraints (max active weight per stock, max sector deviation, minimum liquidity, target total TE). The active weight is scaled by conviction score and *divided* by the stock's volatility so that riskier names get smaller active bets for the same conviction.

5. Worked Examples

Example 1 — Active share and tracking error for a focused fund

A fund holds 4 stocks against a 4-stock benchmark. Weights:

Stock	Benchmark w_b	Portfolio w_p	$\sqrt{w_p - w_b}$
A	40%	30%	10%
B	30%	45%	15%
C	20%	25%	5%
D	10%	0%	10%
Sum	100%	100%	40%

Active share $= \frac{1}{2} \times 40\% = \mathbf{20\%}$.

Interpretation: despite dropping stock D entirely and over/under-weighting the rest, this is a *low* active-share portfolio — a near-closet-indexer. If it charges 1.2%, it charges 6% *on the active portion*. Verification: the overweights (B +15%, C +5% = +20%) exactly offset the underweights (A –10%, D –10% = –20%), confirming the deviations sum to zero and the sum of absolute deviations (40%) is exactly twice the active share — internally consistent.

Now suppose over 4 quarters the active returns (portfolio minus benchmark) are +1.5%, –0.5%, +2.0%, –1.0%. Mean active return $\bar{R}_A = (1.5 - 0.5 + 2.0 - 1.0)/4 = 0.5\%$ per quarter. Deviations from mean: +1.0, –1.0, +1.5, –1.5. Squared: 1.0, 1.0, 2.25, 2.25; sum = 6.5. $TE = \sqrt{6.5/(4-1)} = \sqrt{2.1667} = \mathbf{1.47\%}$ per quarter. Annualised: $0.5\% \times 4 = 2.0\%$ active return; $TE = 1.47\% \times \sqrt{4} = 2.94\%$. **Information ratio** $= 2.0\% / 2.94\% = \mathbf{0.68}$ — respectable skill, but note this fund achieved it with only 20% active share, meaning small bets, moderate TE. A closet-indexer profile.

Example 2 — Brinson attribution: sectors vs stocks

Two sectors, IT and Financials. Benchmark and portfolio data:

Sector	w_b	w_p	r_b (sector bench return)	r_p (my stocks' return)
IT	50%	70%	10%	12%
Financials	50%	30%	4%	3%

Benchmark total return $r_b = 0.5(10\%) + 0.5(4\%) = 7.0\%$. Portfolio total return $r_p = 0.7(12\%) + 0.3(3\%) = 8.4\% + 0.9\% = 9.3\%$. **Active return** $= 9.3\% - 7.0\% = \mathbf{+2.3\%}$.

Now decompose (Brinson-Hood-Beebower):

Allocation $(w_p - w_b)(r_{b,i} - r_b)$: - IT: $(0.70-0.50)(10\%-7\%) = 0.20 \times 3\% = +0.60\%$ - Fin: $(0.30-0.50)(4\%-7\%) = (-0.20)(-3\%) = +0.60\%$ - Allocation total $= +1.20\%$

Selection $w_b(r_{p,i} - r_{b,i})$: - IT: $0.50(12\%-10\%) = +1.00\%$ - Fin: $0.50(3\%-4\%) = -0.50\%$ - Selection total $= +0.50\%$

Interaction $(w_p - w_b)(r_{p,i} - r_{b,i})$: - IT: $\$0.20(2\%) = +0.40\%$ - Fin: $\$(-0.20)(-1\%) = +0.20\%$ - Interaction total $= +0.60\%$

Reconciliation: $\$1.20\% + 0.50\% + 0.60\% = \mathbf{\$2.30\%}$ — exactly the active return computed directly above. ✓

Interpretation: over half the outperformance (+1.20% of 2.30%) came from the *allocation* call — overweighting IT (which beat the index) and underweighting Financials (which lagged). Stock selection added a modest +0.50%, dragged by poor financial-stock picks. This manager is a better *sector allocator* than *stock picker* — a top-down profile. If they market themselves as a bottom-up stock picker, the attribution says otherwise.

Example 3 — GARP screen and the diversification floor

You compare two stocks for a GARP sleeve:

	Stock X	Stock Y
P/E	30	18
Expected earnings growth	25%	12%
PEG	$30/25 = \mathbf{1.20}$	$18/12 = \mathbf{1.50}$

Despite the higher headline P/E, **Stock X is the better GARP buy** — its PEG of 1.20 means you pay less per unit of growth than Stock Y's 1.50. This is the classic GARP counter-intuition: the "expensive" stock is cheaper once growth is priced in.

Now the diversification math. Assume every stock has $\sigma_i = 30\%$, average pairwise correlation $\rho = 0.4$, equal weights $w = 1/N$. Portfolio variance simplifies to:

$$\sigma_p^2 = \frac{\sigma^2}{N} + \frac{N-1}{N}\rho\sigma^2$$

- $N = 5$: $\sigma_p^2 = \frac{0.09}{5} + \frac{4}{5}(0.4)(0.09) = 0.018 + 0.0288 = 0.0468 \rightarrow \sigma_p = \mathbf{21.6\%}$
- $N = 20$: $\sigma_p^2 = \frac{0.09}{20} + \frac{19}{20}(0.4)(0.09) = 0.0045 + 0.0342 = 0.0387 \rightarrow \sigma_p = \mathbf{19.7\%}$
- $N = 100$: $\sigma_p^2 = 0.0009 + 0.03564 = 0.03654 \rightarrow \sigma_p = \mathbf{19.1\%}$
- $N \rightarrow \infty$: $\sigma_p^2 \rightarrow \rho\sigma^2 = 0.4 \times 0.09 = 0.036 \rightarrow \sigma_p = \mathbf{18.97\%}$

Going from 5 to 20 stocks cuts risk from 21.6% to 19.7% (−1.9 points). Going from 20 to *infinite* stocks cuts it only to 19.0% (−0.7 points more). **Most diversification benefit is captured by ~20 names; the systematic floor of ~19% cannot be diversified away** because it is driven by the 0.4 average correlation. This is the quantitative justification for concentrated portfolios: past ~20–25 names you add negligible risk reduction while diluting your best ideas. The floor also shows why "add more stocks" stops helping — you hit the covariance term.

6. Connections

- **CAPM and beta (earlier chapters):** Jensen's alpha and Treynor rest on CAPM; a portfolio's beta *is* its systematic sector/market tilt. Factor models generalise CAPM's single beta into value/size/momentum

betas.

- **Modern Portfolio Theory:** the diversification variance formula and the efficient frontier underlie the concentration-vs-diversification trade-off and risk-budgeted position sizing.
- **Market efficiency:** the whole active/passive debate is a referendum on the Efficient Market Hypothesis. Semi-strong efficiency implies factor premia (compensation for risk) can persist while pure stock-picking alpha is competed away.
- **Behavioural finance:** value and momentum premia are usually justified by behavioural biases (over-reaction, under-reaction, herding) — the "structural edge" that active managers claim.
- **Fixed income & asset allocation:** the top-down macro layer connects to the rates/credit chapters; equity is one sleeve of a total portfolio whose overall risk budget determines how much active equity risk you can afford.
- **Performance measurement:** Sharpe/Treynor/IR/attribution feed directly into GIPS-compliant reporting and fee justification.

7. Key Terms

- **Active return / excess return:** portfolio return minus benchmark return.
- **Active share:** half the sum of absolute active weights; holdings-based measure of how different you are from the index.
- **Tracking error:** standard deviation of active return; return-based measure of active risk.
- **Information ratio:** active return ÷ tracking error; risk-adjusted skill.
- **Information coefficient (IC):** correlation between forecasts and realised returns; raw per-bet skill.
- **Breadth:** number of independent bets per year.
- **Closet indexer:** a fund with high fees but low active share — mostly the index in disguise.
- **PEG ratio:** P/E divided by growth rate; the GARP valuation tool.
- **Factor tilt:** systematic exposure to a return-earning characteristic (value, size, momentum, quality, low-vol).
- **Allocation vs selection effect:** attribution split between sector-weight bets and stock-picking within sectors.
- **Jensen's alpha:** return above CAPM's prediction for the beta taken.
- **Effective number of stocks (N_{eff}):** inverse Herfindahl of weights; true concentration.
- **Value trap:** a statistically cheap stock that is cheap because the business is deteriorating.

8. Common Confusions

1. **"Tracking error and active share are the same thing."** No. TE is return-based and measures *volatility* of active return; AS is holdings-based and measures *how much* of the portfolio differs from the index. A diversified stock-picker can have high AS but low TE (bets wash out); a sector-bettor can have modest AS but high TE (few concentrated bets). You need both.
2. **"Higher active share always means better."** Active share measures *how different*, not *how good*. High AS just means more opportunity for the manager's skill (or lack of it) to show. It is necessary for outperformance but not sufficient.

3. **"Growth means high-quality; value means low-quality."** Style axes are independent. Quality is orthogonal to value/growth — you can have cheap junk (deep value trap), expensive junk (hyped growth), cheap quality (the holy grail), or expensive quality (quality compounder). GARP and quality both cut across the value/growth line.
4. **"A low P/E stock is cheap."** Only relative to *sustainable* earnings. A cyclical at trough earnings shows a *high* P/E when it's actually cheap, and a low P/E at peak earnings when it's expensive. Value requires normalising earnings.
5. **"More diversification is always safer, so hold hundreds of stocks."** Past ~20–30 names, marginal risk reduction is trivial (Example 3) while you dilute conviction and drift toward closet indexing. Over-diversification is a real cost, not a free lunch.
6. **"Alpha and factor tilt are the same excess return."** No — the whole point of Fama-French regression is to *strip out* factor tilt (which is buyable cheaply via smart beta) and isolate true selection alpha. A "star" who is really a small-cap-value bet has factor beta, not alpha.
7. **"Sharpe and Treynor should agree on ranking."** They agree only for fully diversified portfolios. Sharpe uses total risk (σ); Treynor uses systematic risk (β). For an *undiversified* portfolio, Sharpe penalises the idiosyncratic risk that Treynor ignores, so rankings can diverge — a red flag that the portfolio isn't diversified.

9. Recap

Equity portfolio management is the discipline of taking *deliberate, sized, measurable* deviations from a benchmark. The core identity is Portfolio = Benchmark + Active bets, and everything you get paid for — and every risk you take — lives in those active bets. Active management is a negative-sum game after costs (Sharpe's arithmetic), so you win only with a structural edge; the Fundamental Law ($IR \approx IC \sqrt{BR}$) shows that edge can come from deep skill on few bets or modest skill across many independent ones, which is why both focused stock-pickers and diversified quants exist.

Styles — value, growth, GARP, quality — describe *what kind* of stocks your bets favour; process — top-down vs bottom-up — describes *how* you generate them; sector and factor tilts are the *sized* expressions of those views. Two rulers measure your active-ness: tracking error (return-based volatility of active return) and active share (holdings-based difference from the index), and they capture different things. The concentration-vs-diversification trade-off is quantitative — ~20–30 stocks capture most diversification, past which you dilute your best ideas for negligible risk reduction. Finally, attribution (Brinson) tells you *whether your allocation calls or your stock picks* drove returns, exactly reconciling to total active return — the ultimate test of whether the process did what it claimed.

10. Quick-Reference / Interview Points

Formulas to have cold: - Active share $= \frac{1}{2} \sum |w_p - w_b|$ - Tracking error $= \sigma(R_p - R_b)$; Information ratio $= (R_p - R_b)/TE$ - Fundamental Law: $IR \approx IC \sqrt{BR}$ - Sharpe $= (R_p - R_f)/\sigma_p$; Treynor $= (R_p - R_f)/\beta_p$ - Jensen's alpha $= R_p - [R_f + \beta_p(R_m - R_f)]$ - PEG $= (P/E)/g$ - Brinson: Allocation $= (w_p - w_b)(r_b - r_{bench})$; Selection $= w_b(r_p - r_b)$; Interaction $= (w_p - w_b)(r_p - r_b)$ — these sum to active return. - $R_{eff} = 1/\sum w_i^2$

Interview soundbites: - "Active management is zero-sum before costs, negative-sum after — so I need a *named*, structural edge, not just optimism." - "TE and active share are complements, not substitutes: one measures active *risk*, the other active *positioning*. High fee + low active share = closet indexer, the worst product in the market." - "Quality is orthogonal to value and growth — that's why GARP and quality-value can coexist." - "The PEG ratio can make a P/E of 30 cheaper than a P/E of 18 — always price growth, don't just look at the multiple." - "Diversification is nearly free up to ~25 names and nearly useless past that; concentration is a bet that your IC justifies giving up breadth." - "Attribution is the lie detector: it tells the client whether I was actually right about sectors or about stocks, and it must reconcile to the total to the basis point." - "Before crediting a manager with alpha, regress on Fama-French — most 'alpha' is just a factor tilt you can buy for 30 bps."

Red flags to spot in a fund: - High fee + active share below 40% (closet indexer). - Sharpe and Treynor rankings diverge sharply (undiversified). - IR consistently negative (no skill, paying for nothing). - "Alpha" that vanishes after factor regression (it was beta). - N_{eff} far below stated stock count (hidden concentration).

Q&A — Equity Portfolio Management

Practice bank for Chapter 09. Work each question before reading the answer. Weights and returns are in percentages unless stated; betas and ratios are unit-free.

Section A — Concept Checks

A1. State the core identity of equity portfolio management in one line. $\text{Portfolio} = \text{Benchmark} + \text{Active positions}$. Everything you get paid for — and every active risk you take — lives entirely in the deviations of your weights from the index. If every weight equals the index weight, active return is mathematically zero before fees and negative after.

A2. Why is active management a "loser's game"? Sharpe's arithmetic: before costs, active managers *in aggregate* are the market — one manager's overweight is another's underweight, so they net to the index return. After fees and trading costs, the average active rupee must therefore underperform the average passive rupee. It is accounting, not a claim about efficiency. To win you must beat the average participant, who is a professional.

A3. What are the four linked choices the chapter breaks the craft into? (1) Active vs passive — do you take active bets at all, and how aggressively; (2) Style — value, growth, GARP, or quality; (3) Process — top-down macro or bottom-up stock picking; (4) Structure — how concentrated the bets are and how far they push you from the benchmark (tracking error, active share).

A4. Write the Fundamental Law of Active Management and say what it teaches. $IR \approx IC \sqrt{BR}$, where IR is the information ratio, IC is the information coefficient (per-bet skill, the correlation between forecasts and outcomes), and BR is breadth (independent bets per year). Lesson: a strong track record can come from deep skill on a few bets *or* modest skill spread across many independent bets — which is why concentrated stock-pickers and diversified quants are both viable.

A5. Distinguish tracking error from active share. Tracking error is the standard deviation of active return: return-based, backward-looking, measures the *volatility* of the active bet. Active share is $\frac{1}{2} \sum |w_p - w_b|$: holdings-based, forward-looking, measures *how much* of the portfolio differs from the index. They capture different things — a diversified stock-picker can have high active share but low TE (idiosyncratic bets wash out); a sector bettor can have modest active share but high TE.

A6. Why the factor of one-half in the active-share formula? Because in a fully invested portfolio every overweight is matched by an equal-sized underweight, so the sum of absolute active weights double-counts the true difference. Halving it gives the fraction of the portfolio genuinely deviating from the index, on a 0%–100% scale.

A7. What is a closet indexer, and which tool exposes it? A fund charging active fees while holding mostly the index — e.g. 1% fee for 20% active share means you pay 5x the index fee on the active portion (or ~1% for 20% of a differentiated product). Active share exposes it: high fee plus active share below ~40% is the tell.

A8. Define the PEG ratio and the GARP counter-intuition. $PEG = (P/E)/g$, where g is expected earnings growth in percent. PEG near 1.0 is "fair," below 1.0 attractive. The counter-intuition: a high-P/E, fast-growing stock can have a *lower* PEG than a low-P/E, slow grower — so the "expensive" stock is cheaper once you price the growth.

A9. Why is quality "orthogonal" to value and growth? Value/growth is one axis (cheap vs fast-growing); quality (durable ROE, low leverage, stable margins) is an independent axis. You can have cheap junk (value trap), expensive junk (hyped growth), cheap quality (the holy grail), or expensive quality (the compounder). Quality cuts across the value/growth line.

A10. What does the Fama-French-Carhart regression let you catch? Regressing fund returns on market, SMB (size), HML (value), and WML (momentum) strips out return that is merely a buyable factor tilt, isolating the intercept α as true selection skill. It catches a "star manager" who is really a leveraged small-cap-value bet — factor beta masquerading as alpha.

A11. Why does the diversification benefit have a floor? With equal weights, portfolio variance is $\sigma_p^2 = \frac{\sigma^2}{N} + \frac{N-1}{N}\rho\sigma^2$. As $N \rightarrow \infty$ the own-variance term vanishes but the covariance term tends to $\rho\sigma^2$ — the systematic floor set by average correlation. Adding stocks past ~20–30 chips at almost nothing while diluting your best ideas.

A12. Name the SAMURAI criteria for a valid benchmark. Specified in advance, Appropriate, Measurable, Unambiguous, Reflective of the manager's opinions, Accountable (owned by the manager), Investable. Benchmark a small-cap fund against Nifty 50 and every downstream metric is meaningless.

Section B — Numerical Problems (full working)

B1. Active share of a focused fund. Benchmark and portfolio weights: A (40% / 30%), B (30% / 45%), C (20% / 25%), D (10% / 0%). Absolute deviations: $|30-40|=10$, $|45-30|=15$, $|25-20|=5$, $|0-10|=10$; sum = 40%. $AS = \frac{1}{2}(40\%) = \mathbf{20\%}$. **Reconcile:** overweights (+15 B, +5 C = +20) exactly offset underweights (−10 A, −10 D = −20), so deviations net to zero and their absolute sum (40%) is exactly twice the active share. Despite dropping D entirely, this is a *low* active-share book — a near-closet-indexer. ✓

B2. Tracking error and information ratio. Quarterly active returns: +1.5%, −0.5%, +2.0%, −1.0%. Mean $\bar{R}_A = (1.5-0.5+2.0-1.0)/4 = 0.5\%$. Deviations: +1.0, −1.0, +1.5, −1.5; squared: 1.0, 1.0, 2.25, 2.25; sum = 6.5. $TE = \sqrt{6.5/(4-1)} = \sqrt{2.1667} = \mathbf{1.47\%}$ per quarter. Annualise: active return $0.5\% \times 4 = 2.0\%$; $TE = 1.47\% \times \sqrt{4} = 2.94\%$. $IR = 2.0\%/2.94\% = \mathbf{0.68}$. **Reconcile:** IR of 0.68 is respectable, but achieved with only 20% active share (B1) — small bets, moderate TE, a closet-indexer profile. ✓

B3. Brinson-Hood-Beebower attribution. Two sectors: IT (w_b 50%, w_p 70%, r_b 10%, r_p 12%), Financials (w_b 50%, w_p 30%, r_b 4%, r_p 3%). Benchmark return $= 0.5(10)+0.5(4) = 7.0\%$. Portfolio return $= 0.7(12)+0.3(3) = 8.4+0.9 = 9.3\%$. Active $= +2.3\%$. **Allocation** $(w_p-w_b)(r_{b,i}-r_b)$: IT $0.20(10-7)=+0.60\%$; Fin $(-0.20)(4-7)=+0.60\%$; total $+1.20\%$. **Selection** $w_b(r_{p,i}-r_{b,i})$: IT $0.50(12-10)=+1.00\%$; Fin $0.50(3-4)=-0.50\%$; total $+0.50\%$. **Interaction** $(w_p-w_b)(r_{p,i}-r_{b,i})$: IT $0.20(2)=+0.40\%$; Fin $(-0.20)(-1)=+0.20\%$; total $+0.60\%$. **Reconcile:** $1.20+0.50+0.60 = \mathbf{2.30\%}$ = active return computed directly. ✓ Over half came from allocation (overweighting IT, which beat the index) — a top-down profile, not a stock-picker.

B4. GARP screen via PEG. Stock X: P/E 30, growth 25% → $PEG = 30/25 = \mathbf{1.20}$. Stock Y: P/E 18, growth 12% → $PEG = 18/12 = \mathbf{1.50}$. **X is the better GARP buy** despite the higher headline P/E — you pay less per unit of growth. **Reconcile:** the "expensive" stock is cheaper once growth is priced in — the classic GARP counter-intuition. ✓

B5. Diversification floor. Each stock $\sigma = 30\%$ ($\sigma^2 = 0.09$), average correlation $\rho = 0.4$, equal weights. Use $\sigma_p^2 = \frac{\sigma^2}{N} + \frac{N-1}{N}\rho\sigma^2$. - $N=5$: $\frac{0.09}{5} + (4/5)(0.4)(0.09) = 0.018 + 0.0288 = 0.0468 \rightarrow \sigma_p = \mathbf{21.6\%}$. - $N=20$: $\frac{0.09}{20} + (19/20)(0.36) = 0.0045 + 0.0342 = 0.0387 \rightarrow \sigma_p = \mathbf{19.7\%}$. - $N \rightarrow \infty$: $\sigma_p^2 \rightarrow \rho\sigma^2 = 0.036 \rightarrow \sigma_p = \mathbf{18.97\%}$. **Reconcile:** 5→20 stocks cuts risk 1.9 points; 20→infinite cuts only 0.7 more. Most benefit is captured by ~20 names; the ~19% floor is the covariance term you cannot diversify away. ✓

B6. Effective number of stocks (hidden concentration). A 6-stock portfolio: top name 40%, then 20%, 15%, 10%, 10%, 5%. $\sum w_i^2 = 0.16 + 0.04 + 0.0225 + 0.01 + 0.01 + 0.0025 = 0.245$. $N_{\text{eff}} = 1/0.245 = \mathbf{4.08}$. **Reconcile:** a raw count of 6 stocks behaves like only ~4 equally weighted names because 40% sits in one position — the Herfindahl gauge exposes concentration the headline count hides. (An equal-weighted 6-stock book would give $N_{\text{eff}} = 1/(6 \times (1/6)^2) = 6$.) ✓

B7. Sector active-return contribution. IT is 15% of the index; you hold 20%. The IT sector returned 18% while the benchmark returned 9%. Active weight $= 20\% - 15\% = +5\%$. Contribution $= (w_p - w_b)(r_i - r_b) = 0.05(18\% - 9\%) = 0.05 \times 9\% = \mathbf{+0.45\%}$. **Reconcile:** overweighting a sector that beat the benchmark by 9 points adds 0.45% to active return — positive because both the active weight and the relative sector return are positive. Had IT lagged the index, the same overweight would have subtracted. ✓

B8. Information ratio target via the Fundamental Law. A quant has $IC = 0.05$ and takes 400 independent bets a year. Estimate IR. Then find the IC a 4-bet macro manager needs for the same IR. Quant: $IR = IC\sqrt{BR} = 0.05 \times \sqrt{400} = 0.05 \times 20 = \mathbf{1.0}$. Macro for $IR = 1.0$ with $BR=4$: $IC = IR/\sqrt{BR} = 1.0/2 = \mathbf{0.50}$. **Reconcile:** the macro manager needs 10x the per-bet skill (0.50 vs 0.05) to match the quant, because breadth (400 vs 4) does the heavy lifting on the quant side. Same equation, opposite ends of the skill-breadth curve. ✓

Section C — Interview-Style Questions (model answers)

C1. "How do you think about active vs passive?" It is a continuum, not a switch. Sharpe's arithmetic says active management is zero-sum before costs and negative-sum after, so passive — harvesting the equity premium at near-zero fee — beats most active managers over long horizons. I only pay for active management where I can name a structural edge: informational, analytical, or behavioural (patience, contrarianism, holding through drawdowns). Absent a named edge, I index the core and spend the risk budget only where the edge is real.

C2. "A fund has 1.2% fees and 25% active share. Your verdict?" Closet indexer — the worst product in the market. Three-quarters of the book is the index, so the client pays a full active fee on a portfolio that is mostly free beta in disguise. The effective fee on the genuinely active quarter is roughly 5%. I'd either demand far higher active share for that fee or replace it with a cheap index fund plus a small high-conviction satellite.

C3. "Tracking error and active share — aren't they the same?" No. Tracking error is return-based: the volatility of active return, how bumpy the ride relative to the index has been. Active share is holdings-based: how much of the portfolio's *positions* differ from the index right now. They can diverge sharply — a diversified stock-picker holding 100 idiosyncratic bets has high active share but low TE because the bets wash out; a manager making two big sector bets has low active share but high TE. You need both to characterise a manager: one is active *positioning*, the other active *risk*.

C4. "A manager beat the benchmark by 3% last year. How do you tell skill from luck?" Two moves. First, Brinson attribution — split the 3% into allocation (sector-weight bets) and selection (stock picks) and confirm it reconciles to the total; that tells me *what* they were right about and whether it matches their stated process. Second, a Fama-French-Carhart regression — if the 3% collapses once I control for size, value, and momentum, it was a factor tilt I can buy for 30 bps, not alpha. Real skill is a positive intercept that survives the factor controls, repeated across enough independent periods to clear the luck threshold.

C5. "Why not just hold 200 stocks to be safe?" Because diversification is nearly free up to ~25 names and nearly useless past that. Portfolio variance falls like $1/N$ on the idiosyncratic term but converges to the average-covariance floor set by correlation — beyond ~30 stocks you cut almost no risk while diluting your best ideas toward the index. Over-diversification is a real cost: you drift into closet indexing and pay active fees for index performance. Concentration is a deliberate bet that your per-bet skill justifies giving up breadth.

C6. "Value or growth — which do you prefer, and is that even the right question?" It is an incomplete question, because quality is an independent axis. I care less about the value/growth label than about paying a sensible price for durable earnings — which is why GARP and quality-value strategies cut across the divide. Value works when the market over-extrapolates bad news, but its risk is the value trap: cheap because the business is dying. Growth works when the market under-appreciates the duration of growth, but its risk is multiple compression when growth disappoints. I'd rather own cheap quality than argue the label.

C7. "What's the single most important number on a fund fact sheet to you?" Active share relative to fee, closely followed by the information ratio. Active share tells me whether I'm actually buying a differentiated product or the index in disguise; the IR tells me whether the active risk has been rewarded with skill. A fund with high active share and a consistently positive IR is taking real, well-sized bets and getting paid for them — that is the entire job described in two numbers.

C8. "Explain the Fundamental Law of Active Management to a client." Your information ratio — skill per unit of active risk — is roughly your accuracy per bet times the square root of how many independent bets you make. So there are two honest ways to be good: be very right on a handful of calls, or be slightly right across hundreds of uncorrelated ones. A focused fund of 20 names is betting on depth; a systematic fund of 400 is betting on breadth. Neither is inherently better; what fails is being mediocre on few bets — low skill and low breadth.

Section D — MCQs (with reasoning)

D1. Active share is calculated as: (a) $\sigma(R_p - R_b)$ (b) $\frac{1}{2} \sum |w_p - w_b|$ (c) $\sum w_i^2$ (d) $(R_p - R_b) / TE$ **Answer: (b).** Half the sum of absolute active weights, a holdings-based measure. (a) is tracking error, (c) relates to the Herfindahl/ N_{eff} , (d) is the information ratio.

D2. Before costs, the average actively managed rupee earns: (a) more than the index (b) exactly the index return (c) the risk-free rate (d) less than the index **Answer: (b).** Sharpe's arithmetic — active managers in aggregate are the market, so they net to the index *before* costs. Only *after* fees does the average active rupee fall below passive (option d describes the after-cost result).

D3. A fund with high active share but low tracking error is best described as: (a) a closet indexer (b) a concentrated stock picker (c) a diversified stock picker (d) a factor bettor **Answer: (c).** Many small, idiosyncratic bets differ from the index (high AS) yet wash out at the portfolio level (low TE). Closet indexer = low AS; concentrated picker and factor bettor = high TE.

D4. Stock X has P/E 30 and growth 25%; Stock Y has P/E 18 and growth 12%. On PEG: (a) Y is cheaper (b) X is cheaper (c) equal (d) PEG cannot compare them **Answer: (b).** $PEG\ X = 1.20 < PEG\ Y = 1.50$. X pays less per unit of growth despite the higher multiple — the GARP counter-intuition.

D5. In Brinson attribution, the allocation effect for a sector is: (a) $w_p(r_p - r_b)$ (b) $(w_p - w_b)(r_p - r_b)$ (c) $(w_p - w_b)(r_{b,i} - r_b)$ (d) $w_p(r_p - r_b)$ **Answer: (c).** Allocation isolates the sector-weight bet against the sector's benchmark return relative to the total benchmark. (a) is selection, (b) is interaction.

D6. As the number of equally weighted stocks grows without limit, portfolio variance tends to: (a) zero (b) the average variance (c) the average covariance $\rho\sigma^2$ (d) the market variance **Answer: (c).** The own-variance term dies like $1/N$; what remains is the average covariance $\rho\sigma^2$ — the systematic floor diversification cannot remove.

D7. The information ratio is defined as: (a) $(R_p - R_f)/\sigma_p$ (b) $(R_p - R_b)/TE$ (c) $(R_p - R_f)/\beta_p$ (d) $IC \times BR$ **Answer: (b).** Active return per unit of tracking error. (a) is Sharpe, (c) is Treynor, (d) misstates the Fundamental Law (it uses $\sqrt{BR}$, not BR).

D8. A "star" manager's excess return disappears after a Fama-French-Carhart regression. This means: (a) genuine alpha (b) the return was a factor tilt (c) negative skill (d) a data error **Answer: (b).** The return was explained by size, value, or momentum exposure — buyable cheaply as smart beta — leaving no intercept alpha. It was beta in disguise, not selection skill.

D9. Which best captures hidden concentration in a portfolio? (a) raw stock count (b) tracking error (c) $N_{\text{eff}} = 1/\sum w_i^2$ (d) active share **Answer: (c).** The effective number of stocks (inverse Herfindahl) falls far below the raw count when weights are lopsided, exposing concentration a headline count misses.

D10. The Fundamental Law implies a manager can achieve a high information ratio by: (a) high skill on few bets only (b) many independent bets only (c) either high per-bet skill or high breadth (d) raising fees **Answer: (c).** $IR \approx IC\sqrt{BR}$ — a strong record can come from deep skill on few bets or modest skill across many independent ones, which is why focused pickers and diversified quants both exist.

End of Chapter 09 Q&A. Drill Section B until active share, tracking error, IR, Brinson attribution (reconciling to the basis point), PEG, and the diversification floor are reflexive — and be ready to name a fund's profile from its active-share/TE quadrant on demand.

Chapter 10 — Fixed Income Portfolio Management

1. The Problem / Need

Everything up to this point — Markowitz, CAPM, multifactor models — was told through the language of *equities*, where "risk" mostly means the standard deviation of an uncertain price. Bonds break that framing. A plain-vanilla bond's cash flows are **contractually fixed**: you know the coupon, you know the redemption, you know the dates. If nothing defaults and you hold to maturity, there is, in one sense, *no uncertainty at all* about the rupee amount you receive.

And yet fixed income is where some of the largest portfolios in the world live — insurance companies, pension funds, central banks, banks' treasury books, debt mutual funds — and where enormous sums are won and lost. So what is the risk that a bond manager is actually paid to manage?

Three things, mainly:

1. **Interest-rate risk.** The *fixed* cash flow is exactly the problem. When market yields rise, the present value of those locked-in cash flows falls. A 10-year bond can lose 15% of its price on a 2% rate move — without any default. This is the dominant risk in a high-grade portfolio.
2. **Reinvestment risk.** Coupons arrive along the way and must be reinvested at *whatever* rate then prevails. If rates fall, your realised compound return disappoints even though the bond "did what it promised."
3. **Credit risk.** The issuer may not pay. Spreads over the risk-free curve compensate for default probability and loss given default — and those spreads move, inflicting mark-to-market pain long before any actual default.

Layer on **liquidity risk**, **call/prepayment risk**, and **inflation risk**, and the fixed-income manager's job becomes clear: not to pick "cheap stocks," but to **shape the portfolio's sensitivity to the yield curve and to credit spreads** so that it meets a specific objective — often a *liability* the fund must fund years from now.

This chapter is the toolkit for that job. It is the most quantitative, most "engineering" corner of asset management, and interviewers for AMC debt desks, insurance ALM teams, and bank treasuries probe it hard: *duration, convexity, immunisation, curve trades, spread positioning*. We build each idea from the cash-flow mechanics up.

2. The Core Idea

A bond portfolio is a package of interest-rate exposure and credit exposure. Managing it means deliberately choosing (a) how sensitive the portfolio's value is to changes in the level, slope, and curvature of the yield curve — summarised by duration and convexity — and (b) how much compensated credit and liquidity risk to hold, then aligning that exposure with the fund's objective, whether that objective is to track a benchmark, beat a benchmark, or fund a known stream of liabilities.

Two organising axes run through the whole discipline:

- **Passive vs active.** Passive strategies take the market's view as given — index the bond market, or ladder maturities, or immunise against a liability — and minimise the need for forecasts. Active strategies bet

that *your* forecast of rates, curve shape, or spreads beats the market's, and position duration/credit accordingly.

- **Total-return vs liability-driven.** A total-return manager (typical debt mutual fund) wants the highest risk-adjusted return versus a benchmark. A liability-driven manager (insurer, pension) does not care about beating an index — they care about having enough money on the exact dates the liabilities fall due.

Immunisation is the bridge that makes a bond portfolio behave like a liability.

Once you internalise that a bond is just a fixed stream of cash flows discounted at a curve, every technique in this chapter becomes a variation on one question: *how does my portfolio's value change when that curve moves, and is that the change I want?*

3. Why / How It Works

Price is a discounted cash-flow, and that is the whole game

A bond's price is nothing but the present value of its cash flows at the market yield y :

$$P = \sum_{t=1}^N \frac{C_t}{(1+y)^t}$$

Because y sits in the denominator, price and yield move **inversely** — the first law of fixed income.

Everything else (duration, convexity, immunisation) is calculus performed on this one equation. If you can differentiate P with respect to y , you can measure and control interest-rate risk.

Why duration is *the* risk number

Take the first derivative of price with respect to yield and scale by price. That gives the percentage price change per unit yield change — the portfolio's interest-rate "beta." Formalised, **Macaulay duration** is the cash-flow-weighted average time to receipt:

$$D_{\text{Mac}} = \frac{\sum_{t=1}^N t \cdot \frac{C_t}{(1+y)^t}}{P}$$

and **modified duration** converts it into a price sensitivity:

$$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+y} \quad \rightarrow \quad \frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y$$

Duration is powerful because it is **additive across a portfolio** (value-weighted), so a manager can steer a 300-bond portfolio's entire rate exposure with a single number. Want less rate risk before an expected hike? Cut portfolio duration. That is the master lever.

Why convexity exists and why it is a gift

Duration is a *linear* approximation to a *curved* price–yield relationship. The true relationship bows toward the origin (convex). The second-order term corrects it:

$$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y + \frac{1}{2} C \Delta y^2$$

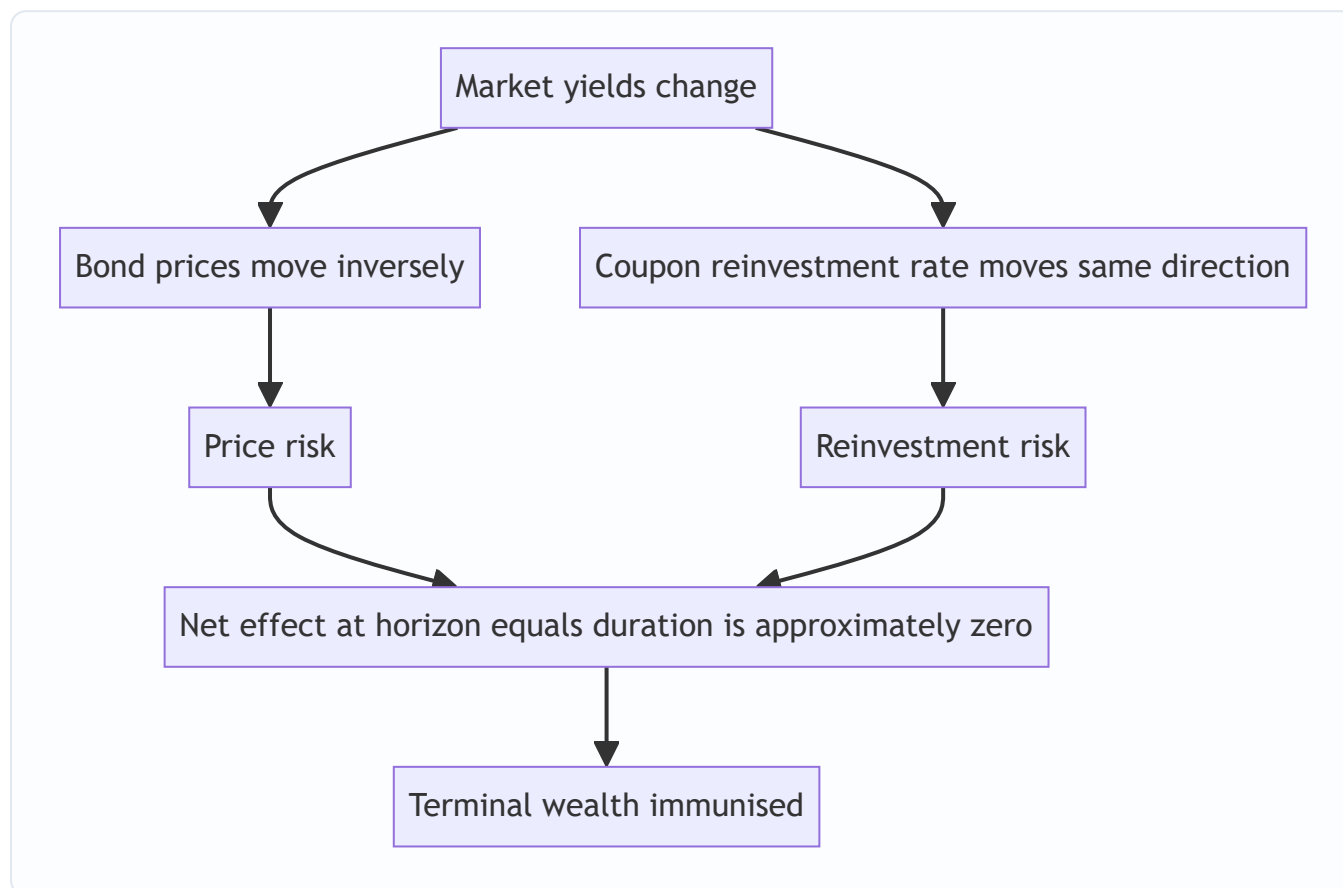
where C is convexity. Positive convexity means the bond **gains more when yields fall than it loses when yields rise by the same amount** — a favourable asymmetry. All else equal, investors pay for convexity (accept lower yield), and it matters most for large rate moves and for barbell structures.

Why reinvestment risk offsets price risk — the immunisation insight

Here is the elegant core of the chapter. When yields *rise*: bond prices *fall* (bad) but coupons reinvest at *higher* rates (good). When yields *fall*: prices *rise* (good) but reinvestment suffers (bad). These two effects push

in opposite directions. There exists a holding horizon where they **exactly cancel** — and that horizon is the **duration**. Set your portfolio's duration equal to your investment horizon (or your liability's duration) and, to a first approximation, you are **immunised**: your terminal wealth is locked in regardless of which way rates move. This is why duration is not just a risk metric but the foundation of liability matching.

Figure 10.1 — the two components of interest-rate risk offset each other, and duration is the horizon where they balance.



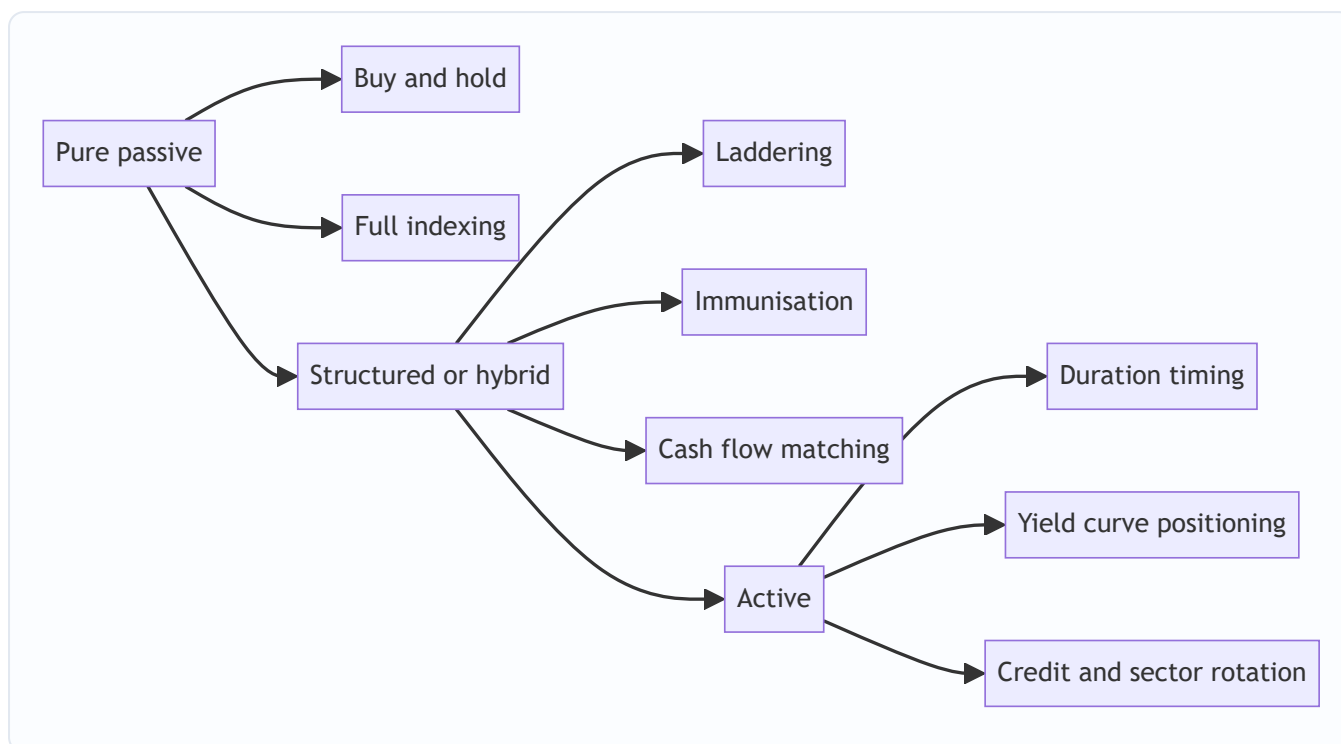
4. Full Content — Formulas, Strategies and Models

4.1 Setting the objective

Before any strategy, the manager fixes the **investment policy**: return requirement, risk tolerance, and constraints (liquidity, time horizon, tax, legal, regulatory). For debt funds the objective is usually *total return vs a benchmark index* (e.g., CRISIL Composite Bond Index). For insurers and pensions it is *asset–liability matching* — the surplus (assets minus liabilities) must be protected. This objective determines everything downstream.

4.2 The strategy spectrum

Figure 10.2 — fixed-income strategies from purely passive to fully active.



4.3 Passive strategies

(a) Indexing. Replicate a bond index's risk/return. Unlike equity indexing, full replication is impractical — a bond index can hold thousands of illiquid issues that trade rarely. Managers use **stratified sampling (cellular) matching**: partition the index into cells by duration, sector, credit quality, and coupon, then hold a small basket that matches each cell's weight and, crucially, **matches the index's duration**. Tracking error is the risk metric.

(b) Laddering. Spread holdings evenly across maturities (e.g., equal amounts maturing every year from 1 to 10 years). Each year a bond matures and is reinvested at the long end. Benefits: automatic diversification across the curve, steady reinvestment that averages over rate cycles, predictable liquidity, and no forecasting required. Contrast with a **bullet** (all maturities clustered at one point) and a **barbell** (concentrated at short and long ends).

Structure	Maturity profile	Convexity	Curve view it expresses
Bullet	Clustered at one horizon	Lower	Curvature stable; belly cheap
Barbell	Short + long ends	Higher	Curvature to increase; wings cheap
Ladder	Evenly spread	Medium	Agnostic; reinvestment averaging

(c) Buy-and-hold. Purchase and hold to maturity, ignoring interim price moves. Eliminates transaction costs and reinvestment-timing decisions but forgoes active opportunity.

4.4 Active strategies

(a) Duration / rate anticipation. If you expect yields to fall, *extend* duration to maximise price gains; if you expect a hike, *shorten* duration (even go to cash). This is the highest-impact active bet and the riskiest — it depends entirely on forecasting rate direction.

(b) Yield-curve positioning. Even with the average level unchanged, the curve's *shape* moves. Three canonical moves and the trades that exploit them:

Curve move	Description	Trade to profit
Parallel shift	Whole curve up/down equally	Adjust total duration
Steeper / flatter	Slope changes	Long-short across two maturities (duration-neutral)
Butterfly (curvature)	Belly moves vs wings	Barbell vs bullet

A **flattener** (expecting the gap between long and short yields to shrink) is typically expressed by going long the long end and short the short end, dollar-duration-weighted so the position is neutral to parallel shifts and profits only from the slope change.

(c) Credit and sector rotation. Move down (or up) the credit-quality ladder and across sectors (sovereign, PSU, corporate, structured) to capture spread. When the economy is strengthening and spreads are wide, rotate *into* credit (lower quality, higher spread) to earn carry and ride spread compression. When a downturn looms, rotate *up* in quality ("flight to quality"). **Credit spread** compensation:

$$\text{Yield}_{\text{corp}} = \text{Yield}_{\text{govt}} + \text{Credit spread} + \text{Liquidity premium}$$

(d) Riding the yield curve (rolldown). In an upward-sloping, stable curve, buy a bond longer than your horizon; as time passes the bond "rolls down" to lower yields (higher price), earning capital gain on top of coupon. Works only if the curve does not shift up.

(e) Sector/security selection & relative value. Identify individual bonds trading cheap to their peers (rich/cheap analysis on the spread curve).

4.5 The risk-measurement toolkit

Dollar duration (DV01 / PVBP) — the money change in value for a 1 bp yield move:

$$\text{DV01} = D_{\text{mod}} \times P \times 0.0001$$

This is what traders actually hedge with, because it is additive in currency terms and lets you neutralise a book.

Portfolio duration — market-value-weighted average of component durations:

$$D_P = \sum_i w_i D_i, \quad w_i = \frac{\text{MV}_i}{\sum_j \text{MV}_j}$$

Key-rate (partial) durations. A single duration assumes a parallel shift. Key-rate durations measure sensitivity to a move at *specific* points (2y, 5y, 10y, 30y) so a manager can see and control non-parallel (curve-reshaping) risk that a single duration hides.

Spread duration. Sensitivity of price to a change in the *credit spread* (holding the govt curve fixed) — the credit analogue of duration, and the key risk number for a corporate book.

Convexity for a portfolio is likewise value-weighted and adds a second-order correction that becomes material for big moves.

4.6 Immunisation — the mathematics

To immunise a **single liability** of known amount at a known horizon H :

1. Set **portfolio Macaulay duration = liability horizon H** .

2. Set **PV(assets) = PV(liability)** (fund it fully).

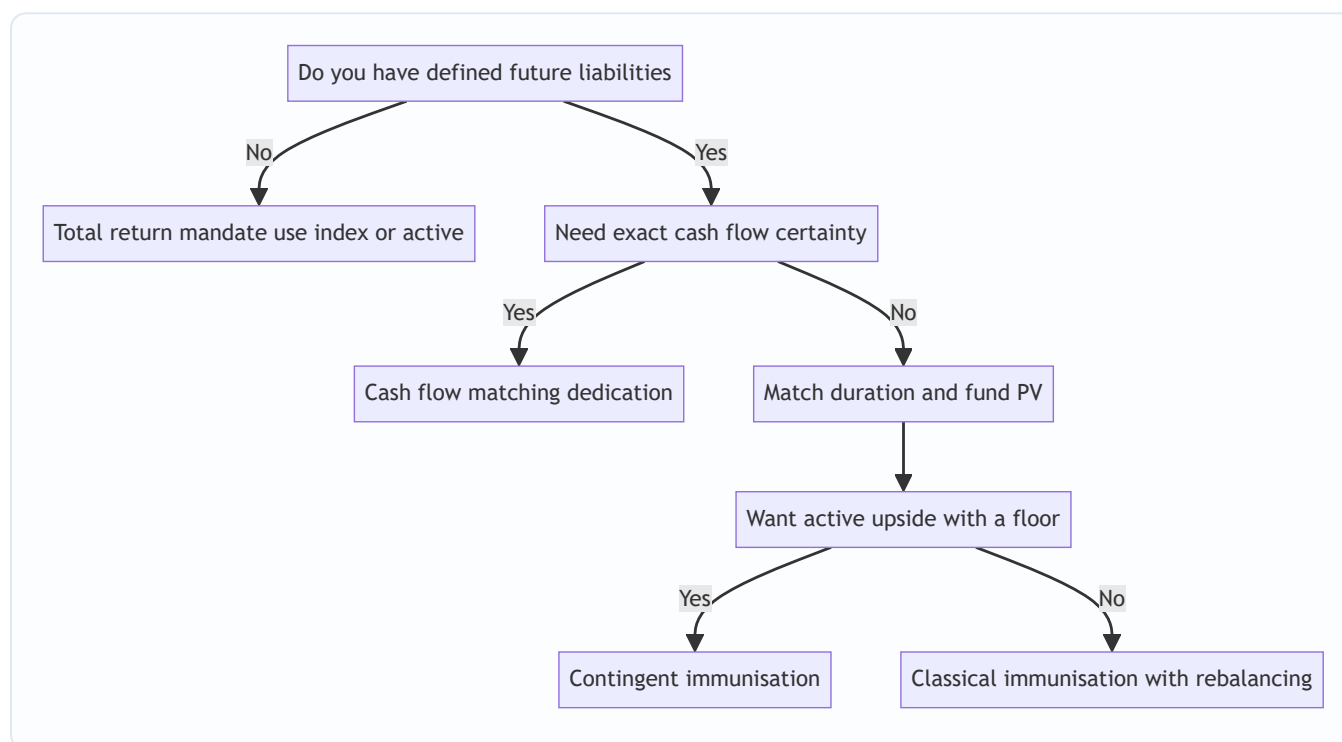
3. **Rebalance** periodically, because duration drifts as time passes and as yields move (duration falls slower than time, so the match degrades).

For **multiple liabilities**, conditions generalise: (i) $PV(\text{assets}) = PV(\text{liabilities})$; (ii) asset duration = liability duration; (iii) assets have **greater convexity/dispersion** than liabilities so the asset value dominates the liability value for *any* rate move (this is the *Fong–Vasicek* / dispersion condition). Immunisation protects against *parallel* shifts; residual risk from twists is managed by keeping convexity close and using key-rate matching.

Cash-flow matching (dedication) is the stricter alternative: buy bonds whose coupons and maturities *exactly* replicate the liability stream, so no rebalancing or rate forecast is ever needed. It is more expensive (fewer degrees of freedom, gives up yield) but eliminates reinvestment and rebalancing risk entirely.

Contingent immunisation is a hybrid: manage actively as long as the surplus cushion exceeds the cost of locking in an immunised return; if the cushion erodes to that trigger, switch to pure immunisation.

Figure 10.3 — choosing a liability-driven approach as a decision tree.



5. Worked Examples

Example 1 — Duration, DV01, and a convexity-corrected price move

A 3-year annual-coupon bond, face ₹1,000, coupon 8%, yielding 8% (so it trades at par). Compute price, Macaulay duration, modified duration, DV01, and estimate the price change for a **+100 bp** yield move, first with duration alone, then with a convexity correction. Verify against full repricing.

Step 1 — Cash flows and PV at $y = 8\%$.

t	C_t	$DF = (1.08)^{-t}$	$PV = C_t \cdot DF$	$t \cdot PV$	$t(t+1) \cdot PV$
1	80	0.92593	74.074	74.074	148.148
2	80	0.85734	68.587	137.174	411.523
3	1080	0.79383	857.339	2572.017	10288.07
Sum			1000.00	2783.265	10847.74

Price $P = ₹1,000.00$ (par, as expected). ✓

Step 2 — Macaulay duration. $D_{\text{Mac}} = \frac{2783.265}{1000} = 2.7833 \text{ years}$

Step 3 — Modified duration. $D_{\text{mod}} = \frac{2.7833}{1.08} = 2.5771$

Step 4 — DV01 (1 bp). $\text{DV01} = 2.5771 \times 1000 \times 0.0001 = ₹0.2577 \text{ per bond}$

Step 5 — Convexity. With annual compounding, $C = \frac{\sum t(t+1) \cdot C_t (1+y)^{-t}}{P(1+y)^2} = \frac{10847.74}{1000 \times 1.08^2} = \frac{10847.74}{1166.4} = 9.300$

Step 6 — Estimate ΔP for $\Delta y = +0.01$ (rates rise 100 bp).

Duration only: $\frac{\Delta P}{P} \approx -2.5771 \times 0.01 = -2.5771\% \Rightarrow \Delta P \approx -₹25.77$

With convexity: $\frac{\Delta P}{P} \approx -2.5771(0.01) + \frac{1}{2}(9.300)(0.01)^2 = -0.025771 + 0.000465 = -0.025306 \Rightarrow \Delta P \approx -₹25.31$

Step 7 — Full repricing at $y = 9\%$ (the truth). $P_{9\%} = \frac{80}{1.09} + \frac{80}{1.09^2} + \frac{1080}{1.09^3} = 73.394 + 67.331 + 833.958 = ₹974.69$ Actual change = $₹974.69 - ₹1000 = -₹25.31$.

Reconciliation. Duration alone predicted $-₹25.77$ (overstates the loss). Adding convexity gives $-₹25.31$, matching the exact repricing to the paise. This is the payoff of the second-order term: for a 100 bp move it closed essentially the entire error, and the *positive* convexity made the real loss smaller than the linear estimate. ✓

Example 2 — Immunising a single liability with a two-bond barbell

An insurer owes **₹10,00,000 in exactly 3 years**. The flat yield curve is at 8%. The manager will fund this using two zero-ish instruments approximated by two bonds: **Bond A** with duration 1.5 years and **Bond B** with duration 5.0 years. Find the weights that immunise the liability, and show it works for a rate shock.

Step 1 — PV of the liability. $PV_L = \frac{1,000,000}{1.08^3} = \frac{1,000,000}{1.259712} = ₹793,832$ So the manager invests ₹7,93,832 today. (Liability Macaulay duration = 3.0 years, since it is a single payment.)

Step 2 — Match duration. Need portfolio duration = 3.0: $1.5w_A + 5.0(1-w_A) = 3.0 \Rightarrow 1.5w_A + 5.0 - 5.0w_A = 3.0 \Rightarrow -3.5w_A = -2.0 \Rightarrow w_A = 0.5714, w_B = 0.4286$

Step 3 — Rupee allocation. - Bond A: $0.5714 \times 793,832 = ₹453,618$ - Bond B: $0.4286 \times 793,832 = ₹340,214$

Check duration: $0.5714(1.5) + 0.4286(5.0) = 0.8571 + 2.1429 = 3.000$ ✓

Step 4 — Stress test: yields jump to 9% immediately. Approximate each asset's and the liability's price change by modified duration ($D_{\text{mod}} = D_{\text{Mac}}/1.08$).

- Liability: $\$D_{\text{mod}} = 3.0/1.08 = 2.778\$$; $\Delta = -2.778 \times 0.01 = -2.778\%$. New PV $\approx \$793,832 \times (1 - 0.02778) = ₹771,780\$$.
- Bond A: $\$D_{\text{mod}} = 1.5/1.08 = 1.389\$$; $\Delta = -1.389\% \rightarrow 453,618 \times 0.98611 = ₹447,316\$$.
- Bond B: $\$D_{\text{mod}} = 5.0/1.08 = 4.630\$$; $\Delta = -4.630\% \rightarrow 340,214 \times 0.95370 = ₹324,462\$$.
- New asset value $\$ = 447,316 + 324,462 = ₹771,778\$$.

Reconciliation. After the shock, assets = ₹7,71,778 and liability PV = ₹7,71,780 — matched to within ₹2 (rounding). Because asset and liability durations were equal, the parallel shift moved both sides by the same percentage, preserving the surplus. That is immunisation working. The tiny residual is convexity: a real barbell (Bond B far out at 5y) has **more convexity than the single 3-year liability**, so for larger shocks the assets would actually pull slightly *ahead* — the favourable dispersion condition. The manager must still **rebalance** as time passes, because after one year the liability duration falls to 2.0 while the bonds' durations fall more slowly, breaking the match. ✓

Example 3 — A duration-neutral flattener (curve positioning)

A trader believes the **2s10s** spread will flatten (10-year yield to fall relative to 2-year). She wants a position that profits from flattening but is neutral to a parallel shift. Setup: go long the 10-year, short the 2-year, sized by DV01.

- 10-year bond: price ₹100, $\$D_{\text{mod}} = 8.0 \rightarrow \$ \text{DV01 per } ₹100 \text{ face } \$ = 8.0 \times 100 \times 0.0001 = ₹0.080\$$.
- 2-year bond: price ₹100, $\$D_{\text{mod}} = 1.9 \rightarrow \$ \text{DV01 } \$ = 1.9 \times 100 \times 0.0001 = ₹0.019\$$.

Step 1 — DV01-neutral hedge ratio. To make the book insensitive to a parallel shift, equalise dollar duration: $\text{Face}_{2y} = \text{Face}_{10y} \times \frac{\text{DV01}_{10y}}{\text{DV01}_{2y}} = 100 \times \frac{0.080}{0.019} = ₹421.05 \text{ face of 2-year short per } ₹100 \text{ face of 10-year long.}$

So: long ₹100 face of the 10y, short ₹421 face of the 2y. Net DV01 ≈ 0 .

Step 2 — Parallel shift of +10 bp on both. - 10y long P/L: $\$-0.080 \times 10 = -₹0.80\$$. - 2y short P/L: $\$+0.019 \times 10 \times 4.2105 = +₹0.80\$$. - Net $\approx ₹0$. The book ignores level moves. ✓

Step 3 — Flattening: 10y falls 10 bp, 2y unchanged. - 10y long P/L: $\$-0.080 \times (-10) = +₹0.80\$$. - 2y short P/L: 2y yield unchanged $\Rightarrow ₹0$. - Net = **+₹0.80** profit from the curve reshaping alone.

Reconciliation. The DV01 weighting stripped out the parallel-shift risk (Step 2 nets to zero) while leaving pure exposure to the *slope* (Step 3 pays off). This is exactly how a curve view is isolated from a level view — the essence of yield-curve positioning. ✓

6. Connections

- **To CAPM / portfolio theory (Ch 03–05):** duration is the bond world's *beta* — a single sensitivity to the dominant systematic factor (the yield curve). Where equities have one market beta, bonds have a term-structure exposure captured by duration and refined by key-rate durations.
- **To multifactor models (Ch 06):** modern fixed-income risk decomposes returns into **level, slope, and curvature** factors (the first three principal components of yield-curve moves explain ~95%+ of variation).

Bullet/barbell/butterfly trades are literally bets on these factors — an APT-style factor structure applied to rates.

- **To bond valuation fundamentals:** this chapter is applied DCF. Every duration and convexity number is a derivative of the price = $\sum PV(\text{cash flows})$ equation.
- **To derivatives / hedging:** DV01 is how desks size interest-rate futures and swaps to hedge a cash book. A manager who wants to cut duration fast sells bond futures rather than dumping illiquid bonds.
- **To ALM and insurance:** immunisation and cash-flow matching are the theoretical spine of asset-liability management and regulatory solvency frameworks. Surplus-at-risk is duration-gap analysis.
- **To macro:** duration positioning is a direct expression of a monetary-policy view. A rate-cut thesis = extend duration; a rate-hike / inflation thesis = shorten and up-quality.

7. Key Terms

Term	Meaning
Macaulay duration	Cash-flow-weighted average time to receipt of a bond's payments, in years.
Modified duration	$-D_{\text{Mac}}/(1+y)$; approximate % price change per 1% (100 bp) yield change.
Effective duration	Duration computed by repricing under shifted curves; used for bonds with embedded options where cash flows change.
DV01 / PVBP	Rupee change in price for a 1 bp yield move; the trader's hedging unit.
Convexity	Second-order (curvature) term correcting the duration estimate; positive convexity is favourable.
Key-rate duration	Sensitivity of price to a yield change at one specific maturity point; captures non-parallel risk.
Spread duration	Sensitivity of price to a change in credit spread.
Immunisation	Setting portfolio duration = liability horizon so price and reinvestment risk offset, locking terminal wealth.
Cash-flow matching / dedication	Building a portfolio whose cash flows exactly meet liabilities; no rebalancing needed.
Contingent immunisation	Active management while a surplus cushion exists, switching to immunisation if the cushion hits a trigger.
Bullet / Barbell / Ladder	Maturity structures: clustered / short-plus-long / evenly spread.
Riding the yield curve (rolldown)	Earning gains as a bond ages down a stable upward-sloping curve.
Credit spread	Extra yield over the risk-free curve compensating for default and liquidity risk.
Tracking error	Standard deviation of a portfolio's return difference vs its benchmark index.

8. Common Confusions

- **"Higher yield means higher return."** No — yield to maturity is only *realised* if every coupon is reinvested at that yield and the bond is held to maturity. Reinvestment risk breaks the equality; horizon (holding-period) return can differ sharply.
- **Duration is *not* just "years to maturity."** For a coupon bond, duration is always *less* than maturity (early coupons pull the weighted average in). Only a zero-coupon bond has duration equal to maturity.
- **Duration vs modified duration.** Macaulay is a time (years); modified is a sensitivity (%/%). Confusing the two mis-sizes every hedge. DV01 is modified duration expressed in rupees per bp.
- **Immunisation is not "set and forget."** It protects only against parallel shifts *at an instant* and drifts immediately — it demands periodic **rebalancing**. Cash-flow matching is the true set-and-forget approach.
- **Convexity is not always good for you.** Positive convexity helps; but **callable bonds and MBS exhibit negative convexity** near the strike — price gains are capped as rates fall (issuer calls / borrowers prepay), so you get the downside without the upside.
- **Barbell always beats bullet?** A barbell has more convexity (good for big moves) but usually *less yield* than a duration-matched bullet — you *pay* for convexity. Which wins depends on how much the curve actually moves.
- **Credit spread widening $\neq$ default.** You can lose money on a corporate bond that never defaults, purely because its spread widened and marked the price down. Spread duration, not just default probability, is the live risk.
- **Falling rates are unambiguously good for a bond fund?** Only for *price*. They hurt *reinvestment* and, for callable/MBS books, trigger negative convexity. Net effect depends on horizon vs duration.

9. Recap

A bond is a fixed stream of cash flows discounted at a market curve, so **price moves inversely with yield**, and the whole discipline is calculus on that identity. **Duration** measures first-order interest-rate sensitivity and is additive across a portfolio, making it the master risk lever; **convexity** corrects the estimate for large moves and is a favourable asymmetry (except in callable/MBS books, where it turns negative). **DV01** turns duration into rupees-per-basis-point for hedging.

Strategies run from **passive** (indexing via stratified sampling, laddering, buy-and-hold) to **active** (duration timing, yield-curve positioning via bullets/barbells/flatteners, and credit/sector rotation to harvest spread). For funds with defined liabilities, **immunisation** — setting portfolio duration equal to the liability horizon so price and reinvestment risk cancel — locks in terminal wealth, with **cash-flow matching** as the stricter no-rebalance alternative and **contingent immunisation** as the active-with-a-floor hybrid.

The worked examples showed the machinery reconciling: convexity closed the duration approximation error to the paise on a 100 bp move; a duration-matched barbell held surplus flat through a rate shock; and a DV01-weighted flattener neutralised parallel risk while isolating the slope bet. Managing a bond portfolio, in one line, is **choosing your exposure to the level, slope, and curvature of the yield curve, and to credit spreads, and aligning it with your objective.**

10. Quick-Reference / Interview Points

Core formulas

Quantity	Formula
Price	$P = \sum_t C_t (1+y)^{-t}$
Macaulay duration	$D_{\text{Mac}} = \frac{\sum_t t C_t (1+y)^{-t}}{P}$
Modified duration	$D_{\text{mod}} = \frac{D_{\text{Mac}}}{1+y}$
Price approximation	$\frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y + \frac{1}{2} C (\Delta y)^2$
DV01	$D_{\text{mod}} \times P \times 0.0001$
Portfolio duration	$\sum_i w_i D_i$ (value-weighted)
Convexity (annual)	$\frac{\sum_t t(t+1) C_t (1+y)^{-t}}{P(1+y)^2}$
Immunisation condition	$D_{\text{assets}} = D_{\text{liab}}$ and $PV_{\text{assets}} = PV_{\text{liab}}$

Rapid-fire answers

- *Why does price fall when yields rise?* Yield is the discount rate in the denominator of the PV; higher discount rate $\Rightarrow$ lower PV. Inverse by construction.
- *What's the single best measure of a bond portfolio's rate risk?* Modified duration (or DV01 in rupee terms); refine with key-rate durations for curve risk.
- *Bullet vs barbell?* Same duration, but barbell has higher convexity (wins on big moves) and lower yield (you pay for it). Bullet wins if the curve stays put.
- *How do you immunise?* Set portfolio duration = liability horizon, fund the PV, ensure asset convexity $\geq$ liability convexity, and rebalance as duration drifts.
- *Immunisation vs cash-flow matching?* Immunisation matches duration and needs rebalancing and protects mainly against parallel shifts; dedication matches exact cash flows, needs no rebalancing, but costs more yield.
- *Expecting a rate cut — what do you do?* Extend duration (buy long-dated bonds / receive fixed) to maximise price gains.
- *Expecting a curve flattener?* Long the long end, short the short end, DV01-weighted to be neutral to parallel shifts.
- *What is negative convexity and where?* Callable bonds and MBS: as rates fall, calls/prepayments cap the upside, so price gains flatten — you hold the downside without the upside.
- *How do you express a credit view without a rate view?* Buy/sell corporate vs duration-matched govt (asset swap), isolating spread; size by spread duration.
- *Biggest risk in a high-grade bond fund?* Interest-rate (duration) risk. In a high-yield fund, credit/spread risk dominates.
- *Why can a bond fund lose money when rates fall?* Reinvestment drag, and in callable/MBS books, negative convexity — the price upside is capped while carry deteriorates.

Q&A — Fixed Income Portfolio Management

Scope: Investments — Chapter 10 (Fixed Income Portfolio Management). Every question is followed by a full model answer. All rates are annual and in percent unless stated; compounding is annual unless noted. Work every numerical problem yourself before checking the solution. Sections: **A** concept-check · **B** numerical (full step-by-step, reconciling) · **C** interview-style with model answers · **D** MCQs with reasoning.

The chapter in one line

$$P = \sum_{t=1}^N \frac{C_t}{(1+y)^t} \quad \rightarrow \quad \frac{\Delta P}{P} \approx -D_{\text{mod}} \Delta y + \frac{1}{2} C \Delta y^2$$

One-line statement: A bond is a fixed stream of cash flows discounted at a market curve, so price moves inversely with yield; duration measures that first-order sensitivity, convexity corrects it, and the whole discipline is choosing your exposure to the level, slope and curvature of the curve (and to credit spreads) and aligning it with your objective.

Section A — Concept Check

A1. If a plain bond's cash flows are contractually fixed, what risk is the manager actually paid to manage? Three main risks. **Interest-rate risk** — the *fixed* cash flows are the problem: when market yields rise, the present value of those locked-in flows falls, so price drops even with no default (the dominant risk in a high-grade book). **Reinvestment risk** — coupons arrive along the way and must be reinvested at whatever rate then prevails. **Credit risk** — the issuer may default, and the spread that compensates for it moves, inflicting mark-to-market pain before any actual default. Liquidity, call/prepayment and inflation risk sit on top.

A2. Why do price and yield move inversely — is it an empirical fact or a definition? A definition, forced by the maths. Yield y is the discount rate sitting in the denominator of $P = \sum C_t(1+y)^{-t}$. A higher discount rate shrinks every present value, so price falls. It is inverse *by construction*, the first law of fixed income.

A3. Distinguish Macaulay from modified duration — which is a time and which is a sensitivity?

Macaulay duration is a **time**: the cash-flow-weighted average number of years to receipt, $D_{\text{Mac}} = \sum_t t \frac{C_t(1+y)^{-t}}{P}$. Modified duration is a **sensitivity**: $D_{\text{mod}} = D_{\text{Mac}}/(1+y)$, giving the approximate % price change per 100 bp yield change, $\Delta P/P \approx -D_{\text{mod}} \Delta y$. Confusing the two mis-sizes every hedge.

A4. Why is duration "the" risk number rather than one metric among many? Because it is **additive across a portfolio** on a value-weighted basis, so a single number steers the entire rate exposure of a 300-bond book. Want less rate risk before a hike? Cut portfolio duration. It is the master lever.

A5. Why does a coupon bond always have duration less than its maturity? Because the early coupons are cash received *before* maturity, and they pull the weighted-average time to receipt below the final date. Only a **zero-coupon bond** — one single payment at maturity — has duration exactly equal to its maturity.

A6. What is convexity and why is positive convexity a "gift"? Duration is a linear approximation to a curved price–yield relationship. Convexity is the second-order correction, $\frac{1}{2} C(\Delta y)^2$. Positive convexity means the bond **gains more when yields fall than it loses when yields rise by the same amount** — a favourable asymmetry. Investors pay for it (accept lower yield), and it matters most for large moves and barbell structures.

A7. Explain the immunisation insight: how do two opposite risks cancel? When yields *rise*, prices fall (bad) but coupons reinvest at higher rates (good); when yields *fall*, prices rise (good) but reinvestment suffers (bad). Price risk and reinvestment risk push in opposite directions. There is a holding horizon where they exactly offset — and that horizon is the **duration**. Set portfolio duration equal to the investment horizon (or the liability's duration) and terminal wealth is locked in to a first approximation.

A8. Why is bond indexing harder than equity indexing? A bond index can hold thousands of illiquid issues that trade rarely, so full replication is impractical. Managers use **stratified sampling (cellular matching)**: partition the index into cells by duration, sector, credit quality and coupon, hold a small basket matching each cell's weight, and crucially match the index's overall duration. The risk metric is tracking error.

A9. Contrast bullet, barbell and ladder maturity structures. A **bullet** clusters maturities at one horizon (lower convexity). A **barbell** concentrates at the short and long ends (higher convexity, expresses a view that curvature will rise). A **ladder** spreads holdings evenly across the curve — automatic diversification, steady reinvestment that averages over rate cycles, and no forecasting required.

A10. What is DV01 (PVBP) and why do traders prefer it to duration? DV01 is the rupee change in a position's value for a 1 bp yield move: $\text{DV01} = D_{\text{mod}} \times P \times 0.0001$. Traders hedge with it because it is **additive in currency terms** — you can neutralise a whole book by summing DV01s across positions and offsetting with futures or swaps.

A11. A single duration assumes a parallel shift. What captures non-parallel risk? Key-rate (partial) durations — sensitivity to a yield move at *specific* maturity points (2y, 5y, 10y, 30y). They reveal and let the manager control curve-reshaping (twist) risk that one aggregate duration hides.

A12. What is spread duration and why is it the key number for a corporate book? Spread duration is the sensitivity of price to a change in the *credit spread*, holding the government curve fixed. A corporate bond can lose money purely from spread widening with no default, so spread duration — not just default probability — is the live risk for a credit portfolio.

A13. State the classical immunisation conditions for a single liability. (1) Set **portfolio Macaulay duration = liability horizon** $\$H$; (2) set **PV(assets) = PV(liability)** (fund it fully); (3) **rebalance** periodically, because duration drifts as time passes and yields move (duration falls slower than calendar time, degrading the match).

A14. How do the conditions generalise for multiple liabilities? (i) $\text{PV}(\text{assets}) = \text{PV}(\text{liabilities})$; (ii) asset duration = liability duration; (iii) assets have **greater convexity/dispersion** than the liabilities (the Fong–Vasicek dispersion condition) so asset value dominates liability value for any parallel shift. Residual twist risk is handled with key-rate matching.

A15. Immunisation vs cash-flow matching (dedication) — one-line each. Immunisation matches *duration*, protects mainly against parallel shifts, and needs rebalancing. Cash-flow matching buys bonds whose coupons and maturities *exactly* replicate the liability stream, so no rebalancing or forecast is ever needed — but it costs more yield (fewer degrees of freedom).

A16. What is contingent immunisation? A hybrid: manage the portfolio **actively** as long as the surplus cushion exceeds the cost of locking in an acceptable immunised return; if the cushion erodes to that trigger, switch to pure immunisation. Active upside with a floor.

A17. Where does convexity turn *negative*, and why is that dangerous? In **callable bonds and mortgage-backed securities**. As rates fall, the issuer calls or borrowers prepay, capping price gains at the strike — you hold the downside when rates rise but forfeit the upside when they fall. Falling rates are therefore not unambiguously good for such books.

A18. Why can "higher yield to maturity" not be equated with "higher realised return"? YTM is realised only if every coupon is reinvested at that same yield and the bond is held to maturity. Reinvestment risk breaks the equality, so horizon (holding-period) return can differ sharply from the quoted YTM.

Section B — Numerical Problems

B1. Duration, DV01 and a convexity-corrected price move. A 3-year annual-coupon bond, face ₹1,000, coupon 8%, yields 8% (trades at par). Find price, Macaulay and modified duration, DV01, and estimate the price change for **+100 bp**, first with duration alone then with convexity. Verify against full repricing.

Solution. Cash-flow table at $y=8\%$:

t	C_t	$DF=(1.08)^{-t}$	SPV	$t \cdot PV$	$t(t+1) \cdot PV$
1	80	0.92593	74.074	74.074	148.148
2	80	0.85734	68.587	137.174	411.523
3	1080	0.79383	857.339	2572.017	10288.07
Σ			1000.00	2783.265	10847.74

Price $P = ₹1,000$ (par ✓). Macaulay $D_{\text{Mac}} = 2783.265/1000 = 2.7833$ yr. Modified $D_{\text{mod}} = 2.7833/1.08 = 2.5771$. DV01 $= 2.5771 \times 1000 \times 0.0001 = ₹0.2577$. Convexity $C = 10847.74/(1000 \times 1.08^2) = 10847.74/1166.4 = 9.300$.

For $\Delta y = +0.01$: duration only $\Rightarrow \Delta P/P \approx -2.5771\% \Rightarrow \Delta P \approx -₹25.77$. With convexity $\Rightarrow -0.025771 + \frac{1}{2}(9.300)(0.0001) = -0.025771 + 0.000465 = -0.025306 \Rightarrow \Delta P \approx -₹25.31$.

Full repricing at 9%: $80/1.09 + 80/1.09^2 + 1080/1.09^3 = 73.394 + 67.331 + 833.958 = ₹974.69$, so actual change $= -₹25.31$.

Reconciliation. Duration alone overstated the loss ($-₹25.77$); adding the positive convexity term shrank it to $-₹25.31$, matching exact repricing to the paise. Positive convexity made the real loss *smaller* than the linear estimate. ✓

B2. Portfolio duration and rebalancing to a target. A ₹100 crore book holds ₹60 cr of Bond X ($D_{\text{mod}} = 3.0$) and ₹40 cr of Bond Y ($D_{\text{mod}} = 7.0$). (a) Find portfolio duration. (b) The manager expects a rate cut and wants duration 6.0. What rupee shift from X to Y achieves it?

Solution. (a) $D_P = 0.60(3.0) + 0.40(7.0) = 1.8 + 2.8 = 4.6$. (b) Let new weight in Y be w : $3.0(1-w) + 7.0w = 6.0 \Rightarrow 3.0 + 4.0w = 6.0 \Rightarrow w = 0.75$. So Y must be ₹75 cr and X ₹25 cr — **shift ₹35**

cr from X into Y. Check: $\$0.25(3.0) + 0.75(7.0) = 0.75 + 5.25 = 6.0\$$ ✓. Extending duration ahead of an expected cut maximises the price gain.

B3. Immunising a single liability with a two-bond barbell. An insurer owes **₹10,00,000 in exactly 3 years**; the flat curve is 8%. Fund with Bond A ($D=1.5\$$) and Bond B ($D=5.0\$$). Find weights, rupee allocation, and stress-test at 9%.

Solution. $PV_L = 1,000,000 / 1.08^3 = 1,000,000 / 1.259712 = ₹793,832$; liability duration $= 3.0\$$ (single payment). Match: $1.5w_A + 5.0(1-w_A) = 3.0 \Rightarrow -3.5w_A = -2.0 \Rightarrow w_A = 0.5714, w_B = 0.4286$. Allocation: A $= 0.5714 \times 793,832 = ₹453,618$; B $= 0.4286 \times 793,832 = ₹340,214$. Duration check $0.5714(1.5) + 0.4286(5.0) = 0.857 + 2.143 = 3.00\$$ ✓.

Stress to 9% ($D_{\text{mod}} = D/1.08$): Liability $\Delta = -2.778\% \Rightarrow 793,832 \times 0.97222 = ₹771,780$. Bond A $\Delta = -1.389\% \Rightarrow 453,618 \times 0.98611 = ₹447,316$. Bond B $\Delta = -4.630\% \Rightarrow 340,214 \times 0.95370 = ₹324,462$. New assets $= 447,316 + 324,462 = ₹771,778$.

Reconciliation. Assets ₹7,71,778 vs liability ₹7,71,780 — matched within ₹2 (rounding/convexity). Equal durations meant the parallel shift moved both sides by the same %, preserving surplus. Because the barbell has more convexity than the single-payment liability, for larger shocks assets pull slightly ahead — the favourable dispersion condition. Rebalancing is still needed as time passes. ✓

B4. DV01-neutral flattener. A trader expects the 2s10s to flatten. 10y bond: price ₹100, $D_{\text{mod}} = 8.0\$$. 2y bond: price ₹100, $D_{\text{mod}} = 1.9\$$. (a) Size a duration-neutral long-10y/short-2y trade per ₹100 of 10y. (b) Show it is flat to a +10 bp parallel shift. (c) Show the P/L if the 10y falls 10 bp and the 2y is unchanged.

Solution. $DV01(10y) = 8.0 \times 100 \times 0.0001 = ₹0.080$; $DV01(2y) = 1.9 \times 100 \times 0.0001 = ₹0.019$. (a) Face of 2y short $= 100 \times 0.080 / 0.019 = ₹421.05$. Long ₹100 face 10y, short ₹421 face 2y. (b) +10 bp both: 10y long $\$ -0.080 \times 10 = -₹0.80$; 2y short $\$ +0.019 \times 10 \times 4.2105 = +₹0.80$; net $\approx ₹0$ ✓. (c) 10y -10 bp: long $\$ -0.080 \times (-10) = +₹0.80$; 2y unchanged $\Rightarrow ₹0$; net **+₹0.80** from the slope alone. The DV01 weighting stripped out parallel-shift risk and isolated the pure curve bet. ✓

B5. Riding the yield curve (rolldown). The curve is stable and upward-sloping: a 3-year zero yields 7%, a 2-year zero yields 6%. A manager with a 1-year horizon buys the 3-year zero (face ₹1,000). If the curve is unchanged in one year, compute the 1-year holding return and compare with simply holding a 1-year bond at 5%.

Solution. Buy price $\$ = 1000 / 1.07^3 = 1000 / 1.225043 = ₹816.30$. After 1 year it is a 2-year bond; with the curve unchanged it now yields 6%: value $\$ = 1000 / 1.06^2 = 1000 / 1.1236 = ₹889.99$. Holding return $\$ = 889.99 / 816.30 - 1 = 0.0903 = \text{9.03}\%$. The 1-year bond returns just 5%. The extra $\approx 4.03\%$ is the **rolldown**: as the bond ages it "rolls down" to a lower yield (6% instead of 7%), gaining price on top of carry — but only because the curve did not shift up. ✓

B6. Credit spread decomposition and spread-duration P/L. A 5-year corporate yields 9.2%; the matched 5-year government yields 7.0%; the estimated liquidity premium is 0.3%. (a) What is the pure credit spread? (b) The bond's spread duration is 4.2. If the credit spread widens 40 bp (govt curve unchanged), what is the % price change?

Solution. (a) $\text{Yield}_{\text{corp}} = \text{Yield}_{\text{govt}} + \text{credit spread} + \text{liquidity premium} \Rightarrow 9.2 = 7.0 + \text{CS} + 0.3 \Rightarrow \text{CS} = 1.9\%$. (b) $\Delta P/P \approx -\text{SpreadDur} \times \Delta \text{spread} = -4.2 \times 0.0040 = -0.0168 = \text{-1.68}\%$. The bond loses

1.68% purely from spread widening — no default needed, no move in the govt curve. This is exactly why spread duration is the live risk on a corporate book. ✓

Section C — Interview-Style Questions with Model Answers

C1. "Walk me through what duration really measures and its main limitation." Duration is the first derivative of price with respect to yield, scaled by price — the % price change per 100 bp move, i.e. the portfolio's interest-rate beta. Its power is additivity: one number steers a whole book. Its limitation is that it is a *linear* approximation to a *convex* price–yield curve, so it is accurate only for small moves, and a single aggregate duration assumes a **parallel** shift. For large moves I add convexity; for curve-resaping (twist) risk I use key-rate durations.

C2. "You expect the RBI to cut rates. Position a debt fund." Extend duration — buy longer-dated bonds (or receive fixed via swaps / go long bond futures) to maximise price gains from falling yields. I would size the extension to my conviction and the fund's mandate risk limits, because duration timing is the highest-impact and riskiest active bet; if my rate call is wrong, the same lever amplifies losses. I would watch reinvestment: falling rates hurt coupon reinvestment, and if I hold callable/MBS paper, negative convexity caps the upside.

C3. "An insurer must pay a fixed sum in seven years. How would you fund it, and what can still go wrong?" Classical immunisation: set the asset portfolio's Macaulay duration to 7 years, fund the full present value of the liability, ensure asset convexity/dispersion at least matches the liability, and rebalance periodically as duration drifts. What can still go wrong: immunisation protects mainly against *parallel* shifts, so a curve **twist** (short rates and long rates moving differently) can break it — I mitigate with key-rate matching. Duration also drifts immediately, so neglecting rebalancing degrades the match. If certainty is paramount and yield give-up is acceptable, I would instead cash-flow match (dedication) and remove reinvestment and rebalancing risk entirely.

C4. "Bullet or barbell — which do you buy?" Same duration, different convexity. A barbell has higher convexity, so it outperforms a duration-matched bullet on **large** rate moves in either direction; but you pay for that convexity with lower yield. A bullet earns more carry and wins if the curve **stays put**. So it is a bet on realised volatility: if I expect a big move or rising curvature, barbell; if I expect a quiet, stable curve, bullet to harvest the yield pickup.

C5. "How do you express a credit view without taking a rate view?" Buy (or sell) the corporate bond against a **duration-matched government** position — an asset swap. Matching duration neutralises the level of rates, leaving pure exposure to the credit spread. I size it by **spread duration**, not modified duration, because spread duration is what governs the P/L per basis point of spread move. That isolates the compensated credit/liquidity risk I actually want.

C6. "Why can a bond fund lose money even when interest rates fall?" Two reasons. **Reinvestment drag**: coupons and maturities now reinvest at lower rates, so realised compound return disappoints. **Negative convexity**: if the book holds callable bonds or MBS, falling rates trigger calls and prepayments that cap the price upside while carry deteriorates — the classic "heads you lose, tails you don't win" of embedded short optionality. Whether the fund nets a gain depends on its horizon versus duration and the option content of its holdings.

C7. "What is the single biggest risk in a high-grade bond fund versus a high-yield fund?" In a high-grade (sovereign/AAA) fund it is **interest-rate / duration risk** — spreads are thin and stable, so the yield

curve dominates the P/L. In a high-yield fund it is **credit / spread risk** — defaults and spread widening drive returns, and duration matters less because prices are governed more by issuer fundamentals and recovery than by the risk-free curve. The right risk metric shifts from modified duration to spread duration and default probability.

Section D — MCQs with Reasoning

D1. A 10-year coupon bond and a 10-year zero-coupon bond, same yield. Which has the higher duration? (a) The coupon bond (b) The zero (c) Equal (d) Depends on coupon size **Answer: (b).** The zero has a single payment at year 10, so its Macaulay duration equals 10. The coupon bond's early coupons pull its weighted-average time to receipt below 10. A zero always has the maximum duration for a given maturity.

D2. Modified duration is 6.0 and the bond is priced at ₹95. DV01 per ₹100 face is approximately: (a) ₹0.057 (b) ₹0.57 (c) ₹5.70 (d) ₹0.0095 **Answer: (a).** $\text{DV01} = D_{\text{mod}} \times P \times 0.0001 = 6.0 \times 95 \times 0.0001 = ₹0.057$. A 1 bp move changes value by about 5.7 paise per ₹100 of price.

D3. Positive convexity implies that, for equal-sized yield moves: (a) Price falls more than it rises (b) Price rises more than it falls (c) Price change is symmetric (d) Duration is zero **Answer: (b).** The convex price–yield curve bows toward the origin: the second-order term $\frac{1}{2} C(\Delta y)^2$ is always positive, adding to gains when yields fall and cushioning losses when they rise. The favourable asymmetry is why investors pay for convexity.

D4. To immunise a single liability, the manager sets: (a) Coupon = liability payment (b) Maturity = liability horizon (c) Portfolio duration = liability horizon and $PV(\text{assets}) = PV(\text{liability})$ (d) Convexity = 0 **Answer: (c).** Matching duration to horizon makes price and reinvestment risk offset; funding the PV ensures the money is there. Matching *maturity* is not enough — a longer-maturity coupon bond can have the right duration, and a bullet at the horizon still needs full PV funding.

D5. Which structure has the highest convexity for a given duration? (a) Bullet (b) Barbell (c) Ladder (d) All equal **Answer: (b).** Spreading cash flows to the short and long ends (barbell) increases the dispersion of payment times, which raises convexity above a bullet of the same duration. The trade-off is lower yield.

D6. A callable bond exhibits negative convexity primarily when: (a) Rates rise sharply (b) Rates fall toward the call strike (c) The issuer defaults (d) The curve steepens **Answer: (b).** As rates fall, the issuer's call option moves into the money; the market prices in redemption, capping the bond's price appreciation. The price–yield curve bends the "wrong" way near the strike, giving negative convexity.

D7. A DV01-weighted long-long-end / short-short-end trade is designed to profit from: (a) A parallel upward shift (b) A parallel downward shift (c) A flattening of the curve (d) Rising credit spreads **Answer: (c).** DV01 weighting neutralises parallel-shift P/L, leaving exposure only to the *slope*. Long the long end and short the short end gains when the spread between them narrows — a flattener.

D8. Cash-flow matching (dedication) differs from immunisation chiefly because it: (a) Requires more frequent rebalancing (b) Requires no rebalancing or rate forecast, but costs more yield (c) Protects only against parallel shifts (d) Ignores present value **Answer: (b).** Dedication buys bonds whose cash flows exactly meet the liabilities, so no rebalancing or forecasting is ever needed; the cost is a lower yield from the loss of flexibility. Immunisation, by contrast, matches duration, needs rebalancing, and mainly guards against parallel shifts.

D9. Which measure best captures a corporate bond portfolio's risk to a widening of credit spreads with the government curve unchanged? (a) Modified duration (b) Macaulay duration (c) Spread duration (d) Convexity **Answer: (c).** Spread duration is the sensitivity of price to a change in credit spread holding the risk-free curve fixed — precisely the exposure that widening spreads hit. Modified/Macaulay duration measure sensitivity to the risk-free yield, not the spread.

D10. A single aggregate duration fails to capture which risk? (a) Parallel shift risk (b) Non-parallel (twist/curve-reshaping) risk (c) Default risk only (d) Reinvestment risk **Answer: (b).** Aggregate duration assumes the whole curve moves by the same amount. A twist — short and long rates moving differently — is invisible to it; key-rate durations are needed to see and control that non-parallel risk.

D11. Riding the yield curve (rolldown) earns extra return only when: (a) The curve is downward-sloping (b) The curve is upward-sloping and stays put (c) Rates rise (d) Credit spreads tighten **Answer: (b).** On a stable, upward-sloping curve, a bond ages into lower yields and gains price on top of coupon. If the curve shifts up, the roll-up in yield offsets or reverses the gain, so stability of the curve is essential.

D12. Yield to maturity equals realised return only if: (a) The bond is investment grade (b) Every coupon is reinvested at the YTM and the bond is held to maturity (c) Duration equals maturity (d) The curve is flat **Answer: (b).** YTM embeds the assumption that all coupons compound at that same yield to maturity. Reinvestment at any other rate — or an early sale — breaks the equality, which is the essence of reinvestment risk.

End of Q&A — Fixed Income Portfolio Management.

Chapter 11 — Performance Measurement and Attribution

1. The Problem / Need

A portfolio manager tells you, "I made 22% last year." Your first job is to distrust that number — not because she is lying, but because a raw return, on its own, answers almost none of the questions that matter to an investor, an allocator, or an interviewer probing whether you understand what "good performance" actually means.

Consider three managers who all reported 22%:

- Manager A ran a concentrated portfolio of small-cap momentum stocks with 35% volatility and drew down 40% mid-year.
- Manager B tracked the Nifty 50, which itself returned 21%, so she added almost nothing.
- Manager C ran a market-neutral book with 6% volatility, no drawdown, and 22% return.

The same headline number represents wildly different skill. C is a phenomenon; B is a closet indexer charging active fees; A may simply have been lucky with leverage on a beta bet. **Performance measurement is the discipline of converting a raw return into a judgment about skill.** It has to answer four distinct questions:

1. **How much did the portfolio actually earn**, computed in a way that is not distorted by the timing of client cash flows the manager did not control?
2. **How much risk was taken to earn it?** Return without a risk denominator is meaningless.
3. **Did the manager beat a fair alternative** — the benchmark the client could have bought passively for a few basis points?
4. **Where did the value come from?** Was it asset allocation, security selection, or currency? Was it repeatable skill or a single lucky bet?

The stakes are practical. Allocators move billions based on these numbers. Fees — often "2 and 20" in hedge funds or 50–100 bps in mutual funds — are justified (or not) by them. Managers are hired and fired on three-year rolling numbers that are often statistically indistinguishable from noise. Getting the measurement wrong means rewarding luck and punishing skill. This chapter builds the full toolkit: return measurement, risk-adjusted ratios, benchmarking, and attribution — with the actual math and worked examples that reconcile.

2. The Core Idea

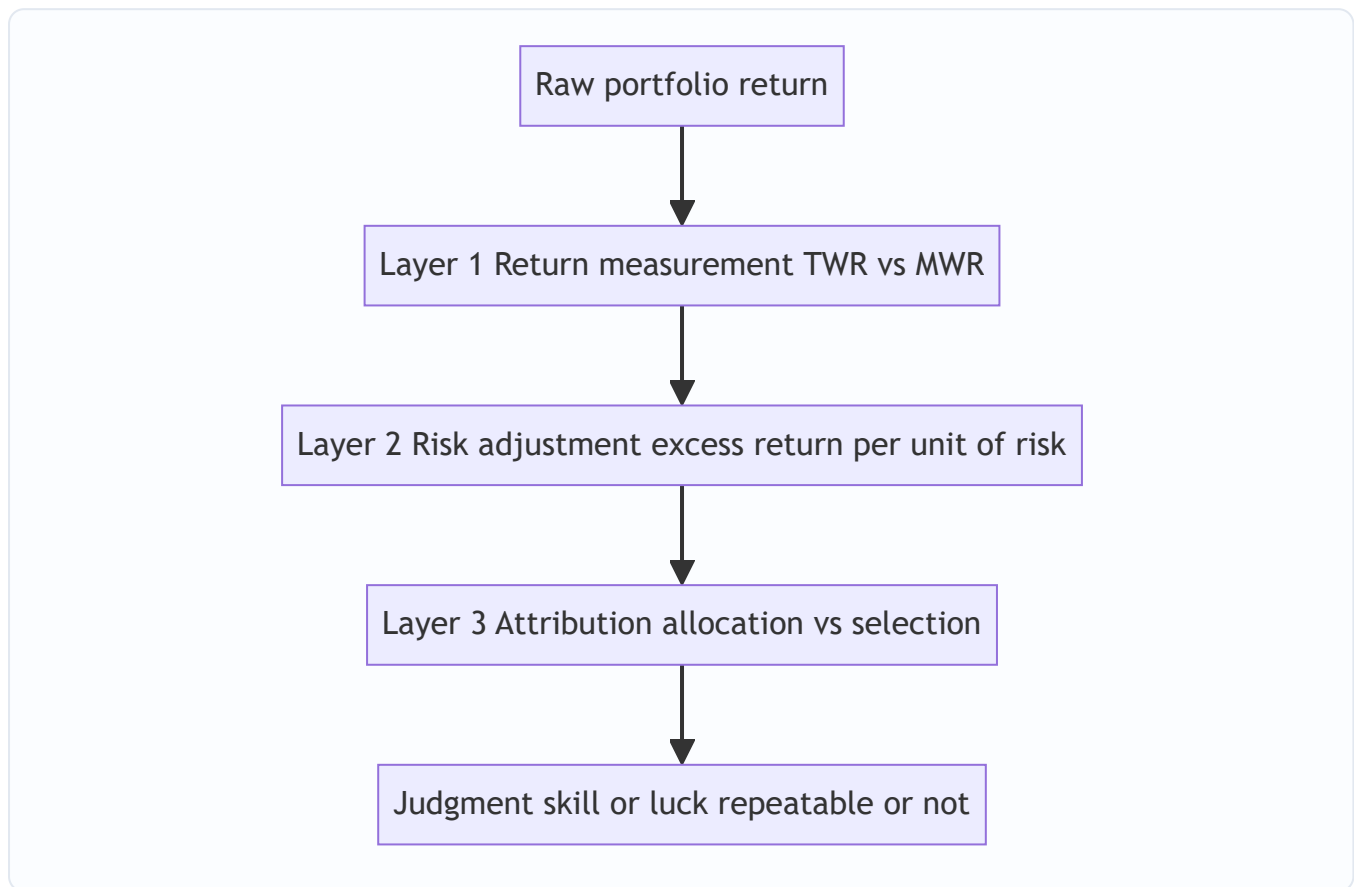
Performance evaluation rests on one organizing principle: **a return is only interpretable relative to (a) the risk taken and (b) a fair alternative.** Everything in this chapter is an elaboration of that sentence.

We break the analysis into three layers that stack on top of each other:

- **Layer 1 — Return measurement.** Get the return number right. The central subtlety is *whose* return we are measuring: the *portfolio's* (the manager's skill, time-weighted) or the *investor's* (their actual wealth outcome, money-weighted). These differ whenever cash flows move in and out.
- **Layer 2 — Risk adjustment.** Divide excess return by a measure of risk. Which risk measure you use in the denominator (total volatility, systematic beta, downside deviation, or tracking error) defines the ratio (Sharpe, Treynor, Sortino, Information Ratio). Each answers a subtly different question.

- **Layer 3 — Attribution.** Decompose the excess return into the *decisions* that produced it — allocation across asset classes/sectors versus selection of securities within them. This is where you separate "the manager was overweight the right sector" from "the manager picked the right stocks."

Figure 11.1 — The three layers of performance evaluation stack from a raw number to a judgment about skill.



The intellectual honesty of the whole exercise depends on comparing like with like: the same currency, the same period, the same risk-free rate, and a benchmark that genuinely represents the manager's opportunity set.

3. Why / How It Works

Why time-weighted and money-weighted returns diverge

Imagine a manager who is genuinely skilled and compounds capital at a steady rate. If a client happens to pour in a large deposit right before a bad quarter and withdraw before a good one, the client's *realized* dollar outcome will be worse than the manager's *strategy* return — through no fault of the manager. Conversely, a lucky client whose deposits land before rallies will earn more than the strategy itself.

The **time-weighted return (TWR)** neutralizes cash-flow timing by breaking the period at each cash flow, computing a sub-period return on each, and geometrically chaining them. Because it removes the size and timing of external flows, TWR measures *the manager's decisions* and is the GIPS (Global Investment Performance Standards) mandated basis for comparing managers.

The **money-weighted return (MWR)**, which is just the internal rate of return (IRR) of the portfolio's cash flows, *does* embed timing. It answers "what did the *investor* actually earn on the money they had at work?" It

is the right measure when the manager (e.g., a private-equity GP calling capital) controls the timing, or when you want the client's true experience.

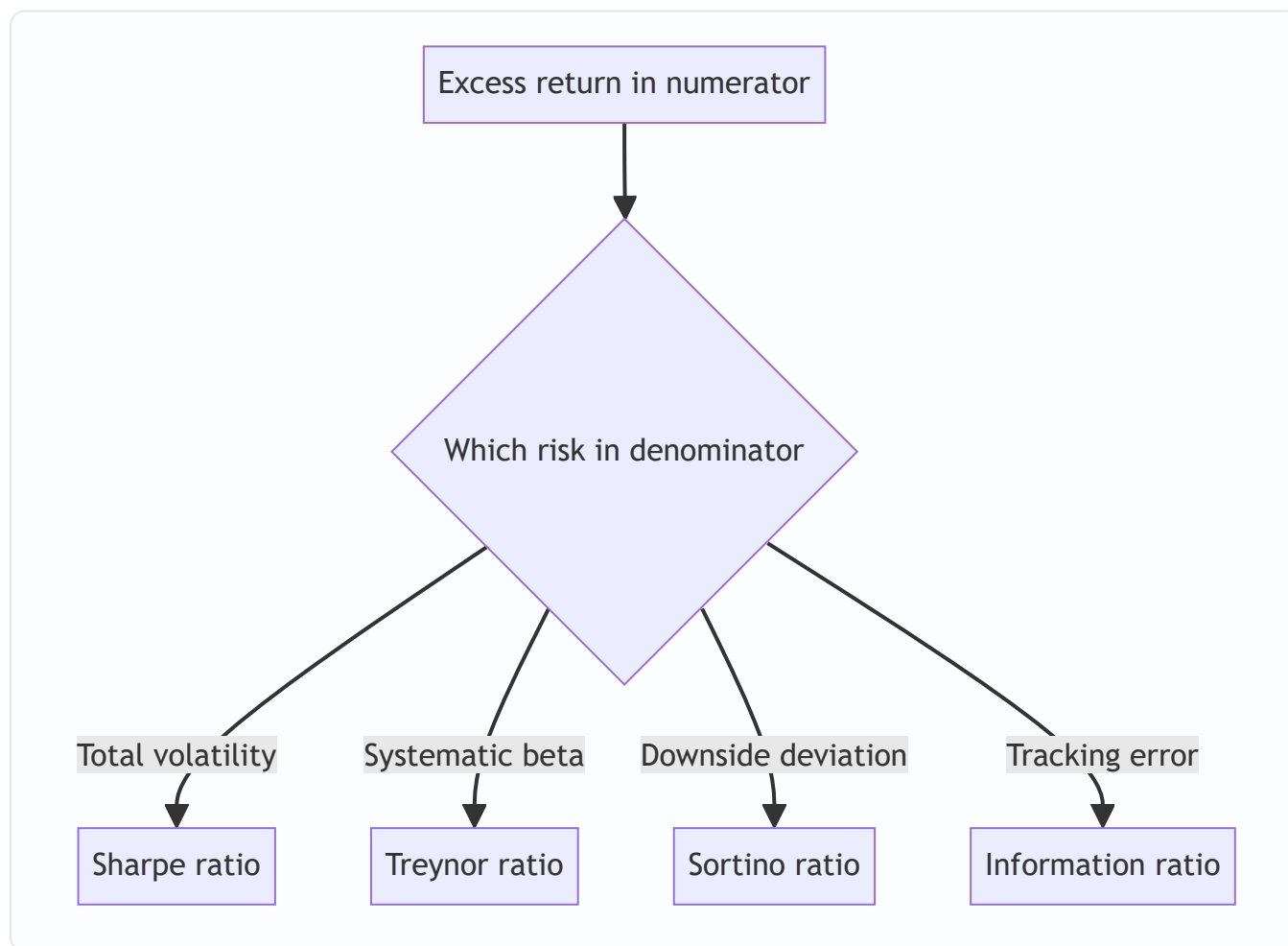
Why you must divide by risk

Financial theory says higher expected return is compensation for bearing risk. So the natural way to detect *skill* is to ask: per unit of risk taken, how much return did you deliver? A manager who doubles her return by doubling her leverage has added no skill — the risk-adjusted ratio stays flat. Only if she raises return *without* proportionally raising risk (or cuts risk without cutting return) does the ratio improve. That is the signature of genuine value-add.

Why the choice of risk denominator matters

- If the portfolio is an investor's *entire* wealth, total risk (standard deviation) is what hurts them → **Sharpe**.
- If the portfolio is *one sleeve* of a diversified whole, only its *systematic* (non-diversifiable) risk matters → **Treynor** and **Jensen's alpha** use beta.
- If investors only fear *downside* (they do not mind upside volatility) → **Sortino** uses downside deviation.
- If the manager is judged against a benchmark and we care about *active* risk → **Information Ratio** uses tracking error.

Figure 11.2 — The risk denominator you choose defines which ratio you compute and which question it answers.



4. Full Content — Formulas and Models

4.1 Return measurement

Holding-period return (single period, no flows):

$$R = \frac{V_1 - V_0 + D}{V_0}$$

where V_0 , V_1 are beginning and ending values and D is income (dividends/coupons).

Time-weighted return (TWR). Break the period into sub-periods at each external cash flow. For each sub-period compute:

$$r_t = \frac{V_{\text{end}} - V_{\text{begin}} - CF_t}{V_{\text{begin}}}$$

(placing the flow appropriately; the clean version values the portfolio just before each flow). Then chain geometrically:

$$1 + R_{\text{TWR}} = (1 + r_1)(1 + r_2) \cdots (1 + r_n)$$

Annualize by $(1 + R_{\text{TWR}})^{1/\text{years}} - 1$ for multi-year.

Money-weighted return (MWR / IRR). Solve for the rate r that sets the net present value of all flows to zero:

$$V_0 = \sum_{t=1}^n \frac{-CF_t}{(1+r)^t} + \frac{V_n}{(1+r)^n}$$

or equivalently the r satisfying $\sum \frac{CF_t}{(1+r)^t} = 0$ with the ending value as a final inflow and the beginning value plus contributions as outflows.

Feature	Time-Weighted (TWR)	Money-Weighted (MWR / IRR)
Removes cash-flow timing	Yes	No
Measures	Manager's decisions	Investor's actual \$ experience
Standard for	GIPS, comparing managers	PE/VC, client-level reporting
Sensitive to large flows	No	Very
Computation	Geometric chaining	Solve polynomial for IRR

Arithmetic vs geometric mean of returns. Arithmetic mean $\bar{R} = \frac{1}{n} \sum R_i$ overstates realized growth; geometric mean $\left(\prod (1 + R_i) \right)^{1/n} - 1$ is the actual compounded rate. Geometric $\leq$ arithmetic always, with the gap widening with volatility (roughly $R_g \approx \bar{R} - \frac{1}{2} \sigma^2$).

4.2 Risk-adjusted performance measures

Let R_p = portfolio return, R_f = risk-free rate, R_m = market/benchmark return, σ_p = portfolio standard deviation, β_p = portfolio beta, $\sigma_{p,d}$ = downside deviation, and $\sigma_{(R_p - R_b)}$ = tracking error.

Measure	Formula	Risk in denominator	Best used when
Sharpe ratio	$S = \frac{R_p - R_f}{\sigma_p}$	Total risk (σ)	Portfolio is the investor's whole wealth
Treynor ratio	$T = \frac{R_p - R_f}{\beta_p}$	Systematic risk (β)	Portfolio is one part of a diversified whole
Jensen's alpha	$\alpha = R_p - [R_f + \beta_p(R_m - R_f)]$	(intercept vs CAPM)	Testing skill vs the CAPM benchmark line
Information ratio	$IR = \frac{R_p - R_b}{\sigma_{(R_p - R_b)}}$	Tracking error (active risk)	Judging active manager vs a benchmark
Sortino ratio	$\text{Sortino} = \frac{R_p - R_{\text{MAR}}}{\sigma_{p,d}}$	Downside deviation	Investor only cares about downside
M² (Modigliani)	$M^2 = R_f + S_p \cdot \sigma_m$	Total risk, rescaled	Expressing Sharpe as a return number

Downside deviation (for Sortino), against a minimum acceptable return (MAR):

$$\sigma_{p,d} = \sqrt{\frac{1}{n} \sum_{i=1}^n \max[\min(R_i - R_{\text{MAR}}, 0), 0]^2}$$

Note we sum squared *shortfalls* only, but still divide by total n .

M² (M-squared) restates the Sharpe ratio as the return a portfolio would have earned if levered/de-levered to the market's volatility, making it directly comparable to the market return in percentage points:

$$M^2 = R_f + \frac{R_p - R_f}{\sigma_p} \cdot \sigma_m = R_f + S_p \cdot \sigma_m$$

Appraisal ratio (a close cousin of IR using CAPM residuals): $\frac{\alpha}{\sigma(\epsilon)}$, α over non-systematic (residual) risk.

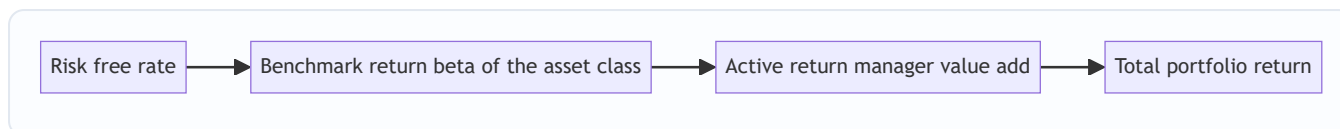
4.3 Sharpe vs Treynor — when do they disagree?

Both share the numerator $R_p - R_f$. They differ only in the denominator: total risk vs systematic risk. For a **fully diversified** portfolio, unsystematic risk ≈ 0 , so $\sigma_p \approx \beta_p \sigma_m$, and the two rankings agree. For a **poorly diversified** portfolio carrying idiosyncratic risk, Sharpe (which penalizes total risk) will rank it lower than Treynor (which ignores diversifiable risk). *A large gap between a portfolio's Sharpe and Treynor rankings is itself a diagnostic — it signals the portfolio is under-diversified.*

4.4 Benchmarking

A benchmark is the passive, investable alternative the client could have held instead of paying the manager. A valid benchmark must be **SAMURAI** (a standard CFA mnemonic): **S**pecified in advance, **A**ppropriate to the style, **M**easurable, **U**nambiguous, **R**eflective of current investment opinions, **A**ccountable (manager accepts it), and **I**vestable. Common failures: style drift (a "large-cap" fund holding mid-caps), survivorship bias in the index, and using a total-return vs price index inconsistently.

Figure 11.3 — Value-add is the gap between the portfolio and a fair benchmark, not the raw return itself.



Active return = $R_p - R_b$. **Active risk** (tracking error) = standard deviation of $(R_p - R_b)$. The Information Ratio is the reward-to-active-risk ratio and is the single most-watched number for active managers; a sustained IR above ~ 0.5 is good, above ~ 1.0 is exceptional.

4.5 Performance attribution — Brinson-Hood-Beebower (BHB)

Attribution decomposes active return into the *decisions* that generated it. The classic Brinson model splits total active return into three effects across sectors/asset classes i :

Let $w_{\{p,i\}}$, $w_{\{b,i\}}$ = portfolio and benchmark weights in segment i ; $R_{\{p,i\}}$, $R_{\{b,i\}}$ = portfolio and benchmark returns *within* segment i ; R_b = total benchmark return.

Allocation effect (did over/underweighting segments help?):

$$A_i = (w_{\{p,i\}} - w_{\{b,i\}})(R_{\{b,i\}} - R_b)$$

Selection effect (did picking better securities within a segment help?):

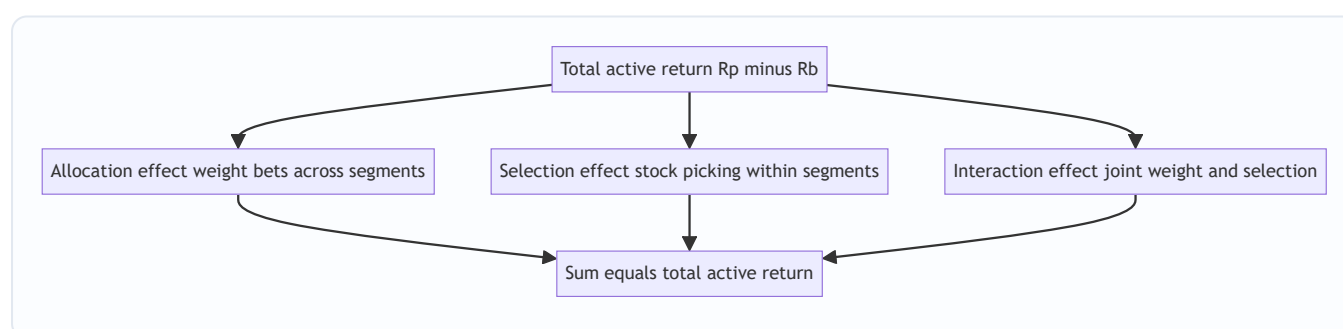
$$S_i = w_{\{b,i\}}(R_{\{p,i\}} - R_{\{b,i\}})$$

Interaction effect (joint term — overweighting a segment where you also picked well):

$$I_i = (w_{\{p,i\}} - w_{\{b,i\}})(R_{\{p,i\}} - R_{\{b,i\}})$$

Total active return $= \sum_i (A_i + S_i + I_i) = R_p - R_b$. Many practitioners fold interaction into selection, giving a two-factor allocation/selection split.

Figure 11.4 — Brinson attribution splits the portfolio-minus-benchmark gap into decision-level effects that sum back to total active return.



The intuition of the allocation term is subtle: you get credit for overweighting a segment only if that segment *beat the overall benchmark* ($R_{\{b,i\}} - R_b > 0$). Overweighting a segment that merely did fine but underperformed the total benchmark is a *negative* allocation decision.

4.6 Evaluating a manager — beyond a single number

- **Statistical significance.** Alpha and IR are noisy. The t-statistic of alpha $\approx IR \times \sqrt{\text{years}}$. To be 95% confident ($t \approx 2$) that an IR of 0.5 is real, you need roughly $(2/0.5)^2 = 16$ years of data. Three-year track records prove almost nothing.
- **Persistence.** Does outperformance repeat, or is it hot-hand luck? Most academic evidence finds weak persistence in equity funds after fees.

- **Style consistency.** Return-based style analysis (Sharpe's method) regresses fund returns on style indices to detect drift.
- **Risk discipline.** Max drawdown, downside capture, and whether volatility stayed within mandate.
- **Capture ratios.** Up-capture / down-capture: what fraction of benchmark up-moves and down-moves the fund captured. A ratio of 110/85 (captures 110% of upside, only 85% of downside) is the ideal asymmetry.
- **Fees.** Everything above should be judged *net of fees*, which is what the client keeps.

5. Worked Examples

Example 1 — TWR vs MWR, and why they differ

A fund starts the year at ₹100. Mid-year (exactly 6 months in) the client *adds* ₹50 immediately after the fund's value had risen to ₹110. The fund ends the year at ₹176.

Step 1 — Sub-period returns for TWR.

- H1: value went ₹100 → ₹110 before the flow. $r_1 = 110/100 - 1 = 10.0\%$
- The ₹50 deposit brings the base to ₹110 + ₹50 = ₹160 at the start of H2.
- H2: value went ₹160 → ₹176. $r_2 = 176/160 - 1 = 10.0\%$

Step 2 — Chain geometrically.

$$R_{\text{TWR}} = (1.10)(1.10) - 1 = 21.0\%$$

Step 3 — MWR (IRR). Cash flows from the *investor's* perspective (outflows negative): −100 at $t=0$, −50 at $t=0.5$, +176 at $t=1$. Solve

$$100 + \frac{50}{(1+r)^{0.5}} = \frac{176}{(1+r)^1}$$

Try $r = 21\%$: LHS = $100 + 50/1.1 = 100 + 45.45 = 145.45$; RHS = $176/1.21 = 145.45$. They match, so **MWR ≈ 21.0%** too.

Reconciliation / interpretation. Here TWR = MWR because the two sub-period returns were *identical* (10% each), so the timing of the deposit didn't matter. Now change H2: suppose instead the fund *fell* in H2 so it ended at ₹144 (a −10% H2). Then $R_{\text{TWR}} = (1.10)(0.90) - 1 = -1.0\%$. But the MWR would be dragged *down more heavily* because the large ₹50 deposit was exposed to the bad H2. Solving $100 + 50/(1+r)^{0.5} = 144/(1+r)$: try $r = -4\%$: LHS = $100 + 50/0.9798 = 100 + 51.03 = 151.03$; RHS = $144/0.96 = 150.0$ — close, MWR ≈ −3.6%, *worse* than the −1.0% TWR. The extra money at work during the loss hurt the investor's dollar return even though the manager's decision-return was only −1%. **That gap — MWR below TWR when big money arrives before bad periods — is exactly what TWR is designed to strip out.**

Example 2 — The full risk-adjusted suite, computed and cross-checked

Given for one year: $R_p = 18\%$, $R_f = 6\%$, $R_m = 13\%$, $\sigma_p = 20\%$, $\sigma_m = 15\%$, $\beta_p = 1.2$.

Sharpe: $S_p = \frac{0.18 - 0.06}{0.20} = \frac{0.12}{0.20} = 0.60$. Market Sharpe = $\frac{0.13 - 0.06}{0.15} = \frac{0.07}{0.15} = 0.467$. The portfolio beats the market on a total-risk basis ($0.60 > 0.467$). ✓

Treynor: $T_p = \frac{0.18 - 0.06}{1.2} = \frac{0.12}{1.2} = 0.10$ (i.e., 10% excess per unit beta). Market Treynor = $\frac{0.07}{1.0} = 0.07$. Portfolio wins on systematic-risk basis too ($0.10 > 0.07$). ✓

Jensen's alpha: CAPM-required return $R_f + \beta_p(R_m - R_f) = 0.06 + 1.2(0.13 - 0.06) = 0.06 + 1.2(0.07) = 0.06 + 0.084 = 0.144 = 14.4\%$. Alpha $= 18\% - 14.4\% = +3.6\%$. Positive alpha confirms genuine outperformance versus the CAPM line. ✓

Cross-check (alpha ↔ Treynor). Alpha and Treynor must tell the same story. Treynor gap over the market $= \$0.10 - 0.07 = 0.03$ per unit beta; multiplied by $\beta_p = 1.2$ gives $\$0.036 = 3.6\%$ — exactly the Jensen alpha. ✓ *This identity ($\alpha = \beta_p (T_p - T_m)$) is a favourite interview gotcha and confirms our arithmetic.*

M²: $M^2 = R_f + S_p \cdot \sigma_m = 0.06 + 0.60 \cdot 0.15 = 0.06 + 0.09 = 0.15 = 15\%$. The portfolio, de-levered to the market's 15% volatility, would have returned 15% vs the market's actual 13% — an M² spread of **+2%**. Consistent with the positive Sharpe advantage. ✓

Information ratio (add data): suppose benchmark return $R_b = 13\%$ and tracking error $\sigma_{(R_p - R_b)} = 5\%$. Active return $= 18\% - 13\% = 5\%$. $IR = 5\% / 5\% = 1.0$ — an excellent active manager. ✓

Every measure agrees the manager added value; they simply quantify it against different risk yardsticks. The consistency is the self-check.

Example 3 — Sortino ratio from monthly data

A fund's excess-over-MAR monthly returns (MAR = 0%) over 8 months: **+4%, −3%, +2%, −5%, +6%, −1%, +3%, +2%**. Mean monthly return $= (4-3+2-5+6-1+3+2)/8 = 8/8 = +1.0\%$.

Downside deviation — square only the negative months, divide by total $n = 8$:

- Negatives: −3, −5, −1 → squares 9, 25, 1 → sum = 35.
- $\sigma_d = \sqrt{35/8} = \sqrt{4.375} = 2.092\%$ per month.

Sortino (monthly) $= \frac{1.0\% - 0\%}{2.092\%} = 0.478$.

Compare to Sharpe on the same data. Full standard deviation: mean 1.0%; deviations squared for all 8 months: $(3)^2 + (-4)^2 + (1)^2 + (-6)^2 + (5)^2 + (-2)^2 + (2)^2 + (1)^2 = 9 + 16 + 1 + 36 + 25 + 4 + 4 + 1 = 96$; sample variance $= 96/7 = 13.71$; $\sigma = 3.70\%$. Sharpe (excess mean/ σ , $R_f \approx 0$) $= 1.0/3.70 = 0.270$.

Interpretation. Sortino (0.478) is nearly double the Sharpe (0.270) because roughly two-thirds of this fund's volatility came from *upside* months (+4, +6, +3, +2, +2), which Sharpe penalizes but Sortino ignores. For an investor who only fears losses, Sortino paints the more relevant — and more flattering — picture. The gap is the signature of a positively-skewed return stream. ✓

Example 4 — Brinson attribution (allocation vs selection)

A two-sector portfolio (Tech and Financials) vs its benchmark. All returns are for the period.

Sector	w_p	w_b	$R_{p,i}$	$R_{b,i}$
Tech	70%	50%	15%	12%
Financials	30%	50%	5%	8%

Total benchmark return $R_b = 0.50(12\%) + 0.50(8\%) = 6\% + 4\% = 10\%$. **Total portfolio return** $R_p = 0.70(15\%) + 0.30(5\%) = 10.5\% + 1.5\% = 12.0\%$. **Total active return** $= 12.0\% - 10.0\% = +2.0\%$. This is the number our effects must sum to.

Allocation effect $A_i = (w_p - w_b)(R_{\{b,i\}} - R_b)$ \$:

- Tech: $\$(0.70 - 0.50)(12\% - 10\%) = 0.20 \times 2\% = +0.40\%$.
- Financials: $\$(0.30 - 0.50)(8\% - 10\%) = (-0.20)(-2\%) = +0.40\%$.
- **Allocation total = +0.80%.** (Overweighting Tech, which beat the benchmark, and underweighting Financials, which lagged — both correct calls.)

Selection effect $S_i = w_b(R_{\{p,i\}} - R_{\{b,i\}})$ \$:

- Tech: $\$0.50(15\% - 12\%) = 0.50 \times 3\% = +1.50\%$.
- Financials: $\$0.50(5\% - 8\%) = 0.50 \times (-3\%) = -1.50\%$.
- **Selection total = 0.00%.** (Great Tech picks exactly offset by poor Financials picks.)

Interaction effect $I_i = (w_p - w_b)(R_{\{p,i\}} - R_{\{b,i\}})$ \$:

- Tech: $\$(0.20)(3\%) = +0.60\%$.
- Financials: $\$(-0.20)(-3\%) = +0.60\%$.
- **Interaction total = +1.20%.**

Reconciliation: $0.80\% + 0.00\% + 1.20\% = +2.00\%$ = total active return. ✓

Interpretation. The headline is that this manager's edge was almost entirely *allocation* (getting the sector weights right) plus a favourable interaction — she overweighted the sector where she *also* picked well (Tech) and underweighted the sector where she picked badly (Financials). Pure selection skill netted to zero. If you fold interaction into selection (the common two-factor convention), you'd report Allocation +0.80% and Selection +1.20%, still summing to +2.0%. The story an allocator hears — "this is an asset-allocation shop, not a pure stock-picker" — changes hiring decisions.

6. Connections

- **CAPM and the SML (Ch. on asset pricing).** Jensen's alpha is literally the vertical distance of the portfolio from the Security Market Line; Treynor is the slope of the line from R_f to the portfolio in beta space. Sharpe is the slope in total-risk (CML) space. The whole risk-adjusted toolkit is CAPM read backwards.
- **Modern Portfolio Theory.** The Sharpe ratio is the gradient of the Capital Market Line; maximizing Sharpe is equivalent to finding the tangency portfolio.
- **Factor models (Fama-French, Carhart).** "Alpha" from single-factor CAPM often shrinks to zero once you control for size, value, and momentum factors — much of what looks like skill is factor exposure (smart beta). Multi-factor alpha is the modern skill test.
- **Fixed income (Ch. on bonds).** Attribution there decomposes return into duration, curve, and spread effects — the same allocation/selection philosophy applied to yield-curve positioning.
- **Behavioral finance.** MWR-vs-TWR gaps quantify the "behavior gap" — the return investors lose by mistiming their own contributions and redemptions (DALBAR studies).
- **Fund rating agencies.** Morningstar's star ratings and risk-adjusted return metrics are institutionalized versions of these ratios.

7. Key Terms

- **Time-weighted return (TWR):** Return that removes the effect of cash-flow timing; the standard for comparing managers.

- **Money-weighted return (MWR / IRR):** The internal rate of return of the portfolio's cash flows; reflects the investor's actual dollar experience.
- **Sharpe ratio:** Excess return per unit of *total* risk (standard deviation).
- **Treynor ratio:** Excess return per unit of *systematic* risk (beta).
- **Jensen's alpha:** Return in excess of the CAPM-required return; the SML intercept.
- **Information ratio (IR):** Active return divided by tracking error; the key active-management metric.
- **Sortino ratio:** Excess return over a minimum acceptable return, divided by *downside* deviation.
- **M² (Modigliani-squared):** Sharpe ratio re-expressed as a benchmark-comparable percentage return.
- **Tracking error (active risk):** Standard deviation of (portfolio – benchmark) return.
- **Benchmark:** The passive, investable alternative; must satisfy SAMURAI properties.
- **Allocation effect:** Active return from over/underweighting segments relative to benchmark.
- **Selection effect:** Active return from picking better-than-benchmark securities within segments.
- **Interaction effect:** Joint term combining weighting and selection decisions.
- **Downside deviation:** Standard deviation computed using only returns below the MAR.
- **Capture ratio:** Fraction of benchmark up-moves (up-capture) or down-moves (down-capture) the portfolio captured.

8. Common Confusions

- **"Higher return = better manager."** No — without the risk denominator and a benchmark, a raw return says nothing about skill. This is the single most common amateur error.
- **TWR vs MWR direction.** MWR is *not* "always higher" or "always lower." It exceeds TWR when the investor's large flows happen to precede good periods, and falls below when they precede bad periods. It equals TWR when there are no flows or when sub-period returns are equal.
- **Sharpe vs Treynor confusion.** Same numerator, different denominator (σ vs β). Use Sharpe for a standalone/total portfolio, Treynor for a sub-portfolio within a diversified whole. They agree only when the portfolio is well diversified.
- **Alpha $\neq$ excess return.** Alpha is excess return *over the CAPM-required (beta-adjusted) return*, not over R_f or over the benchmark. A fund that returned 3% above its benchmark but had beta 1.5 in an up market may have *negative* alpha.
- **Sortino denominator divisor.** Downside deviation squares only the negative shortfalls but still divides by the *total* number of observations n , not the number of negatives. Dividing by the count of negatives is a frequent mistake.
- **Selection vs allocation credit.** You get *allocation* credit for overweighting a segment only if that segment beat the *total* benchmark, not merely if the segment had a positive return. And selection is measured at *benchmark* weights, so it isolates pure picking skill from sizing.
- **Interaction effect sign.** Interaction is often lumped into selection; make sure your effects sum to total active return before interpreting — a common source of "my attribution doesn't reconcile."
- **Geometric vs arithmetic mean.** Reporting the arithmetic average of periodic returns overstates the compounded growth an investor actually experienced, especially for volatile funds.
- **Ignoring statistical significance.** A three-year IR of 0.8 sounds great but is statistically indistinguishable from zero; skill needs a long track record to prove.

9. Recap

Performance evaluation converts a raw return into a judgment about skill through three stacked layers. **First**, measure the return correctly: use time-weighted return to judge the *manager* (it strips cash-flow timing) and money-weighted return to capture the *investor's* actual dollar outcome — the two diverge whenever large flows land before unusually good or bad periods. **Second**, adjust for risk by dividing excess return by the right risk denominator: total volatility gives Sharpe (whole-portfolio view), beta gives Treynor and — as an intercept — Jensen's alpha (systematic-risk view), downside deviation gives Sortino (loss-focused view), and tracking error gives the Information Ratio (active-management view). These measures are all CAPM read in reverse and, on clean data, corroborate one another — we verified in Example 2 that $\alpha = \beta(T_p - T_m)$ and that M^2 tells the same story as Sharpe. **Third**, attribute the active return to decisions via the Brinson model: allocation (weight bets across segments), selection (security picking within segments), and interaction, which must sum exactly to portfolio-minus-benchmark — as they did to +2.0% in Example 4. Finally, wrap all of it in judgment: benchmark validity (SAMURAI), statistical significance (skill needs many years), style consistency, capture ratios, and everything net of fees. The discipline is one sentence repeated at every layer: *a return means nothing except relative to the risk taken and a fair alternative*.

10. Quick-Reference / Interview Points

Formula flash-card:

Metric	Formula	One-line meaning
Sharpe	$(R_p - R_f) / \sigma_p$	Excess return per unit total risk
Treynor	$(R_p - R_f) / \beta_p$	Excess return per unit market risk
Jensen α	$R_p - [R_f + \beta_p(R_m - R_f)]$	Return above the SML
Info ratio	$(R_p - R_b) / TE$	Active return per unit active risk
Sortino	$(R_p - R_{\{MAR\}}) / \sigma_{\{down\}}$	Excess return per unit downside risk
M^2	$R_f + S_p \sigma_m$	Sharpe expressed as a % return
Allocation	$(w_p - w_b)(R_{\{b,i\}} - R_b)$	Weight-bet contribution
Selection	$w_b(R_{\{p,i\}} - R_{\{b,i\}})$	Stock-picking contribution

Interview-ready talking points:

- "Why TWR not MWR to rank managers?" Because TWR removes the timing of client flows the manager doesn't control; MWR is right only when the manager controls the cash flows (PE/VC) or you want the client's true experience.
- "Sharpe vs Treynor — which and when?" Same numerator; Sharpe uses σ (use for a standalone total portfolio), Treynor uses β (use for a sleeve inside a diversified book). A big ranking gap between them flags poor diversification.
- "Is alpha the same as beating the benchmark?" No — alpha is return above the *beta-adjusted* CAPM requirement. High beta in a bull market can produce benchmark outperformance with *negative* alpha.
- "How much data to trust an IR?" t-stat $\approx IR \times \sqrt{\text{years}}$; you need roughly $(2/IR)^2$ years for 95% confidence. An IR of 0.5 needs ~16 years — so short track records prove little.

- *"Allocation or selection — which is 'better'?"* Neither inherently; attribution just tells you *where* the edge is. But allocation skill is harder to sustain and often just factor timing, whereas repeatable selection alpha at benchmark weights is the cleaner signal of stock-picking skill.
- *"What kills a Sharpe ratio unfairly?"* Positive skew — Sharpe penalizes upside volatility. Report Sortino alongside for asymmetric or option-like strategies.
- *"First thing you check on a track record?"* The benchmark's validity (SAMURAI) and whether the numbers are net of fees, time-weighted, and long enough to be significant.
- *Key identity to quote:* $\alpha = \beta_p (T_p - T_m)$ and $\sigma^2 = R_f + S_p \sigma_m$ — knowing these signals you understand the measures are one coherent system, not disconnected formulas.

Q&A — Performance Measurement and Attribution

Practice bank for Chapter 11. Work each question before reading the answer. Returns are in percent unless stated; betas are unit-free. Every numerical answer is worked in full and reconciled against a cross-check.

Section A — Concept Checks

A1. What single sentence underlies the entire chapter? A return is only interpretable relative to (a) the risk taken to earn it and (b) a fair, investable alternative (the benchmark). Every measure here elaborates that one idea.

A2. Why distrust a raw return like "I made 22%"? Because it hides three things that decide whether it was skill: how much risk was taken, whether it beat a passive benchmark the client could have bought cheaply, and whether it was a repeatable decision or a single lucky bet. A 22% from a 6%-vol market-neutral book and a 22% from a 35%-vol leveraged momentum bet are not the same achievement.

A3. Define TWR and MWR in one line each. TWR (time-weighted return) breaks the period at each cash flow and geometrically chains the sub-period returns, stripping out flow timing — it measures the *manager's* decisions. MWR (money-weighted return) is the IRR of the portfolio's cash flows — it embeds flow timing and measures the *investor's* actual dollar experience.

A4. When does MWR fall below TWR, and when does it exceed it? MWR falls below TWR when the investor's large inflows land *before* bad periods (more money is exposed to the loss); it exceeds TWR when large inflows land before good periods. The two are equal when there are no external flows, or when every sub-period return is identical.

A5. Which return should GIPS-compliant managers report for comparison, and why? TWR. It removes the timing and size of client cash flows the manager does not control, so it isolates the manager's investment decisions and makes managers comparable on a like-for-like basis.

A6. Why must you divide return by risk to detect skill? Because higher return is the market's payment for bearing risk. Doubling leverage doubles return but adds no skill — the risk-adjusted ratio is unchanged. Skill shows up only as *more return per unit of risk*: raising return without proportionally raising risk, or cutting risk without cutting return.

A7. Match each ratio to the question it answers. Sharpe — excess return per unit of *total* risk (σ); use when the portfolio is the investor's whole wealth. Treynor — excess return per unit of *systematic* risk (β); use for one sleeve of a diversified whole. Sortino — excess return per unit of *downside* deviation; use when only losses matter. Information ratio — active return per unit of *tracking error*; use to judge an active manager against a benchmark.

A8. Sharpe and Treynor share a numerator. When do their rankings disagree, and what does the gap diagnose? They share $R_p - R_f$ and differ only in the denominator (σ vs β). For a fully diversified portfolio $\sigma_p \approx \beta_p \sigma_m$, so rankings agree. For an under-diversified portfolio carrying idiosyncratic risk, Sharpe (which penalizes total risk) ranks it below Treynor (which ignores diversifiable risk). A large Sharpe-vs-Treynor ranking gap is itself a diagnostic of poor diversification.

A9. Alpha is *not* the same as beating the benchmark — explain. Jensen's alpha is return above the *beta-adjusted CAPM requirement*, $\alpha = R_p - [R_f + \beta_p(R_m - R_f)]$, not return above R_f or above the

benchmark. A fund with beta 1.5 in a strong bull market can beat its benchmark yet still post *negative* alpha, because CAPM required an even bigger return for that much systematic risk.

A10. State the Sortino denominator carefully — where do people go wrong? Downside deviation squares only the shortfalls below the MAR, but still divides by the *total* number of observations n , not the count of negative months: $\sigma_{p,d} = \sqrt{\frac{1}{n} \sum [\min(R_i - R_{\{MAR\}}, 0)]^2}$. Dividing by the number of negatives is the classic error.

A11. What does the SAMURAI mnemonic guarantee about a benchmark? A valid benchmark is Specified in advance, Appropriate to the style, Measurable, Unambiguous, Reflective of current opinions, Accountable (the manager accepts it), and Investable. It ensures the benchmark is a genuine passive alternative and not a moving or unbeatable yardstick.

A12. In Brinson attribution, when do you earn positive allocation credit? You earn allocation credit for overweighting a segment only if that segment *beat the total benchmark* ($R_{b,i} - R_b > 0$), or for underweighting a segment that *lagged* it. Overweighting a segment that had a positive return but still trailed the overall benchmark is a *negative* allocation call.

Section B — Numerical Problems (full working)

B1. TWR vs MWR with a mid-year deposit. A fund starts at ₹200. Exactly 6 months in, after its value has risen to ₹230, the client adds ₹100. The fund ends the year at ₹363. Find TWR and MWR.

- H1 sub-period: ₹200 → ₹230 before the flow. $r_1 = 230/200 - 1 = 15.0\%$.
- Deposit lifts the base to ₹230 + ₹100 = ₹330 to start H2.
- H2 sub-period: ₹330 → ₹363. $r_2 = 363/330 - 1 = 10.0\%$.
- $R_{\{TWR\}} = (1.15)(1.10) - 1 = 1.265 - 1 = 26.5\%$.

MWR (IRR) from the investor's view (outflows negative): −200 at $t=0$, −100 at $t=0.5$, +363 at $t=1$. Solve $200 + \frac{100}{(1+r)^{0.5}} = \frac{363}{1+r}$.

- Try $r = 25.5\%$: $\$(1.255)^{0.5} = 1.1203$, so LHS $\$ = 200 + 100/1.1203 = 200 + 89.26 = 289.26$; RHS $\$ = 363/1.255 = 289.24$. Match → **MWR ≈ 25.5%**.

Reconcile: MWR (25.5%) < TWR (26.5%) because the large ₹100 deposit was exposed to the weaker H2 (10% vs H1's 15%). The extra money at work during the softer half dragged the investor's dollar return below the manager's decision return — exactly the timing effect TWR is built to strip out. ✓

B2. The full risk-adjusted suite, cross-checked. For one year: $R_p = 16\%$, $R_f = 5\%$, $R_m = 11\%$, $\sigma_p = 18\%$, $\sigma_m = 14\%$, $\beta_p = 1.1$.

- **Sharpe:** $S_p = (16-5)/18 = 11/18 = 0.611$. Market Sharpe $= (11-5)/14 = 6/14 = 0.429$. Portfolio wins on total risk ($0.611 > 0.429$). ✓
- **Treynor:** $T_p = (16-5)/1.1 = 11/1.1 = 10.0$ (10% excess per unit beta). Market Treynor $= 6/1.0 = 6.0$. Portfolio wins on systematic risk too. ✓
- **Jensen's alpha:** CAPM required $\$ = 5 + 1.1(11-5) = 5 + 6.6 = 11.6\%$. $\alpha = 16 - 11.6 = +4.4\%$. ✓
- **Cross-check ($\alpha \leftrightarrow$ Treynor):** $\alpha = \beta_p(T_p - T_m) = 1.1(10 - 6) = 1.1 \times 4 = 4.4\%$ — matches the Jensen alpha exactly. ✓
- **M²:** $M^2 = R_f + S_p \sigma_m = 5 + 0.611 \times 14 = 5 + 8.56 = 13.56\%$. De-levered to the market's 14% vol the portfolio returns 13.56% vs the market's 11% — an **M² spread of +2.56%**, consistent with

the Sharpe advantage. ✓

Every measure agrees the manager added value; they just quantify it against different risk yardsticks.

B3. Sortino vs Sharpe from monthly data. Monthly returns in excess of MAR = 0% over 8 months: **+5, -2, +3, -4, +1, -6, +4, +3.**

- Mean $\bar{R} = (5 - 2 + 3 - 4 + 1 - 6 + 4 + 3) / 8 = 4 / 8 = 0.5\%$.
- **Downside deviation:** negatives -2, -4, -6 → squares 4, 16, 36 → sum 56. $\sigma_d = \sqrt{56/8} = \sqrt{7} = 2.646\%$.
- **Sortino** $\bar{S} = (0.5 - 0) / 2.646 = 0.189\%$.
- **Sharpe (same data):** deviations from mean 0.5 are 4.5, -2.5, 2.5, -4.5, 0.5, -6.5, 3.5, 2.5; squares 20.25, 6.25, 6.25, 20.25, 0.25, 42.25, 12.25, 6.25 → sum 114; sample variance $s^2 = 114 / 7 = 16.29\%$; $\sigma = 4.036\%$. Sharpe $\bar{S} = 0.5 / 4.036 = 0.124\%$.

Reconcile: Sortino (0.189) exceeds Sharpe (0.124) because much of this fund's volatility came from *upside* months (+5, +3, +4, +3), which Sharpe penalizes but Sortino ignores. The gap is the signature of a positively-skewed return stream. ✓

B4. Brinson attribution — three segments that must reconcile.

Segment	w_p	w_b	$R_{p,i}$	$R_{b,i}$
Equity	60%	50%	14%	12%
Bonds	30%	40%	6%	5%
Cash	10%	10%	3%	3%

- $R_b = 0.50(12) + 0.40(5) + 0.10(3) = 6 + 2 + 0.3 = 8.3\%$.
- $R_p = 0.60(14) + 0.30(6) + 0.10(3) = 8.4 + 1.8 + 0.3 = 10.5\%$.
- **Total active return** $\bar{A} = 10.5 - 8.3 = +2.2\%$ (the target our effects must sum to).

Allocation $A_i = (w_p - w_b)(R_{b,i} - R_b)$: - Equity: $(0.10)(12 - 8.3) = 0.10 \times 3.7 = +0.37\%$ - Bonds: $(-0.10)(5 - 8.3) = (-0.10)(-3.3) = +0.33\%$ - Cash: $(0)(3 - 8.3) = 0\%$ - **Allocation total = +0.70%.**

Selection $S_i = w_b(R_{p,i} - R_{b,i})$: - Equity: $0.50(14 - 12) = +1.00\%$ - Bonds: $0.40(6 - 5) = +0.40\%$ - Cash: $0.10(3 - 3) = 0\%$ - **Selection total = +1.40%.**

Interaction $I_i = (w_p - w_b)(R_{p,i} - R_{b,i})$: - Equity: $(0.10)(2) = +0.20\%$ - Bonds: $(-0.10)(1) = -0.10\%$ - Cash: 0% - **Interaction total = +0.10%.**

Reconcile: $0.70 + 1.40 + 0.10 = +2.20\%$ = total active return. ✓ The edge here is mostly *selection* (good picks in both equity and bonds). Folding interaction into selection (the two-factor convention) gives Allocation +0.70%, Selection +1.50%, still summing to +2.20%.

B5. Is the track record statistically significant? A manager posts active return 3%, tracking error 4%, over 5 years.

- $IR = 3/4 = 0.75\%$.
- t-stat $\approx IR \times \sqrt{\text{years}} = 0.75 \times \sqrt{5} = 0.75 \times 2.236 = 1.68$.
- Years needed for $t \approx 2$: $(2/IR)^2 = (2/0.75)^2 = 2.667^2 = 7.1$ years.

Reconcile: $t = 1.68 < 2$, so even a healthy IR of 0.75 over 5 years is *not* significant at 95% — you need about 7 years. Short track records prove little; this is why hiring on 3-year numbers rewards noise. ✓

B6. Arithmetic vs geometric mean. Annual returns: +30%, −20%, +30%, −20%.

- Arithmetic mean $\$ = (30 - 20 + 30 - 20)/4 = 20/4 = +5.0\%$.
- Geometric: $\$(1.30 \times 0.80 \times 1.30 \times 0.80)^{1/4} - 1 = (1.0816)^{1/4} - 1 = \sqrt[4]{1.0816} - 1 = 1.0400 - 1 = 0.0400$ → **geometric = +1.98%**.
- Cross-check via σ of the four returns = 25% (deviations ± 25 from mean 5), so $\sigma^2/2 = 0.25^2/2 = 0.03125$; $5\% - 3.125\% = 1.875\% \approx 1.98\%$. ✓

Reconcile: the arithmetic 5% overstates what an investor actually compounded (1.98%); the gap widens with volatility. Report the geometric mean for realized growth. ✓

Section C — Interview-Style Questions (model answers)

C1. "A manager reports 22%. What do you ask next?" Four things. First, is that time-weighted (comparable across managers) or money-weighted (the client's dollar experience)? Second, what was the volatility, beta, and max drawdown — 22% at 6% vol is very different from 22% at 35%. Third, what did the benchmark return; if the Nifty did 21%, she added almost nothing and is a closet indexer charging active fees. Fourth, is it net of fees, and over how long — a one-year number is statistically meaningless. Only after those does 22% become a judgment about skill.

C2. "Why use TWR rather than MWR to rank managers?" Because TWR removes the timing and size of client cash flows, which the manager doesn't control. If a client dumps money in right before a bad quarter, MWR punishes the manager for the client's timing. MWR is the *right* measure only when the manager controls the flows — a private-equity GP calling capital — or when you specifically want the client's realized dollar outcome. For like-for-like manager comparison, GIPS mandates TWR.

C3. "Sharpe or Treynor — which would you use and when?" Same numerator, different risk denominator. Use Sharpe when the portfolio is the investor's entire wealth, because total volatility is what actually hurts them. Use Treynor when the portfolio is one sleeve inside a diversified book, because only its systematic risk survives diversification and matters at the margin. Practically, if the two give very different rankings, that gap itself is telling me the portfolio is under-diversified — it's carrying idiosyncratic risk that Sharpe punishes and Treynor ignores.

C4. "A fund beat its benchmark by 3% — is that alpha?" Not necessarily. Alpha is return above the *beta-adjusted* CAPM requirement, not above the benchmark. If the fund ran a beta of 1.5 in a market that rose strongly, CAPM might have required, say, 5% of outperformance for that much systematic risk — so a 3% beat is actually *negative* alpha. It beat the benchmark by taking more market risk, not by skill. I'd compute $\alpha = R_p - [R_f + \beta_p(R_m - R_f)]$ before calling anything skill.

C5. "Why report a Sortino ratio alongside Sharpe?" Because Sharpe penalizes *all* volatility, including upside. For a positively-skewed or option-like strategy — one whose swings are mostly to the upside — Sharpe unfairly understates quality. Sortino divides only by downside deviation (volatility below a minimum acceptable return), so it matches what investors actually fear: losses. When a fund's Sortino is much higher than its Sharpe, that's the fingerprint of favourable skew, and the Sortino is the more relevant picture.

C6. "Walk me through Brinson attribution." It decomposes active return — portfolio minus benchmark — into the decisions that produced it. Allocation asks: did over/underweighting segments help? You get credit

only if the segment you overweighted beat the *total* benchmark. Selection asks: within each segment, did you pick better securities than the benchmark held? — measured at benchmark weights to isolate pure picking. Interaction is the joint term: overweighting a segment where you also picked well. The three sum exactly to total active return, and that reconciliation is the discipline — if they don't sum, the attribution is wrong. The split tells an allocator whether they're hiring an asset-allocation shop or a stock-picker.

C7. "How much track record do you need to trust an Information Ratio?" The t-stat of the IR is roughly IR times the square root of the number of years. To be 95% confident it's real you want $t \approx 2$, so you need about $(2/IR)^2$ years. An IR of 0.5 needs ~16 years; even a strong IR of 0.75 needs ~7. That's the uncomfortable truth about the industry: three-year track records, which drive most hiring and firing, are statistically indistinguishable from luck.

C8. "First thing you check on any track record?" Benchmark validity and hygiene. Is the benchmark SAMURAI — specified in advance, style-appropriate, investable, and one the manager actually accepts? Then: are the returns net of fees, time-weighted, and long enough to be significant? A great-looking Sharpe against a soft or drifting benchmark, gross of fees, over two years, tells me nothing. Get those foundations right before interpreting a single ratio.

Section D — MCQs (with reasoning)

D1. To compare two managers who each experienced different client cash flows, you should use: (a) MWR (b) TWR (c) IRR (d) arithmetic mean. **Answer: (b).** TWR strips out flow timing, isolating the managers' own decisions; MWR/IRR embed the flows the manager didn't control.

D2. MWR will be *below* TWR when: (a) there are no cash flows (b) large inflows precede a strong period (c) large inflows precede a weak period (d) returns are constant. **Answer: (c).** More money exposed to the loss drags the dollar-weighted return below the strategy return. Inflows before strong periods raise MWR above TWR; no flows or equal sub-period returns make them equal.

D3. The Sharpe and Treynor ratios differ only in: (a) the numerator (b) the risk-free rate (c) the risk in the denominator (d) the benchmark used. **Answer: (c).** Both use excess return $R_p - R_f$ in the numerator; Sharpe divides by total risk σ , Treynor by systematic risk β .

D4. Jensen's alpha measures return in excess of: (a) the risk-free rate (b) the benchmark (c) the CAPM-required (beta-adjusted) return (d) the market return. **Answer: (c).** $\alpha = R_p - [R_f + \beta_p(R_m - R_f)]$ — the vertical distance from the Security Market Line, not a simple benchmark beat.

D5. In computing downside deviation for the Sortino ratio, you divide the sum of squared shortfalls by: (a) the number of negative observations (b) the total number of observations n (c) $n-1$ negatives (d) the number of positive observations. **Answer: (b).** Only shortfalls below the MAR are squared, but the divisor is the *total* count n . Dividing by the number of negatives is the classic error.

D6. A large gap between a portfolio's Sharpe and Treynor rankings signals: (a) high fees (b) poor diversification (c) negative alpha (d) survivorship bias. **Answer: (b).** The gap means total risk (σ) and systematic risk (β) diverge, i.e. the portfolio carries meaningful idiosyncratic risk — it is under-diversified.

D7. In Brinson attribution, allocation effect for a segment is positive when you: (a) overweight any segment with a positive return (b) overweight a segment that beat the *total* benchmark (c) pick better stocks in the segment (d) match the benchmark weight. **Answer: (b).** $A_i = (w_p - w_b)(R_{\{i\}} - R_b)$ — credit comes

from over/underweighting relative to how the segment did versus the *overall* benchmark, not its absolute return.

D8. The Information Ratio is defined as: (a) $(R_p - R_f)/\sigma_p$ (b) $(R_p - R_f)/\beta_p$ (c) $(R_p - R_b)/\text{tracking error}$ (d) $R_f + S_p\sigma_m$. **Answer: (c).** Active return over active risk (tracking error). (a) is Sharpe, (b) is Treynor, (d) is M^2 .

D9. M^2 (Modigliani-squared) re-expresses which ratio as a percentage return? (a) Treynor (b) Sharpe (c) Information ratio (d) Sortino. **Answer: (b).** $M^2 = R_f + S_p\sigma_m$ restates the Sharpe ratio as the return the portfolio would earn if scaled to the market's volatility, making it directly comparable to the market return.

D10. A three-year Information Ratio of 0.8 is best described as: (a) definitive proof of skill (b) statistically strong (c) statistically indistinguishable from luck (d) evidence of negative alpha. **Answer: (c).** $t \approx 0.8 \times \sqrt{3} = 1.39$, well below 2; roughly $(2/0.8)^2 \approx 6$ years are needed for 95% confidence. Short records prove little.

D11. The geometric mean of a return series is: (a) always $\geq$ the arithmetic mean (b) always $\leq$ the arithmetic mean (c) equal to it (d) unrelated. **Answer: (b).** Geometric $\leq$ arithmetic always, with the gap $\approx \sigma^2/2$, widening with volatility. It is the true compounded growth rate.

D12. The identity $\alpha = \beta_p(T_p - T_m)$ serves primarily as: (a) a definition of Sharpe (b) a self-check linking Jensen's alpha and Treynor (c) the M^2 formula (d) the Sortino numerator. **Answer: (b).** Alpha equals beta times the Treynor advantage over the market — a consistency check confirming both live in systematic-risk (beta) space.

Chapter 12 — Market Efficiency and the Active-Passive Debate

1. The Problem / Need

Every rupee an investor allocates rests on a single, unspoken bet: *"I can identify a security whose price is wrong and profit when the price corrects."* An equity research analyst building a discounted-cash-flow model, a fund manager overweighting IT stocks, a retail investor buying on a "hot tip" — all of them are implicitly claiming that the market price is *not* the fair value, and that they can see the gap.

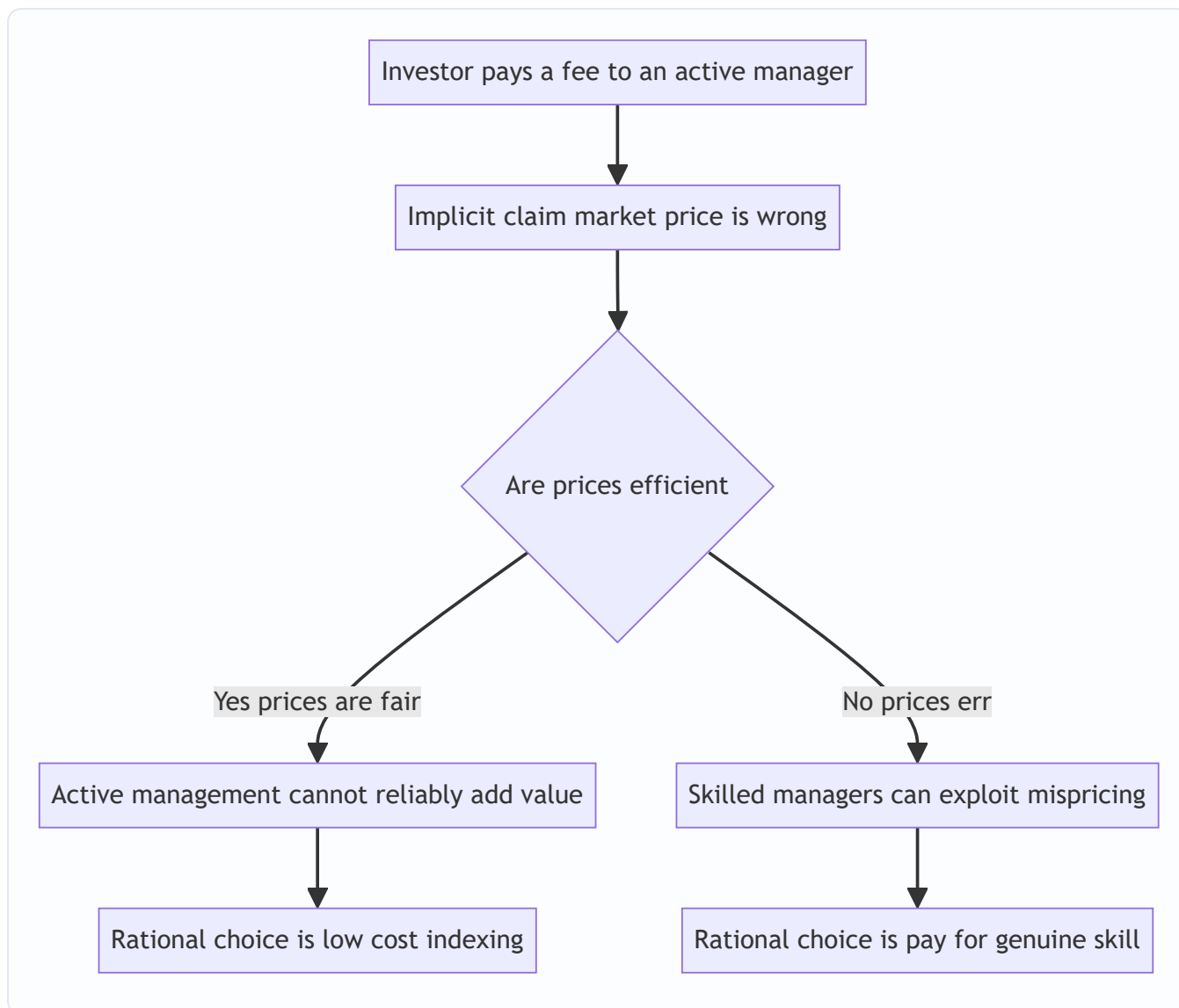
But this raises an uncomfortable question. If markets are populated by thousands of intelligent, well-resourced, profit-hungry participants who all analyse the same public information, why would prices be systematically wrong? And if prices are *right*, what exactly is an active manager being paid to do?

This is not an abstract debate. It decides:

- **Career economics.** Whether the equity research / AMC / portfolio management industry creates value or merely redistributes it (minus fees).
- **Client outcomes.** Whether a saver should pay 1.5–2.5% a year for active management or 0.05–0.20% for an index fund.
- **Fee structures.** Whether "2 and 20" hedge fund economics are justified.
- **Regulation and disclosure.** How much investors need protecting from mispricing versus from their own overtrading.

The **Efficient Market Hypothesis (EMH)** is the theory that formalises "prices are right." The **active–passive debate** is the practical war fought over how right they are. An interview-ready analyst must be able to argue *both* sides with data, because the honest answer — markets are *mostly* efficient, *sometimes* not, and *reliably beating* them net of fees is very hard — is exactly what separates a thoughtful professional from a salesperson.

Figure 12.1 — The central tension that market efficiency resolves.



2. The Core Idea

The Efficient Market Hypothesis, articulated most influentially by **Eugene Fama (1970)**, states:

A market is efficient if security prices fully and instantaneously reflect all available and relevant information.

The intuition is a competition argument, not a claim that investors are individually smart. Suppose a piece of good news arrives about a company. If even a *handful* of profit-seeking traders react, they buy until the price rises to the point where the news is no longer a bargain. Competition among informed traders drives the price to fair value almost immediately. The consequence is startling:

In an efficient market, price changes are unpredictable — they follow a "random walk" — because prices only move on *new* information, and new information is by definition unforecastable.

If tomorrow's price move were predictable from today's information, someone would have already traded on it today, moving the price now. So the very act of smart investors *searching* for mispricing is what *eliminates*

mispricing. This is the deep, almost paradoxical, core of EMH.

Two important refinements:

1. **Efficiency does not mean prices are always correct — it means they are *unbiased* and *not systematically exploitable*.** Prices can be wrong; they just can't be *predictably* wrong in a way you can trade profitably after costs.
2. **The Grossman–Stiglitz paradox (1980):** if prices reflected *all* information perfectly, no one would be paid to gather information, so no one would — and then prices *couldn't* reflect information. Efficiency must therefore be *incomplete*: there must be just enough mispricing to compensate the analysts whose work removes it. Markets are in an "efficiently inefficient" equilibrium.

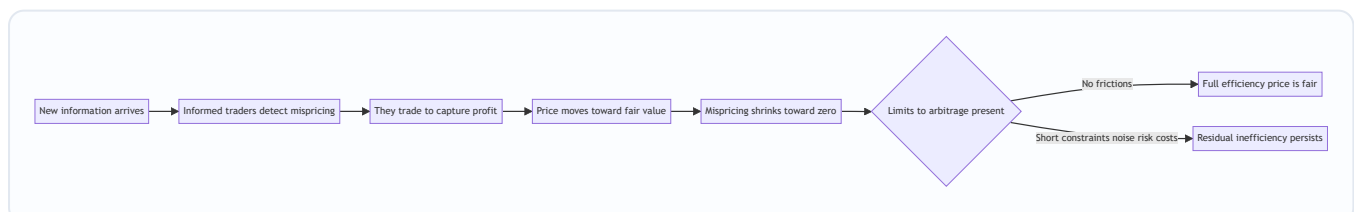
3. Why / How It Works

The mechanism that enforces efficiency is **arbitrage and competition among informed traders**. Walk through the causal chain:

1. **Information is cheap to disseminate but valuable to act on first.** Public information (earnings, RBI policy, oil prices) reaches everyone; the profit lies in reacting fastest and most accurately.
2. **Marginal informed traders set the price.** The price is not the "average opinion" — it is set at the margin by the best-informed, highest-conviction traders willing to transact. A retail investor's ignorance doesn't push the price around if a hedge fund stands ready to take the other side.
3. **Arbitrage removes riskless mispricing.** If two identical claims trade at different prices (e.g., a stock and its ADR, or Nifty spot vs futures), traders buy cheap / sell dear until the gap closes to transaction-cost bounds.
4. **The feedback loop is self-defeating for the searcher.** The more capital chases an anomaly, the smaller it becomes. Published anomalies decay after publication (McLean and Pontiff, 2016, found roughly a 30–58% post-publication decay in documented anomalies).

For this to *fail* — for inefficiency to persist — you need frictions that block the loop: **limits to arbitrage**. Short-selling constraints, funding risk, noise-trader risk (prices can move *against* the arbitrageur before converging, forcing liquidation — the "markets can stay irrational longer than you can stay solvent" problem), and transaction costs. Where these frictions are large, mispricing can survive. This is the single most important idea for a working analyst: **inefficiency lives where arbitrage is hardest, not where analysis is easiest.**

Figure 12.2 — The self-correcting loop that produces efficiency, and the frictions that let it fail.



4. Full Content — Forms, Formulas, and Models

4.1 The Three Forms of Efficiency

Fama classified efficiency by the *information set* that prices reflect. The forms are nested: each stronger form contains the weaker ones.

Form	Information reflected in price	If true, then useless...	Tested by
Weak	All <i>past prices and volume</i> (market data)	Technical analysis	Serial correlation, runs tests, filter rules, momentum
Semi-strong	All <i>publicly available</i> information (financials, news, macro)	Fundamental analysis on public data	Event studies (earnings, splits, M&A)
Strong	All information, including private / insider	Even insider trading	Insider-trade profitability, fund manager records

- **Weak-form** says charting is futile: past prices contain no usable predictive signal. Nested consequence: prices are a random walk.
- **Semi-strong** says that by the time you read the annual report, the price already embeds it. Your careful DCF only pays if your *interpretation* is genuinely superior *and* correct — a much higher bar.
- **Strong-form** is almost certainly *false* in reality: insiders demonstrably earn abnormal returns, which is precisely why insider trading is illegal (SEBI PIT Regulations, 2015, in India; SEC Rule 10b-5 in the US). The empirical consensus: markets are *close* to semi-strong efficient, clearly *not* strong-form efficient, and *mostly* weak-form efficient with documented exceptions (momentum).

4.2 The Random Walk and Martingale Model

Formally, weak-form efficiency implies prices follow a **submartingale**:

$$E[P_{t+1} \mid \Omega_t] = P_t (1 + E[r])$$

where Ω_t is the information set of past prices and $E[r]$ is the required return. Expected excess returns, conditional on past prices, are zero — you cannot use price history to earn abnormal profit. The **random walk with drift** version:

$$P_{t+1} = P_t + \mu + \varepsilon_{t+1}, \quad E[\varepsilon_{t+1} \mid \Omega_t] = 0$$

Returns are serially uncorrelated: $\text{Corr}(r_t, r_{t+k}) = 0$ for $k \neq 0$.

4.3 The Joint-Hypothesis Problem

You can never test efficiency alone. To say a return is "abnormal," you need a model of what "normal" (required) return should be — usually CAPM or a factor model. So every test is a **joint test** of (a) market efficiency *and* (b) the asset-pricing model. If you find "abnormal" returns, you cannot tell whether the market is inefficient *or* your risk model is wrong. This is Fama's own caveat, and it is why the value and size "anomalies" are so contested — are they inefficiencies, or just compensation for risk the CAPM misses?

4.4 Measuring Abnormal Return — Alpha

The workhorse for judging whether skill exists is **Jensen's alpha**, the intercept from regressing a portfolio's excess returns on the market's:

$$R_p - R_f = \alpha_p + \beta_p (R_m - R_f) + \varepsilon_p$$

- $\alpha_p > 0$: outperformance *after* adjusting for market risk — the statistical signature of skill (or luck, or a missing risk factor).
- Extended to the **Fama–French / Carhart** models, alpha is measured net of size (SMB), value (HML), and momentum (WML) factors:

$$R_p - R_f = \alpha_p + \beta_{\text{mkt}}(R_m - R_f) + \beta_s \text{SMB} + \beta_h \text{HML} + \beta_w \text{WML} + \varepsilon_p$$

If a manager's "outperformance" disappears once you control for these factors, they weren't skilled — they were just *tilted* toward known factors, which a cheap smart-beta fund could replicate.

4.5 Risk-Adjusted Performance Ratios

Metric	Formula	Risk measure	Use
Sharpe ratio	$S = \frac{R_p - R_f}{\sigma_p}$	Total risk (SD)	Stand-alone portfolio; the whole wealth
Treynor ratio	$T = \frac{R_p - R_f}{\beta_p}$	Systematic risk (beta)	A sub-portfolio within a diversified whole
Jensen's alpha	$\alpha = R_p - [R_f + \beta_p(R_m - R_f)]$	Beta (via CAPM)	Absolute value added
Information ratio	$IR = \frac{\alpha_p}{\sigma(\varepsilon_p)}$	Tracking error	Skill per unit of active risk

The **Information Ratio** is the professional's yardstick: active return divided by tracking error (the "active risk" taken to get it). A sustained IR above ~0.5 is good; above ~1.0 is elite and rare.

4.6 The Arithmetic of Active Management (Sharpe's Law)

William Sharpe's (1991) "**The Arithmetic of Active Management**" is a near-tautology that every professional must be able to recite:

Before costs, the return on the *average* actively managed dollar equals the return on the *average* passively managed dollar (both hold the market in aggregate). Therefore, *after* costs, the average actively managed dollar must **underperform** the average passively managed dollar. This holds by arithmetic, in any market, efficient or not.

Because active and passive investors together *own the whole market*, and passive investors by construction hold the market portfolio, the active investors as a group *also* hold the market portfolio (the residual). Their gross return is identical to the index. Active managers charge more and trade more, so their *net* return, in aggregate, is lower. Active management can only be a *zero-sum game before costs* and a *negative-sum game after costs* — one manager's outperformance is another's underperformance. Not everyone can be above average.

4.7 The Anomalies — Cracks in the Wall

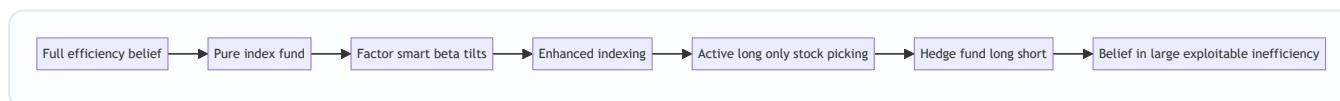
Documented, replicated deviations from EMH:

Anomaly	Description	Efficiency threatened
Momentum	Past 3–12 month winners keep winning (Jegadeesh & Titman 1993)	Weak-form
Value (low P/B, P/E)	Cheap stocks beat glamour stocks long-term	Semi-strong
Size	Small caps historically outperform (weakened post-1980s)	Semi-strong
Post-earnings-announcement drift (PEAD)	Prices keep drifting in the direction of an earnings surprise for weeks	Semi-strong
Low-volatility	Low-beta stocks earn higher risk-adjusted returns — contradicts CAPM	Semi-strong
Calendar (January, turn-of-month)	Seasonal return patterns	Weak-form
Accruals	Firms with high accruals underperform	Semi-strong
IPO / SEO underperformance	New issues underperform for years	Semi-strong

Behavioural finance (Kahneman, Tversky, Thaler, Shiller) supplies the demand-side explanation: overreaction, underreaction, anchoring, herding, and overconfidence generate the mispricings; limits to arbitrage let them survive. Momentum and PEAD look like *underreaction*; long-run reversals and bubbles look like *overreaction*.

4.8 The Active–Passive Spectrum

Figure 12.3 — Investment strategies arranged by the efficiency belief they require.



5. Worked Examples

Example 1 — Random walk and the futility of a filter rule

Setup. A stock's daily log returns are drawn independently with mean $\mu = 0.0004$ (about 10% annualised over 250 days) and daily SD $\sigma = 0.015$. A "chartist" proposes a filter rule: *buy after any up-day, expecting momentum*. Is there an edge?

Analysis. Under weak-form efficiency, returns are serially uncorrelated: $\text{Corr}(r_t, r_{t+1}) = 0$. The expected return tomorrow, *given* an up-day today, equals the unconditional mean:

$$E[r_{t+1} \mid r_t > 0] = \mu = 0.0004$$

The up-day carries **no information** about tomorrow. The rule generates the same expected 0.04% daily gross return as buy-and-hold — but each trade incurs, say, 0.10% round-trip cost. If the rule trades ~120 times a year:

- Buy-and-hold gross $\approx 250 \times 0.0004 = 10.0\%$; costs $\approx$ one trade $\approx 0.10\%$; **net $\approx 9.9\%$** .

- Filter rule gross $\approx 10.0\%$ (same, no edge); costs $\approx \$120 \times 0.10\% = 12.0\%$; **net $\approx 10.0\% - 12.0\% = -2.0\%$.**

Reconciliation / self-check. The gross returns match (as they must under a random walk — the rule can't manufacture edge from noise). The *only* difference is transaction costs, which the churning rule multiplies. The chartist doesn't just fail to win; the trading friction turns a +10% asset into a -2% strategy. This is weak-form efficiency biting in practice. ✓

Example 2 — Alpha, and whether the "star" manager has skill

Setup. A fund returned $R_p = 16\%$ last year. The risk-free rate $R_f = 6\%$ (Indian 10-year G-Sec neighbourhood), the market (Nifty) returned $R_m = 13\%$, and the fund's beta is $\beta_p = 1.4$. Fund SD = 22%, market SD = 15%. Did the manager add value?

Step 1 — CAPM required return. $E[R_p] = R_f + \beta_p (R_m - R_f) = 6\% + 1.4 \times (13\% - 6\%) = 6\% + 1.4 \times 7\% = 6\% + 9.8\% = 15.8\%$

Step 2 — Jensen's alpha. $\alpha = R_p - E[R_p] = 16\% - 15.8\% = +0.2\%$

Step 3 — Sharpe ratios (total-risk view). $S_p = \frac{16 - 6}{22} = \frac{10}{22} = 0.455$, $\quad S_m = \frac{13 - 6}{15} = \frac{7}{15} = 0.467$

Step 4 — Treynor ratios (systematic-risk view). $T_p = \frac{16 - 6}{1.4} = 7.14\%$, $\quad T_m = \frac{13 - 6}{1.0} = 7.00\%$

Reconciliation / self-check. The three metrics tell a *nuanced, consistent* story:

- **Alpha = +0.2%** — a whisper of outperformance, easily within noise.
- **Treynor** ($7.14\% > 7.00\%$) agrees the fund beat the market *per unit of beta* — because alpha is positive, Treynor *must* exceed the market's, and it does. ✓ (Alpha $> 0 \iff$ Treynor $>$ market Treynor, always.)
- **Sharpe** ($0.455 < 0.467$) says the fund *lost* on a total-risk basis. Why the disagreement? The fund's SD (22%) is higher than beta alone predicts. If it were perfectly diversified, its SD would be $\beta_p \times \sigma_m = 1.4 \times 15\% = 21\%$; the actual 22% carries $\sqrt{22^2 - 21^2} = \sqrt{484 - 441} = \sqrt{43} \approx 6.6\%$ of *diversifiable* risk that Treynor and alpha ignore but Sharpe penalises.

Verdict: The "16% vs 13% — beat the market!" headline collapses under risk adjustment. The 0.2% alpha is noise; the fund took extra, poorly diversified risk. This is exactly why an interviewer wants risk-adjusted numbers, not raw returns. ✓

Example 3 — The arithmetic of active management, made concrete

Setup. An index returns 12% gross. The market splits: 70% of assets are passive (fee 0.10%), 30% active (average fee 1.20%, plus 0.40% trading drag). What does each cohort net, and what fraction of active managers *can* beat the index net of fees?

Step 1 — Gross returns. Passive holds the index: gross 12.0%. By Sharpe's arithmetic, active investors *in aggregate* also hold the market residual, so their aggregate gross is *also* 12.0%.

Step 2 — Net returns. - Passive net = $12.0\% - 0.10\% = 11.90\%$. - Active net (average) = $12.0\% - 1.20\% - 0.40\% = 10.40\%$.

Step 3 — The gap. The average active investor underperforms the average passive investor by $11.90\% - 10.40\% = 1.50\%$ per year — exactly the fee-plus-cost differential.

Step 4 — Can *any* active manager win? Yes, individually. Active is zero-sum *gross*: for one manager to earn +2% gross alpha, others must collectively earn −2%. But every active manager pays the 1.6% cost drag. So to beat the 11.90% passive net, a manager needs *gross* alpha above +1.6% (they need 12% + 1.5% ≈ 13.5% gross to net 11.9%).

Reconciliation / self-check. Over 20 years this 1.5% annual gap compounds brutally. ₹100 at 11.90% → $\$100 \times 1.119^{20} = ₹945\$$. At 10.40% → $\$100 \times 1.104^{20} = ₹722\$$. The passive investor ends with **31% more wealth** — from a 1.5% annual edge, purely from lower costs, no forecasting skill required. This is the compounding case for indexing. ✓ (Cross-check: SPIVA data consistently shows ~80–90% of active large-cap funds trail their benchmark over 10–15 years — the arithmetic made empirical.)

6. Connections

- **CAPM and factor models (Ch. on asset pricing).** Efficiency is tested *through* these models — the joint-hypothesis problem. Alpha is the residual after the model explains "normal" return. Fama–French factors reframed many "anomalies" as risk premia, partially rescuing efficiency.
- **Portfolio theory (Markowitz).** If markets are efficient, the market portfolio is mean-variance efficient and the optimal strategy is to hold it — the intellectual foundation of indexing.
- **Behavioural finance.** The demand-side counterweight to EMH: cognitive biases *create* mispricing; limits to arbitrage *preserve* it. EMH and behavioural finance are the thesis and antithesis of modern asset pricing.
- **Fixed income & derivatives.** Arbitrage (put-call parity, spot-futures parity, no-arbitrage bond pricing) is the *enforcement mechanism* of efficiency — the cleanest, near-riskless version.
- **Performance attribution (next material).** Sharpe, Treynor, IR, and alpha are how you *audit* an active manager's claim to skill.
- **Corporate finance.** Semi-strong efficiency justifies event studies used to measure the value impact of M&A, dividends, and buybacks.

7. Key Terms

- **Efficient Market Hypothesis (EMH):** prices fully reflect available information.
- **Weak / Semi-strong / Strong form:** efficiency w.r.t. past prices / all public info / all info including private.
- **Random walk / martingale:** price changes are unpredictable; expected abnormal return conditional on the information set is zero.
- **Joint-hypothesis problem:** any efficiency test is simultaneously a test of the asset-pricing model used.
- **Alpha (Jensen's):** return in excess of the CAPM-required return; the signature of skill (or luck / model error).
- **Information ratio:** alpha ÷ tracking error — active return per unit of active risk.
- **Anomaly:** a persistent, replicable pattern of abnormal returns (momentum, value, size, PEAD).
- **Limits to arbitrage:** frictions (short constraints, noise-trader risk, costs) that stop traders from correcting mispricing.
- **Grossman–Stiglitz paradox:** perfectly efficient prices would leave no incentive to gather information, so some inefficiency must persist.

- **Sharpe's arithmetic:** active management is zero-sum before costs, negative-sum after.
- **Tracking error:** SD of the difference between portfolio and benchmark returns.
- **Smart beta / factor investing:** rules-based tilts toward documented factors — a middle ground between active and passive.

8. Common Confusions

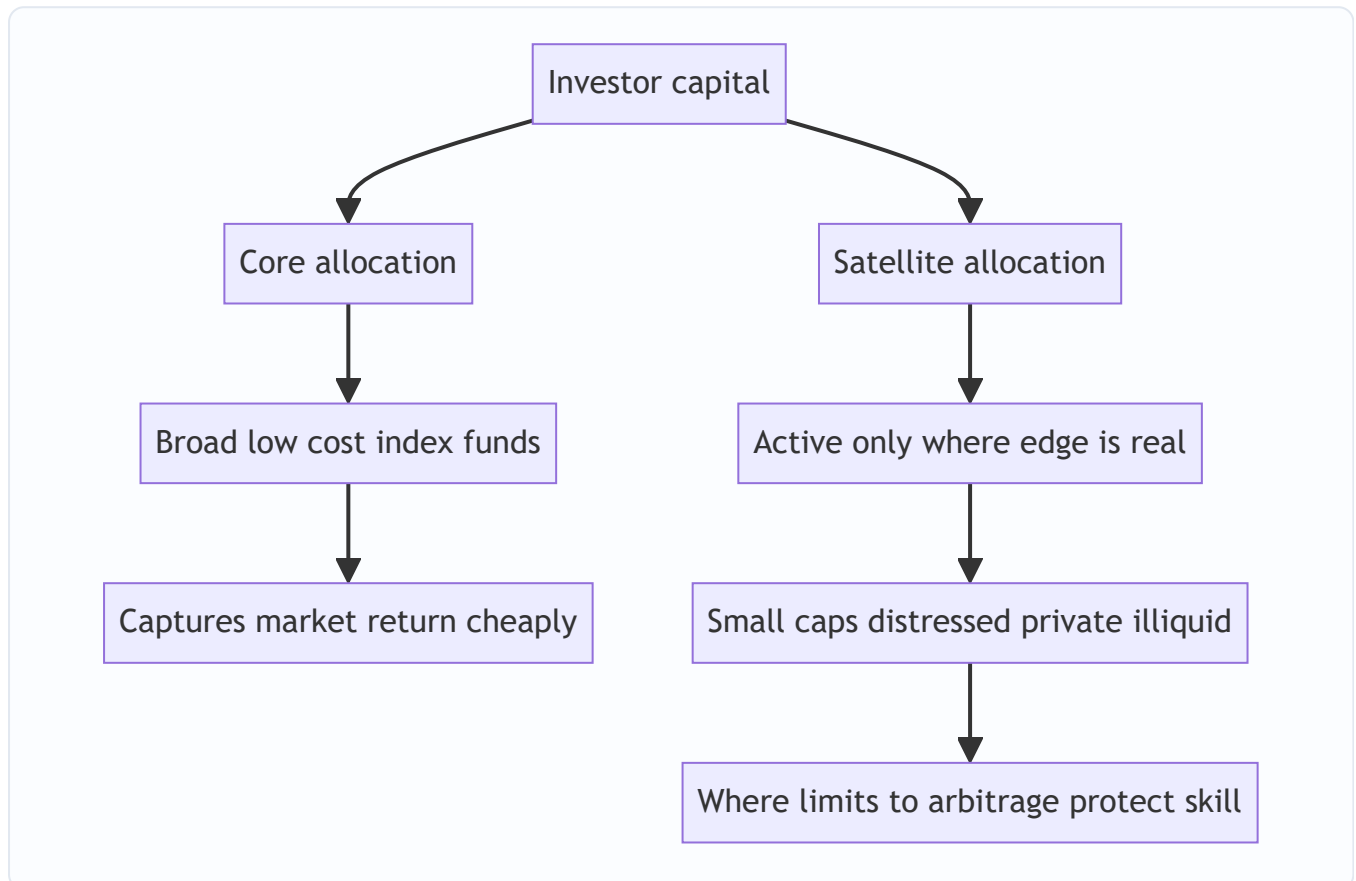
1. **"Efficient means the price is always right / fair value."** No. Efficient means *not predictably exploitable* and *unbiased*. Prices can be wrong; they just aren't wrong in a way you can systematically monetise after costs. Bubbles don't automatically disprove EMH — profiting from spotting one *before* it bursts does.
2. **"EMH says all investors are rational."** No. EMH needs only *enough* rational, well-funded traders at the margin. Plenty of irrational investors can exist; if arbitrageurs can trade against them cheaply, prices stay efficient.
3. **"Some funds beat the market, so EMH is false."** With thousands of funds, pure luck guarantees many winners — the "lucky coin-flippers." The test is *persistence* and *statistical significance* of alpha after fees and factor adjustment, not the existence of past winners.
4. **"Random walk means the market goes nowhere."** No — it's a random walk *with drift*. Prices trend up over time (the equity risk premium); it's the *deviations* around the drift that are unpredictable.
5. **"Passive investing free-rides and would break if everyone indexed."** Correct in the limit — if *everyone* indexed, no one would do price discovery and inefficiency would explode, restoring active's edge. This is self-limiting: it guarantees active management never fully dies. Today passive is ~50% of US equity fund assets, still far from that limit.
6. **"Technical analysis is proven useless by weak-form EMH."** Mostly, but *momentum* is a robust, published exception — a price-based signal with predictive power. The nuance matters: charting patterns fail; systematic momentum survives (though it decays as capital crowds in).
7. **"Alpha proves skill."** Only if it survives (a) factor adjustment — is it *really* alpha or a value/size/momentum tilt? — and (b) a statistical-significance test over a long enough record. Most "alpha" is disguised factor beta.

9. Recap

- The **EMH** says prices reflect available information; competition among informed traders is the enforcing mechanism, and it makes price changes a **random walk**.
- Efficiency comes in **three nested forms** — weak (past prices), semi-strong (public info), strong (all info). Evidence: markets are broadly weak and semi-strong efficient; strong-form is false (insiders profit).
- Every efficiency test is a **joint hypothesis** with an asset-pricing model, so "anomalies" are ambiguous — inefficiency *or* missing risk factors.
- **Anomalies** (momentum, value, size, PEAD, low-vol) and **behavioural finance** are the cracks; **limits to arbitrage** explain why cracks don't close.
- **Sharpe's arithmetic** is decisive: active management is zero-sum before costs, negative-sum after. On average, active must lose to passive by the fee differential — confirmed by SPIVA (80–90% of active funds trail over 10+ years).

- Skill is measured by **risk-adjusted metrics** — Sharpe, Treynor, alpha, and above all the **information ratio** — not raw returns.
- The practical synthesis: markets are **efficient enough** that *most* investors should index for the *core*, while genuine, scarce skill and inefficiency persist at the **margins** — small caps, distressed debt, private markets, and short horizons around information events.

Figure 12.4 — The pragmatic verdict: a barbell of cheap core indexing plus selective active where inefficiency is real.



10. Quick-Reference / Interview Points

One-liner definition: A market is efficient if prices fully reflect available information, making abnormal returns unattainable on that information set after costs.

The three forms — instant recall: - Weak $\Rightarrow$ technical analysis useless (past prices priced in). - Semi-strong $\Rightarrow$ fundamental analysis on public data useless (public info priced in). - Strong $\Rightarrow$ even insider info useless (empirically false).

Killer formulas: - Sharpe $= (R_p - R_f) / \sigma_p$ — total risk. - Treynor $= (R_p - R_f) / \beta_p$ — systematic risk. - Alpha $= R_p - [R_f + \beta_p(R_m - R_f)]$ — value added. - Information ratio $= \alpha / \text{tracking error}$ — the pro's skill metric.

If asked "Are markets efficient?" — Answer with a spectrum, not a yes/no: "Broadly weak- and semi-strong efficient, definitely not strong-form. Efficient enough that the average active manager can't beat the index net of fees — that's Sharpe's arithmetic and SPIVA confirms it. But limits to arbitrage leave persistent pockets — momentum, small caps, distressed debt, illiquid and private markets, and short windows around information

events — where skill can be paid. So: index the core, be active only where you can articulate a specific, structural reason the inefficiency survives."

If asked "Why can't everyone just index?" — Grossman–Stiglitz: someone must do price discovery to be paid; if all indexed, inefficiency would explode and revive active. Passive is self-limiting, so active never fully dies.

If asked "How do you tell skill from luck?" — Persistence over long horizons, statistical significance of alpha, and survival *after* factor adjustment (Fama–French–Carhart). Most "alpha" is disguised factor beta replicable by cheap smart-beta.

The three anomalies to name-drop: momentum (weak-form crack), value and size (Fama–French factors — risk or mispricing?), and post-earnings-announcement drift (underreaction to public news).

The single most important practical takeaway: *Inefficiency lives where arbitrage is hardest, not where analysis is easiest.* Point your active effort at frictions — short-sale constraints, illiquidity, noise-trader risk — not at large, liquid, heavily-covered stocks where a hundred analysts already priced the news.

The compounding punchline: a 1.5% annual fee/cost drag compounds to ~30% less terminal wealth over 20 years. Costs are the one variable an investor controls with certainty; alpha is not. That asymmetry is the entire case for low-cost investing.

Q&A — Market Efficiency and the Active-Passive Debate

Practice bank for Chapter 12. Work each question before reading the answer. Returns are in percentages; betas are unit-free. Recurring theme: efficiency means *not predictably exploitable after costs*, not "always right."

Section A — Concept Checks

A1. State the Efficient Market Hypothesis in one sentence. A market is efficient if security prices fully and instantaneously reflect all available and relevant information, so abnormal returns cannot be reliably earned on that information set after costs.

A2. Why does EMH imply that price changes follow a random walk? Prices already embed all current information, so they can only move on *new* information — which is by definition unforecastable. Hence the resulting price change is unpredictable: a random walk (with an upward drift equal to the required return).

A3. Does "efficient" mean the price is always correct? No. Efficient means the price is *unbiased* and *not systematically exploitable*. Prices can be wrong at any moment; they simply cannot be *predictably* wrong in a way you can trade on profitably after costs.

A4. Name the three forms of efficiency and the analysis each renders useless. Weak form reflects all past prices/volume → technical analysis is useless. Semi-strong reflects all public information → fundamental analysis on public data is useless. Strong form reflects all information including private → even insider trading is useless. The forms are nested: each stronger form contains the weaker.

A5. Which form is almost certainly false in reality, and what is the evidence? Strong form. Insiders demonstrably earn abnormal returns on private information — precisely why insider trading is illegal (SEBI PIT Regulations 2015 in India; SEC Rule 10b-5 in the US). Empirically markets are broadly weak and semi-strong efficient but clearly not strong-form.

A6. Explain the joint-hypothesis problem. To call a return "abnormal" you need a model of the "normal" required return (usually CAPM or a factor model). Every efficiency test is therefore simultaneously a test of (a) efficiency and (b) the asset-pricing model. Finding "abnormal" returns is ambiguous — the market may be inefficient, or your risk model may be wrong.

A7. What is the Grossman–Stiglitz paradox? If prices reflected *all* information perfectly, no one could profit from gathering information, so no one would — and then prices couldn't reflect information. Efficiency must be incomplete: just enough mispricing survives to pay the analysts whose work removes it. Markets are "efficiently inefficient."

A8. What enforces efficiency — smart individuals, or something else? Competition and arbitrage among *marginal* informed traders. Efficiency needs only *enough* well-funded rational traders at the margin, not universally rational investors. Irrational investors can exist; if arbitrageurs cheaply trade against them, prices stay efficient.

A9. Define "limits to arbitrage" and give three examples. Frictions that stop traders correcting mispricing: short-selling constraints, noise-trader risk (prices can move further against the arbitrageur before

converging, forcing liquidation), funding/liquidity risk, and transaction costs. Inefficiency persists where these are large.

A10. State Sharpe's arithmetic of active management. Before costs, the average active dollar earns the same as the average passive dollar (both hold the market in aggregate). So after costs the average active dollar *must* underperform. This holds by arithmetic in any market, efficient or not.

A11. Why is the Information Ratio the professional's preferred skill metric? It measures active return per unit of *active risk*: $IR = \alpha \div \text{tracking error}$. It isolates skill relative to the benchmark rather than rewarding a manager for simply taking more market risk. A sustained IR above ~ 0.5 is good; above ~ 1.0 is elite and rare.

A12. Give two anomalies and the form of efficiency each threatens. Momentum (past 3–12 month winners keep winning) threatens weak-form efficiency — a *price-based* predictive signal. Post-earnings-announcement drift (prices keep drifting in the surprise direction) threatens semi-strong efficiency — underreaction to *public* news.

A13. If some funds beat the market, is EMH refuted? No. With thousands of funds, luck alone guarantees many winners ("lucky coin-flippers"). The real test is *persistence* and *significance* of alpha after fees and factor adjustment — not the mere existence of past winners.

Section B — Numerical Problems (full working)

B1. Alpha, Sharpe, and Treynor — does the "star" have skill? Fund: $\$R_p = 16\%$, $\beta_p = 1.4$, $\sigma_p = 22\%$. Market: $\$R_m = 13\%$, $\sigma_m = 15\%$. Risk-free $\$R_f = 6\%$.

- CAPM required return: $\$6 + 1.4(13-6) = 6 + 1.4(7) = 6 + 9.8 = 15.8\%$.
- Jensen's alpha: $\$16 - 15.8 = +0.2\%$.
- Sharpe: $\$S_p = (16-6)/22 = 0.455$; $\$S_m = (13-6)/15 = 0.467$.
- Treynor: $\$T_p = (16-6)/1.4 = 7.14\%$; $\$T_m = (13-6)/1.0 = 7.00\%$.

Reconcile: Treynor and alpha agree the fund beat the market per unit of *beta* ($\alpha > 0 \Leftrightarrow \text{Treynor} > \text{market Treynor}$, always). But Sharpe says it *lost* on total risk ($0.455 < 0.467$). Why: a perfectly diversified fund with $\beta 1.4$ would have $\sigma = \$1.4 \times 15 = 21\%$; the actual 22% carries $\sqrt{22^2 - 21^2} = \sqrt{43} \approx 6.6\%$ of diversifiable risk that Sharpe penalises but beta ignores. Verdict: the +0.2% alpha is noise, and the fund took uncompensated idiosyncratic risk. The "16% beat 13%" headline collapses under risk adjustment. ✓

B2. The arithmetic of active management, made concrete. Index returns 12% gross. Market split: 70% passive (fee 0.10%), 30% active (fee 1.20% + 0.40% trading drag).

- Gross: passive holds the index = 12.0%; by Sharpe's arithmetic active *in aggregate* also = 12.0%.
- Passive net: $\$12.0 - 0.10 = 11.90\%$.
- Active net (average): $\$12.0 - 1.20 - 0.40 = 10.40\%$.
- Gap: $\$11.90 - 10.40 = 1.50\%$ per year — exactly the fee-plus-cost differential.

Reconcile — compounding over 20 years. ₹100 at 11.90% $\rightarrow \$100 \times 1.119^{20} = ₹945$. At 10.40% $\rightarrow \$100 \times 1.104^{20} = ₹722$. The passive investor ends with $\$945/722 - 1 = 31\%$ more wealth, purely from lower costs, no skill required. This mirrors SPIVA data: $\sim 80\text{--}90\%$ of active large-cap funds trail their benchmark over 10–15 years. ✓

B3. Information Ratio and required gross alpha. A manager delivers 1.8% alpha with 3.0% tracking error. (a) Find the IR. (b) If the fund charges 1.6% in fees/costs, what *gross* alpha is needed just to match a passive net return, given the passive fund charges 0.10%?

- (a) $IR = 1.8 / 3.0 = 0.60$ — good, above the ~ 0.5 threshold.
- (b) To match passive net, gross alpha must cover the cost disadvantage: active 1.6% vs passive 0.10% = 1.5% gap. So gross alpha must be at least $+1.5\%$ before the manager adds any value above indexing.

Reconcile: the 1.8% gross alpha clears the 1.5% hurdle by only 0.3% of net value to the client — despite a respectable IR of 0.60. Costs dominate: even genuine skill is mostly consumed by the fee load. ✓

B4. Filter rule versus buy-and-hold under a random walk. Daily log returns are i.i.d. with mean $\mu = 0.0004$ ($\sim 10\%$ over 250 days) and SD $\sigma = 0.015$. A chartist buys after every up-day, trading ~ 120 times/year; round-trip cost is 0.10%. Buy-and-hold trades once (cost 0.10%).

- Under weak-form efficiency $\text{Corr}(r_t, r_{t+1}) = 0$, so $E[r_{t+1} \mid r_t > 0] = \mu$. The up-day carries no information.
- Buy-and-hold: gross $\$ = 250 \times 0.0004 = 10.0\%$; net $\$ = 10.0 - 0.10 = 9.90\%$.
- Filter rule: gross $\$ = 10.0\%$ (identical — no edge from noise); costs $\$ = 120 \times 0.10 = 12.0\%$; net $\$ = 10.0 - 12.0 = -2.0\%$.

Reconcile: the gross returns *must* match under a random walk — the rule cannot manufacture edge from serially uncorrelated returns. The only difference is transaction costs, which churning multiplies, converting a $+10\%$ asset into a -2% strategy. Weak-form efficiency biting in practice. ✓

B5. Semi-strong efficiency and an event study. A firm announces earnings 20% above consensus at 10:00 a.m. The stock jumps from ₹500 to ₹560 within minutes, then trades flat around ₹560. (a) Consistent with semi-strong efficiency? (b) What pattern would instead signal PEAD?

- (a) Yes. The full $+12\%$ adjustment happens *instantaneously* on the public announcement, then no drift. Investors buying at ₹560 earn only the normal required return going forward — the surprise is already priced. Textbook semi-strong efficiency.
- (b) PEAD would show the price drifting *further* up over the following weeks — e.g., ₹560 $\rightarrow$ ₹580 over a month — as the market *underreacts* and slowly incorporates the surprise. That gradual, predictable drift is the exploitable inefficiency.

Reconcile: the test is *speed and completeness* of adjustment. Instant full jump = efficient; slow continued drift in the surprise direction = semi-strong inefficiency (PEAD). ✓

B6. Two managers with identical raw returns — rank by IR. Both funds return 15% with the benchmark at 12%, so both have 3% active return. Fund P has tracking error 4%; Fund Q has tracking error 8%.

- $IR_P = 3/4 = 0.75$; $IR_Q = 3/8 = 0.375$.

Reconcile: identical headline outperformance, but P generated it with half the active risk — twice the skill per unit of tracking error. Q's higher tracking error suggests its 3% is more likely luck that could reverse. Raw active return alone is misleading; the IR is the correct lens. ✓

Section C — Interview-Style Questions (model answers)

C1. "Are markets efficient?" I'd answer with a spectrum, not a yes/no. Markets are broadly weak- and semi-strong efficient and definitely not strong-form efficient — insiders demonstrably profit. They're efficient enough that the *average* active manager can't beat the index net of fees; that's Sharpe's arithmetic, and SPIVA confirms 80–90% of active funds trail over a decade-plus. But limits to arbitrage leave persistent pockets — momentum, small caps, distressed debt, illiquid markets, short windows around information events — where genuine skill can be paid. So: index the core, be active only where you can name a structural reason the inefficiency survives.

C2. "Walk me through the three forms of efficiency." Fama classified efficiency by the information set prices reflect, and the forms are nested. Weak form: prices embed all past price/volume data, so technical analysis is futile and returns follow a random walk. Semi-strong adds all public information, so fundamental analysis on public data only pays if your *interpretation* is genuinely superior. Strong form adds private information, meaning even insiders couldn't profit — empirically false, which is why insider trading is banned. Consensus: markets are close to semi-strong efficient with documented exceptions like momentum.

C3. "If markets are efficient, why does anyone gather information?" This is the Grossman–Stiglitz paradox. If prices already reflected everything, there'd be no return to research, so no one would do it — but then prices couldn't reflect information. So efficiency must be incomplete: exactly enough mispricing survives to compensate the analysts whose trading removes it — an "efficiently inefficient" equilibrium. This is also why passive is self-limiting: if everyone indexed, price discovery would collapse and inefficiency would explode, reviving active. Active never fully dies.

C4. "How do you tell skill from luck in a fund manager?" Three filters. First, *persistence* — does outperformance repeat across independent periods, or is this one lucky run among thousands of funds where luck guarantees winners? Second, *statistical significance* of alpha over a long record. Third, and most important, does the alpha survive *factor adjustment*? I'd regress the fund's returns on the Fama–French–Carhart factors — market, size (SMB), value (HML), momentum (WML). If the "alpha" disappears once I control for these, the manager wasn't skilled; they were tilted toward known factors a cheap smart-beta fund could replicate. True alpha is what remains.

C5. "Why can't active management beat passive on average?" It's arithmetic, not opinion — Sharpe's insight. Active and passive investors together own the entire market. Passive holds the market portfolio by construction, so active as a group must hold the same residual — its aggregate gross return equals the index. Since active managers charge more and trade more, their aggregate *net* return is lower by exactly the cost differential. Active is zero-sum before costs, negative-sum after. Individuals can win, but only by another's loss, and every one still pays the cost drag.

C6. "Doesn't the existence of bubbles disprove EMH?" Not automatically. EMH says prices aren't *predictably* exploitable, not that they're always fair. Spotting a bubble in hindsight proves nothing; *profiting* from spotting one before it bursts is what would challenge efficiency — and that's extraordinarily hard because of limits to arbitrage. As Keynes put it, markets can stay irrational longer than you can stay solvent: an arbitrageur shorting an overvalued stock risks it climbing further, forcing liquidation before convergence. So bubbles are more evidence *for* limits to arbitrage than *against* tradable efficiency.

C7. "Where would you actually look for inefficiency?" The single most useful principle: inefficiency lives where arbitrage is hardest, not where analysis is easiest. I wouldn't hunt for edge in large, liquid, heavily-covered stocks where a hundred analysts already priced the news. I'd point active effort at *frictions* — short-

sale constraints, illiquidity, transaction costs, noise-trader risk. Concretely: small caps with thin coverage, distressed debt, private/illiquid markets, and short windows around earnings where underreaction (PEAD) creates drift. The friction is what protects the skill from being competed away.

C8. "What's your asset allocation philosophy given all this?" A barbell, or core-satellite. The core holds broad, low-cost index funds that capture the market return cheaply; costs are the one variable I control with certainty, and a 1.5% annual drag compounds to ~30% less wealth over 20 years. The satellite is a smaller active allocation, only where I can name a structural reason the inefficiency survives arbitrage. This respects both truths: markets are efficient enough that most investors should index, yet scarce skill persists at the margins.

Section D — MCQs (with reasoning)

D1. Weak-form efficiency implies that which analysis is useless? (a) fundamental analysis (b) technical analysis (c) insider trading (d) factor analysis **Answer: (b).** Weak form says prices already reflect all past price/volume data, so charting past prices yields no edge. Fundamental analysis is defeated by semi-strong; insider trading by strong.

D2. A random walk with drift means: (a) prices go nowhere (b) prices are perfectly predictable (c) prices trend up but deviations are unpredictable (d) returns are negatively autocorrelated **Answer: (c).** The drift is the equity risk premium (prices trend up); the deviations *around* the drift are unforecastable. "Goes nowhere" ignores the drift.

D3. The joint-hypothesis problem states that any efficiency test is simultaneously a test of: (a) two different markets (b) the asset-pricing model used (c) the risk-free rate (d) investor rationality **Answer: (b).** Judging a return "abnormal" requires a benchmark for "normal" return, so you jointly test efficiency and that asset-pricing model. Abnormal returns could mean inefficiency *or* a wrong model.

D4. According to Sharpe's arithmetic, after costs the average active dollar must: (a) beat the average passive dollar (b) equal the average passive dollar (c) underperform the average passive dollar (d) beat the risk-free rate **Answer: (c).** Active and passive earn the same gross return in aggregate; higher active costs make active's aggregate net return lower by the cost differential. This holds in any market.

D5. The Information Ratio is defined as: (a) $\alpha \div \beta$ (b) $\alpha \div \text{tracking error}$ (c) $\text{excess return} \div \text{standard deviation}$ (d) $\text{excess return} \div \beta$ **Answer: (b).** $\text{IR} = \alpha \div \text{tracking error}$ — active return per unit of active risk. Option (c) is Sharpe; (d) is Treynor.

D6. The Grossman–Stiglitz paradox implies markets must be: (a) perfectly efficient (b) perfectly inefficient (c) efficiently inefficient (d) strong-form efficient **Answer: (c).** Some mispricing must survive to reward the information-gatherers whose trading removes it; otherwise no one would gather information. Hence "efficiently inefficient."

D7. Post-earnings-announcement drift (PEAD) is an anomaly that most directly challenges: (a) weak-form efficiency (b) semi-strong efficiency (c) strong-form efficiency (d) the random walk **Answer: (b).** PEAD is a predictable drift following *public* earnings news — a failure to instantly price public information, which is the semi-strong claim.

D8. Momentum is notable among anomalies because it challenges: (a) semi-strong efficiency using private data (b) weak-form efficiency using past prices (c) strong-form efficiency only (d) no form of efficiency

Answer: (b). Momentum is a *price-based* signal (past winners keep winning), so it contradicts weak-form efficiency, which says past prices carry no usable predictive information.

D9. "Limits to arbitrage" best explain why: (a) all prices are always correct (b) anomalies get instantly eliminated (c) mispricing can persist despite rational traders (d) insiders never profit **Answer: (c).** Frictions — short constraints, noise-trader risk, costs — prevent arbitrageurs from fully correcting mispricing, so it can survive. Inefficiency lives where arbitrage is hardest.

D10. A fund shows positive raw returns above its benchmark, but the outperformance vanishes after Fama–French–Carhart adjustment. This indicates: (a) genuine skill (alpha) (b) disguised factor exposure, not skill (c) a strong-form inefficiency (d) a violation of Sharpe's arithmetic **Answer: (b).** If controlling for size, value, and momentum removes the "alpha," the manager was merely tilted toward known factors replicable by cheap smart-beta — factor beta, not skill.

D11. Which statement about strong-form efficiency is correct? (a) it is empirically well-supported (b) it implies insiders cannot profit, and is empirically false (c) it is the weakest form (d) it only concerns past prices **Answer: (b).** Strong form claims prices reflect even private information, so insiders couldn't profit — but they demonstrably do, which is why insider trading is illegal. It is the strongest and empirically rejected form.

D12. The most reliable practical case for low-cost index investing rests on: (a) markets always being perfectly efficient (b) no manager ever winning (c) the certainty of cost savings compounding, versus the uncertainty of alpha (d) technical analysis being useful **Answer: (c).** Costs are controllable with certainty and compound powerfully (~1.5% drag $\approx$ 30% less wealth over 20 years), whereas alpha is uncertain and scarce. That asymmetry drives the indexing case.

End of Chapter 12 Q&A. Master three reflexes: (1) restate efficiency as "not predictably exploitable after costs," (2) recite Sharpe's arithmetic on demand, (3) always risk- and factor-adjust before crediting a manager with skill.

Chapter 13 — Behavioral Finance

1. The Problem / The Need

Classical finance is built on an elegant fiction: the *homo economicus*, a perfectly rational agent who has stable, well-defined preferences, processes all available information without error, updates beliefs according to Bayes' rule, and acts purely to maximize expected utility. On the backs of such agents, Eugene Fama's Efficient Market Hypothesis (EMH) declares that prices always reflect all available information, so nobody can consistently beat the market except by luck or by taking on more risk.

There is only one problem. Actual human beings do not behave like this — and neither, in aggregate, do markets.

If markets were perfectly efficient and investors perfectly rational, several things simply should not happen. Yet they do, repeatedly and at enormous scale:

- **Bubbles and crashes.** The dot-com mania of 1999–2000 valued companies with no revenue at billions. Pets.com IPO'd and collapsed inside a year. In 2008, subprime assets rated AAA imploded. In 2021, GameStop went from \$4 to \$483 on a wave of Reddit-driven retail buying with no change in fundamentals.
- **Excess volatility.** Robert Shiller showed in 1981 that stock prices swing far more than the present value of subsequent dividends can justify. Fundamentals are smooth; prices are hysterical.
- **Predictable patterns.** Momentum, the January effect, post-earnings-announcement drift, and the value premium are all documented regularities that a truly efficient market should have arbitrated away.
- **The "closed-end fund puzzle."** Identical baskets of assets trade at persistent discounts or premiums to their net asset value depending on investor sentiment.

Something is missing from the rational model. That something is *psychology*. Behavioral finance is the discipline that reinserts the real, cognitively-limited, emotion-driven human being back into the theory of asset prices — and in doing so, explains phenomena that the standard model waves away as "noise."

For an aspiring investment professional, this is not academic garnish. It is a working toolkit. Understanding *why* investors err lets you (a) avoid destroying your own returns, (b) recognize when the crowd has mispriced an asset, and (c) answer the interview question every buy-side firm eventually asks: "*If markets are efficient, why does your fund exist?*"

2. The Core Idea

Behavioral finance rests on a single, testable claim:

Asset prices are set by real humans who are subject to systematic, predictable cognitive and emotional biases, and whose ability to arbitrage those errors away is limited. Therefore prices can deviate from fundamental value — sometimes for a long time.

There are two pillars here, and *both* are needed:

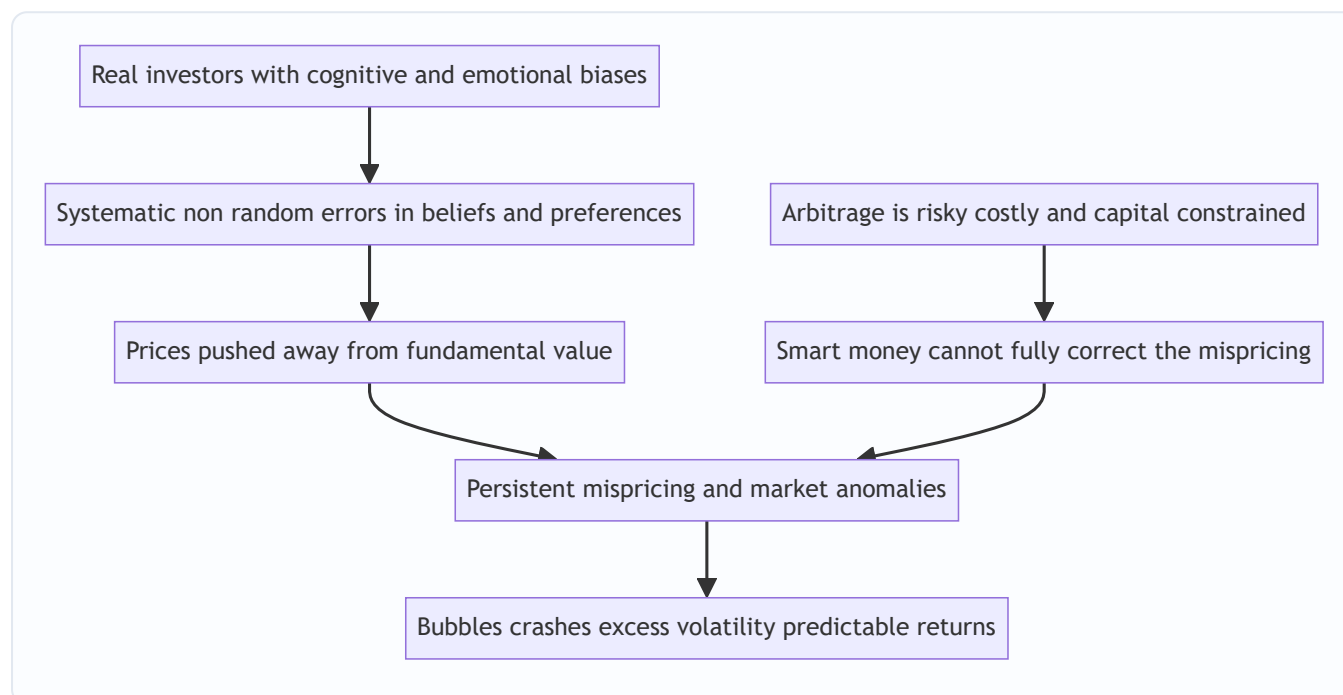
1. **Limited rationality (psychology).** Investors do not process information correctly. Their errors are not random — random errors would cancel out in aggregate. The errors are *systematic*, meaning most people

lean the same wrong way at the same time, so they push prices in a common direction.

2. **Limits to arbitrage.** In classical theory, even if some investors ("noise traders") are irrational, smart "arbitrageurs" will trade against them, correct the price, and pocket a riskless profit. Behavioral finance argues that arbitrage is *risky, costly, and capital-constrained*. Mispricings can persist and even *worsen* before they correct — as Keynes put it, "the market can remain irrational longer than you can remain solvent."

The interplay is the whole game. Biases create the mispricing; limits to arbitrage let it survive. Remove either pillar and the standard model reasserts itself.

Figure 1 — The two-pillar architecture of behavioral finance.



3. Why / How It Works

3.1 Two systems of thinking

Daniel Kahneman (Nobel 2002) popularized the **dual-process** model of cognition:

- **System 1** is fast, automatic, intuitive, emotional, and effortless. It answers "what is $2 + 2$?" and it recoils from a snake. It is also where most biases live.
- **System 2** is slow, deliberate, logical, and effortful. It answers "what is 17×24 ?" and it does careful expected-value math.

The trouble is that System 2 is *lazy*. It consumes glucose and attention, so the brain defaults to System 1 wherever it can. Under time pressure, information overload, or emotional arousal — precisely the conditions of financial markets — System 1 takes the wheel. Fast, intuitive judgments are excellent for dodging predators on the savannah and disastrous for pricing a 30-year cash-flow stream.

3.2 Heuristics: useful shortcuts that misfire

Because the world is complex and cognition is scarce, the mind uses **heuristics** — rules of thumb. Amos Tversky and Kahneman identified three that dominate financial judgment:

- **Representativeness** — judging probability by similarity to a stereotype, ignoring base rates. "This company looks like the next Amazon, so it will be the next Amazon." (It probably won't; most won't.)
- **Availability** — judging probability by how easily examples come to mind. After a plane crash dominates the news, people overestimate crash risk. After a market crash, investors overestimate the odds of another one and flee equities at the bottom.
- **Anchoring and adjustment** — starting from an arbitrary reference number and adjusting insufficiently. (Detailed below.)

Heuristics are not stupidity. They are *adaptive* — they work most of the time, cheaply. Behavioral finance simply maps out the specific, repeatable situations where they fail in markets.

3.3 Why the errors don't cancel out

The crucial insight is **correlation of errors**. If 10,000 investors each made an independent random mistake, the mistakes would average to roughly zero and prices would still be efficient. But biases are *shared*. Human brains are wired similarly, exposed to the same news, and susceptible to the same emotional contagion. When fear grips a market, it grips almost everyone at once. Correlated errors do not cancel — they *accumulate*, moving the whole market in one direction. That is the mechanism by which individual psychology becomes an aggregate price effect.

4. Full Content

4.1 Prospect Theory — the new theory of choice

The intellectual heart of behavioral finance is **Prospect Theory** (Kahneman & Tversky, 1979), which replaced expected utility theory as a *descriptive* model of how people actually choose under risk. Four ideas define it.

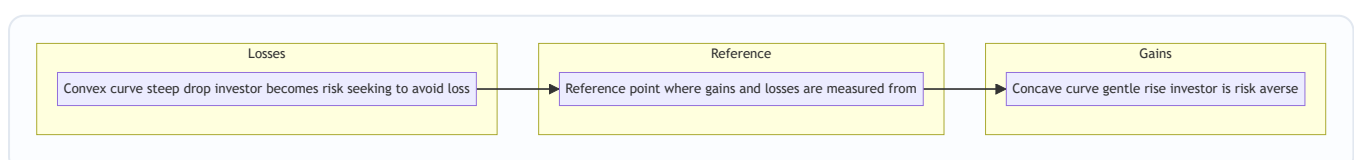
(a) Reference dependence. People do not evaluate outcomes in terms of final wealth (as utility theory assumes). They evaluate *changes* relative to a **reference point** — usually the status quo or the purchase price. A portfolio worth ₹1 crore feels like triumph to someone who started at ₹50 lakh and like catastrophe to someone who started at ₹2 crore. Same wealth, opposite emotion.

(b) Loss aversion. Losses hurt roughly **2 to 2.5 times** more than equivalent gains feel good. Losing ₹10,000 inflicts more pain than winning ₹10,000 delivers pleasure. This asymmetry is the single most important fact in behavioral finance.

(c) Diminishing sensitivity / an S-shaped value function. The value function is **concave for gains** (risk-averse — you'll take a sure ₹900 over a gamble at ₹1,000) but **convex for losses** (risk-seeking — you'll gamble to avoid a sure loss). The difference between +₹100 and +₹200 feels bigger than between +₹1,100 and +₹1,200.

(d) Probability weighting. People do not use true probabilities. They **overweight small probabilities** (which is why they buy both lottery tickets and insurance) and **underweight moderate-to-high probabilities**.

Figure 2 — The S-shaped prospect theory value function versus classical utility.



The value function can be written compactly as:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda(-x)^{\beta} & \text{if } x < 0 \end{cases}$$

where x is the gain or loss relative to the reference point, $\alpha, \beta \approx 0.88$ (diminishing sensitivity), and $\lambda \approx 2.25$ is the **loss-aversion coefficient**. That single parameter $\lambda > 1$ is what bends the curve and drives the **disposition effect**, the **equity premium puzzle**, and much of investor misbehavior.

4.2 The catalogue of biases

Biases fall into two families: **cognitive** (faulty reasoning, often correctable with information) and **emotional** (feelings that override reason, harder to fix). Here are the ones every investor must know.

Bias	What it is	Market consequence
Overconfidence	Overestimating one's knowledge, precision, and control	Excessive trading, under-diversification, ignoring risk
Anchoring	Over-relying on an initial reference number	Sticky valuations, slow reaction to news, "it must come back to my buy price"
Loss aversion	Losses hurt ~2x more than gains please	Holding losers, selling winners, panic selling
Herding	Following the crowd rather than one's own analysis	Bubbles, crashes, momentum, fads
Framing	Decisions change with how a choice is worded	Inconsistent risk-taking, product mis-selling
Recency	Overweighting the most recent events	Chasing performance, extrapolating trends
Confirmation	Seeking evidence that supports prior beliefs	Ignoring red flags, echo chambers, thesis lock-in
Mental accounting	Treating money differently by arbitrary category	Irrational spending, "house money" gambling
Availability	Judging odds by ease of recall	Overreacting to vivid news
Hindsight	"I knew it all along" after the fact	False confidence, poor learning
Status quo / endowment	Overvaluing what one already owns	Inertia, failure to rebalance
Regret aversion	Avoiding action to avoid future regret	Paralysis, staying in cash, refusing to sell

Let us go deeper on the seven the syllabus flags.

Overconfidence. Ask a room of drivers whether they are above-average; roughly 80% say yes. Investors are worse. Overconfidence has three flavors: *overprecision* (confidence intervals too narrow), *overestimation* (thinking you're better than you are), and *overplacement* (thinking you're better than others). Barber and Odean's famous study of 66,000 brokerage accounts (2000) found the most active traders earned **6.5 percentage points per year less** than the market after costs — and that men traded 45% more than women and earned correspondingly less. The mechanism: overconfidence → overtrading → transaction costs and

bad timing eat returns. Overconfidence typically *rises* after a run of gains (people credit skill, not luck), which is exactly when caution is most needed.

Anchoring. In a classic experiment, subjects spun a rigged wheel landing on 10 or 65, then estimated the percentage of African nations in the UN. Those who saw 65 guessed far higher — despite the number being visibly random. In markets, the anchor is often the **purchase price** ("I'll sell when it gets back to what I paid") or a **52-week high** or an analyst's **round-number target**. Anchoring makes valuations sticky and causes **underreaction** to genuine news, because the mind refuses to move far from its anchor.

Loss aversion & the disposition effect. Because losses hurt disproportionately, investors **hold losing stocks too long** (selling would "realize" the painful loss and admit a mistake) and **sell winners too soon** (to lock in the pleasant gain before it evaporates). This is the **disposition effect** (Shefrin & Statman, 1985). It is exactly backwards from optimal tax behavior — you should harvest losses and let winners run — and it degrades returns, because winners tend to keep winning (momentum) and losers to keep losing.

Herding. Humans are social animals; there is safety and status in the crowd. Herding has both an *informational* basis ("everyone else is buying — maybe they know something") and a *reputational* basis (a fund manager who is wrong alongside everyone else keeps her job; one who is wrong alone gets fired — "it is better to fail conventionally"). Herding is the engine of **bubbles**: rising prices attract buyers, whose buying raises prices further, in a self-reinforcing loop detached from fundamentals.

Framing. Tversky and Kahneman's "Asian disease problem": people told a program "saves 200 of 600 lives" overwhelmingly choose it; people told the same program means "400 of 600 die" reject it. Identical facts, opposite choices — driven purely by the gain frame versus the loss frame. In finance, framing explains why "80% of funds beat this benchmark" sells better than "20% underperform," and why a stock quoted as "down 30% from its high" feels different from the same stock quoted as "up 15% this year."

Recency. The mind gives outsized weight to what just happened. After three good years, investors extrapolate them forever and pile in at the top; after a crash, they assume the pain is permanent and sell at the bottom. Recency bias is why retail fund flows are notoriously **procyclical** — money floods into asset classes *after* they have already risen. Morningstar's "behavior gap" studies repeatedly show the *dollar-weighted* return investors actually earn trailing the *time-weighted* return of the very funds they own, precisely because they buy high and sell low.

Confirmation. Once we hold a thesis ("this stock is a winner"), we hunt for supporting evidence and dismiss contradicting evidence. We read the bullish analysts, follow the bullish forums, and rationalize the bad quarter. Confirmation bias creates **thesis lock-in** and is why disciplined investors deliberately seek out the *bear case* and pre-mortem their own ideas.

4.3 From biases to anomalies and mispricing

The payoff of all this psychology is that it *predicts* the market anomalies that embarrass the EMH.

- **Underreaction → Momentum & Post-Earnings-Announcement Drift (PEAD).** Anchoring and conservatism make investors slow to fully incorporate news. So prices drift in the direction of the surprise for weeks or months. Stocks that beat earnings keep rising (Jegadeesh & Titman, 1993 documented that past 3–12 month winners outperform for the next 3–12 months).
- **Overreaction → Long-term Reversal & the Value Premium.** Representativeness makes investors extrapolate too far — they overprice "glamour" growth stocks and underprice beaten-down "value" stocks. Over 3–5 years this reverses: De Bondt & Thaler (1985) showed prior losers outperform prior

winners. Fama and French had to add a *value* (HML) factor to their model to capture returns their efficient-market framework couldn't otherwise explain.

- **Sentiment** → **Excess volatility, bubbles, closed-end fund discounts.** When collective mood swings, prices swing more than fundamentals — Shiller's excess volatility. Closed-end funds trade below NAV when retail sentiment is bearish (Lee, Shleifer, Vishny, 1991), a direct fingerprint of noise traders.
- **The Equity Premium Puzzle.** Stocks have historically returned ~5–6% more than bonds — far more than plausible risk aversion justifies. Benartzi and Thaler's **myopic loss aversion** explains it: loss-averse investors who check their portfolios frequently see many small losses, feel the ~2.25× pain, and demand a fat premium to hold stocks at all.

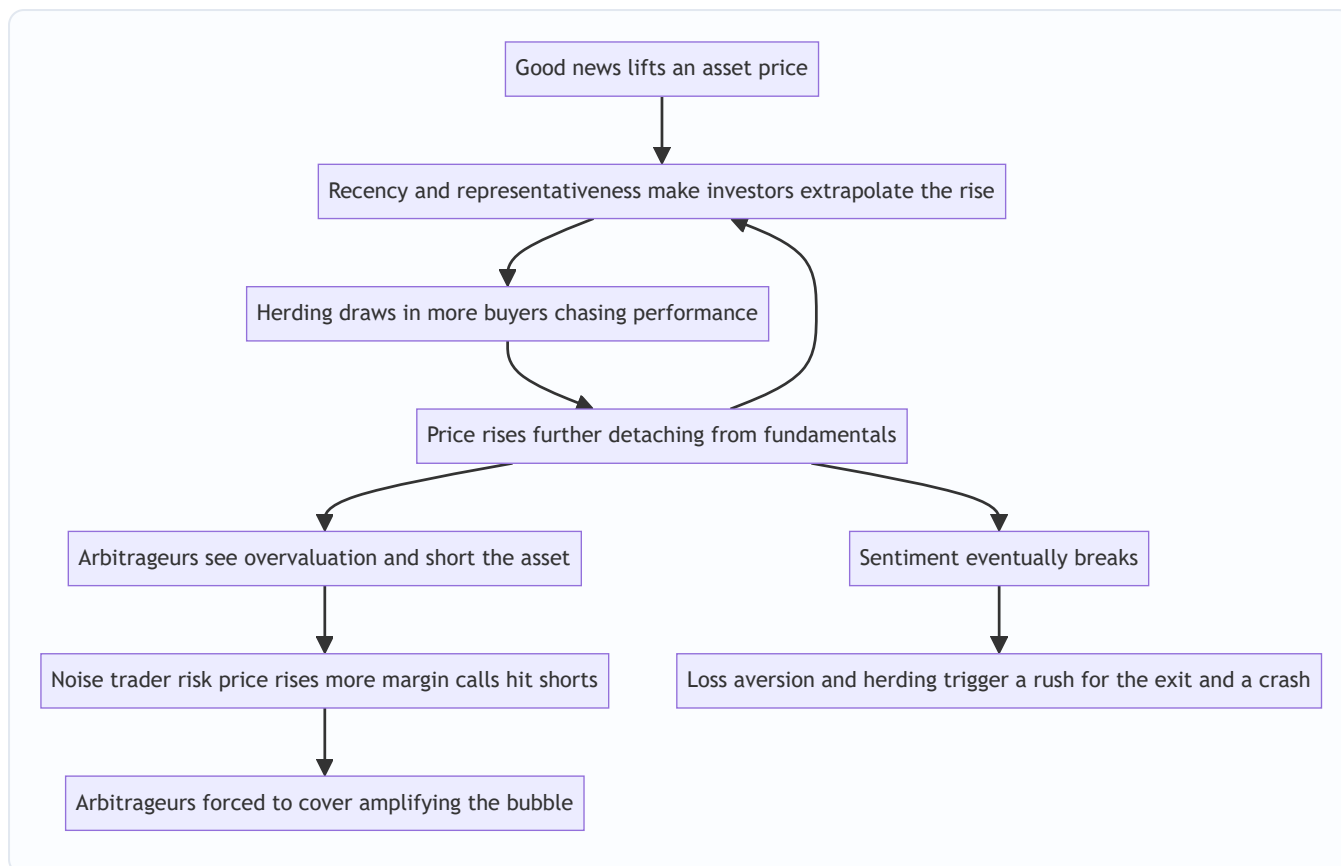
4.4 Limits to arbitrage — why smart money can't fix it

If mispricings are real, why don't hedge funds instantly erase them? Because arbitrage is not the free lunch textbooks imply:

- **Fundamental risk** — the mispriced asset may have no perfect substitute to hedge with; the "cheap" stock might get cheaper because the whole sector falls.
- **Noise-trader risk** (Shleifer & Vishny, 1997) — the mispricing can *widen* before it corrects. An arbitrageur short an overpriced bubble stock can be wiped out by margin calls before being proven right. Julian Robertson's Tiger fund shorted dot-coms too early and was forced to close in early 2000, just before the crash he correctly predicted.
- **Implementation costs** — short-selling is expensive, sometimes impossible (hard-to-borrow), and subject to recall.
- **Capital / horizon constraints** — arbitrageurs manage *other people's money*. When positions move against them, clients redeem exactly when the opportunity is best, forcing liquidation at the worst time.

These frictions are why the EMH is not a law of nature but an *approximation* that breaks down precisely when biases are strongest and arbitrage weakest.

Figure 3 — How a bubble is built and why arbitrage fails to stop it.



5. Worked / Applied Examples

Example 1 — Loss aversion quantified (Prospect Theory math)

You are offered a coin-flip: heads you win ₹1,000, tails you lose ₹800. Expected value = $0.5(1000) + 0.5(-800) = +₹100$, a positive-EV bet a rational agent takes.

Now evaluate it through the prospect-theory value function with $\lambda = 2.25$ (ignore curvature for simplicity):

$$V = 0.5 \times v(1000) + 0.5 \times v(-800) = 0.5(1000) + 0.5(-2.25 \times 800) \\ V = 500 + 0.5(-1800) = 500 - 900 = \mathbf{-400}$$

The *subjective* value is **negative**, so a loss-averse person **rejects** a bet with positive expected value. This single calculation explains why so many investors refuse sensible risks, hoard cash, and demand an outsized equity premium. It also shows *how large* a gain must be to tempt them: to make $V = 0$, the winning payoff W must satisfy $0.5W = 0.5(2.25 \times 800)$, i.e. **$W = ₹1,800$** — you must offer more than *double* the potential loss before the average person will flip the coin.

Example 2 — The disposition effect destroys after-tax returns

Priya holds two stocks, each bought for ₹100:

- **Stock A** is now ₹130 (up ₹30, a winner).
- **Stock B** is now ₹70 (down ₹30, a loser).

She needs ₹70,000 in cash. The disposition effect predicts she sells the **winner (A)** — realizing a gain feels good and "locks in" the profit — while she clings to the **loser (B)**, unwilling to admit the mistake, telling herself "it'll come back to ₹100."

Why this is doubly wrong:

1. **Tax.** In most regimes (including India's capital-gains framework), selling the winner *triggers a taxable gain* while selling the loser would *harvest a deductible loss* to offset other gains. She has chosen the tax-*inefficient* option.
2. **Momentum.** Empirically winners tend to keep outperforming and losers to keep underperforming over 3–12 months. She has sold the stock more likely to rise and kept the one more likely to fall.

The rational move — sell the loser B, harvest the tax loss, retain the momentum winner A — is the exact opposite of what loss aversion whispers. Studies estimate the disposition effect costs individual investors on the order of **1.5–4% per year**.

Example 3 — Framing changes a real investment decision

A financial advisor pitches the same product two ways:

- **Frame X (gain):** "This structured note has a **90% chance** of returning your capital plus 8%."
- **Frame Y (loss):** "This structured note has a **10% chance** of losing money."

The mathematics is identical. Yet clients accept Frame X far more often. Overweighting the salient outcome and the gain framing makes X feel safe and Y feel dangerous. Regulators know this — which is why disclosure rules increasingly *mandate* both frames (and standardized risk labels) to neutralize the manipulation.

Apply the same lesson to your own portfolio: a stock "down 40% from its 52-week high" (loss frame, anchored to the high) and a stock "up 12% over 12 months, trading at 14× earnings" (neutral frame) can be the *same stock*. Always re-frame a decision at least two ways before acting; if your choice flips, System 1 is driving.

Example 4 — Overconfidence and the cost of overtrading

Rahul is confident he can pick stocks. He turns over his ₹10 lakh portfolio 200% per year (buys and sells the whole thing twice), paying 0.5% in brokerage, spread, and impact per round-trip. His timing skill is, honestly, zero — his gross return equals the market's 12%.

- **Gross return:** 12% → ₹1,20,000.
- **Trading costs:** $200\% \text{ turnover} \times 0.5\% \times ₹10,00,000 = ₹10,000/\text{year drag} = 1.0\%$.
- **Net return:** 11.0% → ₹1,10,000.

A disciplined index investor with 10% turnover pays 0.05% and nets 11.95%. Over 20 years, that ~0.95% annual gap compounds: ₹10 lakh at 11.95% grows to ~₹94.6 lakh; at 11.0% to ~₹80.6 lakh — a **₹14 lakh penalty** purely from overconfidence-driven activity, with no skill assumed at all. This is Barber and Odean's finding in miniature: "*Trading is hazardous to your wealth.*"

6. Connections

Behavioral finance does not stand alone; it plugs into nearly every other topic in this guide.

- **Efficient Market Hypothesis (Ch. on market efficiency).** Behavioral finance is the loyal opposition to EMH. It doesn't claim markets are *always* wrong — arbitrage does work much of the time — but that efficiency is bounded and breaks under sentiment and arbitrage limits. The debate crystallized when Fama (EMH) and Shiller (behavioral) *shared* the 2013 Nobel Prize.

- **Modern Portfolio Theory & CAPM.** MPT assumes rational, mean-variance-optimizing investors. Behavioral portfolio theory (Shefrin & Statman) instead describes investors building portfolios in *mental-accounting layers* — a "safety" bucket and a "aspiration/lottery" bucket — which looks nothing like a single efficient frontier and explains why real portfolios are under-diversified.
- **Factor investing & anomalies.** The value and momentum factors that power quant strategies are, in the behavioral reading, *compensation for exploiting other investors' overreaction and underreaction*. Behavioral finance provides the economic story behind the statistics.
- **Risk management.** Value-at-Risk and stress testing exist partly because loss aversion and herding make crises fat-tailed and correlated — normal-distribution models systematically underestimate crash risk.
- **Corporate finance.** Managers are biased too: overconfident CEOs over-invest and over-acquire (Malmendier & Tate); anchoring shows up in M&A reference prices; behavioral signaling underlies dividend "stickiness."
- **Personal financial planning & product design.** "Nudge" architecture (Thaler & Sunstein) — auto-enrolment in pensions, default contribution escalation — uses status-quo bias *for* the investor rather than against them. Richard Thaler won the 2017 Nobel for exactly this bridge from bias to policy.

7. Key Terms

- **Behavioral finance** — the study of how psychological biases and limits to arbitrage cause prices to deviate from fundamental value.
- **Prospect theory** — descriptive theory of choice under risk featuring reference dependence, loss aversion, an S-shaped value function, and probability weighting.
- **Reference point** — the baseline (often purchase price or status quo) against which outcomes are judged as gains or losses.
- **Loss aversion** — losses loom larger than equivalent gains, with coefficient $\lambda \approx 2.25$.
- **Disposition effect** — the tendency to sell winners too early and hold losers too long.
- **Heuristic** — a mental shortcut (representativeness, availability, anchoring) that is efficient but error-prone.
- **System 1 / System 2** — fast intuitive vs. slow deliberate modes of thought.
- **Overconfidence** — overestimating one's knowledge, precision, or control; drives overtrading.
- **Anchoring** — over-reliance on an initial reference number when estimating.
- **Herding** — imitating the crowd rather than acting on independent analysis.
- **Framing effect** — decisions changing based on how equivalent information is presented.
- **Recency bias** — overweighting recent events and extrapolating them.
- **Confirmation bias** — seeking evidence that supports, and ignoring evidence that contradicts, one's prior belief.
- **Mental accounting** — treating money differently depending on its arbitrary source or label.
- **Myopic loss aversion** — frequent evaluation combined with loss aversion, explaining the equity premium puzzle.
- **Limits to arbitrage** — fundamental risk, noise-trader risk, and cost/capital constraints that prevent smart money from correcting mispricing.
- **Noise trader** — an investor trading on sentiment or non-information rather than fundamentals.

- **Anomaly** — an empirical return pattern (momentum, value, PEAD) not explained by the efficient-market/CAPM model.
- **Equity premium puzzle** — the historically large excess return of stocks over bonds, too big for standard risk aversion.
- **Nudge** — a choice-architecture design that steers behavior while preserving freedom of choice.

8. Common Confusions

"Behavioral finance says markets are always wrong / EMH is useless." No. It says markets are *usually roughly* efficient but deviate systematically under specific conditions. Arbitrage genuinely works most of the time. The claim is about *bounds* on efficiency, not its abolition.

"Biases mean investors are stupid." No. Biases are the by-products of *adaptive* mental shortcuts that serve us well in most of life. Highly intelligent professionals exhibit the same biases — sometimes more, because overconfidence scales with perceived expertise.

"Loss aversion is the same as risk aversion." Different. *Risk aversion* (classical) is about the curvature of utility over total wealth and applies everywhere. *Loss aversion* is about the asymmetric pain of losses versus gains around a reference point — and it makes people *risk-seeking* in the loss domain (gambling to break even), which pure risk aversion can never do.

"If a bias is known, it disappears / can be arbitrated away." Awareness helps but does not cure emotional biases — you can *know* about loss aversion and still feel the panic. And even fully-recognized mispricings persist because of limits to arbitrage. Knowing the dot-com bubble was a bubble in 1999 did not make it safe to short.

"Momentum and value contradict each other." They operate on different horizons and different biases: *underreaction* drives short-term momentum (3–12 months); *overreaction* drives long-term reversal / value (3–5 years). Both can be true simultaneously.

"Herding is irrational for the individual." Not necessarily — herding can be *individually* rational (protecting your job, inferring others' information) while being *collectively* destabilizing. That gap between private rationality and public outcome is precisely what makes bubbles hard to stop.

"Behavioral finance is just a list of biases." The list is the surface. The theory is the two-pillar structure (biases + limits to arbitrage) and the *predictions* it makes about anomalies. An interviewer wants the mechanism, not the vocabulary.

9. Recap

- Classical finance assumes rational agents and efficient markets; real investors are neither, and their errors are **systematic and correlated**, so they move prices rather than cancel out.
- Cognition runs on a lazy **System 2** and an intuitive, bias-prone **System 1**; under stress and overload, System 1 dominates financial decisions.
- **Prospect theory** — reference dependence, **loss aversion** ($\lambda \approx 2.25$), an S-shaped value function (risk-averse in gains, risk-seeking in losses), and probability weighting — is the descriptive replacement for expected utility.
- Core biases: **overconfidence** (overtrading), **anchoring** (sticky prices, underreaction), **loss aversion** (disposition effect), **herding** (bubbles), **framing** (inconsistent choices), **recency** (performance chasing),

confirmation (thesis lock-in).

- These biases *predict* market **anomalies**: underreaction → **momentum & PEAD**; overreaction → **long-term reversal & the value premium**; sentiment → **excess volatility, bubbles, closed-end-fund discounts**; myopic loss aversion → the **equity premium puzzle**.
- Mispricings survive because of **limits to arbitrage**: fundamental risk, **noise-trader risk**, short-sale costs, and capital/horizon constraints — "the market can stay irrational longer than you can stay solvent."
- The practical edge: control your *own* biases (rules, checklists, index discipline) and, cautiously, exploit *others'* (value, momentum, contrarian sentiment) — while respecting that arbitrage is risky.

10. Quick-Reference / Interview Points

The one-sentence definition. "Behavioral finance explains asset prices using two facts the standard model ignores: investors have systematic psychological biases, and arbitrage is too limited to fully correct the resulting mispricings."

If asked 'Does behavioral finance disprove EMH?' "It refines it. Markets are approximately efficient because arbitrage works most of the time, but efficiency is *bounded*. When sentiment is extreme and arbitrage is constrained, prices detach from value — that's why Fama and Shiller shared the 2013 Nobel."

The three-part bias framework to recite. 1. *Belief biases* (how we form views): overconfidence, anchoring, representativeness, availability, confirmation, recency. 2. *Preference biases* (how we value outcomes): loss aversion, prospect theory, mental accounting, regret aversion. 3. *Social biases* (how we act with others): herding, information cascades.

Numbers to have ready. - Loss-aversion coefficient $\lambda \approx 2.0$ – 2.5 (use 2.25). - Barber & Odean: most active traders underperformed by $\sim 6.5\%$ /yr; men traded $\sim 45\%$ more than women. - De Bondt & Thaler (1985): 3–5 year losers beat winners (overreaction). - Jegadeesh & Titman (1993): 3–12 month winners beat losers (momentum/underreaction). - Equity premium historically ~ 5 – 6% , "puzzlingly" large.

Anomaly → bias mapping (memorize this table).

Anomaly	Driving bias	Horizon
Momentum / PEAD	Anchoring, conservatism (underreaction)	Short (3–12 mo)
Value premium / reversal	Representativeness (overreaction)	Long (3–5 yr)
Excess volatility & bubbles	Herding, recency, sentiment	Variable
Closed-end fund discounts	Noise-trader sentiment	Persistent
Equity premium puzzle	Myopic loss aversion	Structural

The disposition effect in one line. "Investors sell winners too soon and hold losers too long — the opposite of both tax efficiency and momentum."

Why mispricings persist (limits to arbitrage), four bullets. Fundamental risk · noise-trader risk (it can get worse before it gets better) · short-sale/implementation costs · capital and client-redemption constraints.

How a good investor uses this. - *Defense*: rules-based process, pre-commitment, checklists, systematic rebalancing, low turnover, seeking the bear case to fight confirmation bias, ignoring the purchase-price

anchor. - *Offense*: disciplined value and momentum tilts, contrarian positioning against sentiment extremes, patient capital that can outlast noise-trader risk.

Killer closing line for an interview. "The market is a voting machine in the short run and a weighing machine in the long run — behavioral finance is the theory of the voting, and limits to arbitrage is the theory of why the weighing takes so long."

Q&A — Behavioral Finance

Practice bank for Chapter 13. Work each question before reading the answer. Recurring theme: prices deviate from value because biases are **systematic and correlated** (they don't cancel) and because **arbitrage is limited** (mispricing survives). Both pillars are needed — remember which one you are invoking.

Section A — Concept Checks

A1. State the core claim of behavioral finance in one sentence. Asset prices are set by real humans subject to systematic, predictable cognitive and emotional biases, and whose ability to arbitrage those errors away is limited — so prices can deviate from fundamental value, sometimes for a long time.

A2. Behavioral finance has two pillars. Name them and explain why *both* are required. Pillar one is **limited rationality** — investors make systematic (not random) errors, so mispricing is created. Pillar two is **limits to arbitrage** — smart money cannot cheaply and risklessly correct the error, so mispricing persists. Remove pillar one and there is nothing to correct; remove pillar two and rational arbitrageurs instantly erase any mispricing and the standard model reasserts itself. Biases create the gap; limits to arbitrage let it survive.

A3. Why don't individual investors' errors simply cancel out in aggregate? Because the errors are **correlated**, not independent. If 10,000 investors each made an independent random mistake, the mistakes would average to roughly zero. But human brains are wired similarly, exposed to the same news, and susceptible to the same emotional contagion — so most people lean the *same* wrong way at the same time. Correlated errors accumulate and move the whole market in one direction.

A4. Distinguish System 1 and System 2 thinking, and say where biases live. System 1 is fast, automatic, intuitive, emotional, and effortless — and it is where most biases live. System 2 is slow, deliberate, logical, and effortful. System 2 is "lazy" (it burns glucose and attention), so under time pressure, information overload, or emotional arousal — the normal conditions of markets — System 1 takes the wheel and biased intuition drives decisions.

A5. State the four defining features of Prospect Theory. (1) **Reference dependence** — outcomes are judged as gains/losses relative to a reference point (often purchase price or status quo), not in terms of final wealth. (2) **Loss aversion** — losses hurt ~2–2.5× more than equivalent gains please. (3) **Diminishing sensitivity / S-shaped value function** — concave (risk-averse) in gains, convex (risk-seeking) in losses. (4) **Probability weighting** — small probabilities are overweighted (lottery + insurance), moderate-to-high probabilities underweighted.

A6. What is the disposition effect, and which Prospect Theory feature drives it? The tendency to **sell winners too soon and hold losers too long**. It is driven by loss aversion combined with reference dependence: selling a loser forces you to "realize" the painful loss and admit a mistake, while selling a winner locks in the pleasant gain. It is backwards from optimal behavior on two counts — tax (you should harvest losses) and momentum (winners tend to keep winning).

A7. Distinguish loss aversion from ordinary risk aversion. Classical **risk aversion** concerns the curvature of utility over *total wealth* and makes people prefer certainty everywhere. **Loss aversion** concerns the asymmetric pain of losses versus gains *around a reference point* — and crucially it makes people **risk-seeking in the loss domain** (they gamble to break even), which pure risk aversion can never do.

A8. Give the anomaly-to-bias mapping for momentum versus the value premium. Momentum / PEAD

is driven by *underreaction* (anchoring and conservatism make investors slow to price news), over a short horizon of 3–12 months. The **value premium / long-term reversal** is driven by *overreaction* (representativeness makes investors extrapolate too far), over a long horizon of 3–5 years. Different biases, different horizons — they do not contradict.

A9. What is myopic loss aversion and which puzzle does it explain? Myopic loss aversion combines loss aversion with **frequent evaluation** of one's portfolio. An investor who checks often sees many small losses, feels the $\sim 2.25\times$ pain, and therefore demands a large premium to hold stocks. It explains the **equity premium puzzle** — the historically large ($\sim 5\text{--}6\%$) excess return of stocks over bonds that ordinary risk aversion cannot justify.

A10. List the four main limits to arbitrage. (1) **Fundamental risk** — no perfect substitute to hedge with. (2) **Noise-trader risk** — the mispricing can widen before it corrects. (3) **Implementation costs** — short-selling is costly, sometimes impossible, subject to recall. (4) **Capital / horizon constraints** — arbitrageurs run other people's money and face redemptions exactly when the opportunity is best.

A11. Distinguish cognitive biases from emotional biases and say which are easier to fix. Cognitive biases are faulty reasoning (anchoring, confirmation, availability) — often correctable with information, education, and process. **Emotional biases** are feelings that override reason (loss aversion, regret aversion) — harder to fix because awareness does not dissolve the felt panic. This is why you can *know* about loss aversion and still act on it.

A12. Why can herding be individually rational yet collectively destabilizing? Herding has an *informational* basis ("everyone is buying — maybe they know something") and a *reputational* basis ("it is better to fail conventionally" — a manager wrong with the crowd keeps her job; wrong alone she is fired). Each of these can make following the crowd privately sensible, even while the aggregate result — self-reinforcing bubbles and crashes — is collectively destructive.

Section B — Applied / Numerical Problems (full working)

B1. Loss aversion rejects a positive-EV bet. Coin flip: heads win ₹1,000, tails lose ₹800. Evaluate with the value function using $\lambda = 2.25$ (ignore curvature).

- Expected value: $0.5(1000) + 0.5(-800) = +₹100$ — a rational agent takes it.
- Prospect-theory value: $V = 0.5v(1000) + 0.5v(-800) = 0.5(1000) + 0.5(-2.25 \times 800)$.
- $V = 500 + 0.5(-1800) = 500 - 900 = \mathbf{-400}$.

Reconcile: the *subjective* value is negative, so a loss-averse person **rejects** a +EV bet. This is the mechanism behind cash-hoarding and the fat equity premium. ✓

B2. How large must the gain be to tempt the loss-averse player? Same flip, same $\lambda = 2.25$, but solve for the winning payoff W that makes the player indifferent ($V = 0$), holding the ₹800 loss fixed.

- Set $0.5v(W) + 0.5(-2.25 \times 800) = 0$.
- $0.5v(W) = 0.5(1800) \Rightarrow W = 1800$.

Reconcile: the potential win must be **₹1,800** — more than *twice* the ₹800 potential loss — before the average person will flip. Loss aversion demands roughly a 2.25:1 gain-to-loss ratio just to break even psychologically. ✓

B3. The disposition effect and after-tax return. Priya bought Stock A and Stock B at ₹100 each. Now A = ₹130 (winner), B = ₹70 (loser). She needs cash and can sell only one. Capital-gains regime: gains are taxable, realized losses are deductible against other gains. Momentum: winners tend to keep outperforming, losers to keep underperforming over 3–12 months.

- **Disposition prediction:** she sells the **winner A** (feels good, locks the gain) and holds the **loser B** ("it'll come back to ₹100").
- **Tax:** selling A triggers a taxable ₹30 gain; selling B would harvest a deductible ₹30 loss. She chose the *tax-inefficient* option.
- **Momentum:** she sold the stock more likely to rise and kept the one more likely to fall.

Reconcile: the correct move — sell loser B, harvest the loss, let winner A run — is the exact opposite of what loss aversion whispers. Studies estimate the disposition effect costs individual investors on the order of **1.5–4% per year**. ✓

B4. Overconfidence and the cost of overtrading. Rahul turns over his ₹10,00,000 portfolio 200%/year, paying 0.5% all-in per round-trip. His stock-picking skill is genuinely zero: gross return = market's 12%. Compare to an index investor at 10% turnover paying 0.05%.

- Rahul gross: 12% → ₹1,20,000. Trading drag: $200\% \times 0.5\% = 1.0\%$ = ₹10,000. **Net 11.0%.**
- Index net: $12\% - 0.05\% \approx 11.95\%$.
- 20-year compounding: ₹10 lakh at 11.95% → ~₹94.6 lakh; at 11.0% → ~₹80.6 lakh.

Reconcile: a ~0.95% annual gap, driven purely by overconfidence-fueled activity with *no* skill assumed, compounds to a **~₹14 lakh penalty**. This is Barber & Odean in miniature: "trading is hazardous to your wealth." ✓

B5. Framing flips a decision with identical math. An advisor pitches one structured note two ways. Frame X: "90% chance of returning your capital plus 8%." Frame Y: "10% chance of losing money." Are these the same offer, and why do clients respond differently?

- The two frames are **mathematically identical** — a 90%/10% split over the same outcomes.
- Clients accept **Frame X far more often**: the gain frame plus overweighting of the salient positive outcome makes X feel safe and Y feel dangerous.

Reconcile: identical facts, opposite choices — pure framing. This is why disclosure regulation increasingly mandates *both* frames and standardized risk labels to neutralize the manipulation. Personal takeaway: re-frame every decision at least two ways before acting; if your choice flips, System 1 is driving. ✓

B6. Myopic loss aversion and evaluation frequency. Stocks return +1% per month on average with enough volatility that any given month is down ~40% of the time. Investor Ravi checks his portfolio *monthly*; investor Meera checks *once a year*. Using loss aversion ($\lambda = 2.25$), explain who is more likely to abandon equities.

- Ravi sees ~4–5 losing months per year, each triggering the $\sim 2.25\times$ pain. Frequent exposure to losses makes the felt experience of holding stocks net-painful.
- Meera evaluates on an annual horizon, where positive years dominate and losses are rarer and smaller in frequency, so the felt experience is net-positive.

Reconcile: the *same* asset produces very different subjective experiences purely because of **evaluation frequency**. The more often you look, the more loss aversion bites — so Ravi is far more likely to sell equities and demand a higher premium. The practical fix: look less often. ✓

Section C — Interview-Style Questions (model answers)

C1. "If markets are efficient, why does your fund exist?" Because efficiency is *bounded*, not absolute. Behavioral finance says two things the standard model ignores: investors have systematic, correlated psychological biases that push prices away from value, and arbitrage is too risky and capital-constrained to fully correct them. That leaves persistent, exploitable pockets — momentum from underreaction, the value premium from overreaction, sentiment-driven mispricing where short-selling is hard. My fund exists to harvest those, disciplined by the knowledge that arbitrage is risky, so I size positions to survive noise-trader risk.

C2. "Does behavioral finance disprove the Efficient Market Hypothesis?" No — it refines it. Markets are approximately efficient because arbitrage genuinely works most of the time. But efficiency is bounded: when sentiment is extreme and arbitrage is constrained, prices detach from value. Behavioral finance is the loyal opposition, not a wrecking ball. The debate is settled enough that Fama, who built EMH, and Shiller, who documented excess volatility and bubbles, *shared* the 2013 Nobel Prize. The honest position is a spectrum, not a binary.

C3. "Explain Prospect Theory as if I've never heard of it." Classical theory says people value final wealth and maximize expected utility. Prospect Theory, from Kahneman and Tversky, says that's not how humans actually choose. Four facts: we judge outcomes as gains or losses against a **reference point**, not absolute wealth; **losses hurt about 2.25× more** than equal gains please; the value function is **S-shaped** — risk-averse in gains but *risk-seeking* in losses, so we gamble to avoid a sure loss; and we **distort probabilities**, overweighting rare events, which is why the same person buys lottery tickets *and* insurance. Those four features explain the disposition effect, the equity premium puzzle, and most investor misbehavior.

C4. "Give me one bias, its mechanism, and what it costs investors." Overconfidence. The mechanism: investors overestimate their knowledge, precision, and control — roughly 80% think they're above-average — which leads to overtrading and under-diversification. Barber and Odean studied 66,000 brokerage accounts and found the most active traders underperformed the market by about **6.5 percentage points a year** after costs; men traded ~45% more than women and earned correspondingly less. Overconfidence rises *after* a run of gains, because people credit skill rather than luck — exactly when caution is most warranted. The cure is a rules-based, low-turnover process.

C5. "Momentum and the value premium seem to contradict each other. Do they?" No — they operate on different horizons and different biases. Short-term **momentum** (3–12 months) is driven by *underreaction*: anchoring and conservatism make investors slow to fully price news, so prices drift in the direction of the surprise — that's Jegadeesh and Titman, and post-earnings-announcement drift. Long-term **reversal / the value premium** (3–5 years) is driven by *overreaction*: representativeness makes investors extrapolate glamour stocks too far and punish value stocks too hard, then it reverses — that's De Bondt and Thaler. Both can be true at once because they describe different frequencies of the same news-processing error.

C6. "Why don't hedge funds just arbitrage away the mispricings you describe?" Because arbitrage isn't the free lunch textbooks imply. Four frictions: **fundamental risk** — often no perfect substitute to hedge with; **noise-trader risk** — the mispricing can widen before it corrects, so you can be right and still be wiped out by margin calls; **implementation costs** — shorting is expensive, sometimes impossible; and **capital constraints** — you run clients' money, and they redeem exactly when the opportunity is best. Julian Robertson's Tiger fund shorted dot-coms too early and was forced to close in early 2000, right before the crash he correctly foresaw. Being right too soon is indistinguishable from being wrong.

C7. "How would you protect yourself and a client from these biases?" Two fronts. Defense against my own biases: a rules-based process with pre-commitment, checklists, systematic rebalancing, low turnover, seeking the bear case to fight confirmation bias, and ignoring the purchase-price anchor. For clients: choice architecture — "nudges" like auto-enrolment use status-quo bias *for* them, and evaluating portfolios less often blunts myopic loss aversion. Offense, cautiously: disciplined value and momentum tilts and contrarian positioning against sentiment extremes, sized to survive noise-trader risk. Thaler won the 2017 Nobel for exactly this bridge from bias to policy.

C8. "What's the single most important number in behavioral finance and why?" The loss-aversion coefficient, $\lambda \approx 2.25$. It says losses hurt roughly 2.25 times more than equal gains please, and that one parameter bends the whole value function. It drives the disposition effect (why we won't realize losses), the equity premium puzzle (why we over-demand compensation for holding stocks), and the general reluctance to take sensible positive-EV risk. If I can only teach a client one fact, it's that their fear of loss is quantifiably more than double their pleasure from gain — and that asymmetry, not the market, is usually their biggest enemy.

Section D — MCQs (with reasoning)

D1. The two pillars of behavioral finance are: (a) rational agents and efficient markets (b) systematic biases and limits to arbitrage (c) System 1 and System 2 (d) momentum and value **Answer: (b).** Biases create mispricing; limits to arbitrage let it persist. Both are needed — remove either and the standard model reasserts itself.

D2. Individual investor errors move prices in aggregate because they are: (a) random and independent (b) small (c) systematic and correlated (d) always cognitive **Answer: (c).** Independent random errors cancel to zero. Correlated errors — shared wiring, shared news, shared emotion — accumulate and push the whole market one way.

D3. In Prospect Theory, the value function is: (a) concave everywhere (b) convex everywhere (c) concave in gains and convex in losses (d) linear **Answer: (c).** S-shaped: risk-averse (concave) in the gain domain, risk-seeking (convex) in the loss domain — the source of gambling to break even.

D4. The disposition effect is the tendency to: (a) sell winners too soon and hold losers too long (b) sell losers too soon and hold winners too long (c) trade too frequently (d) chase recent performance **Answer: (a).** Loss aversion makes realizing a loss painful, so investors cling to losers and lock in gains early — backwards on both tax and momentum grounds. Option (c) is overconfidence; (d) is recency.

D5. The loss-aversion coefficient $\lambda \approx 2.25$ means: (a) gains feel $2.25\times$ better than losses (b) losses hurt $\sim 2.25\times$ more than equal gains please (c) probabilities are weighted by 2.25 (d) stocks beat bonds by 2.25% **Answer: (b).** Losses loom about 2.25 times larger than equivalent gains — the central asymmetry of behavioral finance.

D6. Momentum and post-earnings-announcement drift are best explained by: (a) overreaction (b) underreaction (c) herding only (d) probability weighting **Answer: (b).** Anchoring and conservatism make investors *slow* to fully price news, so prices drift in the surprise direction for weeks to months.

D7. The value premium / long-term reversal is best explained by: (a) underreaction (b) overreaction via representativeness (c) framing (d) mental accounting **Answer: (b).** Representativeness makes investors extrapolate glamour stocks too far and over-punish value stocks; over 3–5 years this overreaction reverses.

D8. Myopic loss aversion explains which puzzle? (a) the closed-end fund discount (b) momentum (c) the equity premium puzzle (d) the January effect **Answer: (c).** Loss aversion plus frequent evaluation makes investors demand a large premium to hold stocks — explaining the puzzlingly large historical equity premium.

D9. Which is NOT one of the limits to arbitrage? (a) fundamental risk (b) noise-trader risk (c) perfect substitutes and free shorting (d) capital/horizon constraints **Answer: (c).** That option is the *absence* of limits. The real limits are fundamental risk, noise-trader risk, implementation/short-sale costs, and capital constraints.

D10. A person who buys both lottery tickets and insurance is displaying: (a) loss aversion (b) probability weighting (overweighting small probabilities) (c) anchoring (d) herding **Answer: (b).** Overweighting rare events makes both the tiny jackpot chance and the tiny disaster chance feel larger than they are — driving lottery buying and insurance buying simultaneously.

D11. Loss aversion differs from classical risk aversion because loss aversion: (a) applies only to wealthy investors (b) makes people risk-seeking in the loss domain (c) is always cognitive (d) disappears once you know about it **Answer: (b).** Risk aversion never produces risk-seeking; loss aversion does, in the loss domain, where people gamble to avoid realizing a sure loss.

D12. An advisor pitches "90% chance of gaining" versus "10% chance of losing" for the same product and gets different acceptance. This is: (a) anchoring (b) the framing effect (c) confirmation bias (d) hindsight bias **Answer: (b).** Identical mathematics, opposite choices, driven purely by gain-versus-loss presentation — the definition of framing.

D13. "The market can remain irrational longer than you can remain solvent" is the practical statement of: (a) the equity premium puzzle (b) noise-trader risk / limits to arbitrage (c) mental accounting (d) System 1 dominance **Answer: (b).** It captures noise-trader risk: an arbitrageur can be correct yet be forced to liquidate before the mispricing corrects.

End of Chapter 13 Q&A. Master three reflexes: (1) always name which pillar you're invoking — biases create mispricing, limits to arbitrage let it survive; (2) keep $\lambda \approx 2.25$ and the anomaly $\rightarrow$ bias $\rightarrow$ horizon table on instant recall; (3) frame behavioral finance as refining, not disproving, EMH — bounded efficiency, Fama and Shiller sharing one Nobel.

Chapter 14 — The Investment Policy Statement

1. The Problem / The Need

Picture a wealthy family that has just sold a business for ₹50 crore. They hire a portfolio manager. On day one, three questions crowd the room. The husband, 58, wants to "not lose money — we already made it." The wife, 55, wants "at least 12% because inflation is killing us." Their financial advisor mutters that they'll need ₹3 crore in eighteen months for their daughter's overseas education, that the money sits in a trust with a specific deed, and that half the corpus is still in unlisted shares that can't be sold for two years.

Every one of these statements is a *constraint* or an *objective*, and they contradict each other. You cannot simultaneously "not lose money" and earn 12%. You cannot lock up capital for growth and also produce ₹3 crore of liquidity next year without planning for it. If the manager simply starts buying stocks, the first bad quarter will end the relationship — the family will accuse him of recklessness, and he will have no document to point to that says "this is the risk *you* agreed to run."

This is the problem the **Investment Policy Statement (IPS)** solves. Before a single rupee is invested, the manager and client must convert vague hopes ("grow the money, but safely") into a precise, written specification: *how much* return is needed, *how much* risk can be tolerated, *when* the money is needed, *what* rules of tax and law apply, and *what* the client's idiosyncrasies are. The IPS is the contract between the investor's real life and the portfolio's structure.

Without it, three failures are almost guaranteed:

1. **Objective drift.** The portfolio ends up reflecting the *manager's* views or the *market's* fashions rather than the client's needs.
2. **Panic at the bottom.** With no pre-agreed risk budget, the client discovers their true risk tolerance only during a crash — the worst possible moment to learn it — and sells.
3. **No accountability standard.** When performance is reviewed, there is no benchmark of what "success" even means, so the conversation degenerates into "you should have bought Bitcoin."

The IPS is not paperwork. It is the *engineering specification* that every downstream decision — asset allocation, security selection, rebalancing, manager evaluation — must satisfy.

2. The Core Idea

An Investment Policy Statement is a written document that specifies a client's investment objectives (return and risk) and constraints (liquidity, time horizon, taxes, legal and regulatory, and unique circumstances), and thereby governs how the portfolio is built and managed.

The mental model is a funnel. At the top are messy human facts — a person's age, wealth, fears, goals, tax bracket, and legal situation. The IPS is the machine that distills those facts into a small set of formal parameters. Out of the bottom comes the single most important output: the **strategic asset allocation** — the long-run policy mix of equities, bonds, cash, and alternatives that is the portfolio's home base.

Two clusters define the IPS, and it is worth burning them into memory:

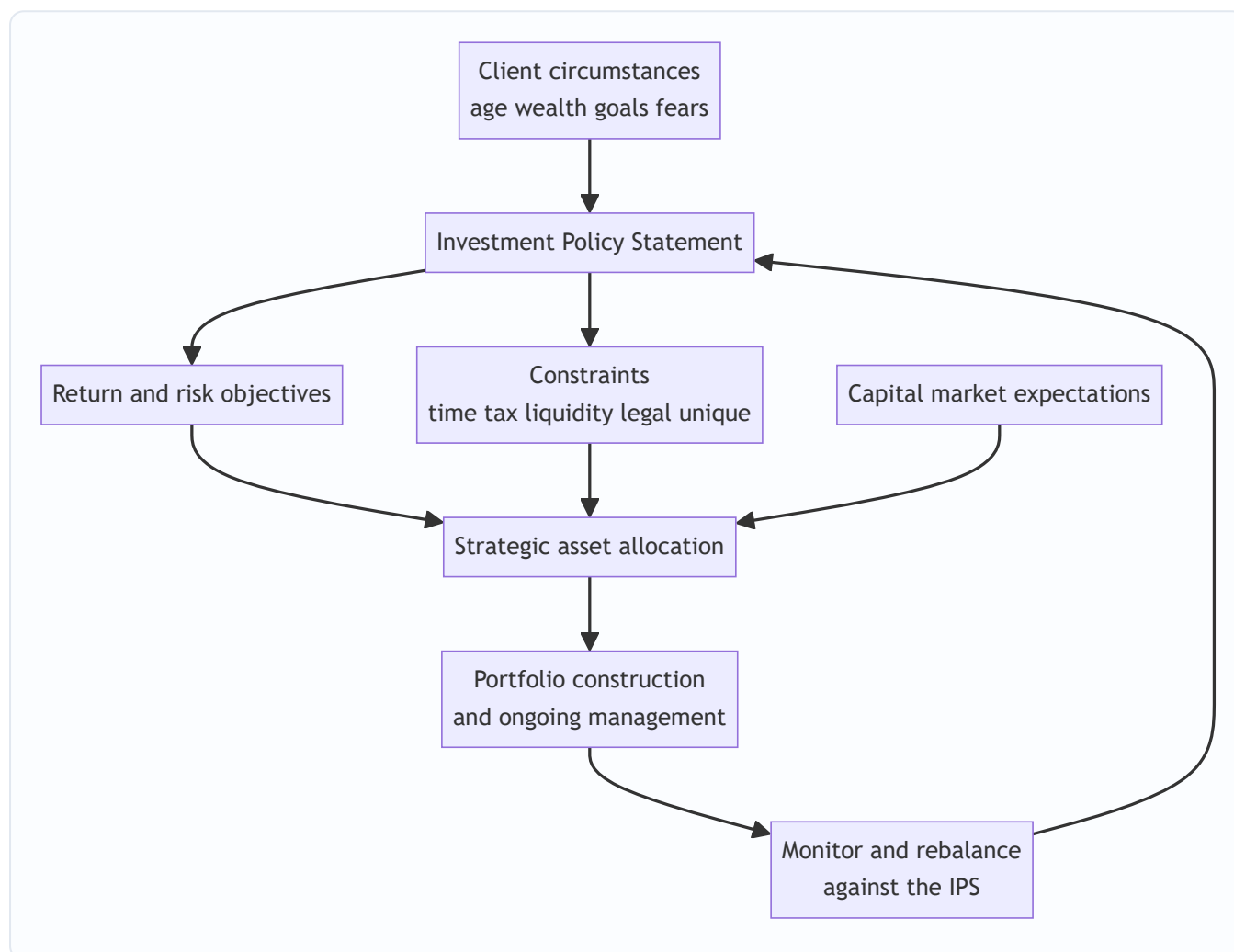
- **Objectives = Return + Risk.** What the portfolio must *achieve* and what volatility or loss it can *endure* on the way.

- **Constraints = the "TTLU" list: Time horizon, Taxes, Liquidity, Legal/regulatory, Unique circumstances.** The *boundaries* inside which the objectives must be pursued.

The IPS says: *Here is the return we must earn, here is the risk we may take to earn it, and here are the five walls of the box we must stay inside while doing so.*

Crucially, the IPS describes the *investor*, not the *market*. Capital-market expectations (what returns asset classes are likely to deliver) are a separate input. The IPS + capital-market expectations together produce the asset allocation. The IPS is durable and changes rarely (a divorce, a windfall, retirement); market expectations change constantly. Keeping them separate is what stops a good policy from being torn up in every panic.

Figure 1 — How the IPS converts client circumstances into a portfolio, combining with market expectations to set the strategic asset allocation.



3. Why / How It Works

Why does a written statement of preferences translate so reliably into a well-run portfolio? Three mechanisms.

First, it forces quantification. "Safe" and "good returns" are useless to a portfolio engineer. The IPS process pins them down: the return objective becomes a *number* (say, "beat inflation by 4% net of fees"), and the risk

objective becomes a *number* (say, "no worse than a 15% drawdown in a bad year, standard deviation under 12%"). Once both are numbers, the feasible asset allocation follows almost mechanically from capital-market assumptions — you simply find the mix on the efficient frontier that meets the return need at or below the risk limit. This connects directly to Markowitz portfolio theory (Chapter 03) and the asset-allocation machinery (Chapter 07): the IPS supplies the *inputs* — the target return and the risk constraint — that the optimiser needs.

Second, it separates two kinds of risk that are constantly confused. *Ability* to take risk is objective — it depends on horizon, wealth relative to needs, and the stability of income. *Willingness* to take risk is psychological — it depends on temperament. A young software engineer with a huge horizon and a stable salary has high *ability* but may have low *willingness* because a 2020-style crash terrifies him. The IPS makes the manager confront both. The governing rule: **when ability and willingness conflict, the portfolio is generally set to the lower of the two, and the manager documents the mismatch and educates the client.** You never force more risk on a nervous client than they can stomach, because they will bail at the bottom and destroy the plan — even if they "could afford" it on paper.

Third, it creates a pre-commitment device against behavioural error. The single largest source of underperformance for real investors is not fees or bad stocks — it is buying high and selling low, driven by greed and fear. The IPS, written in calm times, is a Ulysses-and-the-mast contract: it says *in advance* what the policy mix is and when we rebalance, so that in the panic the answer is already decided. The document out-argues the amygdala.

Finally, the IPS **works because it is dynamic without being fragile.** It is reviewed periodically and after major life events, but it is *not* rewritten every time the market moves. This asymmetry — stable enough to anchor behaviour, flexible enough to reflect genuine changes in the client's life — is exactly what a portfolio needs.

4. Full Content

4.1 Return Objectives

The return objective answers: *what does this portfolio need to earn?* It comes in two flavours that must be reconciled:

- **Required return** — the return the investor *needs* to meet their goals (fund retirement, preserve real capital, meet a spending rule). This is derived from the facts, not preference.
- **Desired return** — the return the investor *wants*. Often higher than required, and often infeasible given the risk constraint.

Returns must be specified with precision on several dimensions:

Dimension	Question it answers	Example
Nominal vs real	Before or after inflation?	"Inflation + 4%" (real) vs "10%" (nominal)
Pre- vs post-tax	Before or after tax?	Pre-tax 12% may be 8.5% post-tax at a 30% rate
Total vs income	Do we count capital gains or only yield?	An endowment spending 4% needs total return; a retiree living on coupons needs income
Gross vs net of fees	Before or after costs?	Always specify net — fees are certain, alpha is not

A well-formed return objective might read: *"The portfolio must achieve a long-term total return, net of fees and after tax, sufficient to grow the corpus at the rate of inflation plus 3.5%, to preserve real purchasing power while funding a 4% annual withdrawal."*

There is an iron discipline here: the return objective must be **consistent with the risk objective**. If a client needs 15% real returns but can only tolerate a 6% standard deviation, the plan is infeasible — the manager must go back and either lower the goal, extend the horizon, increase savings, or accept more risk. The IPS is where this reconciliation happens on paper *before* it happens painfully in reality.

4.2 Risk Objectives

Risk has the two components introduced above — **ability** and **willingness** — and both must be assessed.

Ability to take risk rises with:

- A **longer time horizon** (more time to recover from drawdowns).
- **Greater wealth relative to needs** (a large surplus over required spending absorbs losses).
- **Lower, more flexible spending needs** (discretionary spending can be cut in bad years).
- **Stable, secure income** from outside the portfolio (a tenured salary vs commission income).
- **Low liquidity needs** (no imminent large withdrawals).

Willingness is drawn out through conversation, questionnaires, and history — how did the client behave in 2008 or March 2020? The output is a risk objective expressed as some combination of: a maximum acceptable standard deviation, a maximum tolerable drawdown or loss in a bad year, a shortfall-risk statement ("no more than a 5% chance of the corpus falling below ₹X"), or a value-at-risk limit.

The **reconciliation rule** bears repeating: build to the *lower* of ability and willingness, document the gap, and educate. A client with high ability but low willingness should not be pushed; a client with low ability but high willingness (the dangerous case — a retiree who "loves stocks") must be *restrained*, because their ability is the binding, real-world limit.

4.3 The Five Constraints (TTLU)

Objectives say what to achieve; constraints define the box.

1. Time Horizon. How long until the money is needed? Horizons can be **single-stage** (one goal at a known date) or **multi-stage** (an accumulation phase then a decumulation phase — the classic pre- and post-retirement split). Horizon is the master variable of risk: a long horizon expands the ability to hold volatile, high-return assets like equities, because there is time to ride out cycles. A short horizon compresses the portfolio toward cash and short bonds. A common error is treating a retiree as "short horizon" — a 65-year-old may have a 25-year horizon, and part of the corpus is genuinely long-term.

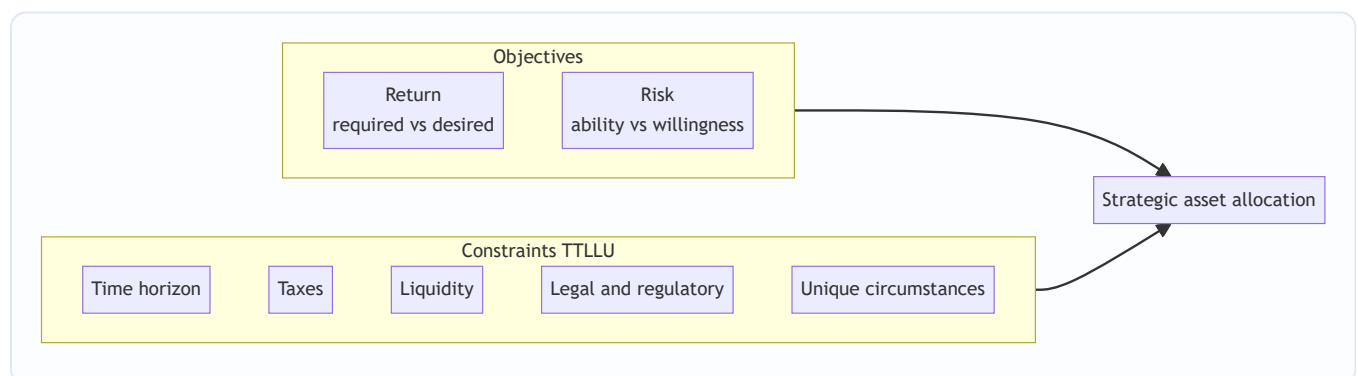
2. Taxes. Taxation shapes both the return target (specify pre- or post-tax) and the *instruments* and *strategies* used. Relevant levers include income tax on interest and dividends, capital-gains tax (and the split between short- and long-term rates), and wealth or estate taxes. Tax-aware management uses tactics such as holding assets long enough to qualify for lower long-term gains rates, harvesting losses to offset gains, and locating tax-inefficient assets (bonds, high-turnover strategies) inside tax-sheltered accounts. In India, for instance, the distinction between long-term and short-term capital gains, and vehicles like ELSS or the tax status of debt vs equity funds, directly change the optimal holding.

3. Liquidity. What cash outflows will the portfolio face, and when? Liquidity needs include planned withdrawals (a child's education, a house purchase, an endowment's annual spending) and an emergency reserve. High liquidity needs force an allocation to cash and highly marketable securities and *penalise* illiquid assets like private equity, real estate, or lock-in instruments. The manager must also distinguish *liquidity requirements* (predictable, plannable) from *liquidity risk* (the danger of being forced to sell illiquid assets at a discount in a crisis). Ignoring liquidity is how leveraged or illiquid portfolios blow up when a withdrawal lands at the wrong time.

4. Legal and Regulatory. What rules bind this specific investor? For individuals, this includes trust documents, prudent-investor standards, and restrictions on the sale of concentrated founder stock. For institutions, it is far heavier: pension funds face funding and fiduciary rules (ERISA in the US; PFRDA/EPFO norms in India); insurers face solvency-capital and reserve regulations (Solvency II, IRDAI rules) that dictate how much can go into equities; banks face capital rules. The IPS must encode every binding legal wall.

5. Unique Circumstances. The catch-all for idiosyncratic constraints that don't fit elsewhere: ESG or religious screens (no alcohol, tobacco, or interest-bearing instruments under Shariah), a large concentrated position in the family business that cannot be diversified, restrictions imposed by a donor on an endowment, ethical exclusions, or a governance requirement to avoid certain countries or sectors. These are as binding as any regulation and must be written down.

Figure 2 — The anatomy of an IPS: two objectives and five constraints feeding the allocation.



4.4 Individual vs Institutional Investors

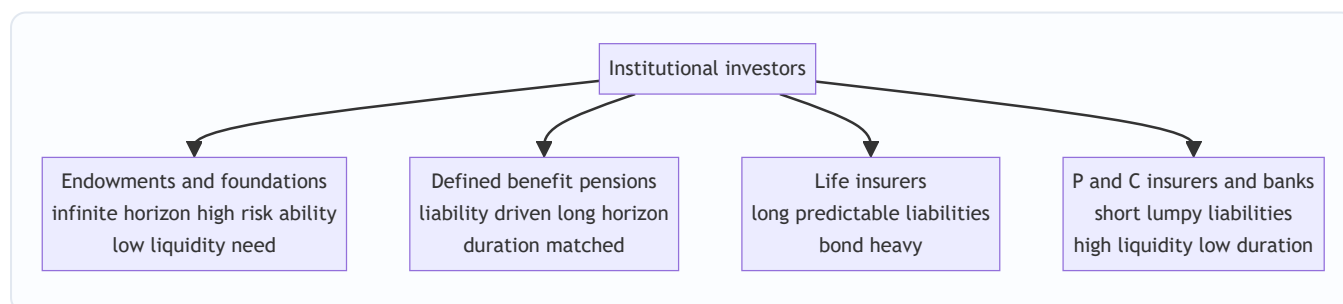
The IPS framework is universal, but the *content* differs sharply between an individual and each type of institution. The differences flow from horizon, liquidity, taxes, and legal status.

Individuals are defined by a **finite, life-cycle horizon** (accumulation then decumulation), **taxable** status (tax is a first-order concern), often **modest ability but variable willingness**, and highly personal unique circumstances. Their goals are typically retirement funding, education, and bequests. Behavioural factors dominate — the manager is as much a coach as an engineer.

Institutions are perpetual or long-horizon pools, usually **tax-exempt** (or tax-advantaged), governed by heavy regulation and formal fiduciary duty. Their willingness is set by governance boards, not emotion. But the three great institutional types differ profoundly from *each other*:

- **Defined-benefit pension funds** promise fixed future payments to retirees. Their true objective is to fund a stream of **liabilities**, so their horizon and risk depend on the age profile of the workforce (a young workforce = long horizon, higher equity; a mature, retiree-heavy plan = shorter, more bonds). This drives **liability-driven investing (LDI)** — matching asset duration to liability duration. Liquidity needs are the net benefit payments. They are the archetype of the institution that must think in *surplus* (assets minus liabilities) terms.
- **Endowments and foundations** (a university endowment, a charitable foundation) are **perpetual** and exist to fund spending *forever* — typically a fixed percentage of assets each year (say 4-5%). Their return objective is famously demanding: **spending rate + inflation + fees**, all in real terms — often 7-8% nominal — because they must both fund current spending *and* preserve real purchasing power for future generations. With an infinite horizon and modest annual liquidity needs, they have very high risk ability, which is why the "Yale model" endowments hold large slugs of illiquid equities, private equity, and real assets.
- **Insurance companies** invest the premiums (the "float") they collect before paying claims. **Life insurers** have long-duration, fairly predictable liabilities and behave somewhat like pensions, emphasising duration-matched fixed income with a bond-heavy core. **Property-casualty (P&C / general) insurers** face short, lumpy, unpredictable liabilities (a catastrophe can trigger huge claims at once), so they demand **high liquidity and short duration**, with only surplus capital going into equities. Insurers are the most *heavily regulated and most liquidity- and solvency-sensitive* of all, with return objectives tightly tied to the yield needed to fund policy reserves.
- **Banks** invest their securities portfolio to manage liquidity and interest-rate risk around a deposit base; they prioritise liquidity, credit quality, and duration management, subject to capital regulation.

Figure 3 — The four institutional archetypes ranked by two defining axes.



The single comparison every interview loves:

Feature	Individual	DB Pension	Endowment	Life Insurer	P&C Insurer
Time horizon	Finite, life-cycle	Long, tied to workforce	Infinite / very long	Long	Short
Risk tolerance	Low-moderate, behavioural	Moderate, surplus-driven	High	Low-moderate	Low
Liquidity need	Moderate	Low-moderate (benefits)	Low (spending %)	Low	High
Taxes	Taxable	Tax-exempt	Tax-exempt	Taxed on operations	Taxed
Return objective	Fund life goals	Fund liabilities	Spending + inflation + fees	Fund reserves / spread	Fund reserves + surplus
Dominant driver	Behaviour & life cycle	Liabilities (LDI)	Perpetuity & spending	Duration matching	Liquidity & solvency

4.5 How the IPS Drives Asset Allocation

This is where the IPS earns its keep. The chain runs:

1. **Return objective** sets the *required* expected return the portfolio must produce.
2. **Risk objective** sets the *ceiling* on volatility or drawdown the portfolio may run.
3. **Constraints** rule out or force in particular asset classes — a large near-term liquidity need forces cash and short bonds; a legal cap forces an equity ceiling; an ESG screen removes whole sectors; a long horizon *permits* illiquid, high-return assets.
4. **Capital-market expectations** (from outside the IPS) provide the expected returns, volatilities, and correlations of each asset class.
5. **Optimisation** finds the mix on the efficient frontier that hits the required return at or below the risk ceiling *while respecting every constraint* — producing the **strategic asset allocation (SAA)**, the policy portfolio.

So a young accumulator with a 30-year horizon, stable income, high risk ability, and low liquidity needs lands at something like 80% equities / 15% bonds / 5% cash. A near-retiree with a short horizon, high liquidity needs, and low willingness lands at 30% equities / 55% bonds / 15% cash. Same machinery, different IPS inputs, radically different portfolios. The IPS is literally the *input file* for the allocation optimiser — and because allocation drives the vast majority of a portfolio's return variance (Chapter 07), the IPS is, indirectly, the single most consequential document in the whole process.

5. Worked / Applied Examples

Example 1 — Reconciling ability and willingness for an individual

Ravi, 35, is a salaried IT professional with a secure ₹40 lakh annual income and ₹1 crore invested for retirement at 60. He needs the corpus to grow at inflation + 5% to retire comfortably. But in the 2020 crash he panic-sold and re-entered late, and he tells you a 20% loss would "keep me awake for months."

Assess ability: 25-year horizon, stable income, no near-term liquidity need, wealth modest relative to goal → **high ability**. *Assess willingness:* history of panic-selling, low stated loss tolerance → **low willingness**.

Reconciliation: Ability says 80% equity; willingness says maybe 45%. The rule is to build to the **lower** — but the *required* return of inflation + 5% ($\approx 11\%$ nominal) cannot be met at 45% equity. This is the crux the IPS exists to surface.

The manager's response is documented in the IPS: set the allocation at a moderate 55–60% equity (a compromise that gives the plan a *chance* of meeting the goal), and explicitly write in the mismatch: "*Client's willingness is below the level required to meet the stated goal. Manager will (a) educate the client on long-horizon volatility, (b) use rupee-cost averaging and pre-committed rebalancing to reduce behavioural error, and (c) revisit the goal or savings rate if willingness cannot rise.*" Ravi may need to save more or retire slightly later rather than take risk he'll abandon. The IPS turns an invisible conflict into an explicit, agreed plan.

Example 2 — An endowment's return objective and allocation

The Meridian University Endowment holds ₹500 crore. Its spending policy funds 4.5% of assets annually toward scholarships; long-run inflation is 5%; investment fees run about 0.5%. It is tax-exempt, perpetual, and needs only the 4.5% for liquidity each year.

Required nominal return (additive approximation):

Spending 4.5% + Inflation 5.0% + Fees 0.5% = **10.0%**

To fund spending *and* preserve real purchasing power forever, the endowment must earn roughly 10% nominal. That is a demanding target — unreachable with bonds alone.

Constraints: infinite horizon (very high risk ability), only 4.5% annual liquidity need (low), tax-exempt, no unusual legal limits.

Allocation implication: high risk ability + low liquidity need + demanding return → the classic endowment tilt toward growth and illiquidity. A plausible SAA: 30% global equities, 20% private equity, 15% real assets (real estate, infrastructure), 20% hedge funds/absolute return, 15% fixed income and cash. The endowment *deliberately* accepts illiquidity because its horizon lets it harvest the illiquidity premium — an allocation that would be reckless for a P&C insurer is optimal here. Notice how directly the IPS constraints (perpetual horizon, low liquidity) *permit* this; change the horizon and the whole allocation collapses toward bonds.

Example 3 — A defined-benefit pension and liability-driven investing

The Kaveri Industries Pension Plan has ₹800 crore of assets and ₹750 crore of liabilities (present value of promised pensions), giving a **surplus of ₹50 crore** and a funded ratio of $\sim 107\%$. The workforce is mature — average member age 52 — so benefit payments are large and near-term, and the liabilities have a duration of about 12 years.

Objective: the real goal is not "maximise return" but **fund the liabilities without letting the surplus erode**. Return required $\approx$ the plan's actuarial discount rate (say 7%).

Risk: risk is measured against the *liabilities*, not against cash. If the manager holds all equities and rates fall, liabilities balloon (their present value rises) while equities may not — the surplus can vanish even if the portfolio "made money." So the risk objective is framed in **surplus** terms: keep the funded ratio from falling below, say, 100%.

Allocation implication (LDI): match a large fixed-income core to the liabilities' 12-year duration to immunise the funded status against rate moves, and hold a smaller "return-seeking" equity sleeve funded largely by

the surplus to earn the extra return. A plausible mix: 65% duration-matched bonds, 30% equities, 5% cash. Contrast this with the endowment above — same "institution" label, completely different portfolio, because the IPS drivers (liabilities, mature workforce, moderate horizon) are entirely different. This is the whole point of the IPS: the *document*, not the label, determines the portfolio.

6. Connections

- **Chapter 03–04 (Portfolio Theory, Efficient Frontier):** The IPS supplies the two inputs the Markowitz optimiser needs — a target return and a risk limit. The SAA is literally the point on the efficient frontier that satisfies the IPS.
 - **Chapter 07 (Asset Allocation):** The IPS is the front end of asset allocation. The strategic asset allocation / policy portfolio is the IPS's primary output; rebalancing rules are usually written *into* the IPS.
 - **Chapter 08 (Portfolio Construction Process):** The IPS is Step 1 (the planning step) of the three-stage planning–execution–feedback loop. Execution (security selection) and feedback (monitoring, rebalancing) both measure themselves against the IPS.
 - **Chapter 11 (Performance Measurement):** The IPS defines the *benchmark* and the notion of success. Attribution asks whether the manager added value *relative to the policy portfolio the IPS specified*.
 - **Behavioural finance:** The IPS is the primary institutional defence against loss aversion, recency bias, and panic-selling — a pre-commitment device.
 - **Capital-market expectations:** The IPS's essential partner. $\text{IPS (the investor)} + \text{CME (the market)} = \text{asset allocation}$. Keeping them separate is what makes the policy durable.
-

7. Key Terms

- **Investment Policy Statement (IPS):** The written document specifying a client's objectives and constraints, governing all portfolio decisions.
- **Return objective:** The specified return the portfolio must achieve; split into *required* (needed) and *desired* (wanted), and specified as nominal/real, pre/post-tax, gross/net.
- **Risk objective:** The specified tolerance for volatility/loss, combining *ability* (objective) and *willingness* (psychological) to take risk.
- **Ability to take risk:** Capacity to absorb losses, driven by horizon, wealth-vs-needs, spending flexibility, and income stability.
- **Willingness to take risk:** Psychological comfort with volatility and loss.
- **Constraints (TTLLU):** Time horizon, Taxes, Liquidity, Legal/regulatory, Unique circumstances.
- **Liquidity requirement:** Predictable, plannable cash outflows the portfolio must fund.
- **Liquidity risk:** The danger of being forced to sell illiquid assets at a discount.
- **Strategic asset allocation (SAA) / policy portfolio:** The long-run target mix of asset classes; the IPS's primary output.
- **Liability-driven investing (LDI):** Managing assets to fund a specific liability stream, matching duration — central to DB pensions.
- **Surplus:** Assets minus the present value of liabilities; the risk lens for pensions.
- **Spending rate:** The percentage of assets an endowment/foundation withdraws annually.

- **Capital-market expectations (CME):** Forecast returns, volatilities, and correlations of asset classes — the market-side input paired with the IPS.
-

8. Common Confusions

"The IPS is just a compliance form." No. It is the engineering spec that drives the allocation and defines success. Treating it as boilerplate is exactly how portfolios end up mismatched to their owners.

Confusing ability and willingness. They are different and often conflict. A young investor can have high ability but low willingness; a retiree can have low ability but dangerously high willingness. Assess both, build to the lower, document the gap.

Required vs desired return. Clients state what they *want* (desired); the manager must derive what they *need* (required) and check feasibility against the risk limit. Chasing desired return beyond what the risk budget allows is how plans blow up.

Treating all institutions alike. A pension, an endowment, and a P&C insurer have almost nothing in common in horizon, liquidity, or risk. The "institution" label tells you little; the *IPS drivers* (liabilities, horizon, liquidity) tell you everything. Examples 2 and 3 land on opposite portfolios despite both being "institutions."

Liquidity requirement vs liquidity risk. The first is a planned outflow you allocate cash for; the second is the crisis danger of forced illiquid sales. Both must be managed; they are not the same thing.

Horizon of a retiree. A retiree is *not* automatically "short horizon." A 65-year-old may have 25+ years; only the near-term spending is short-horizon. Multi-stage horizons are the norm.

Putting the IPS and market views in the same box. The IPS describes the *investor* and changes rarely; capital-market expectations describe the *market* and change constantly. Blurring them means rewriting policy in every panic — the exact behaviour the IPS exists to prevent.

Nominal vs real return targets. "Grow at 8%" is meaningless without saying whether inflation is included. Endowments and long-horizon investors almost always target *real* returns (inflation-plus), because preserving purchasing power is the goal.

9. Recap

The Investment Policy Statement is the foundation of professional portfolio management: a written contract that converts a client's messy human circumstances into a precise specification the portfolio must satisfy. It has two halves. **Objectives** = return (required vs desired; specified nominal/real, pre/post-tax, net of fees) and risk (ability vs willingness, reconciled to the lower, gap documented). **Constraints** = the TTLU list: time horizon, taxes, liquidity, legal/regulatory, and unique circumstances.

The IPS describes the *investor*; combined with capital-market expectations about the *market*, it produces the **strategic asset allocation** — the policy portfolio that is the single most consequential output, because allocation dominates return variance. The framework is universal but its content diverges sharply across investor types: individuals are finite-horizon, taxable, and behaviour-driven; DB pensions are liability-driven; endowments are perpetual, high-risk-ability real-return machines; insurers are liquidity- and solvency-constrained. The same machinery, fed different IPS inputs, produces radically different portfolios — which is precisely the point. Finally, the IPS is durable but not static: reviewed periodically and after major life events, it anchors behaviour through market storms while still adapting to genuine change in the client's life.

10. Quick-Reference / Interview Points

- **One-line definition:** An IPS is a written document specifying a client's return and risk objectives and their constraints (time horizon, taxes, liquidity, legal/regulatory, unique), which governs the portfolio and drives the strategic asset allocation.
- **The structure:** Objectives (Return + Risk) and Constraints (TTLU — *Time, Taxes, Liquidity, Legal, Unique*). Memorise the mnemonic.
- **Return objective:** distinguish *required* vs *desired*; always specify nominal vs real, pre- vs post-tax, gross vs net.
- **Risk objective:** *ability* (objective — horizon, wealth vs needs, income stability, spending flexibility) vs *willingness* (psychological). **Build to the lower; document the mismatch; educate.**
- **Endowment required return shortcut:** Spending rate + Inflation + Fees. (e.g., 4.5% + 5% + 0.5% = 10%.)
- **Pension key idea:** liability-driven; risk measured in *surplus* terms; duration-match assets to liabilities (LDI). Workforce age sets horizon.
- **Endowment key idea:** infinite horizon, low liquidity need, high risk ability, demanding real return → illiquid growth tilt ("Yale model").
- **Insurer key idea:** life = long/predictable/bond-heavy; P&C = short/lumpy/high-liquidity. Most regulated, most solvency-sensitive.
- **The chain:** IPS (return + risk + constraints) + capital-market expectations → efficient-frontier optimisation → strategic asset allocation → construction → monitor/rebalance vs IPS.
- **Why it matters:** allocation drives most of return variance, so the IPS is indirectly the most consequential document in the process; it is also the primary behavioural pre-commitment device against panic-selling.
- **Classic exam trap:** don't confuse liquidity *requirement* (planned outflow) with liquidity *risk* (forced-sale danger); don't call a retiree "short horizon"; don't blur the IPS (investor, stable) with market expectations (market, changing).

Q&A — The Investment Policy Statement

Scope: Investments — Chapter 14 (The Investment Policy Statement). Every question is followed by a full model answer. All rates are annual and in percent unless stated. Work every numerical problem yourself before checking the solution. Sections: **A** concept-check · **B** applied/numerical (full step-by-step, reconciling) · **C** interview-style · **D** MCQs with reasoning.

The chapter in one line

$$\text{IPS} = \underbrace{\text{Return} + \text{Risk}}_{\text{Objectives}} + \underbrace{\text{Time} + \text{Taxes} + \text{Liquidity} + \text{Legal} + \text{Unique}}_{\text{Constraints (TTLLU)}}, \quad \text{IPS} + \text{CME} \rightarrow \text{SAA}$$

One-line statement: The Investment Policy Statement is the written contract that converts a client's messy human circumstances into a precise return-and-risk specification bounded by five constraints, which — combined with capital-market expectations — produces the strategic asset allocation and defines what "success" even means.

Section A — Concept Check

A1. Define the Investment Policy Statement and state its single most important output. The IPS is a written document that specifies a client's investment objectives (return and risk) and constraints (time horizon, taxes, liquidity, legal/regulatory, unique circumstances), and thereby governs how the portfolio is built, managed, and judged. Its single most important output is the **strategic asset allocation (SAA)** — the long-run policy mix of asset classes that is the portfolio's home base. The IPS describes the *investor*; combined with capital-market expectations about the *market*, it yields the SAA.

A2. What three failures occur when a portfolio is run with no IPS? (1) **Objective drift** — the portfolio reflects the manager's views or market fashions rather than the client's needs. (2) **Panic at the bottom** — with no pre-agreed risk budget, the client discovers their true risk tolerance during a crash and sells. (3) **No accountability standard** — with no benchmark, performance reviews degenerate into "you should have bought Bitcoin." The IPS is the engineering spec that prevents all three.

A3. Distinguish required return from desired return. *Required* return is what the investor *needs* to meet their goals (fund retirement, preserve real capital, meet a spending rule) — it is derived from facts, not preference. *Desired* return is what the investor *wants* — often higher and frequently infeasible given the risk constraint. The manager's job is to derive the required return, check it against the risk limit, and reconcile any gap between what the client wants and what the risk budget allows.

A4. Along what four dimensions must a return objective be specified, and why? Nominal vs real (is inflation included?), pre- vs post-tax, total return vs income only, and gross vs net of fees. Each dimension changes the number materially — a 12% pre-tax nominal target may be only 8.5% post-tax, and an endowment preserving purchasing power thinks in *real* terms while a retiree living on coupons needs *income*. A target stated without these tags ("grow at 8%") is operationally meaningless.

A5. Distinguish ability and willingness to take risk, and give the reconciliation rule. *Ability* is objective capacity — it rises with a longer horizon, greater wealth relative to needs, flexible spending, stable outside income, and low liquidity needs. *Willingness* is psychological comfort with volatility, drawn out through questionnaires and behavioural history. **When they conflict, build to the lower of the two, document the gap, and educate the client.** A portfolio the client abandons in a panic is worthless even if they "could afford" the risk on paper.

A6. Which conflict is the *dangerous* one — high ability/low willingness, or low ability/high willingness? Low ability with high willingness is the dangerous case — e.g., a retiree who "loves stocks." Here willingness would push toward more risk, but ability is the binding real-world limit, so the client must be *restrained*. In the reverse case (young investor, high ability, low willingness), you simply respect the lower willingness; the cost is only a possibly under-aggressive portfolio, not ruin.

A7. Name the five constraints (TTLLU) and give a one-line role for each. Time horizon — the master variable of risk; long horizon permits volatile high-return assets. Taxes — shape the return target and the choice of instruments/strategies. Liquidity — planned cash outflows that force cash and penalise illiquid assets. Legal/regulatory — the binding legal walls (trust deeds, prudent-investor rules, ERISA, IRDAI/solvency norms). Unique circumstances — idiosyncratic constraints (ESG/Shariah screens, concentrated founder stock, donor restrictions).

A8. Distinguish a liquidity requirement from liquidity risk. A liquidity *requirement* is a predictable, plannable cash outflow (a child's education next year, an endowment's 4.5% annual spend) that the manager funds with cash and marketable securities. Liquidity *risk* is the danger of being *forced* to sell illiquid assets at a discount during a crisis. The first is planned and budgeted; the second is a tail danger. Both must be managed, but they are not the same thing.

A9. Why is a retiree not automatically a "short-horizon" investor? Because a 65-year-old may live 25+ more years, so much of the corpus is genuinely long-term; only the *near-term spending* is short-horizon. Retirees typically have **multi-stage** horizons — a short bucket for imminent spending and a long bucket that can still hold equities. Treating the whole corpus as short-horizon forces excessive de-risking and fails to preserve real value over a long retirement.

A10. Why must the IPS and capital-market expectations be kept in separate boxes? The IPS describes the *investor* and is durable — it changes only on major life events (divorce, windfall, retirement). Capital-market expectations describe the *market* and change constantly. IPS + CME together produce the asset allocation. Blurring them means rewriting policy every time the market moves — which is exactly the panic-driven, buy-high/sell-low behaviour the IPS exists to prevent.

A11. State the endowment required-return shortcut and why the objective is "demanding." Required nominal return $\approx$ **Spending rate + Inflation + Fees**. It is demanding because the endowment must both fund current spending *and* preserve real purchasing power in perpetuity — so a 4.5% spend, 5% inflation, and 0.5% fees imply $\sim 10\%$ nominal, unreachable with bonds alone. This is why perpetual, low-liquidity endowments tilt toward growth and illiquid assets (the "Yale model").

A12. Why is a DB pension's risk measured in *surplus* terms rather than against cash? Because the plan's true objective is to fund a stream of *liabilities*, not to beat cash. If rates fall, the PV of liabilities rises; an equity portfolio may "make money" yet the funded ratio (assets $\div$ liabilities) can still deteriorate. Framing risk as the danger of the surplus (assets – PV of liabilities) eroding, and duration-matching assets to liabilities (LDI), keeps the funded status stable against rate moves.

Section B — Applied / Numerical Problems

B1. Endowment required return. The Meridian University Endowment holds ₹500 crore. Its spending policy funds 4.5% of assets annually; long-run inflation is 5.0%; fees run 0.5%. It is tax-exempt and perpetual. (a) Find the required nominal return using the additive approximation. (b) What is the one annual liquidity requirement in rupees? (c) Comment on feasibility with bonds alone.

(a) Required nominal return = Spending + Inflation + Fees = 4.5 + 5.0 + 0.5 = **10.0%**. (b) Liquidity requirement = $4.5\% \times ₹500 \text{ cr} = ₹22.5 \text{ crore}$ per year. (c) A ~10% nominal target is far above bond yields, so bonds alone cannot fund spending *and* preserve real value. With an infinite horizon and only 4.5% annual liquidity need, the endowment has very high risk ability and can accept illiquidity — hence a growth/illiquid tilt (equities, private equity, real assets), which the IPS constraints *permit*.

B2. Reconciling ability and willingness for an individual. Ravi, 35, earns a secure ₹40 lakh salary, has ₹1 crore invested, and retires at 60. He needs the corpus to grow at inflation + 5% ($\approx 11\%$ nominal). He panic-sold in 2020 and says a 20% loss would "keep me awake for months." Ability suggests ~80% equity; willingness suggests ~45% equity. Resolve.

Step 1 — Assess ability. 25-year horizon, stable income, no near-term liquidity need → **high ability** (~80% equity capacity). **Step 2 — Assess willingness.** Panic-selling history, low stated loss tolerance → **low willingness** (~45% equity). **Step 3 — Apply the rule.** Build to the *lower* (45%). But a 45% equity mix cannot plausibly deliver ~11% nominal — the required return is infeasible at that risk level. This is the conflict the IPS exists to surface. **Step 4 — Document the resolution.** Set a compromise ~55–60% equity that gives the plan a *chance*, and write the mismatch into the IPS: educate on long-horizon volatility, use rupee-cost averaging and pre-committed rebalancing, and revisit the goal/savings rate/retirement age if willingness cannot rise. **Reconciliation:** the numbers do not both fit, so the honest output is an explicit trade-off — take slightly more risk than willingness alone, *and* adjust the goal or save more.

B3. DB pension surplus and funded ratio. The Kaveri Pension Plan has ₹800 crore assets and ₹750 crore liabilities (PV of promised pensions). Liability duration ≈ 12 years. (a) Compute surplus and funded ratio. (b) If interest rates fall and liability PV rises 8% while a 30%-equity/70%-bond portfolio rises only 4%, does the surplus survive? (c) What does this imply for allocation?

(a) Surplus = $800 - 750 = ₹50 \text{ crore}$; funded ratio = $800/750 = 106.7\%$. (b) New liabilities = $750 \times 1.08 = ₹810 \text{ cr}$. New assets = $800 \times 1.04 = ₹832 \text{ cr}$. New surplus = $832 - 810 = ₹22 \text{ cr}$; new funded ratio = $832/810 = 102.7\%$. The surplus *shrank* by ₹28 cr and the funded ratio fell ~4 pp even though the portfolio "made money" — because liabilities grew faster than assets. (c) Risk must be measured against liabilities, not cash. Duration-match a large fixed-income core to the ~12-year liability duration (LDI) so the asset PV moves with the liability PV, and fund a smaller return-seeking equity sleeve from the surplus — e.g., ~65% duration-matched bonds, 30% equities, 5% cash. **Reconciliation:** had the bond core been duration-matched, its PV would have risen alongside liabilities, protecting the funded ratio — which is the whole point of LDI.

B4. Post-tax return target. A taxable individual states a desired 12% *pre-tax nominal* return. Interest/short-term gains are taxed at 30% and long-term equity gains at 12.5%. If the portfolio's return splits as 4% (taxed at 30%) and 8% (taxed at 12.5%), what is the post-tax return, and why does this matter for the IPS?

Post-tax = $4\%(1 - 0.30) + 8\%(1 - 0.125) = 2.80 + 7.00 = 9.80\%$. So the "12%" the client cited is really **9.8% net of tax**. It matters because the IPS return objective must be stated on a consistent basis: if the client actually *needs* 12% to fund goals, the portfolio must earn ~14.5%+ pre-tax — a far more demanding, riskier target. Confusing pre- and post-tax targets is a classic way plans quietly fall short.

B5. Liquidity-forced allocation floor. A family portfolio of ₹4 crore must fund ₹60 lakh of school fees in 12 months and hold a ₹40 lakh emergency reserve. What minimum fraction should sit in cash/near-cash, and how does this constrain the equity sleeve?

Planned near-term outflow + reserve = 60 + 40 = ₹100 lakh = **₹1 crore**. As a fraction: $1/4 = 25\%$ must be in cash and highly marketable short instruments. That caps risky assets at $\leq 75\%$ of the portfolio *before* any risk-tolerance consideration — the liquidity constraint binds first. Placing this ₹1 crore in equities would risk having to sell into a downturn exactly when the fees fall due (liquidity risk), so the constraint is non-negotiable regardless of the client's risk appetite.

B6. Same label, opposite portfolios. Contrast the SAA that the IPS drivers produce for the Meridian endowment (B1) versus a P&C insurer facing short, lumpy catastrophe claims. State each dominant driver and a plausible equity/illiquid weight.

Endowment: infinite horizon, low (4.5%) liquidity need, high risk ability → **dominant driver = perpetuity & spending** → high equity + large illiquid sleeve (e.g., ~50% equity-like + 35% private/real assets). P&C insurer: short, unpredictable liabilities, high liquidity need, heavy solvency regulation → **dominant driver = liquidity & solvency** → short-duration bonds, minimal illiquidity, only surplus capital in equities (e.g., ~15% equity, near-zero illiquid). **Reconciliation:** both are "institutions," yet the IPS drivers (horizon and liquidity) point in opposite directions — proving the *document*, not the label, determines the portfolio.

Section C — Interview-Style Questions

C1. "A client says: just grow my money safely. Walk me through how you turn that into a portfolio."

Model answer: "Grow safely" is two vague words I quantify through the IPS. First I pin the *return objective* to a number — required vs desired, tagged nominal/real, pre/post-tax, net of fees — say "beat inflation by 4% net." Then the *risk objective*: separate ability (horizon, wealth vs needs, income stability) from willingness (temperament and behavioural history), and anchor to the lower, documenting the gap. Next the five constraints — TLLU — which rule asset classes in or out: a near-term liquidity need forces cash; a legal cap limits equity; an ESG screen removes sectors; a long horizon permits illiquid growth. That IPS, plus capital-market expectations, feeds an optimiser that finds the efficient-frontier point meeting the required return at or below the risk ceiling while respecting every constraint — the SAA. Finally I write in benchmarks and a rebalancing rule so we can judge success and resist panic. "Grow safely" becomes an engineered specification.

C2. "How do individuals and institutions differ in their IPS, and why?" Model answer: The framework is universal — objectives plus TLLU — but content diverges because horizon, liquidity, taxes, and legal status differ. Individuals have finite, life-cycle horizons, are *taxable* (a first-order concern), have modest ability but variable, emotion-driven willingness; behaviour dominates, so the manager is part coach. Institutions are perpetual or long-horizon, usually tax-exempt, governed by boards and fiduciary rules — but they differ profoundly from *each other*: DB pensions are liability-driven and think in surplus terms (LDI, duration-matching); endowments are perpetual real-return machines with high risk ability and an illiquidity tilt; life insurers hold long, predictable liabilities and run bond-heavy; P&C insurers face short, lumpy claims and demand high liquidity and short duration. The label tells you little — the IPS drivers tell you everything.

C3. "Your client's ability and willingness disagree. What do you actually do?" Model answer: I never split the difference blindly. I build to the *lower* of the two, then document and educate. If ability is high but willingness low — a young, secure earner who fears volatility — I respect the low willingness and use rupee-

cost averaging and pre-committed rebalancing to reduce the emotional cost, while showing long-horizon data so willingness can rise. The more dangerous case is low ability, high willingness — a retiree who loves stocks — where I must *restrain* them, because ability is the hard limit and a bad drawdown could be unrecoverable. If the required return simply cannot be met at the lower risk level, I surface that conflict and offer real trade-offs: lower the goal, save more, or extend the horizon. The IPS turns an invisible conflict into an agreed plan before the market forces the issue.

C4. "In 2022 both stocks and bonds fell and many 60/40 plans dropped double digits. Did the IPS fail?"

Model answer: No — a bad year is not a failed policy, provided the IPS pre-agreed the risk budget. The IPS is written precisely so that in calm times the client accepts, in advance, that a bad drawdown can happen; that pre-commitment is its whole behavioural value. What 2022 tests is not the IPS concept but the *inputs* — the correlation and capital-market assumptions behind the allocation. The disciplined response is to check whether anything in the client's *life* changed (usually not) — if not, the IPS stands and we rebalance against it rather than tearing it up. If the episode revealed that the client's true willingness is lower than stated, that is genuine new information and we revise the risk objective. The failure mode the IPS guards against is exactly the temptation to rewrite policy mid-panic.

C5. "Explain liability-driven investing to someone who thinks a pension should just maximise returns."

Model answer: A DB pension does not exist to beat the market — it exists to pay promised pensions. Those promises are a liability whose present value moves with interest rates: when rates fall, the liability balloons. If the plan chased returns with all equities and rates dropped, liabilities could grow faster than assets, and the plan could become *underfunded* even though the portfolio "made money." LDI reframes the goal as protecting the *surplus* — assets minus the PV of liabilities. You hold a large bond core whose duration matches the liabilities' duration, so their present values move together and the funded ratio is immunised against rate moves; a smaller equity sleeve, funded largely by the surplus, earns extra return. Maximising return in isolation ignores the liability side — how well-funded plans quietly slip into deficit.

Section D — Multiple Choice (with reasoning)

D1. The single most important output of the IPS is: (a) a list of approved securities · (b) the strategic asset allocation · (c) the fee schedule · (d) the tax return **Answer: (b).** The IPS distils client circumstances into objectives and constraints that, with capital-market expectations, produce the policy portfolio — the SAA. Allocation dominates return variance, so this is the consequential output.

D2. "Ability to take risk" rises with all of the following EXCEPT: (a) a longer time horizon · (b) greater wealth relative to needs · (c) higher near-term liquidity needs · (d) stable outside income **Answer: (c).** High near-term liquidity needs *reduce* ability, because money may have to be withdrawn before drawdowns can recover. The other three all raise capacity to absorb loss.

D3. When ability and willingness to take risk conflict, the standard rule is to build to: (a) the higher of the two · (b) the average of the two · (c) the lower of the two, document the gap, and educate · (d) whatever the optimiser outputs **Answer: (c).** A portfolio the client abandons at the bottom is worthless; anchor to the lower, record the mismatch, and coach the client.

D4. The mnemonic TTLU for IPS constraints stands for: (a) Total, Tax, Level, Liquidity, Utility · (b) Time horizon, Taxes, Liquidity, Legal/regulatory, Unique circumstances · (c) Timing, Trading, Leverage, Lending, Utility · (d) Term, Tax, Loss, Limit, Underwriting **Answer: (b).** Time, Taxes, Liquidity, Legal/regulatory, Unique — the five walls of the box inside which the objectives are pursued.

- D5. An endowment spends 5% of assets a year, expects 4% inflation, and pays 0.5% fees. Its approximate required nominal return is:** (a) 5.0% · (b) 8.5% · (c) 9.5% · (d) 4.5% **Answer: (c).** Spending + Inflation + Fees = 5 + 4 + 0.5 = 9.5%, the real-preservation-plus-spending target.
- D6. A DB pension frames its risk objective primarily in terms of:** (a) tracking error vs the S&P 500 · (b) the surplus (assets minus PV of liabilities) · (c) absolute standard deviation of assets · (d) the Sharpe ratio **Answer: (b).** The goal is funding liabilities; risk is the danger of the funded status/surplus eroding, which drives LDI and duration-matching.
- D7. A predictable ₹50 lakh school-fee payment due next year is a:** (a) liquidity risk · (b) liquidity requirement · (c) unique circumstance · (d) legal constraint **Answer: (b).** It is a planned, plannable outflow the portfolio budgets cash for. Liquidity *risk* is the separate danger of forced illiquid sales in a crisis.
- D8. Which pairing of investor and dominant driver is correct?** (a) P&C insurer → perpetuity and spending · (b) Endowment → duration-matched liabilities · (c) DB pension → liabilities (LDI) · (d) Individual → tax-exempt status **Answer: (c).** DB pensions are liability-driven; endowments run on perpetuity/spending, P&C insurers on liquidity/solvency, individuals on tax and behaviour.
- D9. Keeping the IPS and capital-market expectations in separate boxes matters because:** (a) regulators require two documents · (b) the IPS (investor) is durable while CME (market) changes constantly, preventing policy from being rewritten in every panic · (c) it lowers fees · (d) CME is confidential **Answer: (b).** The asymmetry — stable investor policy vs constantly changing market views — is what anchors behaviour through storms.

Self-check log

B1: required return $4.5+5+0.5 = 10\%$ ✓; liquidity $4.5\% \times 500 = ₹22.5$ cr ✓. B2: build-to-lower (45%) infeasible for 11% goal → compromise 55–60% + documented mismatch, internally consistent ✓. B3: surplus 50→22 cr, funded ratio 106.7%→102.7% as liabilities outgrow assets ✓; LDI rationale reconciled ✓. B4: $4(0.70)+8(0.875) = 2.80+7.00 = 9.80\%$ ✓. B5: $(60+40)/400 = 25\%$ cash floor, caps risky $\leq 75\%$ ✓. B6: opposite drivers (perpetuity/spending vs liquidity/solvency) from opposite IPS inputs ✓. MCQ keys D1-b, D2-c, D3-c, D4-b, D5-c, D6-b, D7-b, D8-c, D9-b all trace to chapter text ✓.

Chapter 15 — Alternatives in Portfolios

1. The Problem / Need

A traditional portfolio is built from two ingredients: publicly listed **equities** and **bonds**. For most of the twentieth century that was enough. Stocks delivered growth, bonds delivered income and ballast, and the two tended to move differently enough that a 60/40 mix produced a smooth-ish ride. But three structural problems have pushed serious investors — pension funds, endowments, sovereign wealth funds, family offices, and increasingly wealthy individuals — to look beyond stocks and bonds.

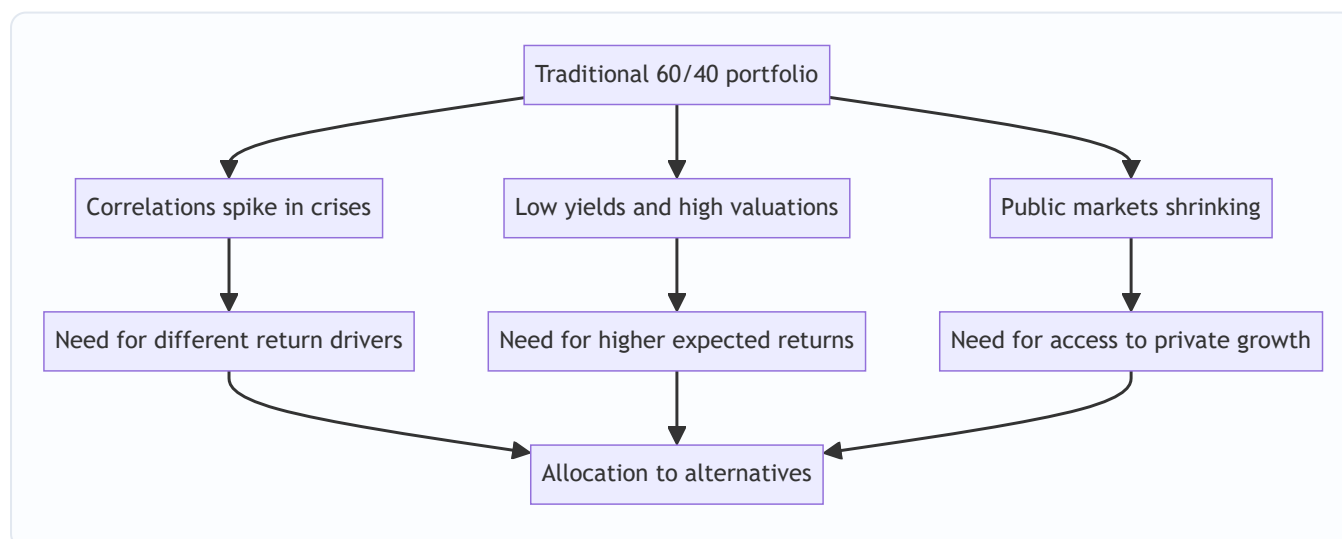
Problem one: the correlation trap. Diversification only works when assets do not fall together. In the crises that matter most — 2008, March 2020, the 2022 rate shock — correlations between stocks and bonds, and between equities across countries, tend to spike toward one. Exactly when you need diversification, it evaporates. Investors began hunting for return streams whose drivers are genuinely *different* from listed equity beta.

Problem two: the low-yield / high-valuation squeeze. After the 2008 crisis, developed-market bond yields collapsed to near zero (and briefly negative in Europe and Japan). A bond portfolio yielding 1% cannot fund a pension promising 7% actuarial returns. Simultaneously, equity valuations climbed. The expected return of the classic 60/40 fell. To hit their return targets, allocators needed sources of return that did not depend on ever-rising public markets.

Problem three: the shrinking public market. The number of US listed companies roughly halved from its 1996 peak. Companies now stay private far longer — a firm that would have IPO'd at a \$500m valuation in 1999 now IPOs at \$30bn, if at all. Much of the *value creation* in the modern economy happens **before** a company ever lists. A public-only investor is structurally locked out of that growth.

Alternatives — private equity, hedge funds, real estate, commodities, infrastructure, private credit, and more — are the answer to these three problems. They promise different return drivers, higher expected returns (partly as compensation for illiquidity), and access to parts of the economy that public markets cannot reach. The catch is that they are complex, expensive, illiquid, opaque, and unforgiving of poor manager selection. This chapter is about using them well.

Figure 15.1 — The three structural forces pushing capital toward alternatives.



2. Core Idea

Alternatives are any investment that is not plain-vanilla listed equity, listed bonds, or cash. That negative definition is deliberately broad, but the category shares a family resemblance built on four features:

1. **Different return drivers.** Their returns come from illiquidity, complexity, active skill, operational control, or exposure to real assets — not simply from public-market beta.
2. **Reduced liquidity.** Many lock your capital up for years. You trade the *option to sell tomorrow* for a higher expected return — the **illiquidity premium**.
3. **Higher and more complex fees.** The industry standard "2 and 20" (2% management fee plus 20% of profits) is far above the few basis points of an index fund.
4. **Manager dispersion matters enormously.** In public equities, the gap between a top-quartile and bottom-quartile index fund is trivial. In private equity or hedge funds, the gap between the best and worst manager can be 15–20 percentage points *per year*. **Access and selection are the whole game.**

The central portfolio idea is this: alternatives earn their place not because they always beat stocks (they often don't), but because they add a return stream that is **imperfectly correlated** with the rest of the portfolio. A modest expected-return boost combined with low correlation improves the portfolio's *risk-adjusted* return — it pushes the whole efficient frontier up and to the left. That is the prize.

3. Why / How It Works

The illiquidity premium

The theoretical heart of alternatives is the **illiquidity premium**: investors demand extra expected return to compensate for the inability to sell quickly. If two assets have identical cash flows and risk, but one can be sold instantly and the other locks you up for ten years, rational investors will pay *less* for the illiquid one — meaning its expected return is *higher*.

Why should a premium exist and persist?

- **Not everyone can bear illiquidity.** A retiree needing income, or a bank needing to meet daily redemptions, cannot lock money away for a decade. Only investors with long, stable liabilities — endowments, sovereign funds, young pension plans — can. Because the buyer pool is small, the illiquid asset must offer more to clear.
- **Illiquidity enforces discipline.** A locked-up investor cannot panic-sell at the bottom. This "forced patience" itself captures returns that skittish public-market investors give up.
- **Complexity and effort.** Sourcing, structuring, and operating private assets requires specialised skill and infrastructure — a barrier that keeps competition (and therefore prices) down.

Estimates of the illiquidity premium vary widely, typically cited at **1–5% per year**, but it is genuinely hard to measure and much of the "premium" observed in private equity can be decomposed into leverage, small-cap tilt, and sector bets rather than pure illiquidity.

The diversification mathematics

Recall the two-asset portfolio variance:

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 w_A w_B \rho_{AB} \sigma_A \sigma_B$$

The cross term contains the correlation ρ_{AB} . When you add an asset with **low or negative correlation** to your equity-heavy portfolio, that cross term stays small, so total portfolio volatility rises less than the new asset's standalone volatility would suggest. If the new asset also has a decent expected return, you get a better return-per-unit-of-risk. That is the mathematical engine behind adding commodities, macro hedge funds, or infrastructure to a stock/bond core.

An important caveat — smoothed returns

Private assets are not priced daily by a market; they are **appraised** periodically. Appraisals lag and smooth true economic values. This makes reported volatility *artificially low* and reported correlations *artificially low* — a statistical illusion sometimes called "**volatility laundering**." A private real-estate fund that reports 6% annual volatility may have true economic volatility closer to public REITs at 18%. So part of the beautiful diversification you see in the data is an artifact of measurement, not real economics. A sophisticated allocator **de-smooths** the return series before trusting the correlation numbers.

Figure 15.2 — How the illiquidity premium arises from a restricted buyer pool.



4. Full Content — The Major Alternative Asset Classes

4.1 Private Equity (PE)

What it is. Ownership of companies that are not publicly listed. The umbrella covers several strategies:

- **Buyouts (LBOs):** Acquiring mature, cash-generative companies using significant debt (leverage), improving operations, and selling in 3–7 years. The largest slice of PE by capital.
- **Growth equity:** Minority stakes in fast-growing, profitable companies needing capital to scale.
- **Venture capital (VC):** Early-stage equity in startups. High failure rate, driven by a few enormous winners (the "power law" — one Uber pays for a hundred flops).

The fund structure. PE is organised as a **closed-end limited partnership** with a ~10-year life. The manager is the **General Partner (GP)**; investors are **Limited Partners (LPs)**. LPs *commit* capital up front but it is **called** ("drawn down") gradually as deals are found, and **distributed** back as companies are sold. Uncalled money is a **commitment**, not cash sitting idle — a key operational complexity.

The J-curve. Early in a fund's life, fees and losers depress returns before winners mature, so cumulative returns trace a "J": down first, then up. Cash flow to LPs is negative for the first several years, positive later.

How PE creates value: (1) **operational improvement** — better management, margins, growth; (2) **financial leverage** — debt magnifies equity returns; (3) **multiple expansion** — buying at a low earnings multiple and selling at a higher one; (4) **buy-and-build** — rolling up small firms into a larger, more valuable platform.

Return metrics. Because cash flows are irregular, PE uses **IRR** (internal rate of return) and **MOIC / TVPI** (multiple of invested capital / total value to paid-in). A common benchmark comparison is the **PME (Public Market Equivalent)** — what you would have earned putting the same cash flows into an index. If PE's PME > 1, it beat the public market.

4.2 Hedge Funds

What they are. Actively managed, lightly regulated pools using tools mutual funds usually can't: **short selling, leverage, and derivatives**. The name is historical — the first fund (A.W. Jones, 1949) "hedged" long positions with shorts. Today "hedge fund" describes a *structure and fee model*, not a single strategy.

Major strategy buckets:

Strategy	What it does	Primary return driver
Long/Short Equity	Long undervalued, short overvalued stocks	Stock-picking skill, reduced net market exposure
Global Macro	Bets on rates, currencies, commodities from macro views	Directional macro calls; often crisis-friendly
Event-Driven / Merger Arb	Profit from M&A, spin-offs, restructurings	Deal completion; corporate events
Relative Value / Fixed-Income Arb	Exploit small pricing gaps between related securities	Convergence; heavy leverage
Managed Futures / CTA	Trend-following across futures markets	Momentum; famous "crisis alpha"
Distressed	Debt of troubled companies	Restructuring / recovery

The portfolio role. The best hedge funds aim for **absolute return** — positive returns regardless of market direction — and **low beta** to equities. Managed-futures/CTA and global-macro funds in particular have historically produced positive returns during equity crashes ("crisis alpha"), making them valuable diversifiers even if their long-run standalone return is modest.

Liquidity. More liquid than PE but still restricted: **lock-ups** (often one year initially), **redemption gates** (a cap on how much can be withdrawn at once), **notice periods** (30–90 days), and occasionally **side pockets** for illiquid positions.

4.3 Real Estate

What it is. Income-producing property — office, retail, industrial/logistics, residential/multifamily, hotels, data centers. Accessed via:

- **Direct ownership** — buying buildings; maximum control, minimum liquidity.
- **Private real-estate funds** — pooled, closed- or open-end, spanning the risk spectrum: **Core** (stabilised, low-leverage, income-driven), **Core-plus, Value-add** (renovate and re-lease), and **Opportunistic** (development, distressed — highest risk/return).
- **REITs (Real Estate Investment Trusts)** — *listed* property companies. Technically a public security, but they give real-estate *exposure*, must distribute ~90% of income as dividends, and trade daily.

Portfolio role: steady income yield, a partial **inflation hedge** (rents and property values often rise with inflation, especially with short leases or CPI-linked rents), and diversification. Beware: *listed* REITs behave much more like equities in the short run than private real estate does — a classic liquidity-vs-diversification trade-off.

4.4 Commodities

What they are. Physical raw materials — energy (oil, gas), metals (gold, copper), agriculture (wheat, coffee). Investors rarely hold barrels of oil; they gain exposure through **futures contracts** or commodity indices (e.g., Bloomberg Commodity Index, S&P GSCI).

The crucial subtlety — you don't earn the spot price. A commodity futures investor's total return has three parts:

$$\text{Total return} = \text{Spot return} + \text{Roll yield} + \text{Collateral return}$$

- **Spot return:** change in the physical price.
- **Roll yield:** because a futures position must be "rolled" to a later contract before expiry, you gain if the curve is **backwardated** (near contract priced *above* far, so you roll into cheaper contracts) and lose if it is in **contango** (far above near). Persistent contango has quietly destroyed returns for many long-only commodity investors.
- **Collateral return:** commodity futures need only margin, so the cash backing them earns the risk-free rate.

Portfolio role: commodities are the classic **inflation hedge** and often have **low or negative correlation** with stocks and bonds, since they respond to supply shocks (wars, droughts, OPEC) rather than corporate earnings. **Gold** is a special case — a monetary/safe-haven asset that tends to rise in crises and when real rates fall.

4.5 Infrastructure

What it is. The physical backbone of the economy — toll roads, airports, ports, utilities, power grids, water systems, pipelines, telecom towers, and increasingly renewable energy and data centers.

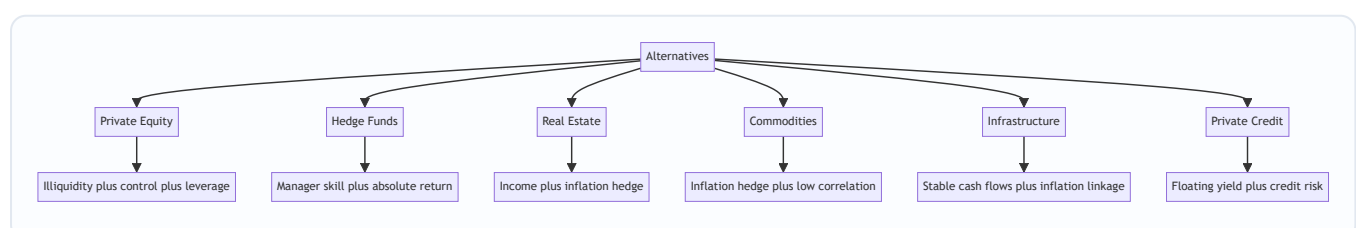
Why it's attractive: infrastructure assets typically have **monopolistic or quasi-monopolistic positions** (one airport per city), **long-lived contracts**, **regulated or inflation-linked revenues**, and **stable, predictable cash flows**. This gives them a **bond-like income profile with equity-like inflation protection** — a rare and prized combination for pension funds matching long-dated liabilities.

Split into: **Brownfield** (existing, operating assets — lower risk, income-focused) versus **Greenfield** (build-from-scratch — construction and demand risk, higher return).

4.6 Private Credit / Private Debt

The fastest-growing alternative of the 2020s. **Direct lending** to mid-sized companies, bypassing banks. Since post-2008 regulation pushed banks out of mid-market lending, funds stepped in, earning **floating-rate yields plus illiquidity premium** (often SOFR + 5–7%). Attractive in a rising-rate world because coupons float up, but carries **credit risk** concentrated in leveraged, unrated borrowers — largely untested through a severe default cycle.

Figure 15.3 — Map of the alternatives universe by liquidity and return driver.



5. Worked / Applied Examples

Example 1 — The illiquidity premium in a buyout deal

A PE fund buys a company for an **enterprise value of \$500m**, financed with **\$300m of debt** and **\$200m of equity**. The company generates **\$50m of EBITDA**, so the entry multiple is 10× EBITDA.

Over five years, the GP: - Grows EBITDA from \$50m to \$75m (operational improvement, +50%). - Pays down \$150m of debt using the company's cash flow (debt falls from \$300m to \$150m). - Sells at 11× EBITDA (multiple expansion from 10× to 11×).

Exit calculation: - Exit enterprise value = $11 \times \$75\text{m} = \825m - Less remaining debt of \$150m → **equity value = \$675m**

Return to LPs (gross): - MOIC = $\$675\text{m} \div \$200\text{m} = 3.375\times$ - IRR $\approx (3.375)^{1/5} - 1 = 27.5\%$ per year (gross, before fees)

Notice how the three levers stacked: EBITDA growth, deleveraging, and multiple expansion together tripled the equity. Now strip out leverage — imagine an all-equity purchase with the same operational improvement and multiple. Equity in = \$500m; exit EV = \$825m; MOIC = 1.65×, IRR $\approx 10.5\%$. **Leverage roughly doubled the equity IRR** — which is exactly why critics argue much of PE's "outperformance" is repackaged leverage and equity beta, not pure alpha. The gap between the 27.5% levered and 10.5% unlevered figures is the leverage contribution, not skill.

Example 2 — Diversification benefit of adding an alternative

You hold a portfolio that is **100% equities**: expected return 8%, volatility 16%. You consider shifting to **80% equities / 20% managed futures (CTA)**. Assume the CTA has expected return 6%, volatility 12%, and **correlation of -0.1** with equities.

New expected return: $E(R_p) = 0.8(8\%) + 0.2(6\%) = 6.4\% + 1.2\% = 7.6\%$

New volatility: $\sigma_p = \sqrt{(0.8)^2(16\%)^2 + (0.2)^2(12\%)^2 + 2(0.8)(0.2)(-0.1)(16\%)(12\%)}$
 $= \sqrt{163.84 + 5.76 - 6.144} = \sqrt{163.456} \approx 12.79\%$

Compare Sharpe-like ratios (assume risk-free 2%): - All equity: $(8 - 2)/16 = 0.375$ - Blended: $(7.6 - 2)/12.79 = 0.438$

We gave up 0.4% of expected return but cut volatility from 16% to 12.8% — the risk fell far more than the return, so **risk-adjusted return improved by ~17%**. The negative correlation did most of the work: the cross-term actually *subtracted* from variance. **This is the entire case for alternatives in one calculation** — you don't need them to out-earn equities, only to diversify them.

Example 3 — Contango destroying a commodity return

An investor buys a front-month oil futures contract at **\$70**. Over the year, the **spot price rises to \$75** (a +7.1% spot gain — seemingly a win). But the curve is in **contango**: each month, to avoid taking delivery, the investor sells the expiring contract and buys the next month at a **\$1.50 premium**. Over 12 rolls that costs roughly **\$18 of roll drag** relative to spot.

Result: despite the spot price rising 7%, the *investor's* futures position may deliver a **negative total return** once roll losses are netted. Add ~2% collateral (risk-free) return and the investor still barely breaks even.

Lesson: in commodities, being right about the physical price is not enough — the **shape of the futures curve** can dominate the outcome. This is precisely why long-only commodity index products disappointed investors through the 2010s.

6. Connections

- **Modern Portfolio Theory (Ch. on MPT):** Alternatives are the practical application of correlation-based diversification — they aim to shift the entire **efficient frontier** outward. The mean-variance math from MPT is exactly what justifies (and constrains) their allocation.
 - **Factor investing:** Much of what alternatives deliver can be decomposed into **factors** — value, size, momentum, illiquidity, credit, term. The "alternative risk premia" industry replicates hedge-fund returns cheaply using liquid factors, challenging the fee model.
 - **The Endowment Model (Yale / David Swensen):** The intellectual blueprint for heavy alternatives use. Swensen argued long-horizon investors are *paid* to bear illiquidity and should tilt hard toward private assets — Yale ran 30%+ in PE and real assets for decades.
 - **Asset-liability management:** Infrastructure and private credit connect directly to **liability matching** for pensions and insurers — long, inflation-linked cash flows against long-dated promises.
 - **Behavioural finance:** The lock-up is a **commitment device** that protects investors from their own panic-selling — a behavioural benefit disguised as a liquidity cost.
 - **Valuation:** PE and real-estate returns depend on **entry and exit multiples**, tying straight back to DCF and relative-valuation techniques.
-

7. Key Terms

- **Illiquidity premium** — extra expected return demanded for holding assets that cannot be sold quickly.
- **GP / LP** — General Partner (the manager) and Limited Partners (the investors) in a fund.
- **Commitment / capital call / drawdown** — money an LP pledges, then hands over in tranches as the GP invests.
- **Distribution** — cash returned to LPs as investments are sold.
- **J-curve** — the early-negative, later-positive shape of a private fund's cumulative return.
- **Vintage year** — the year a fund starts investing; a key comparison unit (you benchmark 2019 funds against other 2019 funds).
- **IRR / MOIC (TVPI) / DPI / PME** — IRR = annualised return; MOIC/TVPI = total value ÷ paid-in; DPI = *cash actually distributed* ÷ paid-in (realised); PME = comparison to a public index.
- **Carried interest ("carry")** — the GP's share of profits, typically 20%.
- **Hurdle rate / preferred return** — minimum return LPs get before the GP earns carry (often ~8%).
- **High-water mark** — a hedge-fund provision ensuring performance fees are paid only on *new* profits above the prior peak.
- **Lock-up / gate / side pocket** — restrictions on withdrawing from a hedge fund.
- **Contango / backwardation** — upward- / downward-sloping futures curve determining roll yield.
- **Roll yield** — return from rolling futures contracts forward.
- **Core / value-add / opportunistic** — the real-estate risk-return spectrum.

- **De-smoothing** — statistically correcting appraisal-based returns to reveal true volatility and correlation.
 - **Vintage diversification** — spreading commitments across years to avoid timing risk.
-

8. Common Confusions

"Alternatives are less risky because they show low volatility." *False and dangerous.* Their reported low volatility is largely a **measurement artifact** — appraisal-based valuations smooth out the swings ("volatility laundering"). The economic risk is real; it's just hidden between valuation dates. Adjust (de-smooth) before trusting the numbers.

"Private equity beats public equity." *Partly true, heavily caveated.* Average PE has beaten public equity *before fees* historically, but (1) much of the excess is **leverage and small-cap beta**, not alpha; (2) **after fees**, median PE has often only *matched* public equity; (3) **dispersion is huge** — only top-quartile GPs reliably add value. Buying "PE" as an asset class without access to good managers can *underperform* an index fund.

"IRR is the return I earned." *Not quite.* IRR assumes interim cash is reinvested at the IRR and can be **flattered by early distributions** or manipulated via **subscription-line financing** (delaying capital calls to boost the reported IRR). Always pair IRR with **MOIC** and **DPI** (actual cash back) for a truthful picture.

"Commodities give me exposure to rising oil prices." *Only partially.* You earn the **futures** return, not the spot return. In persistent **contango**, roll losses can turn a rising spot price into a losing investment (see Example 3).

"Hedge funds are high-risk, high-return." *Misconception.* As a group, hedge funds aim for **lower volatility and low market correlation**, not maximum return. Many deliberately target modest absolute returns with small drawdowns. "Hedge fund" is a fee-and-structure label, not a risk level.

"REITs are just liquid real estate, so they diversify like private real estate." *No.* Listed REITs trade like small-cap equities day to day and carry high short-run equity correlation. They give property *fundamentals* over the long run but *equity* volatility in the short run.

"2 and 20 means I pay 22%." *No.* It's 2% of *assets* annually **plus** 20% of *profits* (usually above a hurdle, subject to a high-water mark). But compounded over a fund's life, total fees can consume **a third or more** of gross gains — fee drag is the single biggest predictor of net-of-fee underperformance.

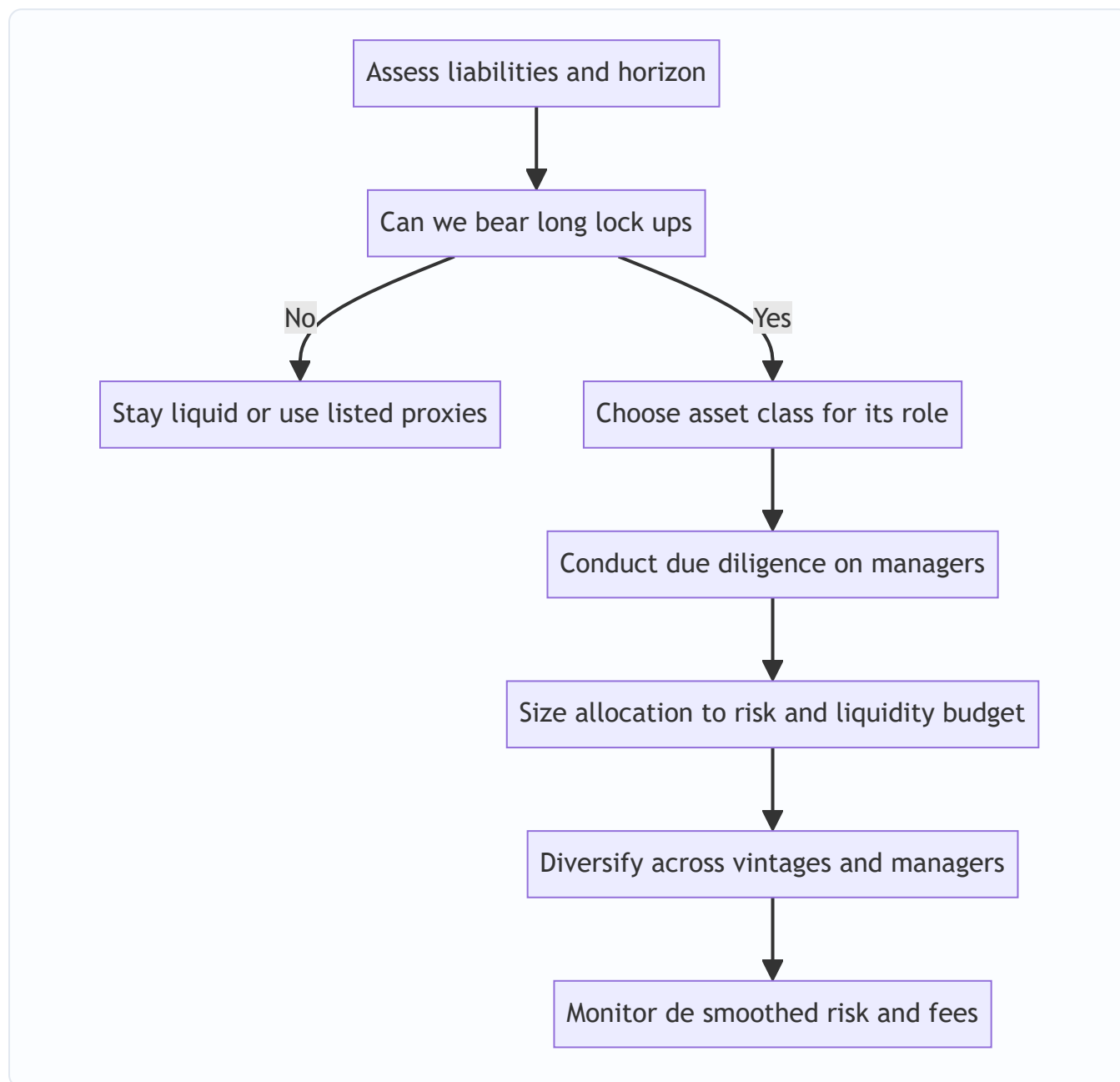
9. Recap

Alternatives exist to solve three problems that plague a stock-and-bond world: **correlations that spike in crises, low expected returns from expensive public markets, and lack of access to a shrinking public universe**. They earn their keep primarily through **diversification** (low correlation to equities) and the **illiquidity premium** (extra return for locking capital up), and secondarily through **manager skill** and **inflation protection**.

The major classes each play a distinct role: **private equity** for return enhancement via operational control and leverage; **hedge funds** for absolute return and crisis alpha; **real estate** for income and inflation hedging; **commodities** for supply-shock diversification and inflation protection; **infrastructure** for stable, inflation-linked, long-dated cash flows; and **private credit** for floating yield.

But the benefits are neither free nor guaranteed. The costs are **illiquidity, high and complex fees, opacity, and enormous manager dispersion**. The reported diversification is partly a **statistical illusion** from smoothed valuations. The whole game is **access and selection** — top-quartile managers add real value; bottom-quartile ones destroy it after fees. Alternatives should be a **deliberate, sized, patient allocation** — not a chase for the last cycle's hot fund.

Figure 15.4 — The decision workflow for adding an alternative allocation.



10. Quick-Reference / Interview Points

How much to allocate? There is no universal number; it flows from **liquidity tolerance and governance capacity**. Rough industry practice:

Investor type	Typical alternatives allocation	Rationale
Large endowment / SWF (Yale model)	40–60%+	Perpetual horizon, no near-term liabilities, strong manager access
Corporate / public pension	15–30%	Long liabilities but some liquidity needs and governance limits
High-net-worth / family office	10–25%	Long horizon but access and diligence constraints
Retail / mass affluent	0–10%	Liquidity needs, limited access, high minimums (loosening via interval funds/ELTIFs)

The binding constraint is the **liquidity budget**: never commit so much to illiquid assets that you'd be a *forced seller* of public assets at the bottom, or unable to fund capital calls or spending during a downturn (the "denominator problem" — when public markets crash, illiquid holdings become an oversized *share* of the shrunken portfolio).

Due-diligence checklist (say this in an interview): 1. **Track record** — through multiple cycles, same team, realised (DPI) not just paper (TVPI) gains. 2. **Team stability and alignment** — is the GP investing its own money ("skin in the game")? Key-person risk? 3. **Repeatable edge** — is the strategy's return driver understood and durable, or luck/leverage? 4. **Fees and terms** — management fee, carry, hurdle, high-water mark, clawback. 5. **Operational due diligence** — independent administrator, auditor, custody, valuation policy (the Madoff lesson: fraud hides in operations, not the investment thesis).

Fee facts: "2 and 20" standard; **hurdle** ~8%; **high-water mark** protects against paying twice; total fee drag can exceed a third of gross returns.

Liquidity terms to name: capital calls, lock-ups, gates, notice periods, side pockets, vintage-year commitments.

The one-line summary for an interview: *"Alternatives don't earn their place by beating equities — they earn it by delivering an imperfectly correlated return stream and an illiquidity premium, which improves the portfolio's risk-adjusted return. But the reported diversification is partly a smoothing illusion, fees are high, and manager dispersion is enormous — so access and due diligence, not the asset-class label, determine whether they actually help."*

Three risks to always cite: (1) **illiquidity risk** — can't exit when you need to, and capital-call/denominator problems in a crisis; (2) **manager/selection risk** — huge dispersion, fraud and opacity; (3) **valuation risk** — smoothed appraisals understate true volatility and correlation, giving false comfort.

Q&A — Alternatives in Portfolios

Scope: Investments — Chapter 15 (Alternatives in Portfolios). Every question is followed by a full model answer. All rates are annual and in percent unless stated. Work every numerical problem yourself before checking the solution. Sections: **A** concept-check · **B** numerical (full step-by-step, reconciling) · **C** interview-style · **D** MCQs with reasoning.

The chapter in one line

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho_{AB} \sigma_A \sigma_B, \quad \text{Commodity return} = \text{Spot} + \text{Roll} + \text{Collateral}$$

One-line statement: Alternatives earn their place not by beating equities but by adding an imperfectly correlated return stream plus an illiquidity premium — yet the reported diversification is partly a smoothing illusion, fees are high, and manager dispersion is enormous, so access and due diligence, not the label, decide whether they help.

Section A — Concept Check

A1. Give the working definition of "alternatives" and the four features that unite the category.

Alternatives are any investment that is not plain-vanilla listed equity, listed bonds, or cash — a broad negative definition. Four features unite them: (1) *different return drivers* — from illiquidity, complexity, active skill, operational control, or real assets rather than public-market beta; (2) *reduced liquidity* — capital often locked up for years, trading the option to sell tomorrow for higher expected return; (3) *higher, more complex fees* — "2 and 20" dwarfs an index fund's few basis points; (4) *enormous manager dispersion* — the top-to-bottom-quartile gap can be 15–20 points a year, so access and selection are the whole game.

A2. What three structural problems pushed serious investors toward alternatives? First, the *correlation trap*: in the crises that matter most (2008, March 2020, the 2022 rate shock) stock-bond and cross-equity correlations spike toward one, so diversification evaporates when needed. Second, the *low-yield / high-valuation squeeze*: post-2008 bond yields collapsed toward zero while equity valuations climbed, cutting the 60/40's expected return below what pensions must earn. Third, the *shrinking public market*: US listed-company counts roughly halved from their 1996 peak and firms stay private longer, so much value creation now happens before a company lists — locking public-only investors out.

A3. Explain the illiquidity premium and why it can persist. It is the extra expected return demanded for assets that cannot be sold quickly: of two assets with identical cash flows and risk, the locked-up one must be cheaper — hence higher expected return. It persists because (a) few investors can bear illiquidity — retirees and banks with daily redemption needs cannot, so the eligible buyer pool is small and the asset must offer more to clear; (b) it enforces discipline — a locked-up investor cannot panic-sell at the bottom; and (c) sourcing and operating private assets needs specialised skill that limits competition. Estimates run 1–5% a year, but much of the observed "premium" decomposes into leverage, small-cap tilt, and sector bets rather than pure illiquidity.

A4. What is "volatility laundering" and why must an allocator de-smooth? Private assets are appraised periodically, not priced daily. Appraisals lag and smooth true economic values, so reported volatility and correlations come out artificially low — the illusion nicknamed volatility laundering. A private real-estate fund reporting 6% volatility may have true volatility near listed REITs' 18%. Part of the diversification in the raw data is a measurement artifact, so a sophisticated allocator *de-smooths* the return series before trusting the numbers used in optimisation.

A5. Describe the PE fund structure and the J-curve. PE is a closed-end limited partnership with a ~10-year life. The manager is the General Partner (GP); investors are Limited Partners (LPs). LPs *commit* capital up front, but it is *called* gradually as deals are found and *distributed* back as companies are sold — uncalled money is a commitment, not idle cash. The J-curve is the shape of cumulative returns: early fees and losers depress returns before winners mature, so returns dip below zero first and rise later; LP cash flow is negative early, positive later.

A6. Distinguish IRR, MOIC/TVPI, DPI, and PME. IRR is the annualised internal rate of return on irregular cash flows. MOIC (or TVPI) is total value — realised plus unrealised — divided by paid-in, a simple multiple. DPI is *cash actually distributed* divided by paid-in, capturing only realised gains. PME compares the fund to putting identical cash flows into a public index; $PME > 1$ means it beat the market. The trap is trusting IRR alone: it assumes reinvestment at the IRR and can be flattered by early distributions or subscription-line financing, so always pair it with MOIC and DPI.

A7. Name the main hedge-fund strategy buckets and their primary return drivers. Long/Short Equity (stock-picking, reduced net exposure); Global Macro (directional rate/currency/commodity bets, often crisis-friendly); Event-Driven / Merger Arbitrage (deal completion); Relative Value / Fixed-Income Arbitrage (convergence of small pricing gaps, heavily leveraged); Managed Futures / CTA (trend momentum, "crisis alpha"); and Distressed (restructuring and recovery in troubled-company debt). "Hedge fund" labels a structure and fee model, not a single strategy or risk level.

A8. Why is infrastructure prized by pension funds? Infrastructure — toll roads, airports, utilities, grids, pipelines, towers, renewables — typically enjoys monopolistic positions, long-lived contracts, and regulated or inflation-linked revenues, producing stable cash flows. That gives a bond-like income profile with equity-like inflation protection, ideal for matching long-dated, inflation-linked liabilities. Brownfield (operating) assets are lower-risk and income-focused; greenfield (build-from-scratch) carries construction and demand risk for higher return.

A9. Decompose a commodity futures investor's total return and explain roll yield. Total return = spot + roll yield + collateral. Spot is the physical price change. Roll yield arises because a position must be rolled to a later contract before expiry: you gain if the curve is *backwardated* (near above far, so you roll into cheaper contracts) and lose in *contango* (far above near). Collateral is the risk-free rate on the cash backing the margin. Persistent contango has quietly destroyed returns for long-only commodity investors — being right about the physical price is not enough.

A10. What is the "denominator problem"? When public markets crash, illiquid alternatives — appraised and slow to reprice — become an oversized *share* of a shrunken portfolio, pushing the alternatives weight above policy with no new commitments. This can force selling public assets at the bottom to rebalance or fund spending and capital calls. Hence the binding constraint is the *liquidity budget*: never commit so much to illiquids that you become a forced seller in a downturn.

A11. Summarise the endowment (Yale/Swensen) model and its core claim. Associated with David Swensen at Yale, it is the blueprint for heavy alternatives use: long-horizon investors with no near-term

liabilities are *paid* to bear illiquidity and should tilt hard toward private assets, harvesting the illiquidity premium and superior manager access. Yale ran 30%+ in private equity and real assets for decades. The caveat: it depends on genuinely top-quartile access, which most investors lack.

Section B — Numerical Problems

B1. Buyout return with the three value levers. A PE fund buys a company at an enterprise value of \$500m (\$300m debt, \$200m equity). EBITDA is \$50m (10x entry). Over five years EBITDA grows to \$75m, \$150m of debt is repaid (to \$150m), and it exits at 11x. Find exit equity value, MOIC, and gross IRR.

Step 1 — Exit enterprise value. $11 \times 75 = \mathbf{\$825m}$. **Step 2 — Exit equity value.** $\$825 - 150 \text{ (remaining debt)} = \mathbf{\$675m}$. **Step 3 — MOIC.** $\$675 / 200 = \mathbf{3.375 \times}$. **Step 4 — Gross IRR.** $3.375^{1/5} - 1 = 0.24328$; $\ln 3.375 = 1.2164$; $1.2164 / 5 = 0.24328$; $e^{0.24328} = 1.2754 \rightarrow \mathbf{27.5\% \text{ per year (gross)}}$.

Reconcile — strip out leverage. All-equity: equity in \$500m, exit EV \$825m $\rightarrow$ MOIC 1.65x, IRR $= 1.65^{1/5} - 1 = 10.6\%$. Leverage lifted the equity IRR from $\sim 10.6\%$ to 27.5% — roughly doubling it. The gap is leverage, not skill, which is why critics call much PE "outperformance" repackaged leverage and equity beta.

B2. Diversification benefit of adding a CTA. A 100% equity portfolio has $E(R)=8\%$, $\sigma=16\%$. Shift to 80% equity / 20% managed futures (CTA: $E(R)=6\%$, $\sigma=12\%$, correlation -0.1 with equities). Risk-free 2%. Find the blend's $E(R)$, σ , and compare Sharpe-like ratios.

Expected return. $0.8(8) + 0.2(6) = 6.4 + 1.2 = \mathbf{7.6\%}$. **Variance** (in $\%^2$): equity $0.8^2 \cdot 16^2 = 0.64 \cdot 256 = 163.84$; CTA $0.2^2 \cdot 12^2 = 0.04 \cdot 144 = 5.76$; cross $2(0.8)(0.2)(-0.1)(16)(12) = -6.144$. Sum $= 163.456 \rightarrow \sigma = \sqrt{163.456} = \mathbf{12.79\%}$. **Sharpe-like ratios.** All-equity $(8-2)/16 = \mathbf{0.375}$; blend $(7.6-2)/12.79 = \mathbf{0.438}$.

Reconcile. We gave up 0.4 pp of return but cut volatility $16\% \rightarrow 12.8\%$; risk fell far more than return, so risk-adjusted return rose $\sim 17\%$ ($0.375 \rightarrow 0.438$). The negative correlation did the work — the cross term is *negative* (-6.14). The whole case for alternatives in one calculation: they need not out-earn equities, only diversify them.

B3. Contango destroys a commodity gain. An investor holds oil futures. Spot rises \$70 $\rightarrow$ \$75 over the year. The curve is in contango: each of 12 monthly rolls costs a \$1.50 premium. Collateral earns 2% on the \$70 notional. Estimate total return.

Spot return. $(75-70)/70 = +7.14\% \rightarrow$ in dollars on \$70, $+\$5.00$. **Roll drag.** $12 \times 1.50 = \mathbf{-\$18}$ (roughly, relative to spot). **Collateral.** $2\% \times 70 = +\$1.40$. **Net dollars.** $5.00 - 18.00 + 1.40 = \mathbf{-\$11.60}$ $\rightarrow$ about **-16.6%** on \$70.

Reconcile. Despite a 7% rise in the physical price, the *investor's* futures position loses heavily once roll losses net out; even the 2% collateral leaves a clear loss. The shape of the futures curve can dominate the spot move — precisely why long-only commodity index products disappointed through the 2010s. (Exact figures depend on how drag compounds; the sign is the point.)

B4. Capital calls and the J-curve. An LP commits \$10m. Calls: Y1 \$3m, Y2 \$3m, Y3 \$2m (\$8m called, \$2m never drawn). Distributions: Y4 \$4m, Y5 \$9m. Year-5 residual NAV \$1m. Compute paid-in, DPI, TVPI, and describe the cash-flow shape.

Paid-in capital. $\$3 + \$3 + \$2 = \mathbf{\$8m}$ (the uncalled \$2m is not paid-in). **Total distributed.** $\$4 + \$9 = \mathbf{\$13m}$. **DPI.** $\$13/8 = \mathbf{1.625\times}$ (realised cash back per dollar in). **TVPI.** $\$(13+1)/8 = 14/8 = \mathbf{1.75\times}$ (adds the \$1m residual NAV).

Reconcile the J-curve. Net LP cash flow: -3, -3, -2, +4, +9 — negative three years, then positive, tracing the J. TVPI (1.75x) exceeds DPI (1.625x) only by the unrealised residual; most value is already distributed, so the paper mark is nearly all realised — healthy versus a fund whose TVPI is mostly NAV.

B5. Sizing to a liquidity budget (denominator problem). A \$100m portfolio holds \$30m alternatives (appraised, flat) and \$70m public equity. Public equity falls 40%. Find the new alternatives weight.

After the crash. Public equity $\$70m \times 0.60 = \$42m$; alternatives still \$30m (appraised, unmoved). Total $= \$72m$. Alternatives weight $= 30/72 = \mathbf{41.7\%}$ — up from 30% with no new commitments. To restore 30% you would sell illiquid stakes (only at a discount, if at all) or accept the drift — the illiquidity bites exactly when flexibility is needed.

Reconcile. The denominator problem in numbers: a 40% equity crash pushed alternatives from 30% to 41.7% purely through the shrinking denominator. Lesson: size illiquid commitments so this over-weighting never forces a bottom-of-market sale of public assets.

B6. Fee drag over a fund's life. A fund earns 15% gross annually over 5 years on \$100, charging 2% of assets a year plus 20% carry on profit above cost (ignore hurdle; carry once at exit on net-of-management-fee profit). Estimate net MOIC and the fee share of gross gains.

Gross. $\$100 \times 1.15^5 = \mathbf{\$201.14}$ → gross profit \$101.14\$. **Management fee.** ~2% of \$100 for 5 years $= \$10$. **After mgmt fee.** $\$201.14 - 10 = \191.14 → profit above cost $= \$91.14$. **Carry.** $20\% \times 91.14 = \$18.23$. **Net to LP.** $\$191.14 - 18.23 = \mathbf{\$172.91}$ → net MOIC $1.729\times$. **Fee share of gross gains.** $\$(10 + 18.23)/101.14 = \mathbf{27.9\%}$.

Reconcile. Even on a strong 15% gross return, ~28% of gross profit went to fees, dragging the LP's compound return from 15% gross to $1.729^{1/5} - 1 = 11.6\%$ net. The chapter's "a third or more of gross gains" is realistic — and larger once a hurdle, catch-up, and higher multiples are modelled.

Section C — Interview-Style Questions

C1. "Why add alternatives at all if, after fees, average private equity has only matched public equity?"

Model answer: Because the case was never that they out-earn equities. It is that they add an *imperfectly correlated* return stream plus an illiquidity premium, improving the portfolio's *risk-adjusted* return and pushing the efficient frontier up and to the left. A CTA returning less than equities but with negative correlation can still raise the portfolio's Sharpe ratio — as in the 80/20 example where 0.4 pp of return bought a 3-point cut in volatility. But "average" hides everything: top-to-bottom-quartile dispersion runs 15–20 points a year, so buying the label without top-quartile access can underperform an index fund. The value is conditional on access and selection.

C2. "A private real-estate fund reports 6% annual volatility and near-zero correlation with equities.

Would you trust those numbers in your optimiser?" Model answer: No — not as reported. Private assets are appraised, not market-priced, so valuations lag and smooth true economic values, mechanically understating volatility and correlation — volatility laundering. A fund reporting 6% may have true volatility closer to listed REITs at 18%. Fed raw into a mean-variance optimiser, it looks like a free lunch and gets massively over-allocated. So I de-smooth the appraisal series first — unsmoothing the return autocorrelation

— to recover realistic inputs, then optimise. The reported diversification is partly a measurement artifact, and treating it as real is one of the most common and expensive mistakes in the field.

C3. "Walk me through your due-diligence checklist for a first-time allocation to a buyout GP." Model answer: Five headings. *Track record* through multiple cycles with the same team, weighting realised DPI over paper TVPI. *Team stability and alignment*: is the GP investing personal money (skin in the game), and is there key-person risk? *Repeatable edge*: did returns come from a durable operational playbook or just leverage, multiple expansion, and a rising market — decompose a past deal. *Fees and terms*: management fee, carry, hurdle, high-water mark, clawback. *Operational due diligence*: independent administrator, auditor, custody, valuation policy — the Madoff lesson is that fraud hides in operations, not the thesis. And insist on vintage diversification so it is not one big timing bet.

C4. "Commodities are up because oil is up — so a long commodity index should be winning, right?" Model answer: Not necessarily, because you earn the *futures* return, not spot. Total return is spot plus roll yield plus collateral. In contango — far contracts above near — every monthly roll sells a cheaper expiring contract and buys a dearer later one, bleeding roll yield. In the worked case, spot rose 7% but 12 monthly rolls at a \$1.50 premium produced a net loss even after collateral. Persistent contango is exactly what made long-only commodity index products disappoint through the 2010s. So the right question is not "is oil rising?" but "what is the shape of the futures curve?"

C5. "How would you size an alternatives allocation for a corporate pension versus a large endowment?" Model answer: The number flows from liquidity tolerance and governance capacity, not a target return. An endowment or sovereign fund — perpetual horizon, no near-term liabilities, strong access — can run 40–60%+, the Yale model. A pension has long liabilities but real liquidity needs (benefit payments) and governance limits, so 15–30% is typical. The binding constraint is the *liquidity budget*: never commit so much to illiquids that a crash makes you a forced seller at the bottom. I would stress-test the denominator problem — model a 40% equity drawdown and check capital calls and spending stay fundable. For the pension I would lean toward infrastructure and private credit, whose long, inflation-linked cash flows match the liabilities.

Section D — Multiple Choice (with reasoning)

D1. The illiquidity premium exists primarily because: (a) illiquid assets are always higher quality · (b) the pool of investors able to bear long lock-ups is small, so the asset must price cheaper to clear · (c) regulators mandate it · (d) illiquid assets have lower risk **Answer: (b).** A small eligible buyer pool forces a lower price and thus higher expected return. Illiquid assets are not lower risk.

D2. "Volatility laundering" refers to: (a) money laundering via hedge funds · (b) appraisal-based valuations smoothing returns so reported volatility and correlation look artificially low · (c) using derivatives to hide leverage · (d) tax evasion on carry **Answer: (b).** Periodic appraisals lag true values, understating volatility and correlation — a measurement artifact that must be de-smoothed.

D3. A commodity futures investor in a persistently contangoed market will, all else equal: (a) earn the full spot return · (b) lose roll yield each time the position is rolled forward · (c) gain roll yield · (d) earn only the collateral return **Answer: (b).** In contango the far contract is dearer than the near, so rolling sells cheap and buys dear — a roll loss. Backwardation gives positive roll yield.

D4. Which metric captures only cash actually returned to LPs? (a) TVPI · (b) IRR · (c) DPI · (d) MOIC **Answer: (c).** DPI = distributions ÷ paid-in, counting realised cash only. TVPI/MOIC include unrealised NAV;

IRR is a time-weighted rate that can be flattered by early cash.

D5. The J-curve in private equity describes: (a) rising then falling leverage · (b) cumulative returns that are negative early (fees and losers) and positive later (as winners mature) · (c) the shape of the futures curve · (d) fund size over time **Answer: (b).** Early fees and write-downs depress returns before winners are realised, tracing a "J."

D6. Adding a low- or negatively-correlated alternative to an equity portfolio improves risk-adjusted return mainly because: (a) it always has higher expected return · (b) the variance cross-term stays small or negative, cutting portfolio volatility by more than it cuts return · (c) it eliminates all risk · (d) it removes fees **Answer: (b).** Low ρ shrinks (or negates) the variance cross term, so volatility falls faster than return — the diversification engine.

D7. The "denominator problem" for an alternatives allocation means: (a) fees grow faster than returns · (b) when public markets crash, slow-to-reprice illiquid assets become an oversized share of a shrunken portfolio · (c) IRR is hard to compute · (d) carry is charged twice **Answer: (b).** The public "denominator" shrinks in a crash, pushing the appraised illiquid weight above target and risking forced sales.

D8. "2 and 20" with a high-water mark means: (a) a flat 22% fee · (b) 2% of assets annually plus 20% of profits, with performance fees paid only on new gains above the prior peak · (c) 2% carry and 20% management · (d) fees only in profitable years **Answer: (b).** The high-water mark charges carry only on profits above the previous high, preventing paying twice for the same recovery.

D9. Much of private equity's historical "outperformance" is best explained by: (a) pure manager alpha only · (b) leverage, small-cap tilt, and sector bets, plus top-quartile selection · (c) low fees · (d) daily liquidity **Answer: (b).** Decomposition shows leverage and beta do much of the work; repeatable alpha is concentrated in top-quartile GPs. Listed REITs are the mirror caution — they trade like small-cap equities short-run despite giving property fundamentals long-run.

Self-check log

B1: exit equity \$675m, MOIC 3.375x, IRR 27.5%; unlevered 10.6% → leverage ~doubles IRR ✓. B2: σ 12.79% via cross term -6.144 ; Sharpe $0.375 \rightarrow 0.438 (+17\%)$ ✓. B3: spot +\$5, roll $-\$18$, collateral $+\$1.40 \rightarrow$ net $-\$11.6$, sign negative despite $+7\%$ spot ✓. B4: paid-in \$8m, DPI 1.625x, TVPI 1.75x, cash flow $-3/-3/-2/+4/+9$ traces J ✓. B5: alternatives $30/72 = 41.7\%$ from a 40% equity crash, denominator problem ✓. B6: net MOIC 1.729x, fees 27.9% of gross profit, net return 11.6% vs 15% gross ✓.

Chapter 16 — Factor Investing and Smart Beta

1. The Problem / The Need

By now you have met two competing worlds. On one side sits **passive indexing**: buy the whole market, cap-weighted, for a few basis points a year, and accept the market return. On the other sits **active management**: pay a manager 60–100 bps (often much more) to pick stocks and time factors, in the hope of beating the index — a hope that, after fees, most managers fail to deliver on (Chapter 12 on market efficiency showed you the grim SPIVA scorecards).

For decades those looked like the only two choices. But Chapter 06 planted a seed that blows this dichotomy apart. If average returns are driven not by one market factor but by a *handful* of systematic factors — size, value, momentum, profitability — then a natural question follows:

When a star active manager "beats the market," how much of that outperformance is genuine skill (alpha), and how much is just a *cheap, mechanical tilt* toward known factors that any rules-based strategy could have replicated for 20 bps?

This is not academic hair-splitting; it is worth billions. In the 1990s, if a value manager returned 3% a year over the S&P 500, investors paid active fees believing they were buying skill. Then research showed that value is a *systematic, harvestable premium* — you can capture it with a transparent rule ("buy stocks with high book-to-market"), no genius required. Suddenly a chunk of what was sold as expensive alpha was revealed to be cheap **factor beta**.

The practical need is threefold:

1. **Decompose returns honestly.** Separate true alpha from factor exposure so investors know what they are actually paying for.
2. **Harvest premia cheaply.** If factor premia are real and persistent, package them in low-cost, rules-based, transparent vehicles — no need to pay 1% for a discretionary manager.
3. **Build better-diversified portfolios.** Diversify not just across *asset classes* (which all crashed together in 2008) but across the underlying *factors* — the true drivers of risk and return.

The answer to all three is **factor investing** — and its retail packaging, **smart beta**. This chapter takes the multifactor theory of Chapter 06 and turns it into an investable discipline.

2. The Core Idea

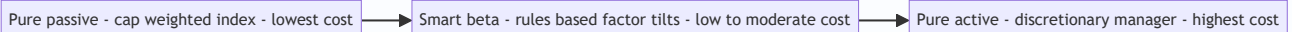
A **factor** is a broad, persistent, quantifiable characteristic that helps explain the risk and return of a group of securities. Think of factors as the *nutrients* in the market's diet, and individual stocks as *foods*: two very different-looking foods (a bank and a utility) may deliver the same underlying nutrient (value exposure).

Factor investing rests on three claims:

- **Returns are driven by factors, not just by individual securities.** A stock's return is mostly its exposures to common factors plus a small idiosyncratic piece.

- **Certain factors have earned a long-run premium** — a persistent excess return above the market — that survives out-of-sample, across countries, and across asset classes.
- **These premia can be captured systematically** through transparent, rules-based portfolio construction, at a fraction of active-management cost.

Smart beta (also "strategic beta" or "factor ETFs") is the *product wrapper*. A smart-beta strategy is any rules-based index that departs from plain market-cap weighting in order to tilt toward one or more factors. It sits on a spectrum between pure passive and pure active:



Smart beta occupies the middle ground - the transparency and low cost of indexing combined with the factor tilts once sold as active skill.

The key mental shift: **cap-weighting is itself a bet**. A market-cap index automatically overweights whatever has already gone up (the biggest, most expensive stocks) and underweights what has fallen. It embeds a momentum-and-size profile whether you intended it or not. Smart beta simply says: *if I am going to make a bet anyway, let me make a deliberate, evidence-based one.*

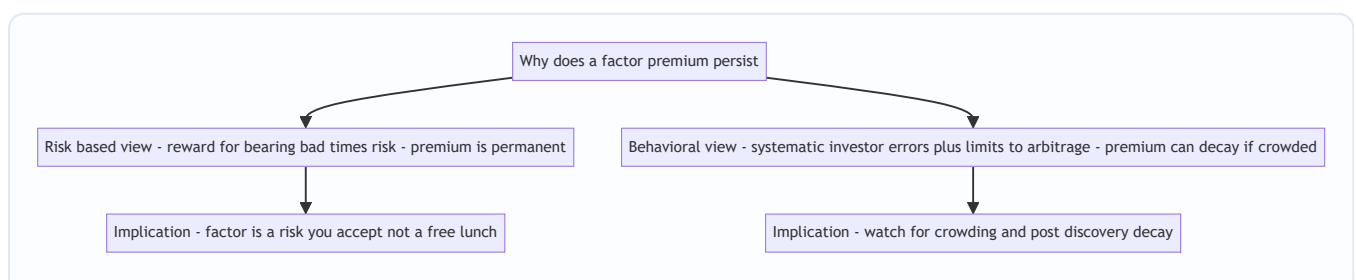
3. Why / How It Works

Why should a factor premium exist at all?

If a strategy reliably earns excess return, efficient-market logic says arbitrageurs should pile in and compete the premium away. That a premium *persists* demands an explanation. There are two rival camps, and mature practitioners hold both simultaneously.

Camp 1 — Risk-based (rational). The premium is *compensation for bearing a risk* that cannot be diversified away. Value stocks are cheap because they are genuinely distressed, cyclically fragile firms that do badly precisely when investors can least afford losses (recessions). You earn the value premium as *payment for holding assets that hurt in bad times*. Under this view the premium is real, permanent, and should *not* disappear — because the risk never disappears. This is the Fama-French position: factors are proxies for systematic risk.

Camp 2 — Behavioral (mispricing). The premium arises from *persistent human errors* that arbitrage cannot fully correct. Investors over-extrapolate growth, so glamorous growth stocks get overpriced and cheap stocks underpriced (value premium). Investors under-react to news, so trends persist (momentum). Limits to arbitrage — short-selling costs, career risk, capital constraints — stop smart money from fully eliminating the mispricing. Under this view premia *can* shrink once discovered and crowded, but frictions keep them alive.



The two economic rationales for factor premia - both can be partly true for the same factor.

Why care which is right? Because it changes expectations. If value is *pure risk compensation*, you should hold it through pain and expect the premium to endure. If it is *pure mispricing*, you should worry that crowding after publication erodes it. Most evidence suggests factors are a *blend*: real risk stories plus behavioral amplification, which is why premia have shrunk but not vanished after discovery.

The three hurdles a "real" factor must clear

Academia has identified *hundreds* of published factors — the so-called "**factor zoo**." Most are data-mining artifacts. A credible factor should be:

1. **Persistent** — works across long periods, not just one lucky decade.
2. **Pervasive** — shows up across countries, sectors, and often other asset classes (bonds, currencies, commodities).
3. **Robust** — survives reasonable changes in definition (value works whether you use book-to-market, earnings yield, or cash-flow yield).
4. **Investable** — survives real trading costs and capacity constraints.
5. **Intuitive** — has a credible risk-based or behavioral *why*, so you are not just fitting noise.

The five factors below (value, size, momentum, quality, low volatility) are the survivors that clear these hurdles.

4. Full Content — The Main Equity Factors

4.1 Value

Definition. Buy stocks that are cheap relative to fundamentals; avoid expensive ones. Common metrics: high book-to-market (B/M), low price-to-earnings (P/E), low price-to-book (P/B), low EV/EBITDA, high dividend or free-cash-flow yield.

Construction. The classic academic proxy is Fama-French **HML** ("High Minus Low"): go long the cheapest 30% of stocks by B/M, short the most expensive 30%, and the return spread is the value factor.

Economic rationale. - *Risk story*: cheap firms are distressed, carry high financial/operating leverage, and have inflexible assets — they suffer most in downturns, so investors demand a premium. - *Behavioral story*: investors extrapolate past growth too far, overpaying for glamour stocks; when the hype fades, mean reversion rewards the patient value buyer.

Historical premium. Roughly 3–5% a year in US large-caps over the very long run (1926–), but with brutal multi-year droughts — notably 2007–2020, when value dramatically lagged growth and many declared it dead, before a sharp 2021–2022 revival.

4.2 Size

Definition. Small-capitalization stocks tend to outperform large-caps over the long run.

Construction. Fama-French **SMB** ("Small Minus Big"): long small-cap, short large-cap.

Economic rationale. - *Risk story*: small firms are less liquid, more fragile, more exposed to funding shocks and business-cycle risk. - *Behavioral/structural*: under-researched, under-owned by institutions, harder to

arbitrage.

Caveats. Of all the classic factors, size is the *weakest and most contested*. The raw premium is small, concentrated in the tiniest illiquid micro-caps, largely disappears after transaction costs, and has been near-zero since the early 1980s. Modern research (e.g., AQR's "Size Matters, If You Control Your Junk") argues size only *works* once you strip out low-quality small "junk" firms — meaning size is really a vehicle that needs a quality filter to shine.

4.3 Momentum

Definition. Stocks that have outperformed over the past 3–12 months tend to keep outperforming over the next few months; recent losers keep losing. "The trend is your friend."

Construction. The standard signal is the **past 12-month return skipping the most recent month** (12-1, skipping month –1 to avoid short-term reversal). Long recent winners, short recent losers. Jegadeesh and Titman documented it in 1993; Carhart added it as the fourth factor (**WML / UMD**, "Winners Minus Losers" / "Up Minus Down") to the Fama-French three-factor model in 1997.

Economic rationale. - *Behavioral*: investors *under-react* to news (information diffuses slowly, prices drift toward fair value), then eventually *over-react* (herding pushes trends too far). Both create autocorrelation. - *Risk story*: weaker and less agreed-upon — momentum's returns are hard to pin to a stable macro risk.

The catch — momentum crashes. Momentum has the highest average premium of any factor (~8%+ historically) *but* suffers rare, violent crashes. Because after a market crash the "loser" leg is full of beaten-down cyclical stocks, a sharp market rebound sends those losers rocketing — and the short leg blows up. March–May 2009 was a textbook momentum crash: as the market V-shaped off the bottom, momentum strategies lost 30%+ in weeks. Its return distribution has fat left-tail skew.

4.4 Quality

Definition. High-quality companies — profitable, stable, low-debt, well-governed — outperform low-quality "junk" on a risk-adjusted basis.

Construction. Composite scores blending: high gross/operating profitability (Novy-Marx's key insight), stable earnings growth, low accruals, low leverage, high margins, conservative asset growth. Fama and French's **five-factor model (2015)** formalized two quality-adjacent factors: **RMW** ("Robust Minus Weak" profitability) and **CMA** ("Conservative Minus Aggressive" investment).

Economic rationale. - *Behavioral*: investors chase exciting, fast-growing but unprofitable stories and underprice boring, cash-generative compounders. - *Risk-related*: somewhat paradoxical — safer firms earning higher returns looks like a *free lunch*, one reason quality is partly explained as a mispricing.

Why it is loved by practitioners. Quality is beautifully *complementary to value*: cheap-and-junky is a value trap; cheap-and-high-quality is the sweet spot. Combining them (buy cheap *good* companies) historically improves both. Quality also tends to be *defensive*, cushioning drawdowns.

4.5 Low Volatility (Low Beta)

Definition. Low-risk stocks — low volatility or low beta — deliver returns as high as, or higher than, high-risk stocks on a risk-adjusted basis. This directly *contradicts CAPM*, which says higher beta must earn higher return. Hence the name "**low-volatility anomaly**."

Construction. Either minimum-variance portfolios (optimize for lowest total risk) or simple low-beta / low-realized-volatility rankings. Frazzini and Pedersen's "**Betting Against Beta**" (BAB) factor formalizes it: long low-beta (levered up), short high-beta.

Economic rationale. - *Leverage constraints (Frazzini-Pedersen):* many investors cannot use leverage, so to chase higher returns they *overpay for high-beta stocks*, bidding down their expected return and leaving low-beta stocks cheap. - *Behavioral "lottery" preference:* investors overpay for volatile, high-upside "lottery ticket" stocks (hoping for the next Tesla), depressing their returns; boring low-vol names are neglected. - *Agency:* benchmarked managers avoid low-beta stocks because they risk large tracking error versus the index.

Character. Low-vol is the classic *defensive* factor — it shines in bear markets and lags in roaring bull markets. It structurally overweights stable sectors (utilities, consumer staples), which introduces sector and interest-rate sensitivity.

Factor summary table

Factor	One-line definition	Classic metric / factor	Primary rationale	Typical character
Value	Cheap beats expensive	Book-to-market (HML)	Distress risk + overextrapolation	Cyclical, deep-drought risk
Size	Small beats large	Market cap (SMB)	Liquidity/fragility risk	Weak, needs quality filter
Momentum	Winners keep winning	12-1 month return (WML)	Under- then over-reaction	High premium, crash risk
Quality	Profitable, stable, low-debt wins	Profitability (RMW), investment (CMA)	Mispricing of boring compounders	Defensive, pairs with value
Low Volatility	Low risk earns high risk-adjusted return	Low beta (BAB)	Leverage constraints + lottery demand	Defensive, bond-like tilt

5. Worked / Applied Examples

Example 1 — Decomposing a "star" manager's alpha (Carhart four-factor attribution)

A value-fund manager, **Meera**, delivers a 14.0% return in a year when the risk-free rate is 5.0%, so her excess return is **9.0%**. Her marketing deck screams "alpha." Let us run a factor attribution using the **Carhart four-factor model**:

$$R_p - R_f = \alpha + \beta_{MKT} \cdot \text{MKT} + \beta_{SMB} \cdot \text{SMB} + \beta_{HML} \cdot \text{HML} + \beta_{WML} \cdot \text{WML} + \epsilon$$

Suppose a regression of her monthly returns gives these loadings, and the realized factor premia for the year were:

Factor	Her loading (β)	Factor premium this year	Contribution = $\beta \times$ premium
Market (MKT)	1.00	6.0%	6.00%
Size (SMB)	0.30	2.0%	0.60%
Value (HML)	0.60	3.0%	1.80%
Momentum (WML)	−0.10	4.0%	−0.40%
Total factor-explained			8.00%

Her total excess return was 9.0%. The factors explain **8.0%** of it. The residual — genuine, unexplained skill — is:

$$\alpha = 9.0\% - 8.0\% = \mathbf{1.0\%}$$

Interpretation. Of Meera's 9% "outperformance," fully 8 points came from *market, size, and value tilts* an investor could have replicated with a cheap smart-beta ETF. Only **1% is true alpha**. If she charges 1.2% in fees, the client is paying more than the entire alpha to access factor beta that costs ~0.20% elsewhere. *This calculation is the beating heart of the factor-investing revolution* — it reframes fee negotiations across the whole industry.

Example 2 — Building a two-factor "quality-value" tilt versus a cap-weighted benchmark

You run a small portfolio and want to tilt toward **value + quality** while staying close to a benchmark of five stocks. The cap-weighted benchmark weights and each stock's factor scores (z-scores, standardized so mean 0) are:

Stock	Cap weight	Value z-score	Quality z-score	Combined score (avg)
A	30%	−1.0	+0.5	−0.25
B	25%	+1.2	+1.0	+1.10
C	20%	+0.5	−0.5	0.00
D	15%	−0.8	+0.8	0.00
E	10%	+0.6	−1.2	−0.30

Tilt rule: new weight = cap weight $\times$ (1 + 0.5 $\times$ combined score), then renormalize so weights sum to 100%.

Step 1 — apply the tilt multiplier: - A: 30% $\times$ (1 + 0.5 $\times$ −0.25) = 30% $\times$ 0.875 = 26.25% - B: 25% $\times$ (1 + 0.5 $\times$ 1.10) = 25% $\times$ 1.55 = 38.75% - C: 20% $\times$ (1 + 0) = 20.00% - D: 15% $\times$ (1 + 0) = 15.00% - E: 10% $\times$ (1 + 0.5 $\times$ −0.30) = 10% $\times$ 0.85 = 8.50%

Step 2 — sum = 26.25 + 38.75 + 20 + 15 + 8.5 = **108.5%**. Renormalize (divide by 1.085):

Stock	Cap weight	Tilted weight	Active bet
A	30.0%	24.19%	−5.81%
B	25.0%	35.71%	+10.71%
C	20.0%	18.43%	−1.57%
D	15.0%	13.82%	−1.18%
E	10.0%	7.83%	−2.17%

Interpretation. The rule mechanically overweights the cheap-and-profitable stock B (+10.7%) and underweights the expensive stock A. Notice this is *fully transparent and rules-based* — no manager judgment. This is exactly how a smart-beta index is constructed. Also note the largest bet landed on B because it scored high on *both* factors — combining factors concentrates conviction where signals agree.

Example 3 — Why factor diversification beats single-factor timing

Two factors, **Value** and **Momentum**, each have an expected premium of 4% with volatility of 12%. Crucially, their returns are **negatively correlated ($\rho = -0.4$)** — a famous empirical regularity, because momentum loads on recent winners (often growth) while value buys losers, so they lean opposite ways.

A 50/50 combination has expected return still 4% (average of two 4%s), but its volatility is:

$$\sigma_p = \sqrt{w_V^2 \sigma_V^2 + w_M^2 \sigma_M^2 + 2w_V w_M \rho \sigma_V \sigma_M}$$

$$\sigma_p = \sqrt{0.5^2(0.12^2) + 0.5^2(0.12^2) + 2(0.5)(0.5)(-0.4)(0.12)(0.12)}$$

$$\sigma_p = \sqrt{0.0036 + 0.0036 - 0.001728} = \sqrt{0.005472} = \mathbf{7.4\%}$$

The combined portfolio's Sharpe-style ratio (premium ÷ vol) rises from **4/12 = 0.33** for either factor alone to **4/7.4 = 0.54** — a *62% improvement* — with no drop in expected return. The negative correlation smooths value's long droughts (when value suffers, momentum often thrives, and vice versa). **This is the single strongest argument for multi-factor investing over betting on one factor:** you get diversification *within* the return-driver dimension, not just across stocks.

6. Connections

- **Chapter 05 (CAPM):** CAPM is the one-factor ancestor — market beta only. Every equity factor here is, in effect, a documented *failure* of CAPM: a source of return CAPM says should not exist. Low-volatility is the most direct rebuke (high beta ≠ high return).
- **Chapter 06 (APT and Multifactor Models):** This chapter is the *applied, investable* face of Chapter 06's theory. Fama-French (HML, SMB, RMW, CMA) and Carhart (WML) supply the regression machinery; factor investing turns those factors into portfolios you can actually own.
- **Chapter 12 (Market Efficiency):** The behavioral rationale for factors is a direct challenge to strong-form EMH. Yet factors are *also* consistent with a "rational risk" reading of efficiency — the debate in Section 3 is the efficiency debate in miniature.
- **Chapter 03 / 04 (Markowitz, Diversification):** Example 3 is pure Markowitz — but diversifying across *factors* rather than *assets*. The insight that asset-class diversification failed in 2008 (everything correlated to 1) while factor diversification held up is a major post-crisis theme.

- **Chapter 11 (Performance Attribution):** Factor models are *the* modern attribution engine — Example 1 shows how they separate skill from style, reshaping how fund performance is judged and fees are set.
 - **Chapter 09 (Equity Portfolio Management):** Smart beta sits precisely on the active-passive spectrum central to equity management strategy.
-

7. Key Terms

- **Factor** — a persistent, quantifiable characteristic that explains the risk/return of a group of securities.
 - **Factor premium** — the long-run excess return earned by tilting toward a factor.
 - **Smart beta / strategic beta** — rules-based indices that deviate from cap-weighting to capture factor tilts, cheaply and transparently.
 - **HML, SMB, WML, RMW, CMA** — the Fama-French/Carhart factor portfolios: High-minus-Low (value), Small-minus-Big (size), Winners-minus-Losers (momentum), Robust-minus-Weak (profitability), Conservative-minus-Aggressive (investment).
 - **Long-short factor portfolio** — long the high-scoring quintile, short the low-scoring quintile; isolates the *pure* factor return, market-neutral.
 - **Long-only tilt** — overweight/underweight within a fully-invested long portfolio; how most retail factor ETFs actually work (no shorting).
 - **Factor zoo** — the 400+ published "factors," most of which are data-mining artifacts.
 - **Betting Against Beta (BAB)** — the low-volatility factor built on leverage constraints.
 - **Factor timing** — attempting to rotate into factors before they outperform; notoriously difficult.
 - **Crowding** — too much capital chasing a factor, compressing its premium and raising crash risk.
 - **Factor cyclicity** — the tendency of factors to have long stretches of under- and out-performance driven by the macro cycle.
-

8. Common Confusions

"Smart beta is passive, so it is safe / low-risk." No. Smart beta is *rules-based* but it is an *active bet* relative to the market. A value ETF that lags growth for a decade (2010–2020) delivers real, painful underperformance. It is passive in *implementation*, active in *exposure*.

"Factor investing = stock picking." No — it is the opposite. Stock picking bets on idiosyncratic, firm-specific stories. Factor investing deliberately *diversifies away* firm-specific risk and bets only on the *systematic* characteristic shared across hundreds of names.

"A factor premium is a free lunch / guaranteed." No. Under the risk view it is *compensation for real pain* in bad times (value crashes in recessions; momentum crashes on rebounds). Under the behavioral view it can *decay* after discovery. Neither is a guarantee; both involve holding uncomfortable positions.

"Value and growth are opposite factors." Subtle. *Value* is a documented factor; "growth" is simply the *absence* of value (the expensive leg), not a separately rewarded premium. There is no robust "growth premium." What people call growth's dominance is usually momentum plus quality plus a low-rate regime, not a factor of its own.

"More factors always means better." No — beyond a handful of robust factors you are mostly buying overlapping exposure and data-mined noise from the factor zoo. Adding a fifth correlated value-variant adds cost, not diversification.

"Cap-weighted indexing is neutral / bet-free." No — Section 2's key point. Cap-weighting is itself a momentum-and-large-size tilt, mechanically buying more of whatever rose. There is no such thing as a bet-free equity portfolio.

"Beta in factor investing means the same as CAPM beta." Watch the word. *Market beta* is CAPM's single sensitivity. *Factor beta* is sensitivity to any factor (a stock has a value beta, a momentum beta, etc.). *Smart beta* uses "beta" loosely to mean "systematic exposure you can buy cheaply."

9. Recap

- Factor investing sits between passive indexing and active management: it harvests **systematic, rules-based tilts** toward characteristics that have earned long-run premia.
 - The five workhorse equity factors are **value, size, momentum, quality, and low volatility**. Each clears the persistence-pervasiveness-robustness-investability-intuition hurdles (size most weakly).
 - Premia persist for two reasons — **risk compensation** (rational) and **behavioral errors plus limits to arbitrage** — and most factors blend both.
 - **Smart beta / factor ETFs** are the cheap, transparent product wrapper. They let investors capture what was once sold as expensive alpha for ~10–40 bps.
 - **Factor attribution** (Carhart four-factor and beyond) reframes performance: much "alpha" is really cheap factor beta (Example 1).
 - Factors are **cyclical and can suffer long, painful droughts** (value 2010–2020) and sharp **crashes** (momentum 2009). The strongest defense is **multi-factor diversification**, exploiting low or negative cross-factor correlations (Example 3).
-

10. Quick-Reference / Interview Points

The one-liner. "Factor investing systematically harvests the handful of persistent return drivers — value, momentum, quality, size, low-vol — through cheap, rules-based portfolios; smart beta is its ETF wrapper, sitting between passive and active."

Name the five factors and their signals. Value (book-to-market), Size (market cap), Momentum (12-1 month return), Quality (profitability/low-debt), Low-vol (low beta). Bonus: Fama-French five-factor adds RMW (profitability) and CMA (investment); Carhart adds WML (momentum).

Give the two rationales for a premium. Risk-based (compensation for bad-times risk — permanent) vs behavioral (mispricing from investor errors plus limits to arbitrage — can decay). Say "most factors are a blend" to sound mature.

The killer insight for a fee discussion. "Much of what active managers charge for is replicable factor beta, not alpha. Run a Carhart four-factor regression; if the loadings explain the excess return, you are paying alpha fees for beta exposure." (Cite Example 1's 8%-of-9% split.)

Why multi-factor over single-factor? Factors are cyclical and their premia are low or negatively correlated (value vs momentum ≈ -0.4). Combining them raises the Sharpe ratio materially with no loss of expected return (Example 3: $0.33 \rightarrow 0.54$). It also spares you the near-impossible task of factor timing.

Know the failure modes. Value's lost decade (2010–2020); momentum crashes on sharp market rebounds (March–May 2009, $\sim 30\%$ drawdown); the low-vol anomaly contradicts CAPM; size largely evaporated post-1980 and needs a quality filter to work.

The "cap-weighting is a bet" line. Market-cap indices overweight the expensive and largest names by construction — a built-in size/momentum tilt. So "the choice isn't bet vs no-bet, it's *which* bet."

Watch out for the factor zoo and crowding. 400+ published factors, most are noise; insist on an economic story and out-of-sample evidence. Post-publication and post-ETF-launch, premia often shrink as capital crowds in — a live risk to monitor.

Real-world anchors. MSCI and FTSE Russell build the factor indices; iShares, Vanguard, Invesco, and DFA/Avantis run the big factor/smart-beta funds; AQR and Research Affiliates ("Fundamental Indexing") popularized the practice. DFA and AQR are essentially "factor premia in a fund" firms.

Q&A — Factor Investing and Smart Beta

Scope: Investments — Chapter 16 (Factor Investing and Smart Beta). Every question is followed by a full model answer. All rates are annual and in percent unless stated. Work every numerical problem yourself before checking the solution. Sections: **A** concept-check · **B** applied/numerical (full step-by-step, reconciling) · **C** interview-style with model answers · **D** MCQs with reasoning.

The chapter in one line

One-line statement: Factor investing harvests a handful of persistent return drivers — value, size, momentum, quality, low-volatility — through cheap, transparent, rules-based portfolios; **smart beta** is its ETF wrapper, sitting between pure passive indexing and expensive discretionary active management.

Section A — Concept Check

A1. What is a "factor," and what distinguishes it from an ordinary stock characteristic? A factor is a broad, **persistent, quantifiable** characteristic that helps explain the risk and return of a *group* of securities. A real factor clears five hurdles: persistence (works over long periods), pervasiveness (across countries, sectors, asset classes), robustness (survives changes in definition), investability (survives real trading costs), and an intuitive economic rationale. A characteristic that sorts returns in one back-test but has no economic story is a data-mining artifact, not a factor.

A2. Define smart beta and place it on the passive–active spectrum. Smart beta (also "strategic beta" or "factor ETFs") is any **rules-based index that deliberately departs from market-cap weighting** to tilt toward one or more factors. It occupies the middle of the spectrum: the transparency and low cost of indexing, with the factor tilts once sold as expensive active skill. It is passive in *implementation* (mechanical rules, no discretion) but active in *exposure* (a real bet against the cap-weighted market).

A3. The chapter claims "cap-weighting is itself a bet." Explain. A market-cap index automatically buys *more* of whatever has risen (the biggest, most expensive names) and *less* of what has fallen, embedding a built-in **large-size and momentum tilt**. There is no bet-free equity portfolio; the real choice is not "bet vs no bet" but "*which* bet, made deliberately or by accident."

A4. Give the two competing rationales for why a factor premium persists. (1) **Risk-based (rational):** compensation for a systematic risk you cannot diversify away — value stocks are distressed, cyclically fragile firms that hurt precisely in recessions, so investors demand payment; the premium is permanent because the risk never disappears. (2) **Behavioral (mispricing):** persistent human errors (over-extrapolation, under-reaction) that limits to arbitrage prevent smart money from fully correcting; the premium *can* decay once discovered and crowded. Mature practitioners treat most factors as a **blend**.

A5. Why does the risk-vs-behavioral debate matter for how you hold a factor? It changes expectations in a drought. If value is pure risk compensation, hold through the pain and expect the premium to endure. If it is pure mispricing, worry that post-publication crowding erodes it. Since the truth is usually a blend, expect real compensation but stay alert to crowding.

A6. Name the five workhorse equity factors and their classic signal. Value (book-to-market / low P/E), Size (market cap), Momentum (past 12-month return skipping the most recent month, "12-1"), Quality (profitability, low debt, stable earnings), Low Volatility (low beta / low realized volatility).

A7. What are HML, SMB, WML, RMW, and CMA? Academic long-short factor portfolios. **HML** (High Minus Low) = value (long cheap, short expensive by B/M). **SMB** (Small Minus Big) = size. **WML/UMD** (Winners Minus Losers) = momentum, added by Carhart (1997). **RMW** (Robust Minus Weak, profitability) and **CMA** (Conservative Minus Aggressive, investment) are the two quality-adjacent factors from Fama-French's 2015 five-factor model.

A8. Why is size the weakest and most contested of the classic factors? The raw size premium is small, concentrated in tiny illiquid micro-caps, largely evaporates after costs, and has been near-zero since the early 1980s. Research (AQR's "Size Matters, If You Control Your Junk") argues size only works once you strip out low-quality "junk" firms — so it needs a **quality filter** to shine and is barely a standalone premium.

A9. Why does momentum have the highest average premium yet frighten practitioners? Momentum has historically earned ~8%+ but suffers rare, violent **crashes** with fat left-tail skew. After a market crash the "loser" leg fills with beaten-down cyclical; when the market V-shapes off the bottom, those losers rocket and the short leg blows up — March–May 2009 saw momentum lose 30%+ in weeks.

A10. Why do quality and value pair so well together? They are complementary. Cheap-and-junky is a **value trap**; cheap-and-high-quality is the sweet spot. A quality filter (profitable, low-debt, stable) screens out distressed value firms that keep sinking. Quality is also defensive, so the combination historically improves both return and risk versus either alone.

A11. Why is the low-volatility factor a direct rebuke to CAPM? CAPM says higher beta *must* earn higher return. The low-vol anomaly shows the opposite: low-beta stocks deliver returns as high as or higher than high-beta stocks on a risk-adjusted basis. Explanations: leverage constraints (Frazzini-Pedersen "Betting Against Beta" — investors who cannot lever overpay for high-beta stocks), lottery-ticket demand, and agency (benchmarked managers shun low-beta stocks for tracking-error reasons).

A12. What is the "factor zoo," and how do you defend against it? The 400+ published "factors," most of which are data-mining artifacts that fail out-of-sample. The defense is discipline: demand a credible economic story, out-of-sample and cross-country evidence, robustness to definition changes, and low correlation with factors you already own. A fifth correlated value-variant adds cost and noise, not diversification.

A13. Distinguish a long-short factor portfolio from a long-only tilt. A **long-short** portfolio goes long the high-scoring quintile and short the low-scoring quintile, isolating the *pure*, market-neutral factor return (how academics measure HML, SMB). A **long-only tilt** overweights high-scoring and underweights low-scoring names within a fully invested portfolio (no shorting). Most retail factor ETFs are long-only tilts, carrying market beta plus a diluted dose of the factor.

A14. Why is "growth" not a factor in the way value is? Value is a documented, rewarded premium; "growth" is simply the *absence* of value — the expensive leg of the sort — not a separately rewarded premium. There is no robust "growth premium." What looks like growth's dominance (e.g., 2010–2020) is usually momentum + quality + a low-rate regime, not a factor of its own.

Section B — Applied / Numerical Problems

B1. Carhart four-factor attribution — how much is real alpha? A manager returns 13.5% when the risk-free rate is 4.0%. A regression gives loadings; the year's realized factor premia are shown. Compute the contribution of each factor and the residual alpha.

Factor	Loading β	Premium	Contribution
Market	1.05	7.0%	?
SMB	0.40	1.5%	?
HML	0.55	3.0%	?
WML	-0.20	5.0%	?

Solution. Excess return = 13.5% – 4.0% = **9.5%**. - Market: $1.05 \times 7.0\% = 7.35\%$ - SMB: $0.40 \times 1.5\% = 0.60\%$ - HML: $0.55 \times 3.0\% = 1.65\%$ - WML: $-0.20 \times 5.0\% = -1.00\%$ - Total factor-explained = 7.35 + 0.60 + 1.65 – 1.00 = **8.60%**

Alpha = 9.5% – 8.6% = **0.9%**. Of the 9.5% "outperformance," 8.6 points are replicable factor beta an investor could buy for ~20 bps; only 0.9% is genuine skill. Note the negative momentum loading cost the manager 1.0% because momentum was positive that year — a value manager who leans away from recent winners bleeds return when momentum works.

B2. Two-factor diversification — the Sharpe uplift. Value and Momentum each have an expected premium of 5% and volatility of 14%, with correlation $\rho = -0.3$. Find the volatility and premium-to-vol ratio of a 50/50 blend and compare with a single factor.

Solution. Premium of blend = $0.5(5\%) + 0.5(5\%) = 5\%$ (unchanged).

$\sigma_p = \sqrt{0.5^2(0.14^2) + 0.5^2(0.14^2) + 2(0.5)(0.5)(-0.3)(0.14)(0.14)}$ - Term 1 = $0.25 \times 0.0196 = 0.0049$ - Term 2 = $0.25 \times 0.0196 = 0.0049$ - Term 3 = $2 \times 0.25 \times (-0.3) \times 0.0196 = -0.00294$ - Sum = $0.0049 + 0.0049 - 0.00294 = 0.00686$ - $\sigma_p = \sqrt{0.00686} = \mathbf{8.28\%}$

Ratio single factor = $5/14 = \mathbf{0.357}$. Ratio blend = $5/8.28 = \mathbf{0.604}$. That is a **69% improvement** with no drop in expected return — the negative correlation smooths value's droughts (momentum often thrives when value suffers). This is the core quantitative case for multi-factor over single-factor investing.

B3. Smart-beta tilt construction. A three-stock benchmark is cap-weighted: X 50%, Y 30%, Z 20%.

Combined value+quality z-scores are X -0.6, Y +1.0, Z +0.4. Tilt rule: new weight = cap weight $\times (1 + 0.5 \times \text{score})$, then renormalize. Find the tilted weights and active bets.

Solution. Step 1 — multipliers: - X: $50\% \times (1 + 0.5 \times -0.6) = 50\% \times 0.70 = 35.0\%$ - Y: $30\% \times (1 + 0.5 \times 1.0) = 30\% \times 1.50 = 45.0\%$ - Z: $20\% \times (1 + 0.5 \times 0.4) = 20\% \times 1.20 = 24.0\%$

Step 2 — sum = 35 + 45 + 24 = **104.0%**. Renormalize ($\div 1.04$): - X: $35.0/1.04 = \mathbf{33.65\%}$ → active -16.35% - Y: $45.0/1.04 = \mathbf{43.27\%}$ → active +13.27% - Z: $24.0/1.04 = \mathbf{23.08\%}$ → active +3.08%

Check: $33.65 + 43.27 + 23.08 = 100.0\%$ ✓. The rule mechanically overweights the high-scoring Y and slashes the low-scoring X — transparent and judgment-free, exactly how a smart-beta index rebalances.

B4. Fee arithmetic — paying alpha fees for beta. Using B1 (gross alpha 0.9%), the manager charges 1.10%; an equivalent smart-beta ETF charges 0.25%. Net alpha after fee = $0.9\% - 1.10\% = \mathbf{-0.20\%}$ — the client is

net negative on skill, and overpays $1.10\% - 0.25\% = 0.85\%$ for beta available cheaply. The heart of the fee-compression argument.

B5. Momentum crash — asymmetric payoff. A momentum long-short book returns $+1.2\%$ per month in 11 months but -28% in a single crash month. Annual arithmetic return?

Solution. $11 \times 1.2\% = +13.2\%$; plus $-28\% \rightarrow -14.8\%$ for the year. Despite a positive premium in most months, one fat-tail crash wipes out more than a year of gains — momentum's **negative skew** means standard deviation understates its true risk.

Section C — Interview-Style Questions

C1. "In one minute, what is factor investing and why did it disrupt the fund industry?" Factor investing systematically harvests the handful of persistent return drivers — value, size, momentum, quality, low-vol — through cheap, rules-based portfolios. It disrupted the industry by revealing that much of what active managers charged $1\%+$ for was not alpha but *replicable factor beta* you could buy for 10–40 bps. Once value and momentum were shown to be harvestable, transparent premia, the discretionary "genius" premium collapsed, and smart-beta ETFs packaged those tilts for retail — compressing fees industry-wide.

C2. "A manager beat the S&P by 4% last year. How would you tell if that's skill?" Run a factor attribution — a Carhart four-factor regression (market, size, value, momentum), ideally extended with quality and low-vol. Estimate the fund's loadings, multiply each by the factor's realized premium, and sum. Whatever the factors explain is *beta*, not skill; the residual is alpha. If 3.5 of the 4 points come from a value-plus-small tilt, the manager delivered only 0.5% of true alpha — and if fees exceed that, the client pays alpha prices for beta. I'd also insist on multi-year data, since one year is noisy.

C3. "Is a factor premium a free lunch?" No. Under the risk view it is compensation for *real pain* in bad times — value crashes in recessions, momentum crashes on sharp rebounds — so you are paid to hold assets that hurt when you can least afford it. Under the behavioral view it is a mispricing that *can decay* once discovered and crowded. Either way you endure uncomfortable positions and long droughts; the premium rewards that discomfort, not a costless edge.

C4. "Why bother with multiple factors instead of just picking the best one?" Two reasons. First, factor timing — knowing which factor will outperform next — is notoriously unreliable. Second, factors are cyclical and their premia are often low or negatively correlated: value and momentum run about -0.3 to -0.4 because value buys losers while momentum buys winners. Combining them raises the risk-adjusted return materially — the premium-to-vol ratio can jump from ~ 0.35 to ~ 0.60 with no loss of expected return. Diversifying across return *drivers* is the strongest, most reliable edge in factor investing.

C5. "What are the main ways factor investing can disappoint an investor?" Long droughts (value lagged growth roughly 2010–2020, prompting many to declare it dead); sharp crashes (momentum lost $\sim 30\%$ in weeks in spring 2009); factor decay from crowding after publication and ETF launch; hidden concentration (low-vol overweights utilities and staples, adding rate sensitivity); and the factor zoo — buying data-mined "factors" with no real premium. Long-only retail ETFs also deliver a diluted dose of the factor plus full market beta, so the tilt is weaker than clients assume.

C6. "A client says smart beta is passive and therefore low-risk. Respond." Smart beta is passive in *implementation* — mechanical rules, no discretion — but an *active bet* relative to the cap-weighted market. A value smart-beta ETF that trails growth for a decade delivers real, painful underperformance. Passive

implementation means transparency and low cost, not low tracking error or low risk. The exposure is deliberately non-market, and non-market exposure is where both the risk and the potential reward live.

C7. "Who are the real-world players, and what should I know about them?" Index providers MSCI and FTSE Russell build the factor indices. Large asset managers — iShares (BlackRock), Vanguard, Invesco — run the big smart-beta/factor ETFs. On the factor-purist end, DFA (Dimensional) and Avantis, plus AQR and Research Affiliates (the latter popularized "Fundamental Indexing"), are essentially "factor premia in a fund" shops. This shows factor investing is not just theory but a large, competitive industry — and that competition is one source of the crowding risk that compresses premia.

Section D — Multiple-Choice Questions (with reasoning)

D1. Which best describes where smart beta sits? A. It is identical to a cap-weighted index fund. B. It is a rules-based index that tilts away from cap-weighting toward factors. C. It is a discretionary active strategy. D. It is a leveraged derivatives overlay.

Answer: B. Smart beta is rules-based (unlike discretionary active, C) but deliberately departs from cap-weighting (unlike A) to capture factor tilts. It requires no leverage or derivatives (D). It occupies the middle of the passive–active spectrum.

D2. The low-volatility anomaly is significant primarily because it: A. Confirms the CAPM prediction that beta drives return. B. Contradicts CAPM by showing low-beta stocks are not rewarded with lower returns. C. Proves markets are perfectly efficient. D. Is only observable in emerging markets.

Answer: B. CAPM predicts higher beta → higher return; the low-vol anomaly shows low-beta stocks earn as much or more on a risk-adjusted basis, directly contradicting CAPM (so A is wrong). It is a well-documented anomaly across developed and emerging markets (D wrong) and is evidence *against* strict efficiency (C wrong).

D3. The standard momentum signal is: A. Past 1-month return. B. Past 12-month return including the most recent month. C. Past 12-month return skipping the most recent month (12-1). D. Forecast next-year earnings growth.

Answer: C. Momentum uses the past 12-month return but *skips the most recent month* to avoid short-term reversal, which contaminates the signal. Including month –1 (B) mixes in reversal; a 1-month signal (A) is reversal, not momentum; D is a growth forecast, unrelated.

D4. Which factor is considered the weakest and most contested? A. Value B. Momentum C. Size D. Quality

Answer: C. The size premium is small, concentrated in illiquid micro-caps, largely gone after costs, and near-zero since the early 1980s. Research suggests it only works with a quality filter to remove "junk" small caps. Value, momentum, and quality all have stronger, more robust evidence.

D5. Value and momentum are frequently combined because their returns are: A. Perfectly positively correlated. B. Negatively (or lowly) correlated, improving diversification. C. Uncorrelated with the market. D. Identical in every period.

Answer: B. Value buys recent losers while momentum buys recent winners, so they tend to be negatively (≈ -0.3 to -0.4) or lowly correlated. Combining them lowers portfolio volatility and raises the risk-adjusted

return without cutting expected return — the classic multi-factor diversification benefit. A and D are false; C misstates the reason.

D6. In a Carhart four-factor attribution, "alpha" is: A. The fund's total return. B. The market's return. C. The residual return not explained by the four factor exposures. D. The largest single factor contribution.

Answer: C. Alpha is the residual — excess return minus the sum of (loading × factor premium) across market, size, value, and momentum. It measures genuine skill relative to the chosen factors. It is neither total return (A), the market leg (B), nor a single factor's contribution (D).

D7. "Cap-weighting is itself a bet" because a cap-weighted index: A. Equally weights every stock. B. Mechanically overweights the largest, best-performing (often most expensive) stocks. C. Rebalances daily into losers. D. Holds only value stocks.

Answer: B. Cap-weighting buys more of whatever has risen and grown largest, embedding a built-in large-size and momentum tilt. It does not equal-weight (A), does not tilt into losers (C), and is not a value strategy (D). Hence even "neutral" indexing is an implicit bet.

D8. A "real" factor should clear all of the following EXCEPT: A. Persistence across long periods. B. Pervasiveness across markets and asset classes. C. Guaranteed positive return every single year. D. A credible economic (risk or behavioral) rationale.

Answer: C. No factor delivers a positive return every year — all suffer droughts and some suffer crashes; that variability is the risk you are paid for. The genuine hurdles are persistence, pervasiveness, robustness, investability, and an intuitive rationale (A, B, D).

D9. The primary danger of the "factor zoo" is: A. Too few published factors to choose from. B. Most published factors are data-mining artifacts that fail out-of-sample. C. Factors are impossible to compute. D. Regulators ban most factors.

Answer: B. With 400+ published factors, most are statistical flukes that vanish out-of-sample. The defense is to require an economic story, out-of-sample evidence, robustness, and low correlation with existing factors. A, C, and D misstate the problem.

D10. The most defensible reason to prefer multi-factor over single-factor investing is: A. It guarantees a higher return every year. B. Factor timing is unreliable, and low/negative cross-factor correlations raise risk-adjusted returns. C. It eliminates all market risk. D. It removes the need to pay any fees.

Answer: B. Because timing factors is notoriously hard and factors are cyclical with low or negative correlations, combining them lifts the risk-adjusted return without sacrificing expected return. It offers no annual guarantee (A), does not remove market risk in long-only form (C), and still carries fees (D).

Self-check summary

- Numerical answers reconcile: B3 tilted weights sum to 100%; B1 alpha = 9.5% excess – 8.6% factor-explained = 0.9%; B2 blend vol 8.28% < 14% single-factor.
- Every question has a full worked answer; coverage maps to chapter sections 2–6 and the interview points in section 10.

Chapter 17 — Passive vs Active Management

1. The Problem / Need

Every rupee or dollar an investor commits to markets faces one unavoidable strategic fork before a single security is bought: **do you try to beat the market, or do you try to be the market?**

This is not an academic curiosity. It is the single decision that most powerfully determines the *distribution* of an investor's long-run outcomes — more than security selection, more than the specific fund manager's brilliance, and often more than asset allocation at the margins. The reason is arithmetic, not opinion. Active management, in aggregate, is a **negative-sum game after costs**: the average actively managed dollar must, by construction, earn the market return *minus* the fees it pays to try to beat that market. Passive management accepts the market return *minus* a tiny fee. Over a 30-year horizon, the gap between a 0.03% index fund and a 0.85% active fund compounds into a difference that can consume a quarter or more of an investor's terminal wealth — before you even ask whether the active manager had any skill.

So the "need" this chapter addresses is threefold:

1. **A decision framework.** An investor — or an analyst advising one — must be able to justify, on evidence, whether to pay for active management at all, and where.
2. **An understanding of the machinery.** Index funds and ETFs are not magic; they are engineered products with tracking mechanisms, replication choices, and hidden frictions. Interview candidates are routinely asked "how does an ETF actually track its index?" and "why is an ETF more tax-efficient than a mutual fund?"
3. **A resolution of a genuinely contested debate.** The passive-vs-active question is often presented as a religious war. It is not. The evidence is nuanced, the arithmetic is airtight, and the sophisticated answer is a *synthesis* — most famously the **core-satellite** approach.

Consider two savers, each investing \$10,000 per year for 30 years into a market returning 8% gross annually. One holds a passive fund costing 0.05%; the other holds an active fund costing 0.85% that, being average, delivers the market return before fees. This is the picture the rest of the chapter unpacks:

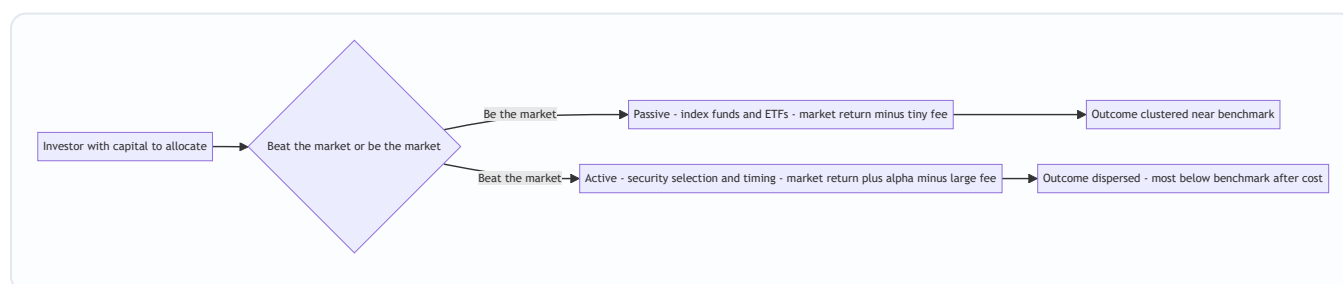


Figure 17.1 — The single strategic fork that shapes the distribution of every investor's outcome.

2. Core Idea

Passive management aims to *replicate* the return of a defined market index (the S&P 500, the Nifty 50, the MSCI World) by holding the index's constituents in the index's weights, and doing as little else as possible. Its philosophy: markets are hard to beat, information is largely priced in, so minimise cost and tracking error

and capture the market's return in full. Its instruments are **index mutual funds** and **exchange-traded funds (ETFs)**.

Active management aims to *beat* a benchmark through skill — selecting securities expected to outperform, avoiding those expected to underperform, timing entry and exit, or tilting toward factors and themes. Its philosophy: markets are imperfectly efficient, mispricings exist, and a skilled analyst can identify them net of the costs of trying. Its instrument is the actively managed fund (or a segregated mandate, hedge fund, etc.).

The core idea that reconciles them is **William Sharpe's "Arithmetic of Active Management" (1991)**, one of the most important and under-appreciated results in finance. It states, as an accounting identity that requires *no assumptions about market efficiency*:

Before costs, the return on the average actively managed dollar equals the return on the average passively managed dollar. After costs, the return on the average actively managed dollar is *less* than the return on the average passively managed dollar.

This is true by definition. The market is the sum of all its participants. Passive investors hold the market portfolio; therefore the *remaining* (active) holdings must also, in aggregate, be the market portfolio. So active and passive earn the same gross return in aggregate — but active pays more to do it. **Active management is a zero-sum game before costs and a negative-sum game after costs.** For every active winner there is an active loser, and the whole active cohort loses the fee bill to the fund industry.

This single result is why the *burden of proof* sits on active management. Passive is the default that must be argued away from, not toward.

3. Why / How It Works

3.1 Why passive works: the cost case and the arithmetic

The passive case rests on three legs.

Leg 1 — Sharpe's arithmetic (above). Costs are the only certainty. A 0.85% annual fee is a *guaranteed* 0.85% headwind; the alpha that is supposed to offset it is uncertain and, on average, absent. The distribution of active returns is roughly symmetric around the benchmark *before* fees, but fees shift the entire distribution left, so the *median* active fund lands below the benchmark.

Leg 2 — the Efficient Market Hypothesis (EMH), in its useful weak form. You do *not* need to believe markets are perfectly efficient to prefer passive. You only need to believe they are efficient *enough* that the marginal mispricing is too small, too rare, or too costly to exploit reliably after fees and trading costs. In large, liquid, heavily-researched markets (US large-cap being the archetype), thousands of well-resourced analysts compete away obvious mispricings, so the *net* opportunity is thin.

Leg 3 — compounding of the cost drag. Fees do not subtract linearly; they compound. A cost drag reduces the base on which all future returns compound. This is the "tyranny of compounding costs" (Bogle). Over decades it is enormous.

The compounding arithmetic, precisely: if the market compounds at gross rate g and the fund charges expense ratio c , terminal wealth multiple after n years is $(1+g-c)^n$ (approximately, ignoring the second-order interaction). The *fraction of the gross-return wealth lost to fees* is:

$$\text{Fee drag fraction} = 1 - \left(\frac{1+g-c}{1+g}\right)^n$$

For $g = 0.08$, $c = 0.0085$, $n = 30$: $\left(\frac{1.0715}{1.08}\right)^{30} = (0.99213)^{30} = 0.789$. So **21% of the wealth you would have had is gone to fees** — and that is *before* the active fund underperforms on skill.

3.2 How index funds and ETFs actually track the index

An index fund's job is to make its return hug the index's return. There are three replication methods:

Method	What it does	Used when	Trade-off
Full replication	Hold every constituent in exact index weight	Concentrated, liquid indices (S&P 500, Nifty 50)	Lowest tracking error; costly for broad indices
Sampling / optimisation	Hold a representative subset chosen to match the index's risk factors	Very broad indices (total-market with 3,000+ names, illiquid small-caps)	Cheaper to run; introduces sampling tracking error
Synthetic (swap-based)	Hold a collateral basket and enter a total-return swap with a bank to receive the index return	Hard-to-access markets, some European ETFs	Low tracking error but adds counterparty risk

The ETF creation/redemption mechanism is what makes ETFs distinctive and is a favourite interview topic. An ETF trades on an exchange like a stock, but its supply is elastic through **authorised participants (APs)** — large institutions (market makers) who can create and redeem shares in large blocks ("creation units", typically 50,000 shares):

- If the ETF's market price rises *above* its net asset value (NAV) — a premium — an AP buys the underlying basket of securities in the open market, delivers it **in-kind** to the fund, and receives newly created ETF shares, which it sells on-exchange. This selling pushes the price back down toward NAV.
- If the ETF trades *below* NAV — a discount — the AP buys cheap ETF shares on-exchange, redeems them with the fund for the underlying basket **in-kind**, and sells the basket. This buying pushes the price back up.

This continuous arbitrage keeps the ETF price tethered to NAV. It also delivers the ETF's celebrated **tax efficiency**: because redemptions are settled *in-kind* (securities out, not cash), the fund can hand its lowest-cost-basis shares to redeeming APs and never realises a capital gain. A traditional mutual fund, by contrast, must sell securities for cash to meet redemptions, realising gains that are distributed to *remaining* shareholders as a taxable event they did not choose.

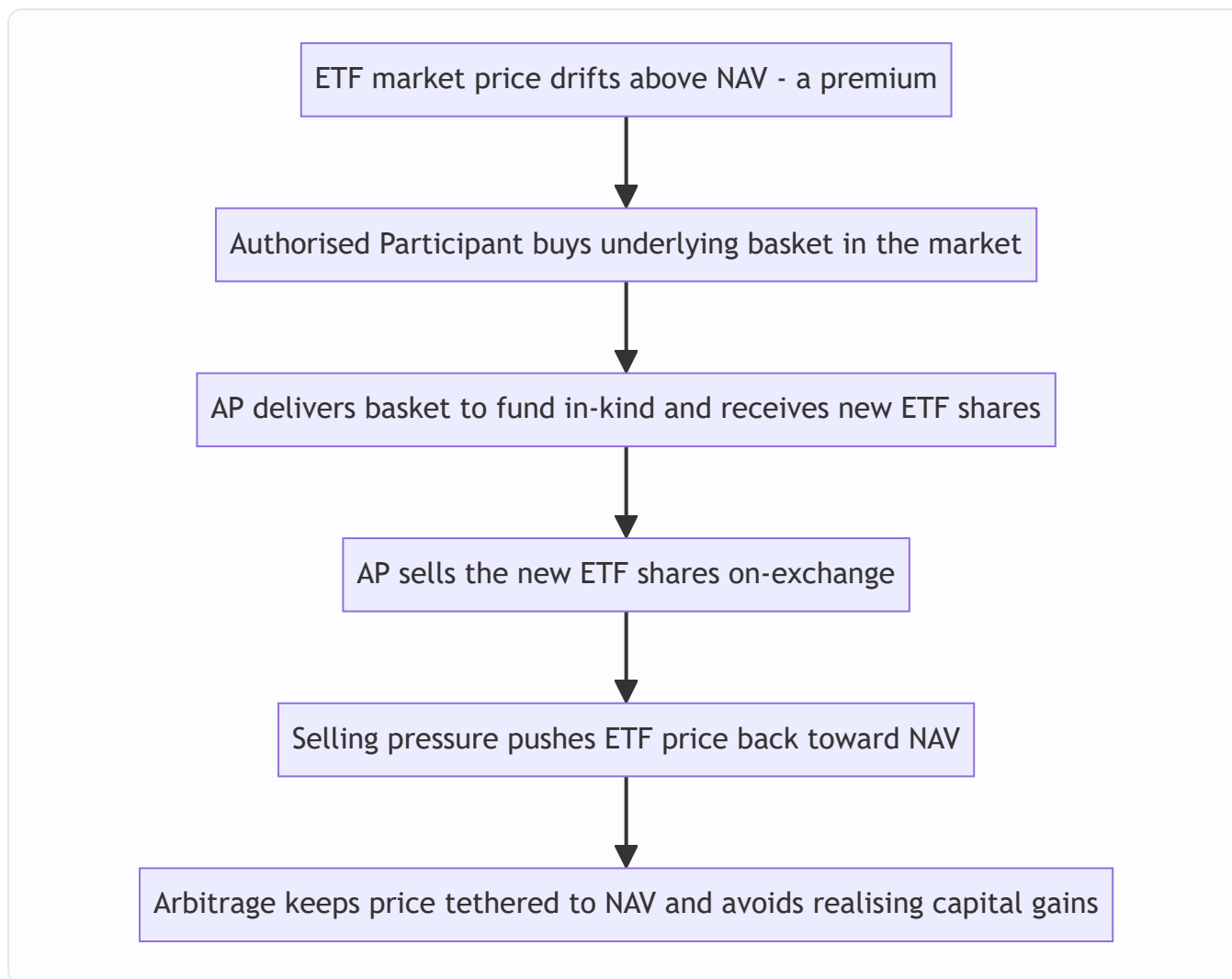


Figure 17.2 — The creation-and-redemption arbitrage loop that keeps an ETF priced at NAV and makes it tax-efficient.

3.3 How active management is *supposed* to work

Active managers seek **alpha** — return in excess of what the benchmark (or the CAPM/factor model) predicts for the risk taken. Sources of purported alpha:

- **Security selection** — fundamental analysis to find under/overvalued securities.
- **Market timing** — shifting between asset classes or cash based on macro views.
- **Factor tilts** — systematically overweighting value, momentum, quality, small-cap (though these are increasingly available *passively* as "smart beta", blurring the line).

The **Grossman-Stiglitz paradox** explains why *some* active management must survive: if markets were perfectly efficient, no one would be paid to gather information, so no one would — and then prices would *not* reflect information and markets would become inefficient. Active managers are, collectively, the mechanism that *makes* markets efficient. They must earn *some* gross return for gathering information, or they would stop. So an equilibrium amount of active management persists. But this equilibrium says nothing about whether the *marginal* fund you can buy is skilled enough to beat its fee — and the evidence (Section 4) says most are not.

4. Full Content

4.1 The evidence: what SPIVA and persistence studies actually show

The empirical case for passive is remarkably robust and consistent across decades and geographies. The gold-standard dataset is **S&P Dow Jones Indices' SPIVA (S&P Indices Versus Active) scorecards**, published semi-annually since 2002, which compare active funds against the appropriate benchmark and — crucially — correct for **survivorship bias** (dead/merged funds are counted, not quietly dropped).

Stylised but representative findings (directionally stable across many reports):

- Over a **1-year** horizon, roughly 50–65% of active US large-cap funds underperform the S&P 500 in a typical year (it varies with the market regime).
- Over **10 years**, roughly **85–90%** underperform.
- Over **15–20 years**, the underperformance rate climbs to **~90–95%**.

Two mechanisms drive the deterioration with horizon: (1) the fee drag compounds, and (2) survivorship — poor funds close, so the longer the window, the more the initial cohort has been culled, and the survivors *still* mostly lag.

The second devastating finding concerns **persistence** — do past winners keep winning? S&P's **Persistence Scorecard** repeatedly shows that a top-quartile fund in one period is roughly *no more likely than chance* (often *less likely*) to remain top-quartile in the next. Of the funds in the top quartile in a given year, only a tiny fraction remain top-quartile four or five years later — fewer than random reshuffling would predict. This is the empirical dagger: even if skilled managers exist, **past performance does not reliably identify them ex ante**, which is exactly the information an investor needs.

4.2 The cost anatomy — why active costs so much more

The headline expense ratio understates the true cost gap:

Cost component	Passive index fund / ETF	Active fund	Note
Management/expense ratio	0.03%–0.20%	0.50%–1.50%+	The visible, quoted number
Trading / transaction costs	Very low (low turnover ~3–15%)	Higher (turnover 50%–150%+)	Not in the expense ratio; drags NAV
Bid-ask spread & market impact	Minimal	Meaningful for large, active traders	Hidden cost of turnover
Cash drag	Minimal	Active funds hold cash for redemptions/timing	Idle cash lags a rising market
Tax (taxable accounts)	Low (ETF in-kind efficiency)	Higher (realised gains from turnover)	Huge in taxable accounts

The often-quoted figure is that all-in active costs run **1–2% per year higher** than passive once turnover and taxes are included. Against an equity risk premium of perhaps 4–5%, that is a colossal, certain handicap.

4.3 When active *can* add value

The passive case is strong but not absolute. Active management can add value under identifiable conditions — this is where a nuanced analyst separates from a dogmatist.

(a) Genuinely inefficient market segments. Sharpe's arithmetic holds in *every* market, but the *dispersion* of active outcomes and the size of exploitable mispricings vary enormously. Efficiency is a spectrum:

More efficient (passive strongly favoured)	Less efficient (active more defensible)
US / developed large-cap equity	Small-cap and micro-cap equity
Sovereign/investment-grade bonds	Emerging & frontier market equity
Highly liquid, heavily-researched names	Distressed debt, high-yield credit
—	Private markets, real assets, catastrophe bonds

In thinly-covered small-caps or frontier markets, fewer analysts compete, information is unevenly distributed, and skilled work can find durable mispricings. Notably, SPIVA data is *less* uniformly damning for some active bond and small-cap categories than for US large-cap — though even there passive usually wins after fees.

(b) Documented, repeatable skill. A minority of managers do appear to possess genuine skill. Academic work (e.g., Fama-French 2010 on the cross-section of fund alphas; Berk-van Binsbergen 2015 on value added) finds the alpha distribution is consistent with *a few* truly skilled managers whose gross alpha is real — but two problems remain: skill is scarce, and it is **hard to distinguish from luck *ex ante*** with a short track record. Berk and van Binsbergen add a subtle point: skilled managers *capture* their alpha in fees (and by growing AUM until returns to scale are exhausted), so the skill accrues to the *manager*, not necessarily the investor.

(c) Non-return objectives. Active management is legitimately used for goals a market-cap index cannot serve: liability-driven investing, ESG/values screening, tax-loss harvesting, downside/tail-risk management, or bespoke exposure. Here "beating the benchmark" is not even the objective.

(d) Structural / capacity constraints and inefficiency by mandate. Some inefficiencies arise from forced behaviour: index reconstitution effects (front-running additions/deletions), forced selling by mandate-constrained holders, or fire sales — exploitable by flexible active capital.

(e) The reflexive limit — how much passive is too much? If passive grows without bound, price discovery weakens and mispricings widen, *increasing* the payoff to active. This self-correcting dynamic means passive can never fully "win"; the two coexist in equilibrium (the Grossman-Stiglitz insight applied at market scale). This is a live debate as passive approaches ~50% of US fund assets.

4.4 The core-satellite approach — the practical synthesis

Rather than choosing *either* passive *or* active, sophisticated investors combine them in a **core-satellite** architecture:

- The **core** (typically 60–90% of the portfolio) is passive: low-cost, broadly diversified index funds/ETFs that capture market beta cheaply and reliably. This anchors the portfolio near the market return and controls overall cost.
- The **satellites** (the remaining 10–40%) are concentrated active positions deployed *only* where the investor has genuine conviction that active can add value — inefficient segments, a manager with demonstrated

skill, a thematic/factor tilt, or a specific objective.

The logic is a cost/conviction budget. You get the market's return cheaply in the core, and you *spend your fee budget and your career risk deliberately* on the few bets where the expected net alpha justifies the cost. It caps the damage active can do (limited to the satellite sleeve) while preserving the upside where active is defensible.

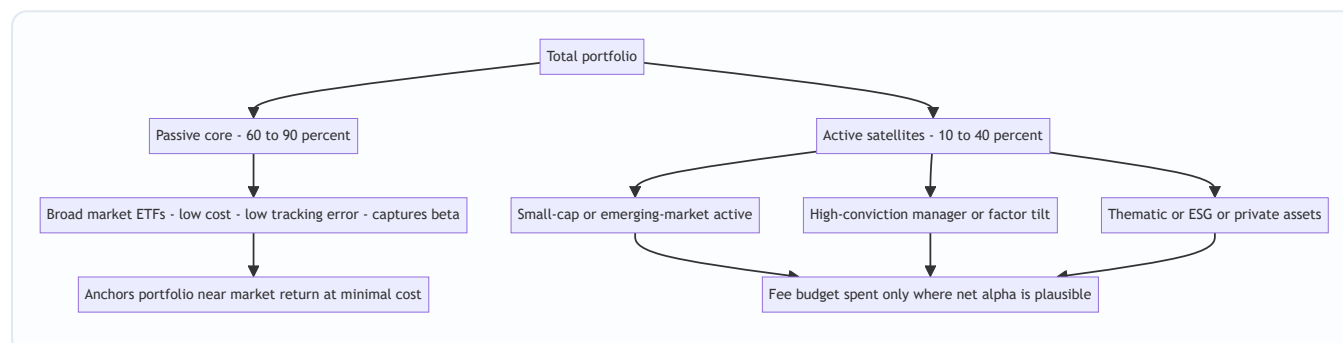


Figure 17.3 — Core-satellite architecture: a cheap passive core plus deliberate active satellites where value-add is defensible.

4.5 Tracking error and closet indexing

Tracking error is the standard deviation of the difference between a fund's return and its benchmark's return, usually annualised:

$$\text{TE} = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (R_{f,t} - R_{b,t})^2} \quad \text{(as std dev of the active return series)}$$

where $R_{f,t}$ is the fund return and $R_{b,t}$ the benchmark return in period t . (Some define TE on the *demeaned* differences — the standard deviation of active returns; others loosely use the mean absolute difference. The standard-deviation-of-active-return definition is the professional one.)

TE means opposite things for the two strategies:

- For a **passive** fund, TE is a *quality defect*. An index fund's whole job is to match the index, so a good index fund has TE of only a few basis points to ~0.20%. High TE means poor replication — from fees, cash drag, sampling error, or securities-lending timing. **Lower is better.**
- For an **active** fund, TE is a *measure of how active the manager is*. To beat the benchmark you must *differ* from it; you cannot outperform an index you are hugging. High TE (say 4–10%) signals genuine active bets. **You cannot get alpha with zero TE.**

This sets up **closet indexing** — the industry's dirty secret. A closet indexer is an "active" fund that charges active fees (0.8%–1.2%) but holds a portfolio that barely deviates from its benchmark (low TE, high overlap). The investor pays for skill and gets, in substance, an expensive index fund — guaranteeing underperformance by roughly the fee.

The tool that exposes this is **Active Share** (Cremers & Petajisto, 2009): the fraction of a fund's holdings that differ from the benchmark, computed as:

$$\text{Active Share} = \frac{1}{2} \sum_{i=1}^N |w_{\text{fund},i} - w_{\text{benchmark},i}|$$

where $w_{\text{fund},i}$ and $w_{\text{benchmark},i}$ are the weights of security i in the fund and benchmark. It ranges from 0% (pure index) to 100% (no overlap). Cremers-Petajisto found:

- Funds with Active Share **below ~60%** are effectively closet indexers.
- Genuinely active funds (Active Share **above ~80–90%**) that also concentrate their bets showed *some* evidence of outperformance in their sample — though this finding is debated and weaker out-of-sample.

The practical rule: if you pay active fees, demand *evidence* of activeness — meaningful Active Share and TE. Otherwise you are paying a Ferrari price for a bus that follows the index bus.

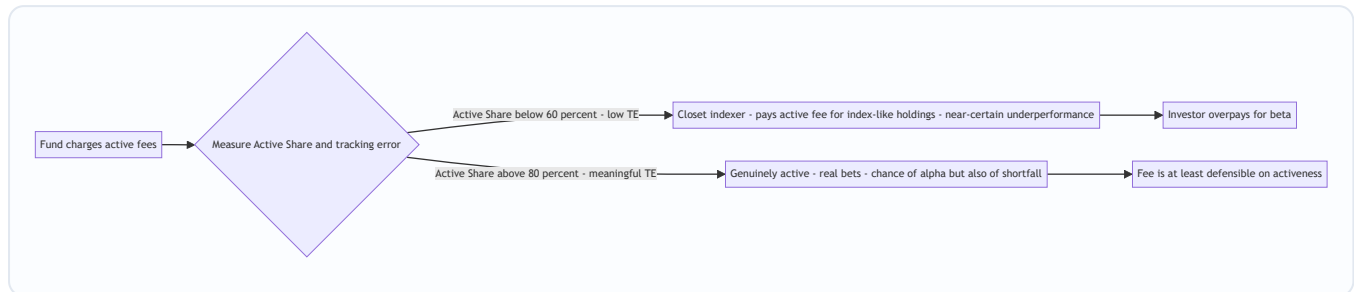


Figure 17.4 — Active Share and tracking error distinguish genuine active management from closet indexing.

5. Worked / Applied Examples

Example 1 — The compounding cost of active fees (the core arithmetic)

Two investors each invest a **lump sum of \$100,000** for **30 years**. The market returns **8% gross** annually. Investor P holds a passive fund with expense ratio **0.05%**; Investor A holds an *average* active fund charging **0.85%** that, being average, earns the market's gross return before fees. Assume net return = gross – expense ratio.

- Investor P net return: $8\% - 0.05\% = 7.95\%$. Terminal wealth $= 100,000 \times (1.0795)^{30}$.
 $\ln(1.0795) = 0.076486$; $\times 30 = 2.29458$; $e^{2.29458} = 9.920$. **Terminal wealth $\approx$ \$992,000.**
- Investor A net return: $8\% - 0.85\% = 7.15\%$. Terminal wealth $= 100,000 \times (1.0715)^{30}$.
 $\ln(1.0715) = 0.069056$; $\times 30 = 2.07168$; $e^{2.07168} = 7.938$. **Terminal wealth $\approx$ \$793,800.**

The 0.80% annual fee difference destroyed about \$198,000 — roughly 20% of the passive investor's terminal wealth — purely to cost, before any skill deficit. And recall: the *average* active fund does *not* even match the gross market return, because Sharpe's arithmetic guarantees the aggregate active pool earns the market gross return, while trading costs push the median further down. So \$793,800 is optimistic; the realistic median active outcome is lower still.

Reconciliation check: fee-drag fraction formula $1 - (1.0715/1.08)^{30}$. Here I used net = gross – fee, so compare \$9.920\$ vs \$7.938\$; ratio $= 0.800$, i.e. 20% lost — consistent with the earlier 21% estimate (small difference because the earlier example used 0.85% against a fee-free 8% base rather than a 0.05% base). Both land at ~20%. ✓

Example 2 — Reading a fund's Active Share and tracking error

An advisor screens three "active" US large-cap funds, all benchmarked to the S&P 500 and all charging **0.90%**:

Fund	Active Share	Tracking error	10-yr net return vs S&P 500
Fund X	35%	1.1%	-0.95% p.a.
Fund Y	72%	4.8%	+0.40% p.a.
Fund Z	88%	8.5%	+1.30% p.a. (but very volatile relative return)

Interpretation. - **Fund X** is a textbook **closet indexer**: 35% Active Share means 65% of it is the index. With TE of only 1.1%, it cannot deviate enough to overcome its 0.90% fee. Its -0.95% shortfall is almost exactly its fee (0.90%) plus a little trading cost — *precisely what the arithmetic predicts*. The client is paying active price for beta. **Recommendation: replace with a 0.05% index fund and pocket ~0.90%/yr.** - **Fund Y** is genuinely active (72% Active Share, 4.8% TE) and has edged out the benchmark by 0.40% net. This is defensible — the manager took real bets that (net of the 0.90% fee) added value. Worth retaining, with monitoring. - **Fund Z** is very active (88% AS, 8.5% TE) with the highest excess return — but the high TE means high relative volatility; the +1.30% could partly be luck, and a bad year could see it lag by many percent. Suitable only as a **satellite** with a small weight and long horizon.

Lesson: Active Share and TE let you see what you are paying for. Never pay active fees for Fund-X-type exposure.

Example 3 — Designing a core-satellite portfolio

A client has **\$1,000,000**, a 20-year horizon, and moderate risk tolerance. The advisor builds core-satellite:

- **Core (75% = \$750,000):** total-world equity ETF at 0.08% + aggregate bond ETF at 0.05%. Blended core cost $\approx 0.07\%$.
- **Satellites (25% = \$250,000):** deployed only where active is defensible —
- \$100,000 emerging-market small-cap active fund (inefficient segment), fee 1.10%.
- \$80,000 high-Active-Share concentrated global manager (documented skill), fee 0.95%.
- \$70,000 thematic clean-energy / private-markets sleeve, fee 1.20%.

Blended cost:
$$\text{Cost} = 0.75(0.07\%) + \frac{100,000}{1,000,000}(1.10\%) + \frac{80,000}{1,000,000}(0.95\%) + \frac{70,000}{1,000,000}(1.20\%)$$

$$= 0.0525\% + 0.110\% + 0.076\% + 0.084\% = 0.3225\%$$

The whole portfolio costs **~0.32%** — a fraction of the 0.90%+ an all-active book would cost — yet retains targeted active exposure precisely where the evidence supports it. The passive core guarantees the bulk of the portfolio tracks the market cheaply; the active downside is *capped at the 25% satellite sleeve*. If every satellite underperformed by 1% net, the drag on the *total* portfolio is only $0.25 \times 1\% = 0.25\%$ — a bounded, deliberate risk.

6. Connections

- **Efficient Market Hypothesis (Ch. on market efficiency):** The passive case is EMH in practical dress. Weak/semi-strong efficiency in large-cap markets is *why* net alpha is scarce. The Grossman-Stiglitz paradox connects active management to the *production* of efficiency.
- **CAPM and factor models (Ch. on asset pricing):** "Beating the market" must be defined relative to a risk model. Alpha is the intercept in a regression of fund excess returns on factor returns. Much apparent

alpha turns out to be *factor beta* (value, size, momentum) now available cheaply as **smart beta** — collapsing the space where traditional active can claim a premium.

- **Portfolio construction & diversification (Ch. on MPT):** The passive core is the market portfolio of MPT — the theoretically optimal risky portfolio under CAPM. Core-satellite is a practical bridge from the theoretical tangency portfolio to a real-world implementable book.
 - **Behavioural finance:** The persistence of active management despite the evidence is itself a behavioural puzzle — overconfidence (both managers and investors believing *they* are the exception), performance-chasing, and the narrative appeal of a "story" over an index.
 - **Fees and compounding (time value of money):** The cost case is an application of compounding; the fee drag is negative compounding.
 - **Taxation:** ETF in-kind efficiency connects to after-tax return analysis; in taxable accounts the passive advantage widens materially.
 - **Market microstructure:** The ETF creation/redemption arbitrage is an application of the law of one price and liquidity provision by market makers.
-

7. Key Terms

- **Passive management** — investing to replicate an index's return at minimal cost.
- **Active management** — investing to beat a benchmark through skill (selection/timing).
- **Index fund** — a mutual fund holding an index's constituents in index weights.
- **ETF (exchange-traded fund)** — an index (usually) fund whose shares trade intraday on an exchange, with supply managed by AP creation/redemption.
- **Authorised Participant (AP)** — an institution licensed to create/redeem ETF shares in large blocks, enforcing the price-NAV link via arbitrage.
- **Creation/redemption (in-kind)** — the mechanism where APs exchange baskets of underlying securities for ETF shares (and vice versa), delivering price efficiency and tax efficiency.
- **NAV (net asset value)** — the per-share market value of a fund's underlying holdings.
- **Premium/discount** — ETF market price above/below NAV.
- **Alpha** — return in excess of the risk-adjusted benchmark expectation.
- **Beta** — exposure to systematic market movement; the "market return" a passive fund captures.
- **Sharpe's arithmetic of active management** — the identity that active = passive gross, and active < passive net, in aggregate.
- **SPIVA** — S&P's survivorship-bias-corrected scorecards of active-vs-benchmark performance.
- **Survivorship bias** — the distortion from excluding funds that closed/merged.
- **Persistence** — whether past outperformers keep outperforming (evidence: barely, if at all).
- **Tracking error (TE)** — standard deviation of a fund's return minus its benchmark's return.
- **Active Share** — the % of holdings that differ from the benchmark; measures genuine activeness.
- **Closet indexing** — charging active fees for near-index holdings (low Active Share/TE).
- **Core-satellite** — a passive core plus targeted active satellites.
- **Smart beta / factor investing** — rules-based strategies capturing factor premia, sitting between passive and active.

- **Expense ratio** — the annual % fee charged by a fund.
 - **Grossman-Stiglitz paradox** — perfectly efficient markets cannot exist because no one would then pay to gather information.
-

8. Common Confusions

"Passive means you never lose money / it's low-risk." No. Passive removes *manager* risk and cost, not *market* risk. An index fund falls exactly as much as the market in a crash. Passive vs active is about *how* you get market exposure, not *how much* market risk you take (that's asset allocation).

"ETFs are always cheaper and better than index mutual funds." Not universally. Broad index mutual funds can have expense ratios as low as top ETFs, and for regular automatic investing (dollar-cost averaging) a no-transaction-cost mutual fund can beat an ETF where you pay spreads/commissions. ETFs win on intraday tradability and (in taxable accounts) tax efficiency; index mutual funds win on simplicity for periodic contributions. Also note: an ETF can itself be *active*.

"Some active funds beat the market, so active works." Survivorship and selection bias. *Some* will beat by chance every period; the question is whether you can identify them *in advance* and whether the *cohort* wins. The evidence says no on both counts.

"High tracking error is bad." Only for a *passive* fund. For an active fund, near-zero TE is the red flag (closet indexing) — you cannot outperform without deviating.

"Active Share above 80% guarantees outperformance." No. High Active Share is *necessary* for outperformance but not *sufficient* — it means the manager is taking real bets, which can be right or wrong. It filters out closet indexers; it does not confer skill.

"Alpha and outperformance are the same." Alpha is *risk-adjusted* excess return. A fund can outperform the index simply by taking more factor risk (more small-cap, more value) — that's beta to a factor, not alpha. Proper attribution separates the two.

"If everyone indexed, markets would break, so passive is self-defeating and you should be active." The first clause is directionally true (price discovery would weaken), but it *strengthens* the case that active and passive coexist in equilibrium — it does not tell *you* that *your* chosen active fund will beat its fee today. The individual decision and the systemic argument are different questions.

"Passive investors are free-riders who contribute nothing to price discovery." Partly true and largely irrelevant to the individual: markets remain efficient as long as *enough* active capital does the price-discovery work. The free-rider can rationally take the efficient prices others produce.

9. Recap

- The choice between beating the market and being the market is the highest-leverage strategic decision in investing.
- **Sharpe's arithmetic** proves, with no efficiency assumption, that active management is **zero-sum before costs and negative-sum after costs**. The average active dollar *must* underperform the average passive dollar by the fee gap.

- **Evidence (SPIVA, persistence studies)** corroborates the arithmetic: ~85–95% of active funds underperform over 10–20 years, and past winners do not reliably repeat. The burden of proof lies on active.
- The **cost case** compounds: an ~0.8% fee gap can consume ~20% of terminal wealth over 30 years — before any skill deficit.
- **Index funds and ETFs** implement passive via full replication, sampling, or synthetic swaps. **ETFs** add intraday trading and superior tax efficiency through the **AP creation/redemption** arbitrage that keeps price $\approx$ NAV.
- **Active can add value** in *inefficient segments* (small-cap, EM, distressed, private), with *documented and repeatable skill* (rare and hard to identify ex ante), or for *non-return objectives* (ESG, tax, liabilities).
- **Tracking error** is a defect for passive but a *requirement* for active. **Closet indexing** — high fee, low Active Share/TE — is the trap to avoid; **Active Share** exposes it.
- The **practical resolution is not either/or but core-satellite**: a cheap passive core for market beta plus deliberate active satellites where net alpha is plausible — capping active downside while retaining defensible upside.

10. Quick-Reference / Interview Points

One-liners to have ready: - "Active management is zero-sum before fees and negative-sum after — that's Sharpe's arithmetic, and it needs no assumption about efficiency." - "You don't have to believe markets are perfectly efficient to prefer passive; you only need to believe net alpha is too scarce to pay for reliably." - "SPIVA: ~90% of active large-cap funds lag over 15 years, survivorship-corrected — and past winners don't persist." - "An ETF stays at NAV because Authorised Participants arbitrage premiums/discounts via in-kind creation and redemption — which is also why ETFs are tax-efficient: in-kind redemptions never realise a capital gain." - "You can't earn alpha with zero tracking error — closet indexers charge active fees for index exposure; check Active Share." - "The sophisticated answer isn't passive-or-active, it's core-satellite: cheap beta in the core, deliberate active bets in the satellites."

Key formulas: - Fee-drag fraction over n years: $1 - \left(\frac{1+g-c}{1+g}\right)^n$. - Tracking error: std dev of $(R_{\text{fund}} - R_{\text{benchmark}})$. - Active Share: $\frac{1}{2} \sum_i |w_{\text{fund},i} - w_{\text{bench},i}|$, from 0% (index) to 100% (no overlap).

Numbers worth memorising: - Typical passive cost 0.03–0.20%; typical active cost 0.5–1.5%+. - All-in active cost drag vs passive $\approx$ 1–2%/yr including turnover and tax. - Active underperformance: ~50–65% over 1yr, ~85–95% over 10–20yrs. - Active Share < 60% $\approx$ closet indexer; > 80% $\approx$ genuinely active. - Passive core typically 60–90% of a core-satellite book.

Likely interview questions and the crisp answer: 1. "Why does the average active fund underperform?" → Sharpe's arithmetic + compounding costs; it's structural, not a skill statement. 2. "How does an ETF track its index and stay at NAV?" → Replication method + AP creation/redemption arbitrage. 3. "Why are ETFs tax-efficient?" → In-kind redemptions let the fund shed low-basis lots without realising gains. 4. "When would you recommend active?" → Inefficient segments, evidence of skill (high Active Share), or non-return objectives — sized as satellites. 5. "What's closet indexing and how do you detect it?" → Active fee, index-like holdings; detect via low Active Share and low tracking error. 6. "If passive keeps growing, does active make a comeback?" → In principle yes — weaker price discovery widens mispricings and raises the payoff to active;

the two coexist in equilibrium (Grossman-Stiglitz). 7. "Isn't tracking error bad?" → For passive yes, for active it's necessary — the right frame is *whose* fund and *what* objective.

The one-sentence resolution of the debate: *For the bulk of a portfolio in efficient markets, pay as little as possible to be the market; spend a bounded, deliberate fee budget on active only where inefficiency or genuine skill gives net alpha a fighting chance — and that synthesis is core-satellite.*

Q&A — Passive vs Active Management

Scope: Investments — Chapter 17 (Passive vs Active Management). Every question is followed by a full model answer. All rates are annual and in percent unless stated. Work every numerical problem yourself before checking the solution. Sections: **A** concept-check · **B** numerical (full step-by-step, reconciling) · **C** interview-style · **D** MCQs with reasoning.

The chapter in one line

$$\begin{aligned} \text{Active Gross} &= \text{Passive Gross}, \quad \text{Active Net} < \text{Passive Net}, \\ \text{Active Share} &= \frac{1}{2} \sum_i |w_{f,i} - w_{b,i}| \end{aligned}$$

One-line statement: Because the market is the sum of its participants, active management is zero-sum before costs and negative-sum after — so passive is the evidence-backed default, and the sophisticated answer is not either/or but a cheap passive core plus deliberate active satellites where inefficiency or genuine skill gives net alpha a fighting chance.

Section A — Concept Check

A1. State Sharpe's arithmetic of active management and explain why it needs no assumption about market efficiency. Sharpe's identity (1991): before costs the average actively managed dollar earns the same as the average passively managed dollar; after costs the active dollar earns *less*. It is accounting, not empirics — passive investors hold the market in market weights, so the *remaining* active holdings must in aggregate also equal the market. Both cohorts earn the same gross return; the active cohort just pays higher fees to do it. No belief about whether prices are "right" is required. This is why the burden of proof sits on active — passive is the default you argue *away* from.

A2. "Active management is a negative-sum game." Reconcile this with some active funds clearly beating the market every year. The claim is about the *aggregate*, not any single fund. Active is zero-sum before costs — every dollar of outperformance is matched by an active dollar of underperformance, because the combined active holdings are just the market — and fees shift the distribution left, making it negative-sum on average. Winners always exist each period; the decisive questions are whether the *cohort* wins (it cannot, by arithmetic) and whether you can identify tomorrow's winners *in advance* (persistence evidence: essentially no).

A3. Distinguish the three index-replication methods and when each is used. *Full replication* holds every constituent in exact weight — for concentrated, liquid indices (S&P 500, Nifty 50); lowest tracking error but costly for broad indices. *Sampling/optimisation* holds a representative subset matched to the index's risk factors — for very broad or illiquid indices; cheaper but adds sampling tracking error. *Synthetic (swap-based)* holds a collateral basket and enters a total-return swap to receive the index return — for hard-to-access markets; low tracking error but adds counterparty risk.

A4. Explain the ETF creation/redemption mechanism and why it keeps price near NAV. Supply is elastic through Authorised Participants (APs) who create and redeem shares in large blocks. At a *premium* to NAV, an AP buys the basket, delivers it in-kind for new ETF shares, and sells those shares — pushing price down.

At a *discount*, the AP buys cheap ETF shares, redeems them in-kind for the basket, and sells it — pushing price up. This continuous arbitrage tethers price to NAV via the law of one price.

A5. Why are ETFs more tax-efficient than traditional mutual funds? ETF redemptions settle *in-kind*: the fund hands its lowest-cost-basis shares to a redeeming AP instead of selling for cash, so it never realises a capital gain. A mutual fund must sell holdings for cash to meet redemptions, realising gains distributed to *remaining* shareholders as an unchosen taxable event. The advantage is largest in taxable accounts, widening the passive edge there.

A6. What does SPIVA measure, why is survivorship-bias correction essential, and what does it show? SPIVA scorecards compare active funds against the correct benchmark, counting funds that closed or merged rather than dropping them — essential because dead funds are disproportionately poor, so ignoring them flatters the survivors. Representative findings: ~50–65% of US large-cap active funds lag over 1 year, ~85–90% over 10 years, ~90–95% over 15–20 years. The deterioration with horizon comes from compounding fee drag plus the culling of weak funds, whose survivors still mostly lag.

A7. What does the persistence evidence add beyond SPIVA, and why is it the "empirical dagger"? SPIVA shows most funds lag; persistence asks whether *past winners keep winning*. S&P's Persistence Scorecard repeatedly finds a top-quartile fund is roughly no more likely than chance — often less — to stay top-quartile next period. The dagger: even *if* skilled managers exist, past performance does not reliably identify them *ex ante* — exactly the information an investor needs to pick a winner.

A8. Contrast what tracking error means for a passive fund versus an active fund. TE is the standard deviation of the fund-minus-benchmark return series. For a *passive* fund it is a quality defect — a good index fund has TE of a few basis points to ~0.20%; high TE means poor replication (fees, cash drag, sampling). Lower is better. For an *active* fund, TE measures how active the manager is — you cannot beat an index you hug, so genuine bets require meaningful TE (say 4–10%). Near-zero TE on an active fund is a red flag.

A9. Define Active Share and explain how it exposes closet indexing. Active Share $= \frac{1}{2} \sum_i |w_{f,i} - w_{b,i}|$, from 0% (pure index) to 100% (no overlap). A *closet indexer* charges active fees (0.8–1.2%) but holds a near-index portfolio (low Active Share, low TE), so the investor gets an expensive index fund and underperforms by roughly the fee. Cremers-Petajisto (2009): funds below ~60% are effectively closet indexers, genuinely active ones sit above ~80%. High Active Share is *necessary* but not *sufficient* for outperformance — it filters out closet indexers, it does not confer skill.

A10. State the Grossman-Stiglitz paradox and its implication for the passive/active balance. If markets were perfectly efficient no one would be paid to gather information, so no one would — and prices would stop reflecting information, making markets inefficient. Hence an equilibrium amount of active management must persist; active managers collectively *make* markets efficient. Implication: passive can never fully "win" — if it grows without bound, price discovery weakens, mispricings widen, and the payoff to active rises, a self-correcting dynamic keeping the two in equilibrium.

A11. What is the core-satellite approach and its underlying logic? It pairs a passive *core* (typically 60–90%) of low-cost broad index funds capturing market beta cheaply with active *satellites* (10–40%) deployed only where the investor has genuine conviction active can add value — inefficient segments, a skilled manager, a factor tilt, or a non-return objective. The logic is a cost/conviction budget: get the market cheaply in the core, and spend your fee budget deliberately on the few bets where expected net alpha justifies the cost — capping active downside to the satellite sleeve while preserving upside.

A12. Name three conditions under which active management is genuinely defensible. (1) *Inefficient segments* — small/micro-cap, emerging and frontier equity, distressed and high-yield credit, private markets — where fewer analysts compete and mispricings survive. (2) *Documented, repeatable skill* — a rare minority whose gross alpha is real, though skill is scarce, hard to separate from luck ex ante, and (Berk-van Binsbergen) tends to be *captured* by the manager in fees. (3) *Non-return objectives* — liability matching, ESG screening, tax-loss harvesting, tail-risk management — where beating the benchmark is not the goal.

Section B — Numerical Problems

B1. The compounding cost of active fees. Two investors each invest a \$100,000 lump sum for 30 years. The market returns 8% gross annually. Investor P holds a passive fund (expense ratio 0.05%); Investor A holds an *average* active fund charging 0.85% that earns the market's gross return before fees. Net return = gross – expense ratio. Find both terminal wealths and the fee cost.

Step 1 — Net returns. P: $\$8 - 0.05 = 7.95\%$. A: $\$8 - 0.85 = 7.15\%$.

Step 2 — Investor P. $\$100,000 \times (1.0795)^{30}$. $\ln 1.0795 = 0.076486$; $\times 30 = 2.29458$; $e^{2.29458} = 9.920$. $\approx$ **\\$992,000.**

Step 3 — Investor A. $\$100,000 \times (1.0715)^{30}$. $\ln 1.0715 = 0.069056$; $\times 30 = 2.07168$; $e^{2.07168} = 7.938$. $\approx$ **\\$793,800.**

Step 4 — Cost & reconcile. Difference $= 992,000 - 793,800 = \mathbf{\$198,200}$: the 0.80% fee gap destroyed ~20% of the passive investor's terminal wealth ($\$793,800 / 992,000 = 0.800$) *before* any skill deficit. And \$793,800 is optimistic — the average active dollar only earns the market gross return, so trading costs push the *median* active outcome lower still.

B2. Diagnosing a closet indexer from its numbers. Fund X is benchmarked to the S&P 500, charges 0.90%, has Active Share 35% and TE 1.1%, and returned –0.95% p.a. vs the index over 10 years. Show that its shortfall is consistent with the arithmetic and state the recommendation.

Solution. Active Share 35% means 65% of the fund is the index, and TE of 1.1% confirms it barely deviates — so it cannot generate enough gross excess return to cover its 0.90% fee. Predicted shortfall $\approx$ fee + trading cost $\approx 0.90\% + 0.05\% \approx 0.95\%$, matching the observed –0.95% almost exactly. Textbook closet indexer.

Recommendation: replace with a ~0.05% index fund and recover ~0.90%/yr.

B3. Computing Active Share. A three-stock fund holds A 50%, B 30%, C 20%. The benchmark holds A 40%, B 40%, D 20% (the fund holds no D; the benchmark holds no C). Compute Active Share.

Solution. Align all four names and take absolute weight differences: - A: $|0.50 - 0.40| = 0.10$ - B: $|0.30 - 0.40| = 0.10$ - C: $|0.20 - 0.00| = 0.20$ - D: $|0.00 - 0.20| = 0.20$

Sum $= 0.60$; Active Share $= \frac{1}{2}(0.60) = \mathbf{30\%}$. **Reconcile:** both weight sets sum to 100%, so overweights equal underweights and the ½ avoids double-counting. At 30% — well below the ~60% threshold — this "active" fund is largely the benchmark.

B4. Core-satellite blended cost. A \$1,000,000 book: core 75% at a blended 0.07%; satellites — \$100,000 EM small-cap active at 1.10%, \$80,000 concentrated global manager at 0.95%, \$70,000 thematic/private sleeve at 1.20%. Find the blended cost and the maximum drag if every satellite underperforms by 1% net.

Solution — blended cost. $\$0.75(0.07\%) + 0.10(1.10\%) + 0.08(0.95\%) + 0.07(1.20\%)$

$= 0.0525\% + 0.110\% + 0.076\% + 0.084\% = \mathbf{0.3225\%}$. A whole-portfolio cost of ~0.32% versus

0.90%+ for an all-active book.

Bounded downside. Satellites are 25% of the book, so if each lags by 1% net the drag on the *total* is $0.25 \times 1\% = \mathbf{0.25\%}$ — the passive core (75%) is untouched by manager risk, fencing the maximum active damage inside the satellite sleeve. That is the point of the architecture.

B5. Turnover, cost, and the true active handicap. A passive fund costs 0.10% all-in. An active fund quotes a 0.85% expense ratio but, at 90% turnover, incurs an extra ~0.40% in trading costs, spreads and tax drag not shown in the expense ratio. Against a 4.5% equity risk premium, express the active fund's total handicap.

Solution. True active cost $= 0.85\% + 0.40\% = 1.25\%$; handicap vs passive $= 1.25\% - 0.10\% = \mathbf{1.15\%}$ /yr — inside the "1–2% all-in" range. As a share of the 4.5% premium, $1.15/4.5 = \mathbf{25.6\%}$: the fund must generate gross alpha equal to over a quarter of the entire equity premium *just to break even*. The visible 0.85% expense ratio understates the true gap by nearly half once turnover is counted — which is why the headline flatters active.

Section C — Interview-Style Questions

C1. "Why does the average active fund underperform its benchmark?" It is structural, not a statement about talent. By Sharpe's arithmetic active investors in aggregate hold the market, so they earn the market's gross return collectively, then pay higher fees, trading costs and taxes to do it — making the *average* active dollar lag the average passive dollar by the cost gap. Compounding turns a ~0.8% annual fee difference into ~20% of terminal wealth over 30 years. It is baked into the accounting before we ask whether any manager has skill.

C2. "How does an ETF track its index and stay at NAV — and why is that tax-efficient?" The fund replicates by full replication, sampling, or a swap; its exchange price stays glued to NAV through Authorised Participants. Above NAV they buy the basket, deliver it in-kind for new ETF shares, and sell those shares, pushing price down; below NAV they buy cheap shares, redeem in-kind for the basket, and sell it, pushing price up. That in-kind redemption is also the tax trick — the fund ships out its lowest-basis lots without selling for cash, so it never realises a gain, unlike a mutual fund that must sell to raise cash and distributes the gain to remaining holders.

C3. "You don't have to believe in the EMH to prefer passive. Explain." Correct — the strong case is Sharpe's arithmetic, an identity independent of efficiency. You need only believe markets are efficient *enough* that the marginal mispricing is too small, rare, or costly to exploit reliably after fees. In large, liquid, researched markets like US large-cap, thousands of analysts compete away obvious mispricings, so net opportunity is thin. Preferring passive is a bet about the *scarcity of net alpha*, not a claim prices are always right.

C4. "When would you actually recommend active management?" Three situations. First, inefficient segments — small/micro-cap, emerging and frontier equity, distressed credit, private markets — where fewer analysts compete and mispricings persist. Second, where there is evidence of genuine activeness and skill (meaningful Active Share and TE, a defensible process), sized as a satellite because skill is hard to identify *ex ante*. Third, non-return objectives a cap-weighted index cannot serve — liability matching, ESG screening, tax-loss harvesting, tail-risk control. In all cases I'd fence it inside a satellite sleeve, not the core.

C5. "A client is proud their active fund beat the index last year. How do you respond?" Congratulate, then reframe. Some active funds beat the market every year — that is guaranteed by the arithmetic, and one

good year is well within luck's range. The decision-relevant question is whether outperformance *persists*, and the Persistence Scorecard shows past top-quartile funds are no more likely than chance to stay there. I'd then check Active Share, TE and fee — genuine bets or closet-indexing beta? — and whether the gain was alpha or simply a factor tilt we could buy more cheaply.

C6. "What is closet indexing and how do you detect it?" A closet indexer charges active fees while holding a near-benchmark portfolio — the investor pays for skill and gets an expensive index fund, guaranteeing underperformance by roughly the fee. Detect it with two numbers: low Active Share (below ~60% means most of the fund is the index) and low tracking error (it can't deviate enough to beat its fee). The rule: if you pay active fees, demand evidence of activeness — otherwise replace it with an index fund and pocket the fee.

C7. "If passive keeps growing, does active eventually make a comeback?" In principle yes — Grossman-Stiglitz at market scale. If passive grows without bound, fewer participants do price discovery, mispricings widen, and the payoff to skilled active work rises, drawing capital back; the two coexist in a self-correcting equilibrium. But the systemic argument doesn't tell *you* that *your* chosen active fund will beat its fee today — that the equilibrium amount of active management survives says nothing about whether the marginal fund you can buy is one of the skilled few.

C8. "Isn't high tracking error a bad thing?" It depends on the fund's job. For a passive index fund, yes — high TE means poor replication, and I'd want only a few basis points. For an active fund, near-zero TE is the red flag: you cannot earn alpha from an index you're hugging, so genuine active management *requires* meaningful TE. The right frame is always "whose fund, what objective" — TE is a defect for the replicator and a prerequisite for the stock-picker.

Section D — Multiple-Choice Questions (with reasoning)

D1. Sharpe's arithmetic of active management depends on which assumption? (a) Markets are perfectly efficient · (b) Markets are weak-form efficient · (c) No assumption about efficiency · (d) Managers are irrational **Answer: (c).** It is an accounting identity — the active cohort's holdings equal the market minus the passive cohort's, so both earn the market gross return regardless of whether prices are "right." (a)/(b) confuse the arithmetic with the separate EMH argument; (d) is irrelevant.

D2. An ETF is trading at a premium to NAV. What does an Authorised Participant do to arbitrage it? (a) Buy ETF shares and redeem them in-kind · (b) Buy the underlying basket, deliver in-kind for new ETF shares, and sell them · (c) Short the basket and buy ETF shares · (d) Do nothing until the discount appears **Answer: (b).** At a premium the AP creates shares — buys the basket, delivers in-kind for new ETF shares, and sells them, and the selling pushes price back to NAV. (a) describes the *discount* (redemption) case. The premium/discount trigger is the key distinction.

D3. Which fund is most likely a closet indexer? (a) Active Share 88%, TE 8.5%, fee 0.95% · (b) Active Share 35%, TE 1.1%, fee 0.90% · (c) Active Share 72%, TE 4.8%, fee 0.90% · (d) Active Share 5%, TE 0.05%, fee 0.05% **Answer: (b).** Low Active Share (35%) and low TE (1.1%) combined with a *high active fee* (0.90%) is the closet-indexer signature — index-like holdings at an active price. (d) has the same low activeness but a passive fee, so it is an honest index fund, not a closet indexer. (a) and (c) are genuinely active.

D4. Over a 15–20 year horizon, SPIVA scorecards typically show what fraction of active US large-cap funds underperform the benchmark? (a) ~25% · (b) ~50% · (c) ~70% · (d) ~90–95% **Answer: (d).**

Underperformance rises with horizon — ~50–65% over 1 year but ~90–95% over 15–20 years — driven by

compounding fee drag and survivorship culling. This is survivorship-bias corrected, so it is not flattered by dropping dead funds.

D5. ETFs are more tax-efficient than mutual funds primarily because: (a) They have lower expense ratios · (b) In-kind redemptions let the fund shed low-basis lots without realising gains · (c) They are exempt from capital-gains tax · (d) They never pay dividends **Answer: (b).** The in-kind creation/redemption mechanism lets the fund hand low-cost-basis shares to redeeming APs without selling for cash, avoiding realised gains. (a) is sometimes but not always true and isn't the tax reason; (c) and (d) are false — ETFs are taxed and do distribute dividends.

D6. For an actively managed fund, a tracking error near zero most likely indicates: (a) Excellent skill · (b) Low risk and high alpha · (c) Closet indexing · (d) High Active Share **Answer: (c).** You cannot outperform an index you hug, so near-zero TE on an *active* fund signals it is barely deviating — closet indexing. (d) is the opposite; high Active Share would come with meaningful TE. (a)/(b) invert the logic — zero TE guarantees you cannot add alpha.

D7. The Grossman-Stiglitz paradox implies that: (a) Passive investing will eventually eliminate all active managers · (b) Markets cannot be perfectly efficient because then no one would pay to gather information · (c) Active managers always beat passive · (d) Index funds cause bubbles **Answer: (b).** If prices already reflected all information, information-gathering would be unpaid and would stop — so some active management must survive to *produce* efficiency, and passive and active coexist in equilibrium. (a) contradicts the paradox; (c) and (d) are not what it states.

D8. In a core-satellite portfolio, if satellites are 25% of the book and each underperforms by 1% net, the drag on the total portfolio is: (a) 1.00% · (b) 0.25% · (c) 0.75% · (d) 0.05% **Answer: (b).** $0.25 \times 1\% = 0.25\%$. The passive core (75%) is unaffected by manager risk, so active downside is fenced inside the satellite sleeve — the whole point of capping the fee/conviction budget.

D9. High Active Share (above ~80%) tells you that a fund: (a) Will outperform its benchmark · (b) Is taking genuine active bets, which may be right or wrong · (c) Has low fees · (d) Has low tracking error **Answer: (b).** High Active Share is *necessary* but not *sufficient* for outperformance — it means real deviation, filtering out closet indexers, but the bets can still be wrong. (a) overclaims; (d) is contradicted (activeness raises TE).

D10. Which is a legitimate reason to choose active management despite the arithmetic? (a) Because active funds are cheaper · (b) Because past winners reliably keep winning · (c) In inefficient segments like small-cap and emerging markets, or for non-return objectives like ESG or liability matching · (d) Because index funds carry more market risk **Answer: (c).** Active is defensible in inefficient segments and for objectives a cap-weighted index cannot serve. (a) is false; (b) is contradicted by persistence evidence; (d) is false — an index fund carries exactly the market's risk, no more.

Self-check

Coverage: **A** 12 concept · **B** 5 worked and reconciled numericals · **C** 8 interview answers · **D** 10 MCQs with distractor reasoning. All figures tie to the source chapter (passive 0.03–0.20% vs active 0.5–1.5%; ~90% underperformance over 15–20yrs; ~20% fee drag over 30yrs; Active Share <60% closet / >80% active; core 60–90%).

Chapter 18 — Portfolio Risk Management

1. The Problem / Need

Every prior chapter taught you to *measure* risk — variance, beta, tracking error, duration, VaR. This chapter is about the harder discipline: what you actually *do* about it, before it does something to you. Measurement is diagnosis; risk management is treatment. And treatment is where most blow-ups actually happen, because the failure is almost never "we could not calculate the number" — it is "we calculated the number, believed it too much, sized the position too big, and had no plan for the day the number was wrong."

Consider three real archetypes an aspiring analyst should carry in their head:

- **Long-Term Capital Management (1998).** Nobel-laureate-run, models everywhere, VaR reported daily. It was leveraged roughly 25-to-1 on the balance sheet and far higher through derivatives. When Russia defaulted, correlations that the model treated as independent all snapped to 1 at once. A portfolio that "should" lose a few percent in a bad month lost ~90% of its capital in weeks. The risk was measured; it was not *managed*.
- **Amaranth Advisors (2006).** A single natural-gas trader accumulated a concentrated calendar-spread position so large it was a material fraction of the entire market's open interest. When the spread moved against him, there was no one to sell to at anything near the marked price. Roughly \$6.6 billion — over half the fund — evaporated in a week. This is a *concentration* and *liquidity* failure, invisible to a naive volatility number.
- **The quant "August 2007" event.** Many market-neutral equity funds ran the same factor bets. When one large fund deleveraged, it pushed down exactly the stocks everyone else was long and up the stocks everyone was short — a crowded-trade unwind. Diversification *within* each fund was real; diversification *across* funds was an illusion.

The common thread: **the danger was not the volatility they saw on a calm Tuesday. It was the tail — the leverage, the concentration, the liquidity, and the correlations that only appear when everyone needs the exit at once.** Portfolio risk management is the set of processes that keeps a portfolio *alive* through those events so that its edge has time to compound. A great investment strategy that occasionally goes to zero has a long-run compound return of zero. Survival is not a constraint on returns; over a long horizon it is the *precondition* for them.

This chapter builds the practitioner's toolkit: managing risk at the *portfolio* level rather than position by position; portfolio VaR and stress testing; hedging with derivatives; position and concentration limits; risk budgeting; drawdown management; and — running through all of it — the crucial distinction between *volatility* and the *true risk of permanent loss*.

2. The Core Idea

Risk management rests on one organizing insight: **the risk of a portfolio is not the sum of the risks of its parts, and the risk that matters is not the wiggle you see on normal days but the loss you suffer on abnormal ones.** Everything below is an elaboration of that sentence.

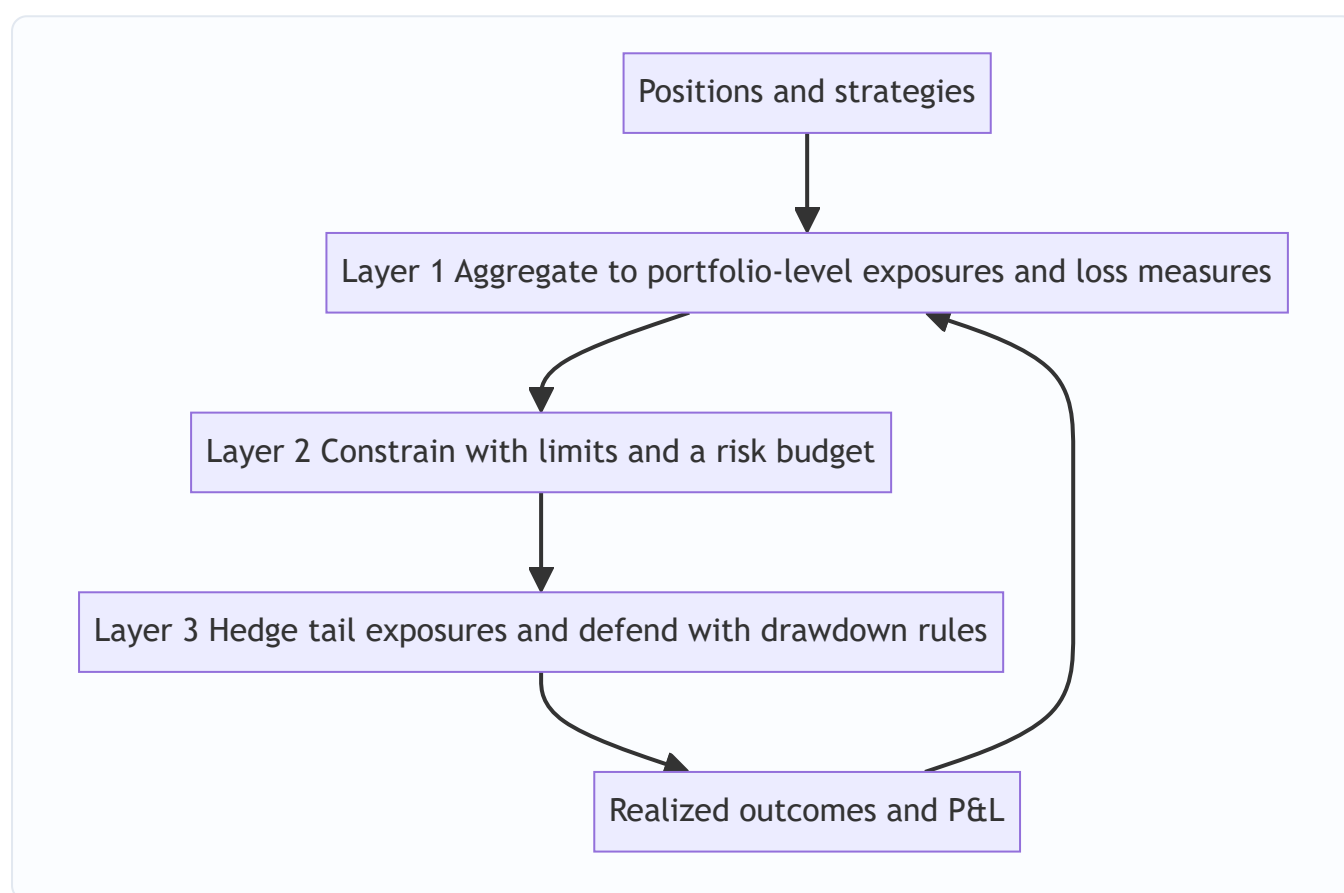
That splits into three layers that build on each other:

- **Layer 1 — Aggregate and see the whole.** Positions interact. Two "hedged" trades can be the same bet in disguise; two "different" desks can be short the same tail. You must roll everything up into portfolio-

level exposures — net beta, factor loadings, currency, duration, sector, single-name — and then into portfolio-level loss measures (VaR, expected shortfall, stress losses). You cannot manage what you cannot see aggregated.

- **Layer 2 — Constrain and budget.** Once you can see it, you impose *ex-ante* discipline: position limits, concentration limits, leverage limits, and a *risk budget* that allocates a scarce quantity — total risk — across strategies the way a capital budget allocates cash. This is the proactive layer that prevents the position from ever getting big enough to kill you.
- **Layer 3 — Hedge and defend.** For the risks you choose to keep exposure to but not fully bear, you buy or build protection — index futures, options, credit hedges — and you set *reactive* rules (drawdown stops, de-grossing triggers) for when the ex-ante controls prove insufficient.

Figure 18.1 — Portfolio risk management is a loop from seeing aggregate risk, to constraining it ex-ante, to hedging and defending, feeding back into what you see.



Underneath sits the theme that gives the chapter its spine: **volatility is a measure of dispersion; true risk is the probability and magnitude of permanent, unrecoverable loss.** They coincide for a diversified, liquid, unleveraged, normally-distributed book. They diverge — sometimes catastrophically — the moment leverage, illiquidity, concentration, or fat tails enter. A risk manager's real job is to manage the gap between the two.

3. Why / How It Works

Why portfolio risk is sub-additive (usually) — and why that can lull you

Recall from portfolio theory that the variance of a two-asset portfolio is

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

Whenever $\rho_{12} < 1$, portfolio volatility is *less* than the weighted average of the components' volatilities. Diversification is a genuine free lunch in *normal* times, and it is why the whole discipline works at the portfolio rather than the position level: risk is not additive.

The danger is that the entire benefit rides on ρ . In a crisis, cross-asset correlations tend toward 1 as everything is sold to raise cash — the "correlations go to one" phenomenon. The diversification you *budgeted* for evaporates precisely when you need it. This is why a competent risk process never trusts a single, calm-period correlation matrix; it stresses the matrix.

Why VaR is useful and simultaneously dangerous

Value at Risk answers a specific, bounded question: *over a given horizon, what loss will not be exceeded with a given confidence?* A 1-day 99% VaR of ₹10 crore means: on 99 of 100 days you lose less than ₹10 crore. It is useful because it is a single, comparable, aggregatable number across a whole firm, and regulators (Basel) built capital rules around it.

But it has three structural blind spots you must be able to articulate in an interview:

1. **It says nothing about the tail *beyond* the threshold.** VaR is the *best* of the bad days at the cutoff, not the average of the worst. A portfolio short deep out-of-the-money options can have a tiny VaR and a catastrophic loss just past the 99th percentile.
2. **It is not sub-additive.** VaR can violate the intuition that merging two portfolios should never increase total risk. This makes it a mathematically "incoherent" risk measure. **Expected Shortfall (ES / CVaR)** — the *average* loss *given* that you breach VaR — fixes both problems and is why Basel is migrating to it.
3. **It is a function of its assumptions.** Feed it a normal distribution and a calm-period covariance matrix and it will systematically understate tail risk, because real returns are fat-tailed and correlations are unstable.

The right mental model: **VaR is a speedometer, not a crash-test.** It tells you your speed on a normal road. Stress testing is the crash-test.

Why stress testing is the necessary complement

Stress testing abandons the probability distribution and asks a *scenario* question instead: "If 2008 happened again, or oil doubled, or the rupee fell 15%, what would this book lose?" It is deliberately non-statistical — it does not care whether the event is "1-in-100" — because the events that kill portfolios are exactly the ones the historical distribution rated as impossible. Stress testing captures the fat tail, the correlation break, and the liquidity freeze that VaR smooths over.

Why limits and budgets beat judgment in the moment

Human risk-taking is pro-cyclical: we feel *safest* right after a long calm (when valuations are stretched and positions are crowded) and *most frightened* at the bottom (when risk is actually cheapest). Pre-committed **limits** and a **risk budget** are a Ulysses-and-the-mast device: they bind the future, panicking or greedy version of you to the discipline of the calm, rational version. That is the deep "why" behind ex-ante controls — they exist because in-the-moment judgment is systematically biased at exactly the wrong times.

4. Full Content

4.1 Managing risk at the portfolio level

The first discipline is *aggregation*. A book is not a list of trades; it is a set of *exposures*, and the same exposure can hide inside many different-looking trades. The risk manager's job is to collapse the position list into a small number of factor exposures and see the *net* bet.

A practical portfolio-level risk report rolls up:

Exposure dimension	What it aggregates	Example limit / view
Net market (beta) exposure	Sum of beta-weighted longs minus shorts	Net beta within [0.8, 1.2] for a benchmark-relative fund
Gross exposure / leverage	Longs + shorts as % of capital	Gross $\leq$ 200% for a long-short book
Factor exposures	Loadings on value, momentum, size, quality, etc.	No single style factor > X% of active risk
Sector / industry	Net weight vs benchmark by sector	$\pm$ 5% active weight per GICS sector
Single-name concentration	Largest positions	No name > 5% of NAV
Currency	Net FX exposure by currency	Hedge $\geq$ 80% of non-base-currency assets
Duration / DV01 (fixed income)	Portfolio interest-rate sensitivity	Active duration within $\pm$ 1 year of benchmark
Liquidity	Days-to-liquidate at X% of ADV	$\geq$ 90% of book liquidatable in 5 days

The essential move is netting. Two portfolio managers each "market-neutral" within their sleeve can, in aggregate, leave the fund net long financials and short technology by a wide margin. Only the aggregated view reveals it. Likewise, a "hedged" convertible-bond position (long bond, short stock) is really a bet on volatility, credit, and interest rates — the risk report must show *those* factor exposures, not just "one long and one short."

4.2 Portfolio VaR

Definition. The Value at Risk at confidence level c over horizon h is the loss L such that $P(\text{loss} > L) = 1 - c$ over horizon h . Common conventions: 1-day or 10-day horizon; 95% or 99% confidence.

There are three standard ways to compute it:

(a) Parametric / variance-covariance (delta-normal). Assume returns are normally distributed. Then

$$\text{VaR} = z_c \cdot \sigma_p \cdot V,$$

where z_c is the standard-normal quantile (1.645 for 95%, 2.326 for 99%), σ_p is the portfolio return standard deviation over the horizon, and V is portfolio value. For a multi-asset book, $\sigma_p = \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$ where Σ is the covariance matrix and $\mathbf{w}$ the value weights. *Fast and analytic; wrong for options and fat tails.*

Time scaling. Under the i.i.d. assumption, volatility scales with the square root of time: $\sigma_h = \sigma_1 \sqrt{h}$. So a 10-day VaR $\approx$ 1-day VaR $\times \sqrt{10}$. (This breaks down when

returns autocorrelate or mean-revert.)

(b) Historical simulation. Take the last N days of actual returns, apply them to today's portfolio, sort the resulting P&L, and read off the percentile. *No distributional assumption; captures real fat tails and correlations — but only the tails that happened to occur in your window, and it weights a 1-in- N event as exactly $1/N$.*

(c) Monte Carlo simulation. Specify a stochastic model for the risk factors, draw thousands of scenarios, revalue the portfolio (including non-linear options) in each, and read the percentile. *Most flexible, handles non-linearity; only as good as the assumed model and the most computationally expensive.*

Component and marginal VaR. The truly useful portfolio outputs are the *decompositions*:

- **Marginal VaR** — how much total VaR changes for a small increase in a position. It is the risk "price" of adding to that position.
- **Component VaR** — each position's *contribution* to total VaR, which sums exactly to total VaR. This tells you *where your risk actually lives*, which is often startlingly different from where your capital lives. A 4% position in a volatile, correlated name can contribute more risk than a 20% position in a stable, diversifying one.

Component VaR is the single most important portfolio-VaR concept for a practitioner, because risk budgeting (Section 4.5) is done in these units.

Expected Shortfall (ES / CVaR). Because VaR is silent about the tail and not sub-additive, mature shops report **Expected Shortfall** — the average loss *conditional on* exceeding VaR:

$$\text{ES}_c = E[L \mid L > \text{VaR}_c]$$

ES is coherent (sub-additive) and tail-sensitive, and it is the measure Basel III's Fundamental Review of the Trading Book moved to (97.5% ES replacing 99% VaR).

4.3 Stress testing

Stress testing revalues the portfolio under *scenarios* rather than distributions. Three families:

1. **Historical scenarios.** Replay actual crises: the 2008 GFC, the 2020 COVID crash, the 1998 LTCM/Russia event, the 2013 taper tantrum, the 2022 rate shock. You apply the actual factor moves from those episodes to today's book.
2. **Hypothetical / forward-looking scenarios.** Construct plausible futures that have *no* historical precedent: "oil to \$150 and a 200bp rate hike simultaneously," "a 20% rupee devaluation," "sovereign downgrade." These probe risks the history has not yet delivered.
3. **Reverse stress testing.** Start from the answer — "what set of moves would wipe out X% of our capital?" — and work backward to find the portfolio's specific vulnerabilities. This is the most revealing exercise because it surfaces hidden concentrations and the exact correlation assumptions the book is silently betting on.

The core value of stress testing is that it **relaxes the two assumptions that make VaR dangerous**: the distribution and the correlation matrix. A good stress test explicitly assumes correlations go to 1 (or flip), liquidity dries up (haircuts on marks), and volatility spikes — the joint regime that VaR's calm-period inputs cannot represent.

4.4 Hedging with derivatives

Hedging is buying or building offsetting exposure so that a risk you do not want to bear is transferred to someone who does. The main instruments:

Index futures — hedging systematic (beta) risk. To neutralize the market exposure of an equity book, short index futures. The number of contracts:

$$N = \beta_p \cdot \frac{V_p}{F}, \text{ where } F = \text{index level} \times \text{multiplier}.$$

To move the portfolio from its current beta β_p to a target beta β^* :

$$N = (\beta^* - \beta_p) \cdot \frac{V_p}{F}.$$

Futures are cheap, liquid, and symmetric — they remove downside *and* upside. Use them when you want to strip out market direction and keep only your stock-selection alpha.

Options — asymmetric protection.

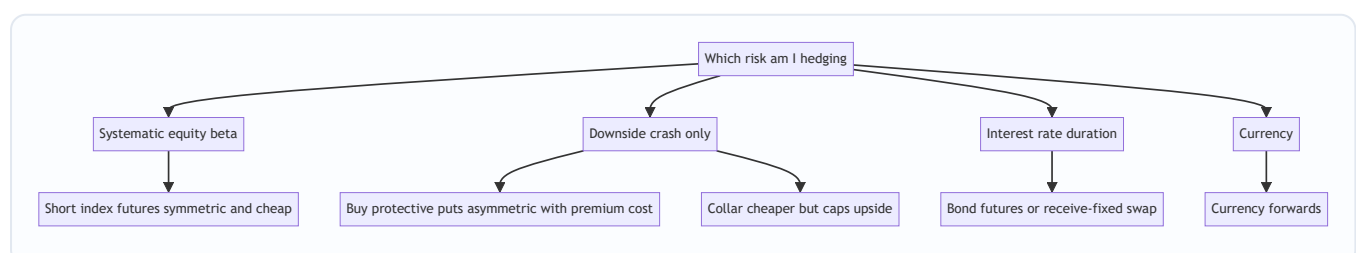
- A **protective put** (long the portfolio + long a put) sets a floor: maximum loss is capped at the strike, while upside is retained, minus the premium paid. It is portfolio insurance — and, like insurance, it has an ongoing cost (premium bleed) that drags returns in calm markets.
- A **collar** (long put financed by a short call) cheapens or eliminates the premium by giving up upside above the call strike. A "zero-cost collar" funds the put entirely with the call — you pay in foregone upside instead of cash.
- Puts are the *only* hedge that protects against a gap/crash without symmetric upside loss, which is why they are the tail-hedge of choice despite their cost.

Other hedges. Interest-rate swaps and bond futures hedge duration; credit default swaps hedge default risk; currency forwards hedge FX; variance/VIX products hedge volatility itself.

Static vs dynamic hedging. A static hedge is set and held. A dynamic hedge (delta-hedging an option book) is continuously rebalanced as exposures change — powerful but vulnerable to gaps and liquidity, because it assumes you can keep trading. Portfolio insurance implemented via dynamic hedging (selling futures as markets fall) was a major amplifier of the 1987 crash: everyone's hedging rule said "sell into the fall" at once.

Basis risk — the catch in every hedge. A hedge is rarely perfect. If you hedge a portfolio of Indian mid-caps with Nifty 50 futures, the hedge covers large-cap market moves but not the mid-cap-vs-large-cap spread. The residual — the imperfect correlation between the exposure and the hedging instrument — is **basis risk**. You have not eliminated risk; you have *exchanged* price risk for basis risk, which you must judge to be smaller and more acceptable.

Figure 18.2 — Choosing a hedge is a decision tree over which risk you are transferring and whether you will pay for symmetric or asymmetric protection.



4.5 Position and concentration limits

Limits are hard, pre-committed ceilings that prevent any single source of risk from becoming existential. They are deliberately *dumb* rules — dumb in the sense of not requiring in-the-moment judgment — because that robustness is the point.

- **Position limits** cap the size of any single position (e.g., no name > 5% of NAV; no more than X% of a security's average daily volume, so the position stays liquidatable).
- **Concentration limits** cap *grouped* exposures — sector, country, issuer, factor, counterparty. The lesson of Amaranth is that a book perfectly within single-name limits can still be lethally concentrated in a *theme* (one calendar spread across many contracts).
- **Leverage limits** cap gross and net exposure as a multiple of capital. Leverage is the single variable that most reliably converts a survivable drawdown into a terminal one, because it forces selling at the bottom (margin calls) and magnifies the loss.
- **Liquidity limits** ensure the portfolio can be unwound within a defined horizon without moving the market against itself — the risk Amaranth and LTCM both ignored.
- **Counterparty / credit limits** cap exposure to any single trading counterparty (the risk that turned Lehman's failure into everyone's problem).

The unifying principle: **limits are how you make a bad outcome survivable in advance, because you cannot rely on being rational during the event.**

4.6 Risk budgeting

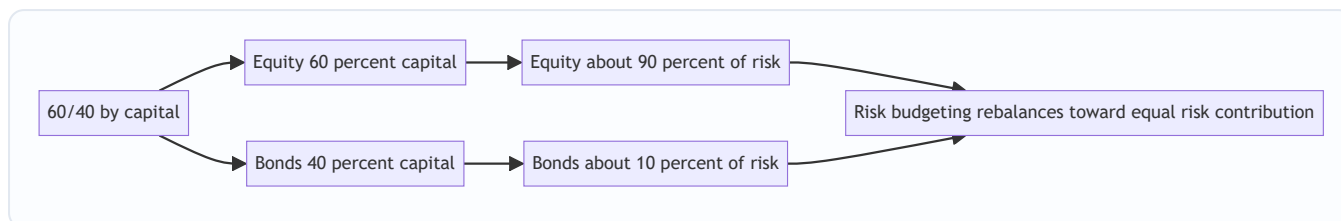
Risk budgeting reframes portfolio construction: instead of allocating *capital* (money) across strategies, you allocate *risk* (typically component VaR or volatility contribution). The insight is that capital weights and risk weights are wildly different — a 60/40 stock/bond portfolio puts 60% of the *money* in equities but roughly **90% of the risk**, because equities are ~3–4× as volatile. A "balanced" fund is, in risk terms, an equity fund with a bond garnish.

Formally, decompose total portfolio volatility into contributions. The **marginal contribution to risk** of asset i is $\partial \sigma_p / \partial w_i$, and its **total risk contribution** is $\text{RC}_i = w_i \cdot \partial \sigma_p / \partial w_i$, with $\sum_i \text{RC}_i = \sigma_p$ exactly. Risk budgeting sets targets for each RC_i .

- **Risk parity** is the special case where every asset (or asset class) contributes *equal* risk. To equalize contributions you must *lever up* the low-volatility assets (bonds) and hold *less* of the high-volatility ones (equities) than capital weighting would suggest. This is the logic behind Bridgewater's All Weather and the broader risk-parity industry.
- More generally, an active manager sets an overall **risk budget** (e.g., 4% tracking error) and *allocates* that budget across bets — so many basis points of risk to sector tilts, so many to stock selection, so many to factor timing — and monitors each strategy's realized risk contribution against its budget.

Why it is powerful: risk, not capital, is the truly scarce and dangerous resource. Budgeting in risk units forces you to notice that a small-looking allocation to a volatile, correlated strategy can consume your entire risk budget — and to size it accordingly.

Figure 18.3 — The same capital allocation and its risk allocation are very different pictures; risk budgeting manages the right one.



4.7 Drawdown management

A **drawdown** is the peak-to-trough decline in portfolio value; **maximum drawdown (MDD)** is the worst such decline over a period. It is the risk measure investors *actually feel*, because it maps directly to the pain of watching wealth fall and to the temptation (or forced requirement) to capitulate at the bottom.

The mathematics of recovery is the reason drawdowns dominate long-run outcomes: **losses and the gains needed to recover them are asymmetric.**

Drawdown	Gain required to recover
–10%	+11.1%
–20%	+25%
–33%	+50%
–50%	+100%
–80%	+400%

A 50% loss requires a 100% gain to break even. This convexity is why drawdown control is not a comfort feature but a *return* feature: avoiding the deep hole is worth more than it looks, because you never have to climb out of it. It is also why leverage is so dangerous — it deepens the hole and, via margin calls, can prevent you from ever participating in the recovery.

Practical drawdown management tools:

- **Stop-losses and de-grossing triggers** — pre-set rules to cut exposure once a drawdown threshold is breached (e.g., "reduce gross by 25% at –10%, by 50% at –15%"). This mechanizes the discipline of not letting a manageable loss become terminal.
- **Volatility targeting** — scale gross exposure inversely to realized/forecast volatility so that risk stays roughly constant; exposure automatically falls as markets get turbulent (turbulence being highly correlated with drawdowns).
- **Trailing stops and drawdown-based fee/redemption terms** — high-water marks in hedge funds tie the manager's fee to recovering past peaks, aligning incentives with drawdown control.
- **Tail hedges** — a standing allocation to long puts or long-volatility positions that pay off precisely in the deep drawdowns, dampening MDD at the cost of a small ongoing premium.

The tension to hold in mind: stop-losses protect against the deep drawdown but can also lock in losses right before a rebound (whipsaw). The art is calibrating triggers to the strategy's natural volatility so you cut *tail* risk without being chopped up by normal noise.

4.8 Volatility versus the true risk of loss

This is the intellectual heart of the chapter and the distinction that separates a sophisticated answer from a textbook one.

Volatility (σ) is the standard deviation of returns — a symmetric measure of *dispersion*. It treats a +8% month and a –8% month as equally "risky." It is convenient because it is well-defined, computable, additive-in-variance, and central to Markowitz, CAPM, Sharpe, and VaR. For a diversified, liquid, unleveraged portfolio with roughly normal returns, volatility is a fine proxy for risk.

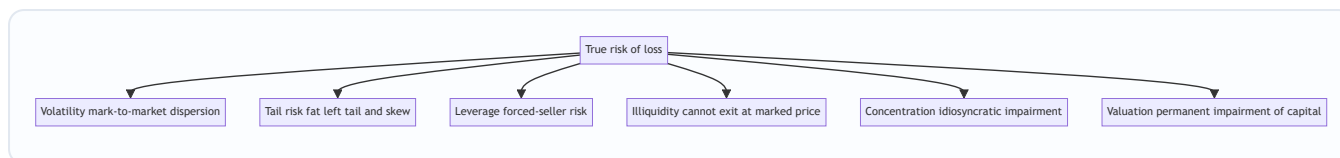
True risk is the probability and magnitude of *permanent, unrecoverable loss of capital* — the outcome from which the portfolio does not come back. The two diverge whenever any of the following are present:

Volatility misses this	Why it matters
Skew / fat tails	Selling deep OTM options gives low volatility and steady income — until a crash delivers a catastrophic left-tail loss. Low σ , huge true risk.
Leverage	A levered position can have modest volatility but face margin calls that force liquidation at the bottom — turning a temporary mark-down into a permanent loss.
Illiquidity	Private assets and small-caps show <i>low reported</i> volatility because they are marked infrequently (stale prices), masking real risk. Illiquidity itself is a risk volatility does not see.
Concentration	A single-name or single-theme bet may have acceptable volatility until an idiosyncratic event (fraud, regulation) impairs it permanently.
Upside vs downside asymmetry	Volatility penalizes upside surprises equally with downside ones — but no investor fears making money. Downside deviation / semi-variance captures the asymmetry volatility ignores.
Valuation / permanent impairment	Overpaying for a great business is a <i>risk of loss</i> that has nothing to do with price wiggle; a stock can be low-volatility on its way to being permanently worth less.

Two framings crystallize it. First, the value-investor's line (Buffett, Marks): **"Risk is the probability of permanent loss, not volatility."** A stock that falls 40% on temporary fear, that you can hold, may be *low* true risk (a chance to buy) despite *high* volatility. A stable, levered, illiquid income strategy may be *high* true risk despite *low* volatility. Second, the practitioner's synthesis: volatility is *a* dimension of risk — the day-to-day, mark-to-market dimension that matters if you can be forced to sell — but it is not *the* risk. The complete risk picture layers volatility *plus* tail risk (ES, stress loss), *plus* liquidity, *plus* leverage, *plus* concentration, *plus* the possibility of permanent impairment.

The managerial takeaway that ties the whole chapter together: **the tools of Sections 4.1–4.7 exist to manage the gap between volatility and true risk.** Stress testing and ES address fat tails; limits address concentration and leverage; liquidity limits address illiquidity; drawdown rules address the forced-seller path from temporary to permanent loss. Volatility tells you how bumpy the ride is; risk management is about making sure you arrive.

Figure 18.4 — Volatility is one slice of true risk; the other slices are exactly what naive volatility measures miss.



5. Worked / Applied Examples

Example 1 — Parametric portfolio VaR and Expected Shortfall

A fund holds ₹100 crore. Its estimated annual return volatility is 20%. Assume ~250 trading days and normally distributed returns.

1-day volatility: $\sigma_{1d} = \frac{20\%}{\sqrt{250}} = \frac{0.20}{15.81} = 1.265\%$

1-day 99% VaR ($z = 2.326$): $\text{VaR} = 2.326 \times 0.01265 \times ₹100 \text{ cr} = ₹2.94 \text{ cr}$

Interpretation: on ~99 of 100 days, the daily loss is less than ₹2.94 crore.

10-day 99% VaR (square-root-of-time): $\text{VaR}_{10d} = ₹2.94 \text{ cr} \times \sqrt{10} = ₹9.30 \text{ cr}$

Expected Shortfall (99%). For a normal distribution, $\text{ES}_c = \sigma \cdot \frac{d\phi(z_c)}{1-c}$. With $\phi(2.326) = 0.0267$ and $1-c = 0.01$: $\text{ES factor} = \frac{0.0267}{0.01} = 2.67$. $\text{ES}_{1d} = 2.67 \times 0.01265 \times ₹100 \text{ cr} = ₹3.38 \text{ cr}$

Note ES (₹3.38 cr) exceeds VaR (₹2.94 cr): the *average* loss on a breach day is larger than the *threshold*, which is exactly the tail information VaR omits. **And this whole calculation assumes normality — if the true distribution is fat-tailed, both numbers understate reality**, which is why stress testing (Example 3) is mandatory.

Example 2 — Hedging beta with index futures

A ₹50 crore equity portfolio has a beta of 1.30 to the Nifty 50. Nifty is at 24,000; the futures multiplier (lot size) is 50, so one contract controls $24,000 \times 50 = ₹12,00,000 = ₹0.12$ crore of index exposure.

(a) Fully hedge to beta 0 (short futures): $N = \beta_p \cdot \frac{V_p}{F} = 1.30 \times \frac{₹50 \text{ cr}}{₹0.12 \text{ cr}} = 1.30 \times 416.7 = 542 \text{ contracts (short)}$

Now suppose the market falls 5%. The portfolio ($\beta = 1.30$) loses $\approx 1.30 \times 5\% \times ₹50 \text{ cr} = ₹3.25 \text{ cr}$. The short futures gain $\approx 5\% \times (542 \times ₹0.12 \text{ cr}) = 5\% \times ₹65.0 \text{ cr} = ₹3.25 \text{ cr}$. They offset — the systematic loss is neutralized, leaving only stock-selection (idiosyncratic) P&L. *This is how a long-short manager isolates alpha from market direction.*

(b) Reduce beta from 1.30 to a target 0.80 instead of fully hedging: $N = (\beta^* - \beta_p) \cdot \frac{V_p}{F} = (0.80 - 1.30) \times \frac{₹50 \text{ cr}}{₹0.12 \text{ cr}} = -0.50 \times 416.7 = -208 \text{ contracts}$. Short 208 contracts to dial market exposure down while keeping a net long tilt.

Basis-risk caveat: if the portfolio is mid-cap heavy, hedging with Nifty 50 leaves the mid-vs-large spread unhedged. In a "flight to quality" where large-caps hold up but mid-caps fall harder, the hedge under-protects and the fund still loses on the basis.

Example 3 — Stress test vs VaR, and the correlation trap

A ₹100 crore book is split ₹50 cr equities and ₹50 cr corporate bonds. In calm-period data, equity–bond correlation is -0.30 (bonds cushion equity falls). Daily vols: equities 1.5%, bonds 0.4%.

Normal-regime 1-day portfolio volatility (weights 0.5/0.5): $\sigma_p = \sqrt{0.5^2(1.5)^2 + 0.5^2(0.4)^2 + 2(0.5)(0.5)(-0.30)(1.5)(0.4)}$. $\sigma_p = \sqrt{0.5625 + 0.04 - 0.09} = \sqrt{0.5125} = 0.716\%$. 1-day 99% VaR $\approx 2.326 \times 0.716\% \times ₹100 \text{ cr} = ₹1.67 \text{ cr}$. Comfortable.

Stress scenario — "risk-off crash": equities -12% in a day, and the correlation *flips to* $+0.70$ as investors dump *both* stocks and credit for cash; assume bonds fall -3% (credit spreads blow out, overwhelming any rate rally). $\text{Loss} = 0.5 \times (-12\%) + 0.5 \times (-3\%) = -6\% - 1.5\% = -7.5\% = -₹7.5 \text{ cr}$.

The stress loss (**₹7.5 cr**) is $\sim 4.5\times$ the VaR (**₹1.67 cr**) — not because the VaR math was wrong, but because its two inputs (the calm-period correlation of -0.30 and the assumption of no fat tail) *both broke at once*. This is the precise mechanism behind LTCM and 2008: **the diversification you budgeted for disappears exactly when you draw on it**. The lesson an interviewer wants: never size a book to survive only its VaR; size it to survive its stress loss.

Example 4 — Risk budgeting: capital weights lie

A pension runs 60% equities ($\sigma = 18\%$), 40% bonds ($\sigma = 5\%$), correlation 0.20.

Portfolio volatility: $\sigma_p = \sqrt{0.6^2(18)^2 + 0.4^2(5)^2 + 2(0.6)(0.4)(0.20)(18)(5)}$. $\sigma_p = \sqrt{116.6 + 4.0 + 8.64} = \sqrt{129.3} = 11.37\%$.

Risk contribution of equities. Marginal contribution $= \frac{\partial \sigma_p}{\partial w_E} = \frac{w_E \sigma_E^2 + w_B \rho \sigma_E \sigma_B}{\sigma_p}$. $= \frac{0.6(324) + 0.4(0.20)(18)(5)}{11.37} = \frac{194.4 + 7.2}{11.37} = \frac{201.6}{11.37} = 17.73$. Total equity risk contribution $= w_E \times 17.73 = 0.6 \times 17.73 = 10.64\%$.

Equities' share of total risk $= 10.64 / 11.37 = 93.6\%$.

So a portfolio that *looks* 60/40 by capital is **$\sim 94\%$ equity risk** — practically an all-equity fund. To move toward **risk parity** (equal risk contribution) you would sharply *reduce* the equity weight and *lever up* the bonds. This is the number that convinces allocators the "balanced fund" label is misleading, and it is why risk budgeting — not capital budgeting — is the correct lens.

6. Connections

- **Portfolio theory (Ch 3–4)**: the covariance matrix that powers diversification is the *same* matrix inside parametric VaR and risk budgeting. Risk management is Markowitz applied *defensively* — and stress testing is the admission that the matrix is unstable.
- **CAPM / beta (Ch 5)**: beta is the exposure you hedge away with index futures to isolate alpha. Risk-adjusted performance (Sharpe, IR) and risk budgets are two sides of the same coin: return *per unit of risk* vs risk *allocated per bet*.
- **Fixed income (Ch 10)**: duration and DV01 are the fixed-income analogues of beta; hedging them with swaps/bond futures is the same logic as equity beta hedging.
- **Performance measurement (Ch 11)**: the Information Ratio's denominator (tracking error) *is* the active risk budget; a manager who blows the risk budget cannot be evaluated fairly. Drawdown and downside deviation feed Sortino and Calmar ratios.
- **Derivatives (options/futures chapters)**: protective puts, collars, and delta-hedging are the mechanical toolkit; here they are deployed at the *portfolio* level.
- **Behavioral finance**: limits and drawdown rules exist because human risk appetite is pro-cyclical — greatest at tops, smallest at bottoms. Pre-commitment fights that bias.

7. Key Terms

- **Value at Risk (VaR):** the loss over a horizon not exceeded with a stated confidence; a threshold, not a worst case.
- **Expected Shortfall (ES / CVaR):** the *average* loss given VaR is breached; coherent and tail-sensitive.
- **Component VaR:** each position's contribution to total VaR; sums to total VaR; the unit of risk budgeting.
- **Marginal VaR:** change in total VaR from a small change in a position — the risk "price" of adding to it.
- **Stress testing:** scenario-based revaluation that ignores the probability distribution to capture tails and correlation breaks.
- **Reverse stress test:** start from a fatal loss and find the scenario that causes it.
- **Basis risk:** residual risk from an imperfect hedge — the exposure and the hedging instrument are not perfectly correlated.
- **Protective put / collar:** option structures giving a downside floor (put) with premium cost, optionally financed by capping upside (collar).
- **Hedge ratio (beta hedge):** $N = \beta_p V_p / F$ contracts to neutralize market exposure.
- **Concentration limit:** cap on grouped exposure (sector, issuer, theme, counterparty), not just single names.
- **Leverage:** ratio of exposure to capital; the primary converter of survivable losses into terminal ones.
- **Risk budget:** allocation of total portfolio risk (not capital) across strategies/bets.
- **Risk parity:** portfolio where every component contributes equal risk, achieved by leveraging low-vol assets.
- **Risk contribution:** $\text{RC}_i = w_i \frac{\partial \sigma_p}{\partial w_i}$, summing to σ_p .
- **Drawdown / maximum drawdown:** peak-to-trough decline; the felt measure of risk.
- **Volatility targeting:** scaling exposure inversely to forecast volatility to keep risk roughly constant.
- **Permanent loss of capital:** impairment from which the portfolio does not recover — the "true risk" volatility can miss.

8. Common Confusions

- **"VaR is the worst case."** No — it is the *best* of the bad tail, the threshold at the chosen percentile. Losses beyond it can be far larger; that is what ES and stress tests are for.
- **"Low volatility means low risk."** Only for a liquid, unlevered, diversified, roughly-normal book. Selling deep OTM options, levered carry, and illiquid private assets all pair low reported volatility with high true risk.
- **"A hedge eliminates risk."** It *transforms* price risk into basis risk (and adds cost/complexity). A poorly matched hedge can even add risk.
- **"Diversification protects me in a crash."** Diversification is a normal-times benefit; correlations converge toward 1 in crises. Budget for the stressed correlation, not the calm one.
- **"60/40 is balanced."** By capital, yes; by risk, it is ~90% equities. Balance must be judged in risk units.
- **"VaR is sub-additive so merging books can only reduce risk."** VaR is *not* sub-additive (that is a known defect); ES is. Do not assume portfolio VaR $\leq$ sum of standalone VaRs.
- **"Stop-losses guarantee protection."** They cap losses in orderly markets but can gap through in a crash and can whipsaw you out before a rebound. They manage, not eliminate, drawdown.

- **"Beta hedging removes all risk."** It removes *systematic* risk; idiosyncratic, basis, and factor risks remain — which is the whole point (you keep your alpha, and its risk).
- **"More hedging is always safer."** Hedges cost premium/carry and drag returns; over-hedging can guarantee underperformance. Risk management optimizes the *trade-off*, it does not minimize risk to zero.

9. Recap

Portfolio risk management is the treatment that follows risk measurement's diagnosis, and its purpose is *survival*, because a strategy that occasionally goes to zero compounds to zero. It operates in a loop: **see** the aggregate exposures and loss measures, **constrain** them ex-ante with limits and a risk budget, then **hedge and defend** the tails you keep. VaR gives a single comparable loss threshold but is blind beyond the tail and dependent on its assumptions; Expected Shortfall fixes its coherence and tail blindness; stress testing supplies the crash-test that VaR's speedometer cannot. Hedging with futures strips systematic beta symmetrically while options buy asymmetric crash protection at a premium — always leaving basis risk behind. Position, concentration, leverage, and liquidity limits are pre-committed rules that keep any one exposure from becoming existential, precisely because in-the-moment judgment is biased at the worst times. Risk budgeting allocates the truly scarce resource — risk, not capital — revealing that a 60/40 book is ~90% equity risk. Drawdown management respects the brutal asymmetry that a 50% loss needs a 100% gain to recover. And threading all of it is the master distinction: **volatility measures the bumpiness of the ride; true risk is the probability of permanent, unrecoverable loss** — and every tool in the chapter exists to manage the gap between the two.

10. Quick-Reference / Interview Points

- **VaR one-liner:** "The loss not exceeded with X% confidence over horizon h — a threshold, not a worst case. Blind beyond the tail, not sub-additive, assumption-dependent." Follow with: "That's why we report Expected Shortfall and stress test."
- **VaR vs ES:** VaR = the cutoff; ES = average loss *beyond* the cutoff. ES is coherent (sub-additive) and Basel FRTB's choice (97.5% ES).
- **VaR methods:** parametric (fast, wrong for options/tails), historical (real tails but only those observed), Monte Carlo (flexible, model-dependent, expensive).
- **Time scaling:** VaR scales with $\sqrt{h}$ under i.i.d.; 10-day $\approx$ 1-day $\times \sqrt{10}$.
- **Component VaR:** where risk *actually* lives; sums to total VaR; the unit of risk budgeting.
- **Beta hedge formula:** $\Delta N = (\beta^* - \beta_p) V_p / F$. Know the sign (short to reduce beta).
- **Protective put vs collar:** put = floor with premium cost; collar = cheaper floor financed by capping upside.
- **Basis risk:** every hedge exchanges price risk for basis risk; name the mid-cap-hedged-with-Nifty example.
- **60/40 is ~90% equity risk** — cite this to demonstrate risk-budgeting fluency; risk parity levers bonds to equalize contributions.
- **Drawdown asymmetry:** -50% needs +100% to recover; this is *why* drawdown control is a return driver, and why leverage kills.
- **"Correlations go to 1"** in crises — the mechanism behind LTCM/2008; always stress the correlation matrix, never trust the calm one.

- **The money quote:** "Volatility is not risk; risk is the probability of permanent loss of capital." Then show you can operationalize it: fat tails → ES/stress; leverage → limits; illiquidity → liquidity limits; concentration → concentration limits.
- **Classic case studies:** LTCM (leverage + correlation), Amaranth (concentration + liquidity), 1987 portfolio insurance (dynamic hedging feedback), quant quake 2007 (crowded trades). Having one crisp example per failure mode signals real understanding.

Q&A — Portfolio Risk Management

Practice bank for Chapter 18. Work each question before reading the answer. Currency is ₹ crore unless stated; volatilities and returns are in percent; betas and correlations are unit-free. Every numerical answer is worked in full and reconciled against a cross-check.

Section A — Concept Checks

A1. State the one sentence the whole chapter elaborates. The risk of a portfolio is not the sum of the risks of its parts, and the risk that matters is not the wiggle you see on normal days but the loss you suffer on abnormal ones. Everything — aggregation, VaR, stress testing, limits, budgets, hedges, drawdown rules — is machinery for acting on that sentence.

A2. What is the difference between measuring risk and managing it? Measurement is diagnosis (variance, beta, VaR, duration); management is treatment (limits, budgets, hedges, drawdown rules). Blow-ups almost never come from failing to calculate the number — they come from calculating it, believing it too much, sizing too big, and having no plan for the day the number is wrong.

A3. Define VaR precisely, and give one thing it does NOT tell you. Value at Risk at confidence c over horizon h is the loss L such that $P(\text{loss} > L) = 1 - c$ over that horizon. A 1-day 99% VaR of ₹10 cr means on 99 of 100 days you lose less than ₹10 cr. It tells you *nothing* about how bad the loss is on the 1 day you breach it — the tail beyond the threshold.

A4. Why is VaR called an "incoherent" risk measure, and what fixes it? Because it is not sub-additive: the VaR of a combined book can exceed the sum of the two standalone VaRs, violating the intuition that merging can only diversify. Expected Shortfall (ES/CVaR) — the *average* loss given VaR is breached — is sub-additive (coherent) and tail-sensitive, which is why Basel III's FRTB migrated to 97.5% ES.

A5. Name the three ways to compute VaR and each one's chief weakness. Parametric/delta-normal (fast, analytic — but wrong for options and fat tails); historical simulation (real correlations and tails — but only the tails that happened to occur in the window); Monte Carlo (flexible, handles non-linearity — but only as good as the assumed model, and computationally expensive).

A6. Distinguish marginal VaR from component VaR. Marginal VaR is how much total VaR changes for a small increase in a position — the risk "price" of adding to it. Component VaR is each position's *contribution* to total VaR, and the components sum exactly to total VaR. Component VaR tells you where your risk actually lives, which is often startlingly different from where your capital lives.

A7. Why is stress testing the necessary complement to VaR? VaR is a speedometer; stress testing is the crash-test. Stress testing abandons the probability distribution and asks a scenario question ("if 2008 happened again, what would we lose?"), deliberately relaxing the two assumptions that make VaR dangerous — the distribution and the calm-period correlation matrix. It captures the fat tail, the correlation break, and the liquidity freeze that VaR smooths over.

A8. What is reverse stress testing and why is it uniquely revealing? It starts from the answer — "what set of moves would wipe out X% of capital?" — and works backward to the scenario. It is revealing because it surfaces hidden concentrations and the exact correlation assumptions the book is silently betting on, rather than testing only scenarios you already thought of.

A9. What is basis risk, and give the canonical example. Basis risk is the residual risk from an imperfect hedge — the exposure and the hedging instrument are not perfectly correlated. Hedging Indian mid-caps with Nifty 50 futures covers large-cap market moves but not the mid-cap-vs-large-cap spread; in a flight to quality large-caps hold up while mid-caps fall harder, so the hedge under-protects. You exchange price risk for basis risk, which you must judge smaller and more acceptable.

A10. Contrast a protective put with a collar. A protective put (long portfolio + long put) sets a downside floor at the strike while keeping upside, at the cost of an ongoing premium bleed. A collar finances that put by selling a call, giving up upside above the call strike; a "zero-cost collar" funds the put entirely with the call — you pay in foregone upside instead of cash.

A11. Why do pre-committed limits beat in-the-moment judgment? Human risk appetite is pro-cyclical: we feel safest right after a long calm (when valuations are stretched and trades are crowded) and most frightened at the bottom (when risk is actually cheapest). Limits and risk budgets are a Ulysses-and-the-mast device — they bind the future, panicking-or-greedy version of you to the discipline of the calm, rational version.

A12. Why is a 60/40 portfolio "not balanced"? Balanced by *capital*, not by *risk*. Equities are roughly 3–4× as volatile as bonds, so a 60/40 book carries about 90% of its risk in equities — practically an equity fund with a bond garnish. Balance must be judged in risk units, which is the whole point of risk budgeting.

A13. State the master distinction of the chapter in one line. Volatility measures the bumpiness of the ride (symmetric, mark-to-market dispersion); true risk is the probability and magnitude of permanent, unrecoverable loss of capital. Every tool in the chapter exists to manage the gap between the two.

Section B — Numerical Problems (full working)

B1. Parametric VaR and Expected Shortfall. A fund holds ₹100 cr with annual return volatility 20%. Assume 250 trading days and normal returns. Find 1-day 99% VaR, 10-day 99% VaR, and 1-day 99% ES.

- 1-day vol: $\sigma_{1d} = 20\% / \sqrt{250} = 0.20 / 15.81 = 1.265\%$.
- 1-day 99% VaR ($z = 2.326$): $\$2.326 \times 0.01265 \times 100 = ₹2.94\$$ cr.
- 10-day VaR ($\sqrt{t}$ -time): $\$2.94 \times \sqrt{10} = 2.94 \times 3.162 = ₹9.30\$$ cr.
- ES factor for normal: $\phi(2.326)/(1-c) = 0.0267/0.01 = 2.67$. ES $\$ = 2.67 \times 0.01265 \times 100 = ₹3.38\$$ cr.

Reconcile: ES (₹3.38 cr) > VaR (₹2.94 cr) always, because the average loss on a breach day exceeds the threshold — exactly the tail info VaR omits. And both numbers assume normality; if returns are fat-tailed, both understate reality. ✓

B2. Beta hedge with index futures. A ₹50 cr equity book has beta 1.30 to the Nifty 50 at 24,000, lot size 50, so one contract = $24,000 \times 50 = ₹12,00,000 = ₹0.12$ cr. (a) Contracts to fully hedge to beta 0? (b) Verify the offset if the market falls 5%. (c) Contracts to move beta to 0.80 instead?

- (a) $\$N = \beta_p V_p / F = 1.30 \times 50 / 0.12 = 1.30 \times 416.7 = 542\$$ contracts short.
- (b) Portfolio loss $\approx 1.30 \times 5\% \times 50 = ₹3.25\$$ cr. Futures gain $\approx 5\% \times (542 \times 0.12) = 5\% \times 65.0 = ₹3.25\$$ cr. They offset — only idiosyncratic P&L remains. ✓
- (c) $\$N = (\beta^* - \beta_p) V_p / F = (0.80 - 1.30) \times 416.7 = -0.50 \times 416.7 = -208\$$ contracts (short 208).

Cross-check: part (c) shorts fewer contracts than part (a) — sensible, since dialing beta from 1.30 to 0.80 removes only 0.50 of beta versus the full 1.30. Ratio $\$208/542 = 0.384 = 0.50/1.30$. ✓

B3. Diversified VaR, then a correlation-break stress. A ₹100 cr book is ₹50 cr equities (daily vol 1.5%) and ₹50 cr bonds (daily vol 0.4%), calm correlation -0.30 . (a) 1-day 99% VaR. (b) Stress: equities -12% , bonds -3% , correlation flips to $+0.70$. Compare.

- (a) $\sigma_p = \sqrt{0.5^2(1.5)^2 + 0.5^2(0.4)^2 + 2(0.5)(0.5)(-0.30)(1.5)(0.4)}$ $= \sqrt{0.5625 + 0.04 - 0.09} = \sqrt{0.5125} = 0.716\%$. VaR $= 2.326 \times 0.716\% \times 100 = ₹1.67$ cr.
- (b) Stress loss $= 0.5(-12\%) + 0.5(-3\%) = -6\% - 1.5\% = -7.5\% = -₹7.5$ cr.

Reconcile: the stress loss (₹7.5 cr) is $\sim 4.5\times$ the VaR (₹1.67 cr) — not because the VaR math was wrong, but because *both* its inputs (the -0.30 correlation and the no-fat-tail assumption) broke at once. This is the LTCM/2008 mechanism: the diversification you budgeted for vanishes exactly when you draw on it. Size to survive the stress loss, not the VaR. ✓ (Note the flipped $+0.70$ correlation does not enter the point-scenario loss — it is a fixed-move revaluation — but it is the reason both legs fall together instead of one cushioning the other.)

B4. Risk budgeting — capital weights lie. A pension runs 60% equities ($\sigma = 18\%$), 40% bonds ($\sigma = 5\%$), correlation 0.20. Find portfolio vol, equities' risk contribution, and equities' share of total risk.

- $\sigma_p = \sqrt{0.6^2(18)^2 + 0.4^2(5)^2 + 2(0.6)(0.4)(0.20)(18)(5)}$ $= \sqrt{116.64 + 4.0 + 8.64} = \sqrt{129.28} = 11.37\%$.
- Marginal contribution of equities $= \frac{w_E \sigma_E^2 + w_B \rho \sigma_E \sigma_B}{\sigma_p} = \frac{0.6(324) + 0.4(0.20)(90)}{11.37} = \frac{194.4 + 7.2}{11.37} = 17.73\%$.
- Equity risk contribution $= w_E \times 17.73 = 0.6 \times 17.73 = 10.64\%$.
- Share $= 10.64/11.37 = 93.6\%$.

Cross-check: bonds must supply the remaining 6.4%. Bond marginal $= \frac{0.4(25) + 0.6(0.20)(90)}{11.37} = \frac{10 + 10.8}{11.37} = 1.83\%$; bond RC $= 0.4 \times 1.83 = 0.73\%$; share $0.73/11.37 = 6.4\%$. The two contributions sum to $\$10.64 + 0.73 = 11.37 = \sigma_p$ exactly. ✓ A "60/40" book is $\sim 94\%$ equity risk; risk parity would lever bonds and cut equities to equalize.

B5. Drawdown recovery asymmetry. A fund falls 40%. What gain recovers it? What if it falls 60%? State the general rule.

- Recovery gain $= \frac{1}{1-d} - 1$. For -40% : $\$1/0.60 - 1 = 1.667 - 1 = 66.7\%$.
- For -60% : $\$1/0.40 - 1 = 2.50 - 1 = 150\%$.

Interpretation: the required gain grows convexly with the drawdown — -50% needs $+100\%$, -80% needs $+400\%$. This asymmetry is why drawdown control is a *return* driver, not a comfort feature, and why leverage (which deepens the hole and can force selling via margin calls before the recovery) is so dangerous. ✓

Section C — Interview-Style Questions (model answers)

C1. "Your VaR model says the fund is safe. Why don't you trust it?" Because VaR answers one narrow question — the loss threshold at a chosen percentile — and is blind to three things. First, it says nothing about the tail beyond the cutoff; a book short deep-OTM options has a tiny VaR and a catastrophic loss just past the 99th percentile. Second, it is not sub-additive, so it can misstate the benefit of combining books. Third, it is only as good as its inputs — feed it a normal distribution and a calm-period correlation matrix and

it systematically understates tail risk, because real returns are fat-tailed and correlations are unstable. So I read VaR as a speedometer, pair it with Expected Shortfall for the tail, and run stress tests as the crash-test. I never size a book to survive only its VaR; I size it to survive its stress loss.

C2. "Walk me through how you'd hedge an equity portfolio's market risk, and what's left afterward."

Short index futures sized as $\$N = \beta_p V_p / F$ to take net beta to zero, or $\$N = (\beta^* - \beta_p)V_p / F$ to dial to a target beta. Futures are cheap, liquid, and symmetric — they strip out market direction so I keep only stock-selection alpha. What's left is idiosyncratic risk (the alpha and its risk — intentional), basis risk if the hedging index doesn't match the book (e.g., mid-caps hedged with Nifty 50), and factor/sector tilts the beta hedge doesn't touch. If I wanted asymmetric crash protection instead of removing upside, I'd buy protective puts and accept the premium bleed, or cheapen them into a collar by capping upside. A hedge never eliminates risk — it transforms* price risk into basis risk plus cost.

C3. "A junior analyst says our 60/40 fund is well diversified because no asset class exceeds 60%. Correct him."

He's judging balance in capital units; risk lives in different units. Equities are 3–4× as volatile as bonds, so decomposing portfolio volatility into risk contributions shows equities supply roughly 90–94% of the total risk — the fund is effectively an equity fund with a bond garnish. If I want genuine balance I budget *risk*, not capital: toward risk parity, where each class contributes equal risk, which means levering the low-vol bonds up and cutting the equity weight down. The lesson is that risk, not capital, is the scarce and dangerous resource, so it's the one you allocate deliberately.

C4. "Buffett says volatility isn't risk. Is he wrong, given everything CAPM and Sharpe rest on?" He's making a real distinction, not denying the math. Volatility is a symmetric measure of mark-to-market dispersion — an excellent risk proxy for a diversified, liquid, unlevered, roughly-normal book, and rightly central to Markowitz, CAPM, Sharpe and VaR. But true risk is the probability of *permanent, unrecoverable* loss, and the two diverge whenever leverage, illiquidity, concentration, fat tails, or overvaluation enter. A stable, levered, illiquid income strategy can show low volatility and carry huge true risk; a quality stock down 40% on temporary fear can be high volatility yet low true risk if you can hold it. So volatility is *a* dimension of risk — the one that bites if you can be forced to sell — but not *the* risk. My job as a risk manager is precisely to manage the gap: ES and stress for fat tails, limits for leverage and concentration, liquidity limits for illiquidity, drawdown rules for the forced-seller path.

C5. "Give me one crisp case study per major failure mode." LTCM (1998) — *leverage plus correlation*: ~25× balance-sheet leverage and models that treated correlations as independent; Russia's default snapped them all to 1 and a "few percent" month became a ~90% loss. Amaranth (2006) — *concentration plus liquidity*: one natural-gas trader held a calendar spread so large it was a material fraction of open interest, and there was no one to sell to when it moved — over half the fund, ~\$6.6 bn, gone in a week. 1987 — *dynamic-hedging feedback*: portfolio insurance rules all said "sell futures as the market falls," so everyone sold at once and amplified the crash. Quant quake (August 2007) — *crowded trades*: many market-neutral funds ran the same factor bets; one deleveraging forced everyone's longs down and shorts up. The common thread: the danger was never the calm-Tuesday volatility — it was leverage, concentration, liquidity, and correlations that only appear when everyone needs the exit at once.

Section D — Multiple Choice (with reasoning)

D1. A 1-day 99% VaR of ₹5 cr means: (a) The most you can lose in a day is ₹5 cr. (b) On about 1 day in 100 you lose more than ₹5 cr. (c) Your average daily loss is ₹5 cr. (d) You will lose ₹5 cr on 99 days out of 100.

Answer: (b). VaR is a threshold not exceeded with 99% confidence, so on ~1 day in 100 the loss is worse. (a) is the "VaR is the worst case" fallacy — losses beyond it can be far larger. (c) describes something closer to Expected Shortfall. (d) misreads the confidence level as a frequency of the loss amount.

D2. Which risk measure is sub-additive (coherent) and captures the tail beyond the cutoff? (a)

Parametric VaR (b) Historical VaR (c) Expected Shortfall (d) Standard deviation **Answer: (c).** ES/CVaR averages losses *given* VaR is breached, making it tail-sensitive, and it is sub-additive — the two defects of VaR it is designed to fix. Any flavour of VaR (a, b) shares VaR's incoherence. Standard deviation (d) is symmetric and says nothing specifically about the tail.

D3. To reduce a ₹50 cr book's beta from 1.20 to 0.60 with futures where one contract = ₹0.10 cr, you should: (a) Short 600 contracts (b) Short 300 contracts (c) Long 300 contracts (d) Short 250 contracts

Answer: (b). $\$N = (\beta^* - \beta_p)V_p/F = (0.60 - 1.20)(50/0.10) = -0.60 \times 500 = -300\$$, i.e. short 300. (a) is the full hedge to beta 0 ($\$1.20 \times 500 = 600\$$). (c) has the wrong sign — you short to reduce beta. (d) uses no clean formula.

D4. In a market crisis, cross-asset correlations typically: (a) Fall toward zero, boosting diversification (b) Stay stable (c) Rise toward 1, eroding diversification (d) Become irrelevant **Answer: (c).** Everything is sold to raise cash, so correlations converge toward 1 and the diversification you budgeted for evaporates precisely when you need it — the LTCM/2008 mechanism. This is why a risk process never trusts a single calm-period correlation matrix and stresses it instead.

D5. A 60/40 (equity/bond) portfolio is best described as: (a) ~60% of risk in equities (b) ~40% of risk in equities (c) ~90% of risk in equities (d) Equal risk in both **Answer: (c).** Because equities are ~3–4× as volatile as bonds, equities contribute roughly 90% of portfolio risk despite 60% of capital. (a) confuses capital weight with risk weight; (d) would require risk parity, achieved by leveraging bonds and cutting equities.

D6. A strategy that sells deep out-of-the-money puts for steady premium shows low reported

volatility. Its true risk is: (a) Low, matching the volatility (b) High — a fat left tail can deliver a catastrophic loss (c) Zero, since premium is collected (d) Equal to the premium received **Answer: (b).** This is the textbook divergence of volatility and true risk: low σ and steady income until a crash delivers a huge left-tail loss. Volatility misses skew and fat tails; ES and stress testing are what catch this exposure.

D7. The chief limitation of historical-simulation VaR is that it: (a) Assumes normally distributed returns (b) Cannot handle correlations (c) Only reflects the tails that occurred in the sample window (d) Requires a full covariance matrix **Answer: (c).** Historical simulation makes *no* distributional assumption and captures real correlations — so (a), (b), (d) describe the parametric method, not this one. Its weakness is that it can only replay events present in the window and weights a 1-in-N event as exactly 1/N; a tail that hasn't happened yet is invisible to it.

D8. A stop-loss rule that cuts exposure at a –10% drawdown primarily protects against , but can . (a)

Turning a manageable loss into a terminal one; whipsaw you out before a rebound (b) Basis risk; increase leverage (c) Inflation; reduce taxes (d) Counterparty risk; raise correlation **Answer: (a).** Drawdown/de-grossing triggers mechanize the discipline of not letting a survivable loss become terminal, respecting the recovery asymmetry (–50% needs +100%). The trade-off is whipsaw — cutting right before a rebound and locking in the loss — so triggers must be calibrated to the strategy's natural volatility.

Self-Verification Notes

- **B1:** $\$0.20/\sqrt{250} = 0.01265\$$; VaR $\$2.326 \times 0.01265 \times 100 = 2.94\$$; $\$ \times \sqrt{10} = 9.30\$$; ES $\$2.67 \times 0.01265 \times 100 = 3.38 > \$ \text{ VaR}$. ✓
- **B2:** $\$1.30 \times (50/0.12) = 542\$$; $\$(0.80-1.30) \times 416.7 = -208\$$; ratio $\$208/542 = 0.50/1.30\$$. ✓
- **B3:** $\sqrt{0.5125} = 0.716\$$; VaR $\$ = 1.67\$$; stress $\$ -7.5\$$; ratio $\$4.5 \times \$$. ✓
- **B4:** $\sqrt{129.28} = 11.37\$$; equity RC $\$10.64\$$ (93.6%), bond RC $\$0.73\$$; sum $\$ = 11.37 = \sigma_p\$$. ✓
- **B5:** $\$1/0.60 - 1 = 66.7\%$; $\$1/0.40 - 1 = 150\%$. ✓ **D3:** $\$ -0.60 \times 500 = -300\$$ (short 300). ✓